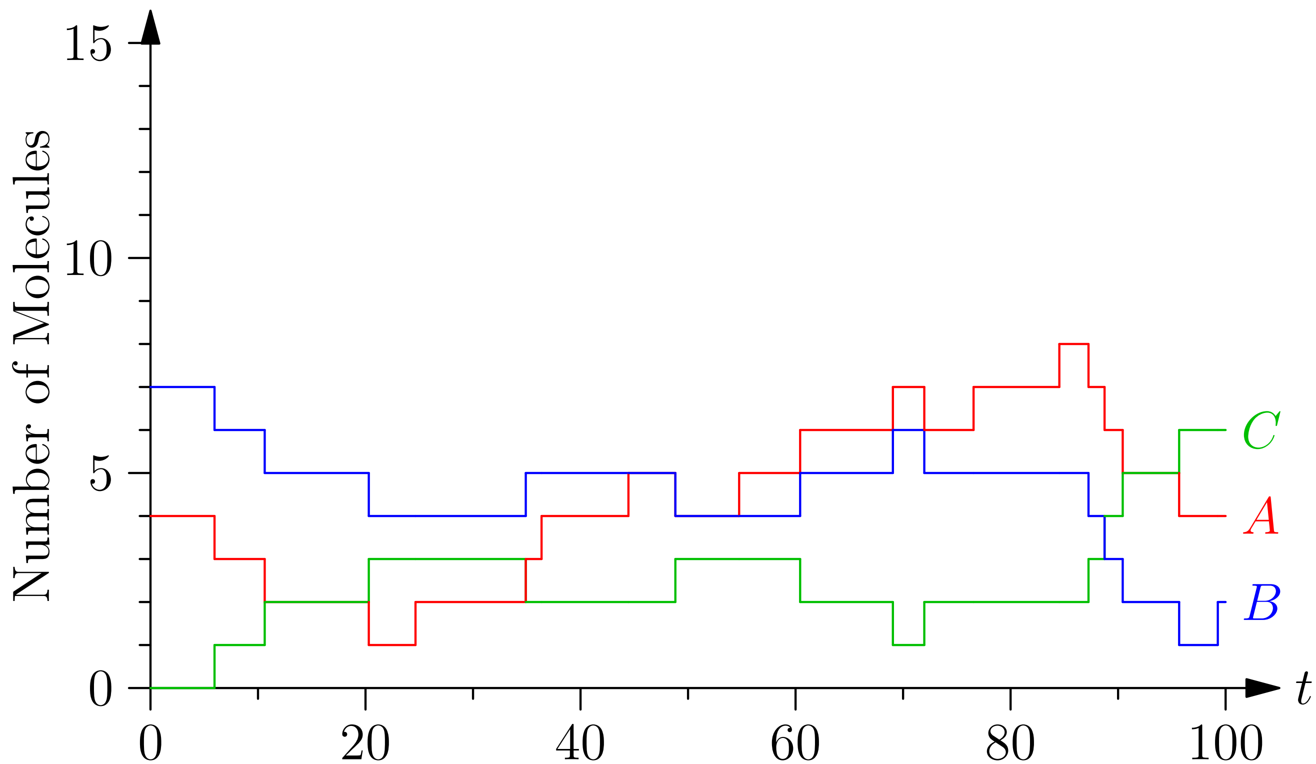
Computational Biology

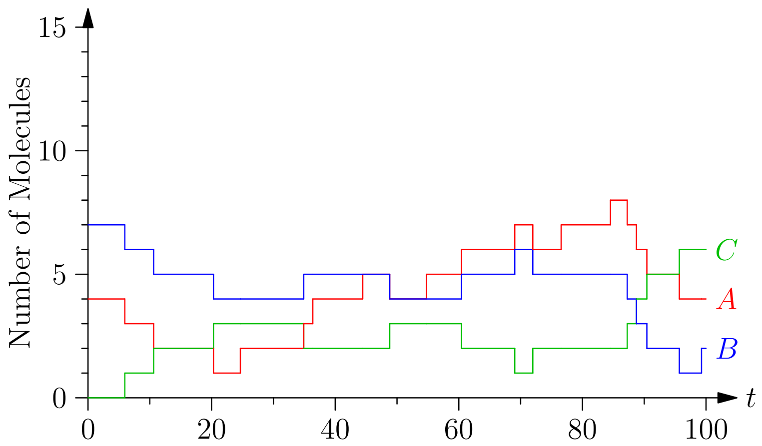
Lecture 5: *Simulating Chemical Equations*



Chemical Equations, Stochasticity, Gillespie Algorithm

Outline

1. **Stochastic Chemical Equations**
2. Gillespie Algorithm



Stochasticity

- A chemical equation

$$A + B \xrightarrow{k_1} C$$

  happens when molecule A by chance hits molecule B with the right momentum

- If the number of molecules are large then we can model this quite accurately by considering an ODE of what happens on average

- But for proteins and other molecules in the cell we often have a rather small number of a particular species of molecule in the cell

- In this case the ODE can be misleading

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Stochasticity in Nature

- Randomness can be problematic in making a system function
- Where this is critical then the cell relies on large molecule number
- However, there are times when randomness can serve a purpose
- In a large population of cells there can be an advantage to cells being in slightly different states
- This can give more robustness when responding to external stimuli

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Sources of Noise

- There are some variations that are common to most reactions coming from e.g.
  - ⋆ number of RNA-polymerase
  - ⋆ number of ribosomes
  - ⋆ size of cell
- This is known as **extrinsic noise**
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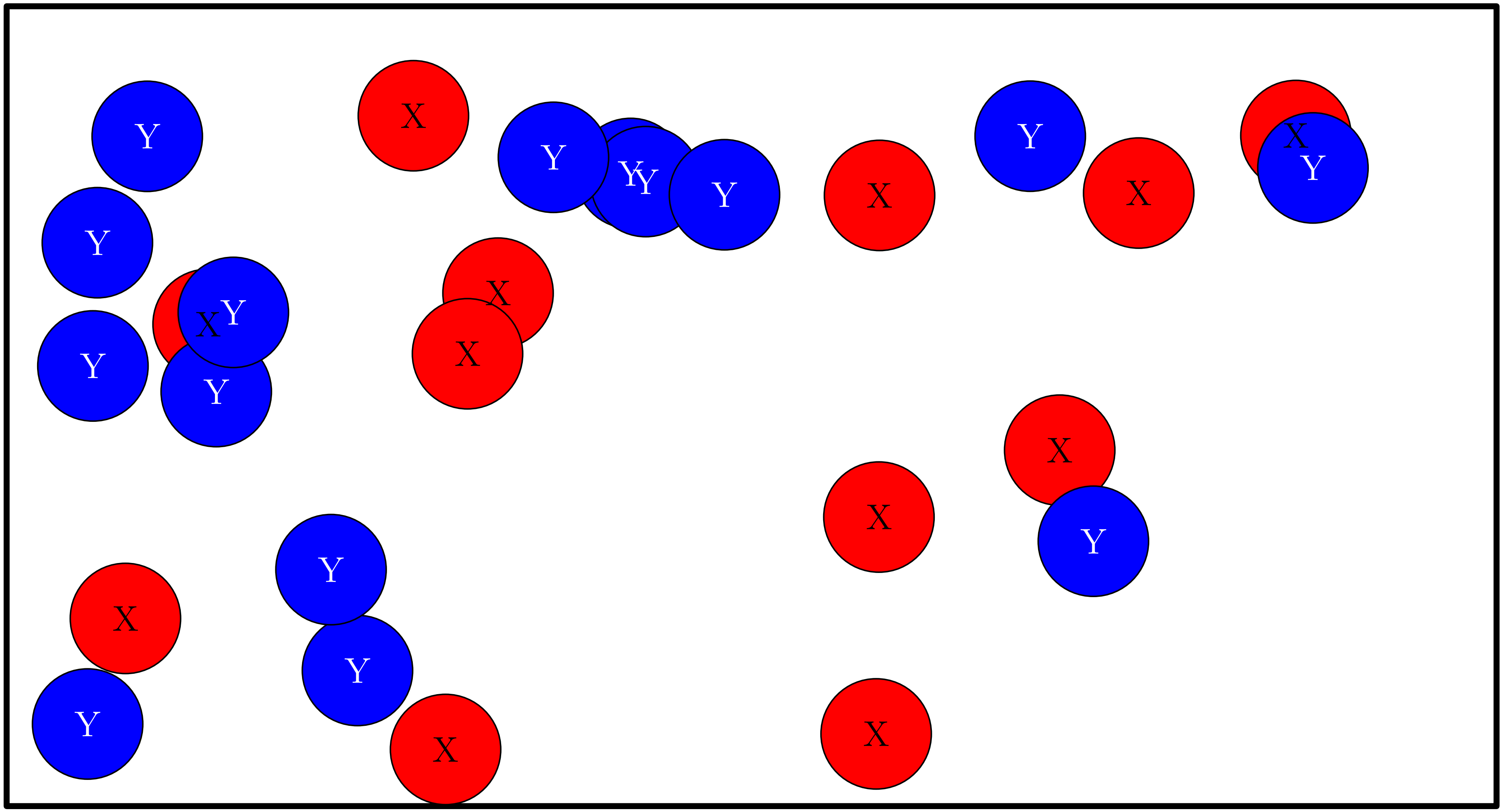

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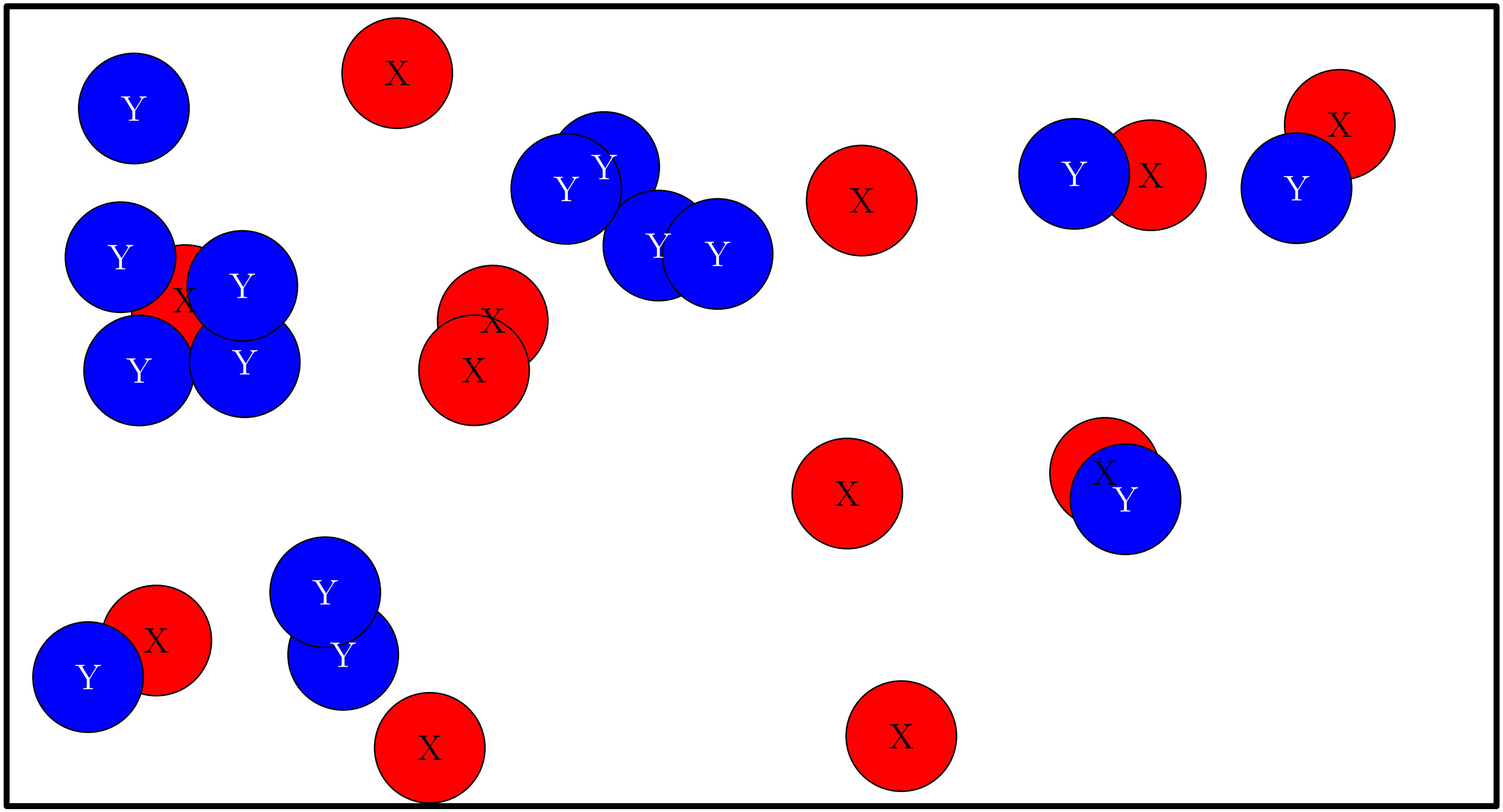

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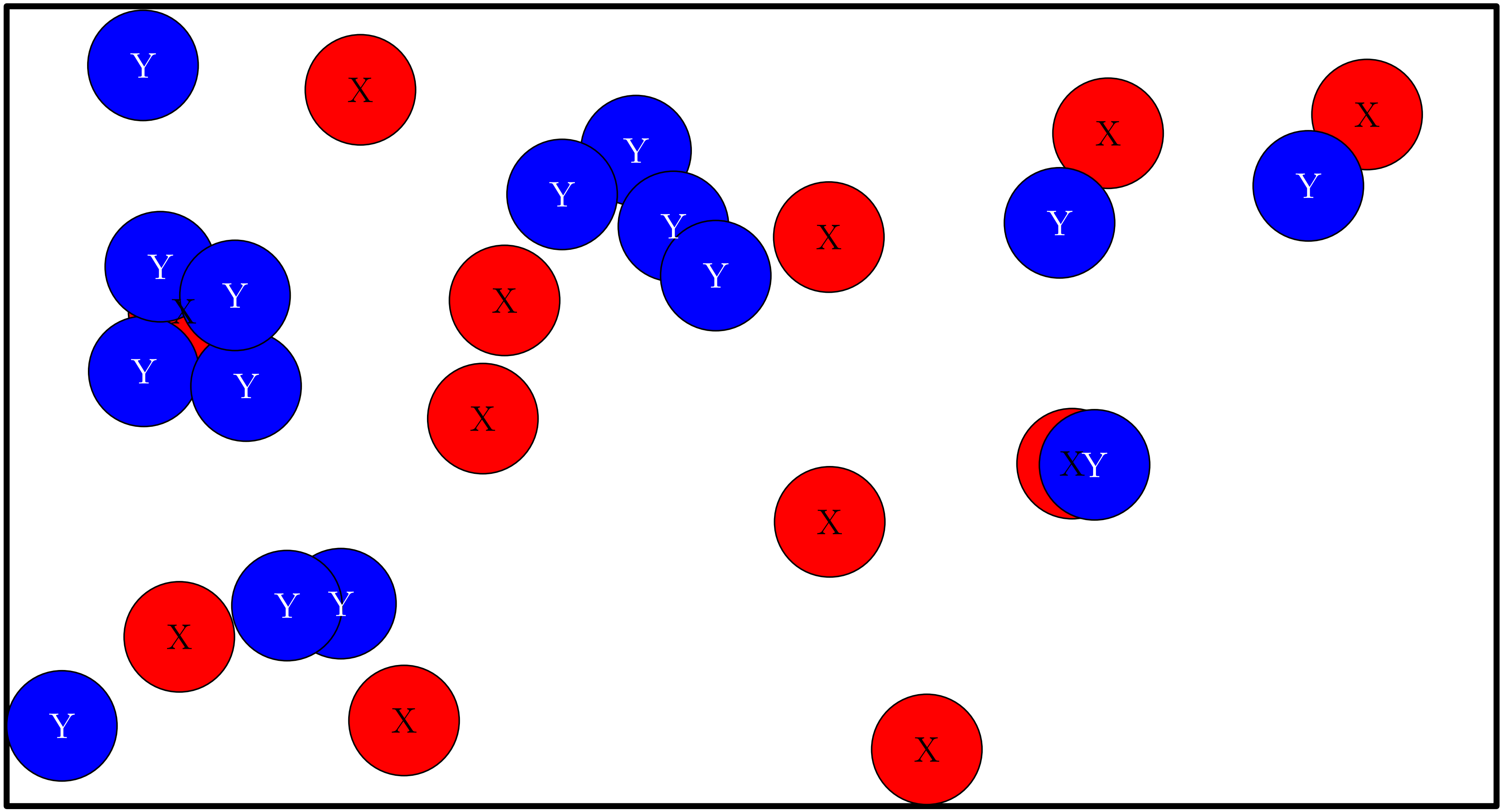

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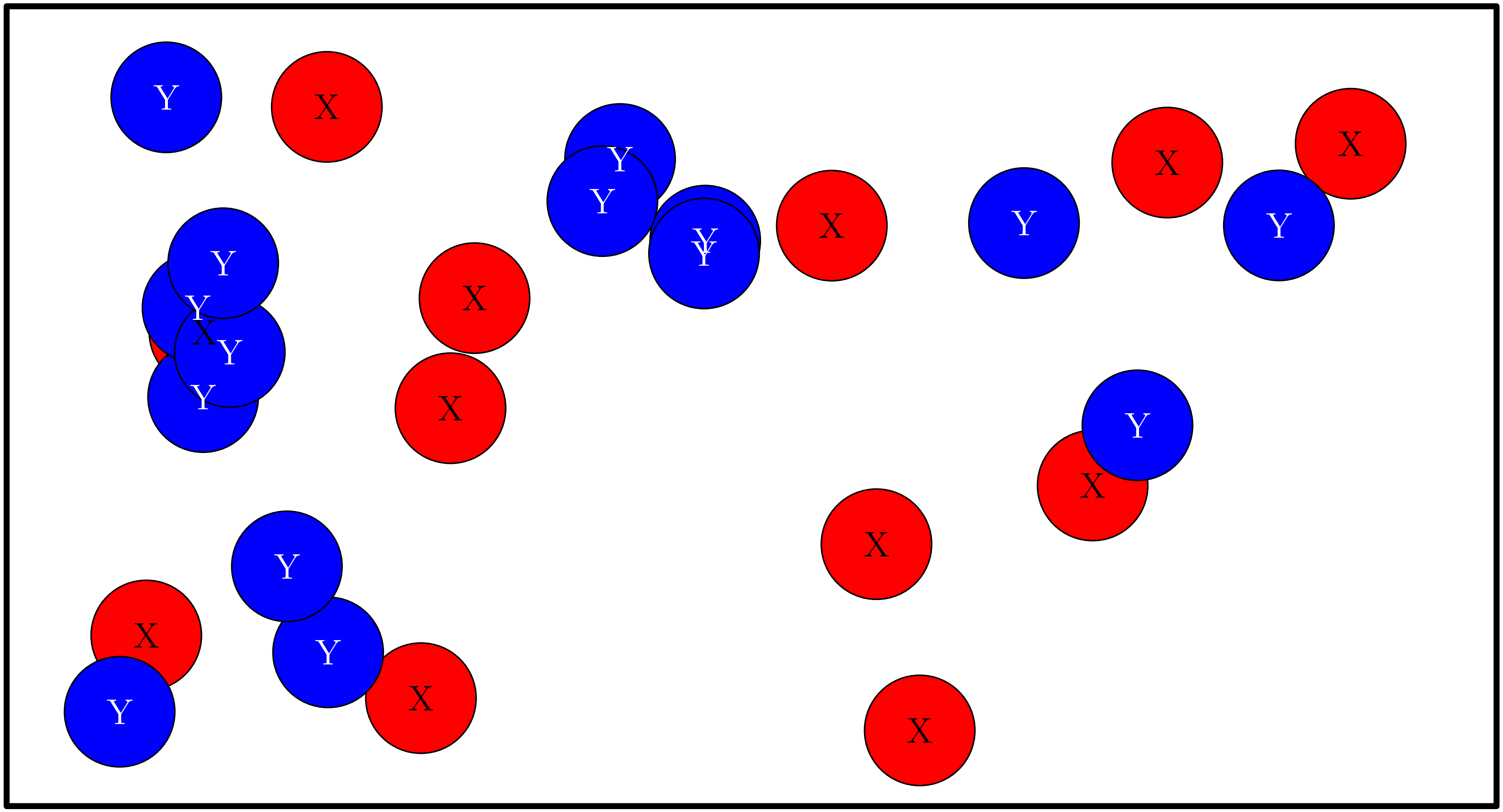

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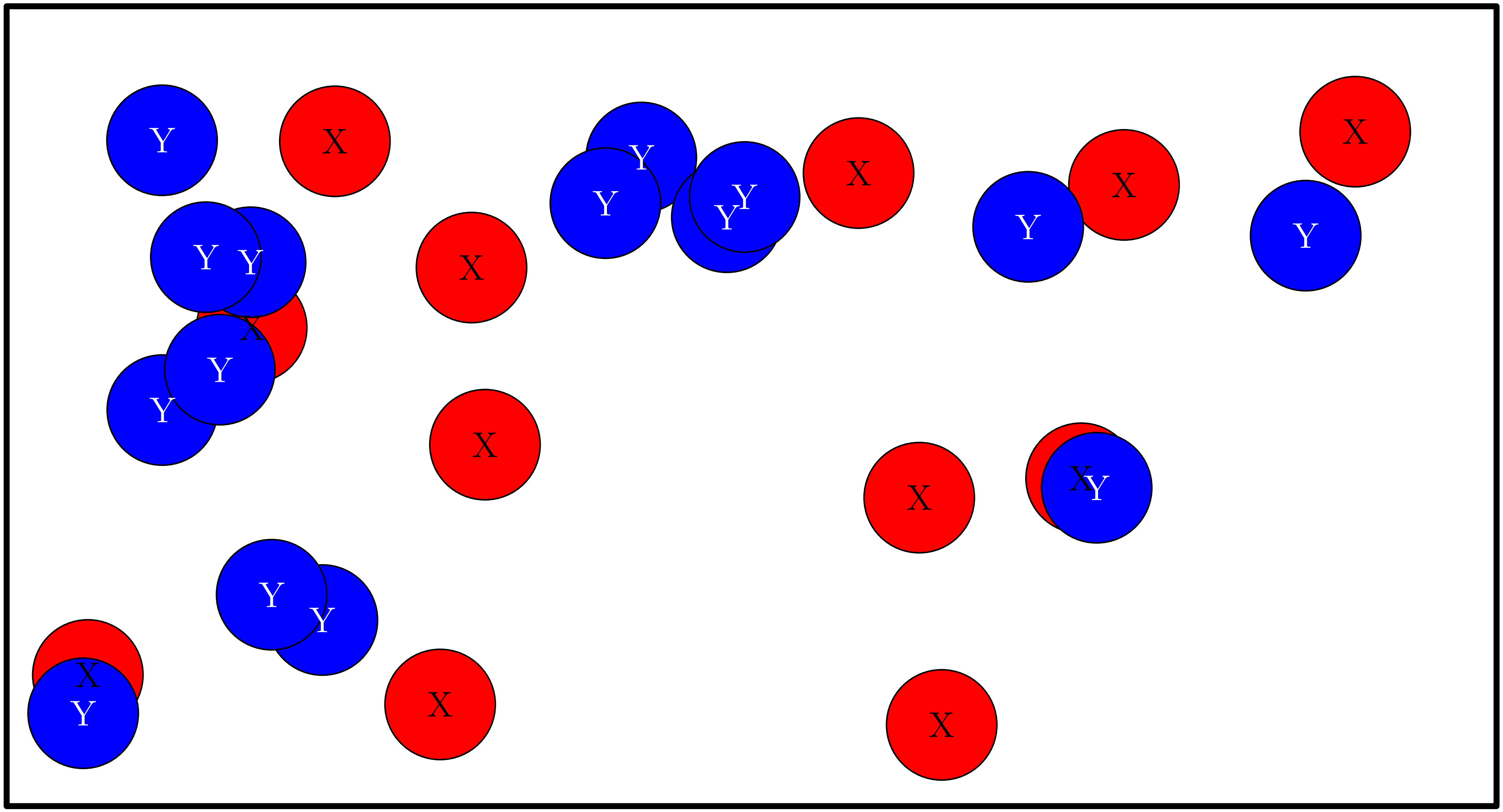

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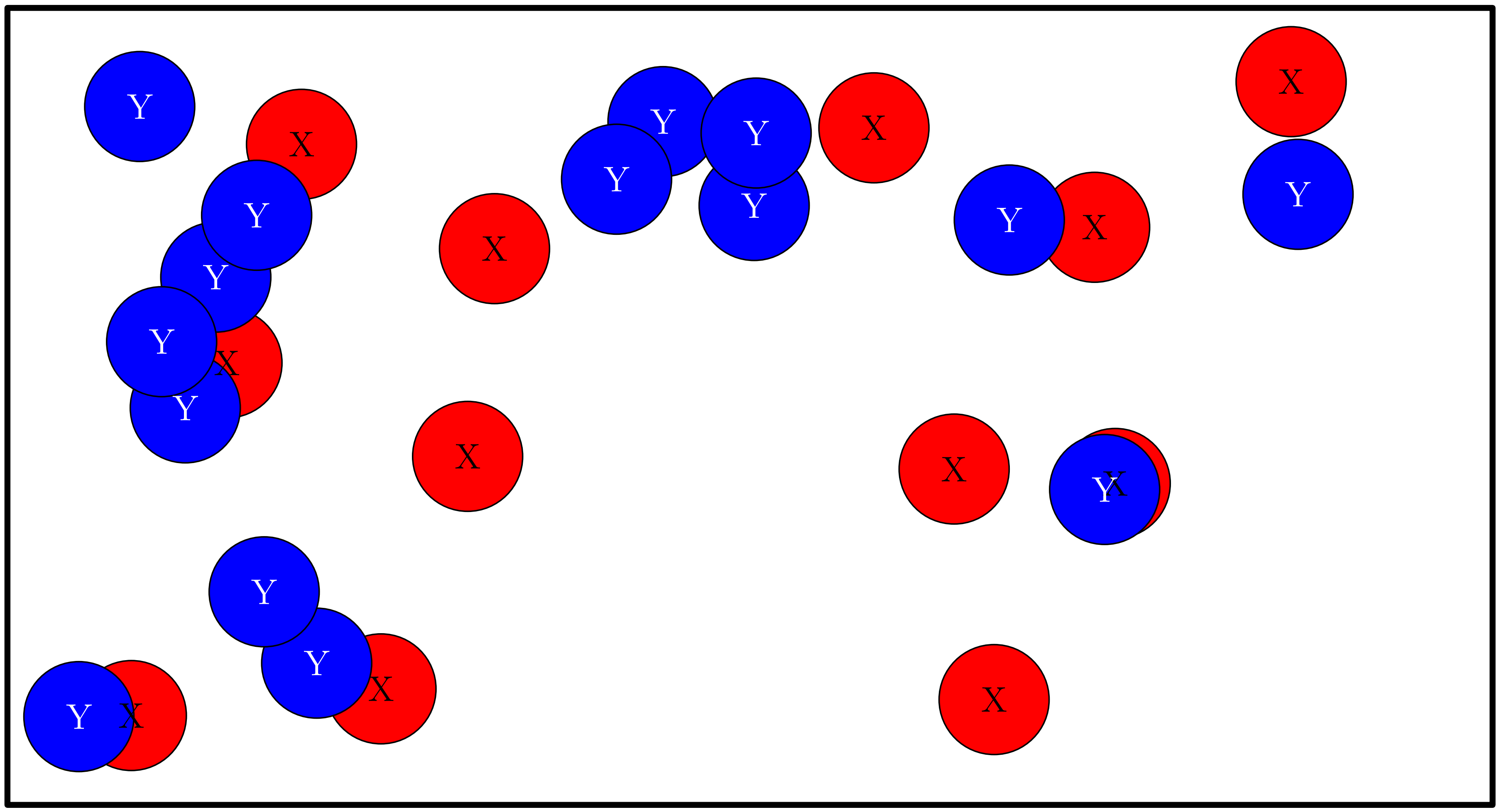

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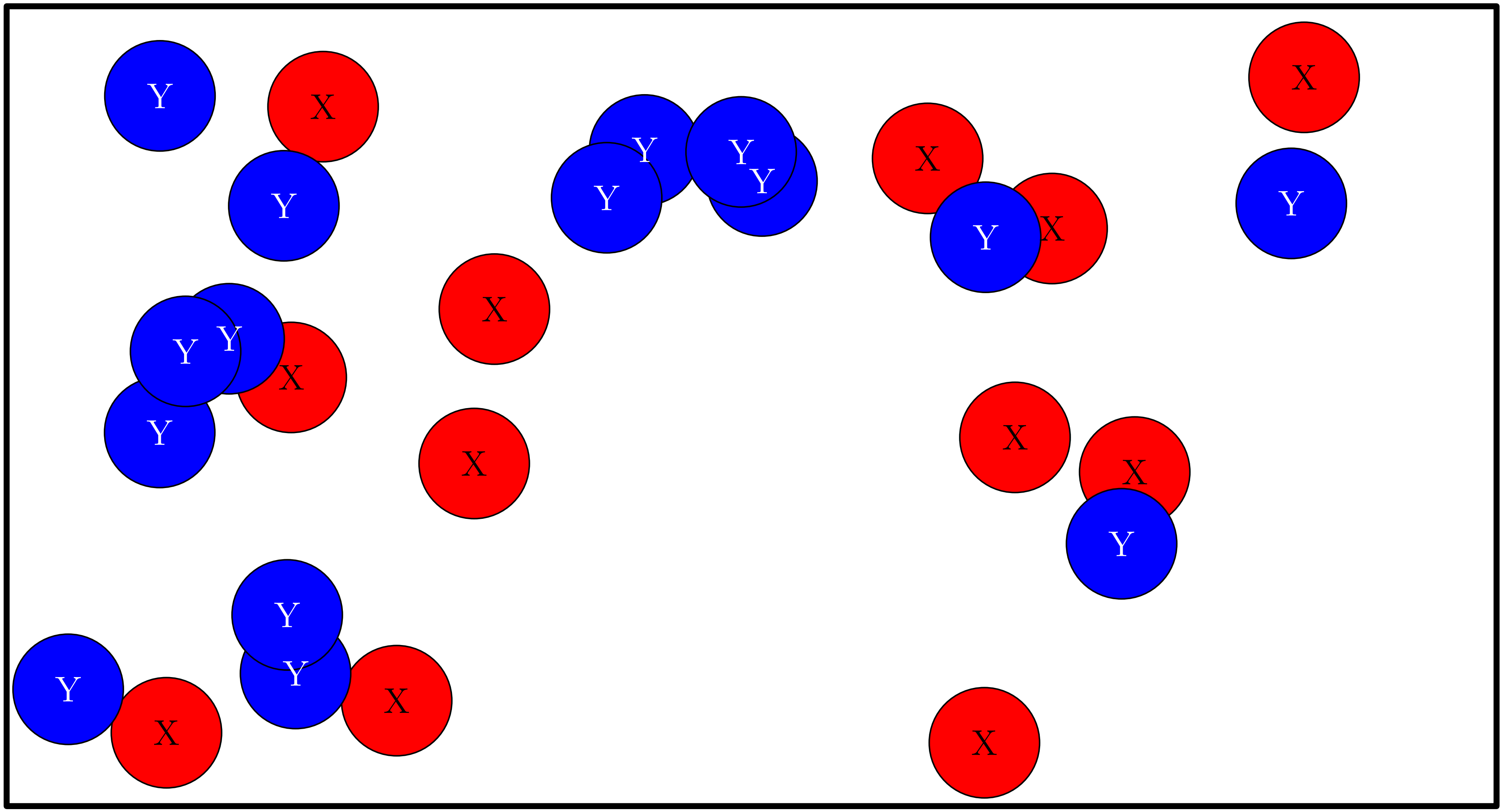

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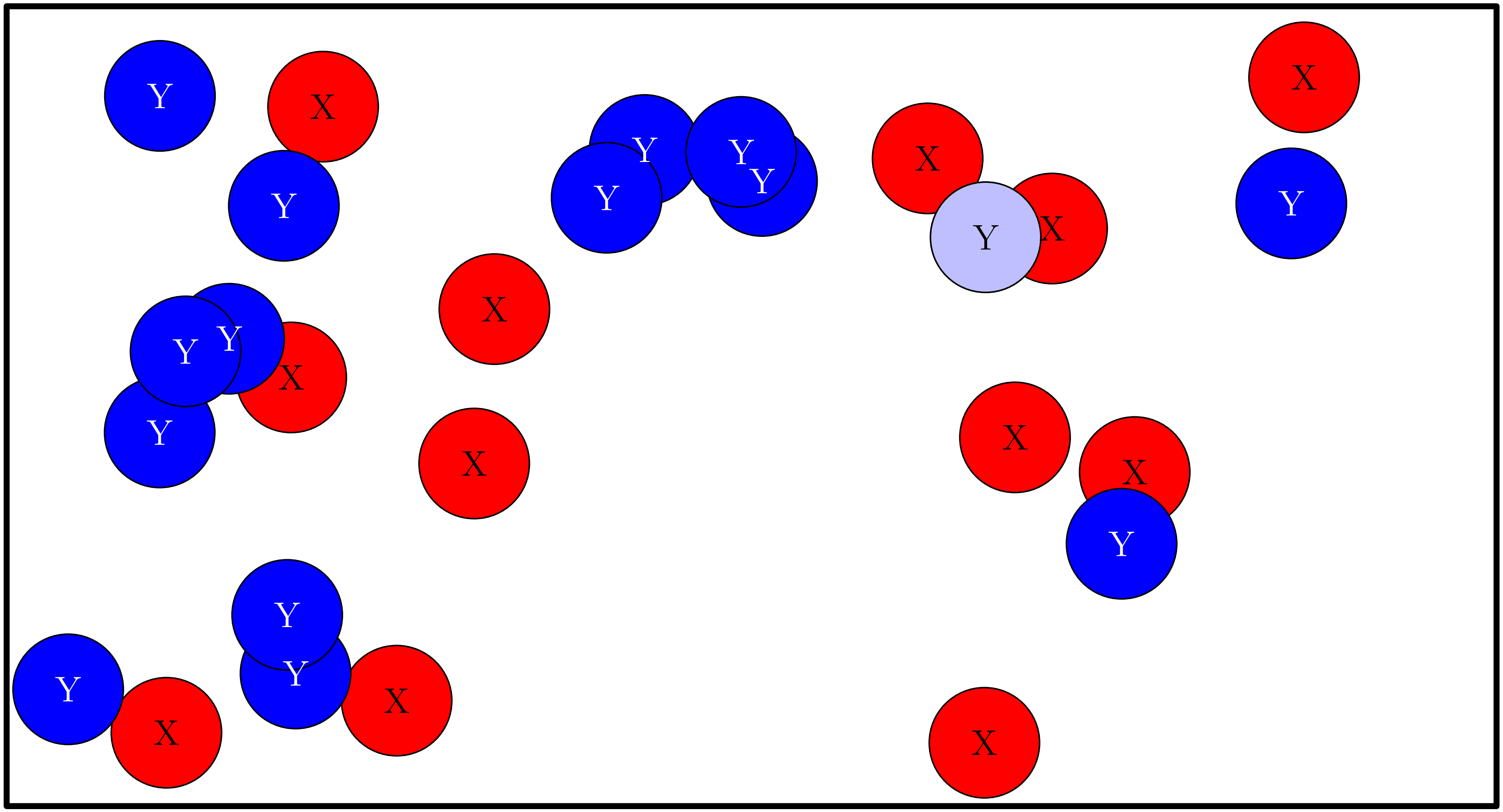

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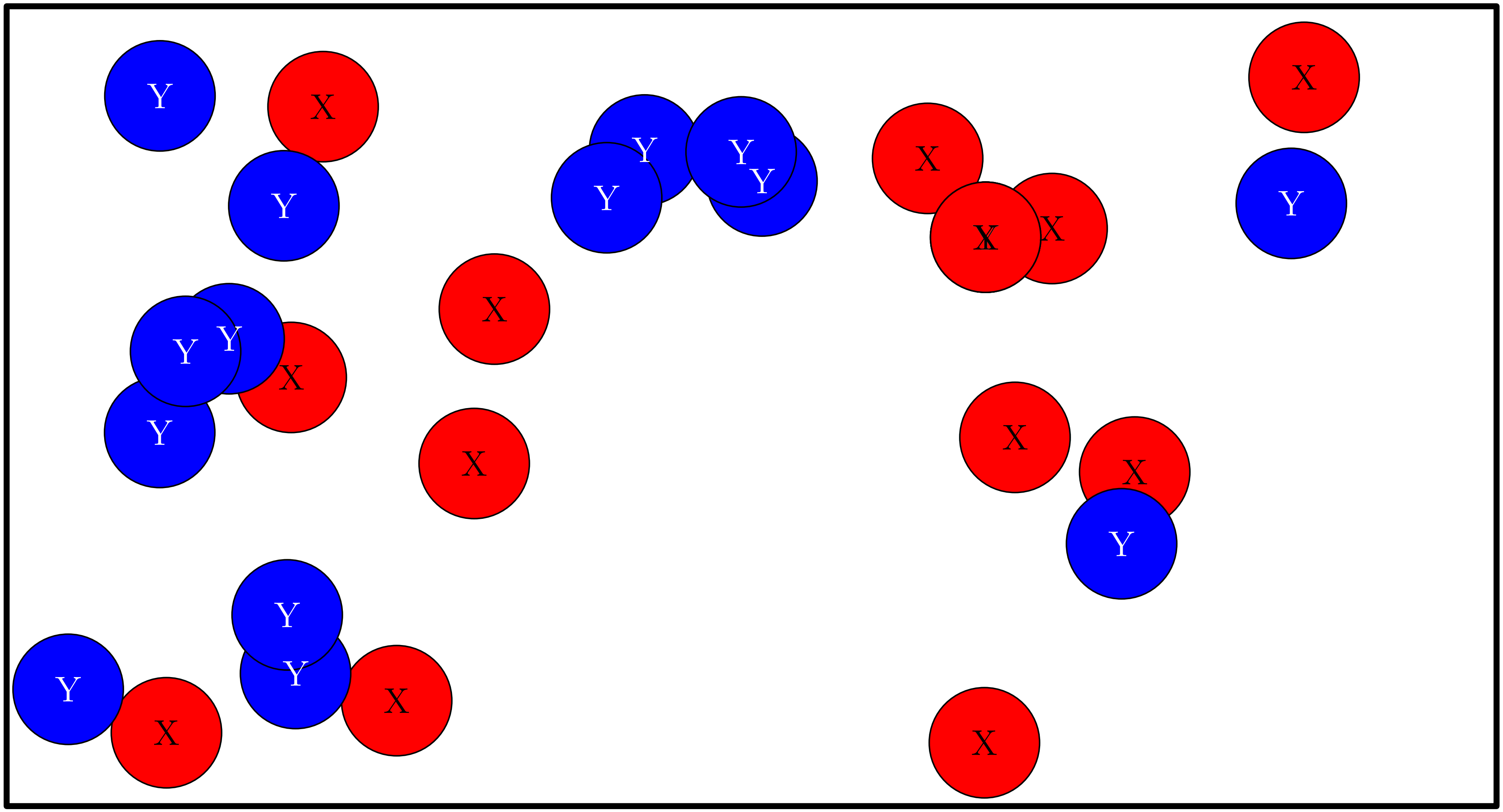

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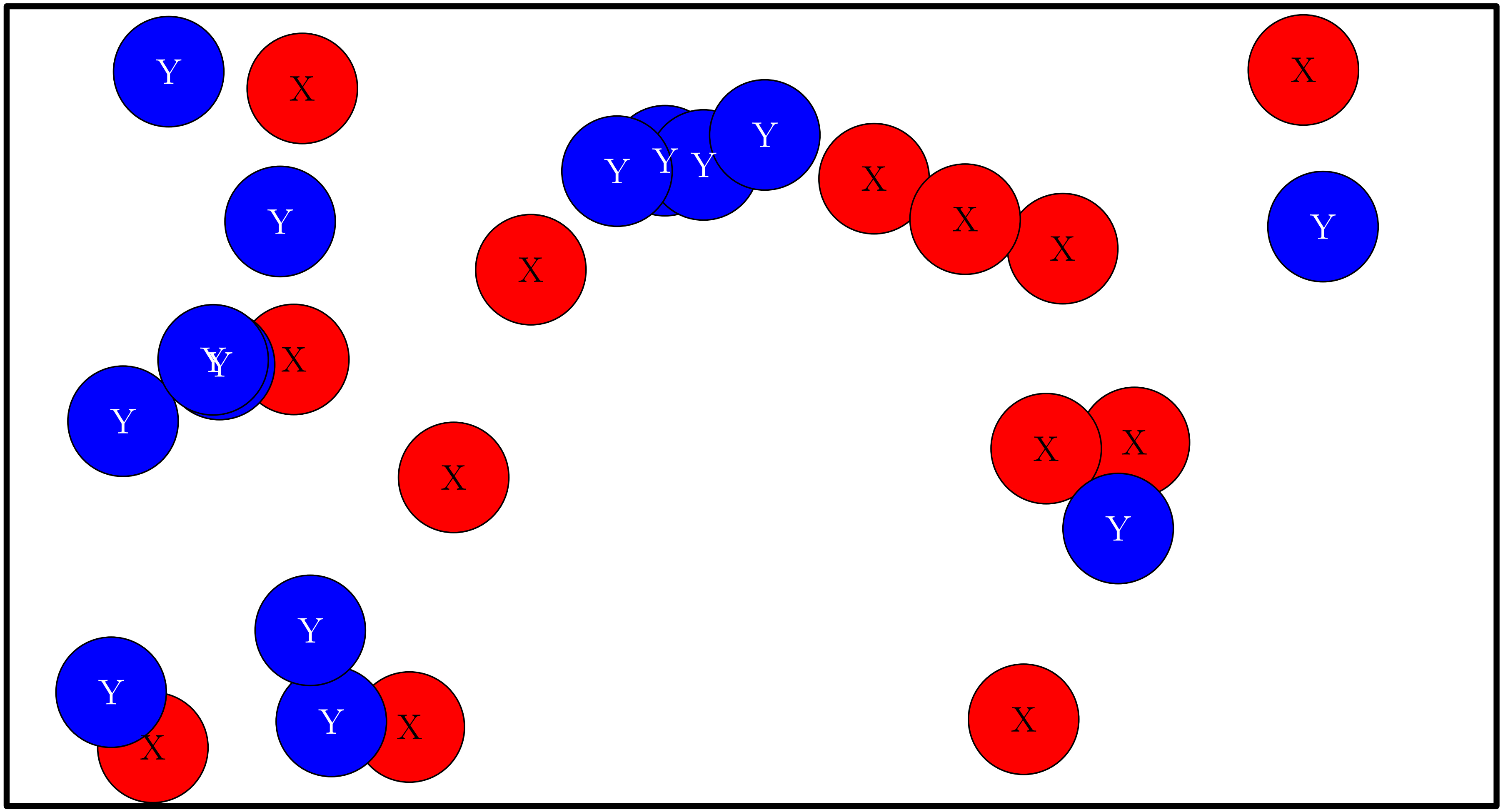

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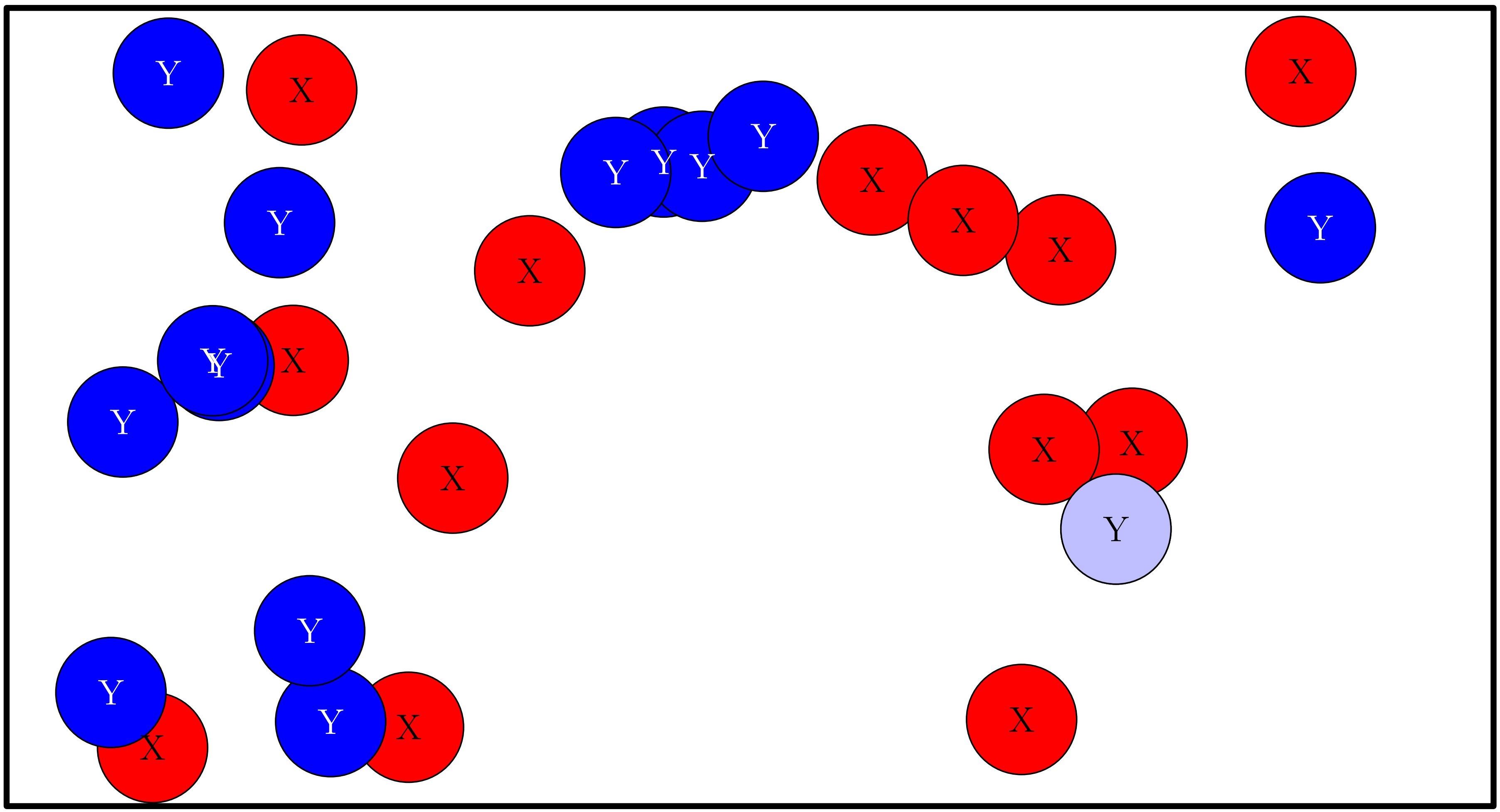

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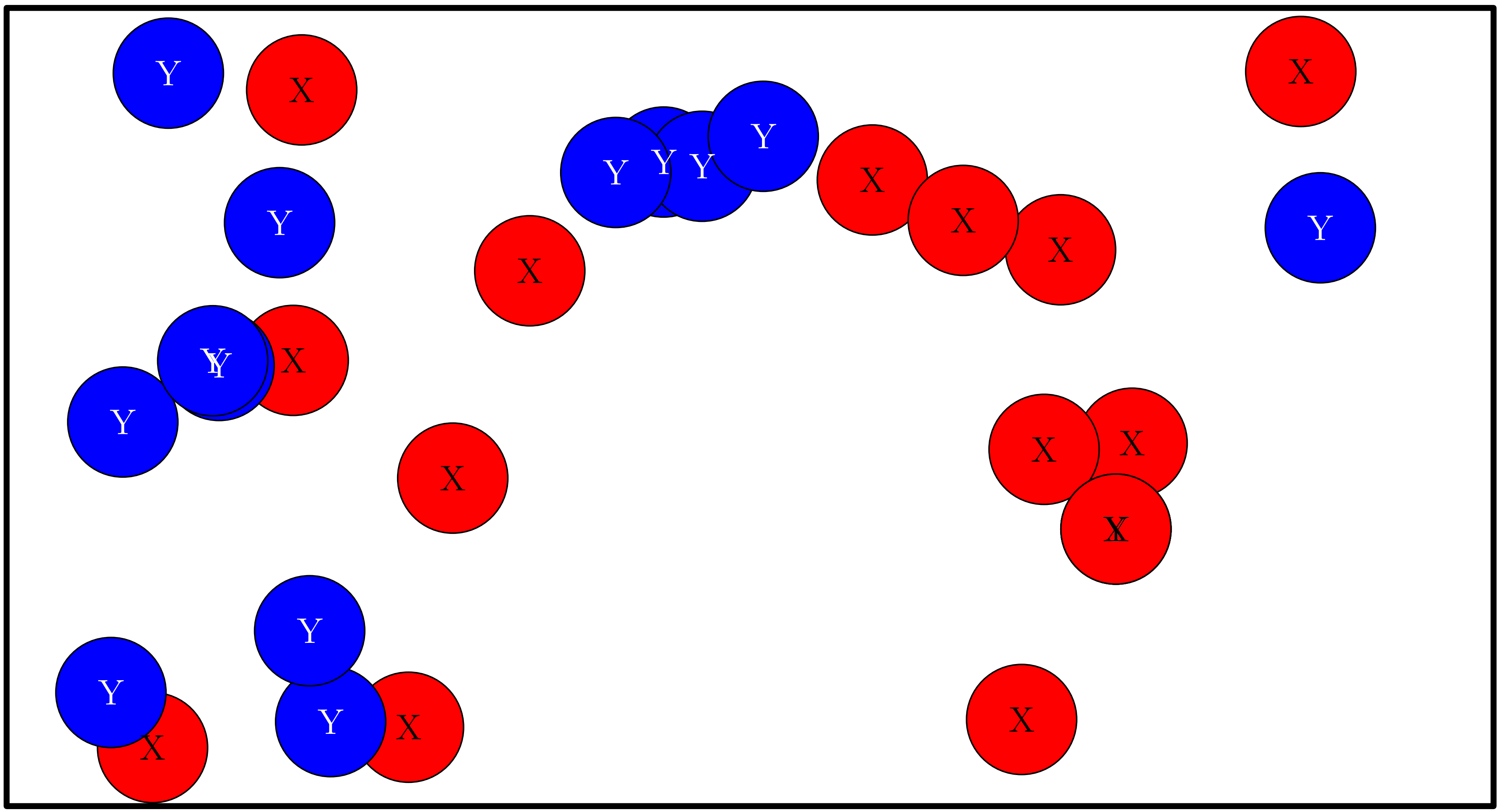

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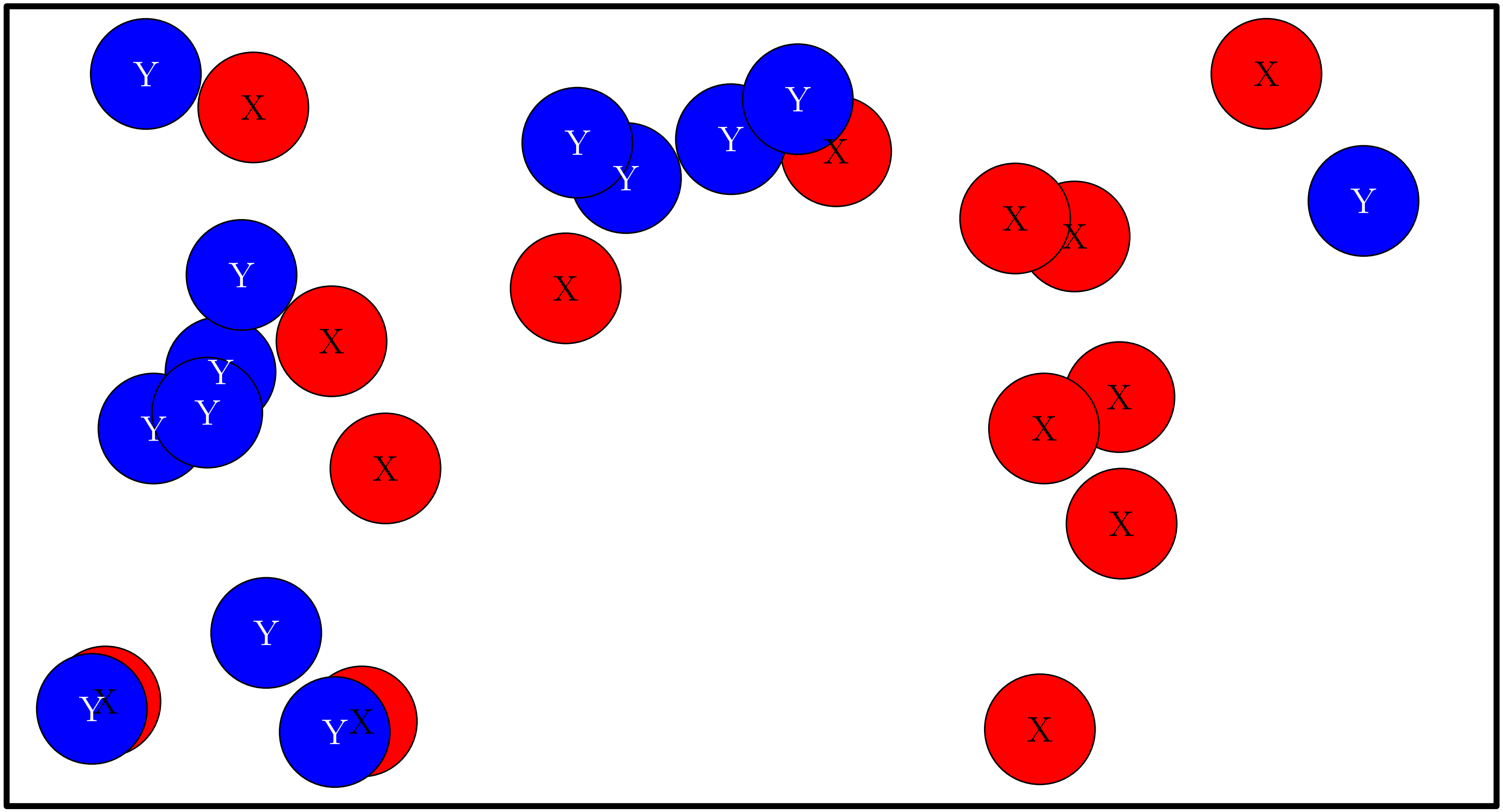

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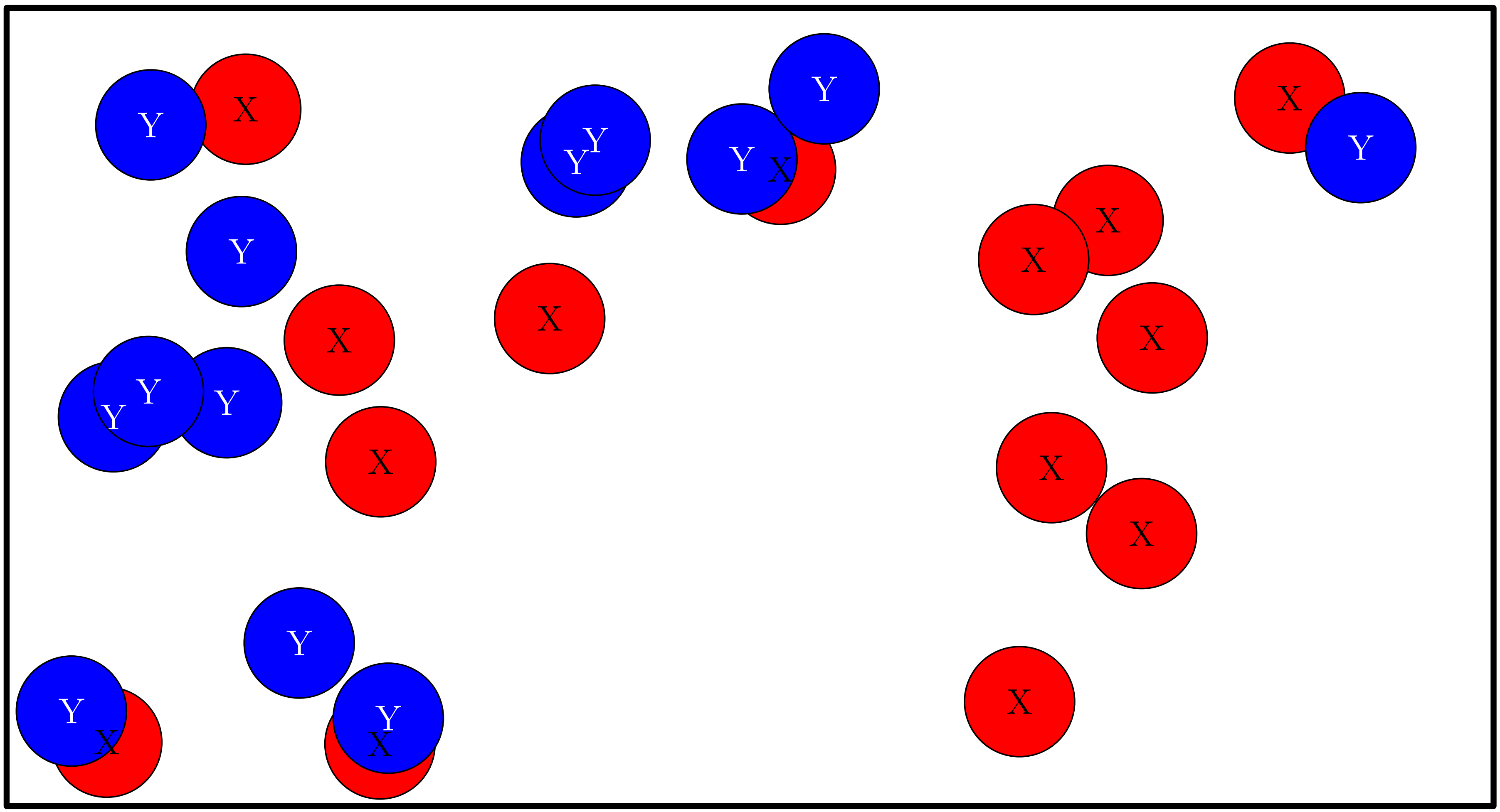

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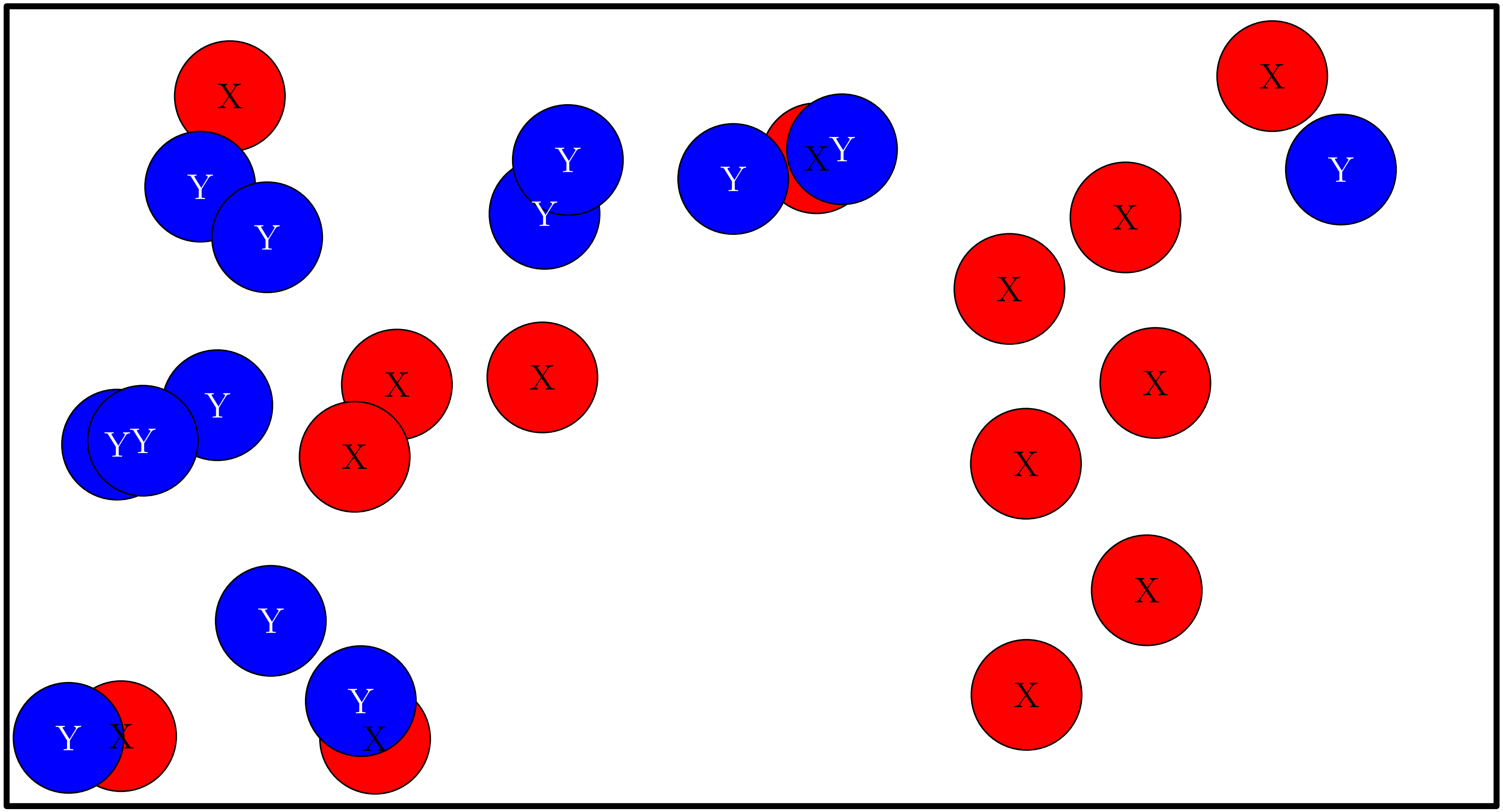

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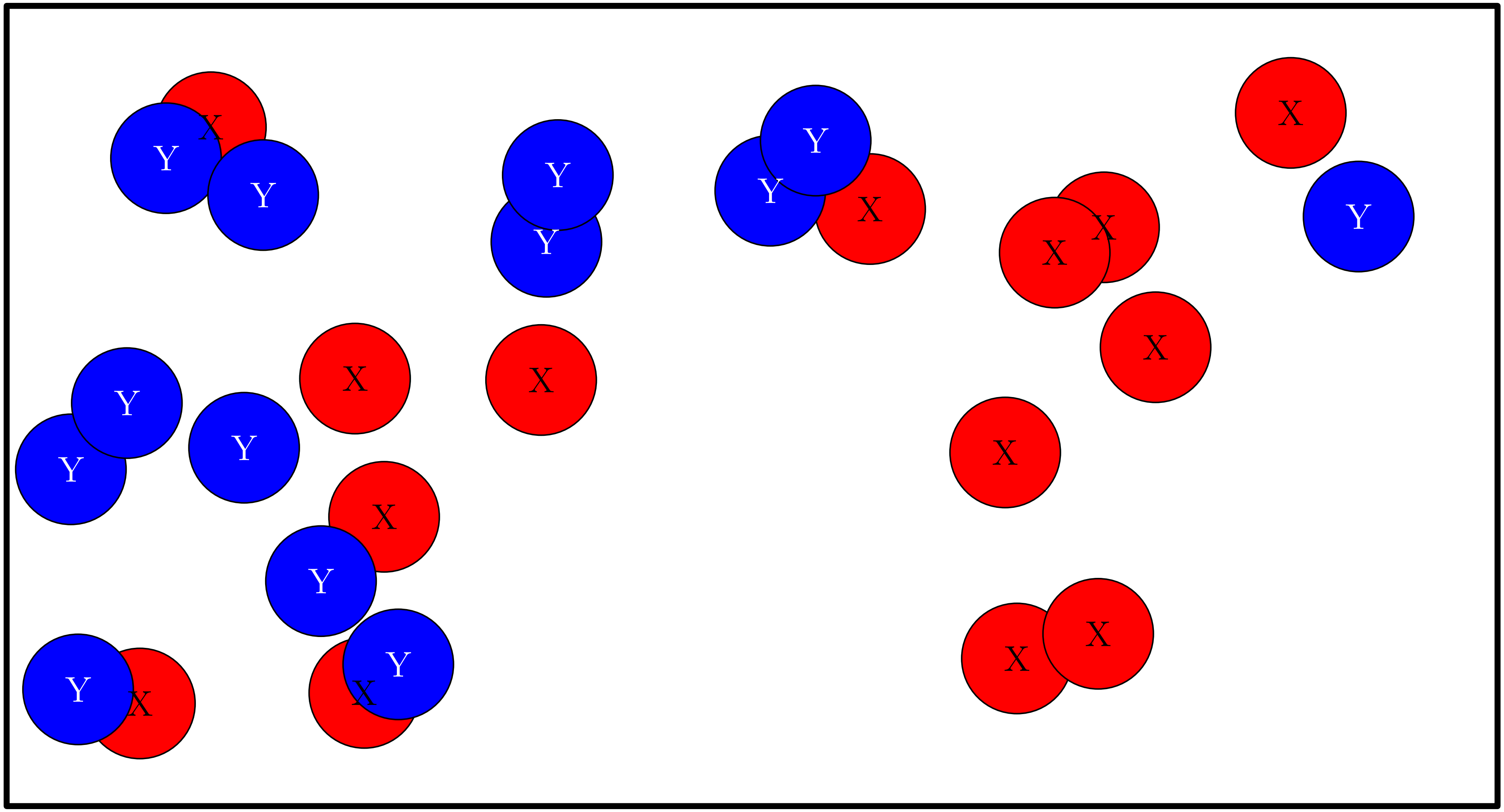

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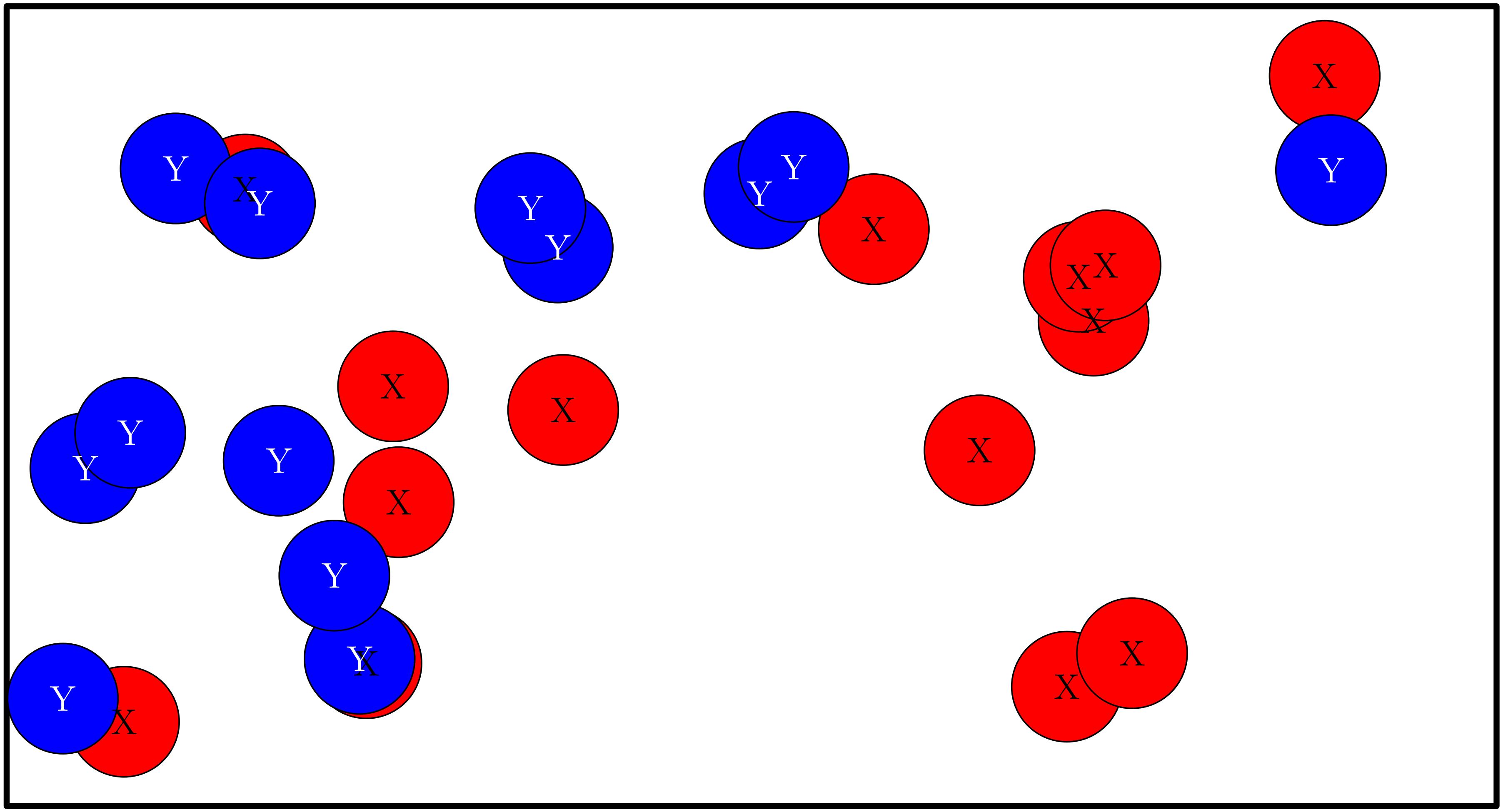

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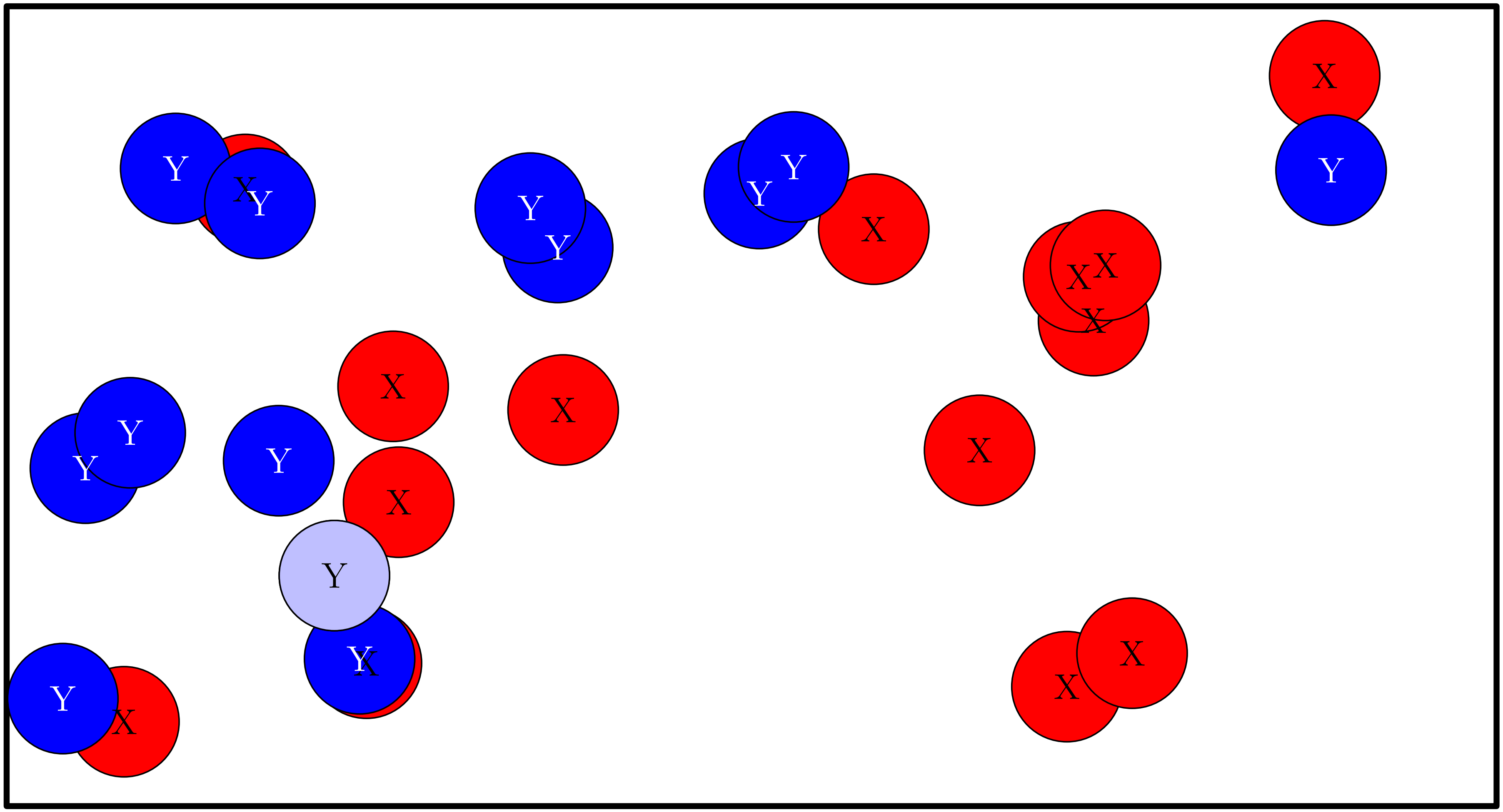

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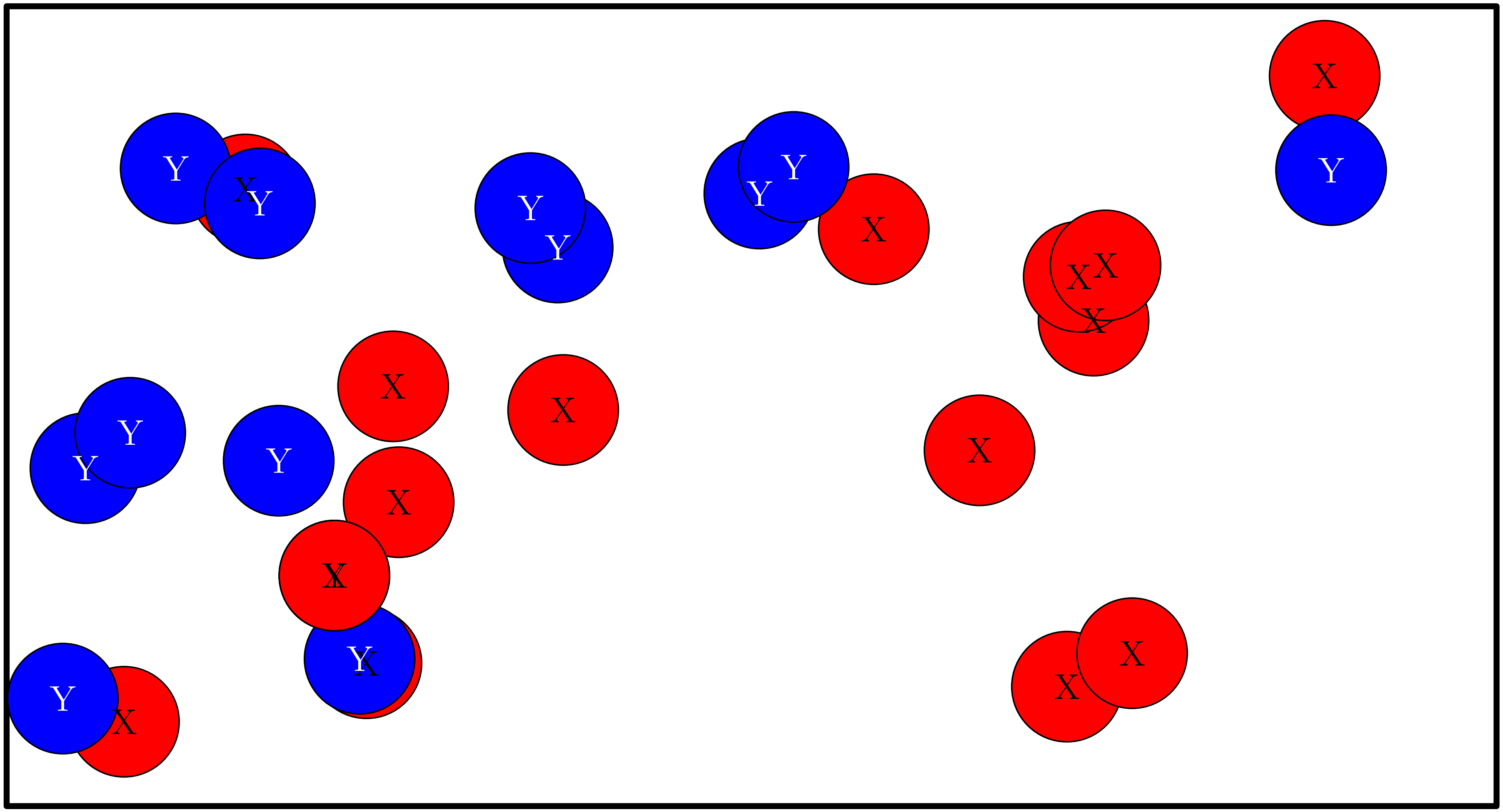

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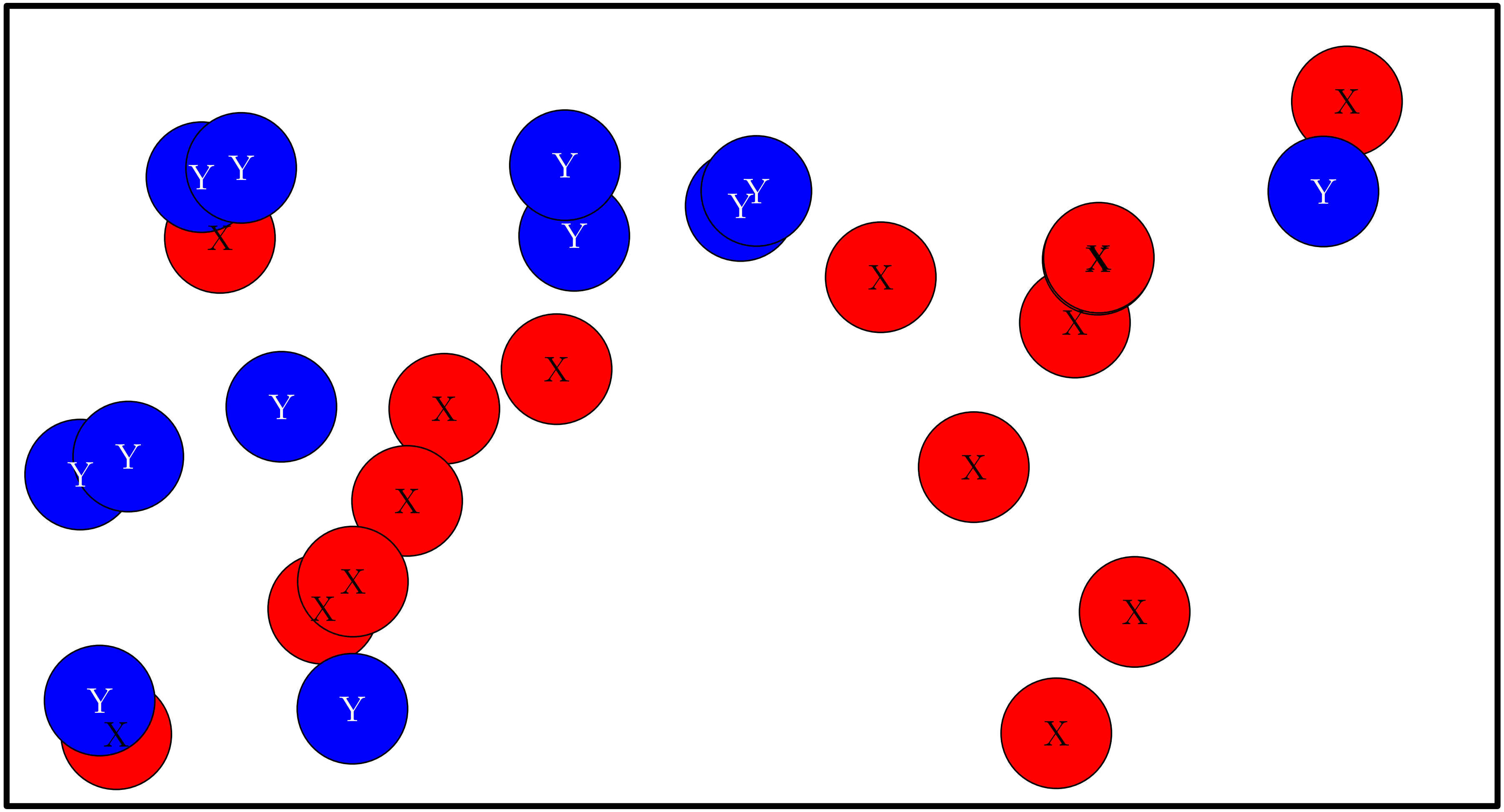

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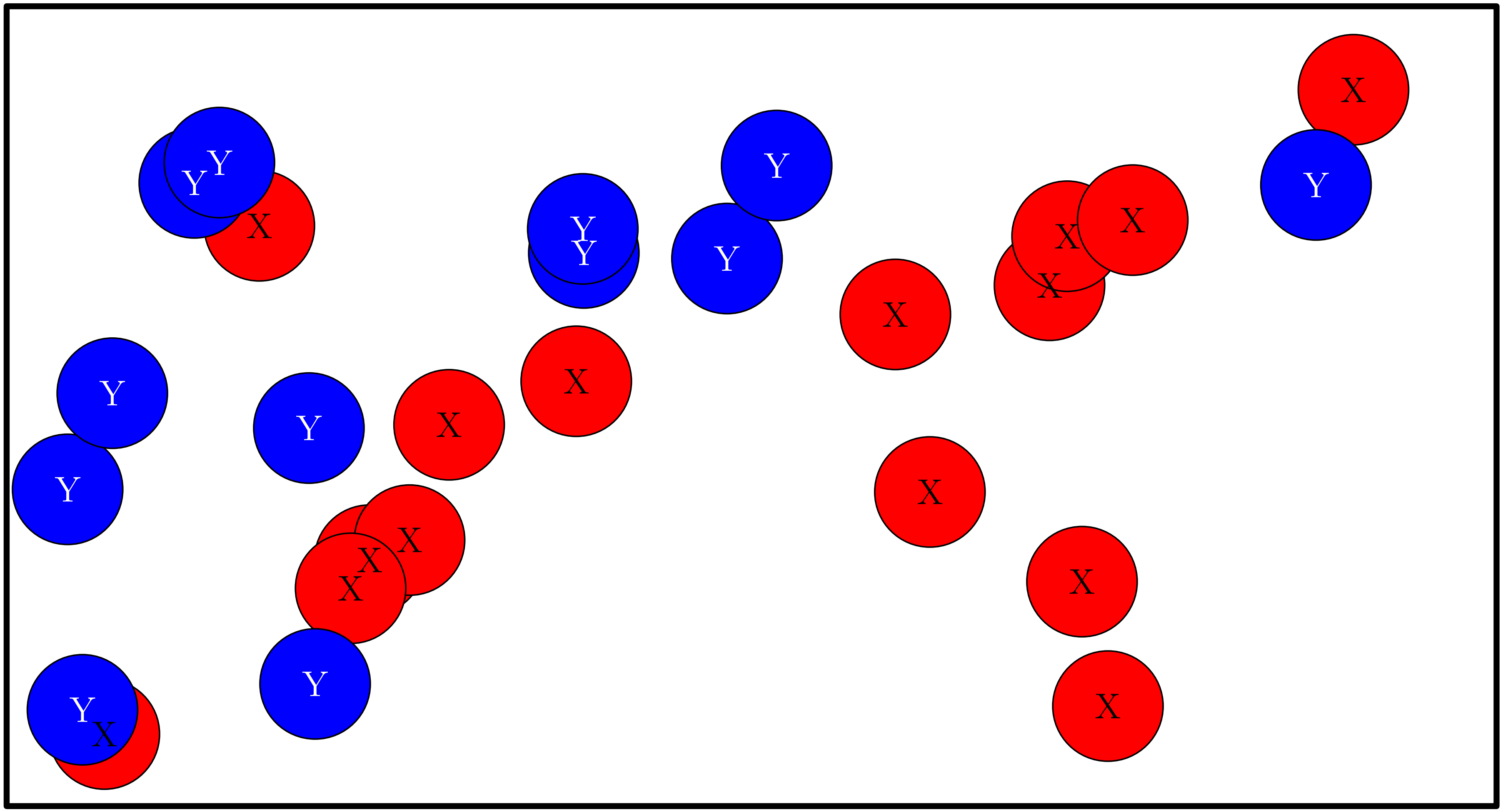

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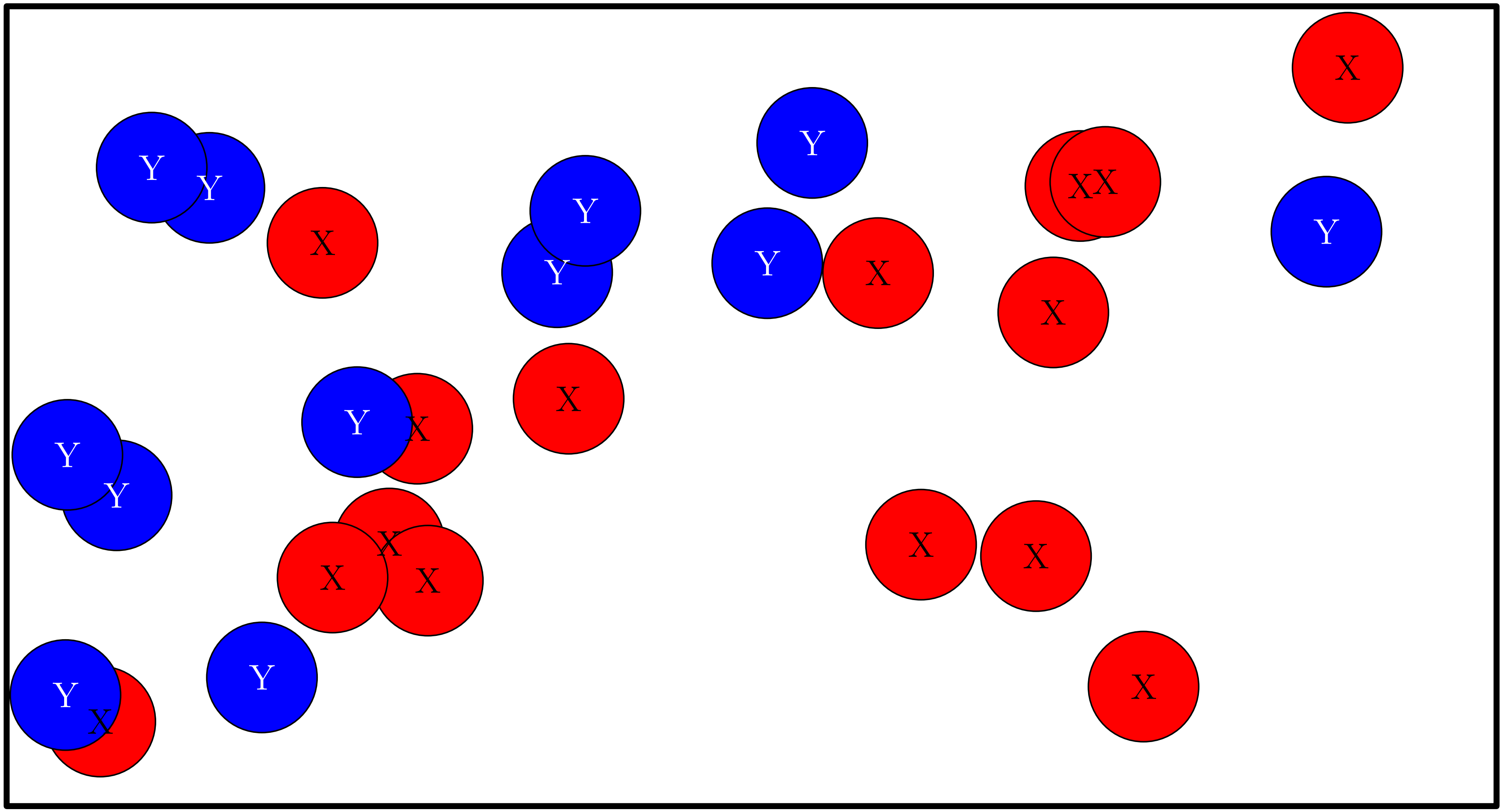

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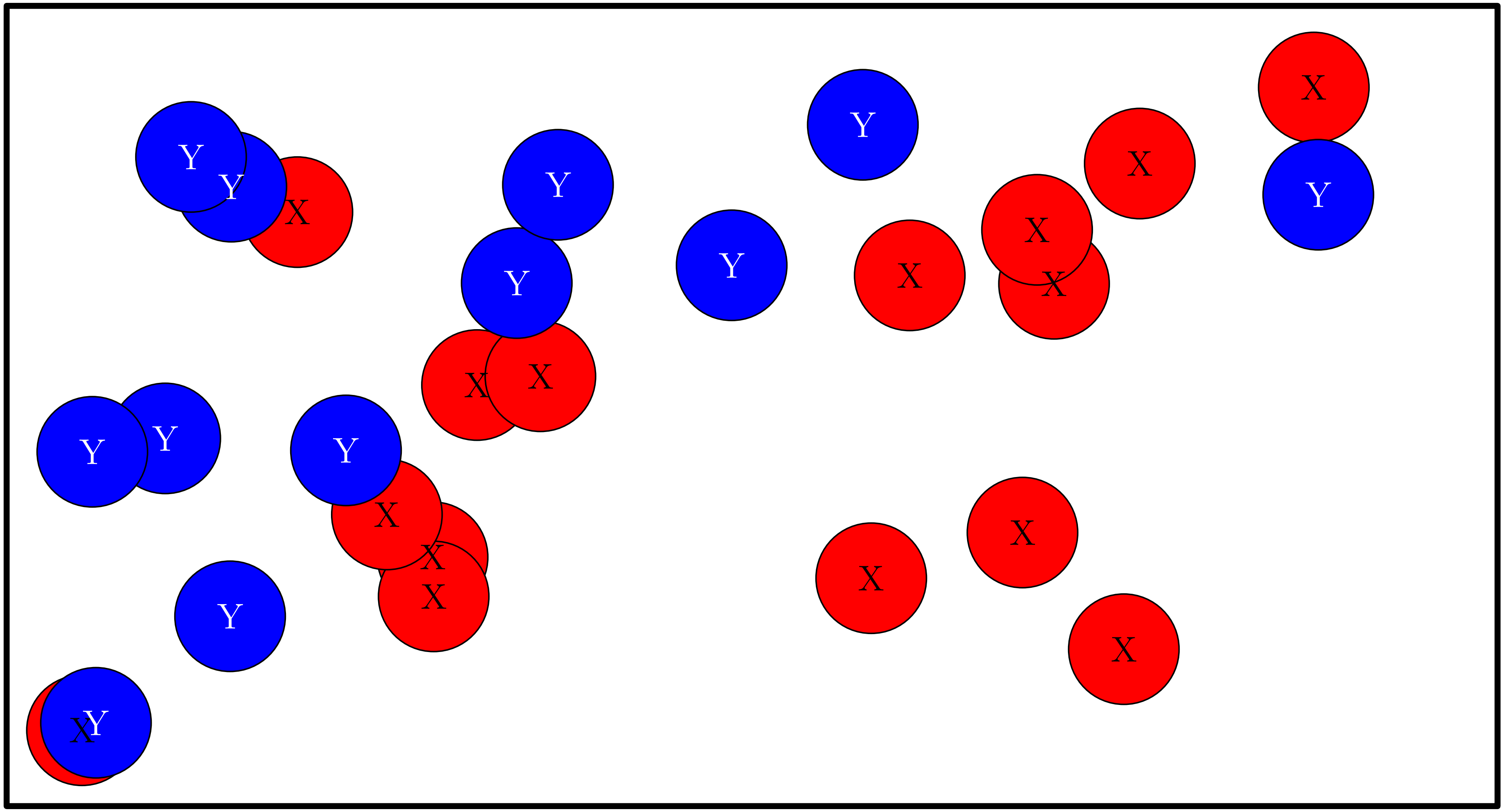

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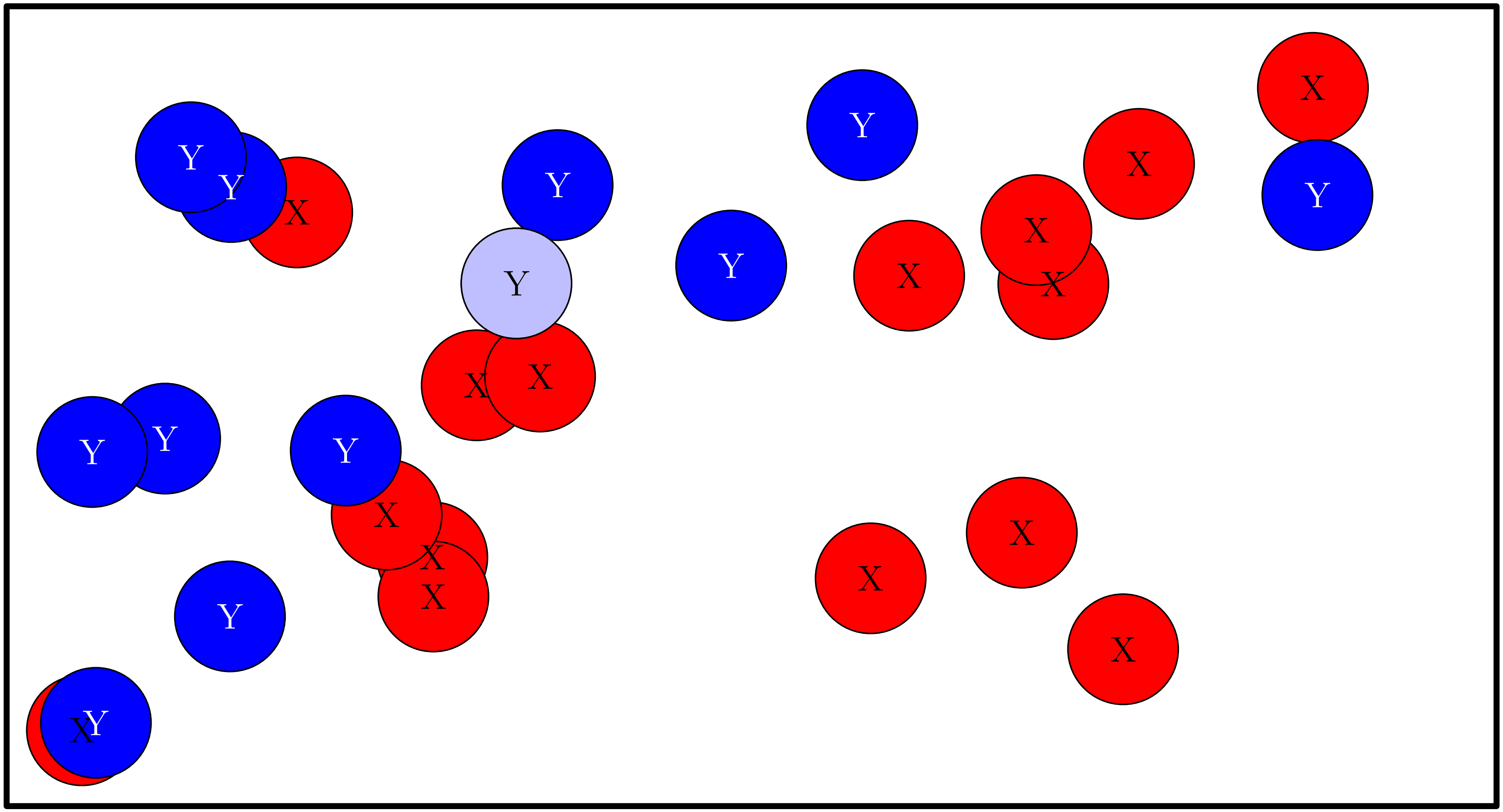

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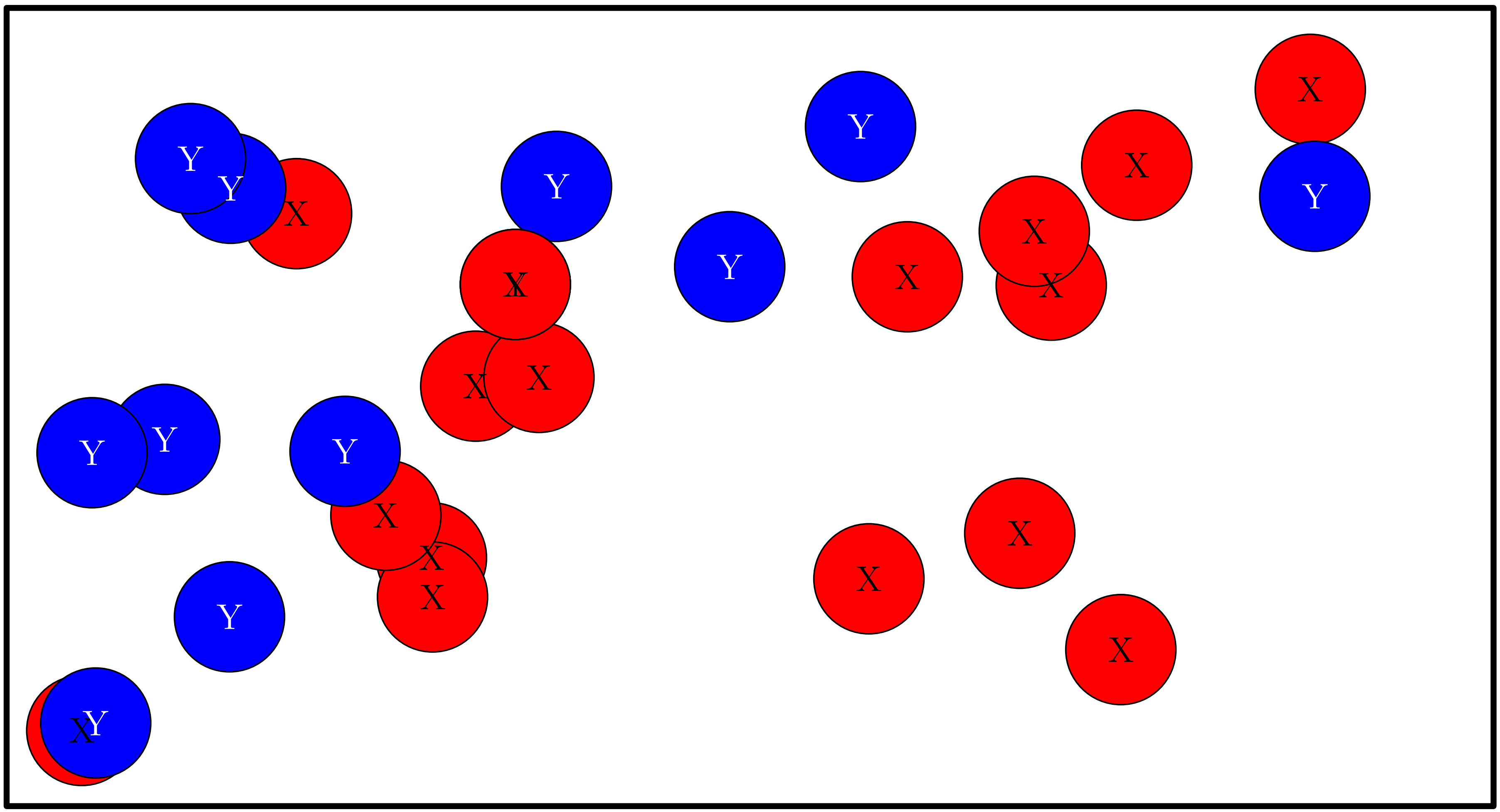

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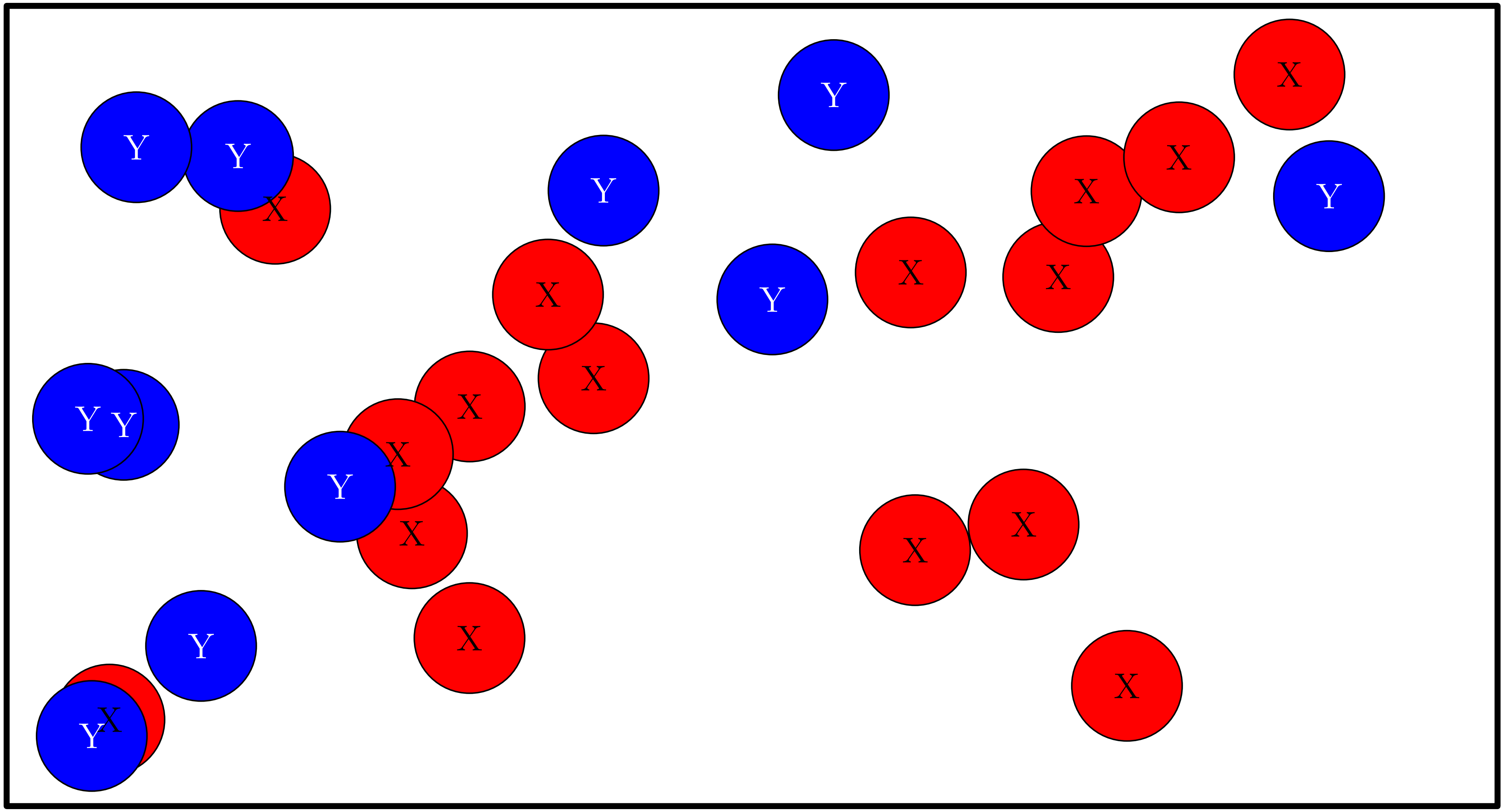

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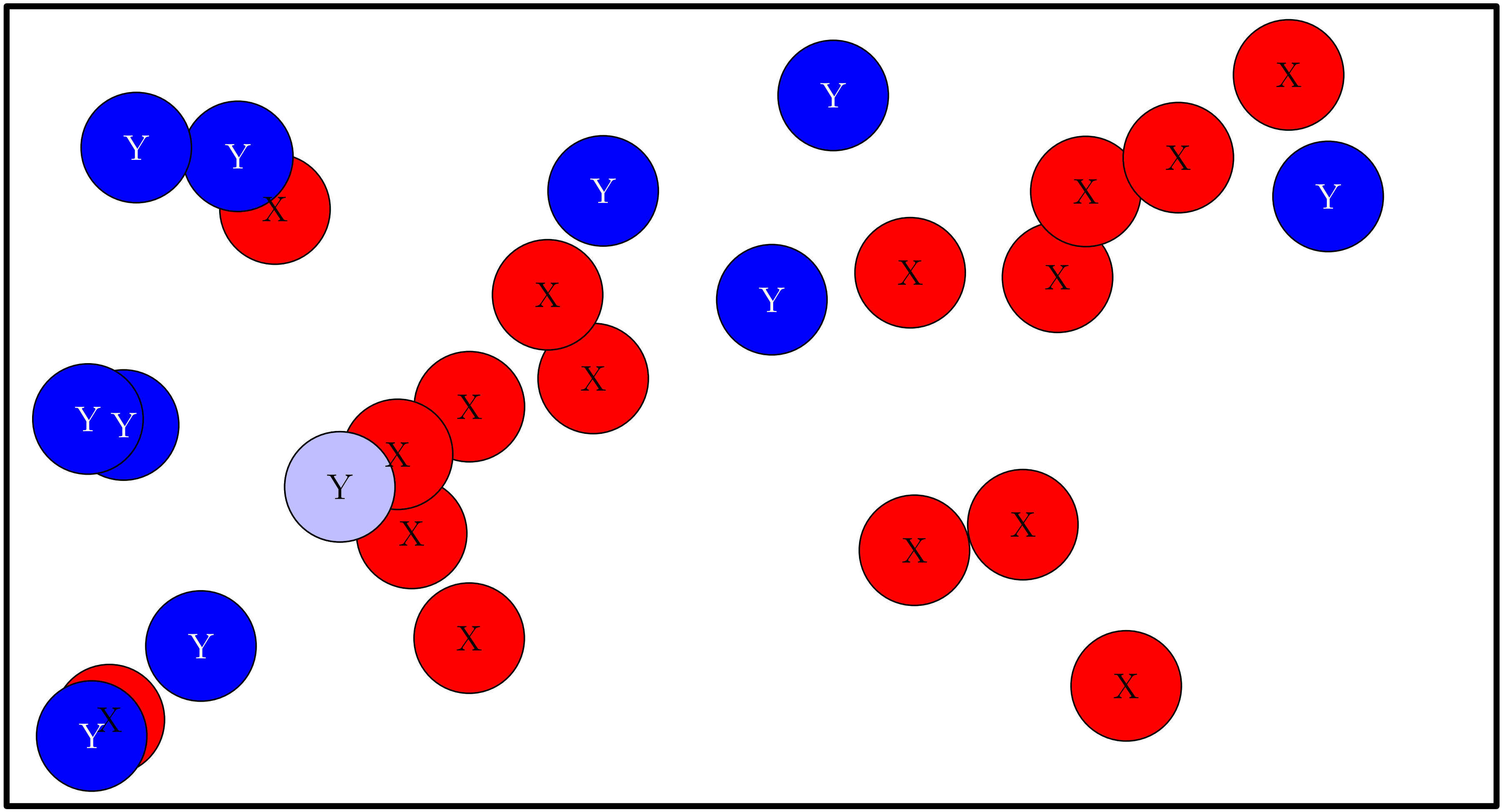

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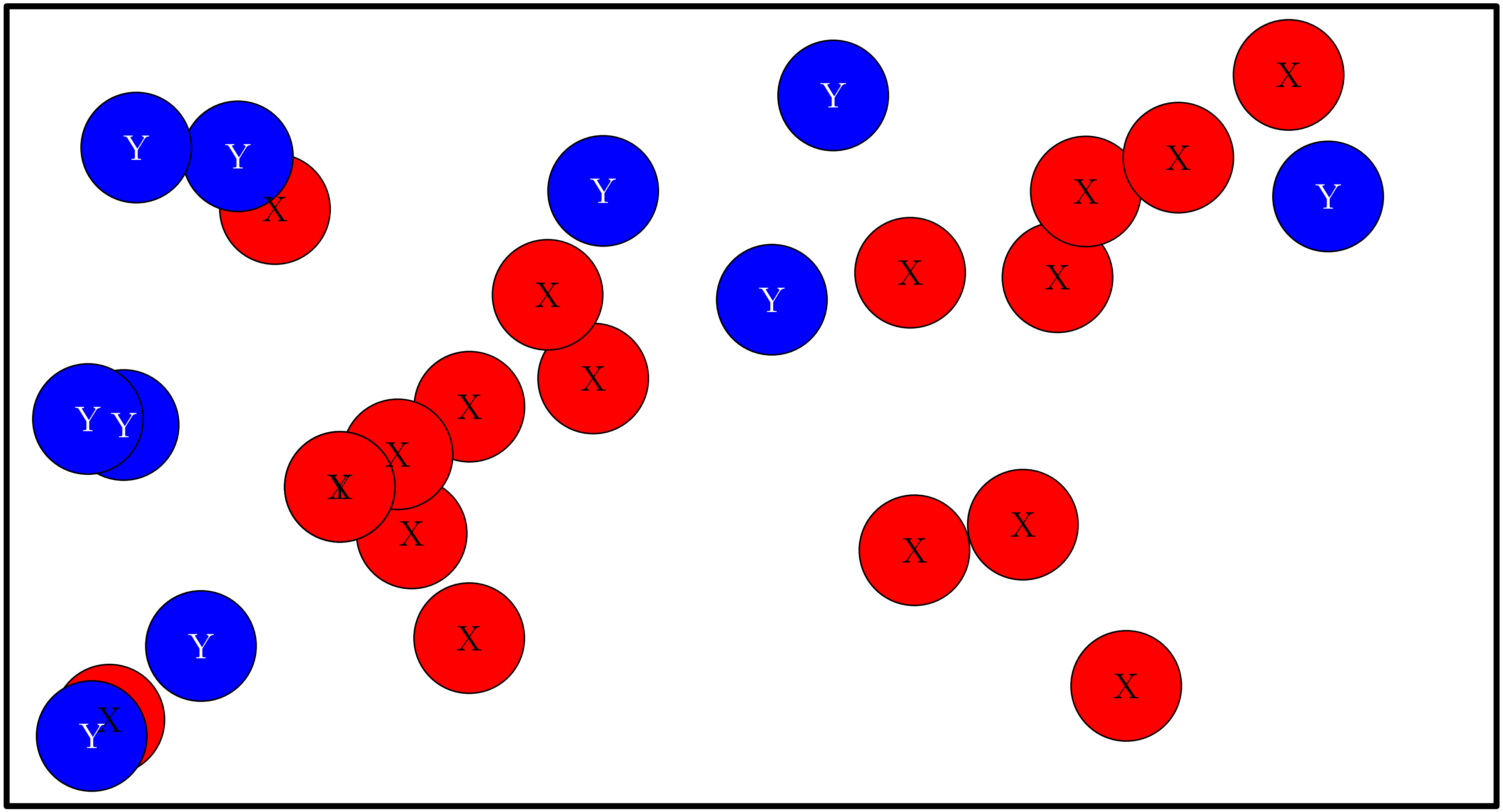

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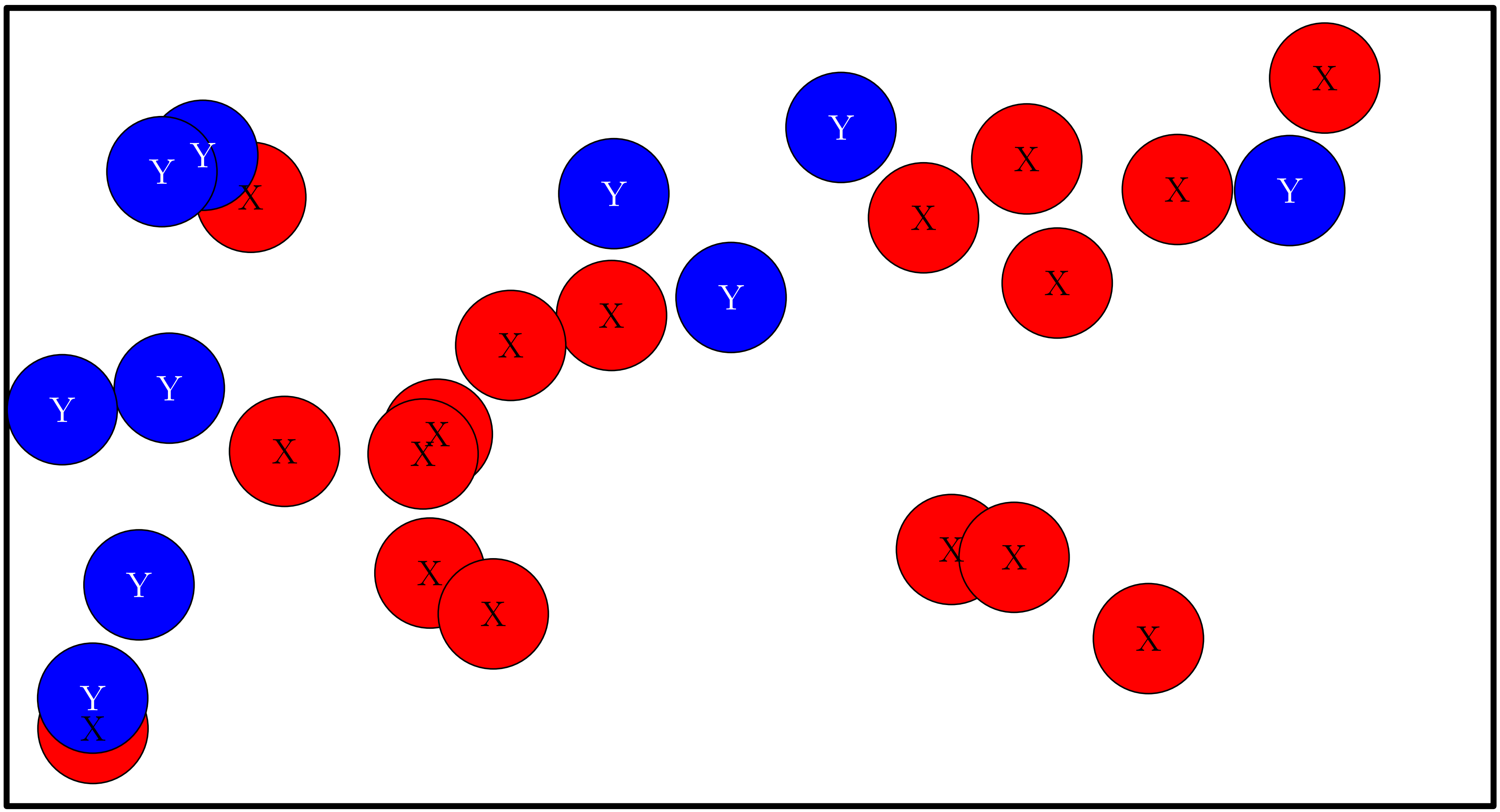

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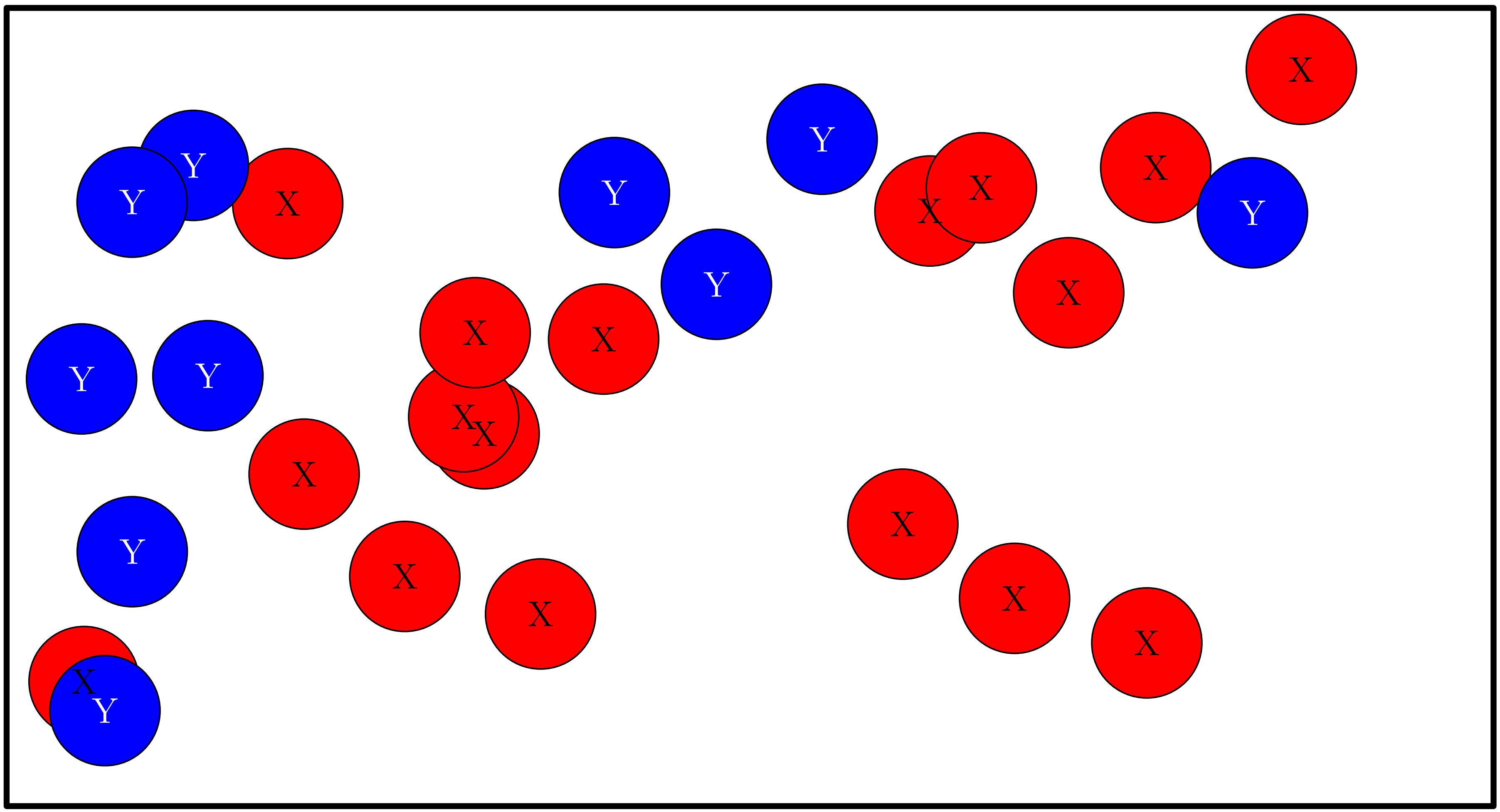
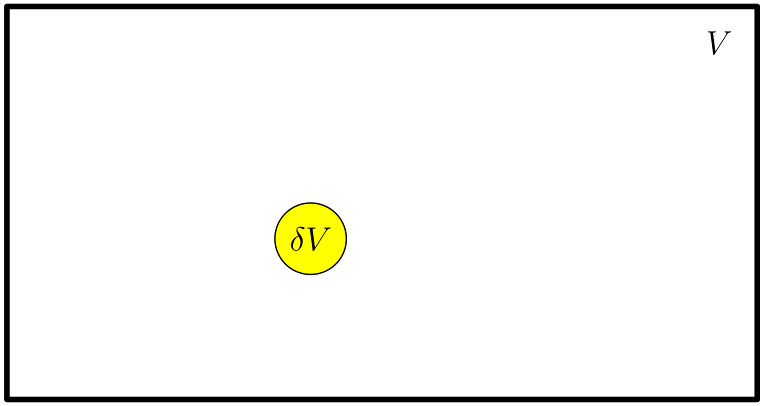


Probability of Reaction $2\,X + Y \xrightarrow{k} 3\,X$



$$\mathbb{P}\left(Y_i \in \delta V\right) = \frac{\delta V}{V}$$

- If we have n particles of type Y the probability of at least one being in δV is $n\,\delta V/V = [Y]\,\delta V$

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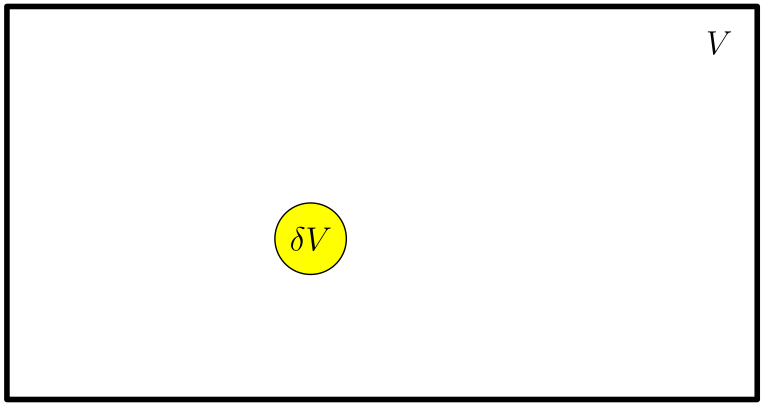

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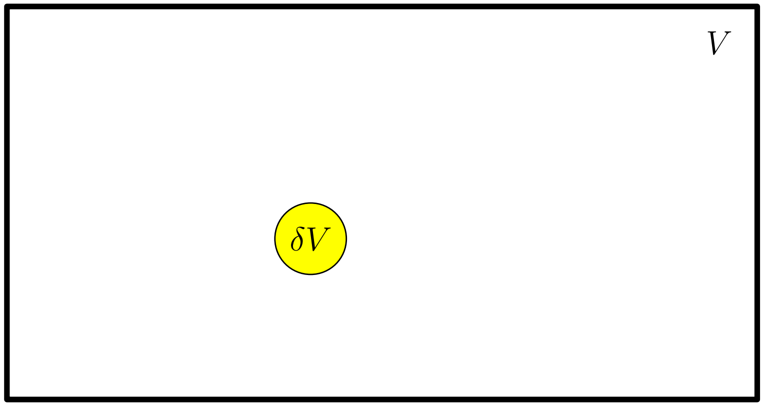

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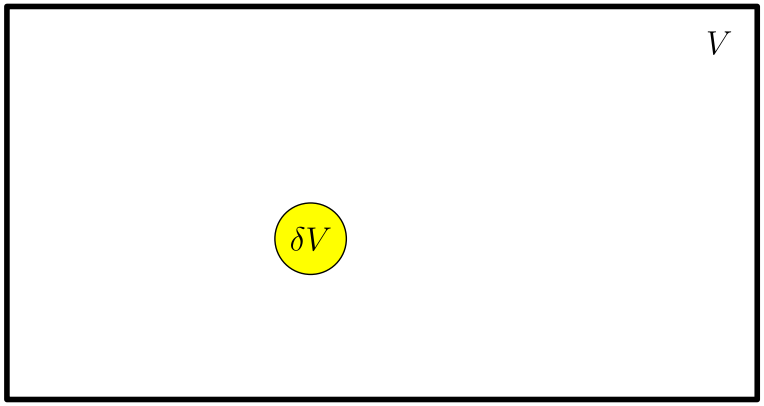

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$$[Y]\,[X]^2(\delta V)^3$$

- The number of δV's in V is $N = V/\delta V$
- The probability of a reaction in V is proportional to

$$N\,[Y]\,[X]^2(\delta V)^3 = V\,[Y]\,[X]^2\,(\delta V)^2$$

- The expected number of reactions per unit volume is proportional to

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Reaction Rate

- A chemist would measure the rate of change of concentration $[X]$ for the reaction and divide by $[Y]\,[X]^2$ to come up with a reaction rate k

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$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = k\,[Y]\,[X]^2$$

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Concentration to Numbers

- To work in numbers of molecules, X, we use $[X] = X/\Omega$
- $\Omega = V \times A$, where V is volume and A Avogadro's number

$$\begin{aligned} \frac{\mathrm{d}[X]}{\mathrm{d}t} &= k[Y]\,[X]^2 \\ \frac{1}{\Omega}\frac{\mathrm{d}X}{\mathrm{d}t} &= k\frac{1}{\Omega^3}Y\,X^2 \\ \frac{\mathrm{d}X}{\mathrm{d}t} &= \frac{k}{\Omega^2}Y\,X^2 \end{aligned}$$

- If we want to be even more accurate the probability of a reaction taking place in time δt is

$$\frac{k}{\Omega^2}Y\,X\,(X-1)$$

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Concentration to Numbers

- To work in numbers of molecules, X, we use $[X] = X/\Omega$
- $\Omega = V \times A$, where V is volume and A Avogadro's number

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = k[Y]\,[X]^2$$

$$\frac{1}{\Omega}\frac{\mathrm{d}X}{\mathrm{d}t} = k\frac{1}{\Omega^3}Y\,X^2$$

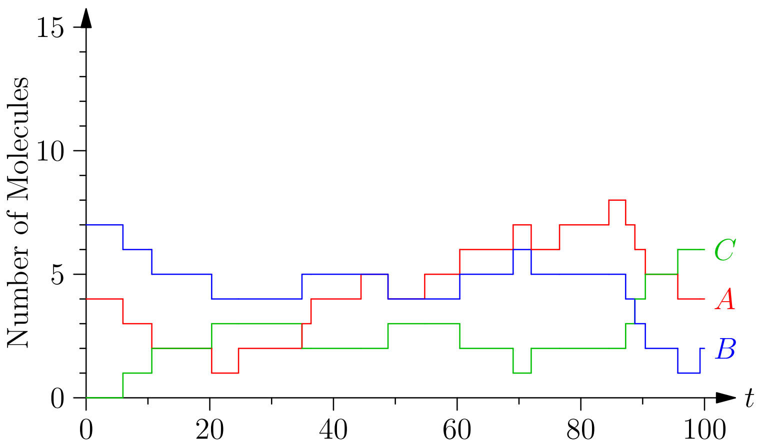
$$\frac{\mathrm{d}X}{\mathrm{d}t} = \frac{k}{\Omega^2}Y\,X^2$$

- If we want to be even more accurate the probability of a reaction taking place in time δt is

$$\frac{k}{\Omega^2}Y\,X\,(X-1)$$

Outline

1. Stochastic Chemical Equations
2. **Gillespie Algorithm**



Discrete Dynamic Models

- We would like to simulate the set of chemical reactions

$$\varnothing \xrightarrow{k_1} A \qquad \varnothing \xrightarrow{k_2} B \qquad A + B \xrightarrow{k_3} C \qquad C \xrightarrow{k_4} A + B$$

- At each time we want to keep note of the number of molecules $A(t)$, $B(t)$, $C(t)$

- These number only change when a reaction happens

- When the numbers change the probability of a reaction changes

- Events that happen at particular points in space or time are known as **point processes**

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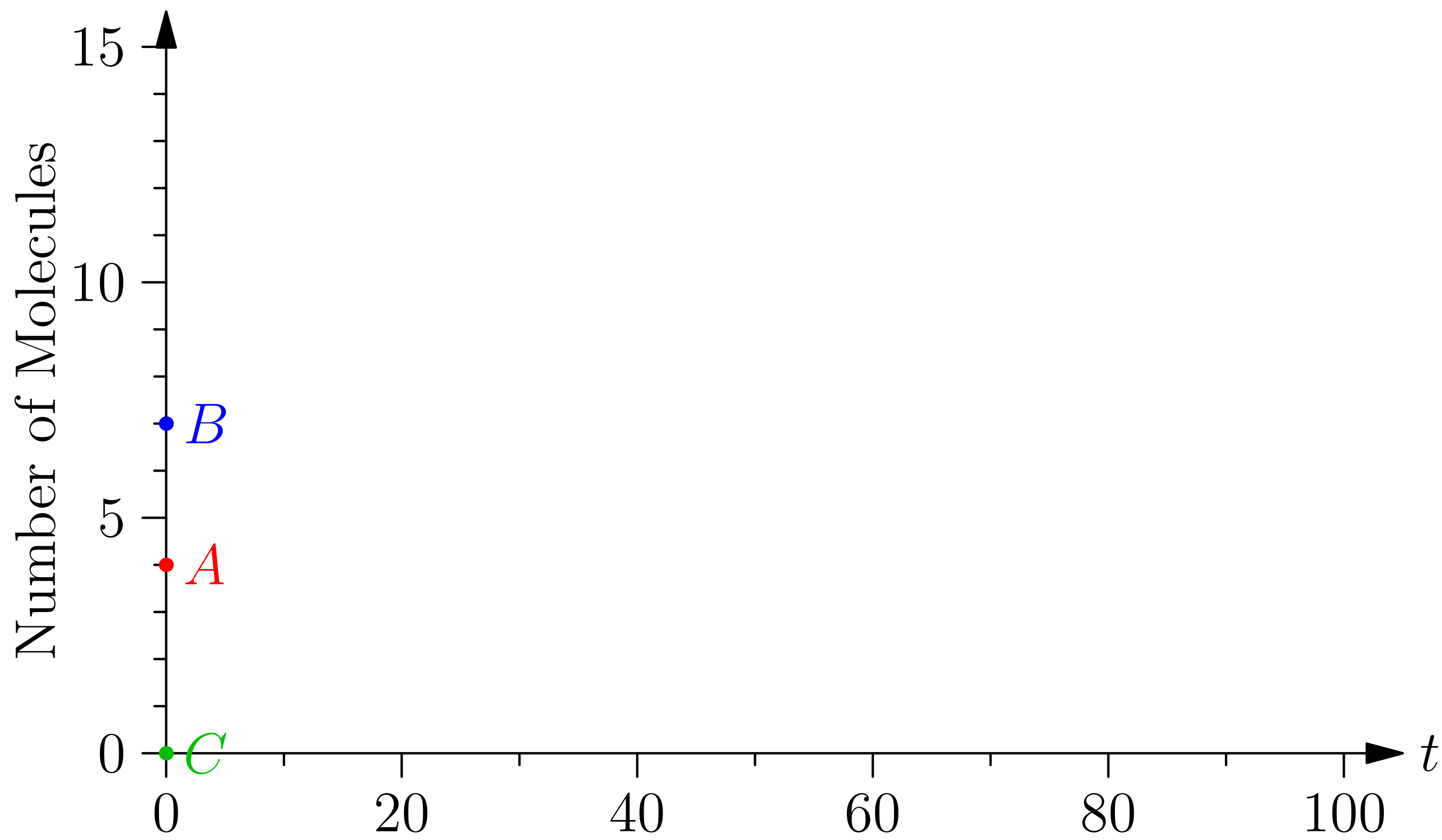

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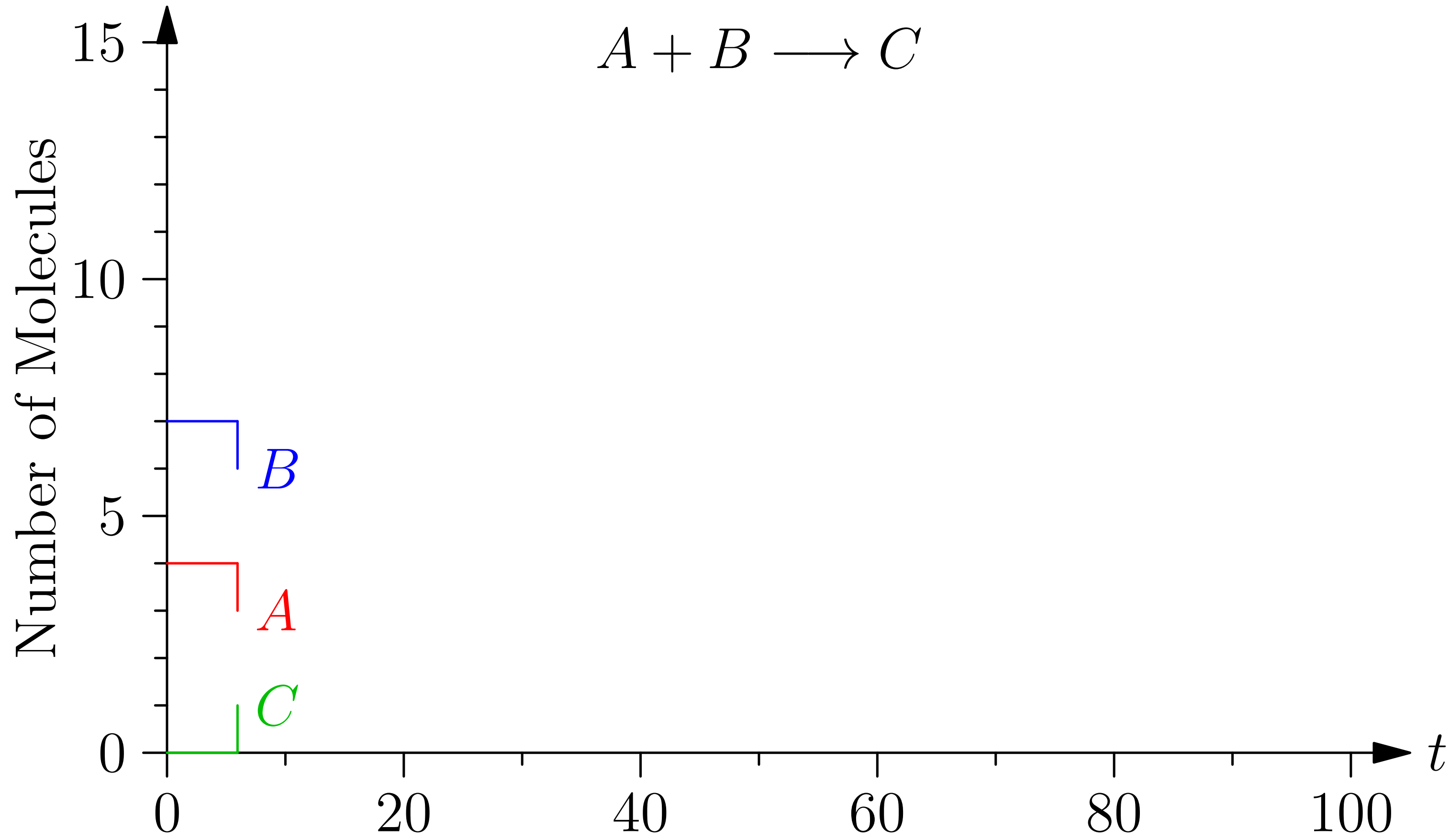

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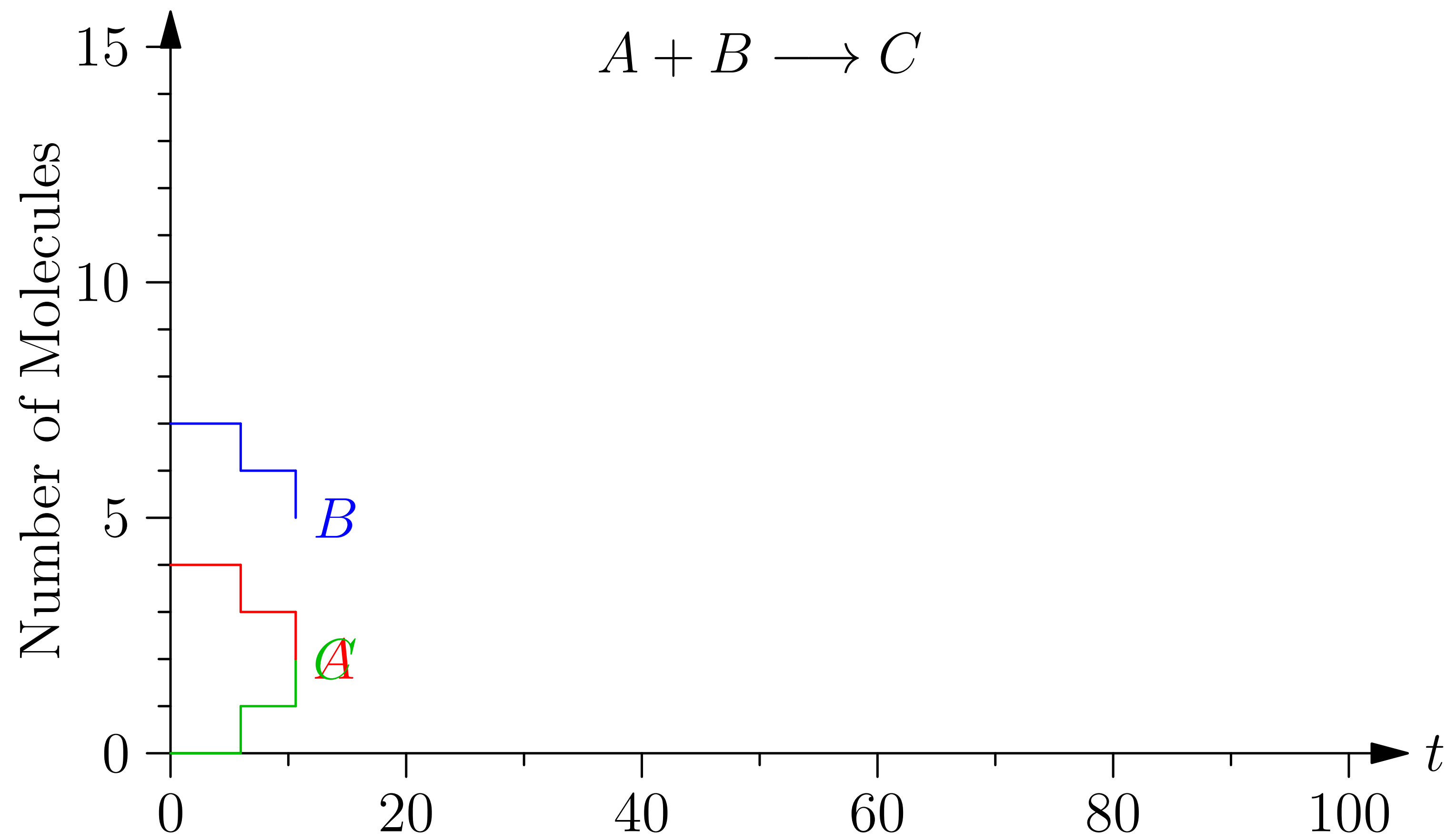

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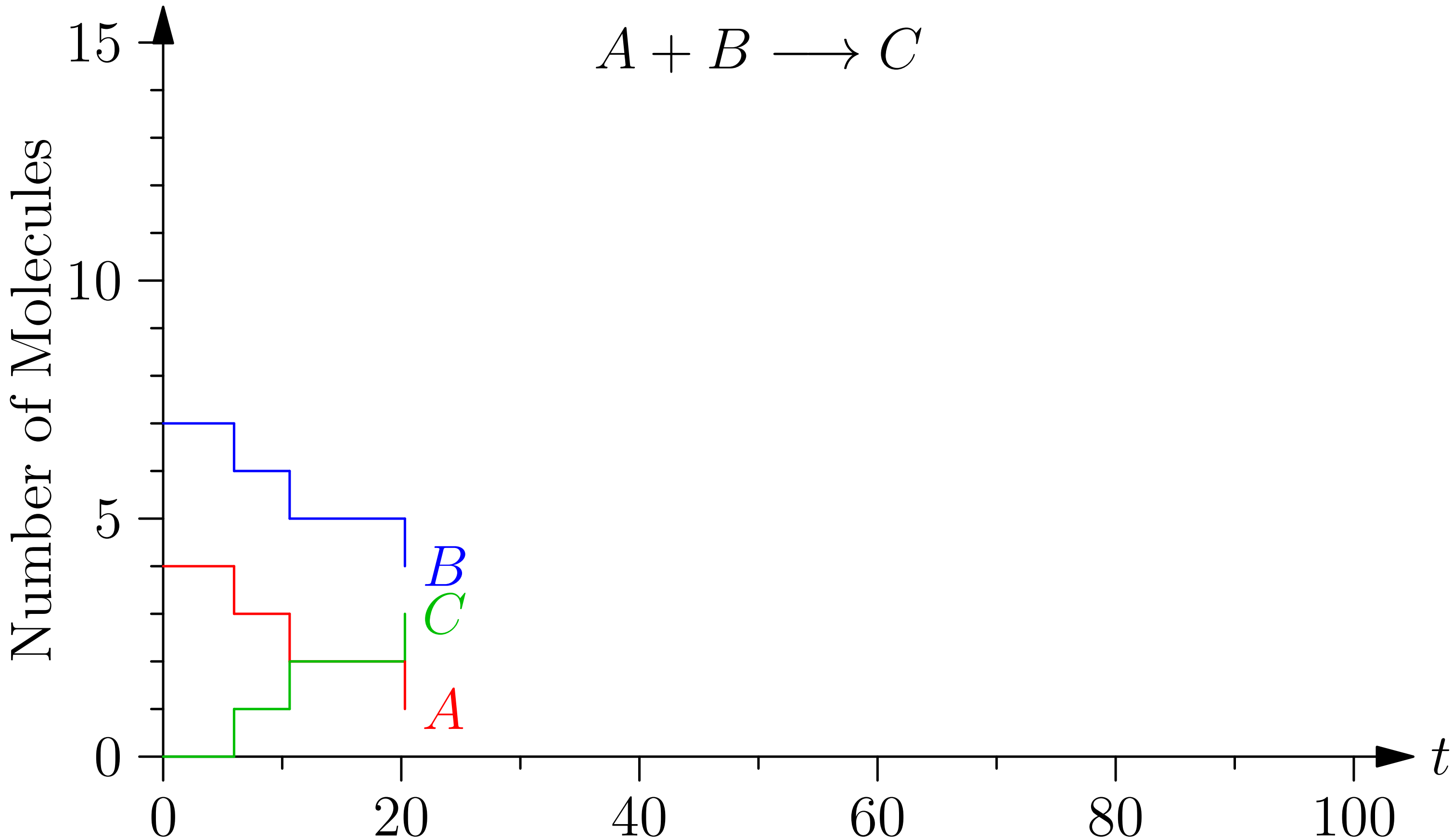

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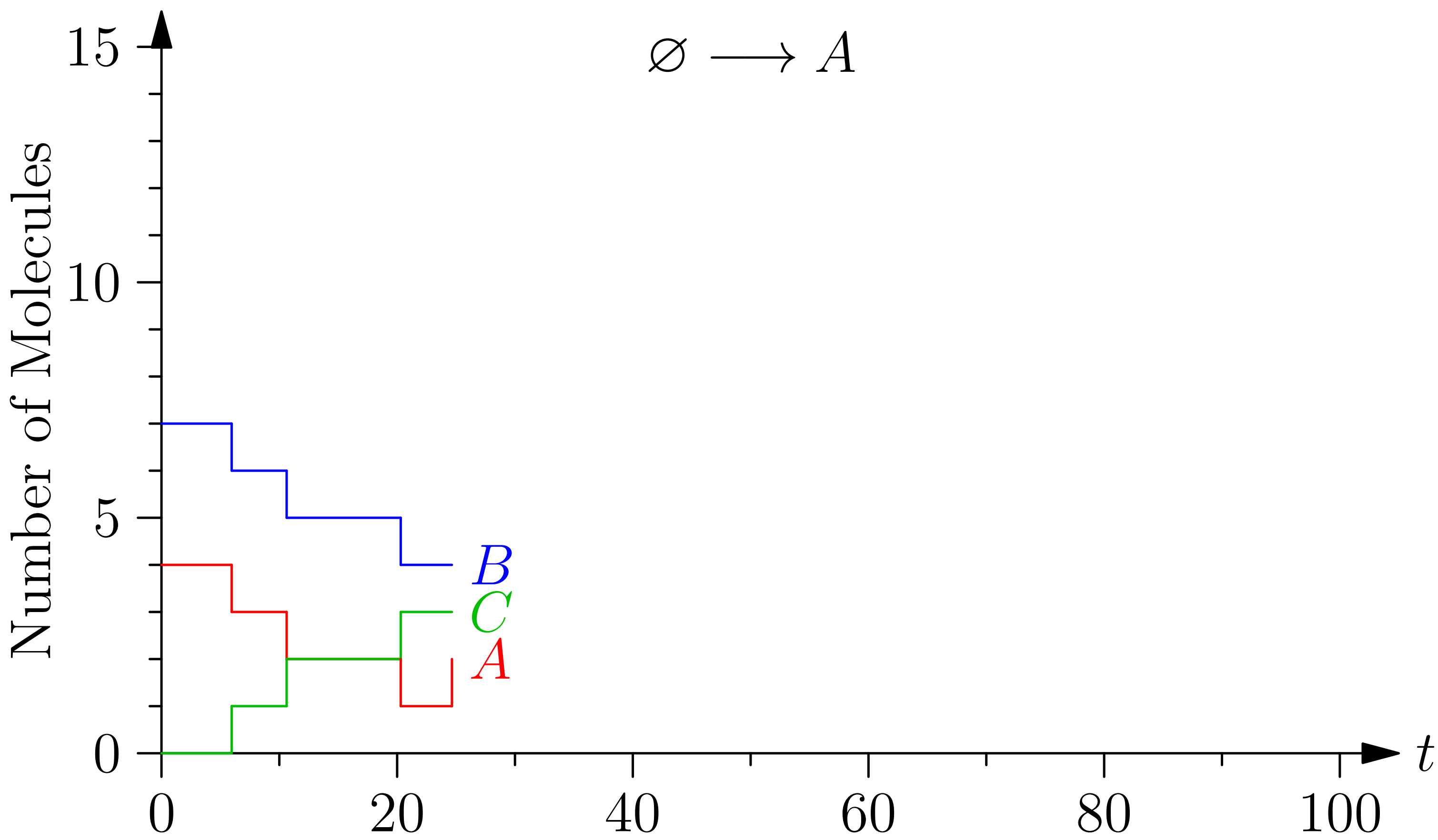

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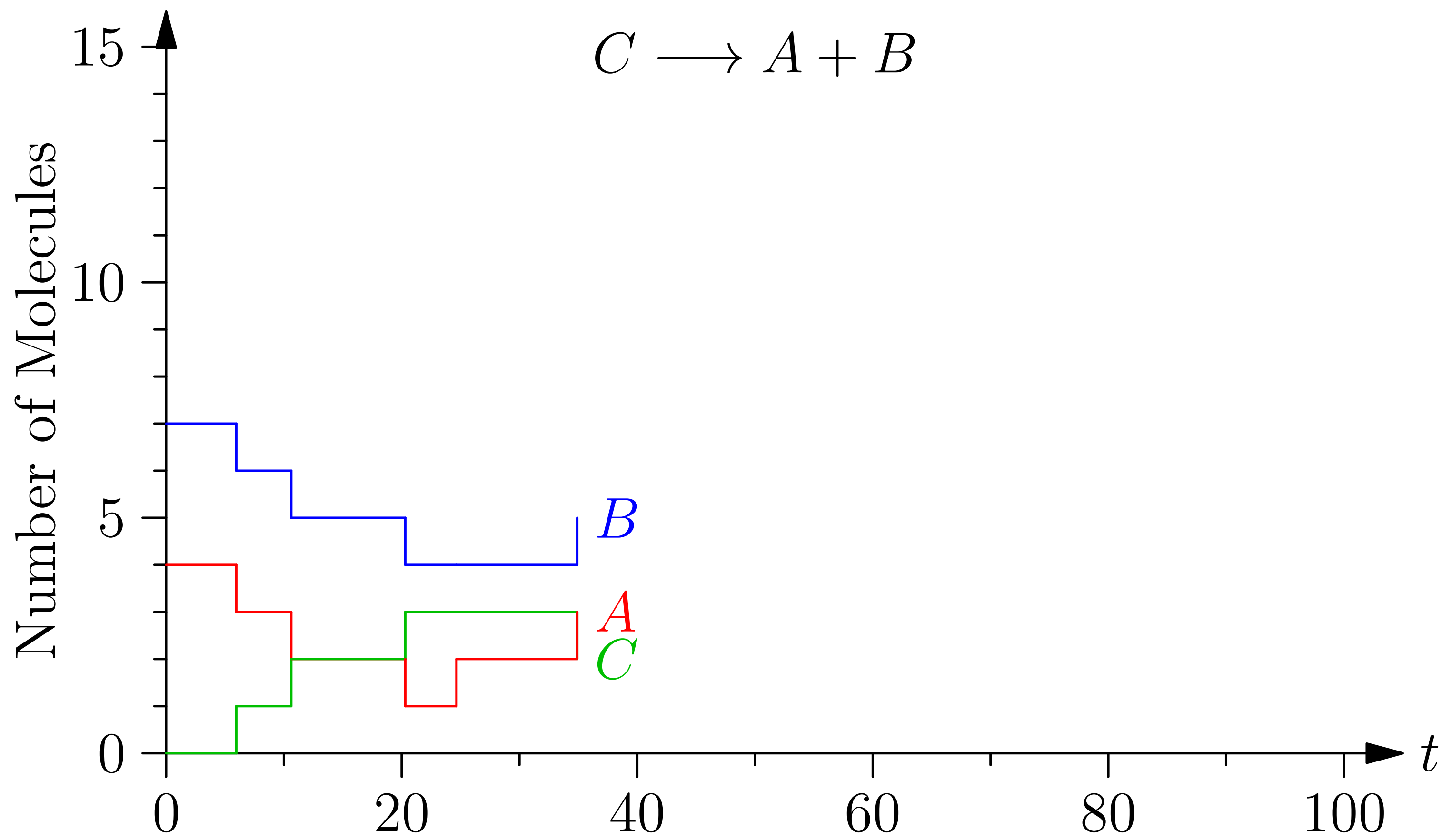

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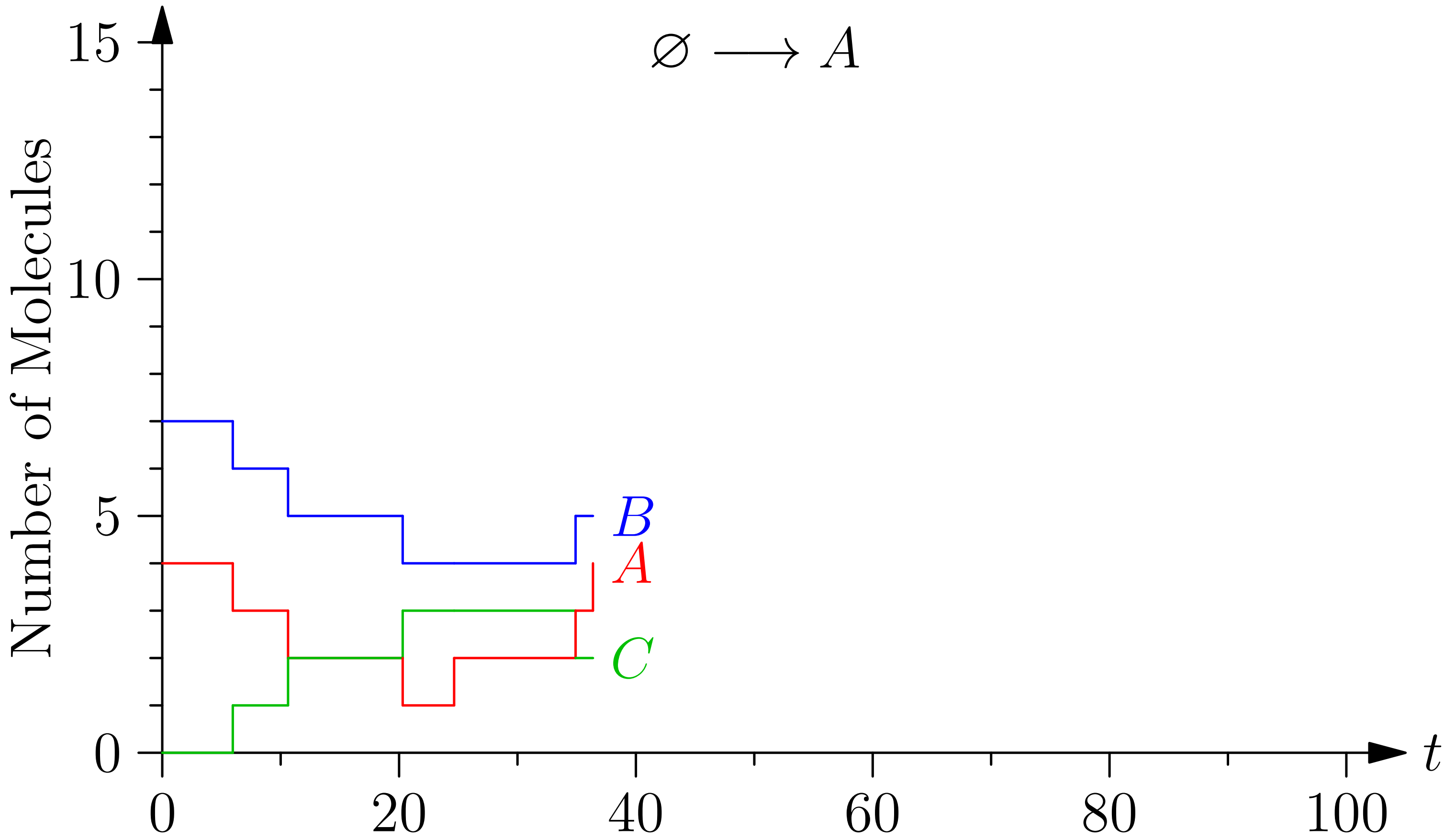

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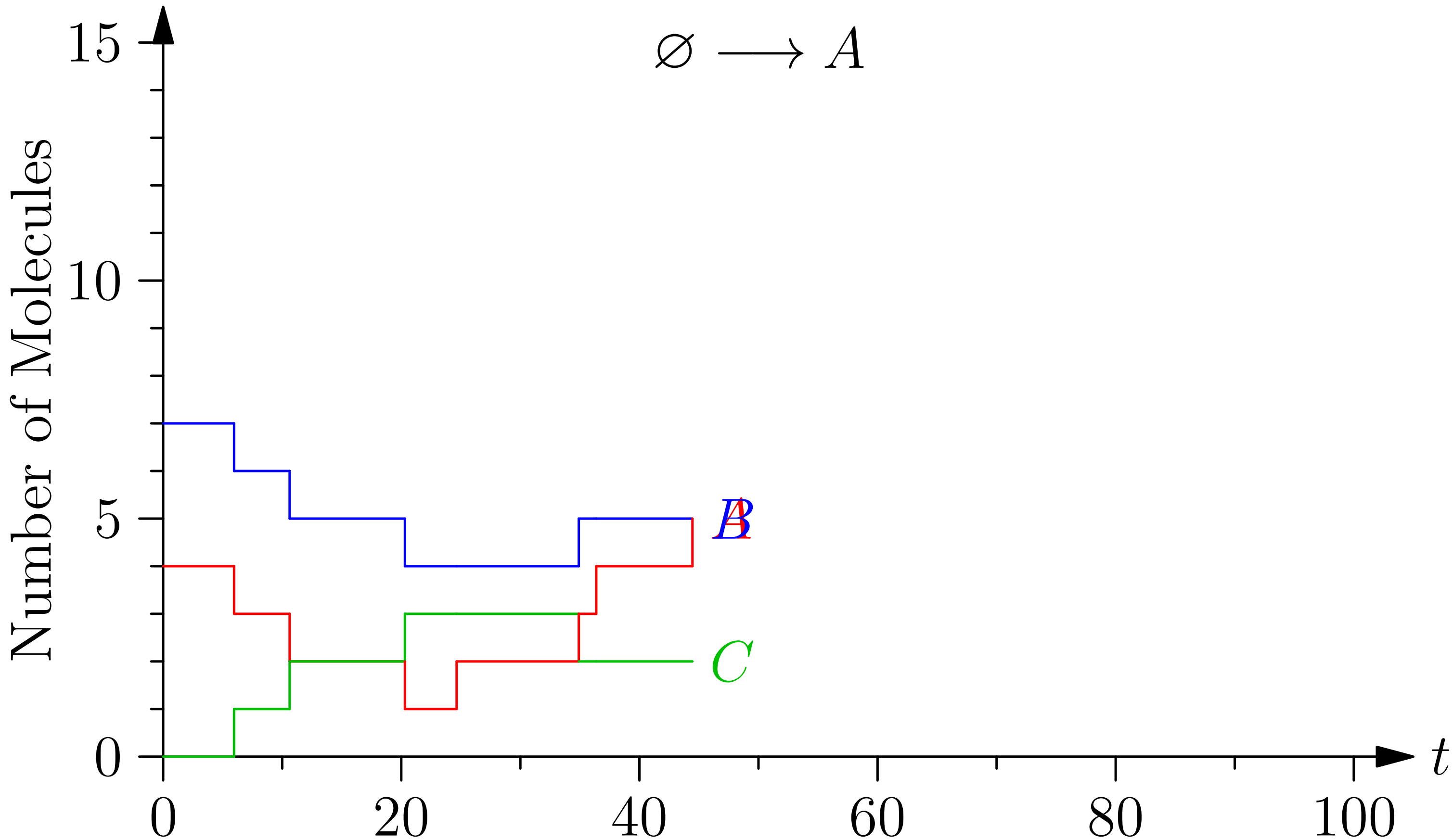

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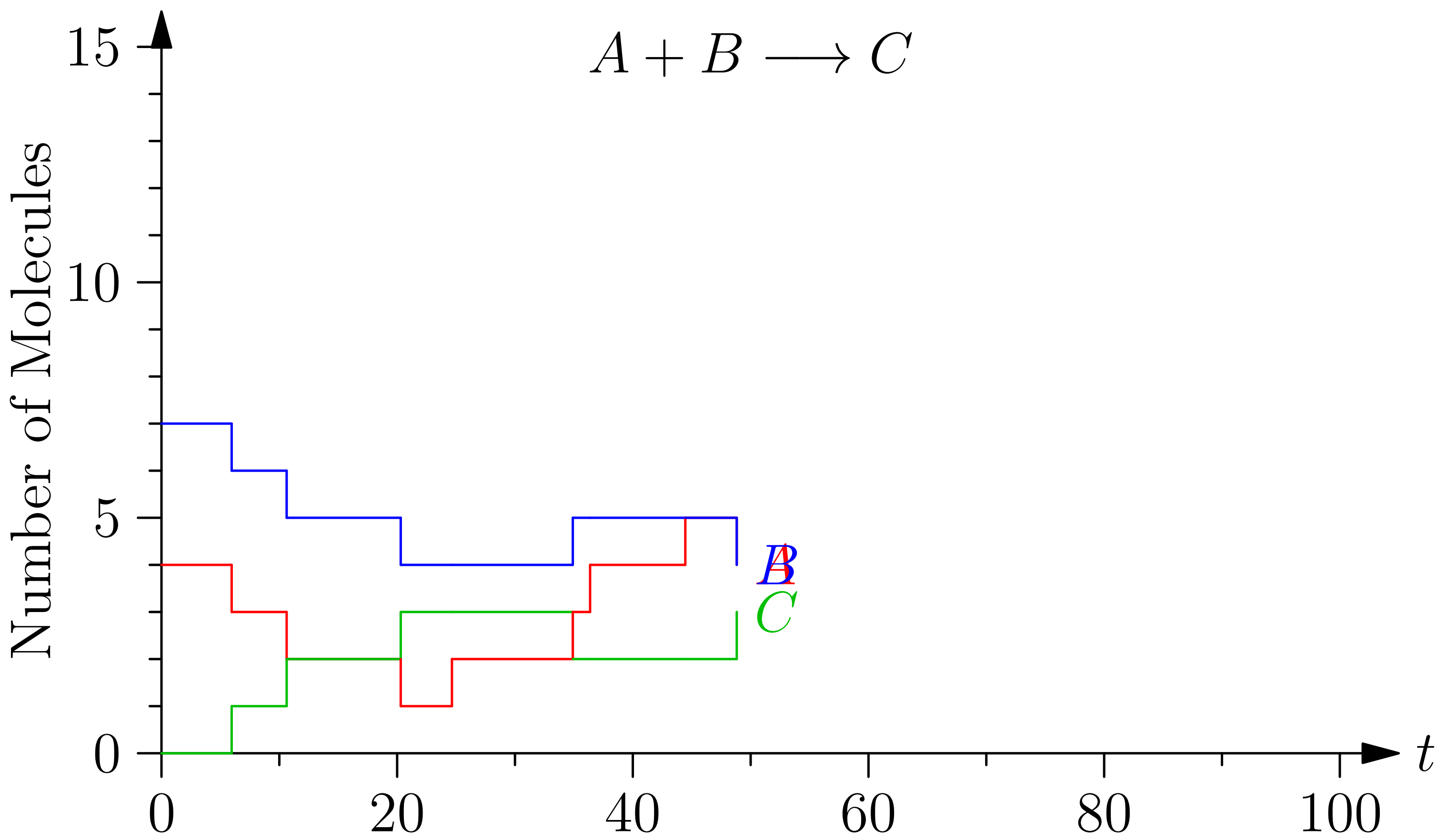

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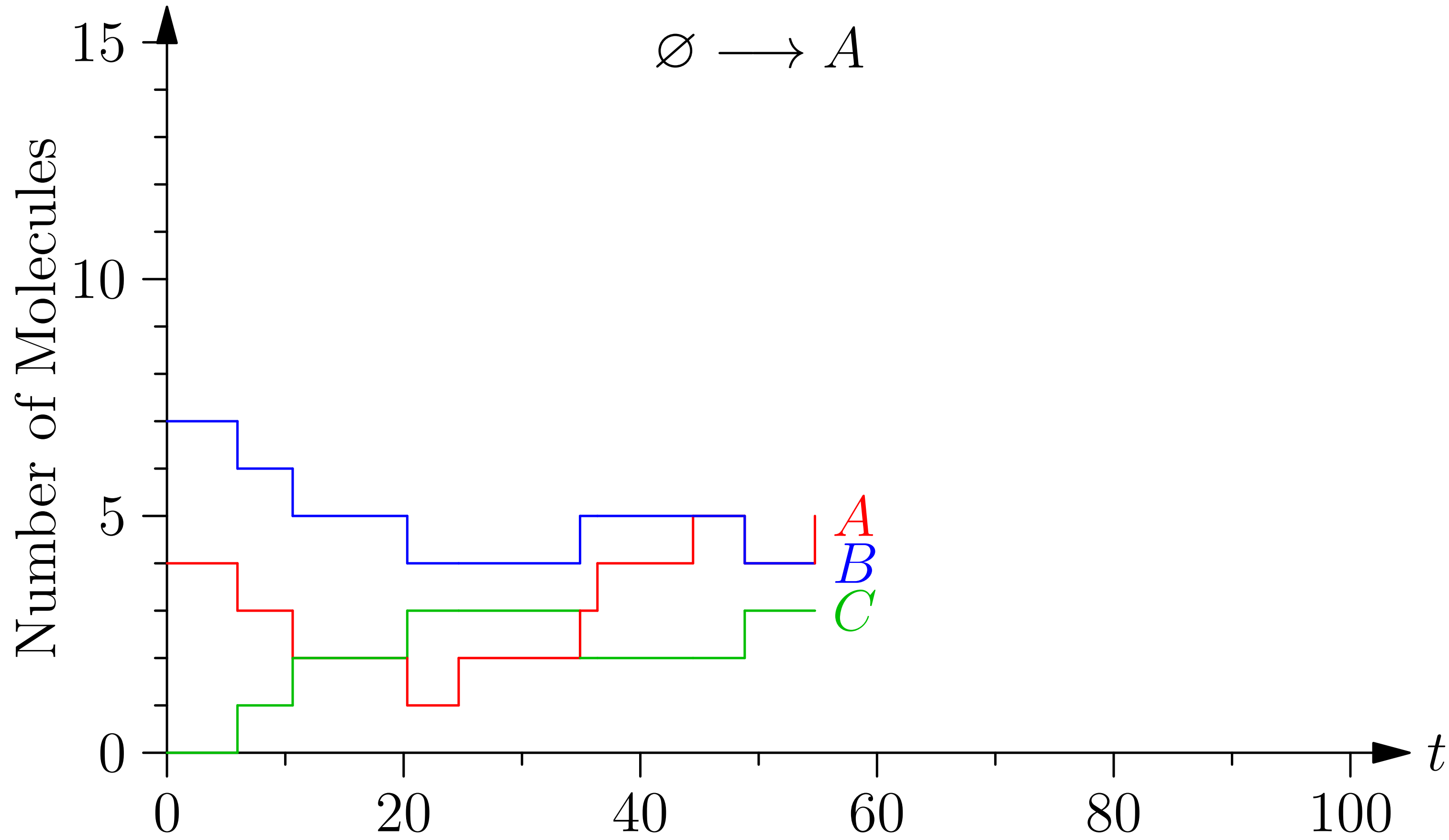

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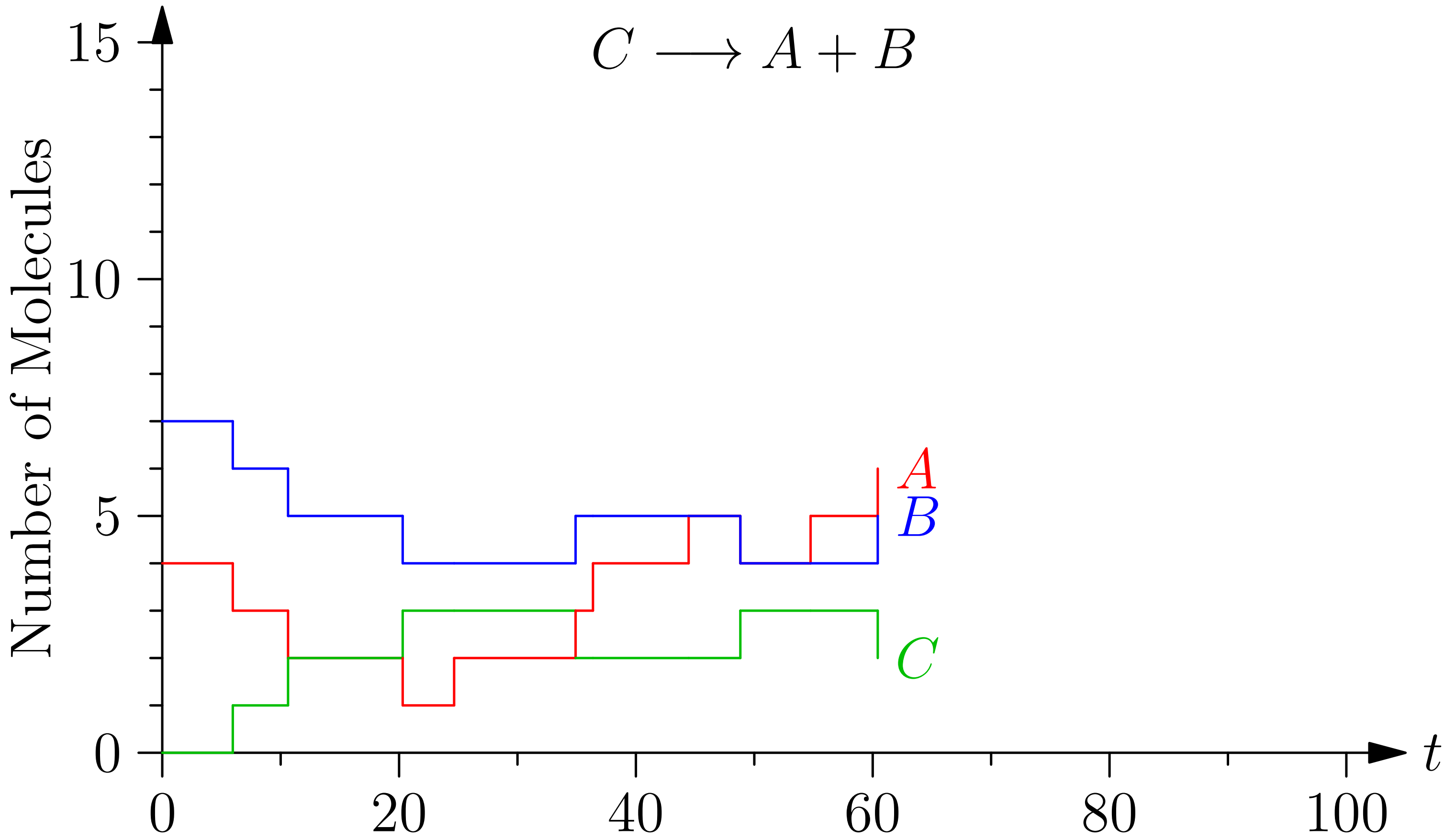

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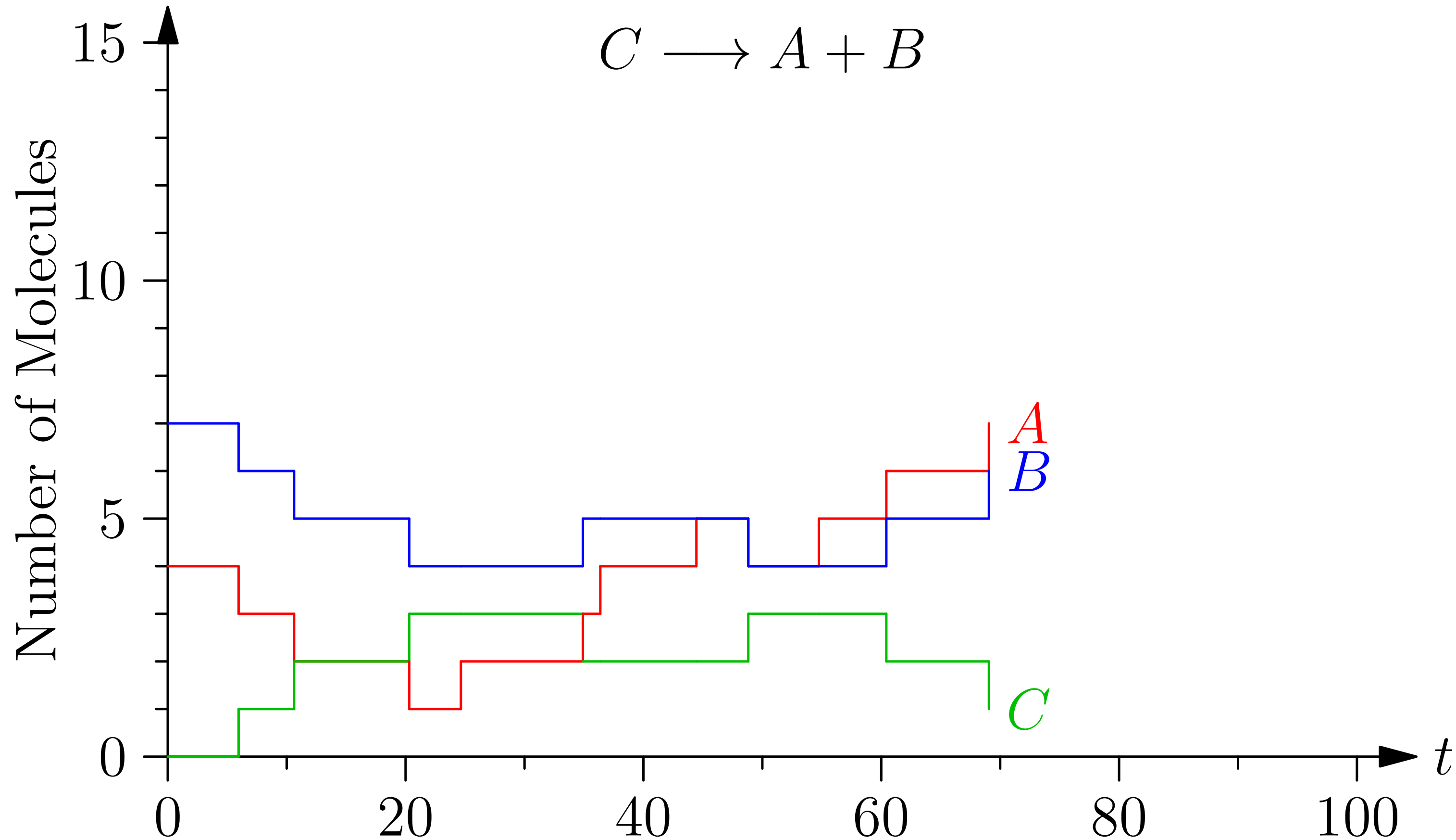

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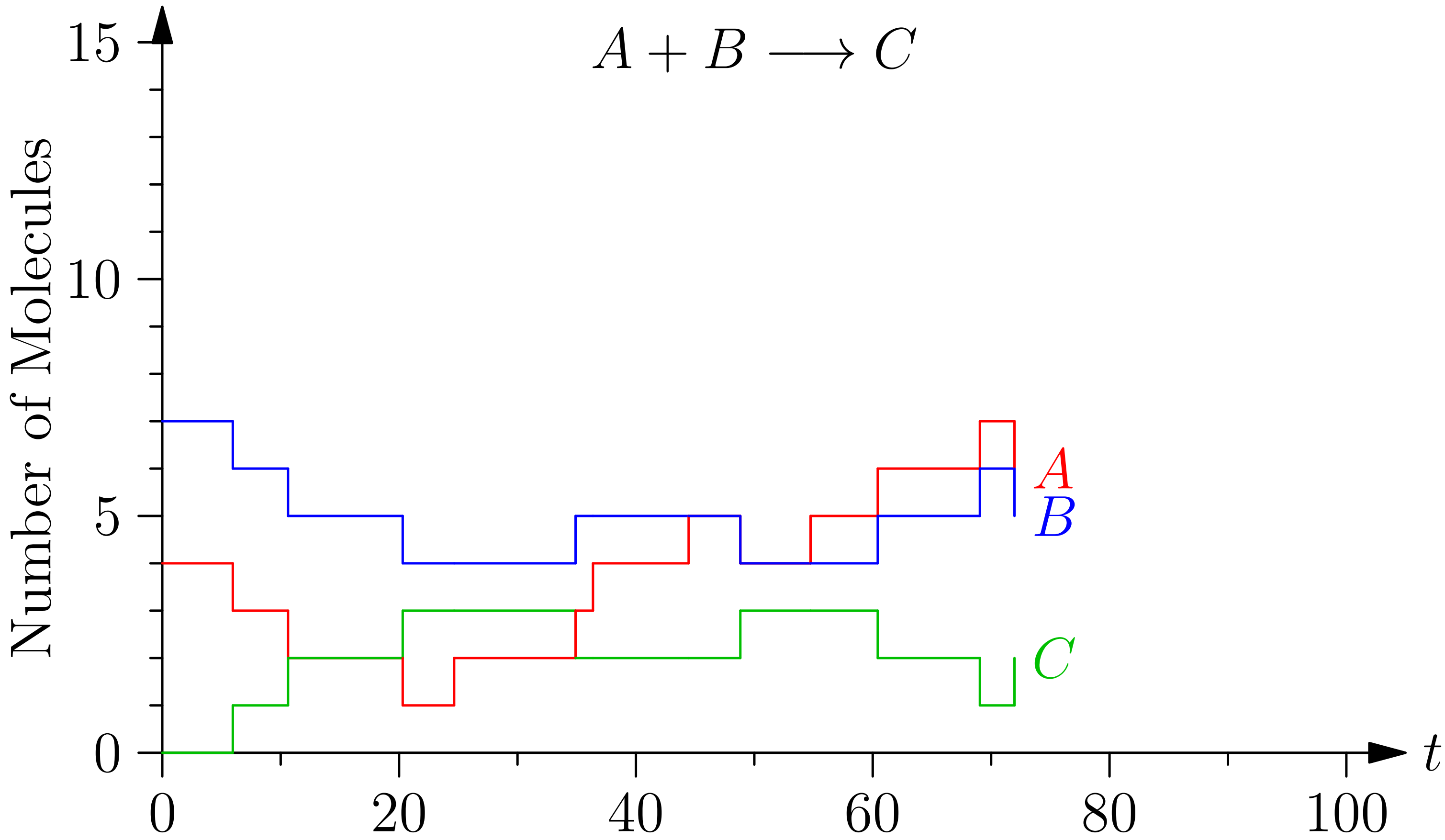

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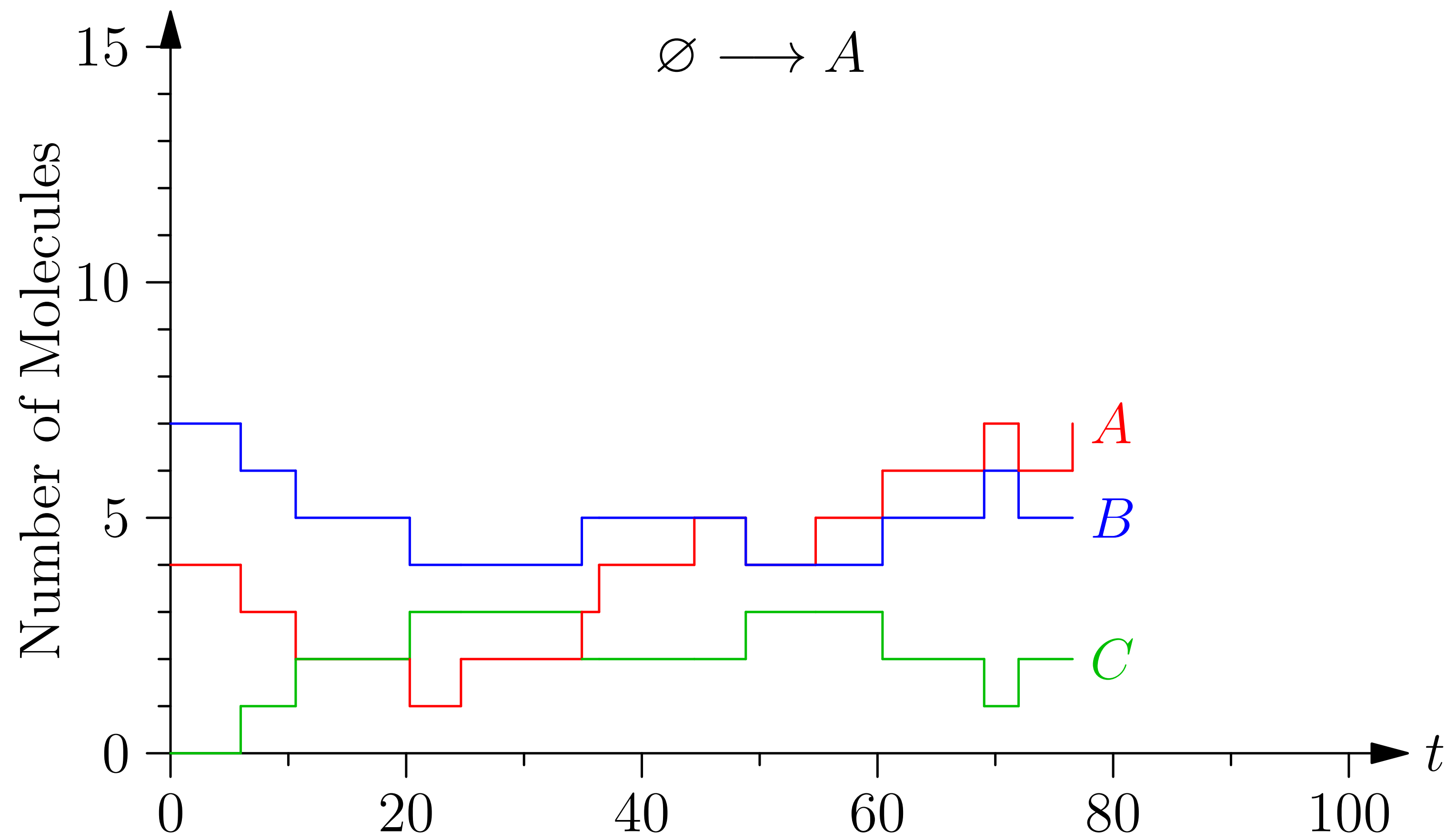

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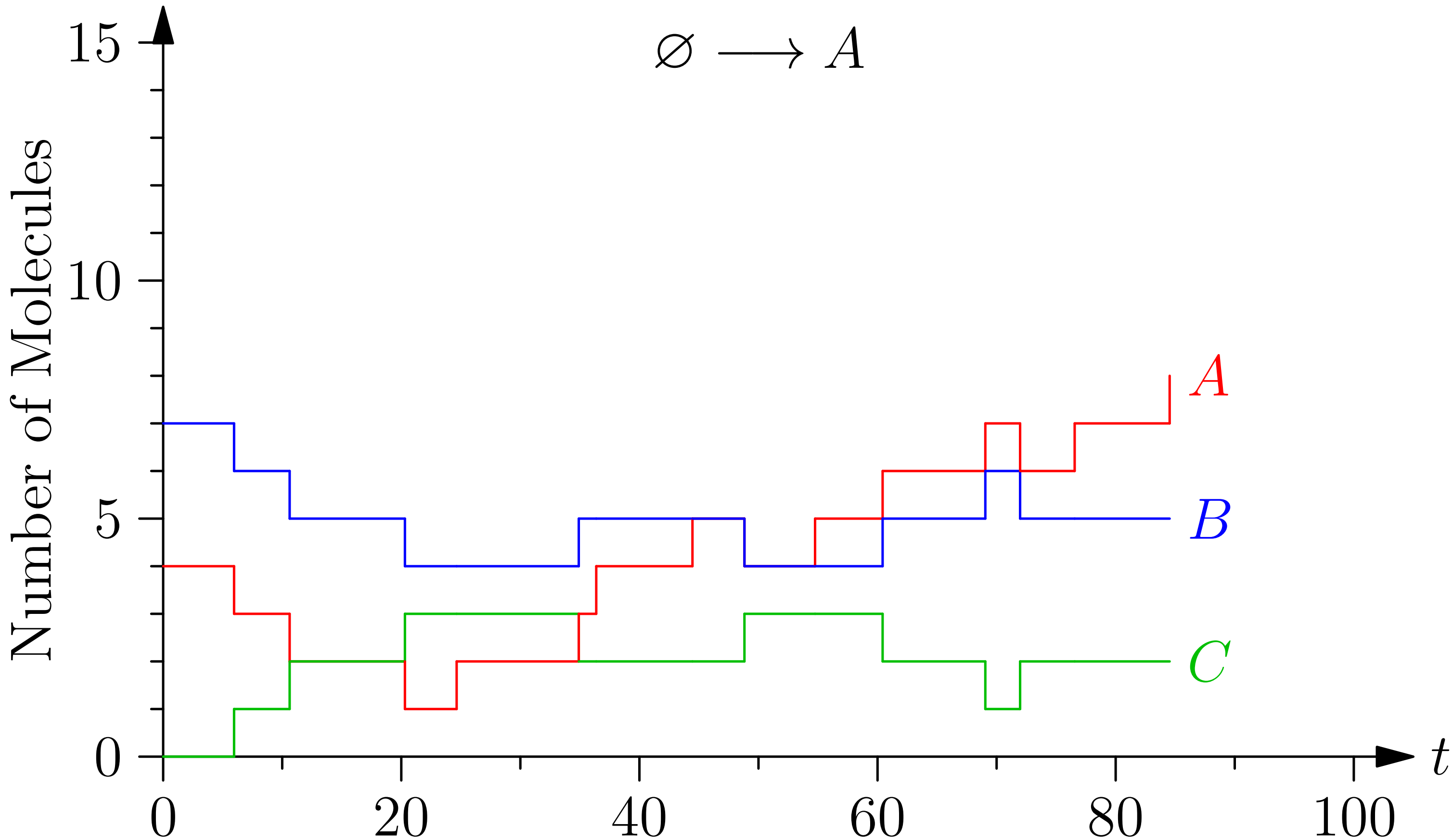

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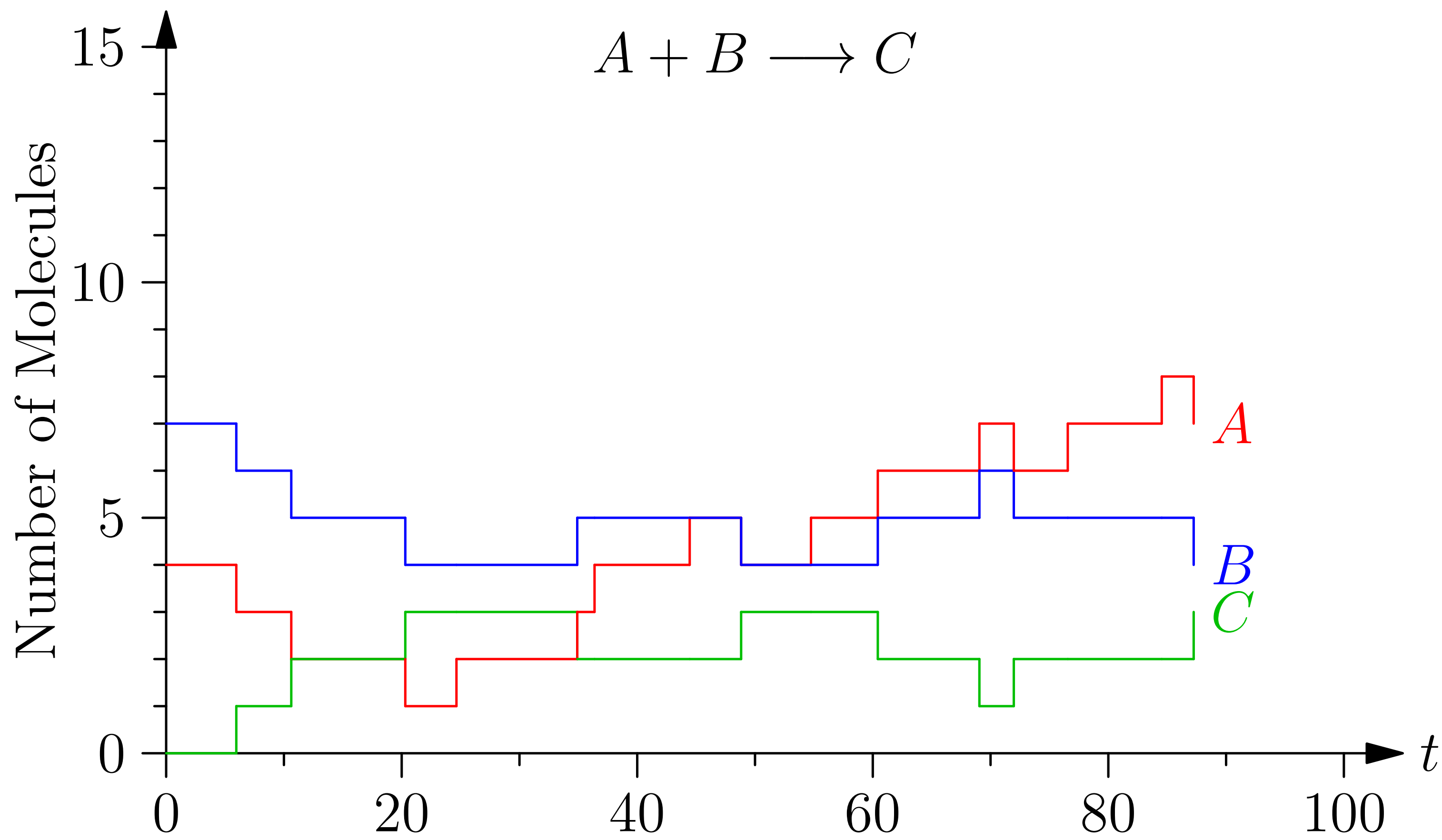

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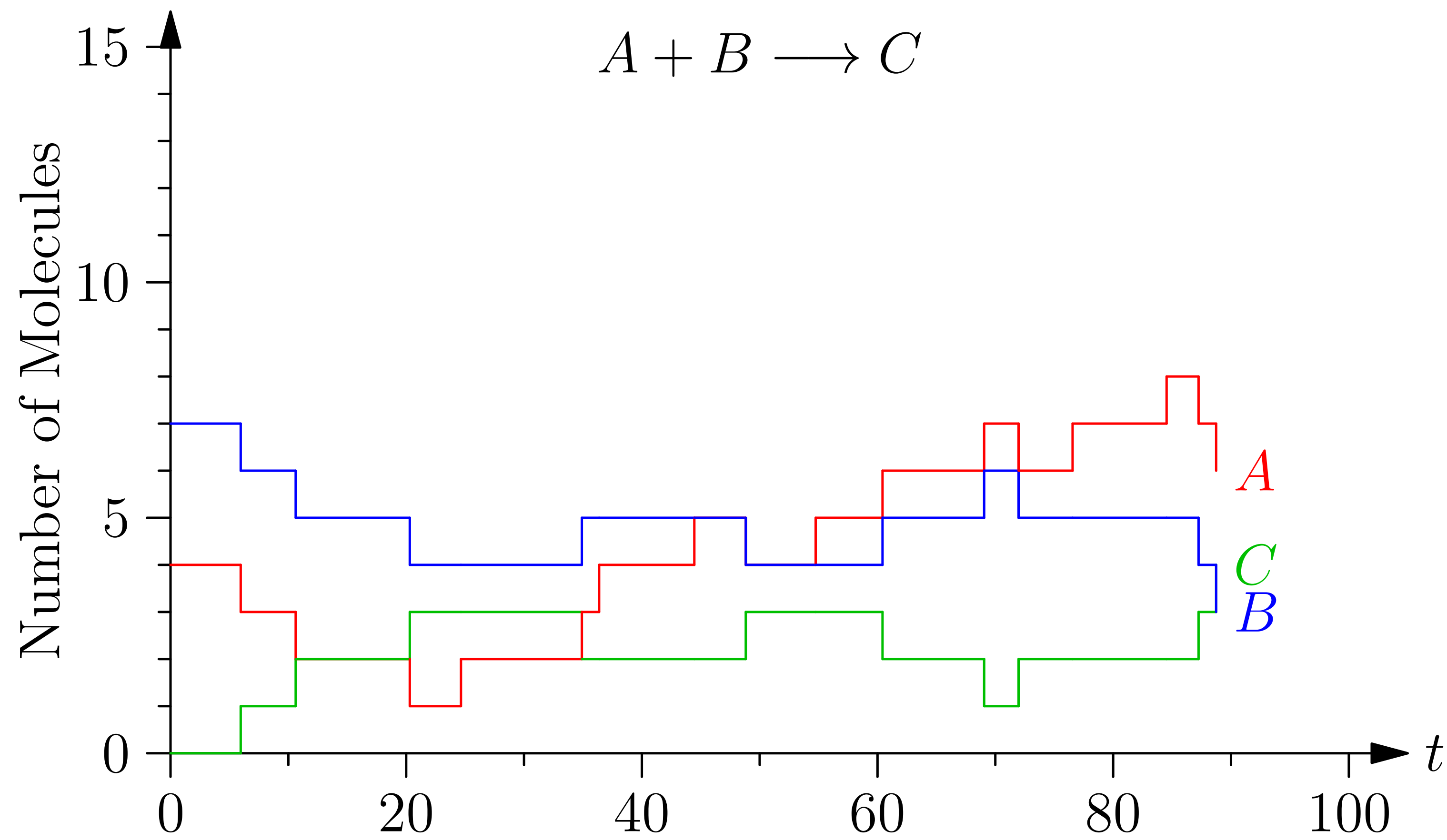

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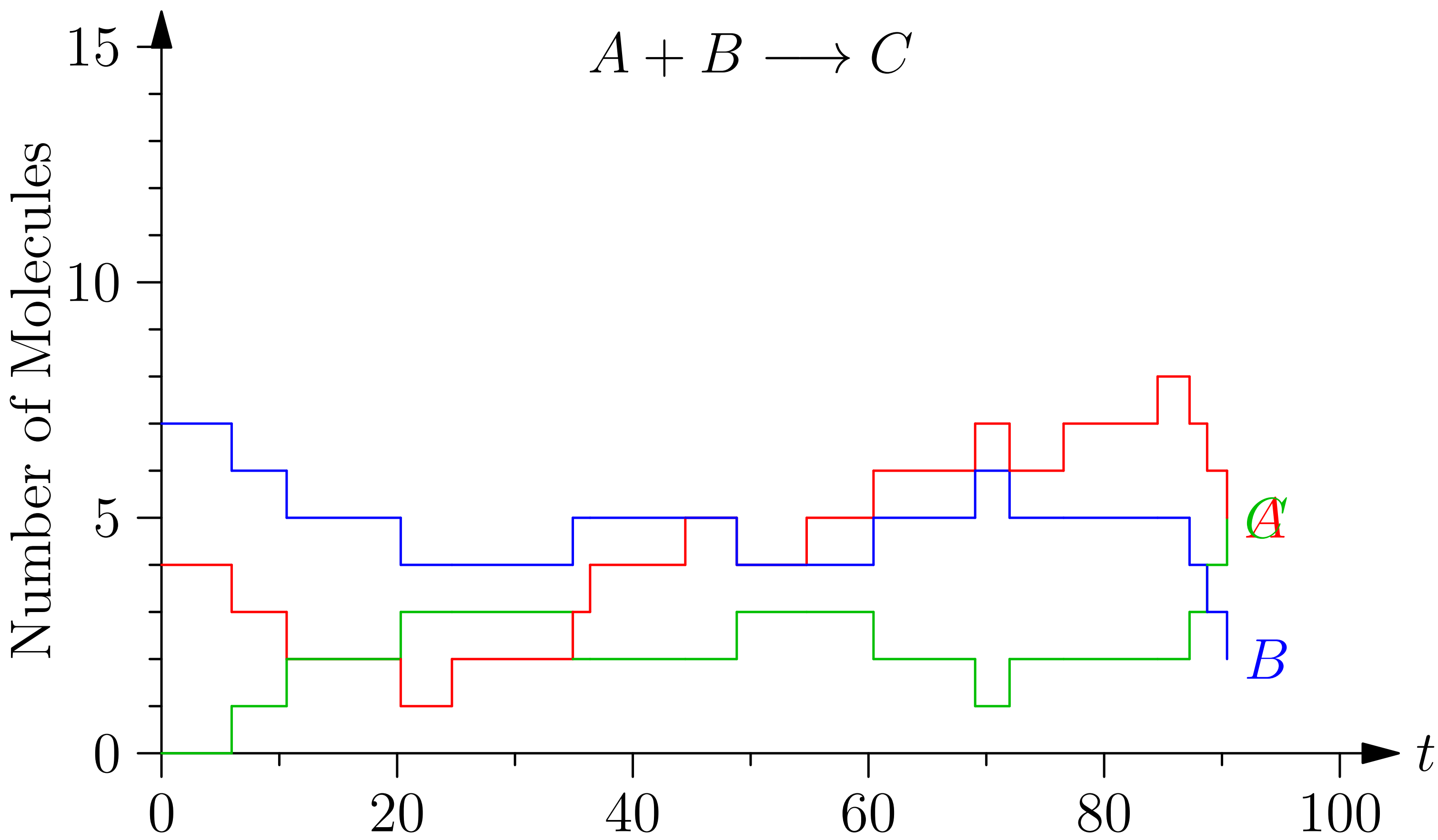

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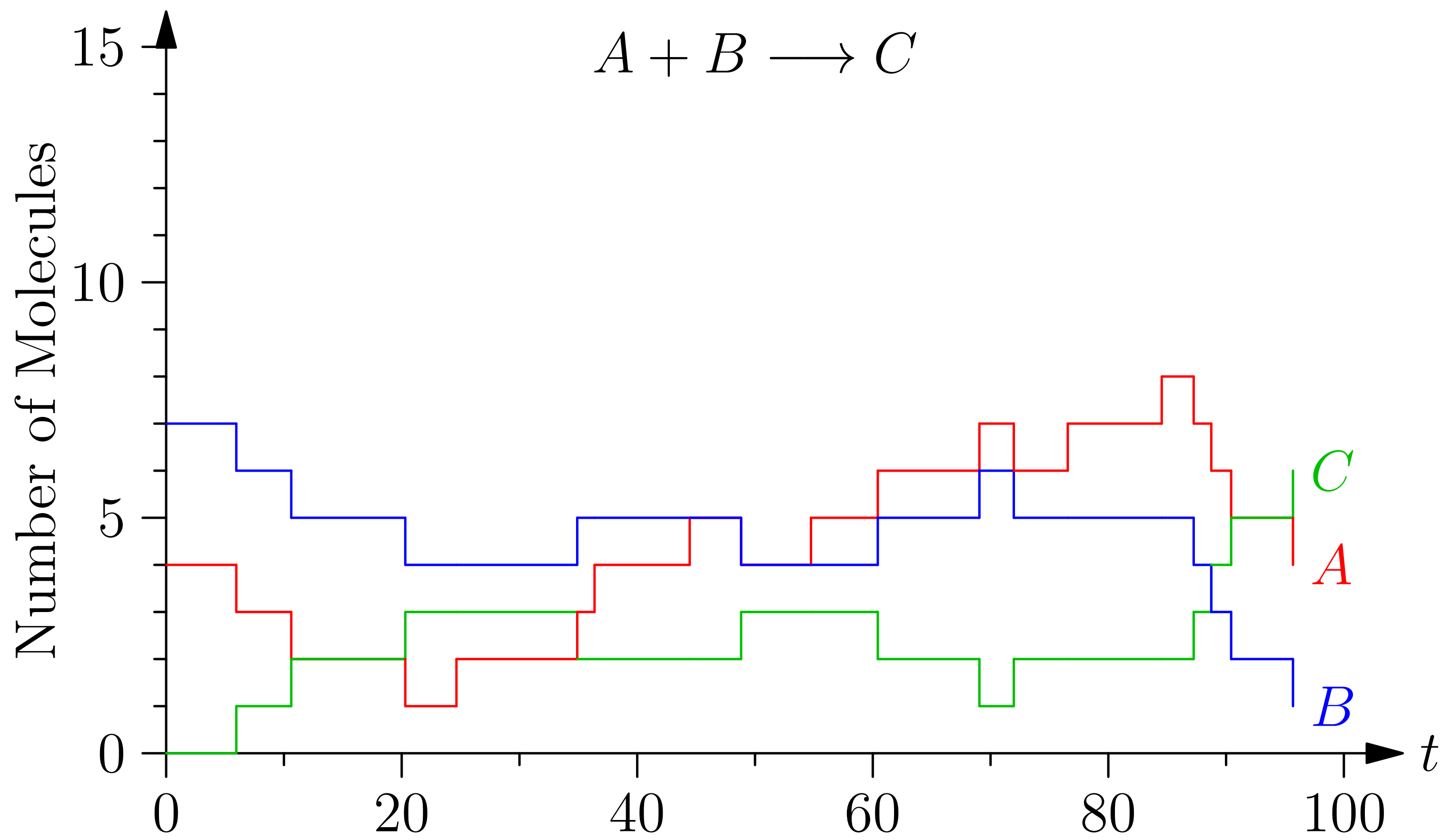

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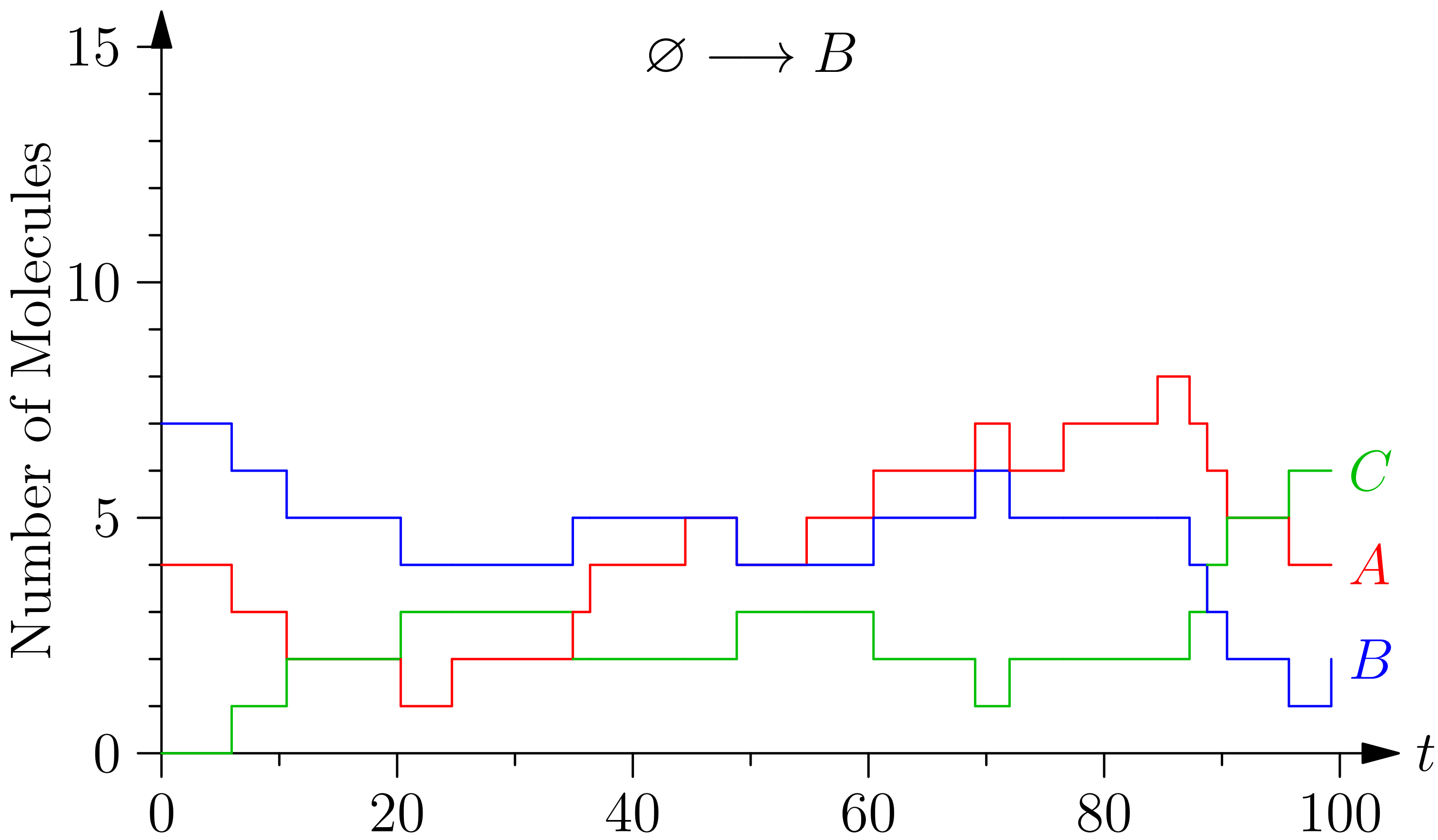

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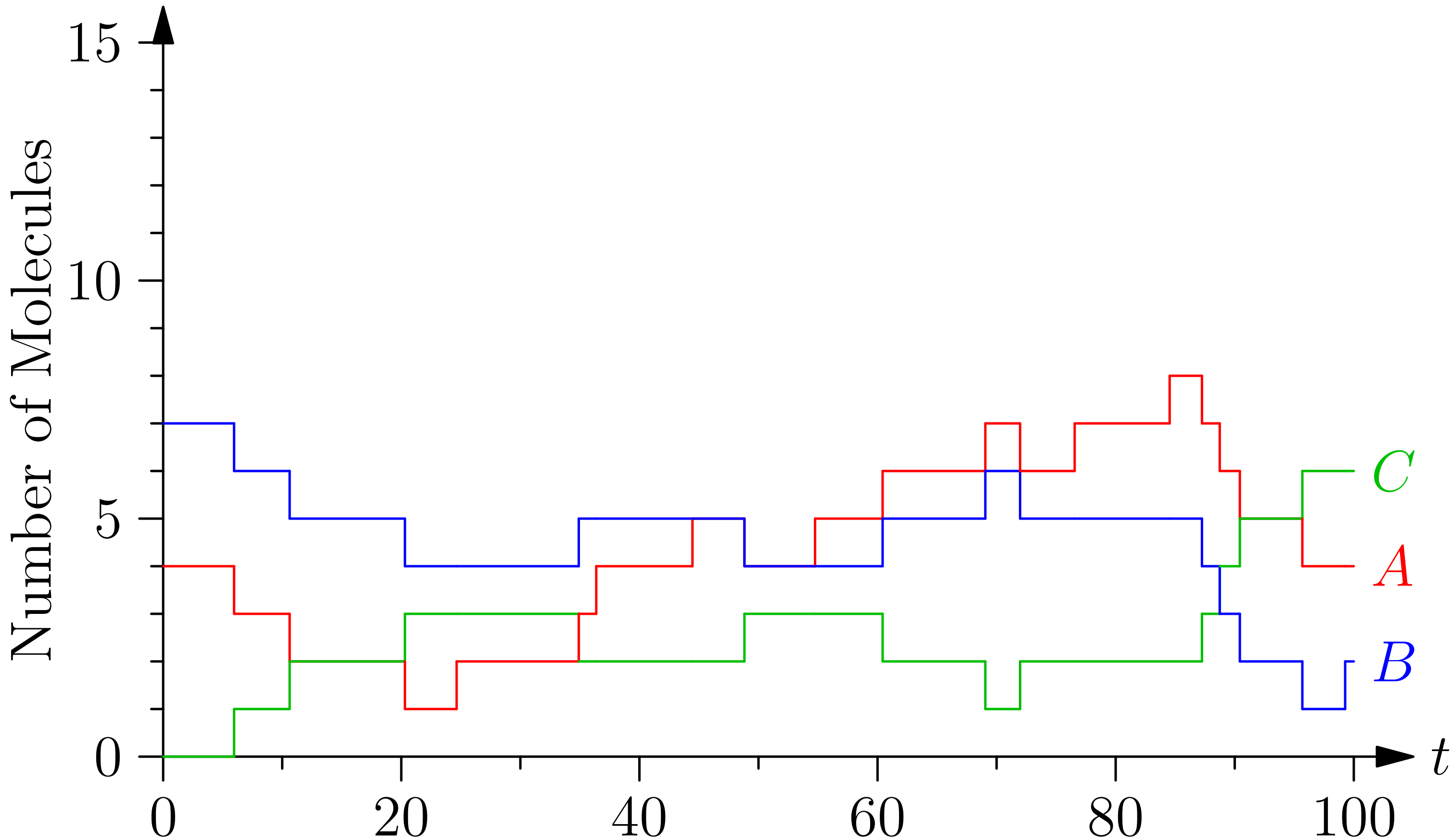

Probabilities of Reaction

- For the reactions

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  where Ω is the volume of reactants

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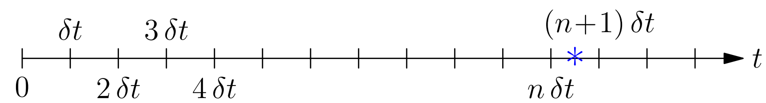
- Suppose we have a point process with a probability of occurring in time δt of $a\,\delta t$

- The probability of the first occurrence happening in time interval $n\,\delta t \leq t \leq (n+1)\,\delta t$ is



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since $\log(1+\Delta) \approx \Delta + O(\Delta^2)$

- If we take the limit $\delta t \to 0$ then

$$f_T(t) = \lim_{\delta t \to 0} \frac{\mathbb{P}\left(n\,\delta t \leq t \leq (n+1)\,\delta t\right)}{\delta t}$$

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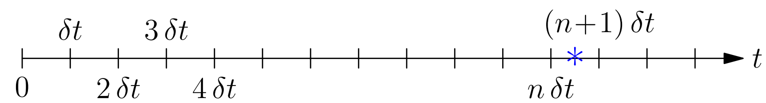

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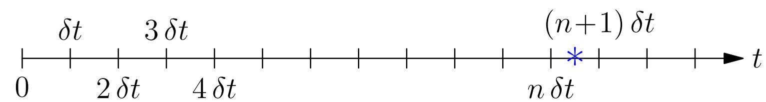

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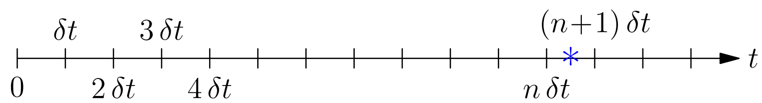

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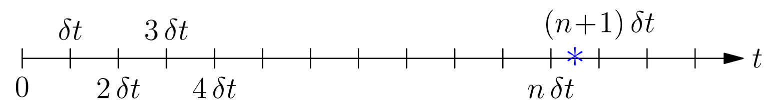

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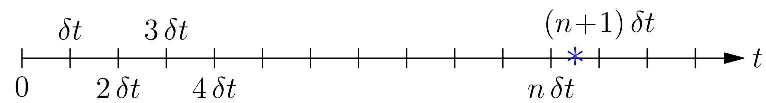

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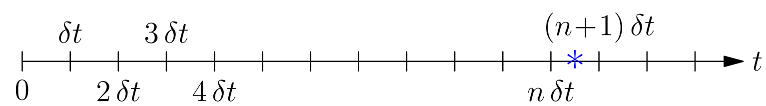

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- If there are K event with probability of occurring in time δt of $a_i\,\delta t$ then the probability of any one of them happening in time δt is $a_0\,\delta t$ where $a_0 = \sum_{i=1}^{K} a_i$

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$$= 1 - \left(1 - \delta t \sum_i a_i + (\delta t)^2 \sum_{i \neq j} a_i\,a_j + \cdots \right) = \sum_i a_i \delta t + O((\delta t)^2) \Bigg)$$

- Thus the distribution of time, T, of the first event is given by

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- The probability of that first even being event i is

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$$\left(1 - \mathbb{P}\left(\forall_i \mathcal{E}_i\right) = 1 - \prod_i (1 - a_i \delta t) = 1 - (1 - a_1\,\delta t)(1 - a_2\,\delta t)\cdots(1 - a_K\,\delta t) \right.$$

$$\left. = 1 - \left(1 - \delta t \sum_i a_i + (\delta t)^2 \sum_{i \neq j} a_i\,a_j + \cdots \right) = \sum_i a_i \delta t + O((\delta t)^2) \right)$$

- Thus the distribution of time, T, of the first event is given by

$$f_T(t) = a_0\,\mathrm{e}^{-a_0\,t}$$

- The probability of that first even being event i is

$$\mathbb{P}\left(\text{first event is event } i\right) = \frac{a_i}{a_1 + \cdots + a_K} = \frac{a_i}{a_0}$$

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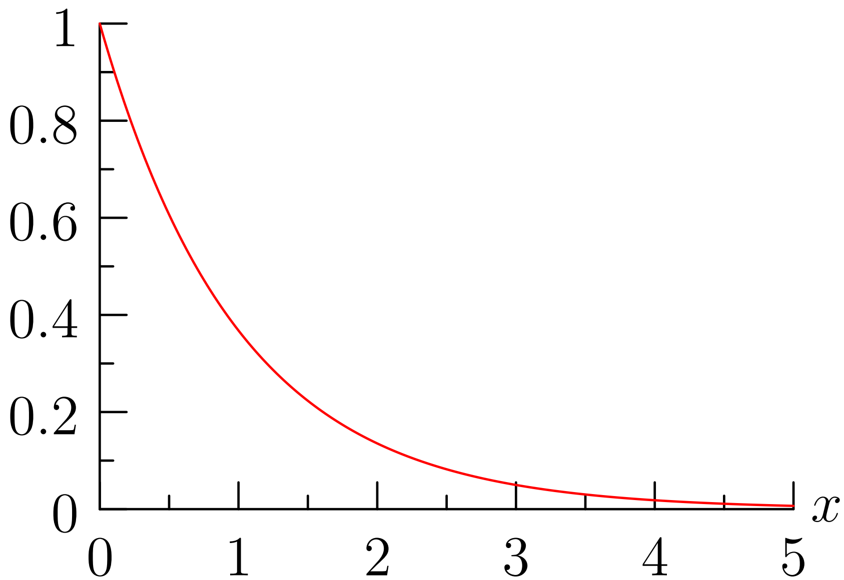

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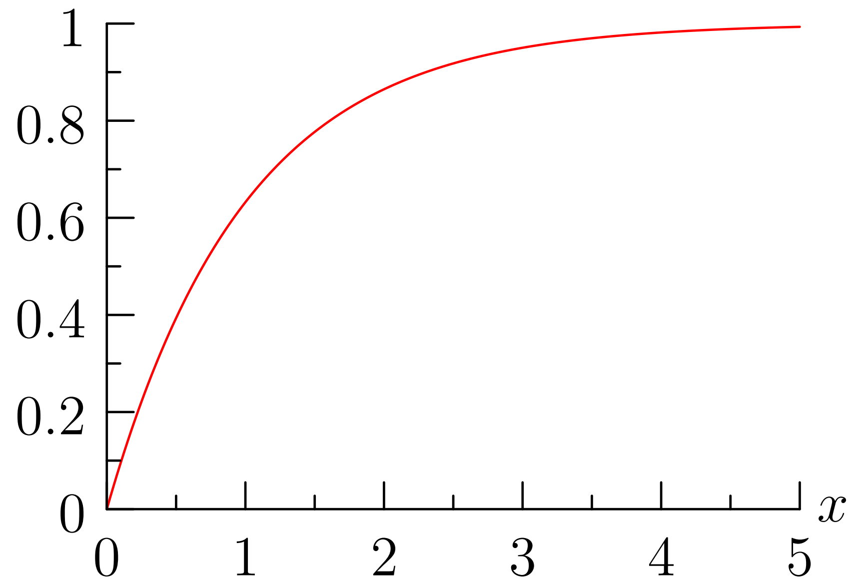

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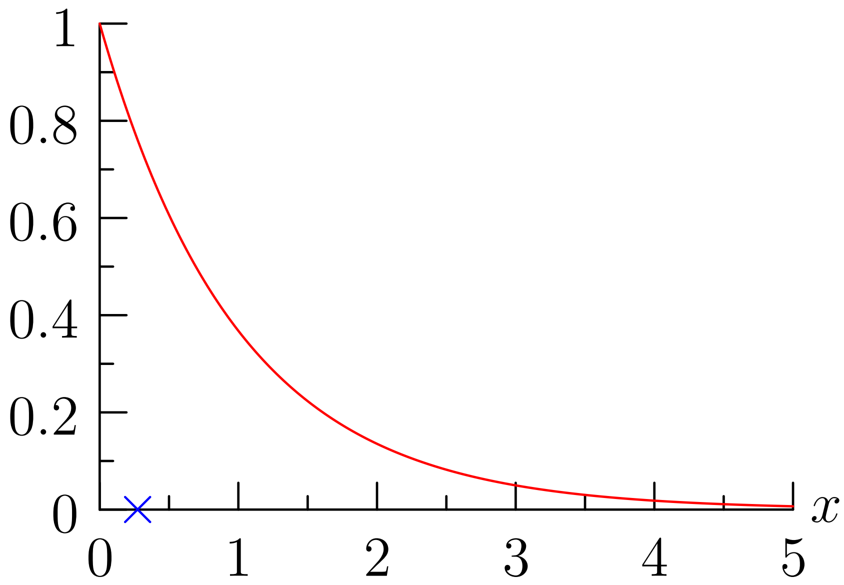

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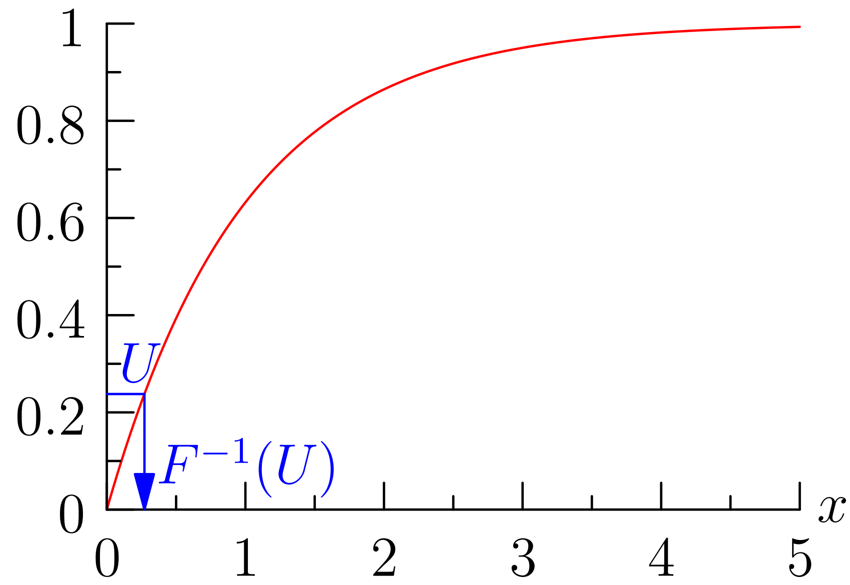

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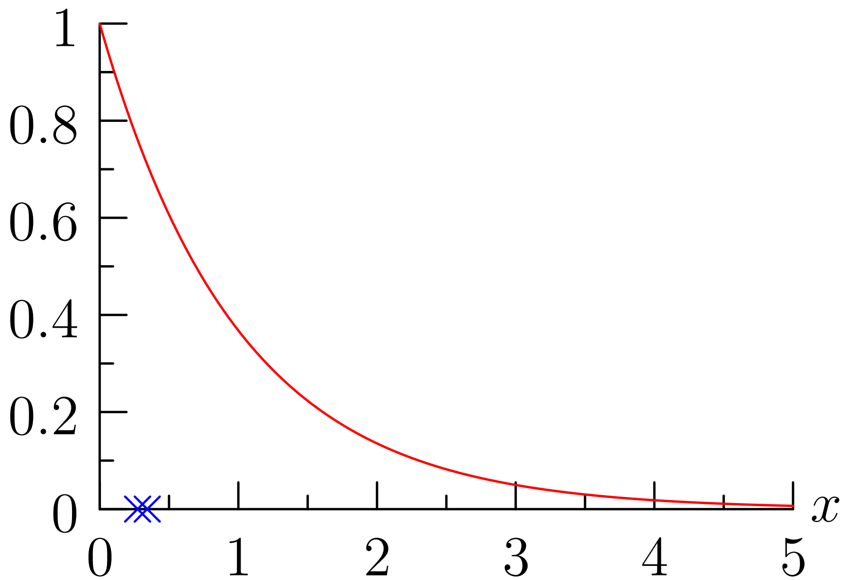

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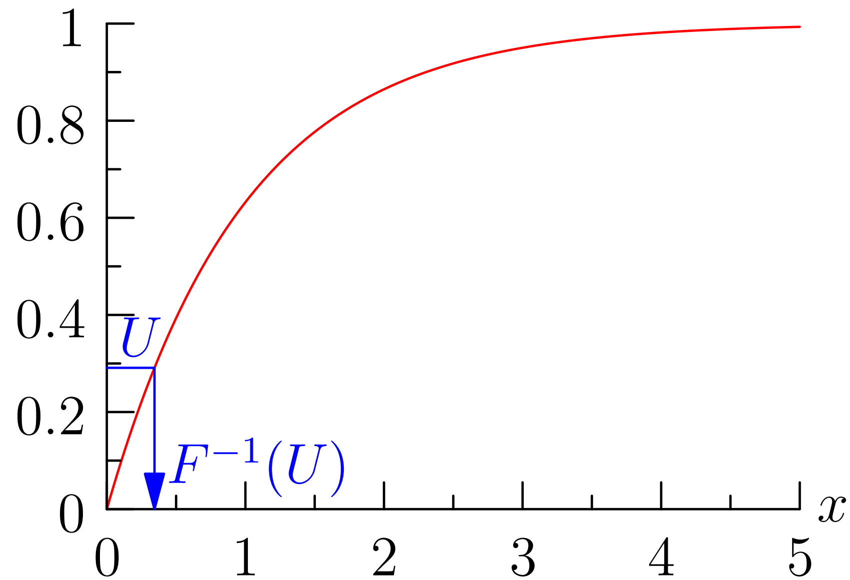

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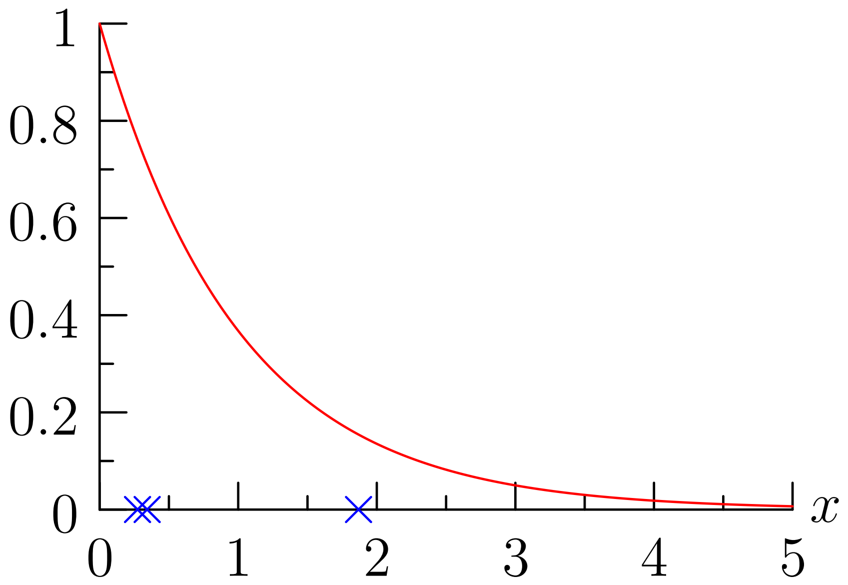

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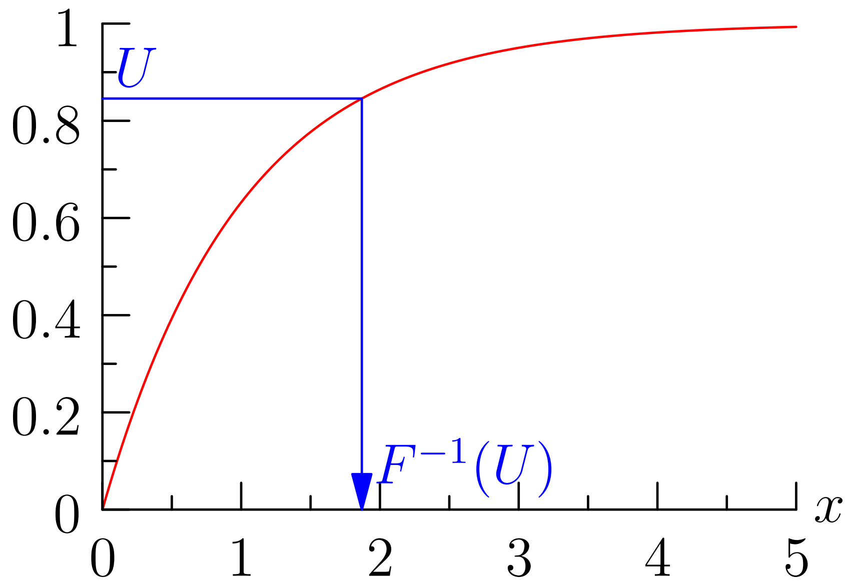

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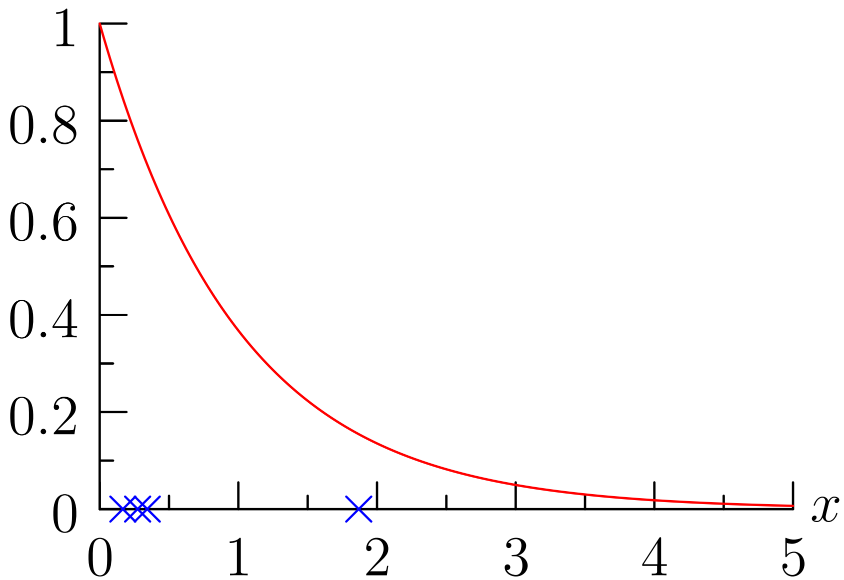

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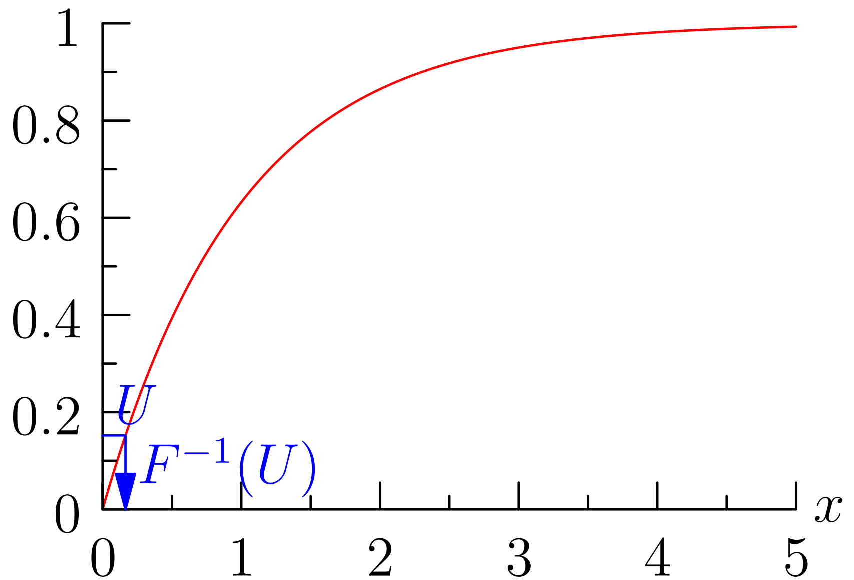

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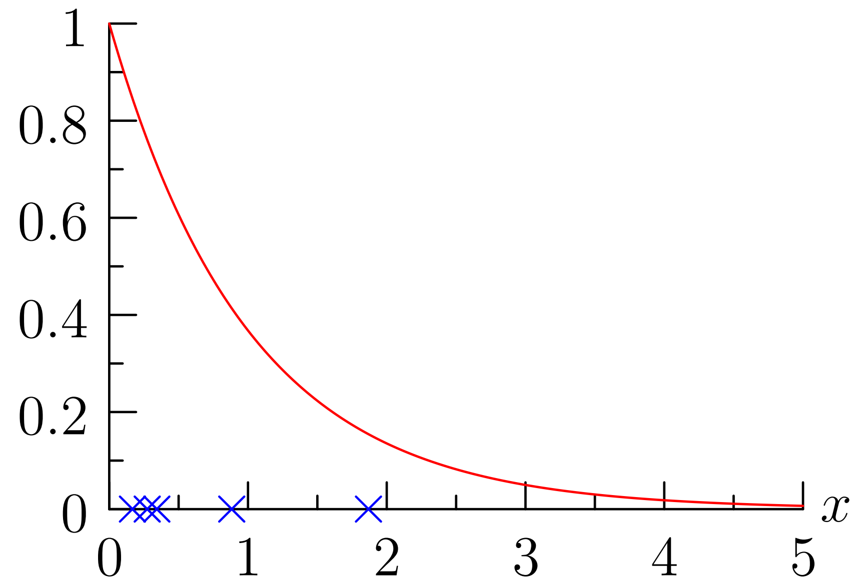

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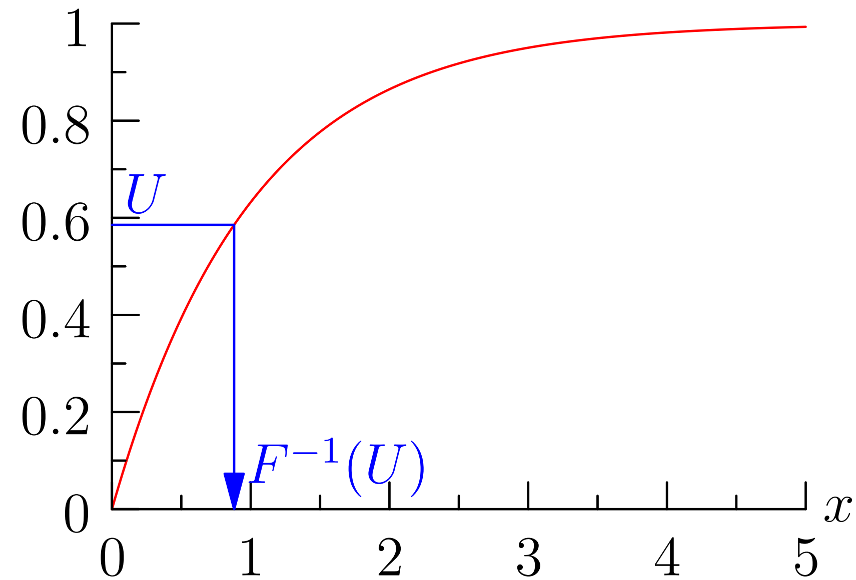

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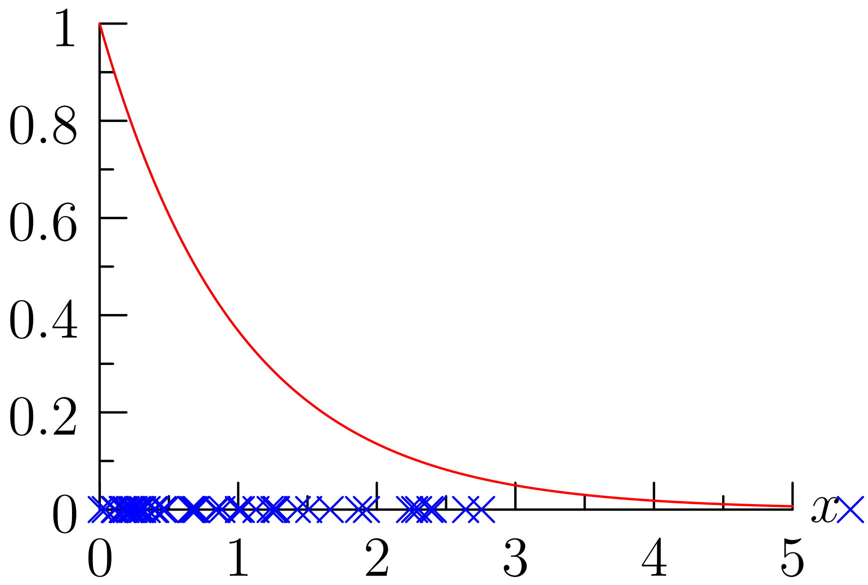

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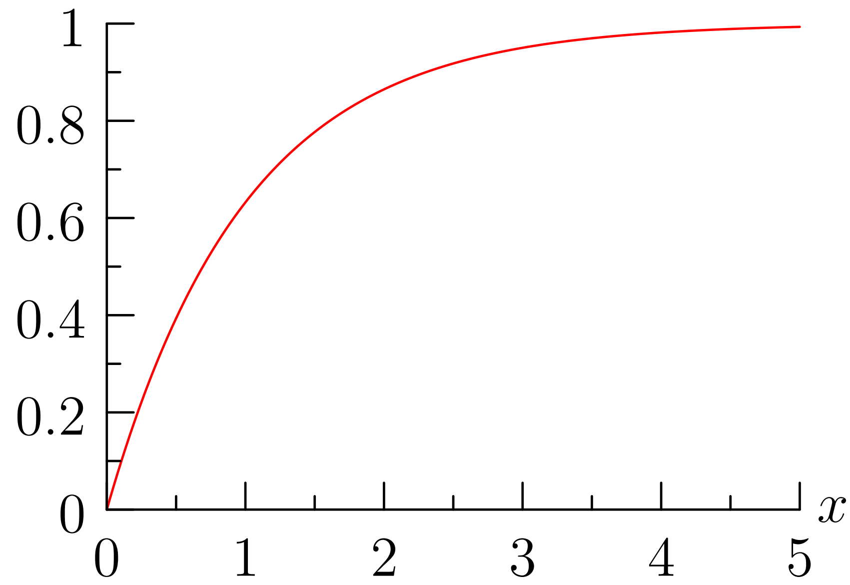

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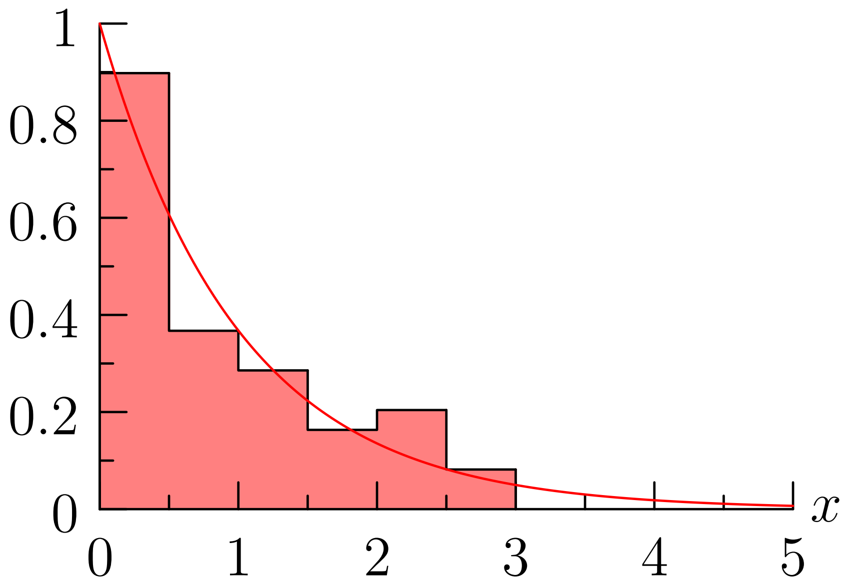

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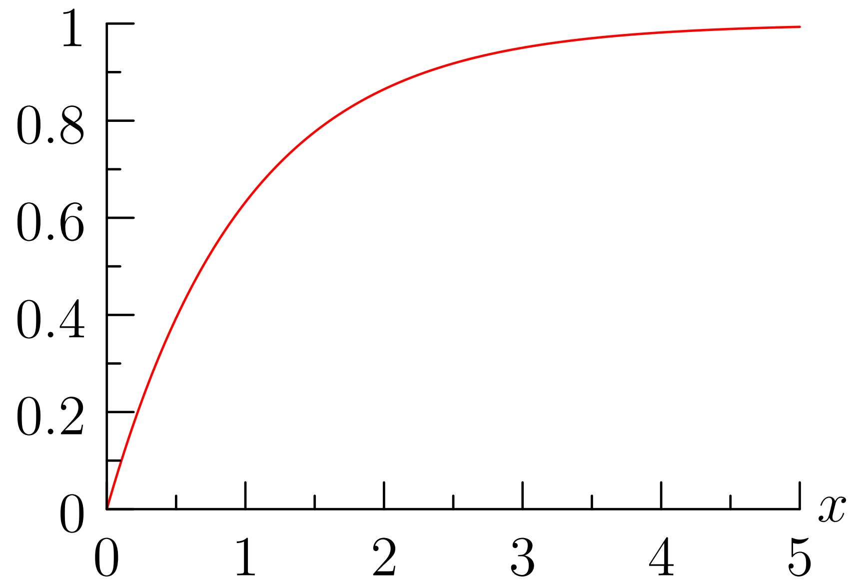

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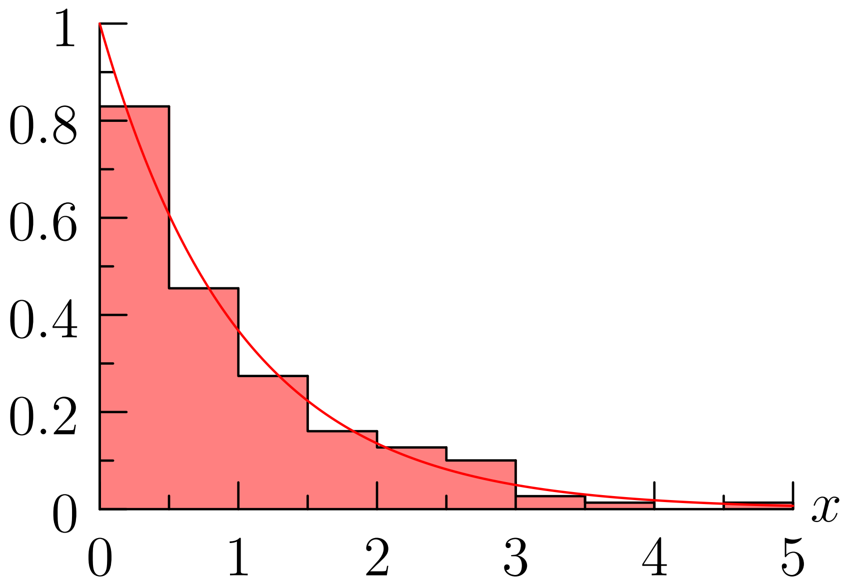

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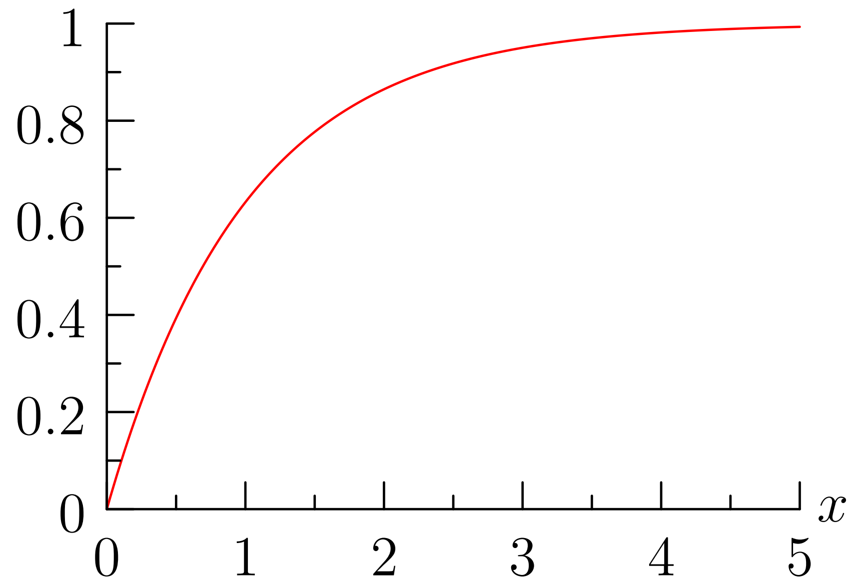

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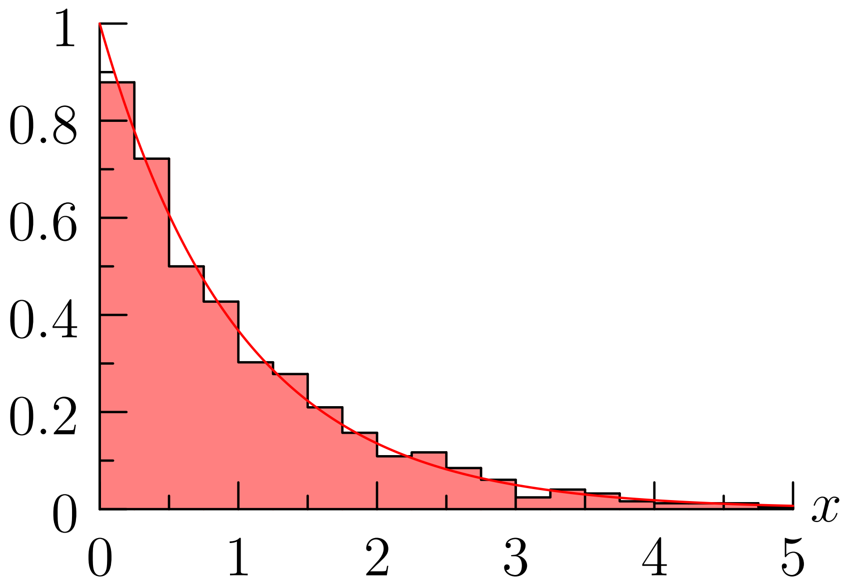

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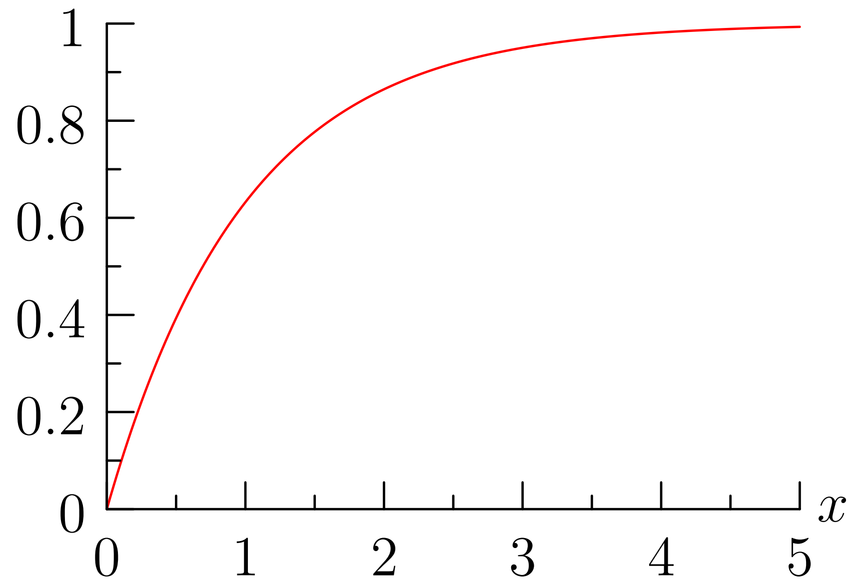

Exponential Deviates

- For exponential deviates we generate a random deviate $U \sim \mathrm{Uni}(0,1)$

- Then to generate $X \sim \mathrm{Exp}(x|1) = \mathrm{e}^{-x}$ (with $F_X(x|1) = 1 - \mathrm{e}^{-x}$) we use

$$X = F_X^{-1}(U) = -\log(1-U)$$

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Gillespie's Algorithm

- To simulate a reaction equation we take $\boldsymbol{X}(t) = (X_1(t), X_2(t), \ldots, X_N(t))$ to be the count of molecules at time t
- We assume we have K reactions which we label R_j
- The probability of a reaction occurring in a volume V in time interval $[t, t + \mathrm{d}t]$ we denote by $a_j(\boldsymbol{X}(t))\,\mathrm{d}t$—this is known as the propensity
- The propensity is equal to the reaction rate times the product of concentrations of chemical species that react together (i.e. on the left-hand side) times the volume

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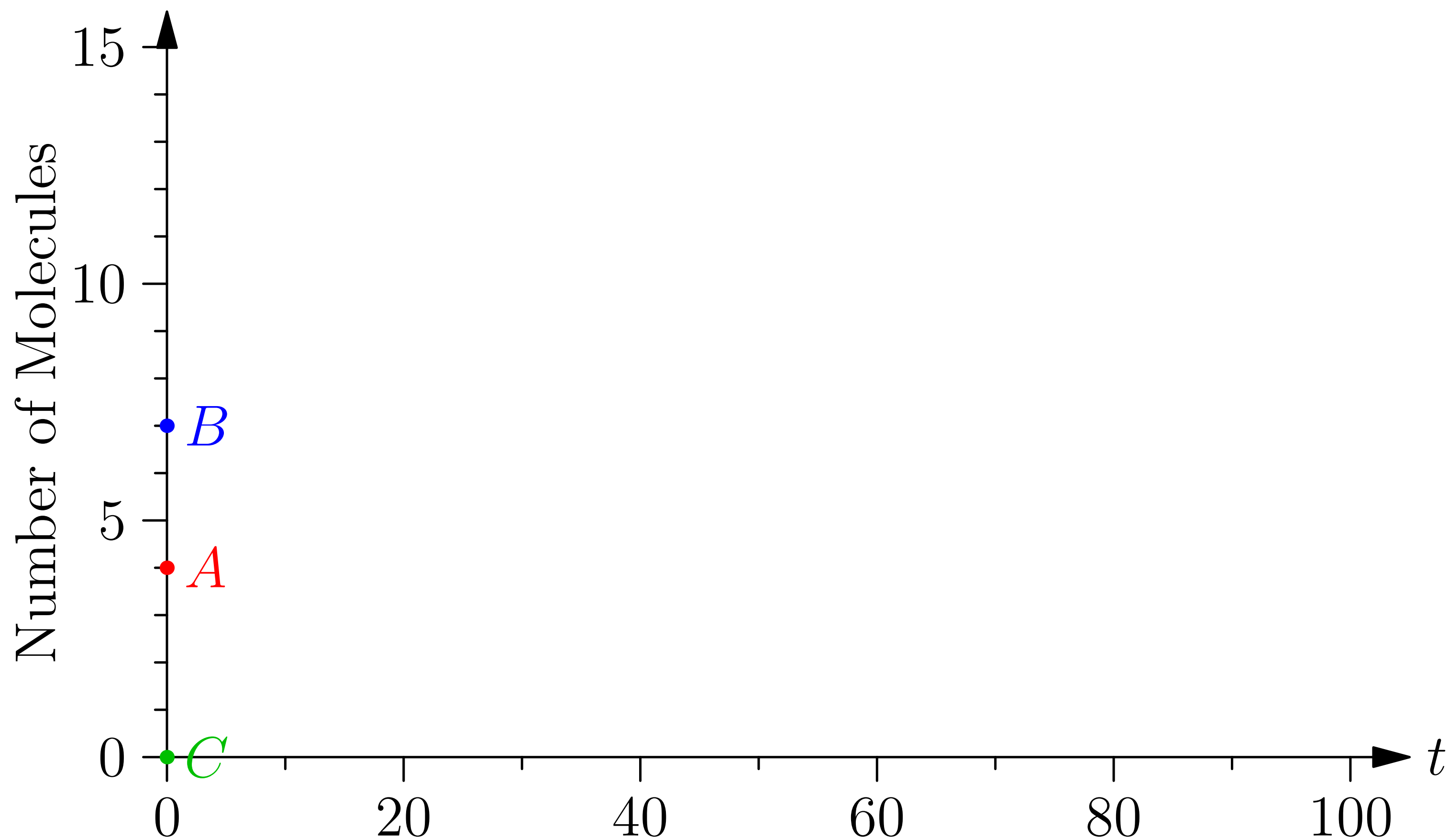

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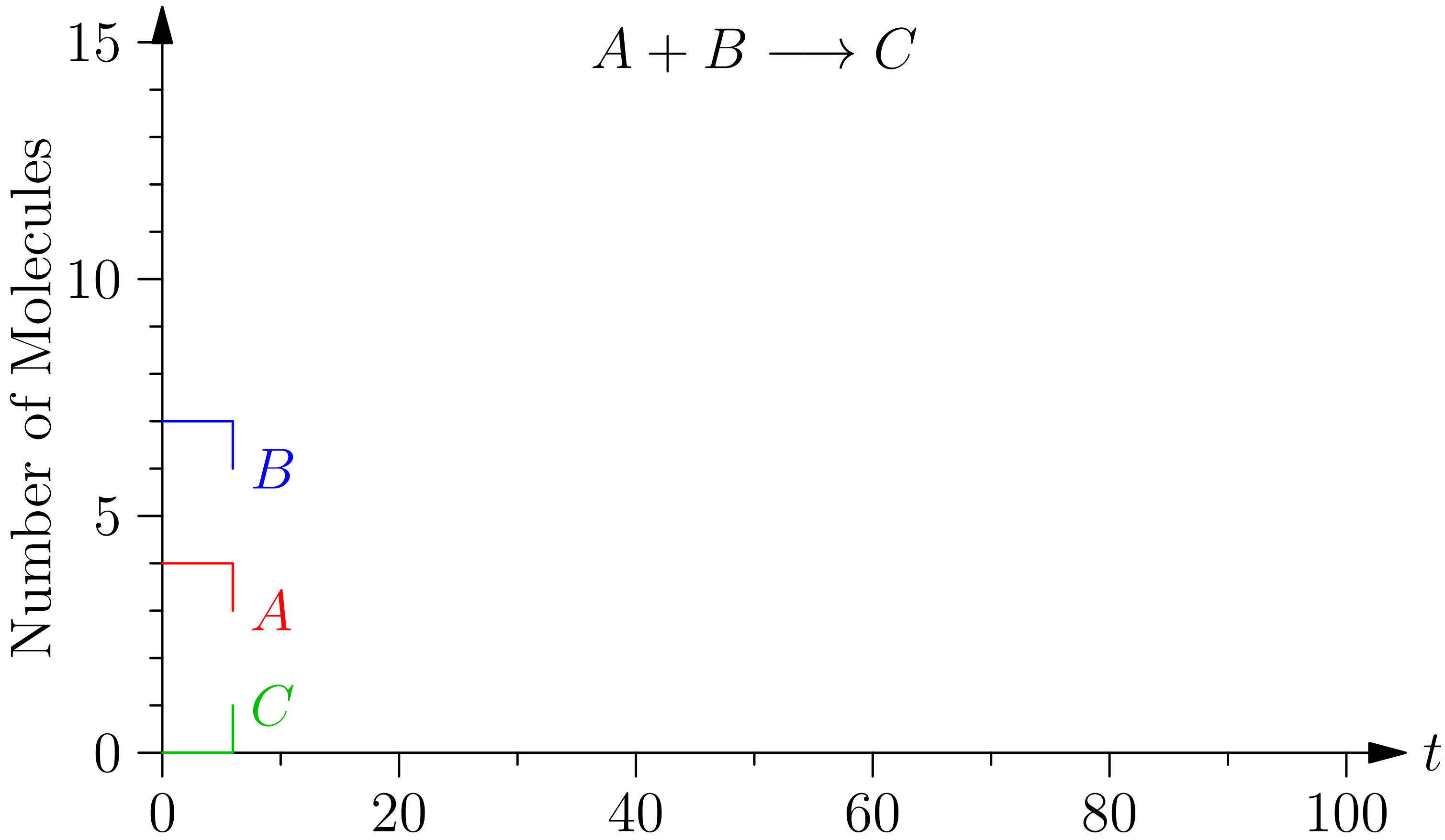

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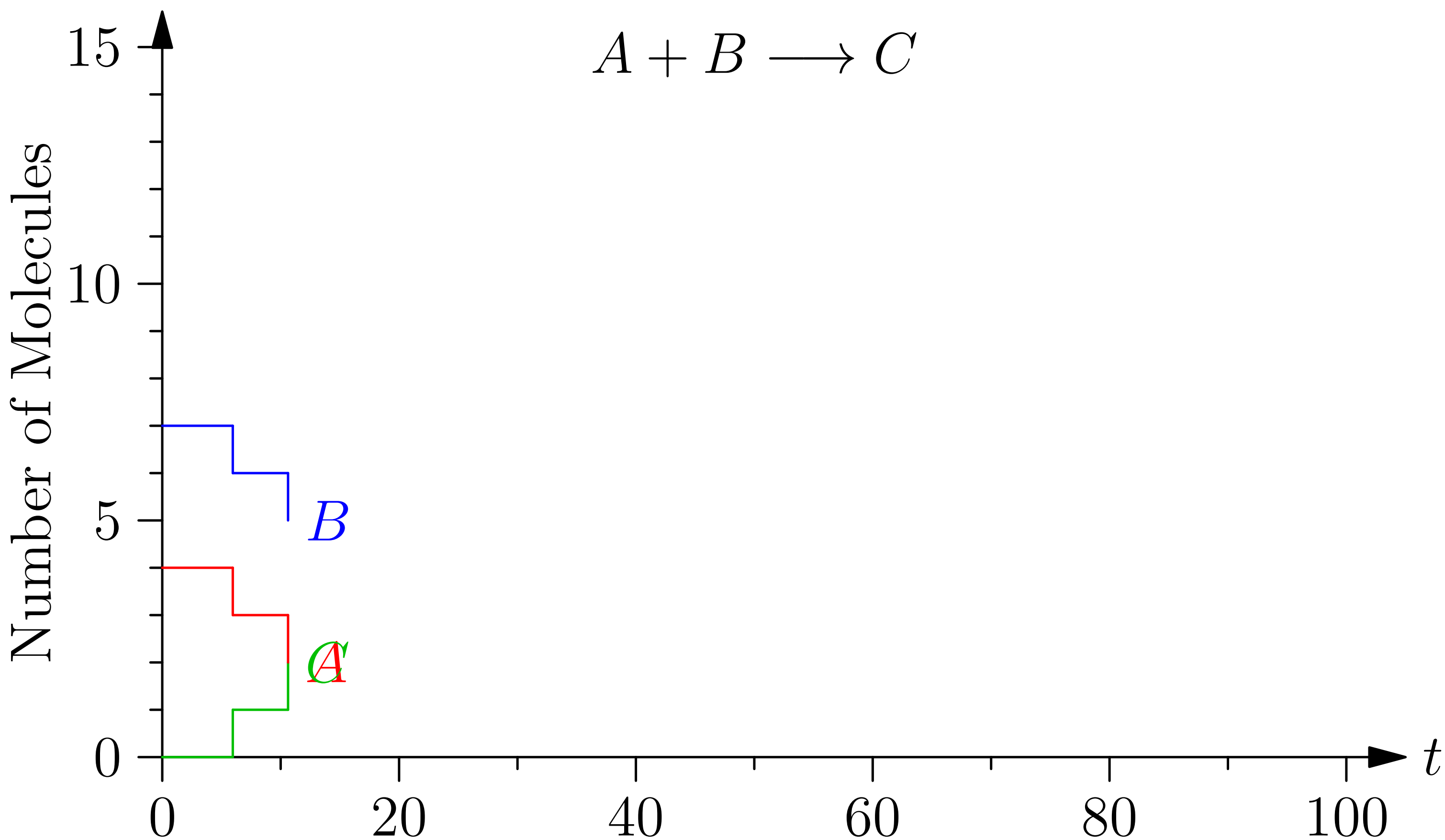

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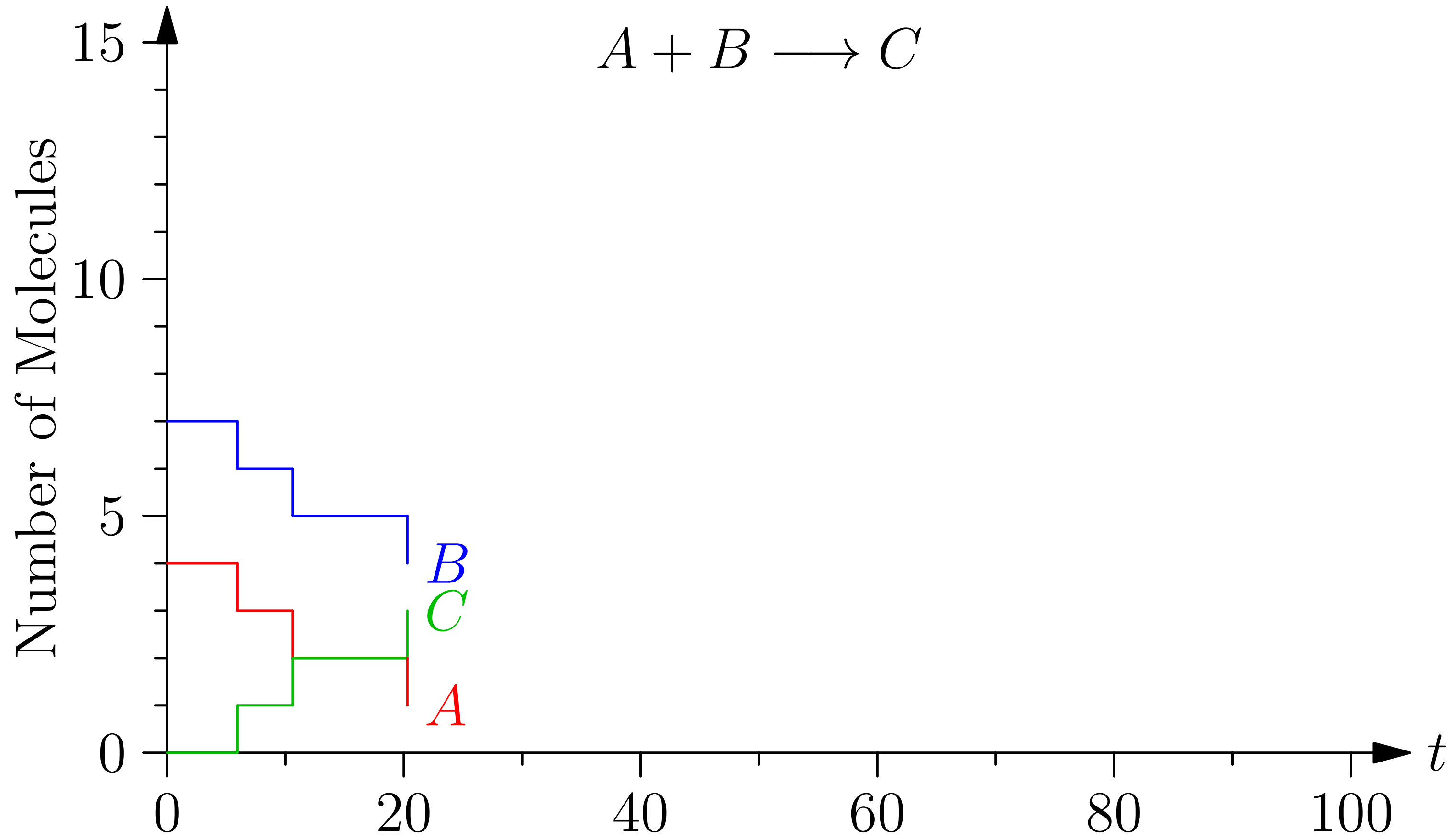

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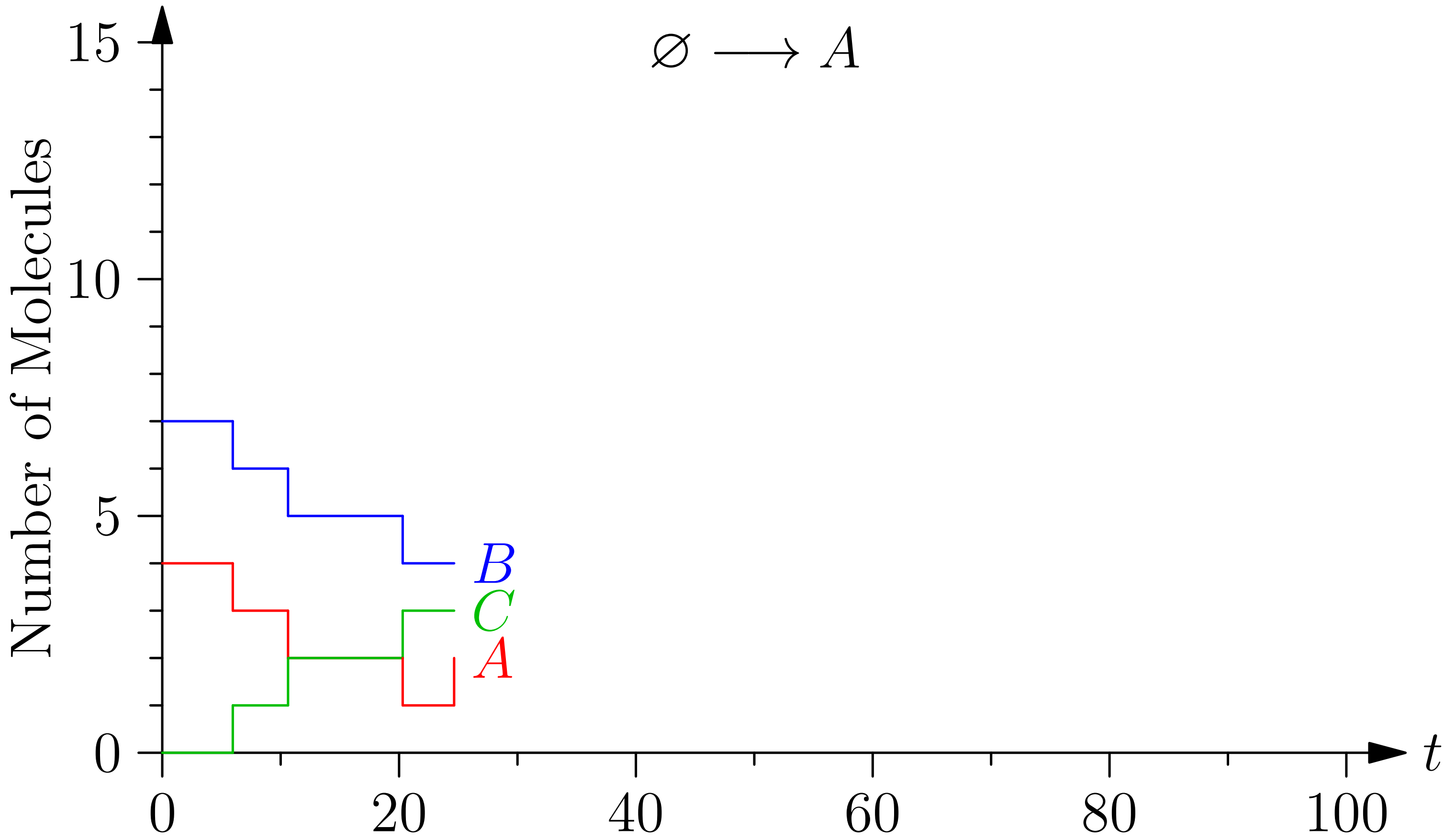

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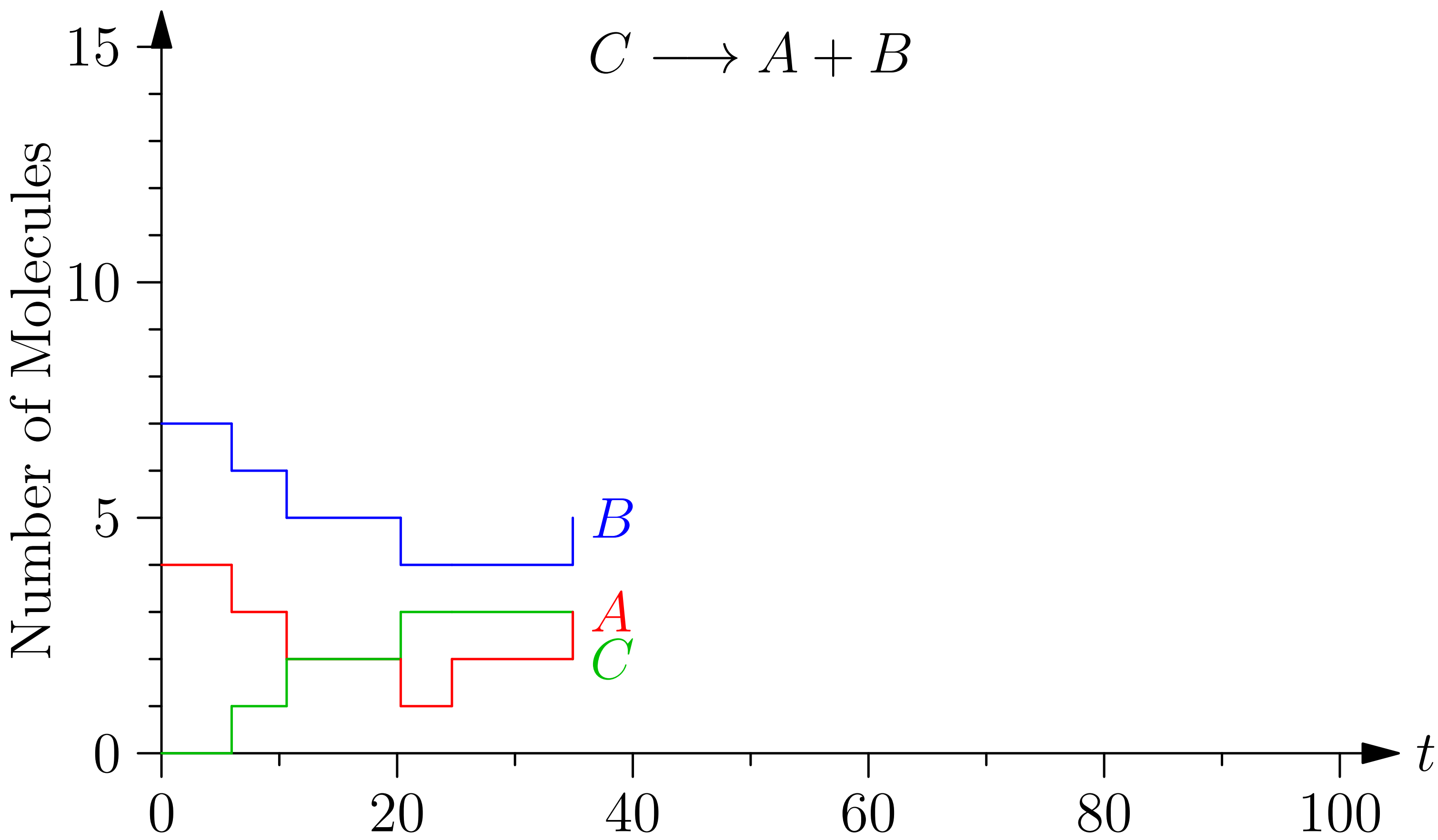

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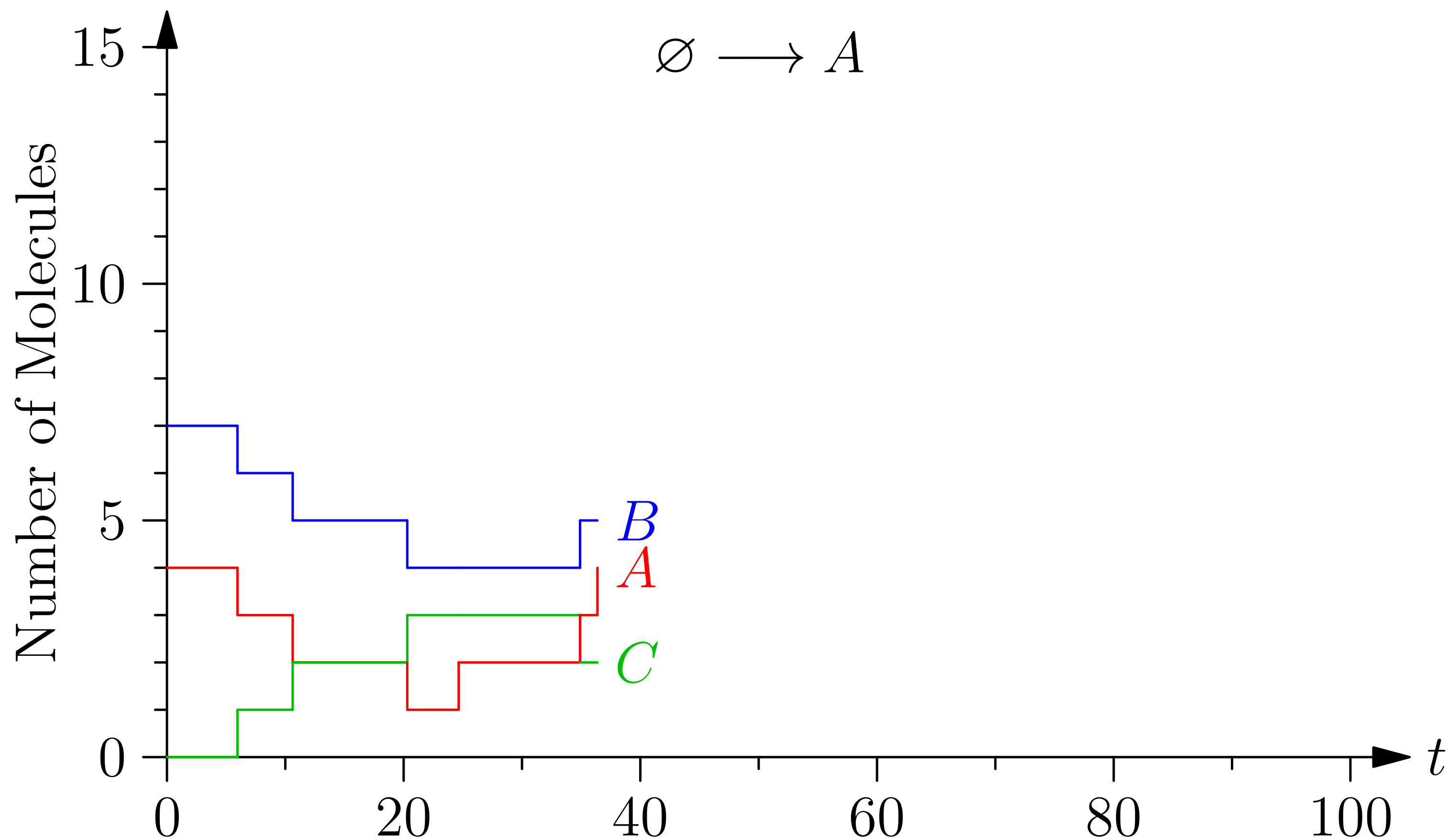

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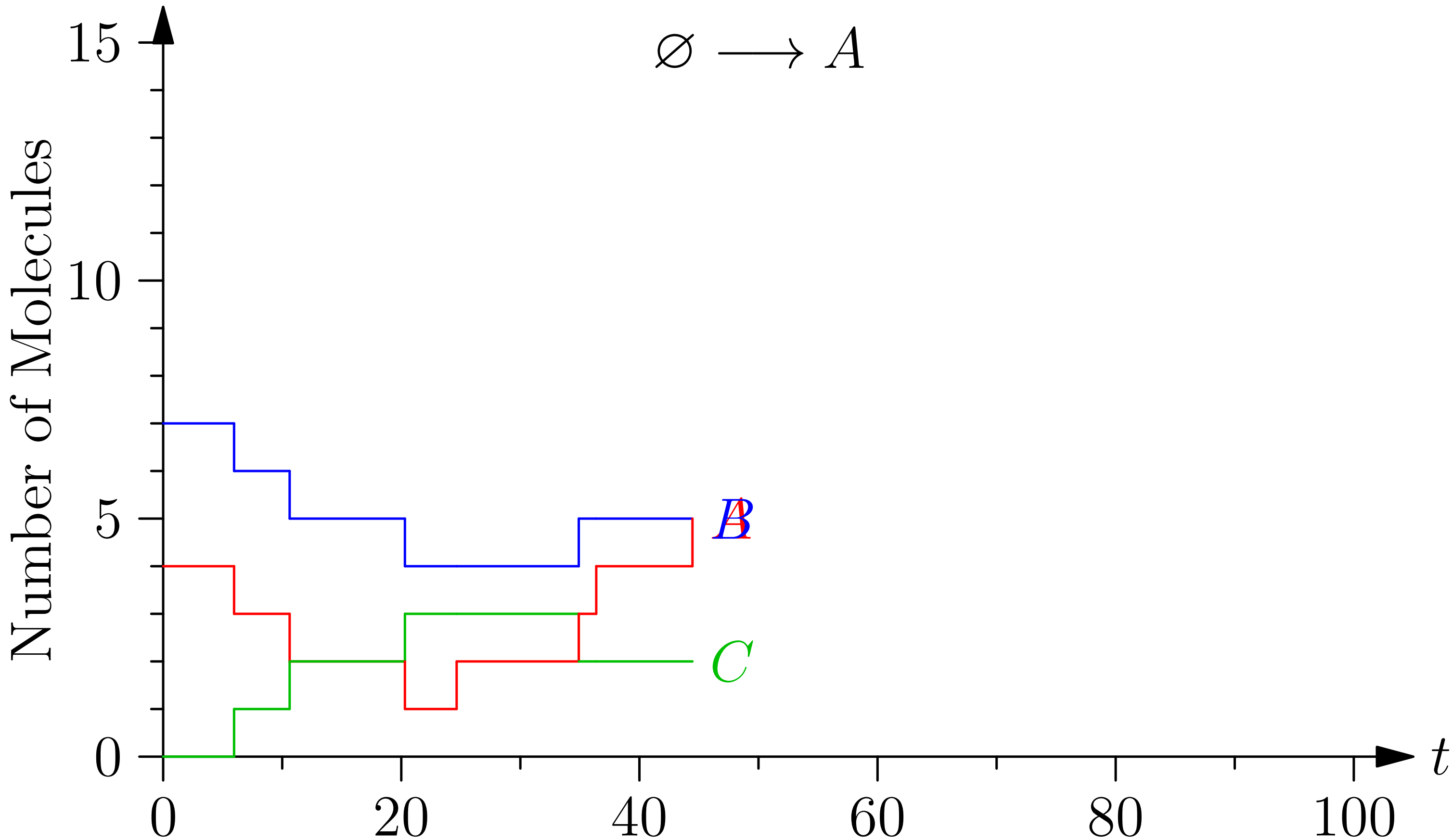

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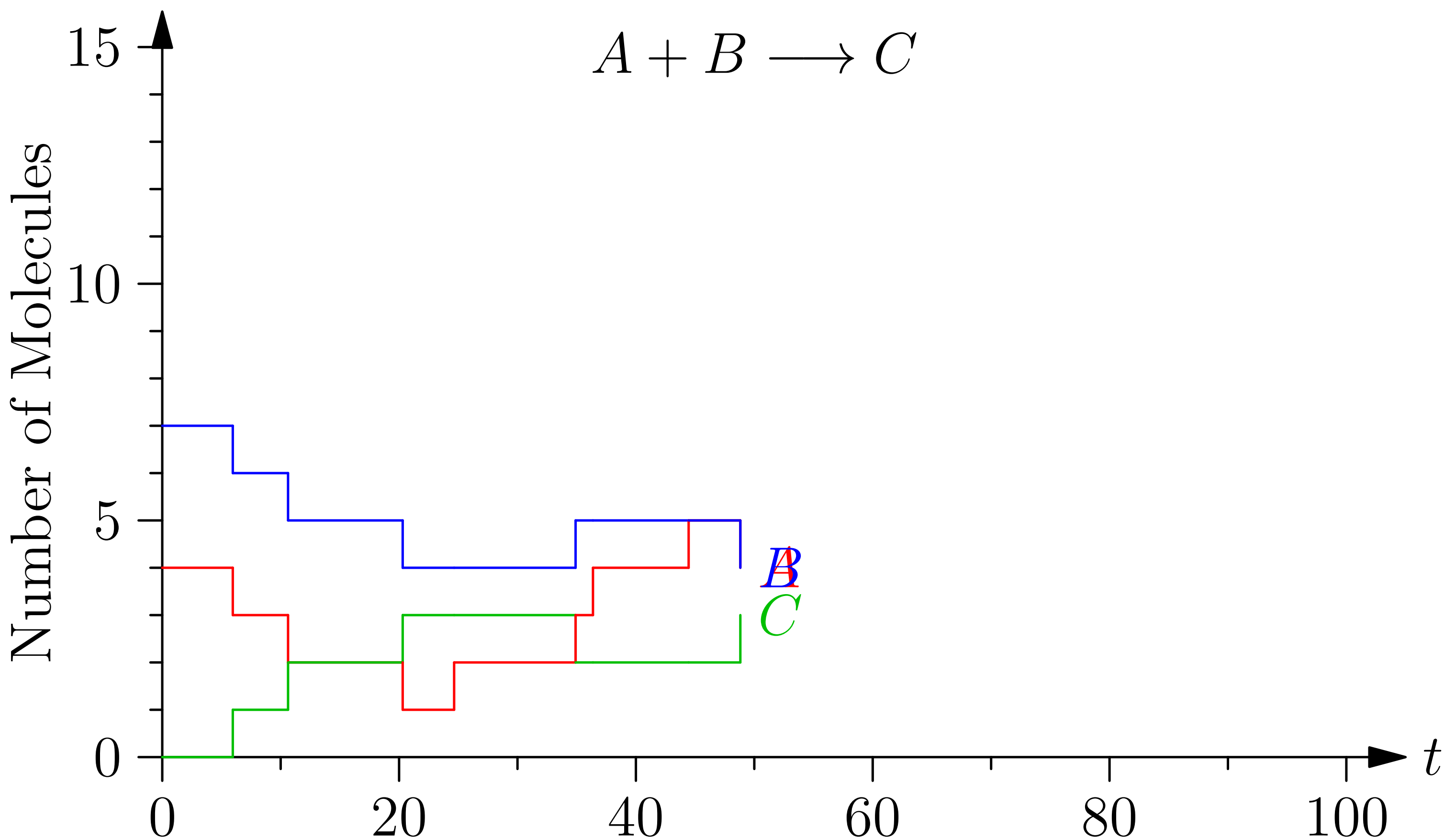

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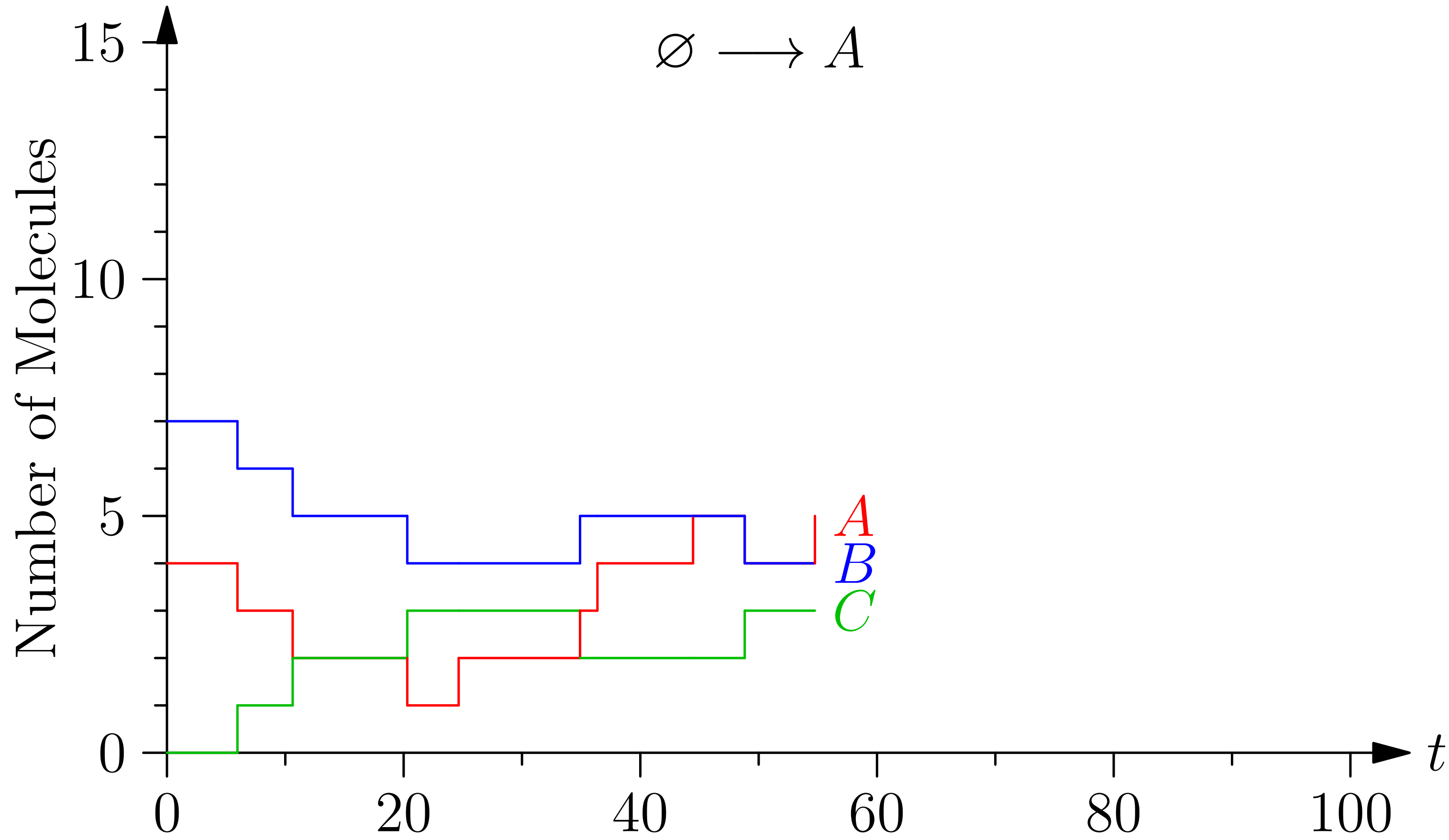

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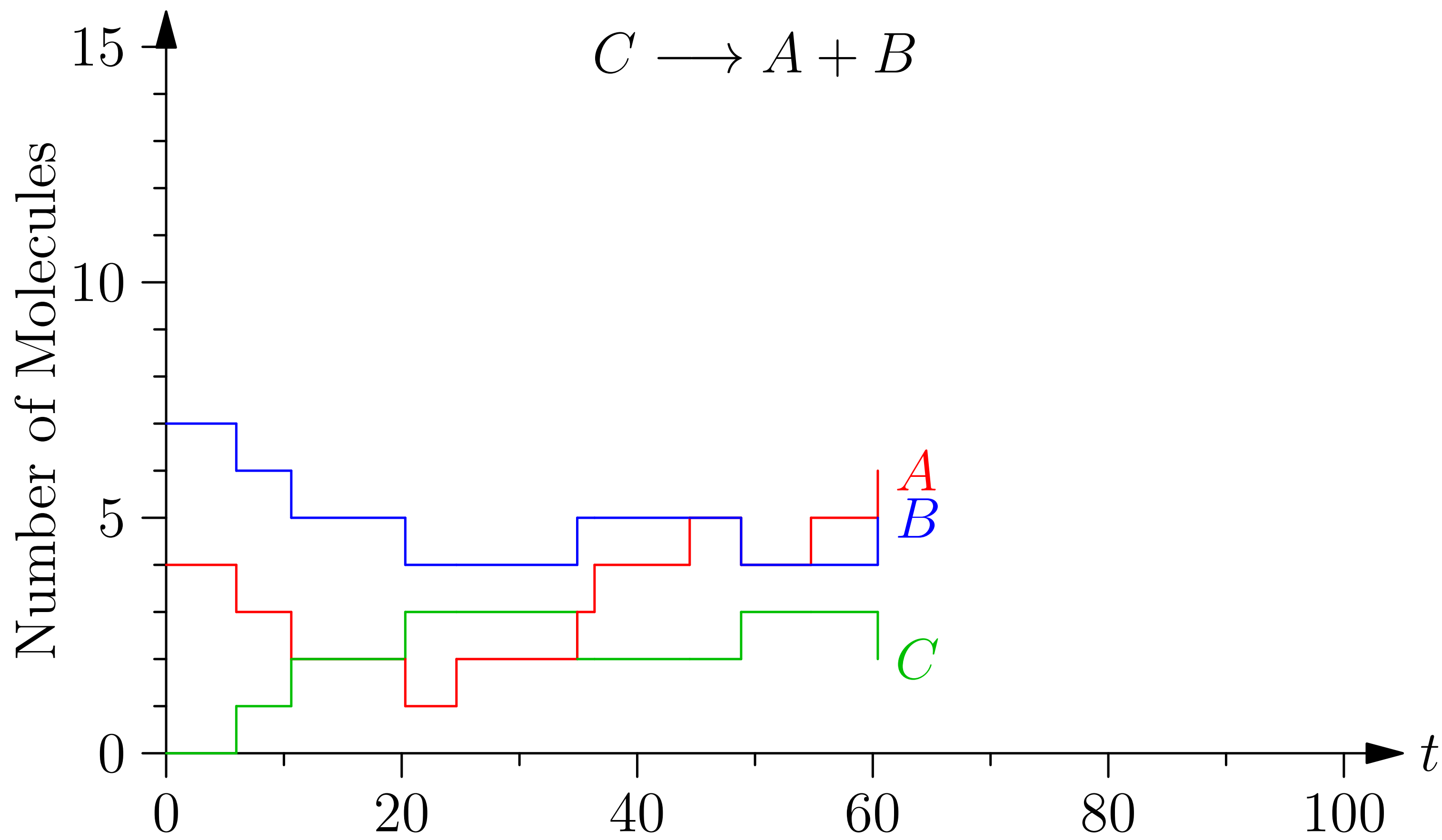

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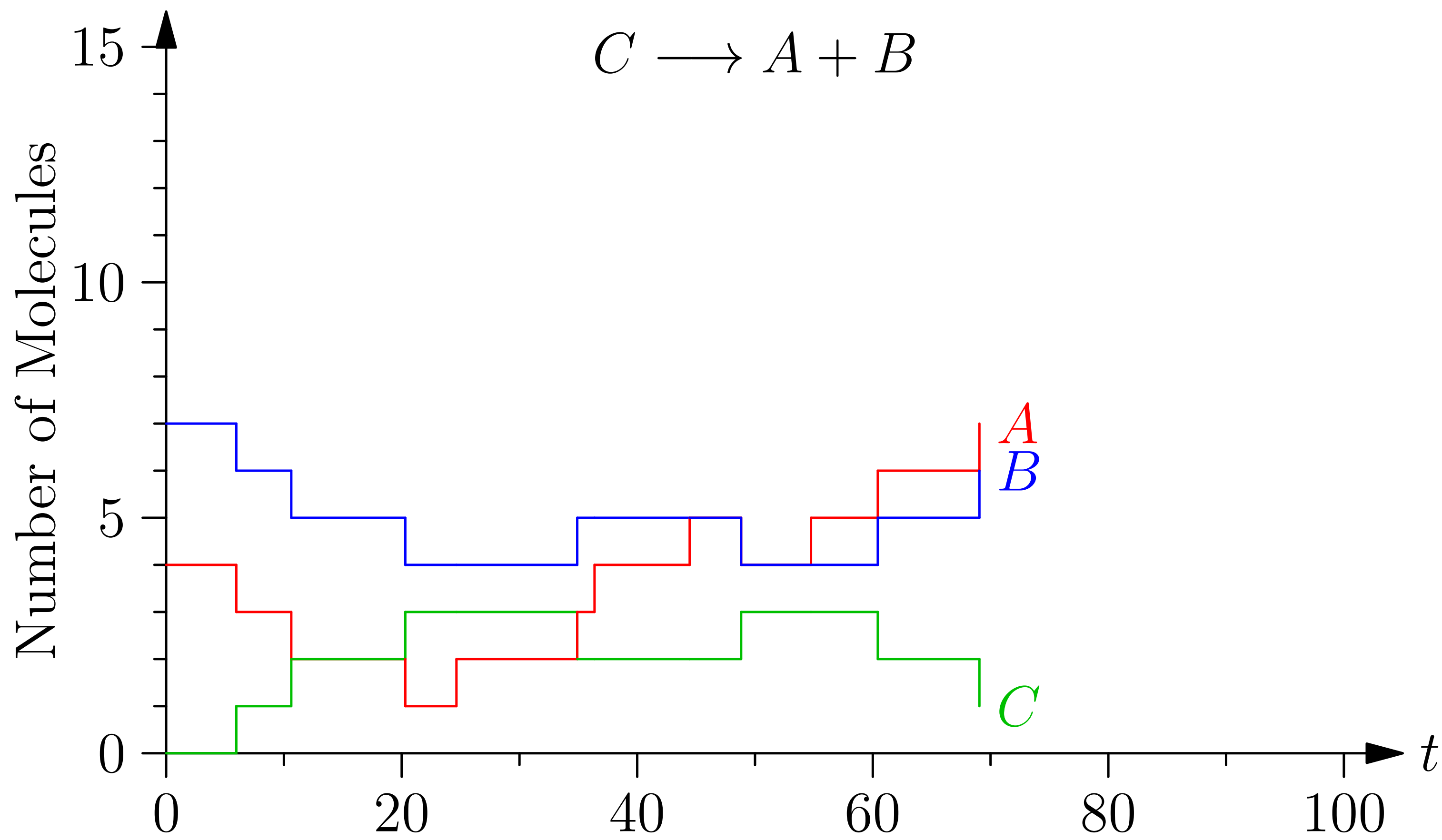

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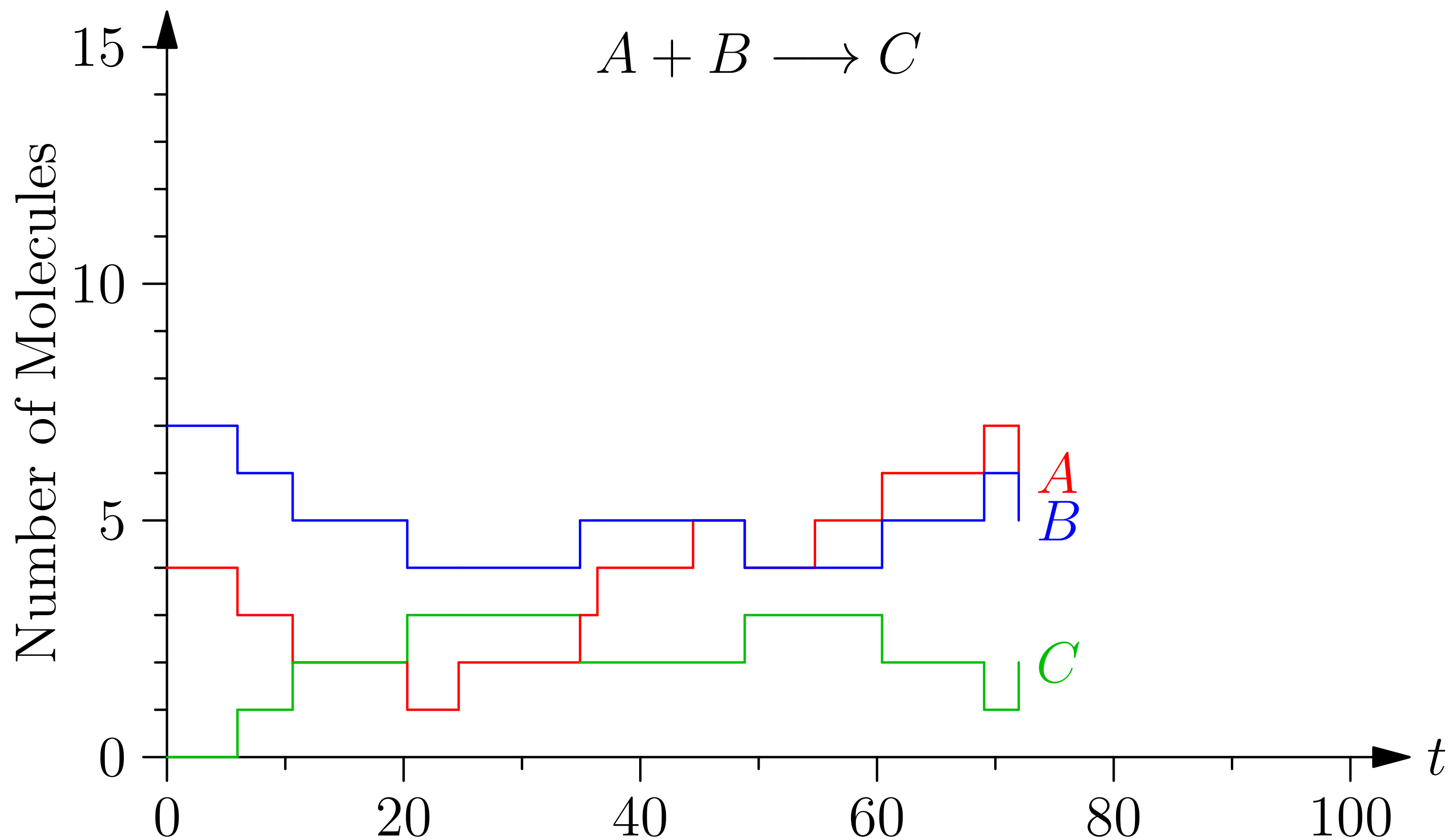

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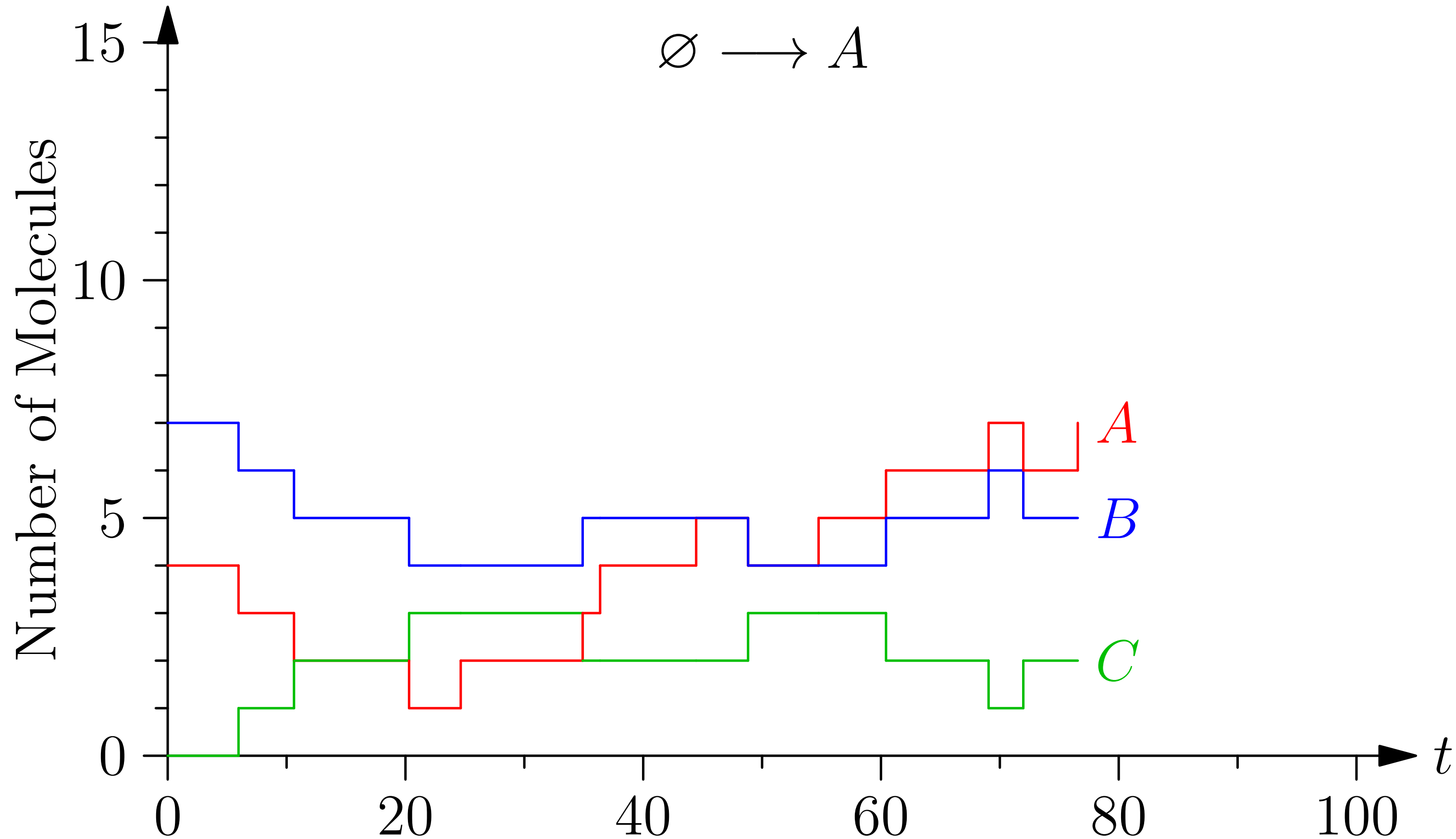

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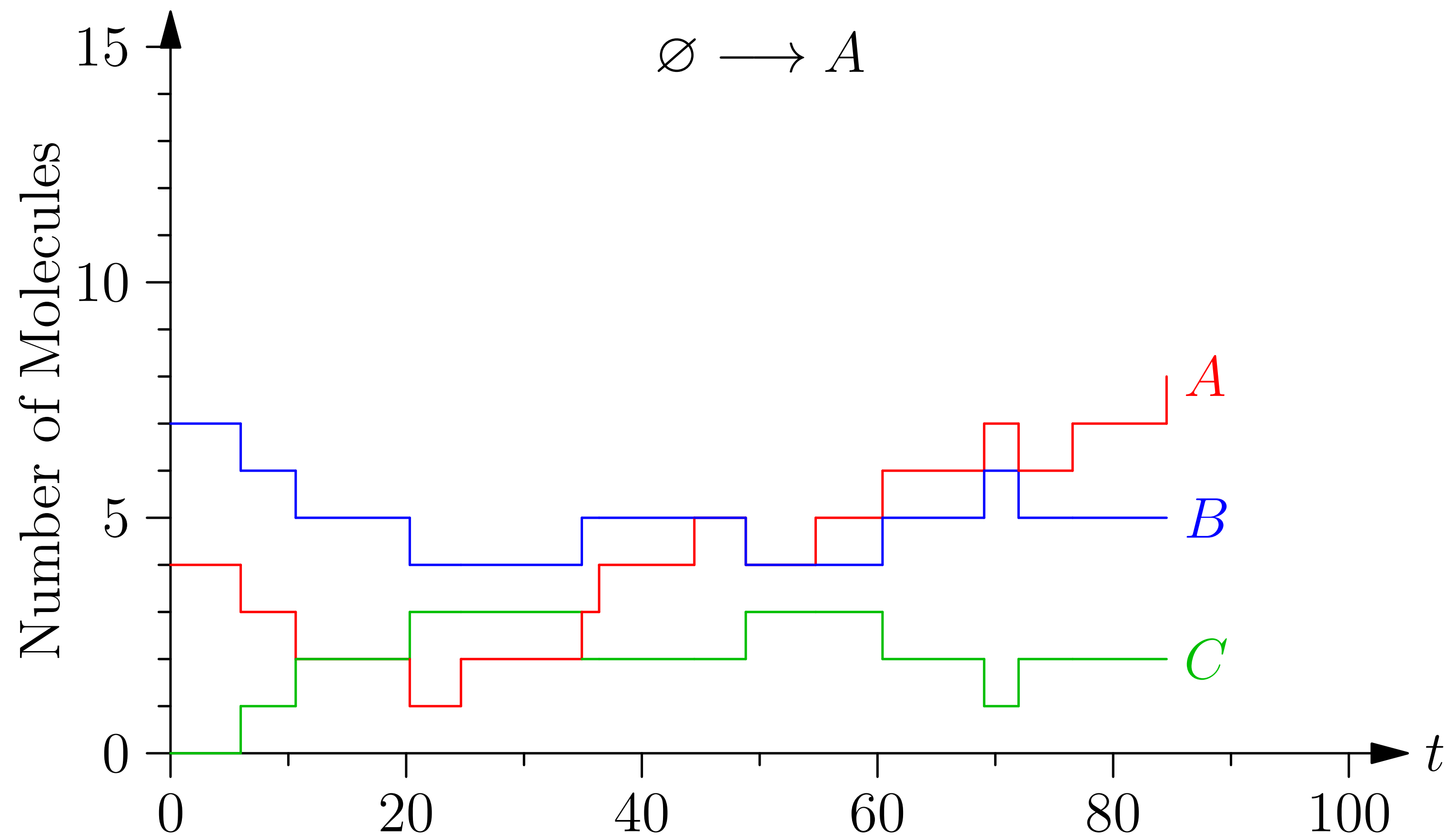

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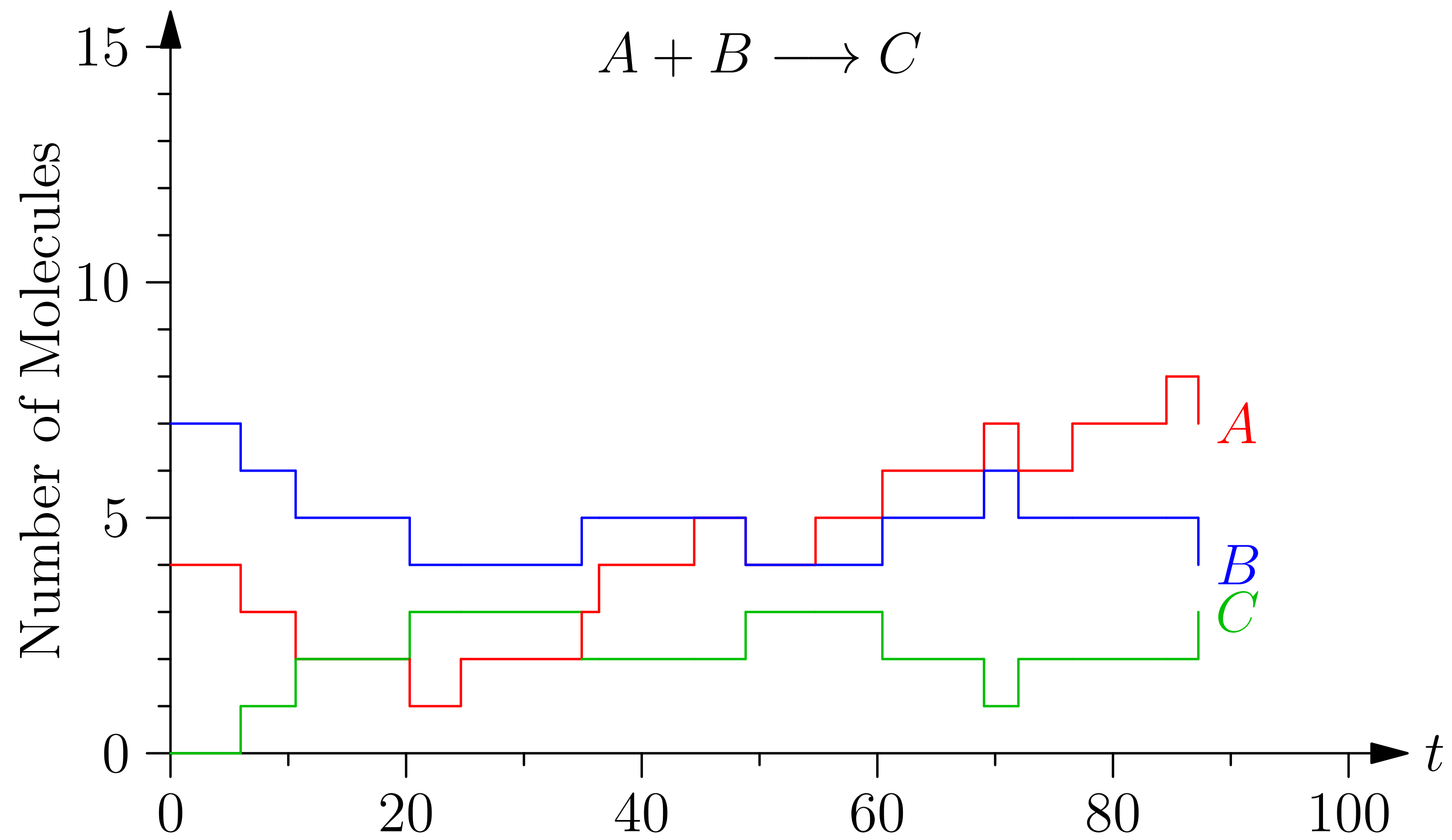

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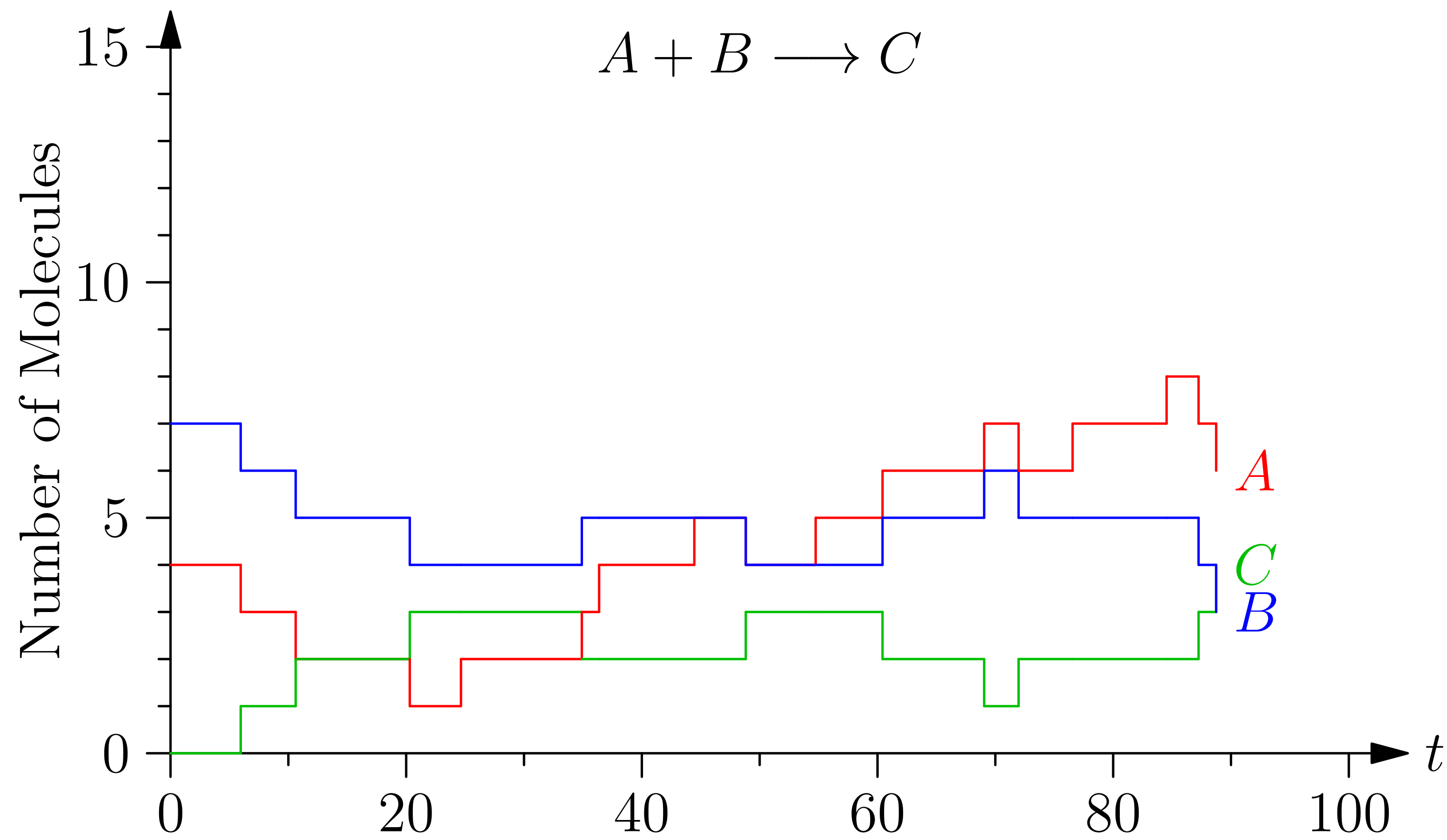

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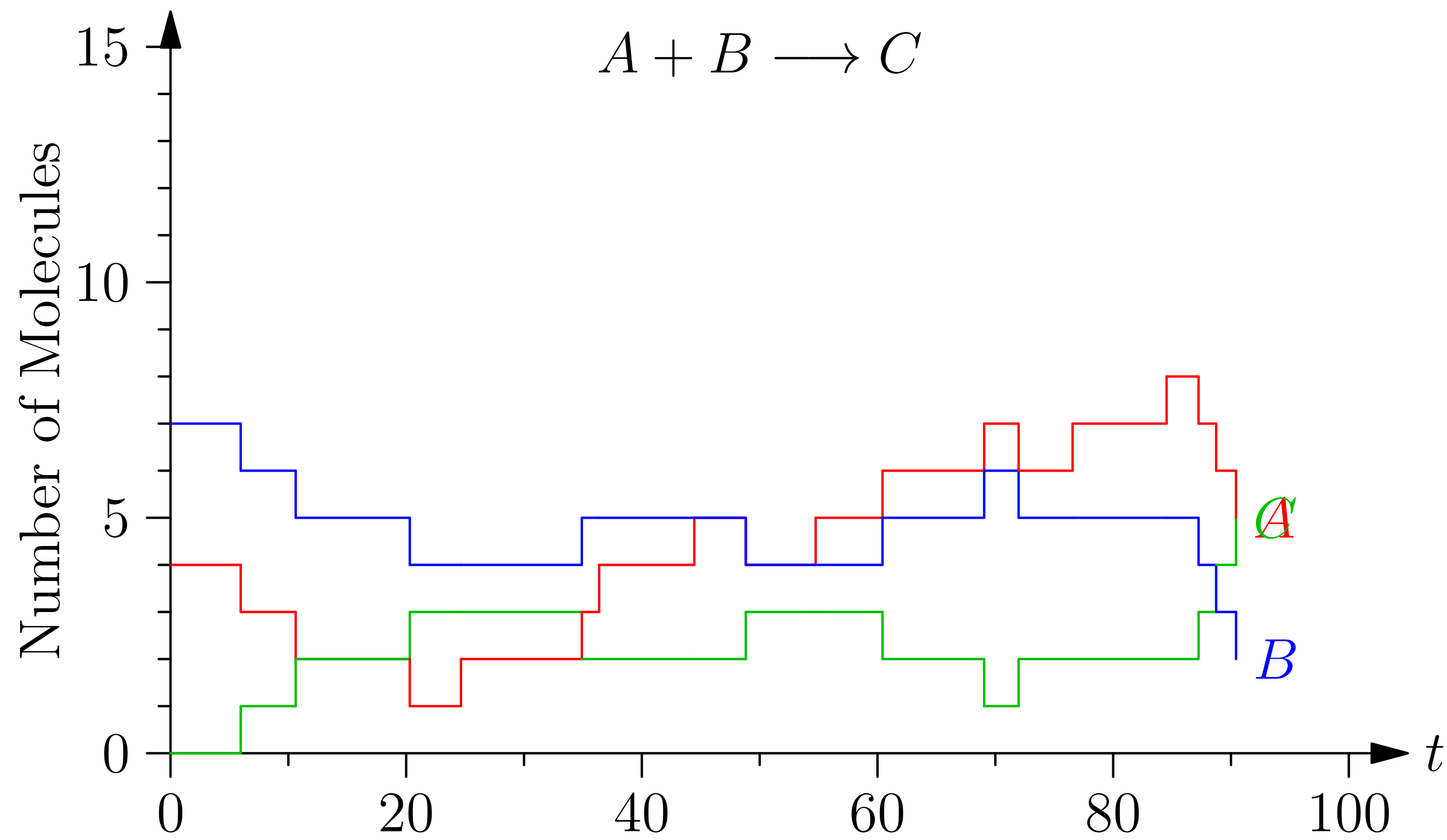

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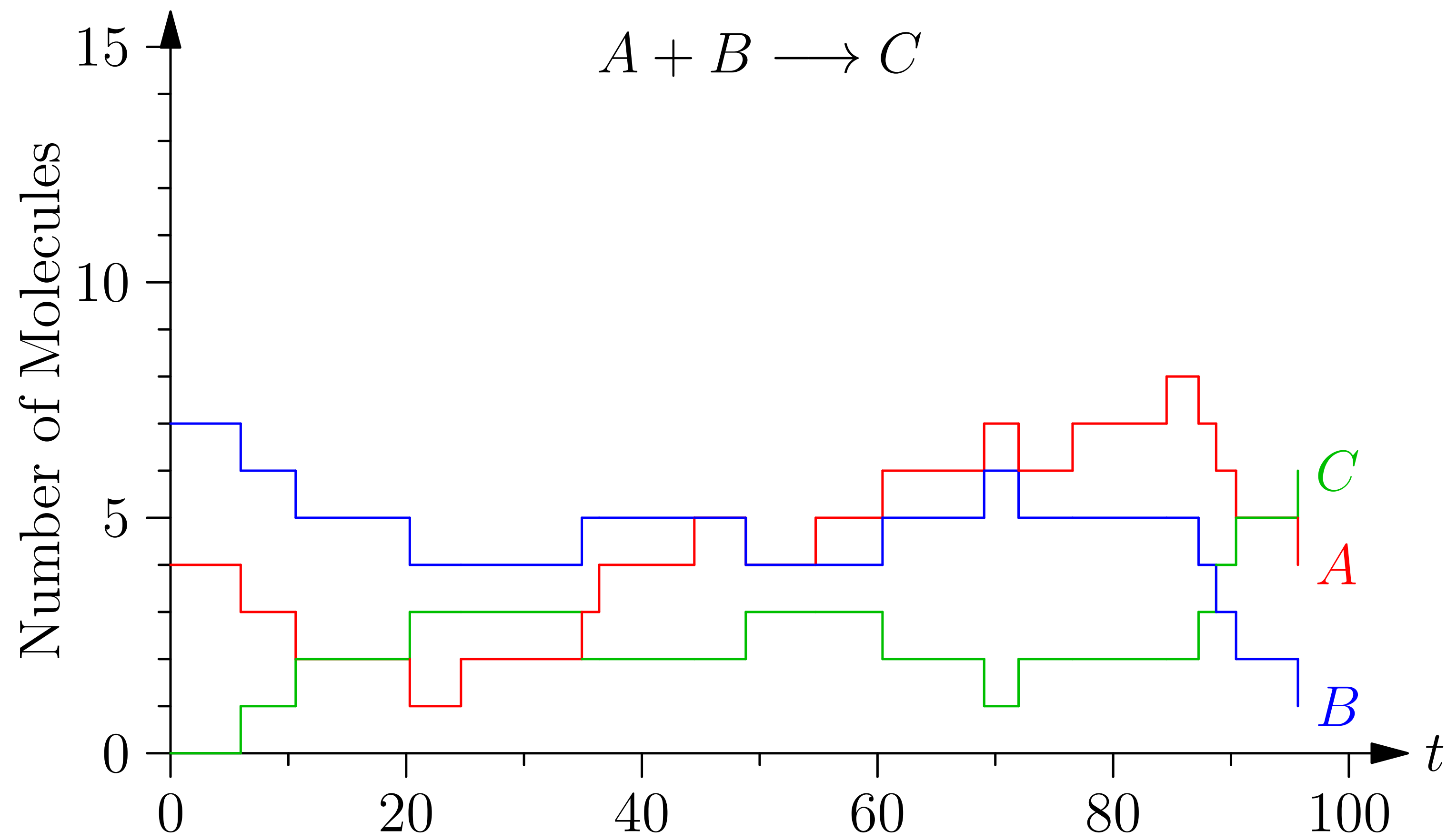

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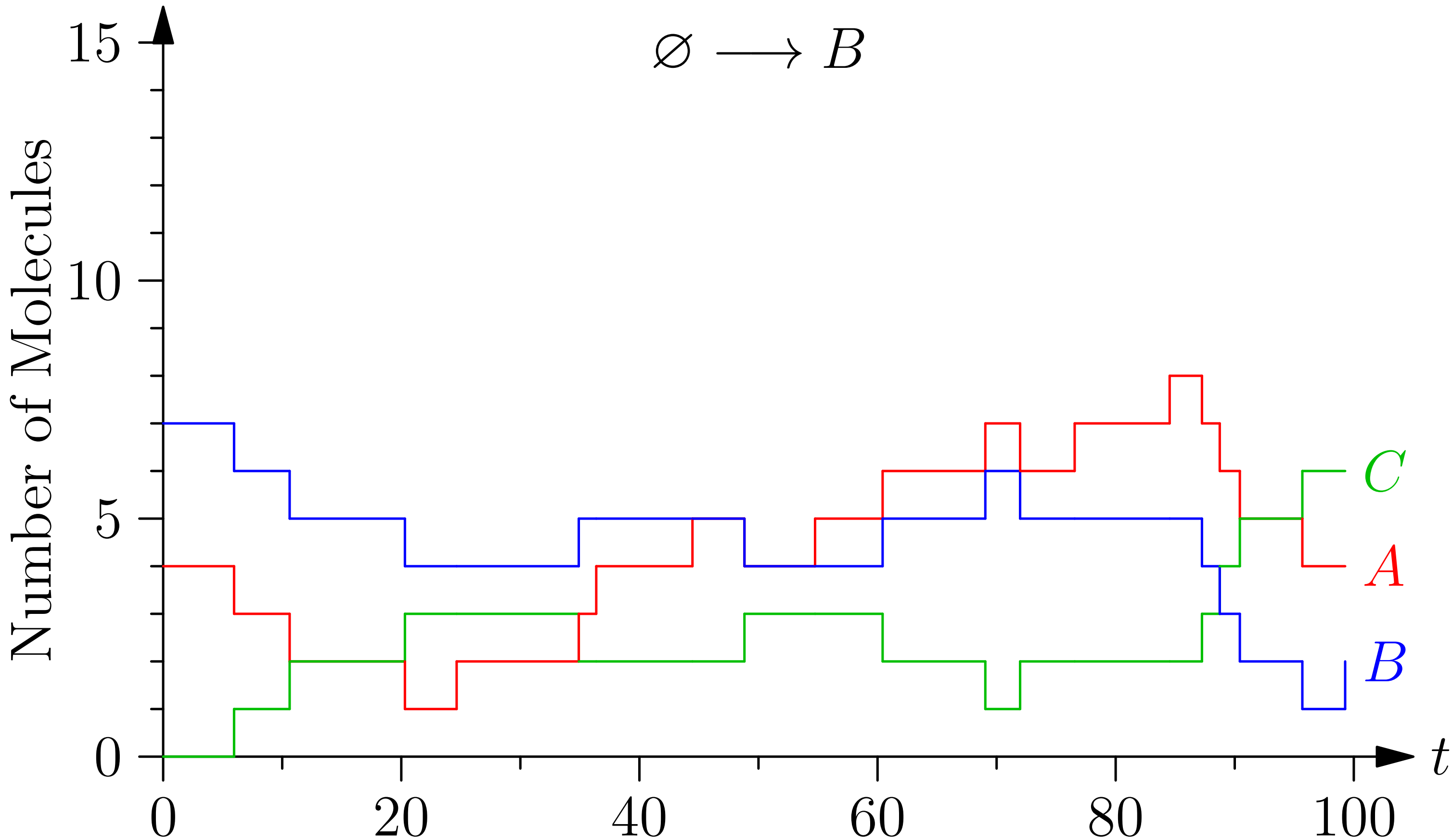

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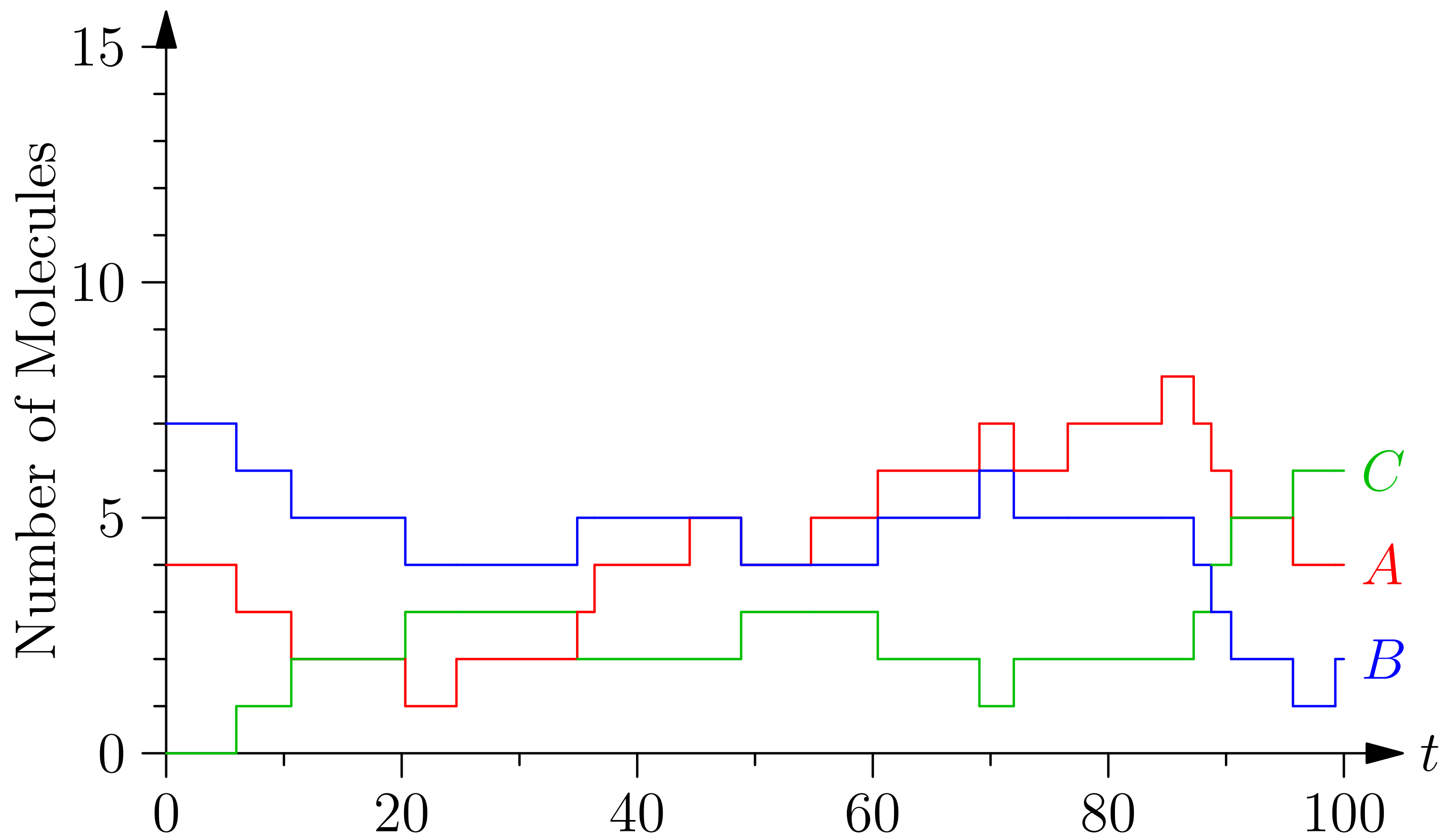

Two Species

- Recall the simple system

$$\varnothing \xrightarrow{f(X_2)} X_1 \qquad X_1 \xrightarrow{k_3} \varnothing \qquad X_1 \xrightarrow{k_5} X_2$$

$$\varnothing \xrightarrow{k_2} X_2 \qquad X_2 \xrightarrow{k_4} \varnothing$$

- Where X_2 represses the expression of X_1

$$f(X_2) = \frac{k_1}{1 + ([X_2]/K)^n}$$

- Leading to the coupled ODEs

$$\frac{\mathrm{d}[X_1(t)]}{\mathrm{d}t} = \frac{k_1}{1 + ([X_2(t)]/K)^n} - k_3\,[X_1(t)] - k_5\,[X_1(t)]$$

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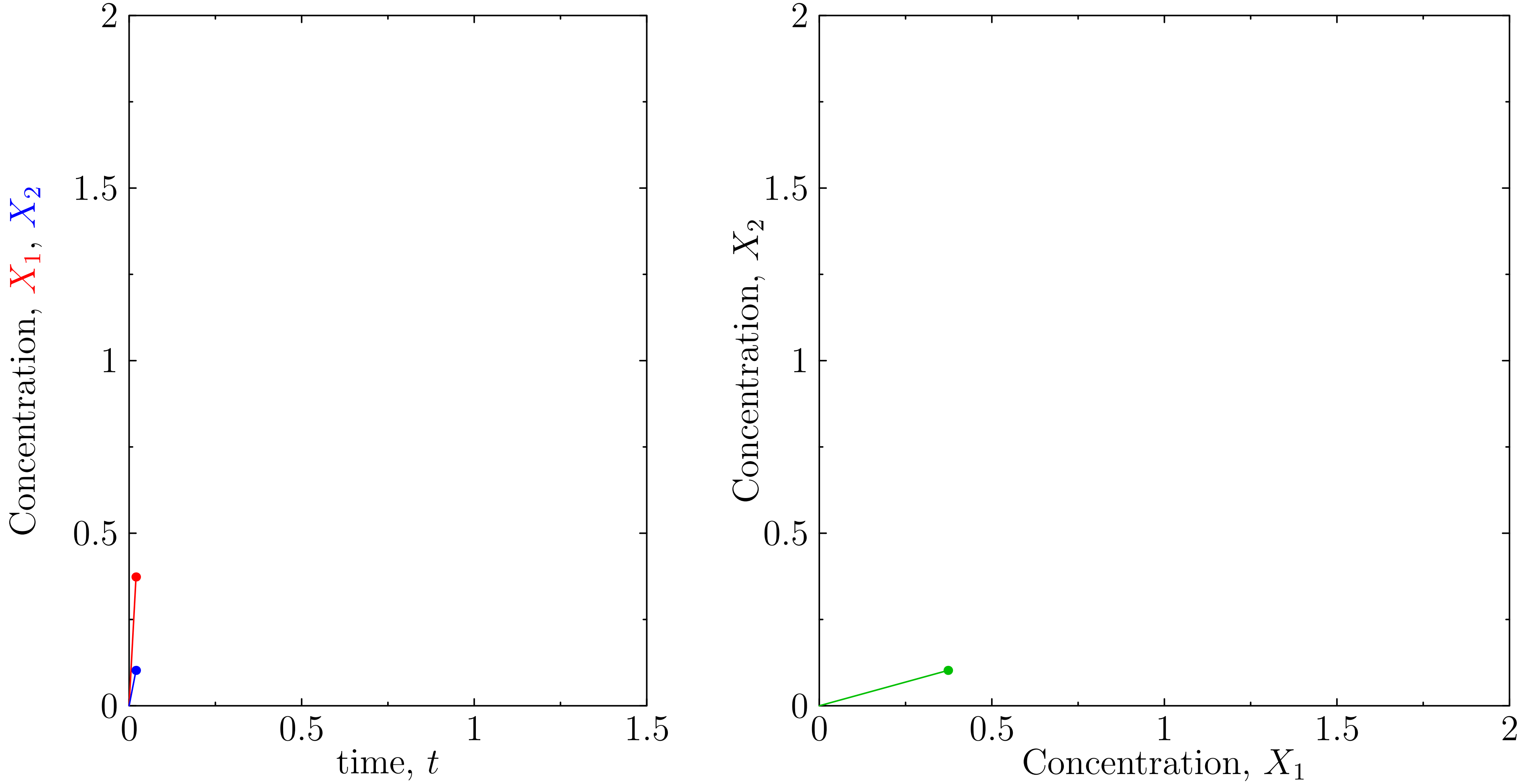

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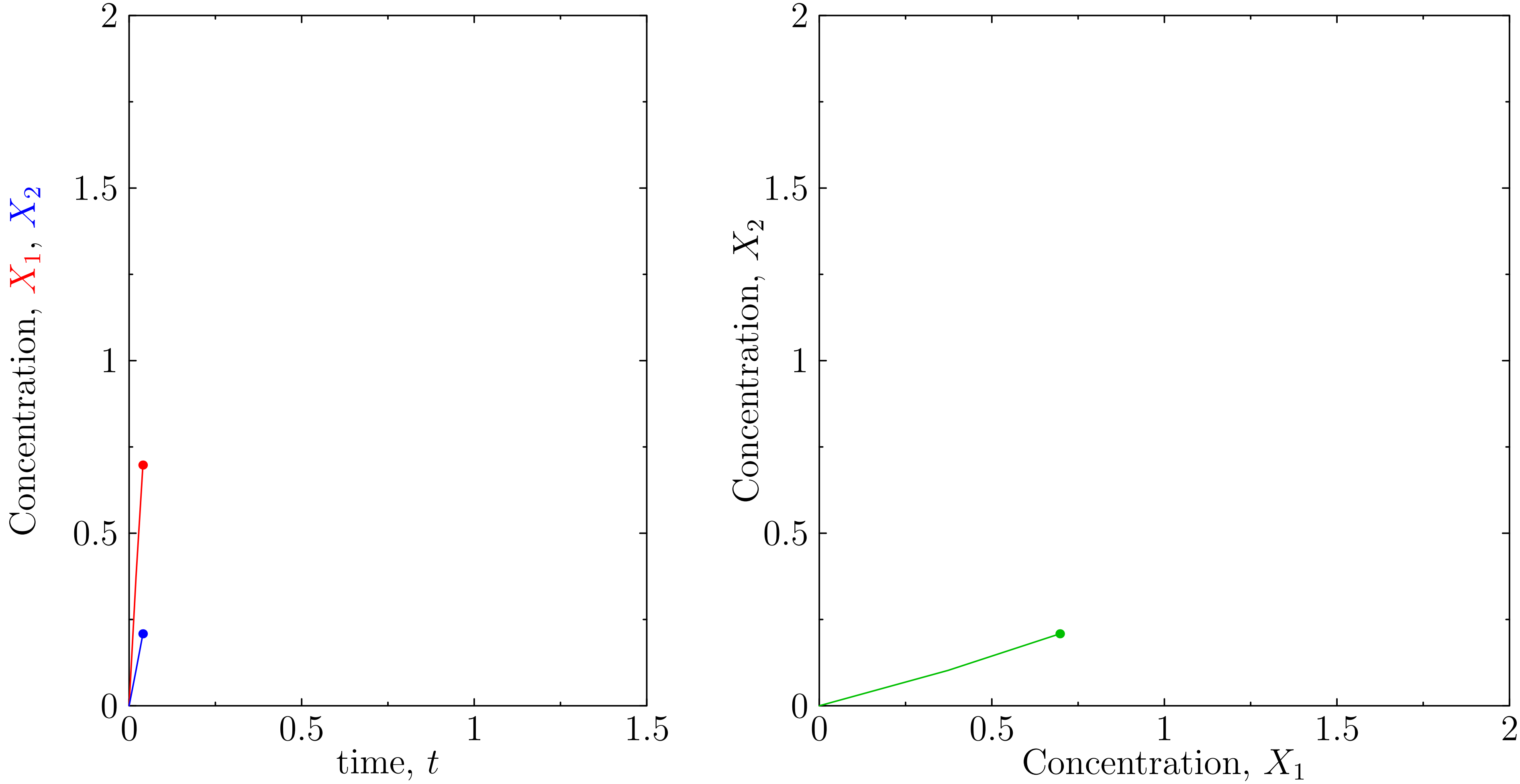

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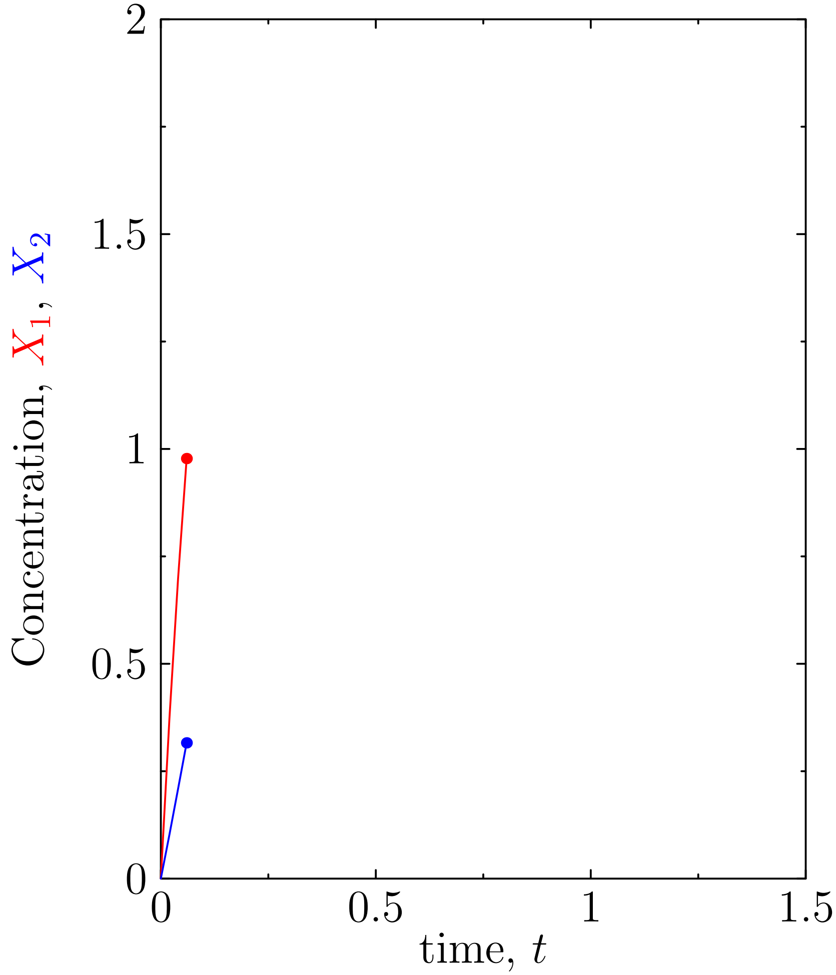

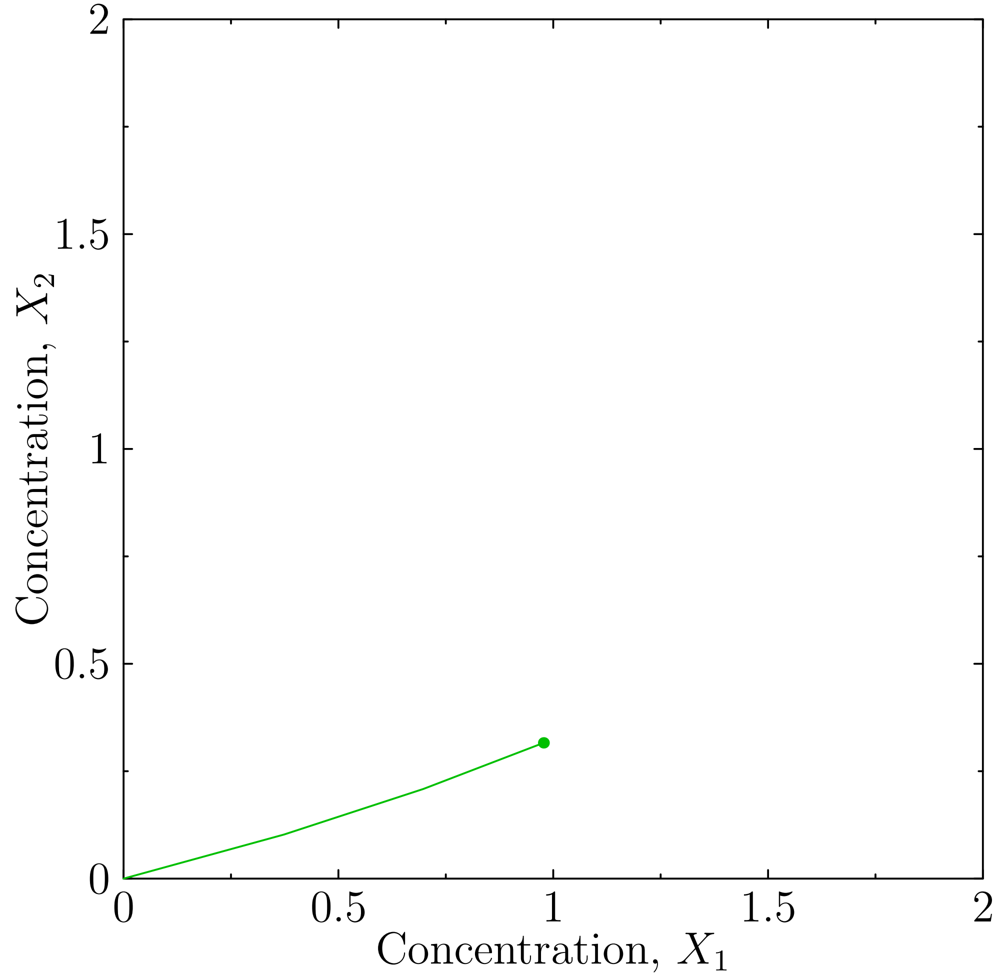

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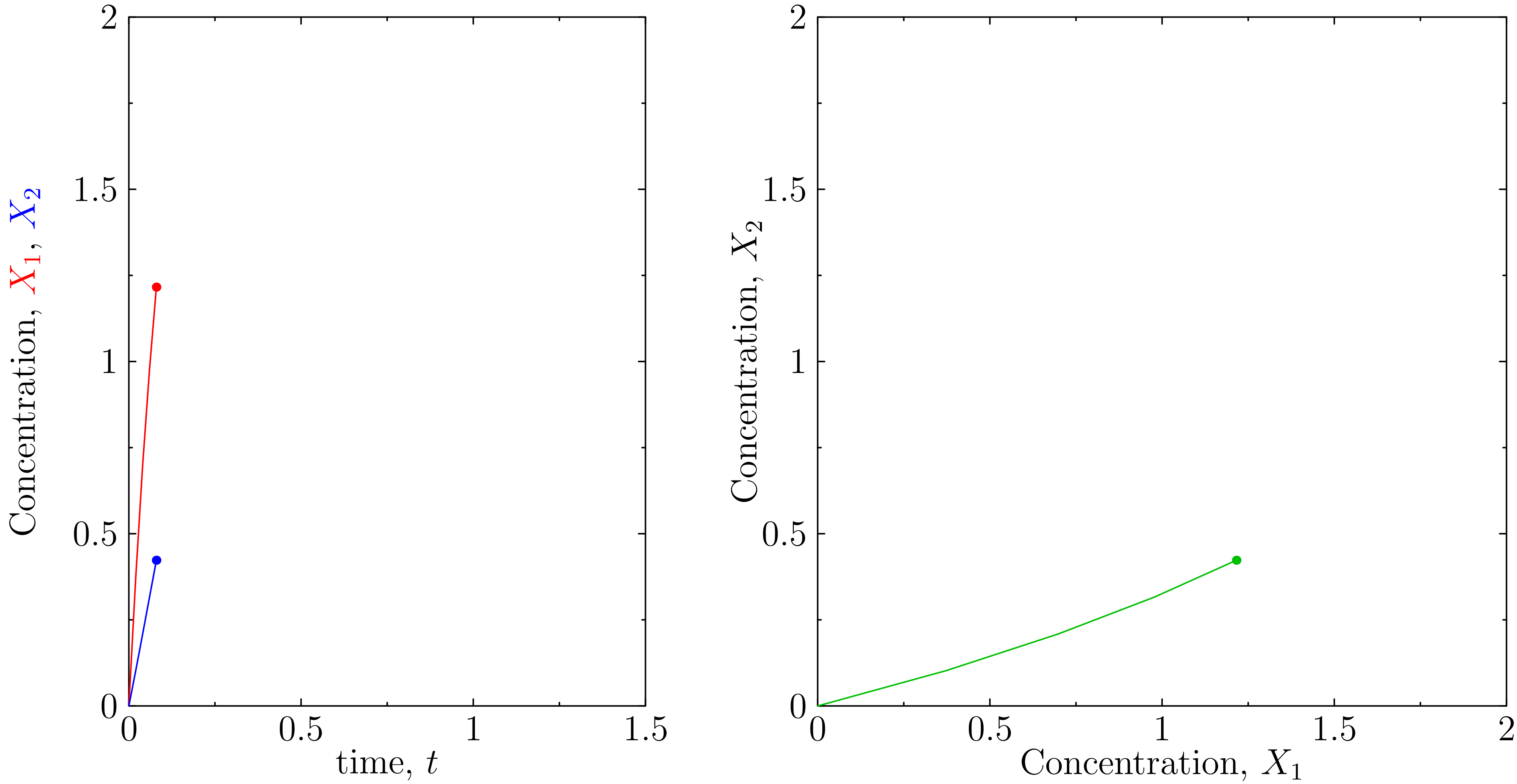

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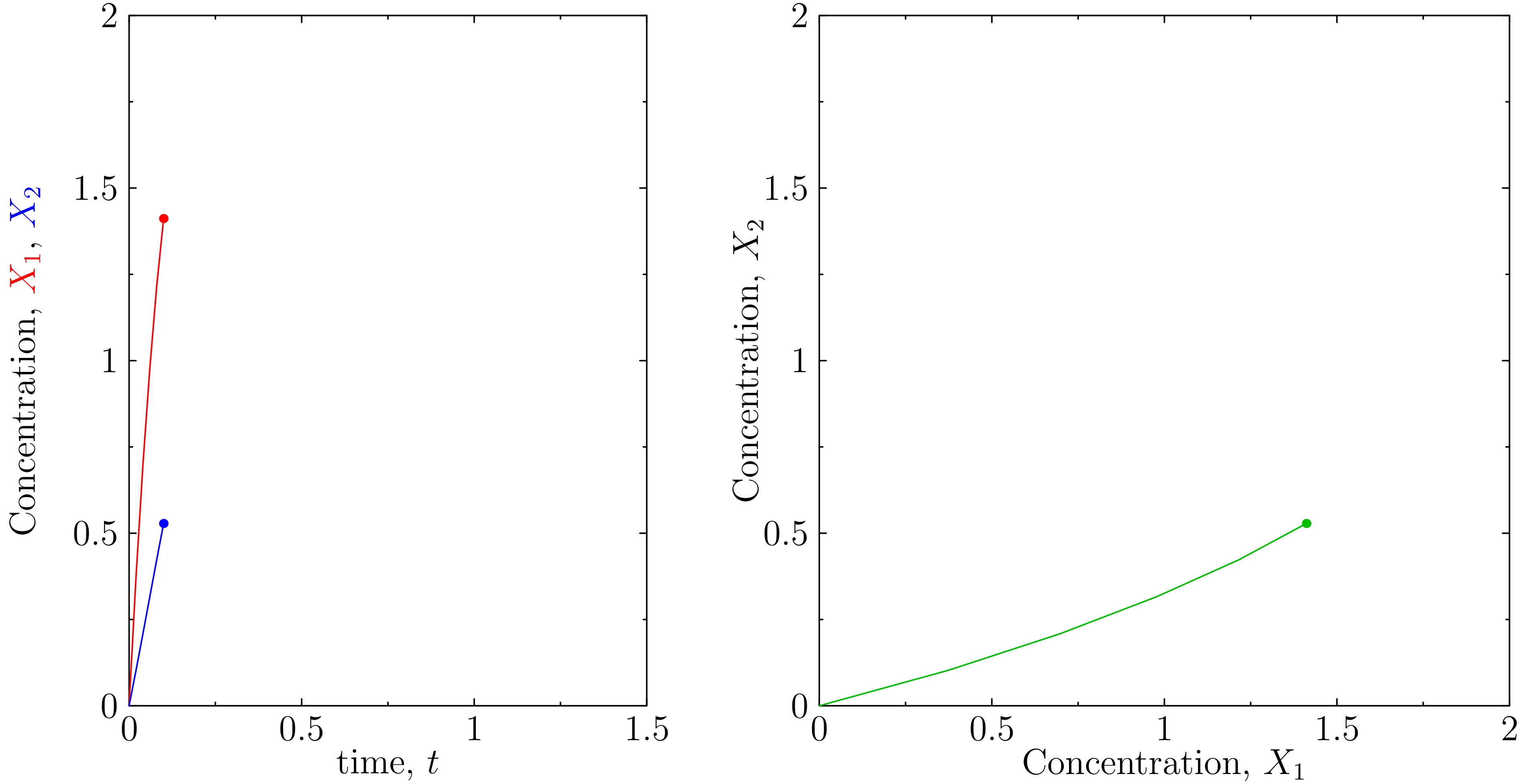

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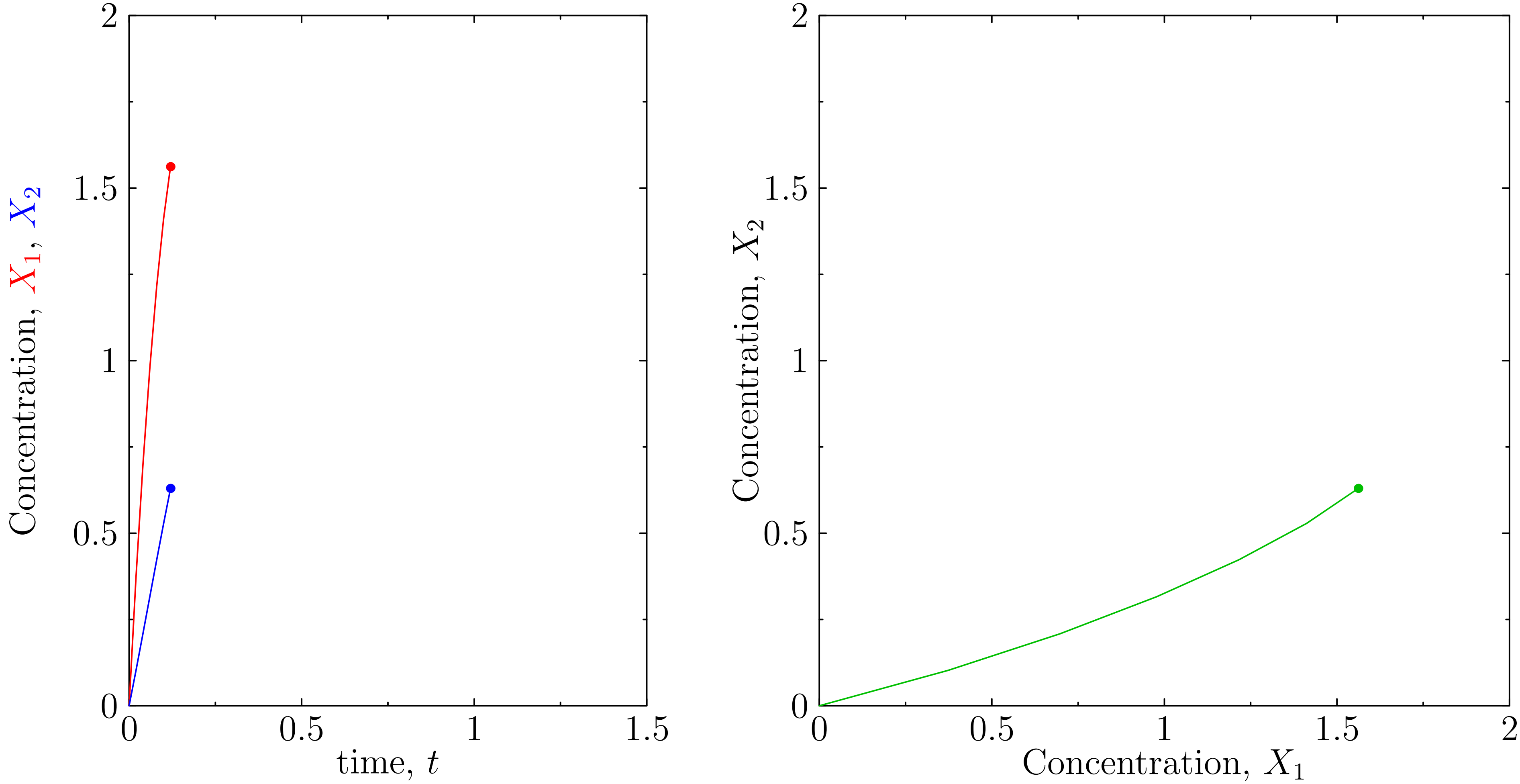

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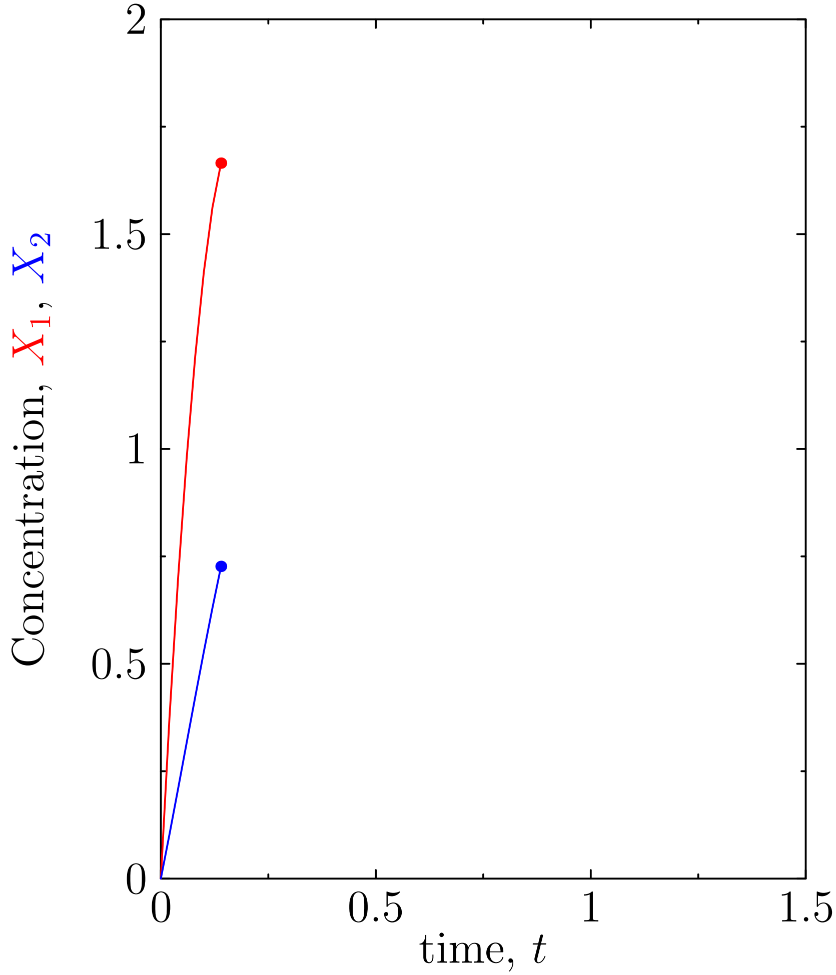

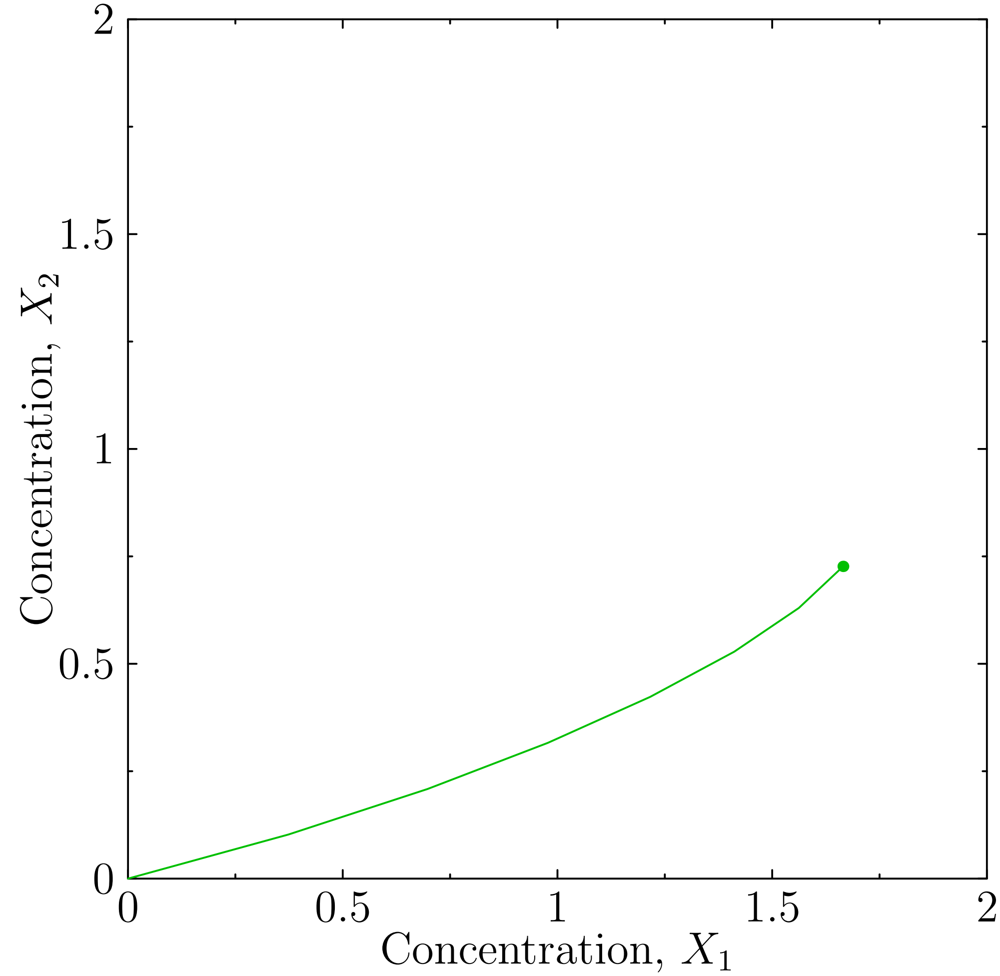

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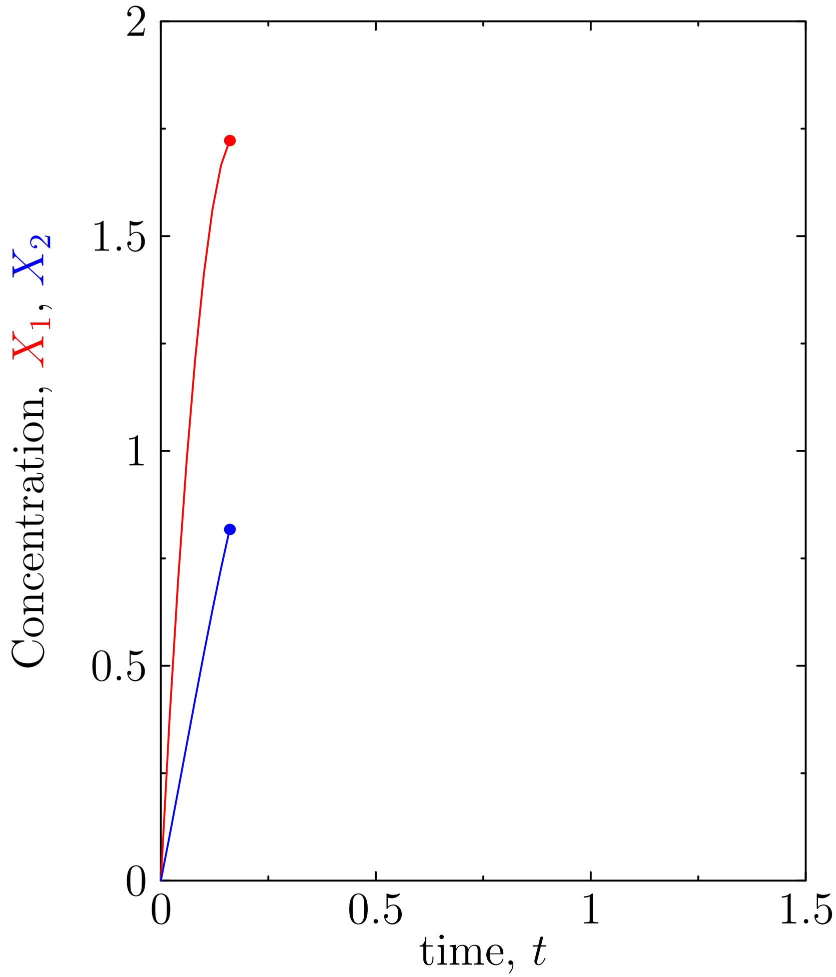

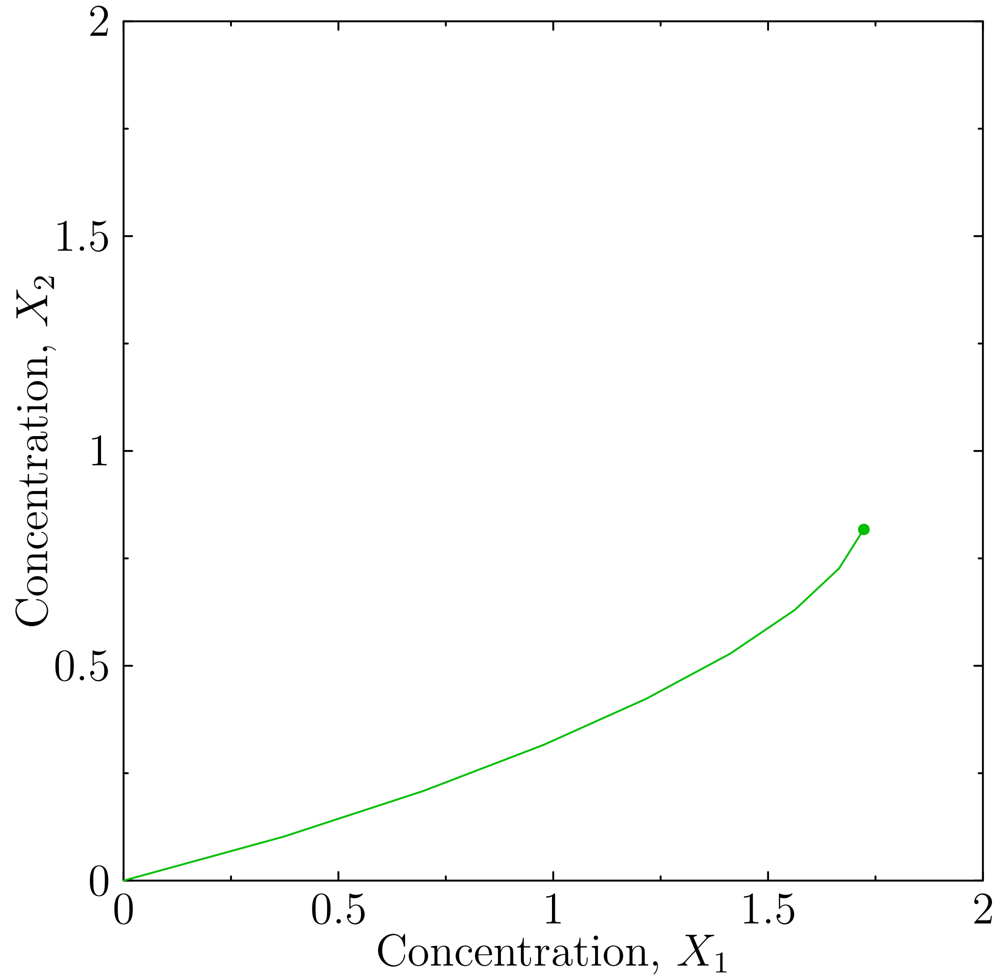

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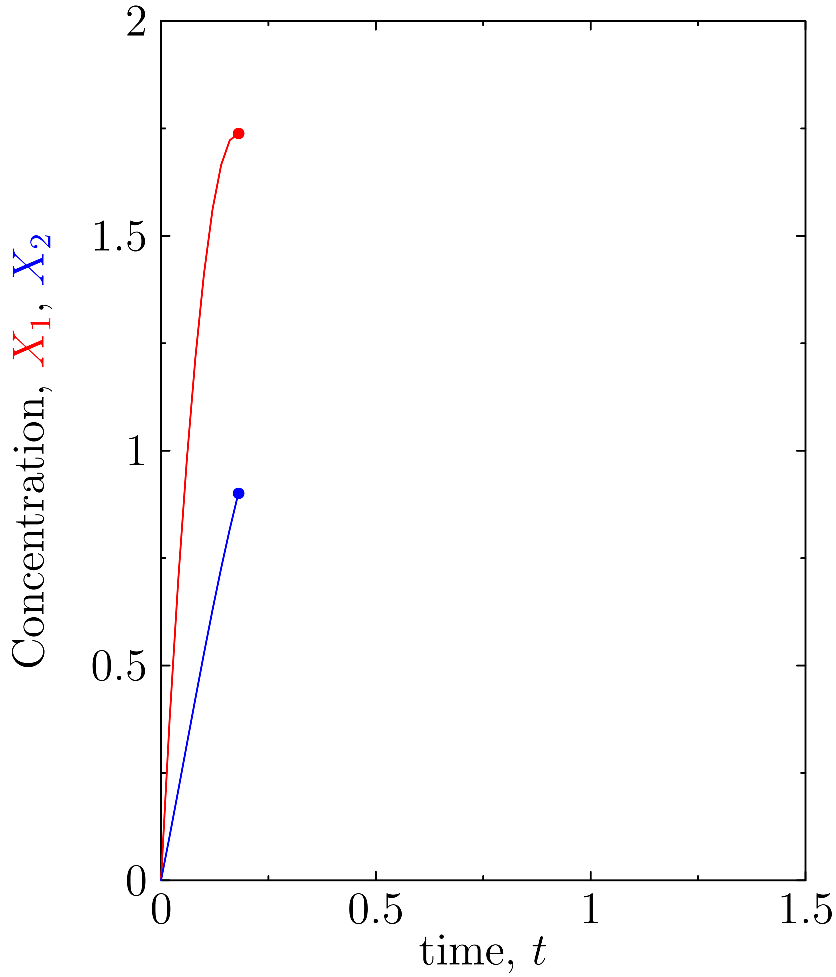

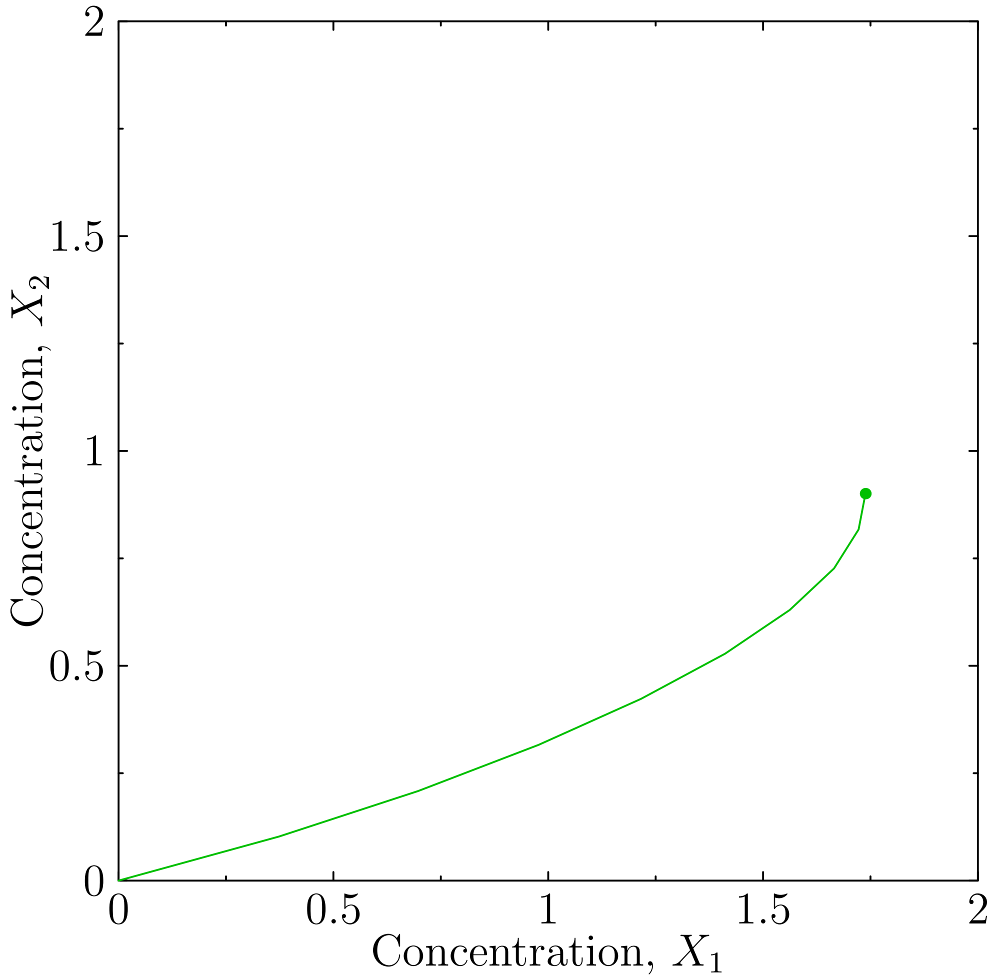

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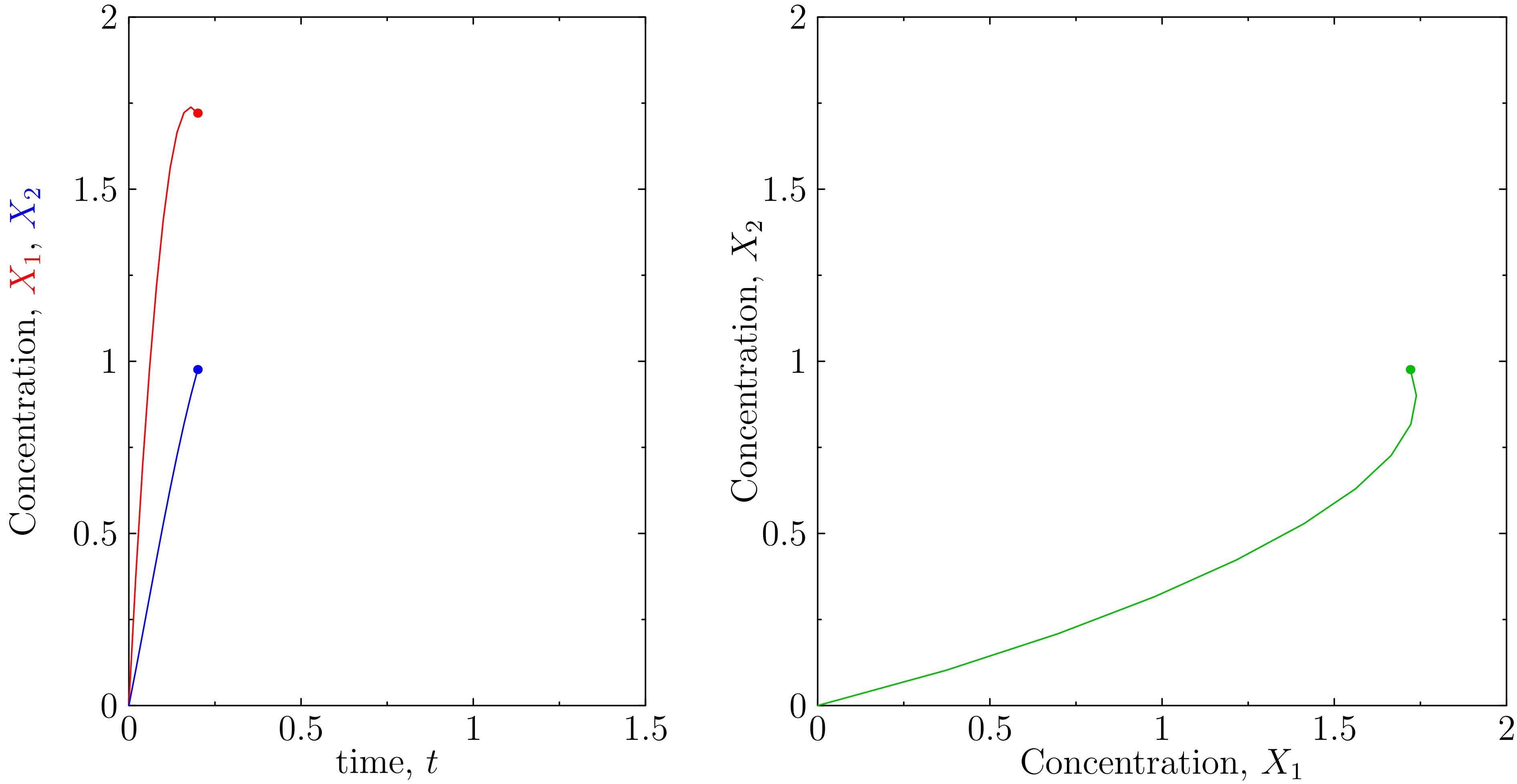

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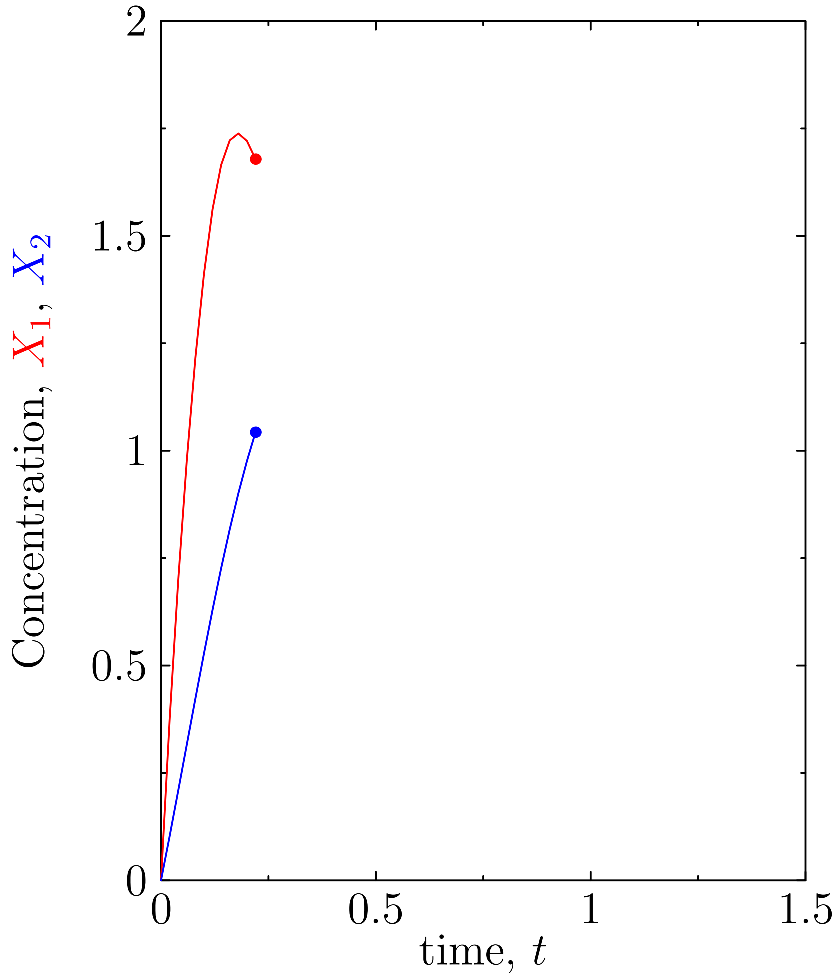

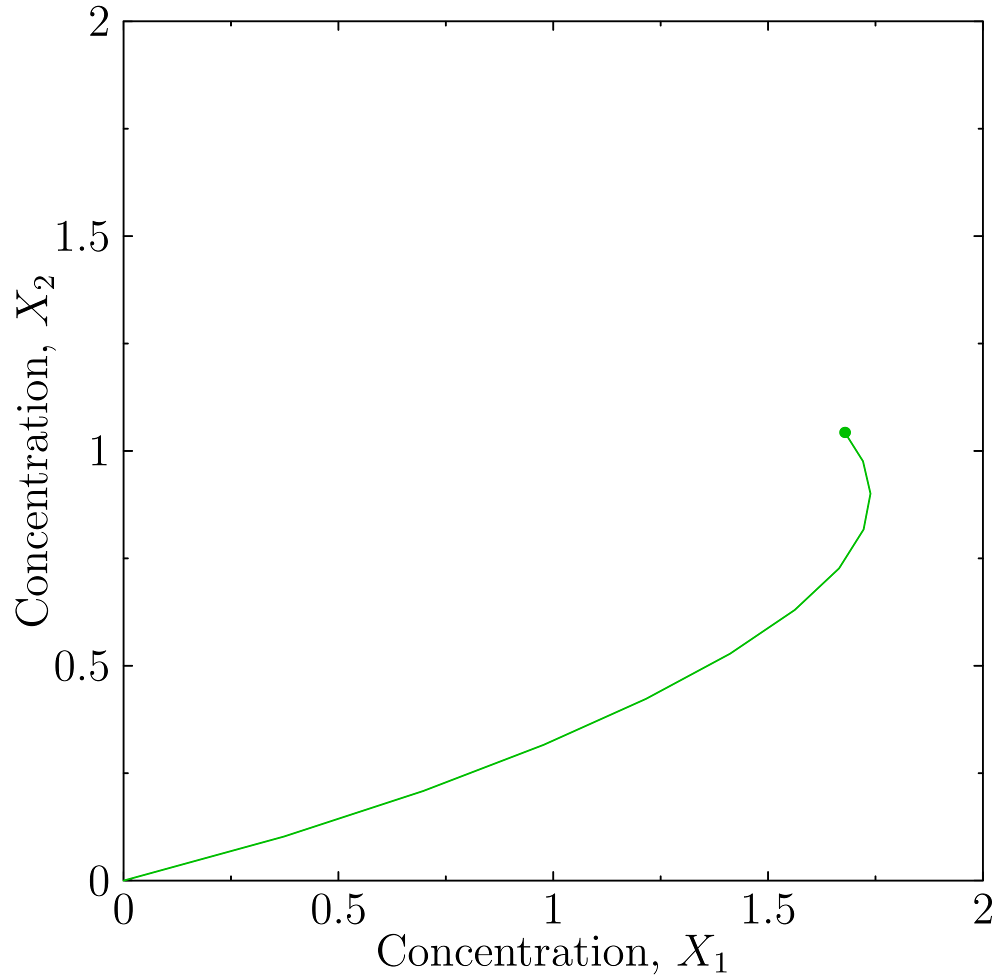

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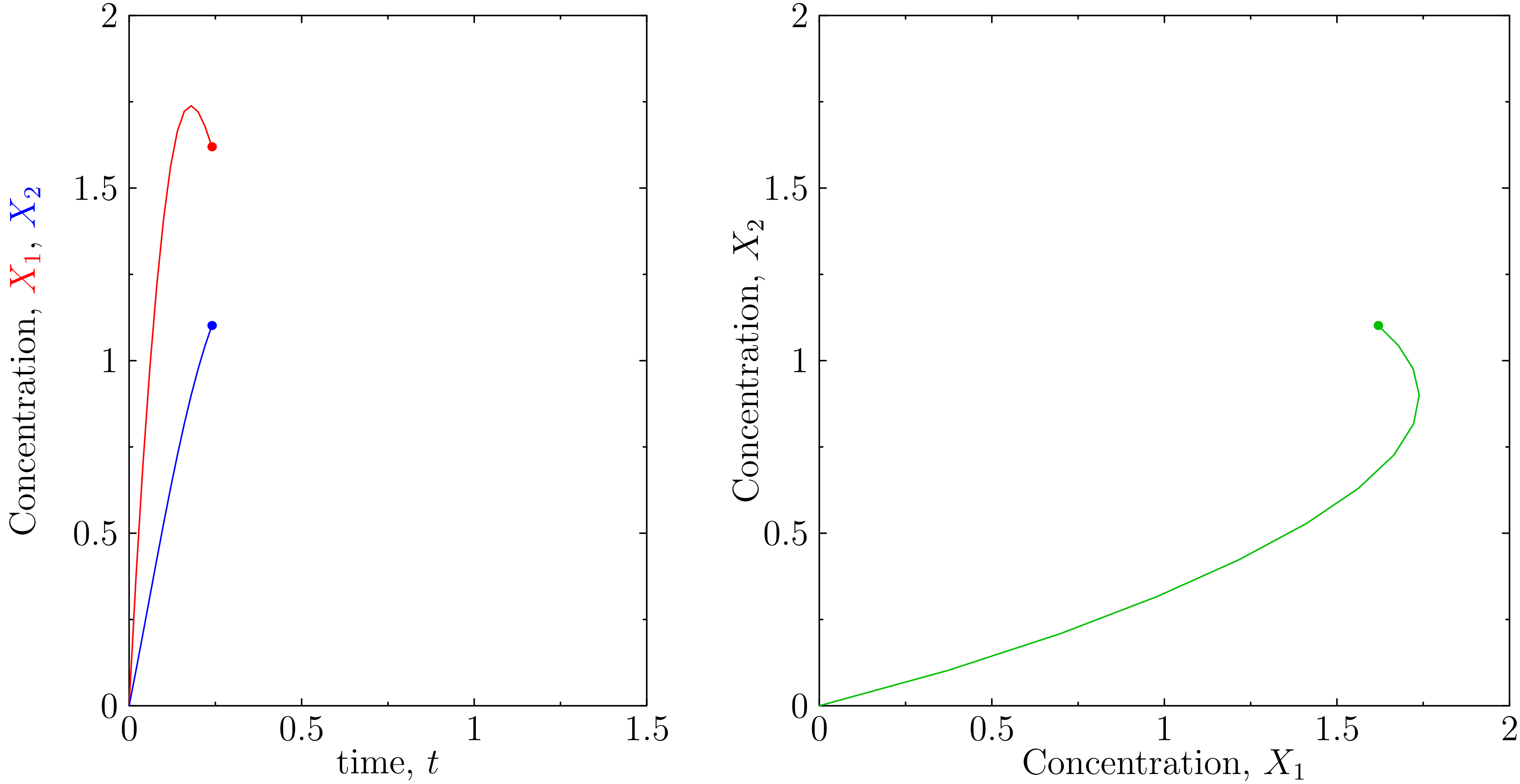

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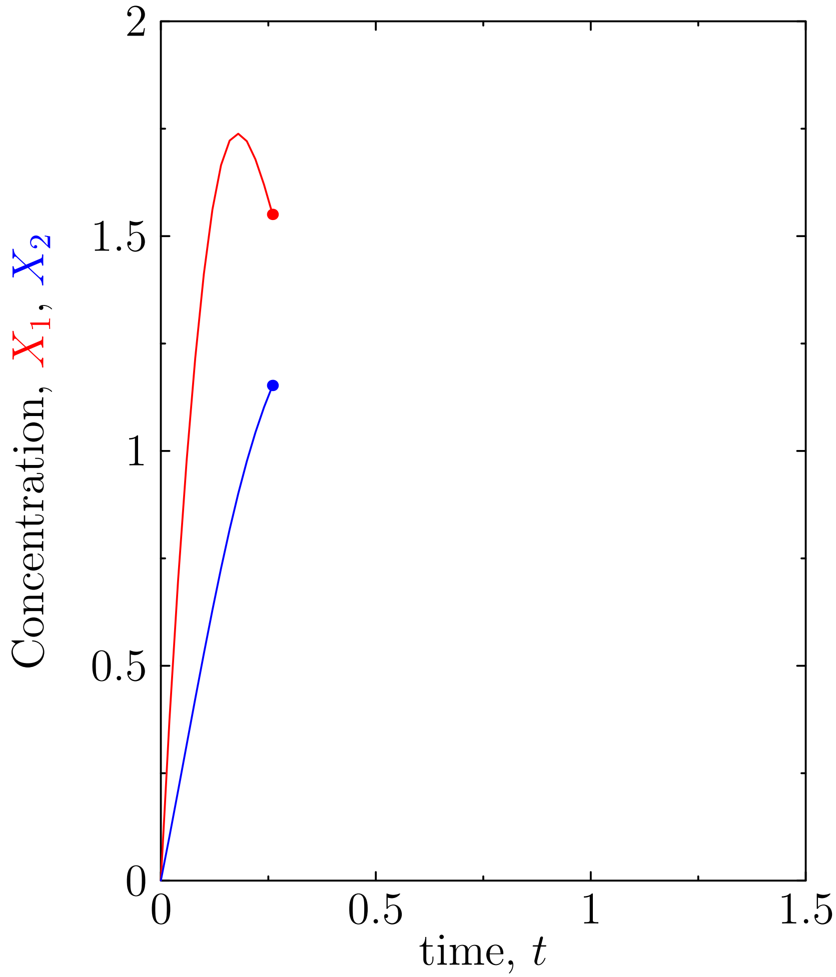

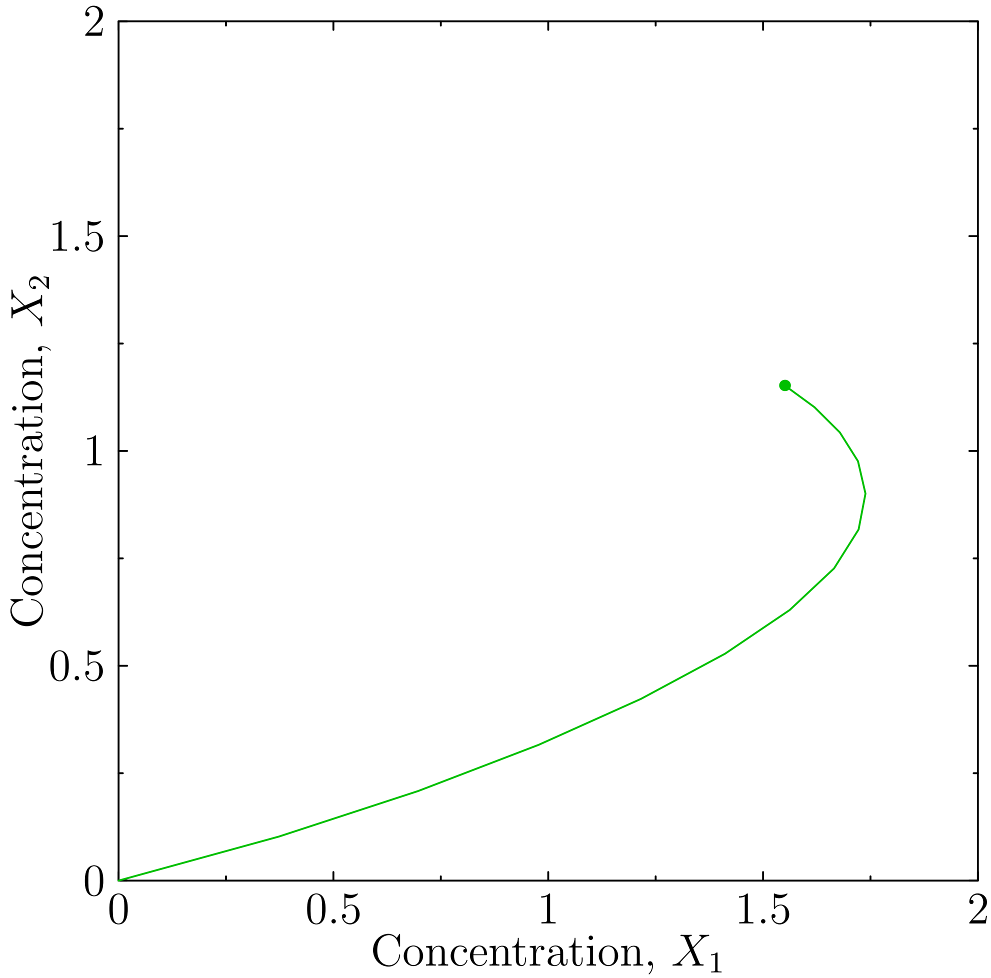

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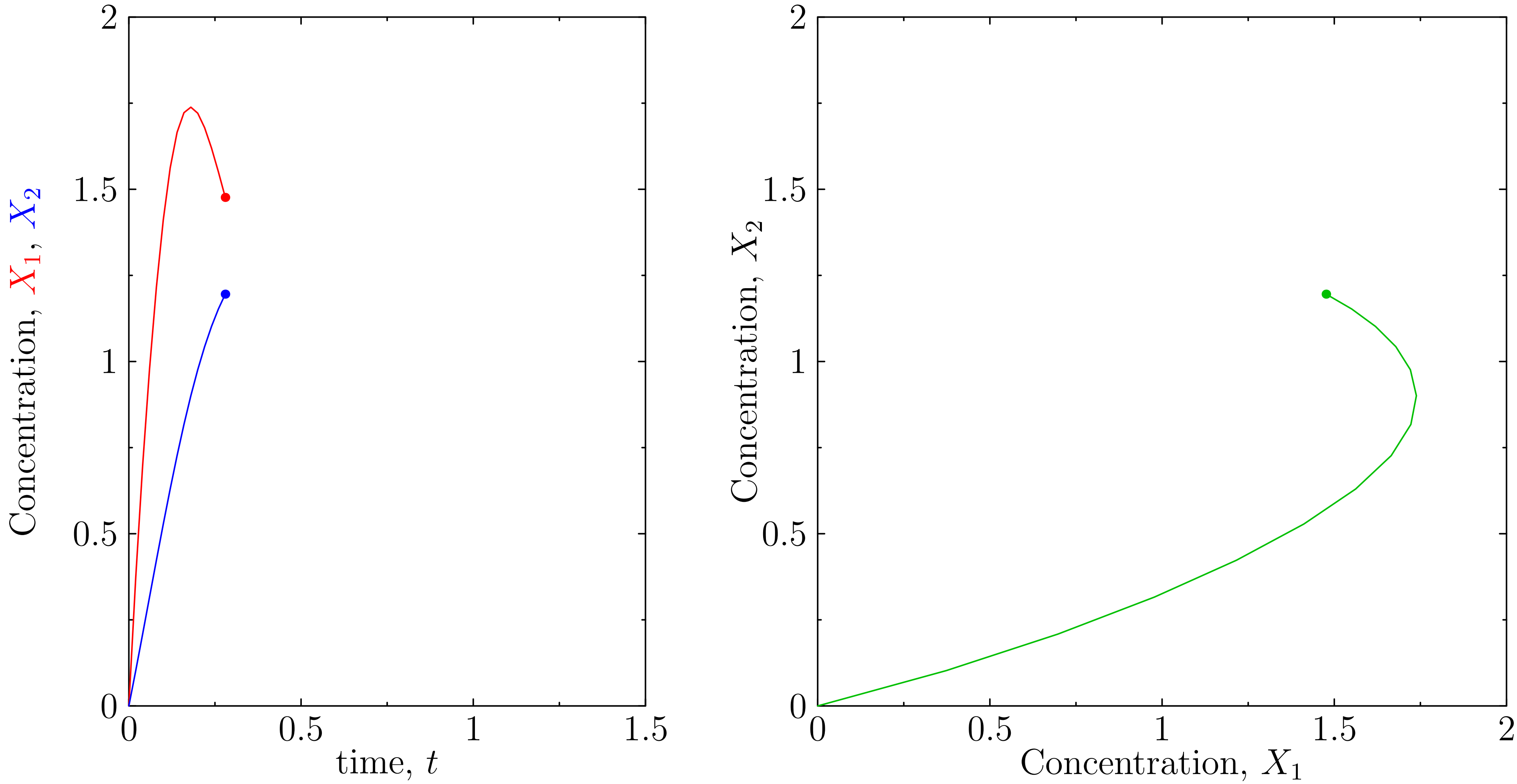

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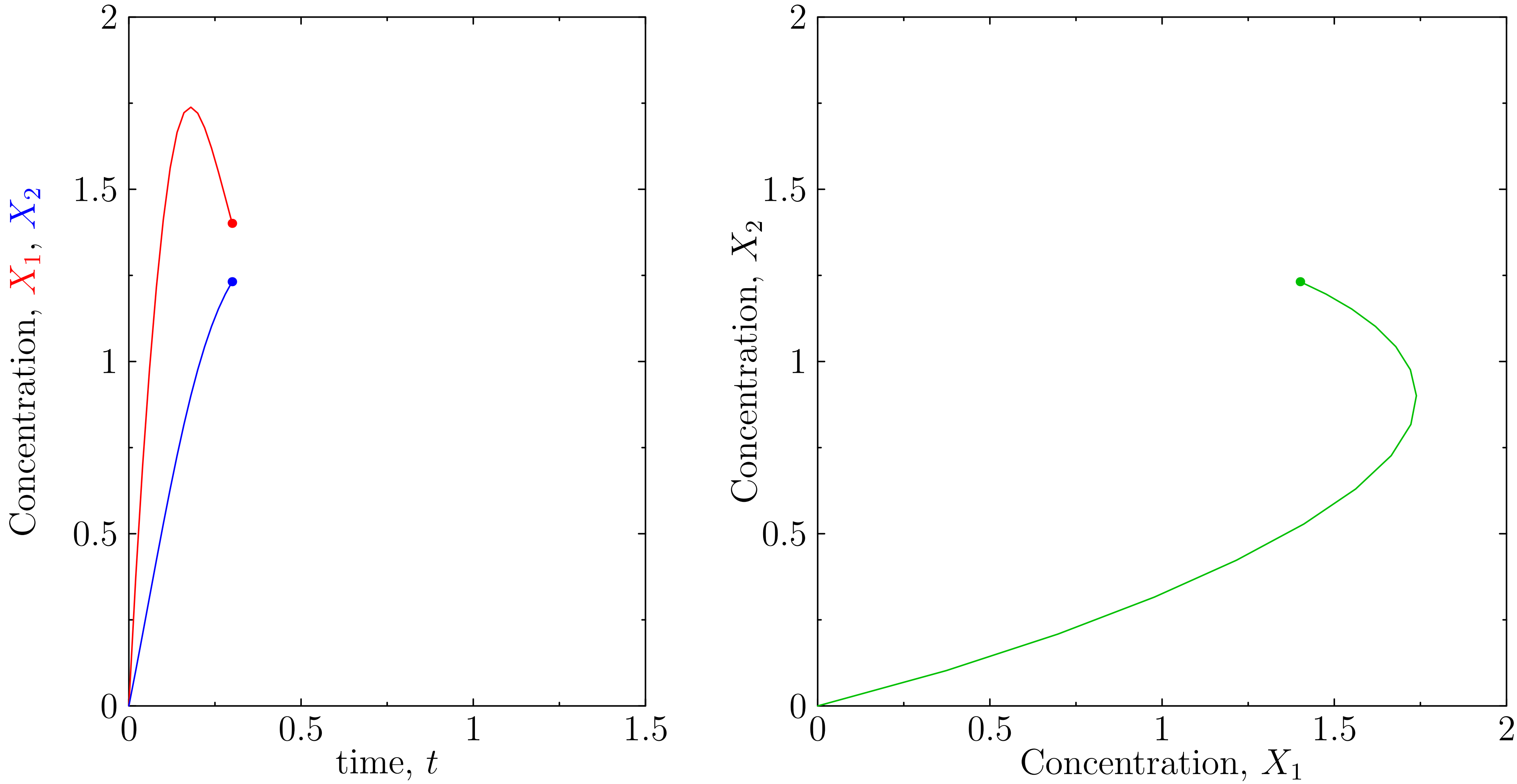

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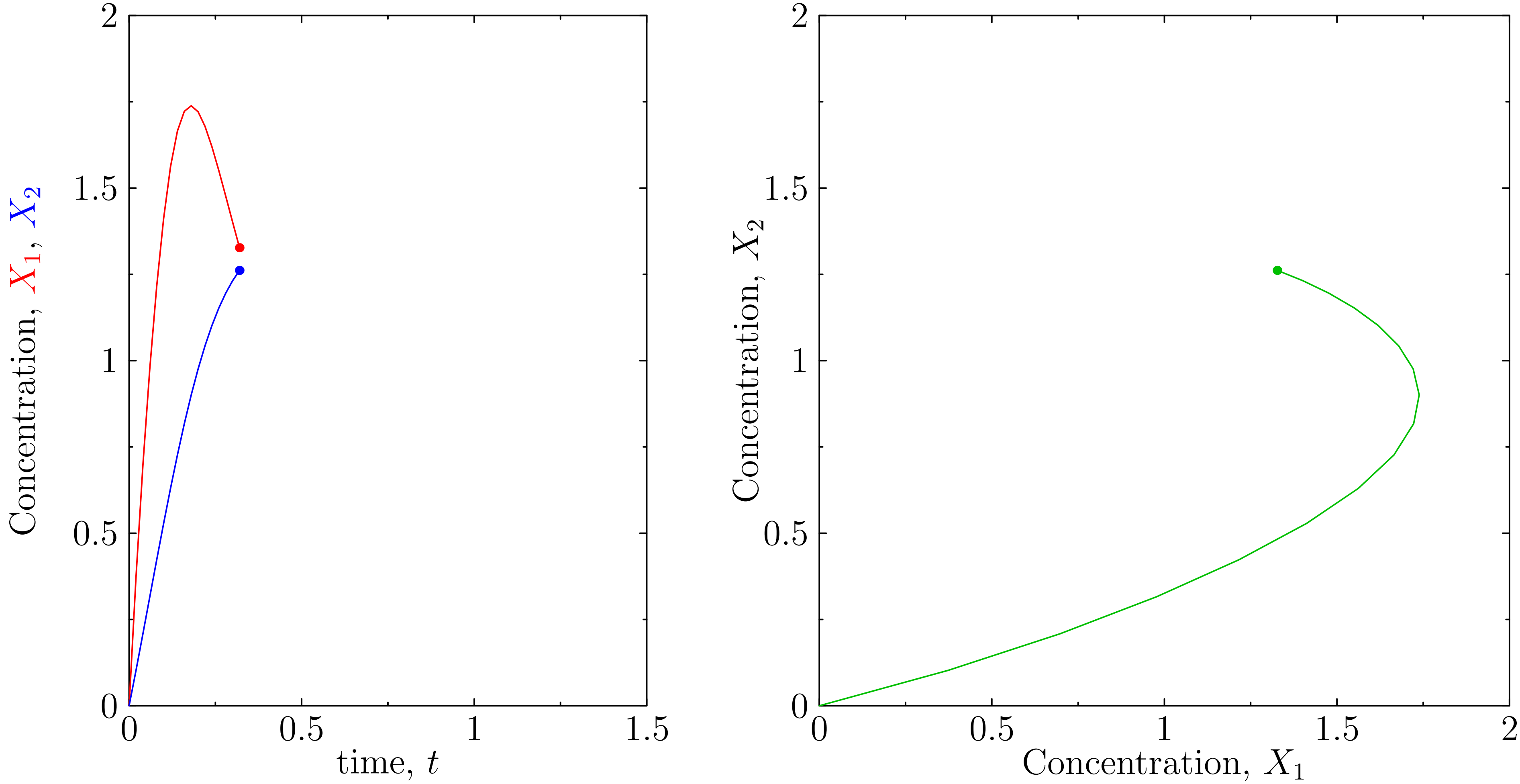

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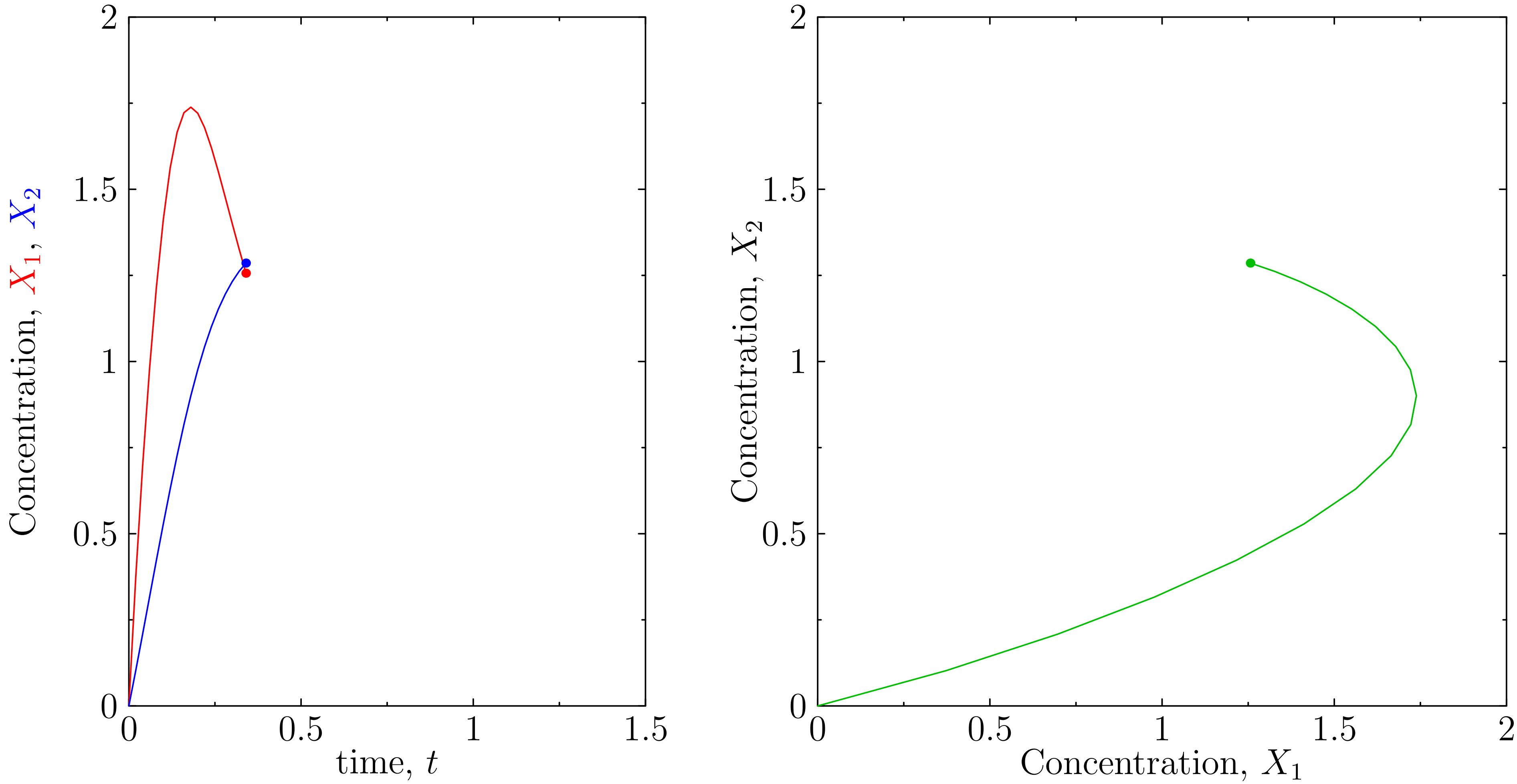

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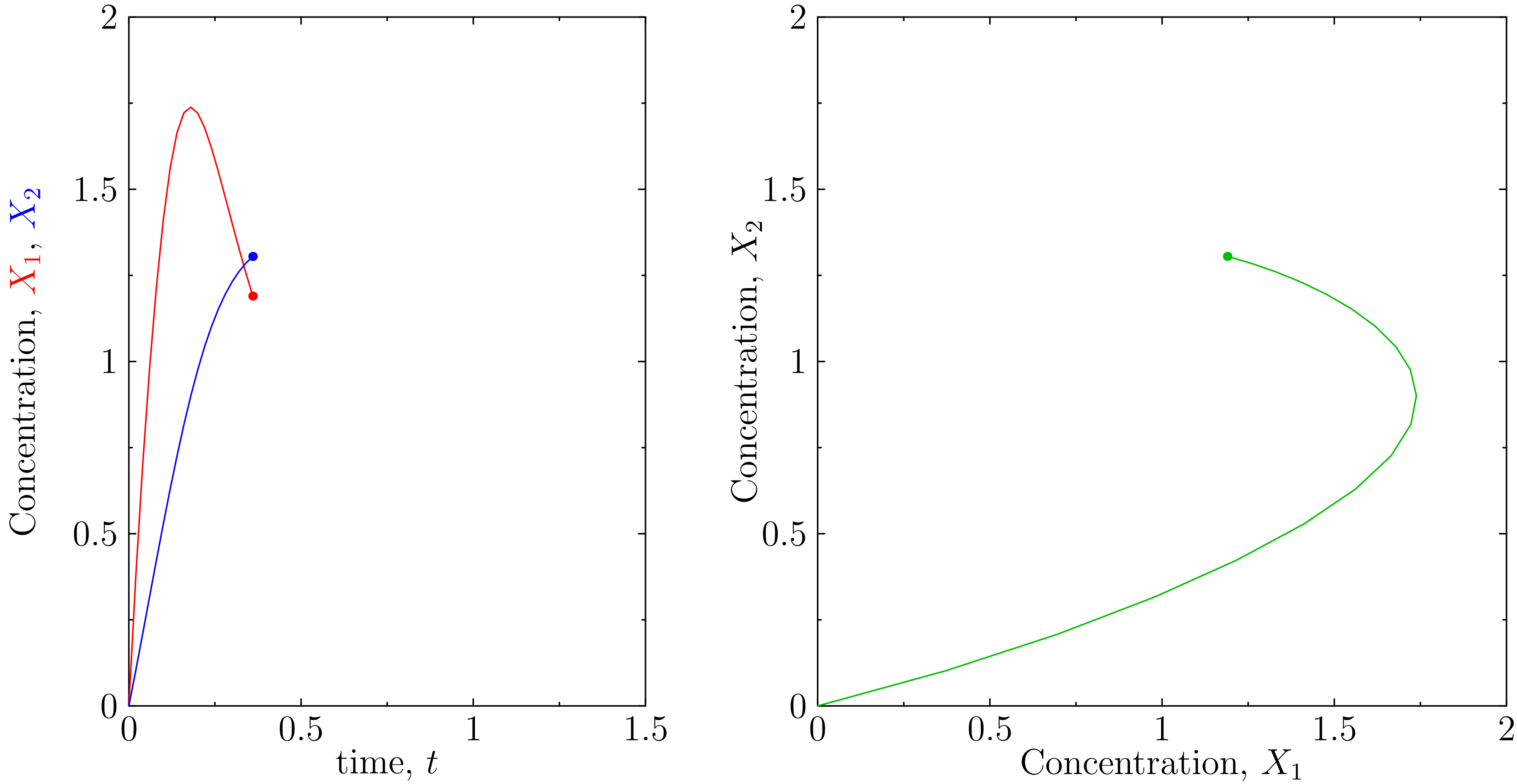

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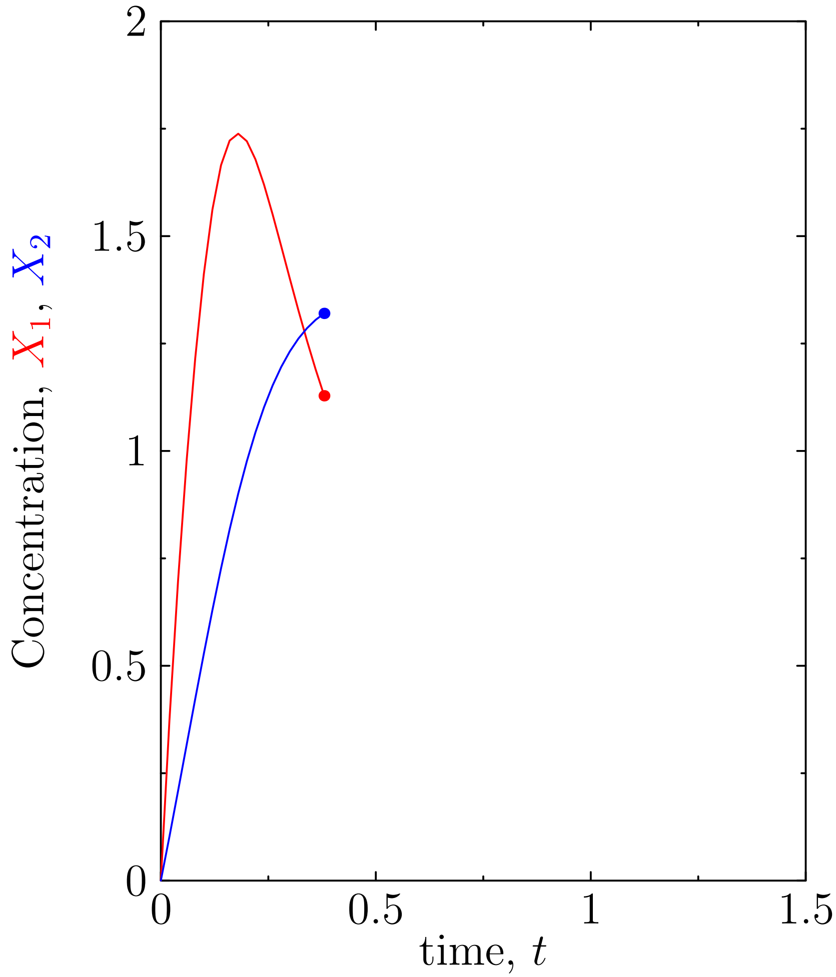

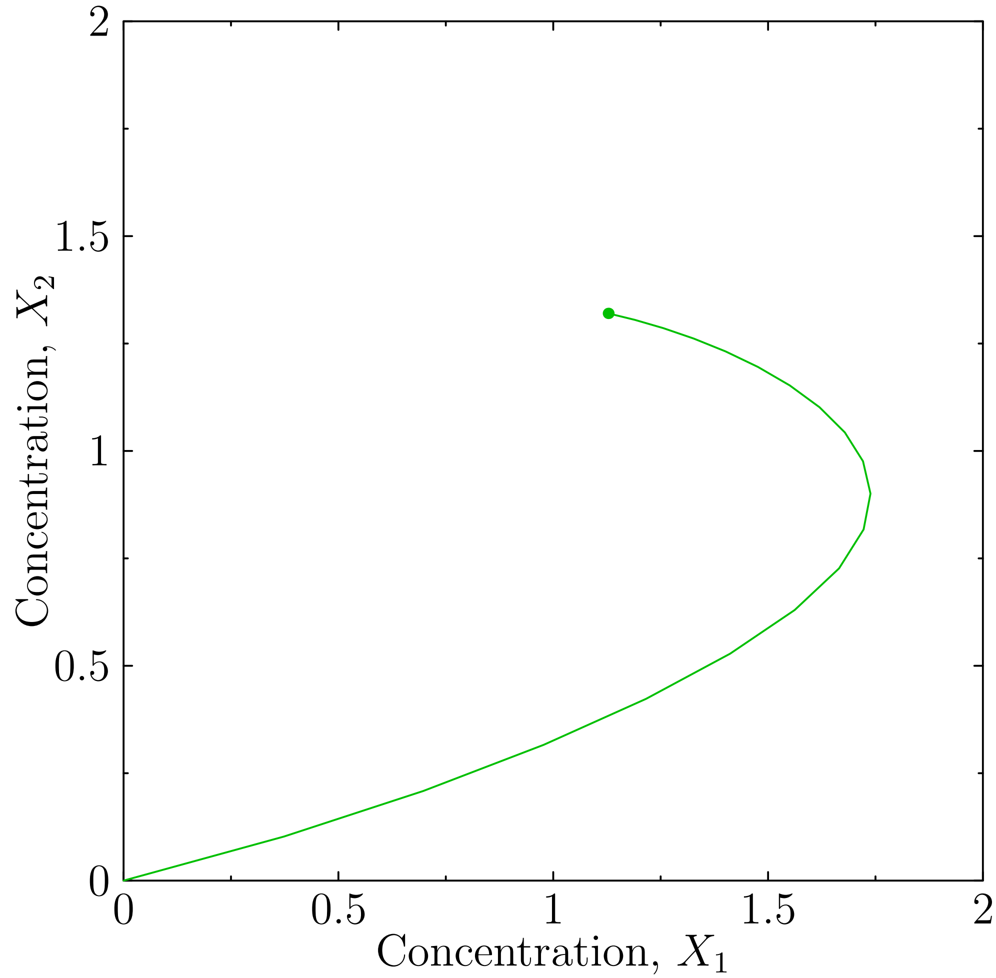

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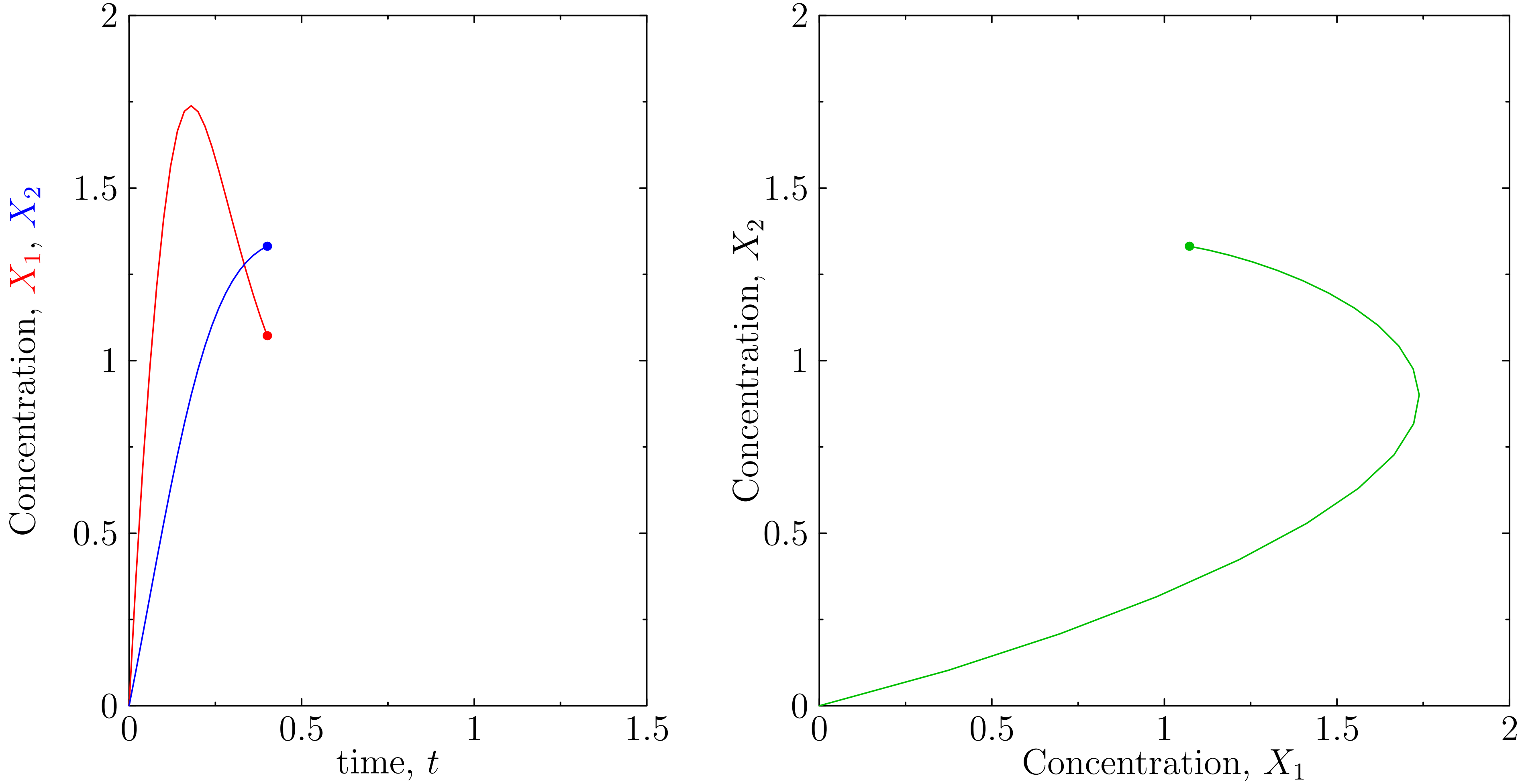

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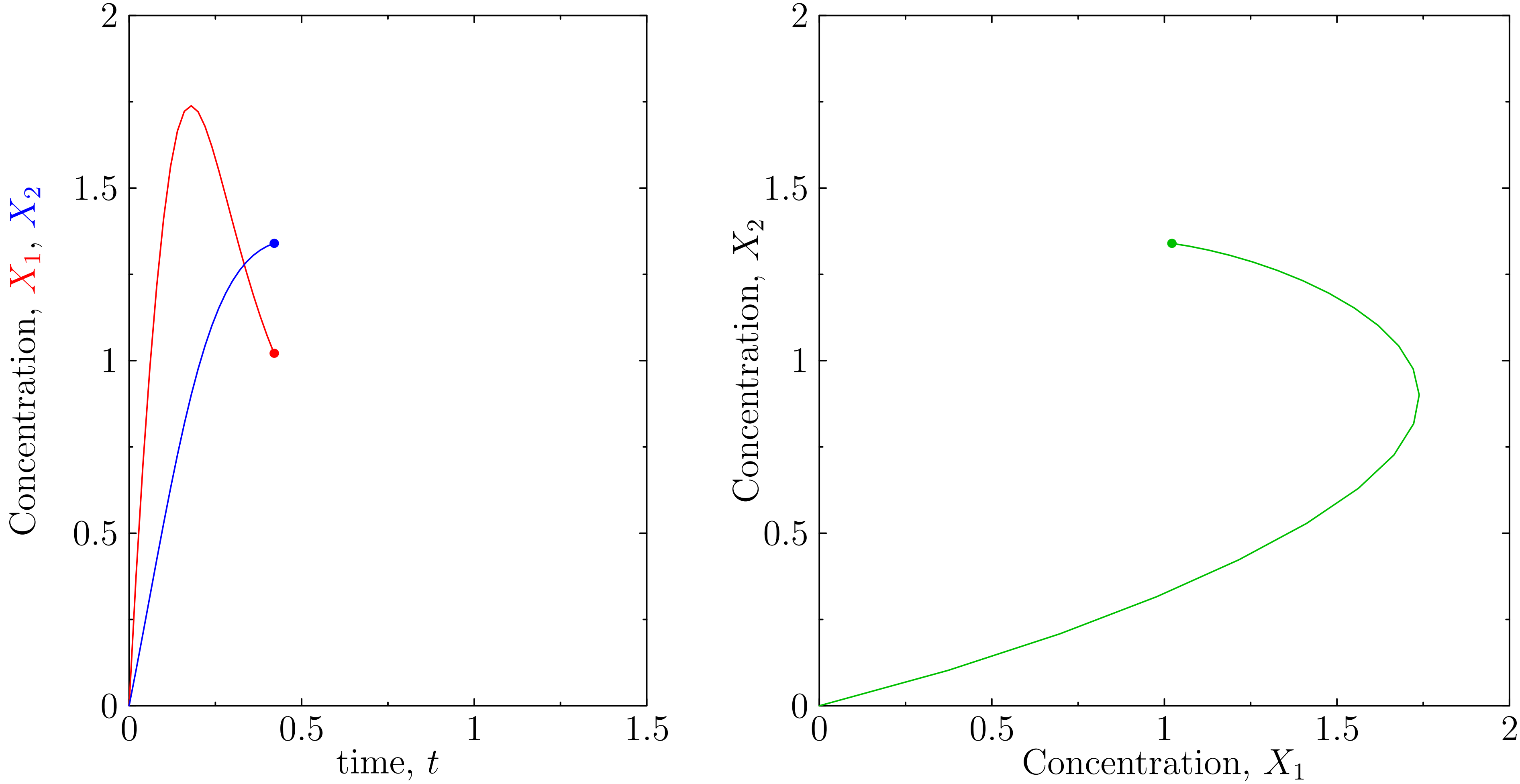

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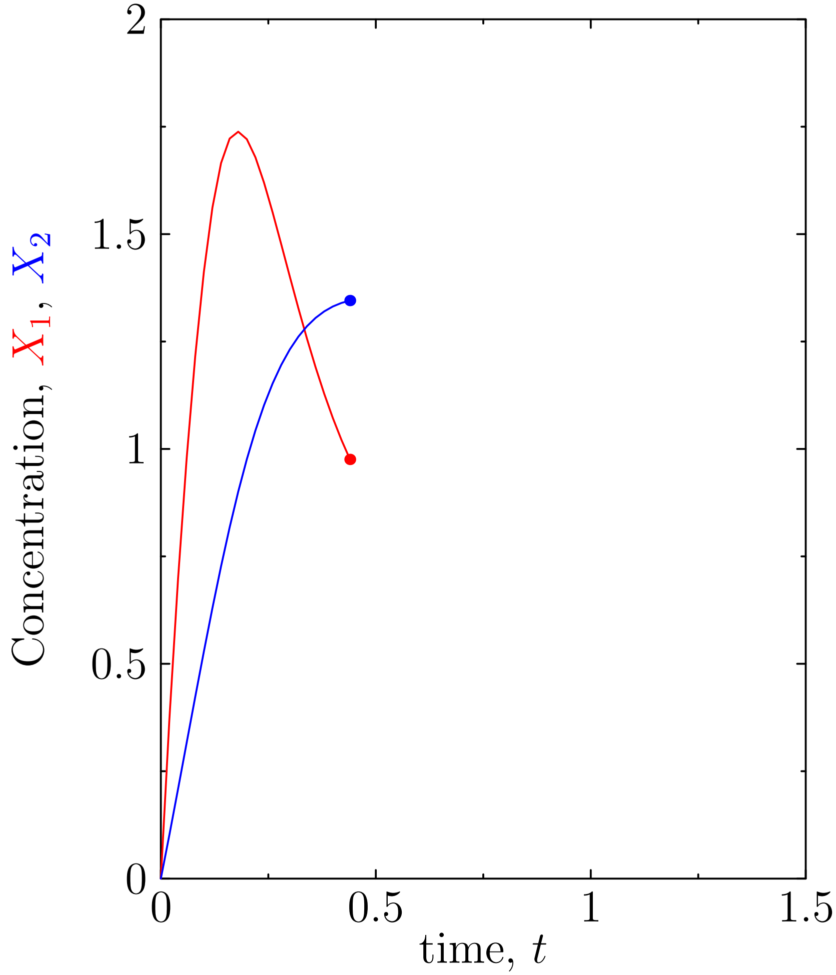

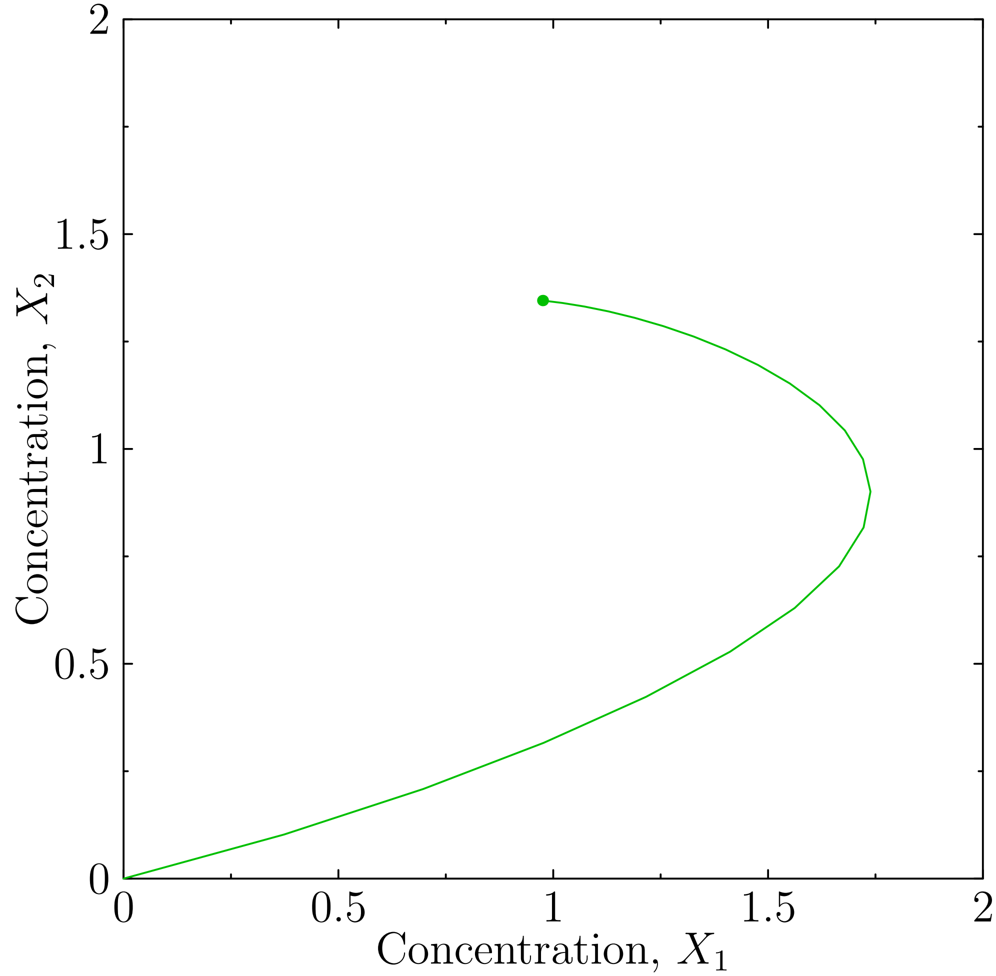

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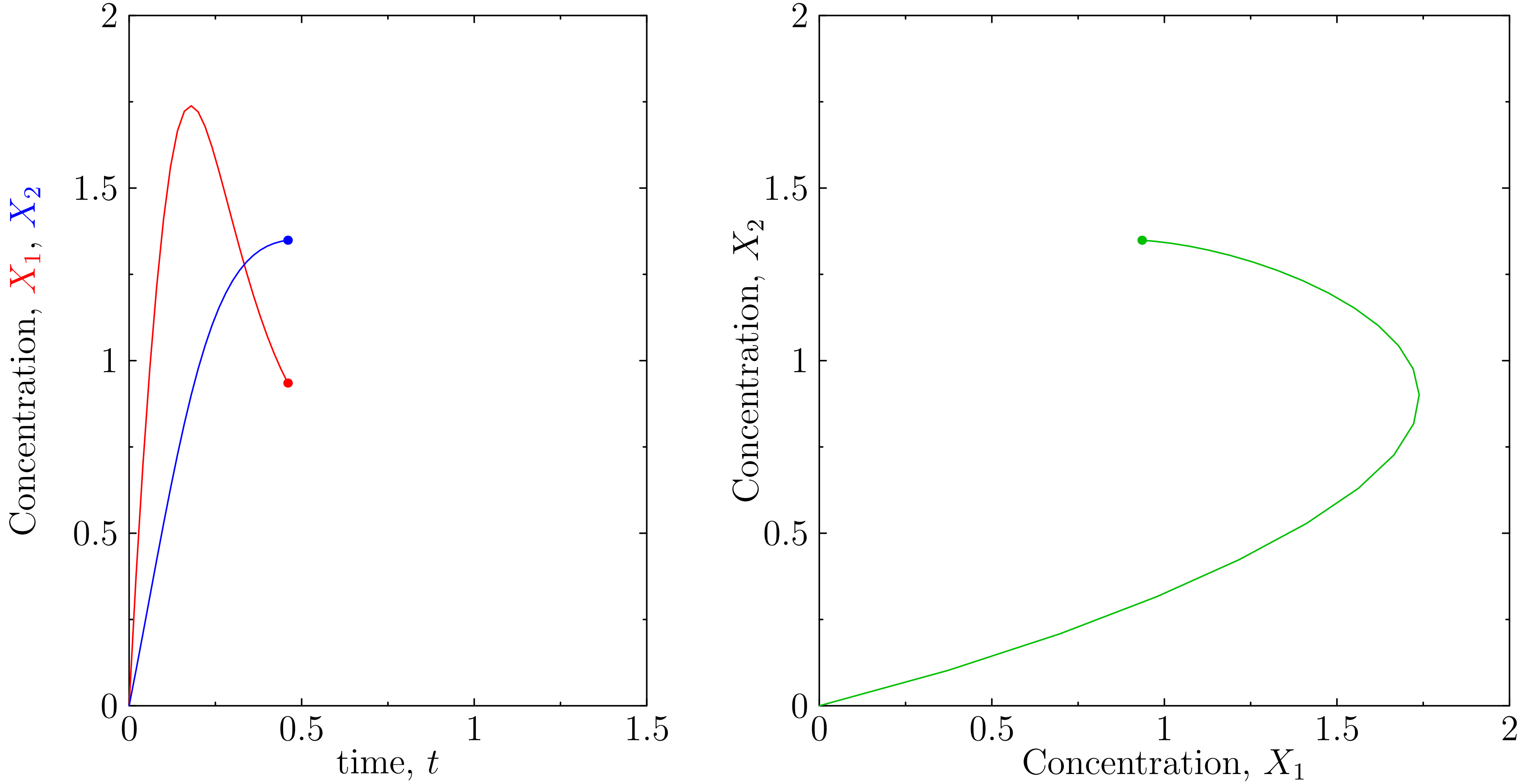

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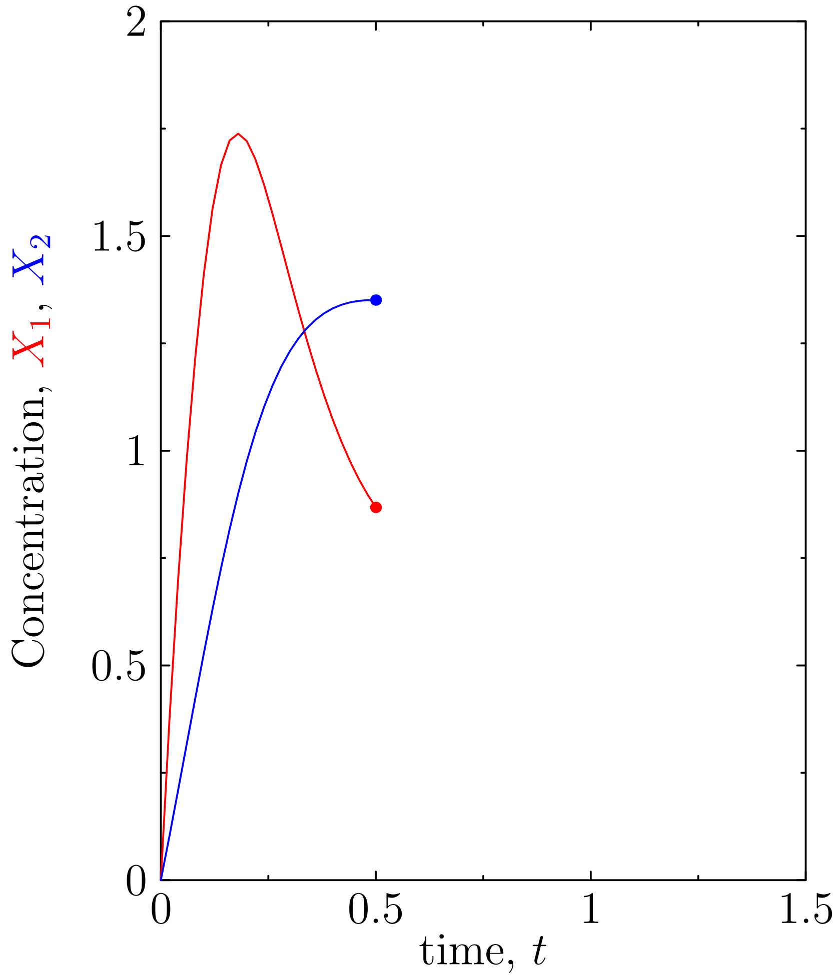

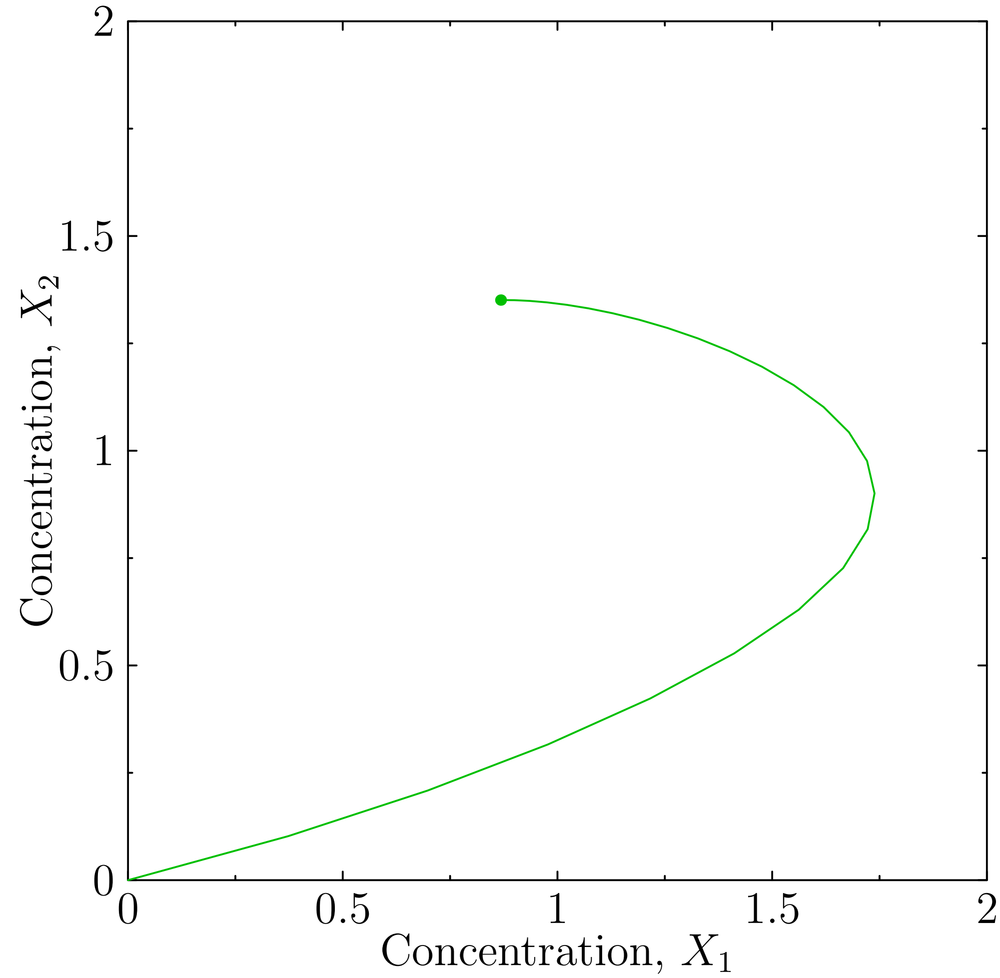

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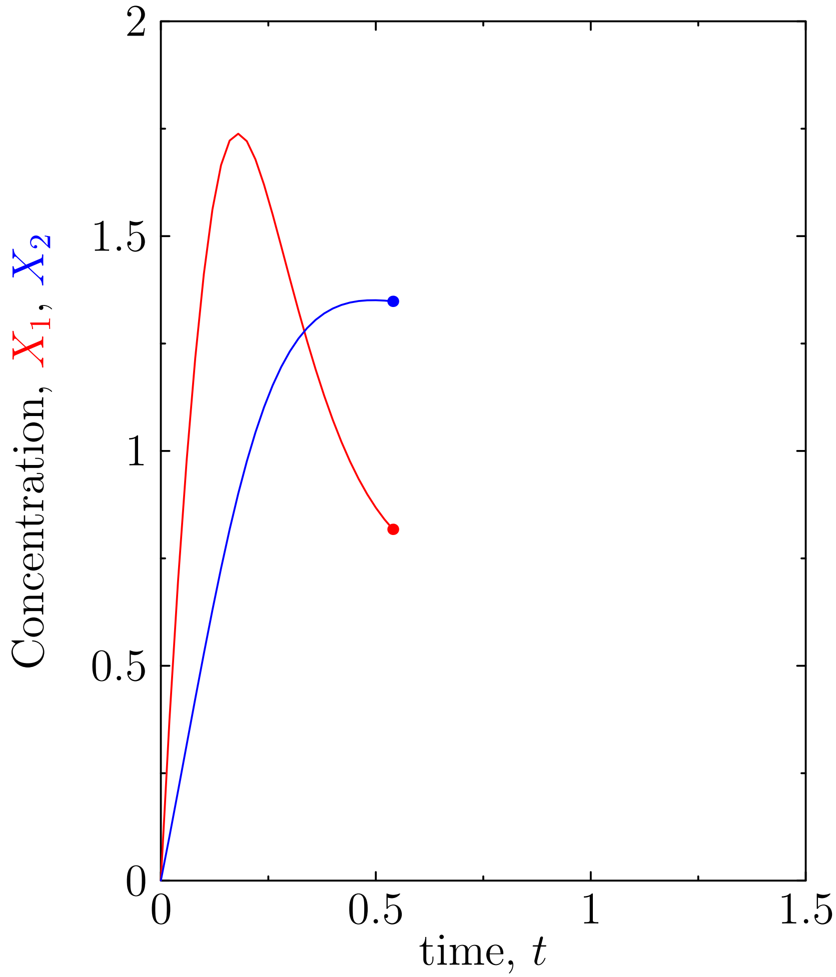

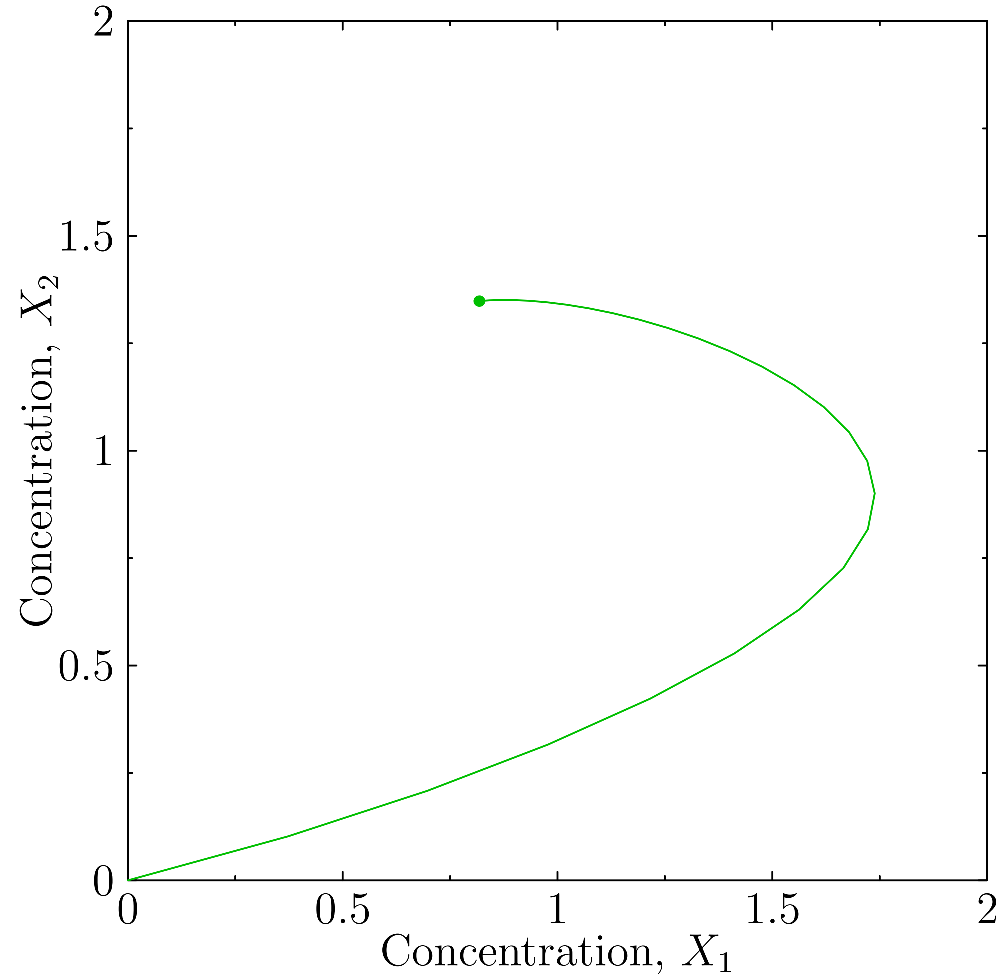

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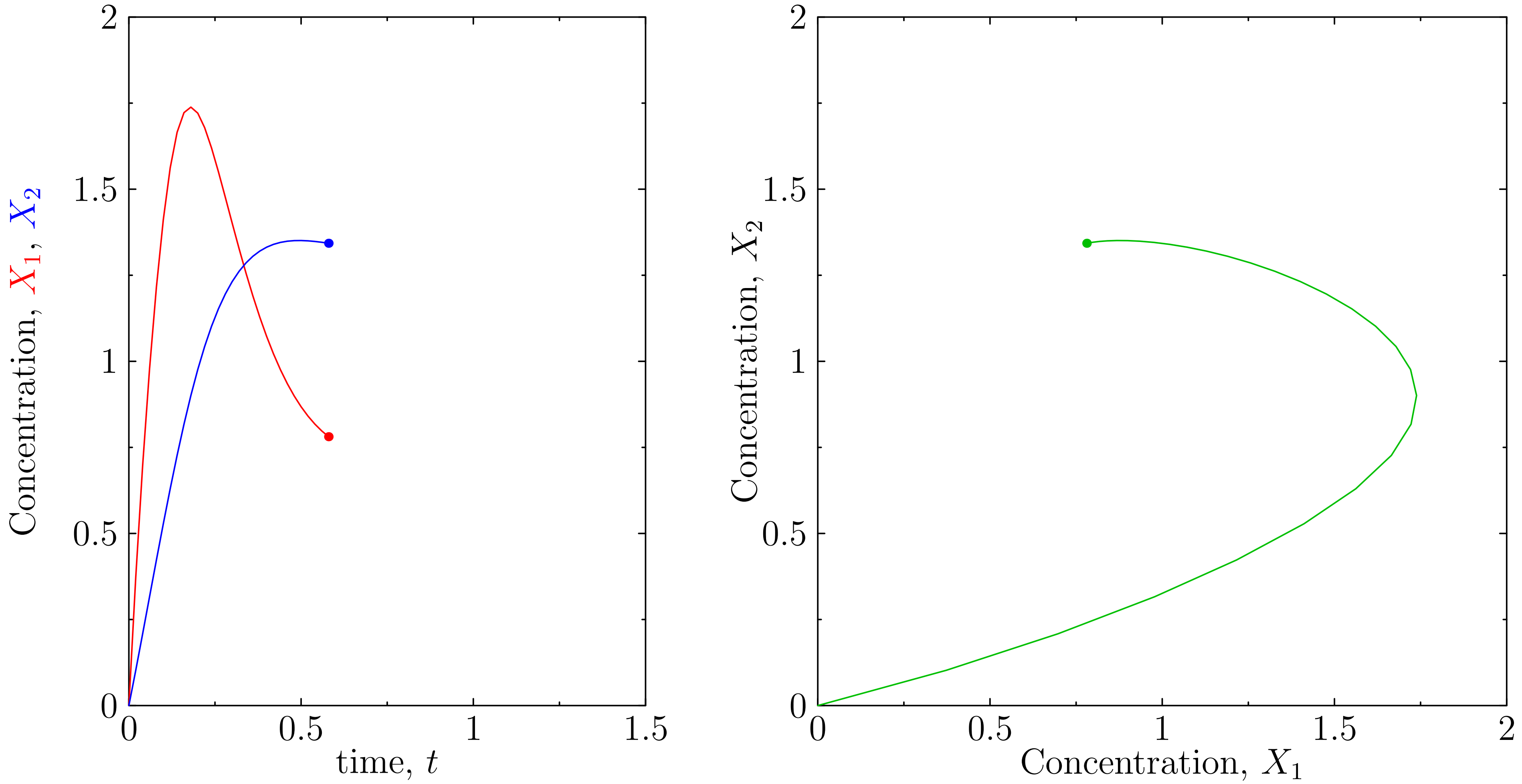

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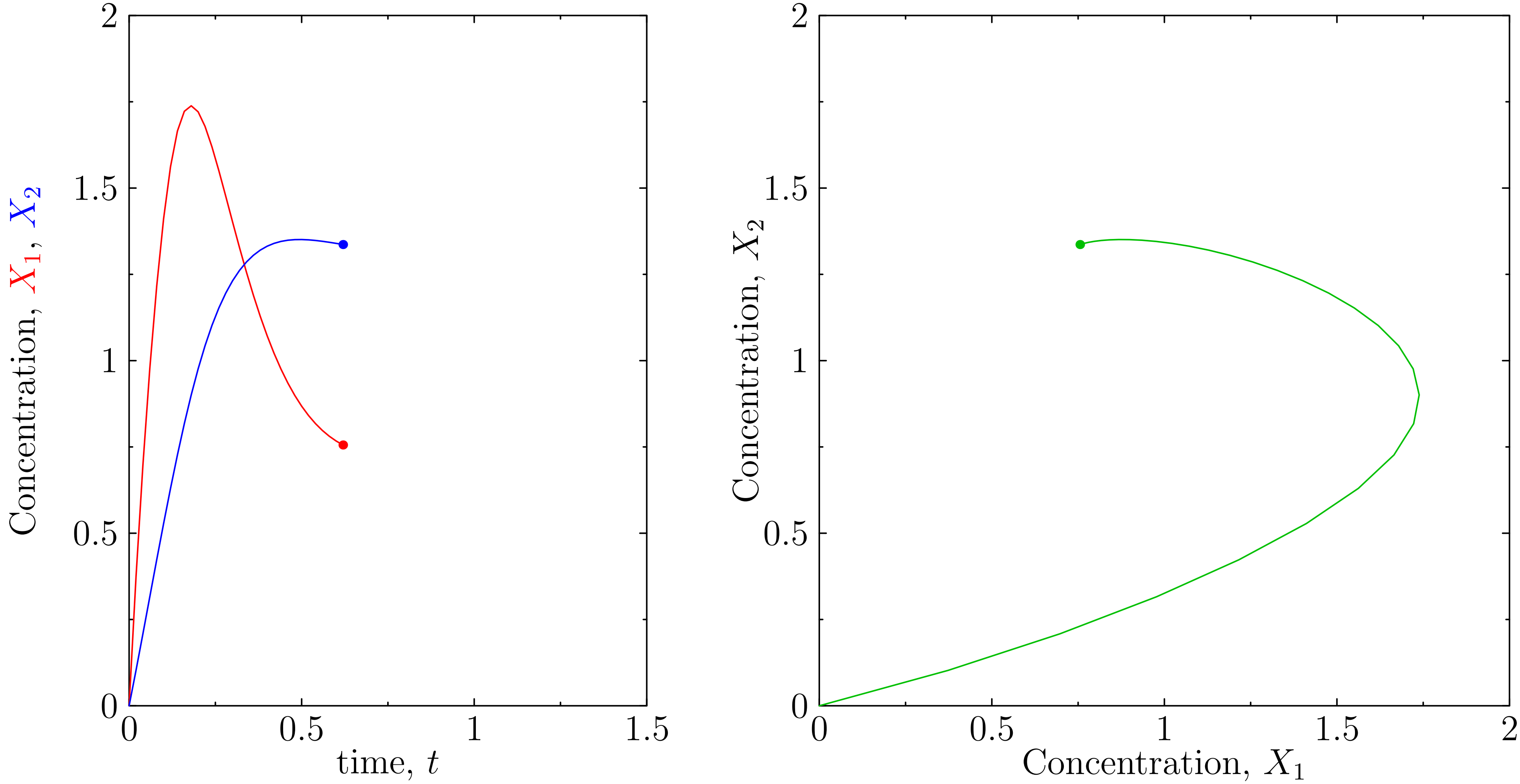

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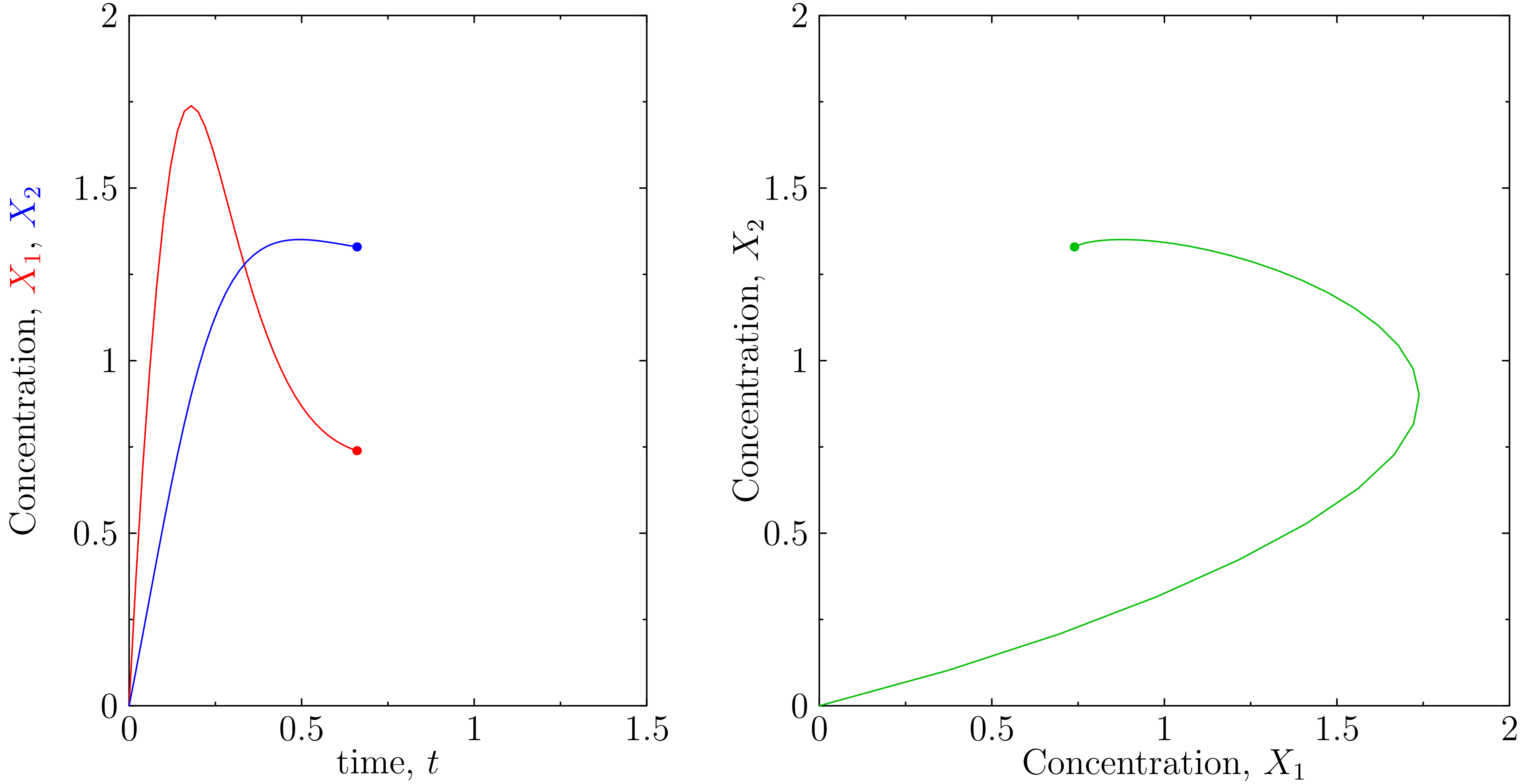

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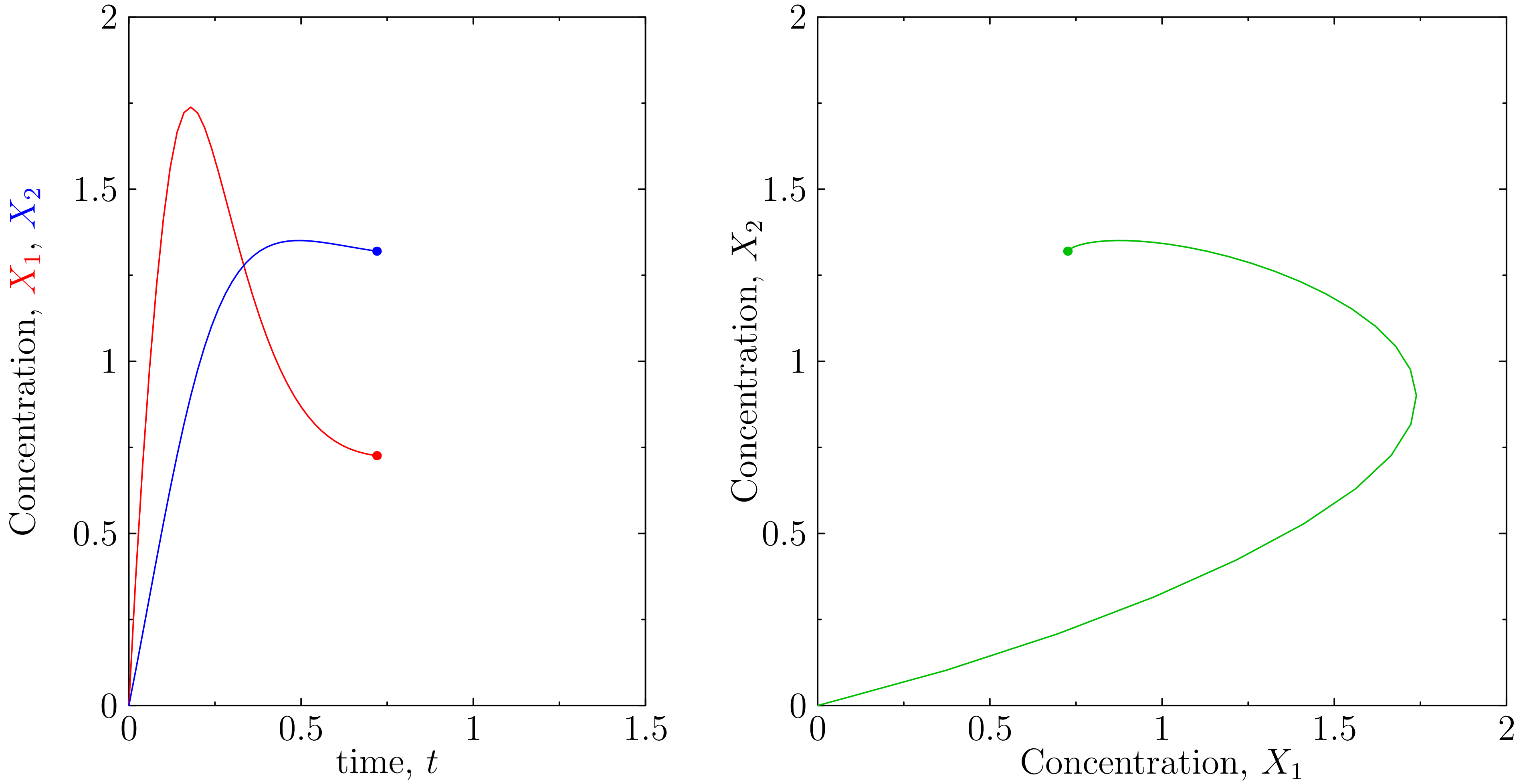

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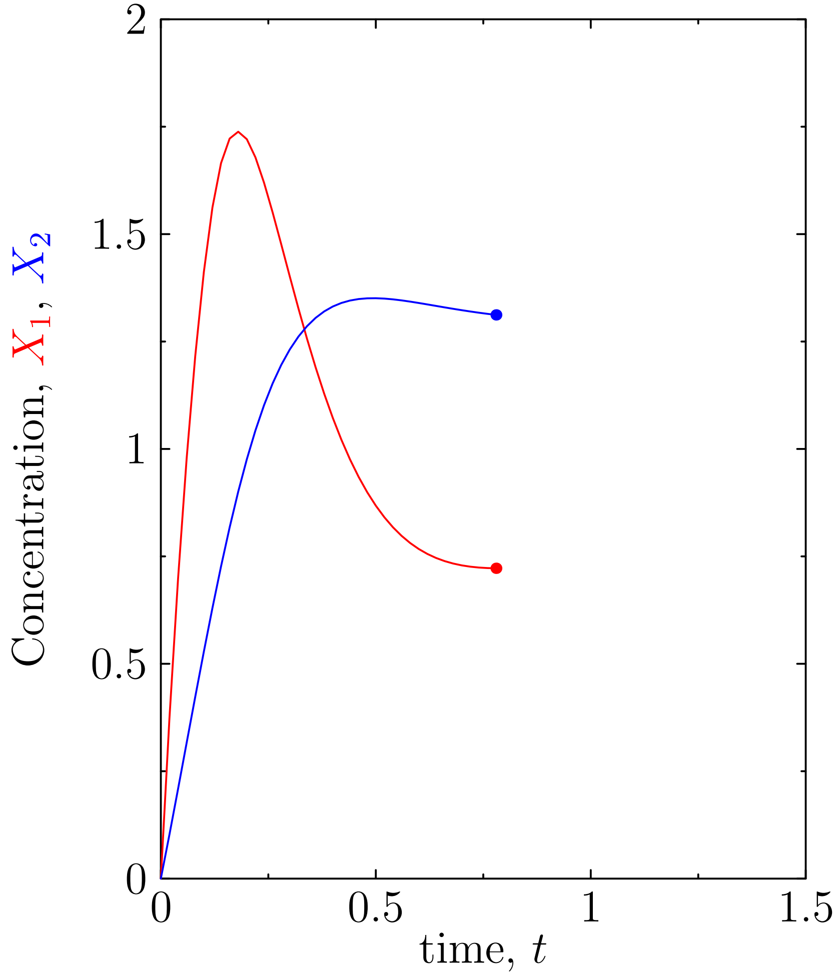

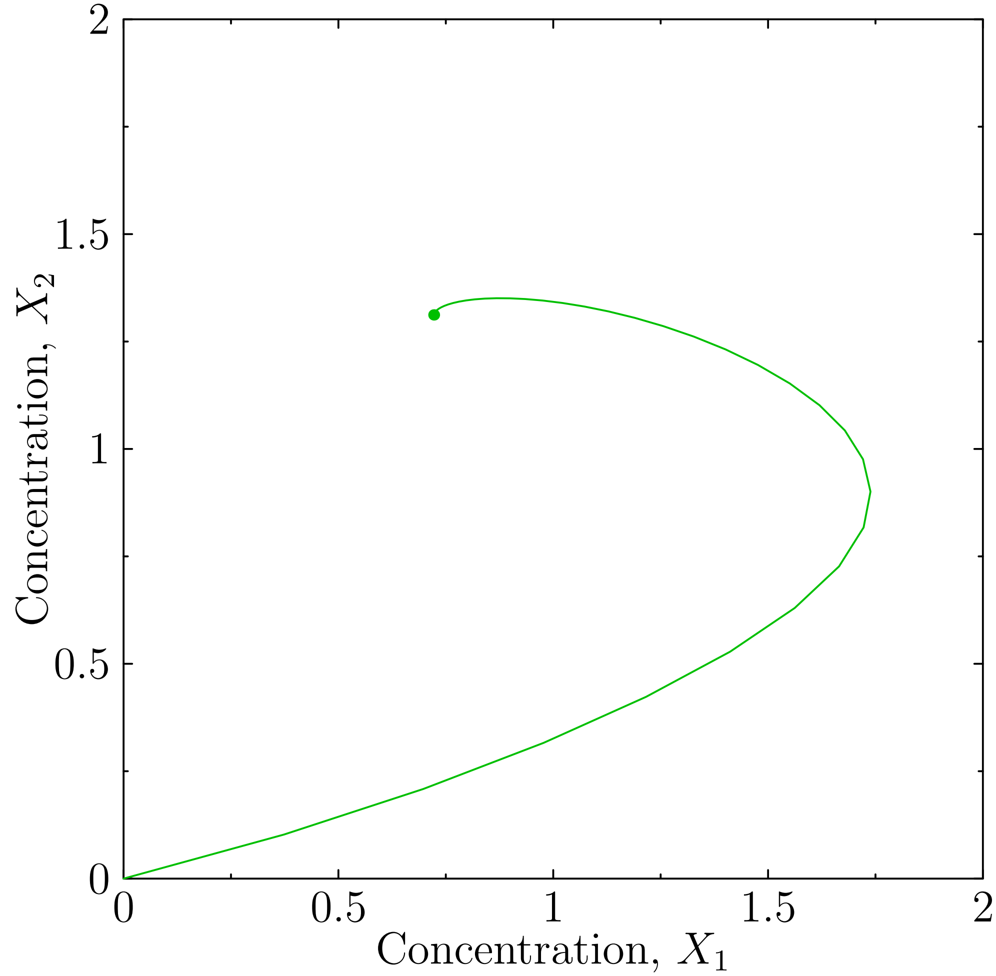

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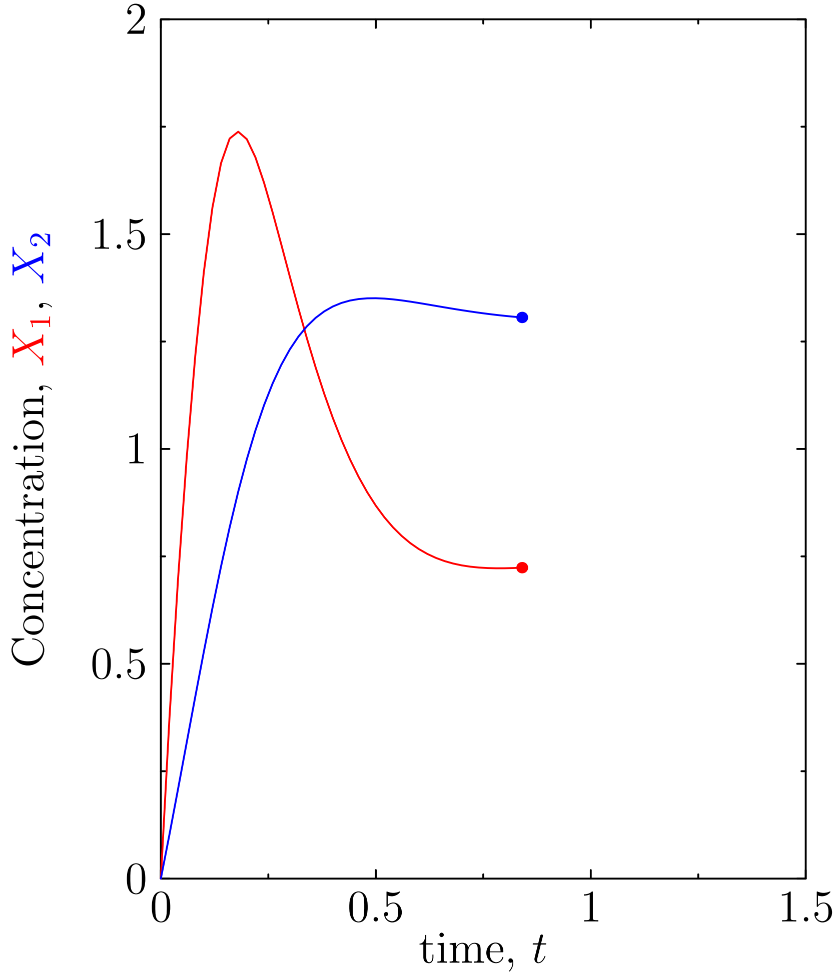

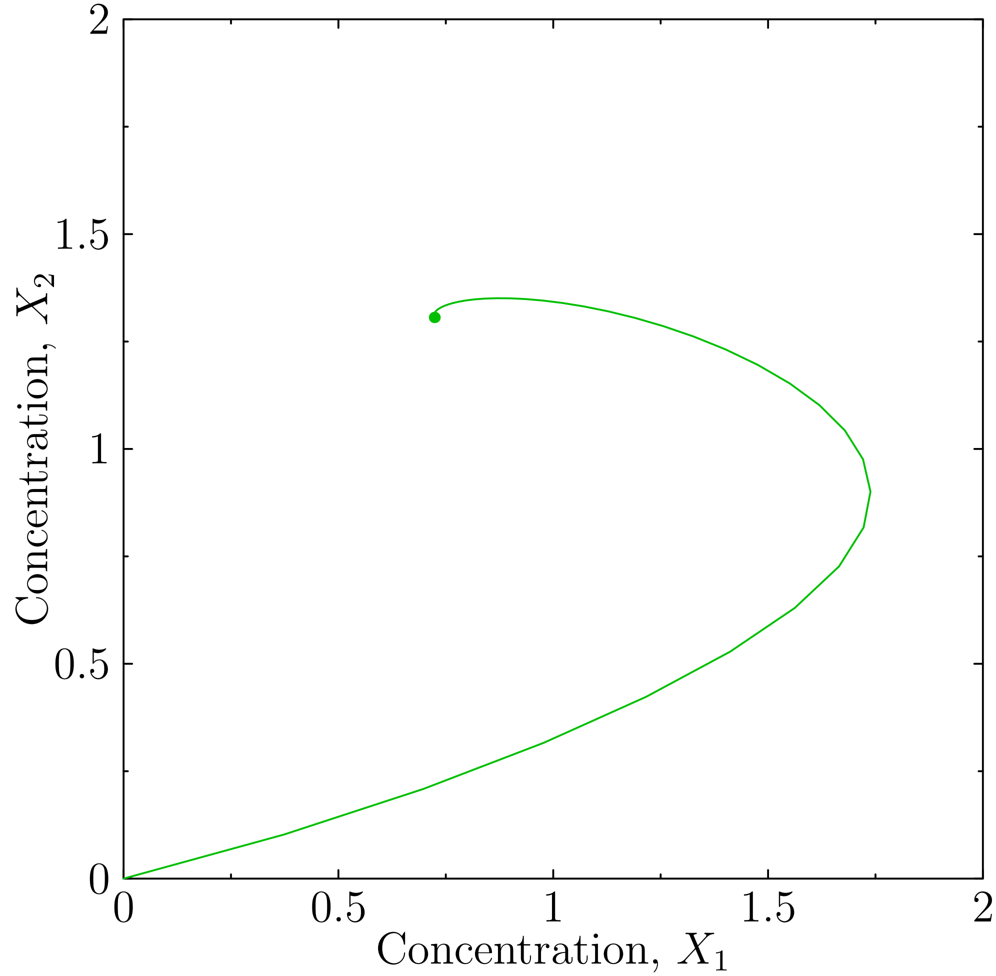

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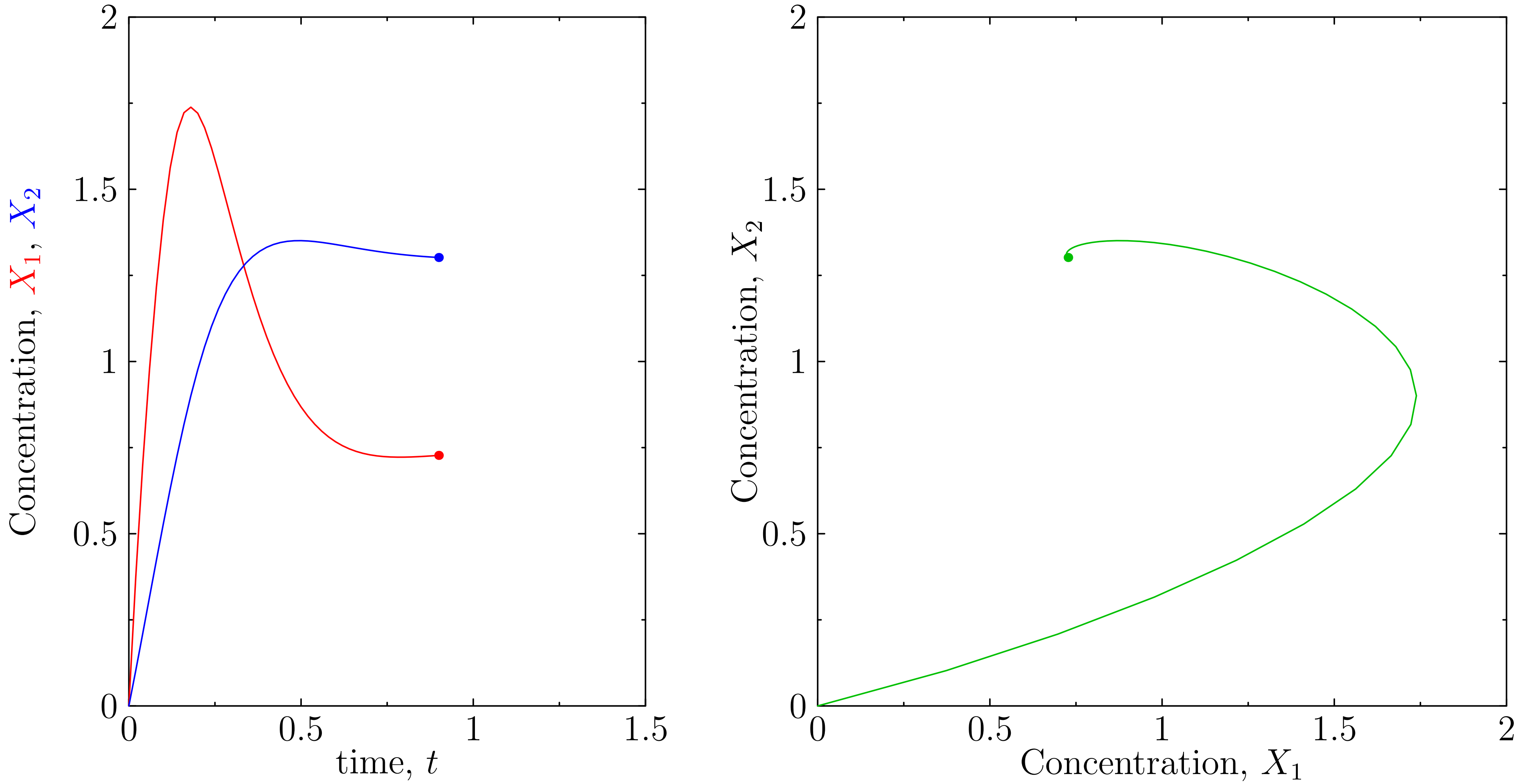

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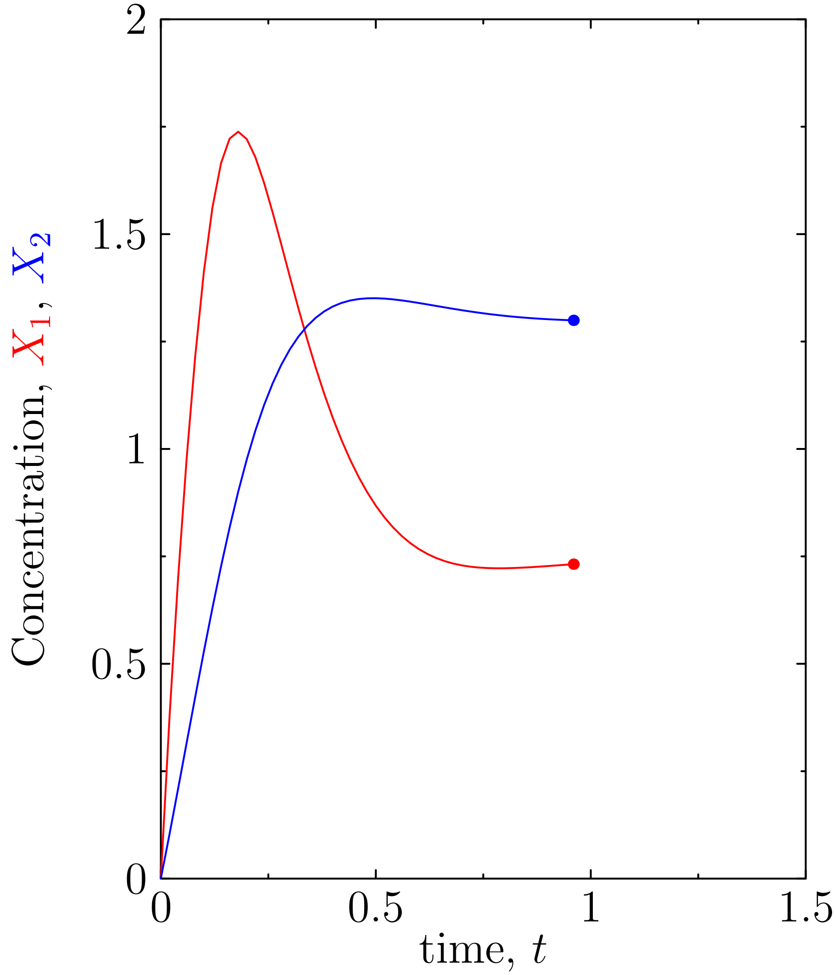

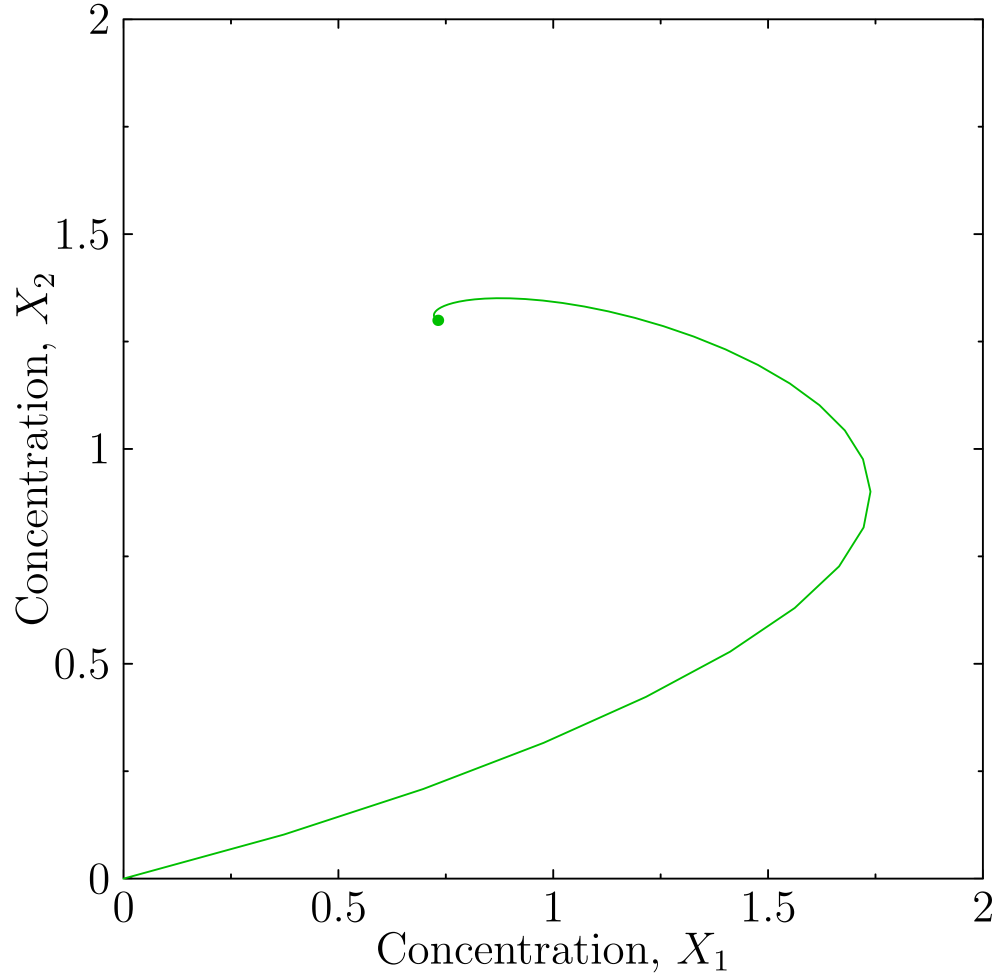

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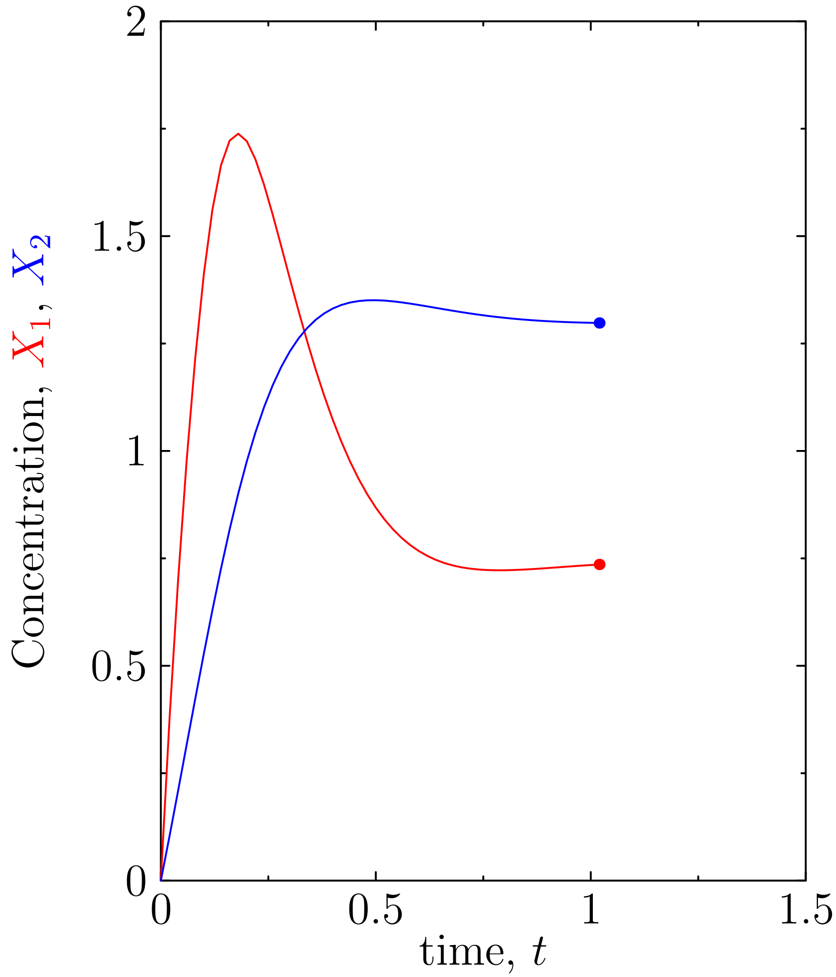
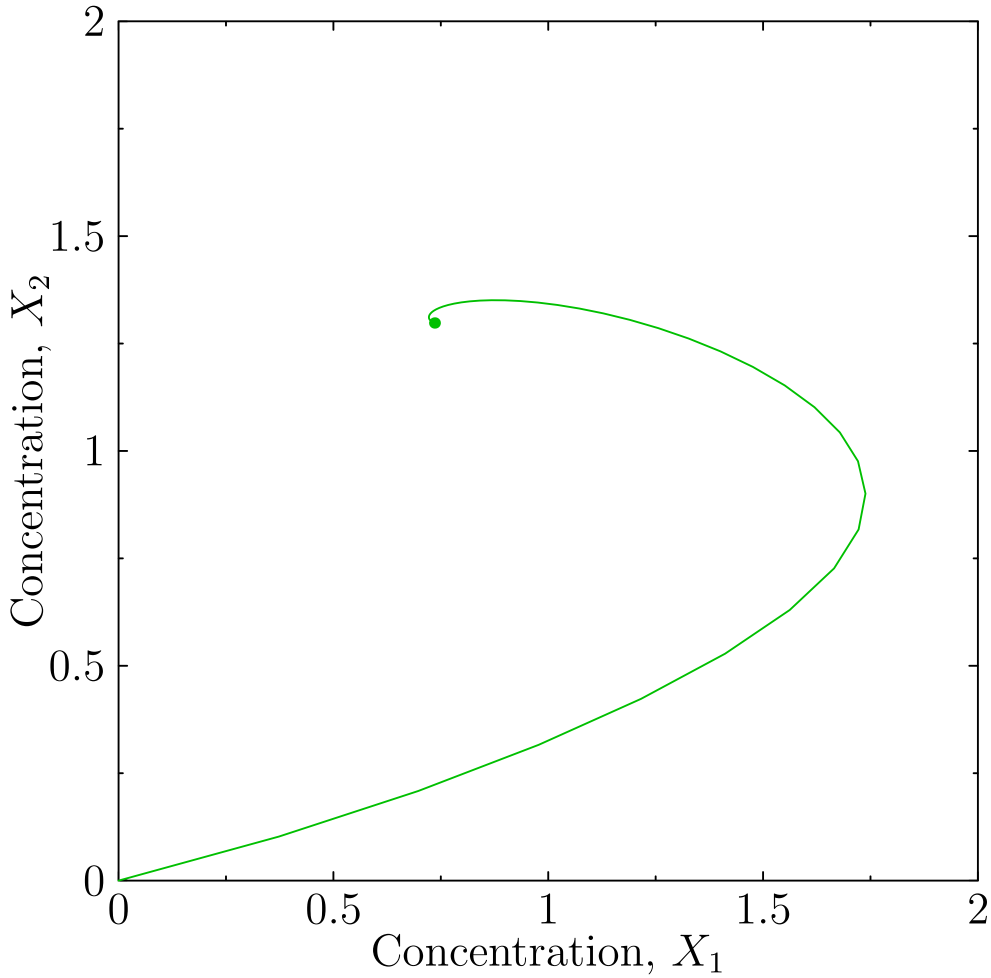
Solving the Dynamics





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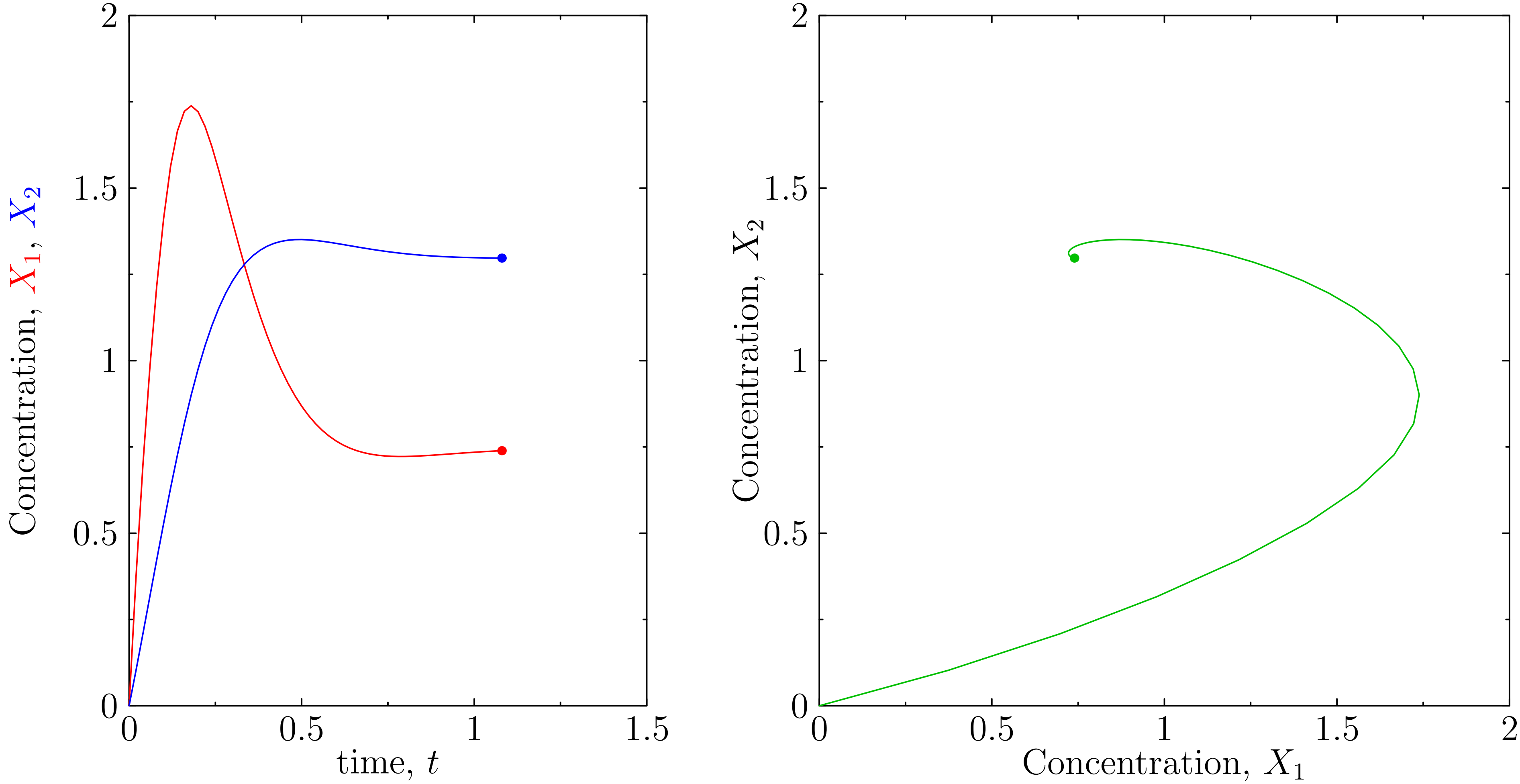

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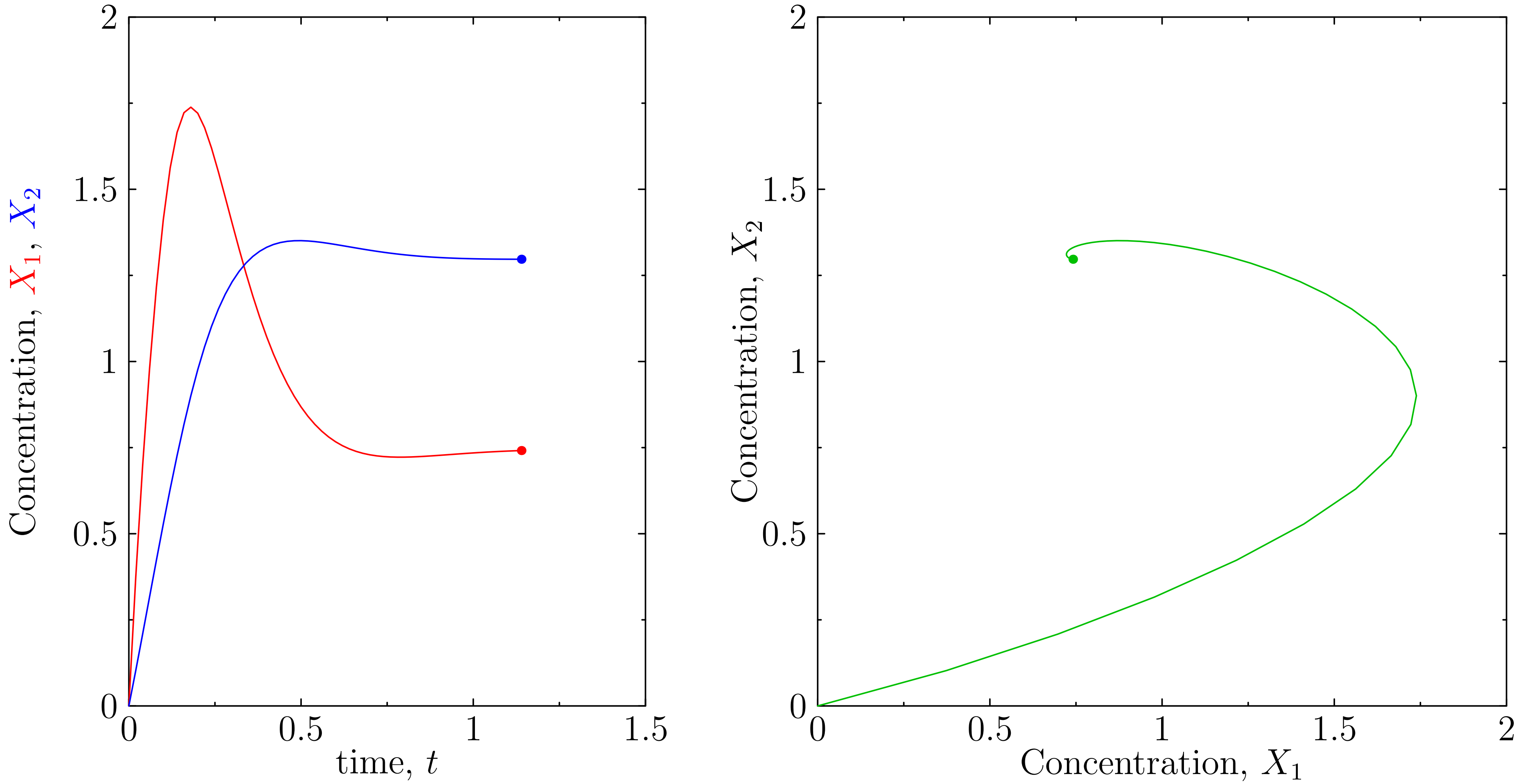

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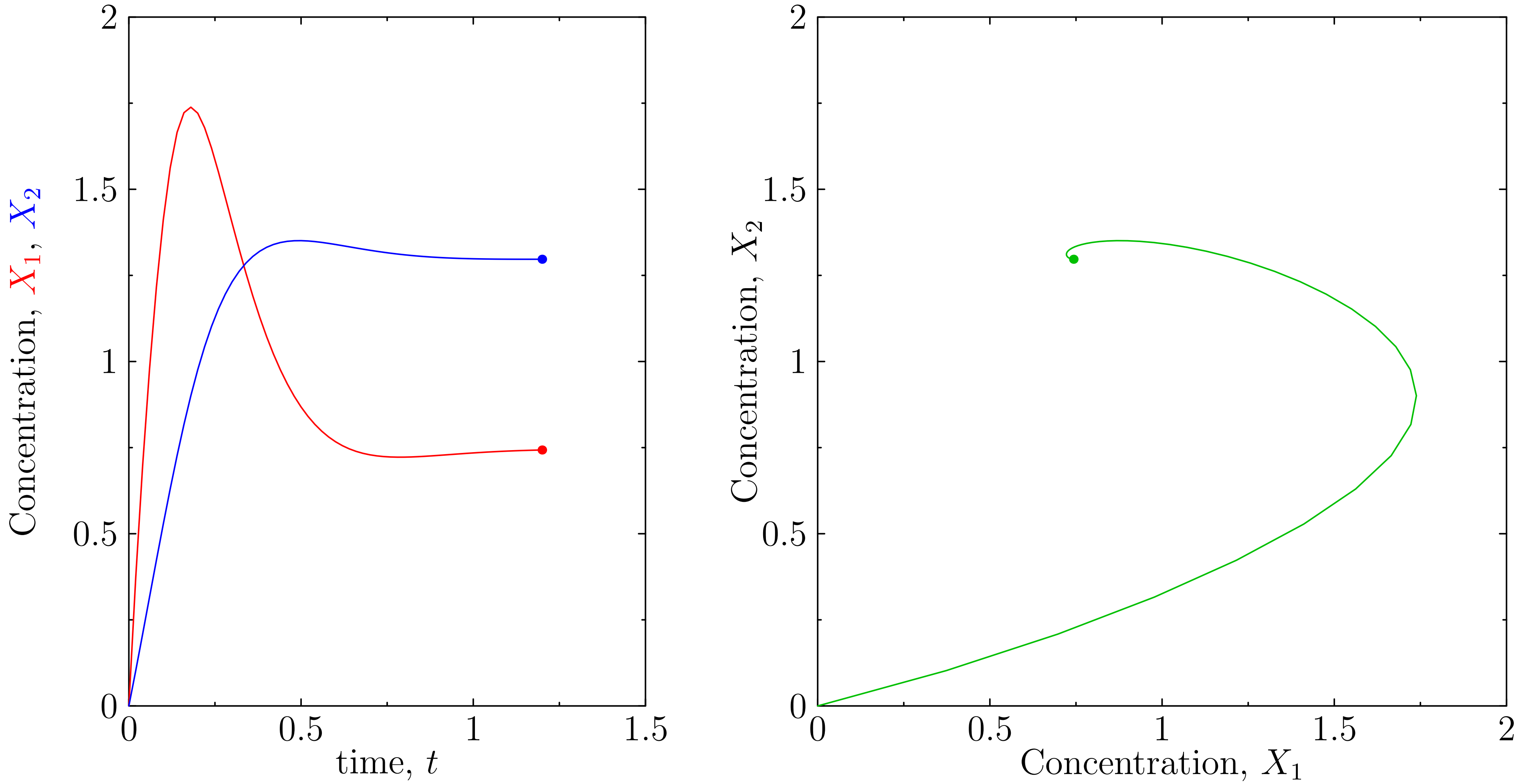

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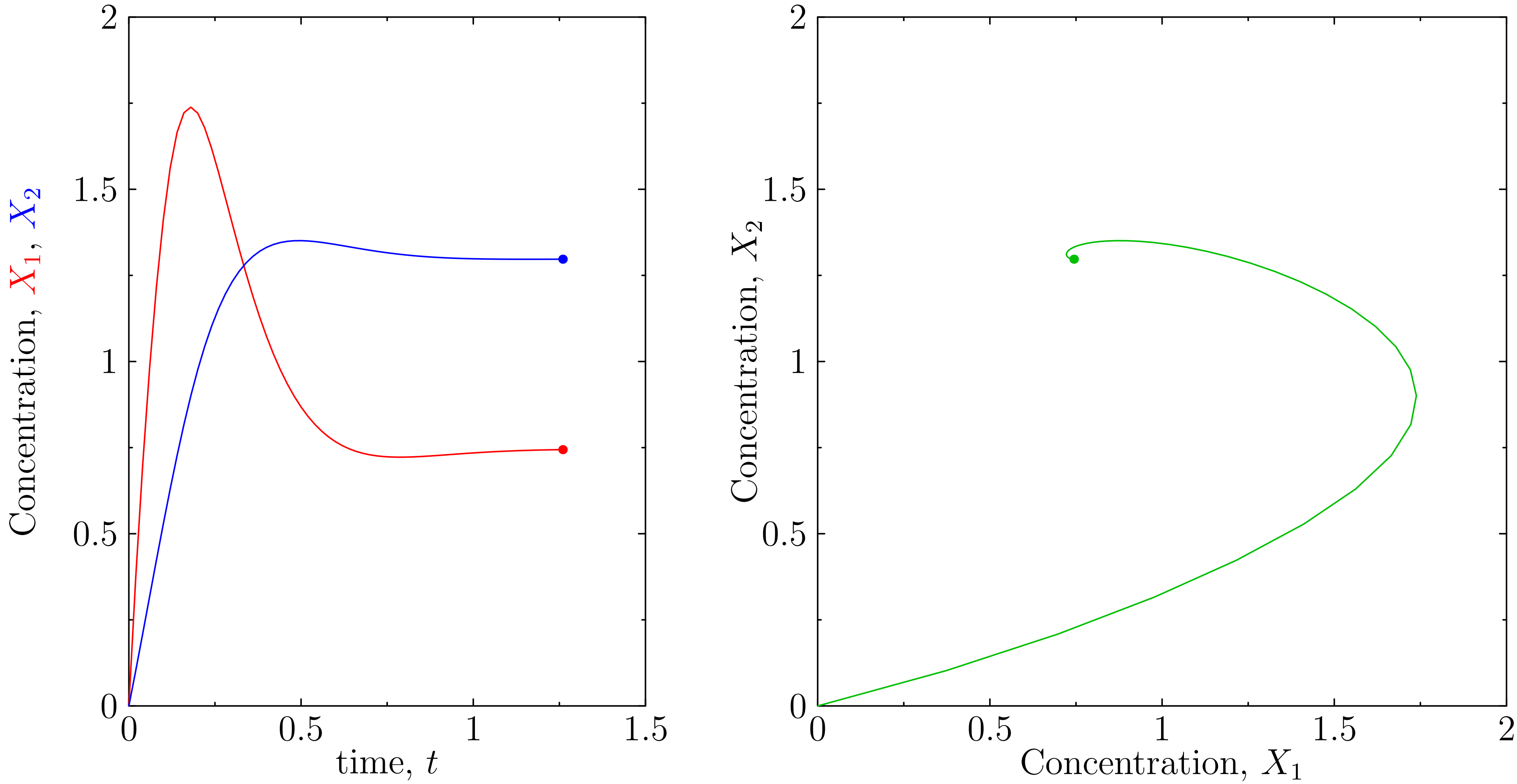

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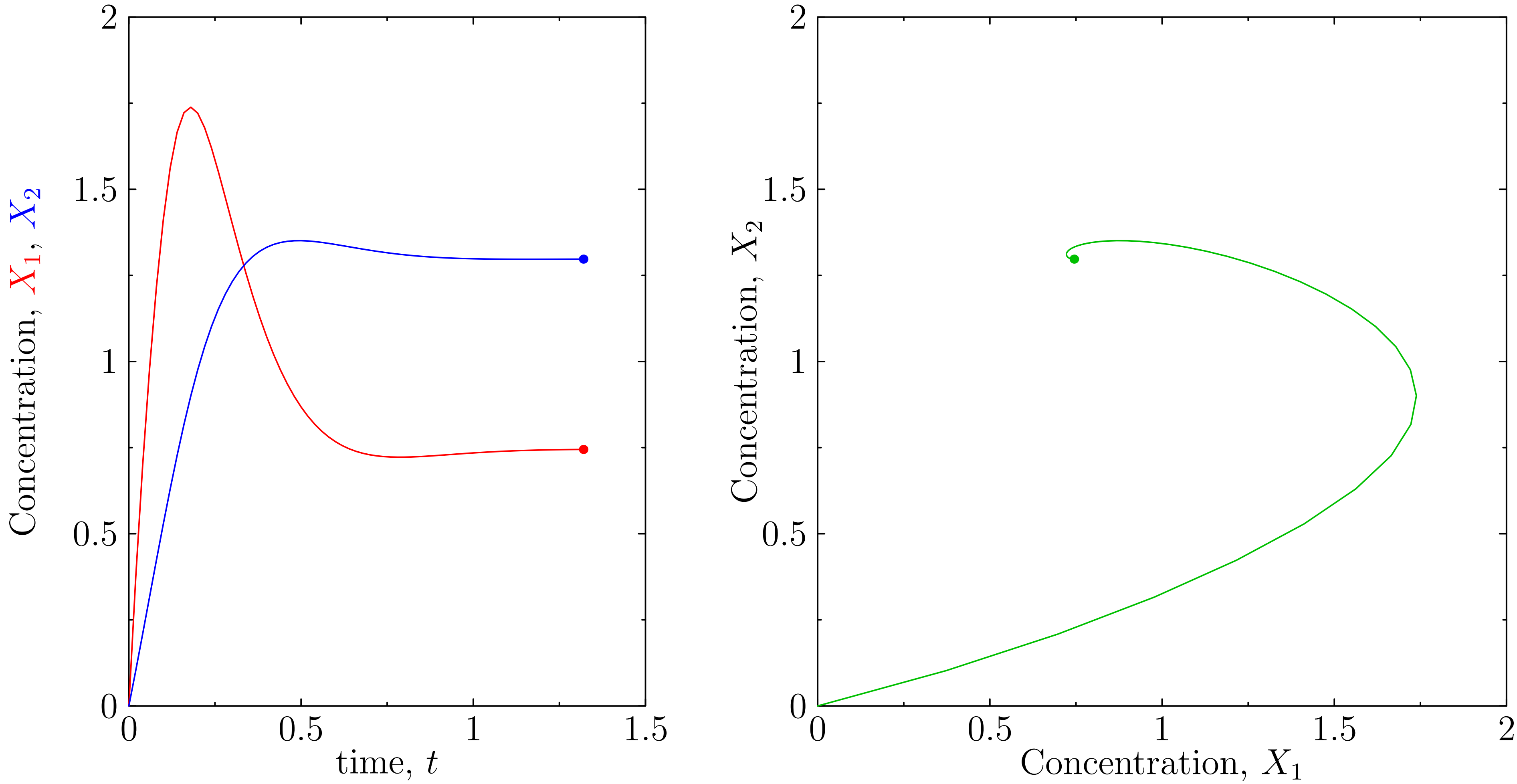

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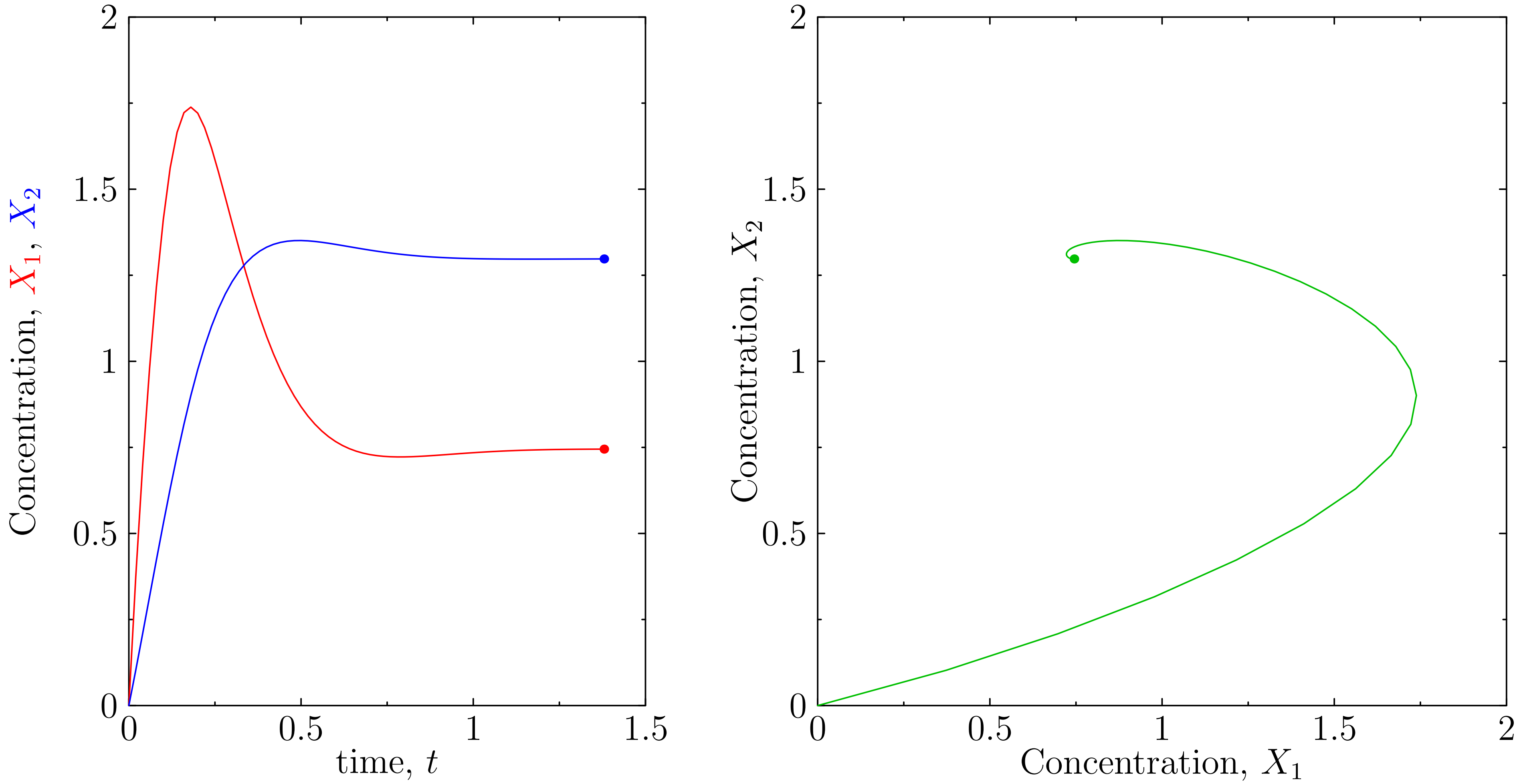

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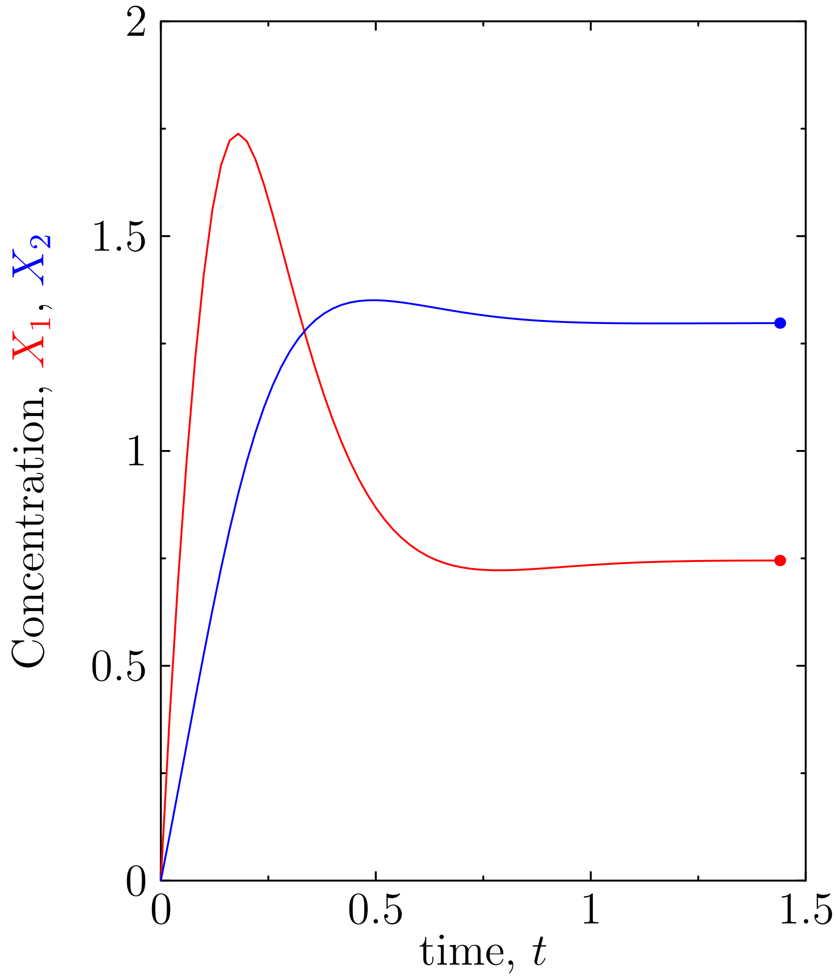

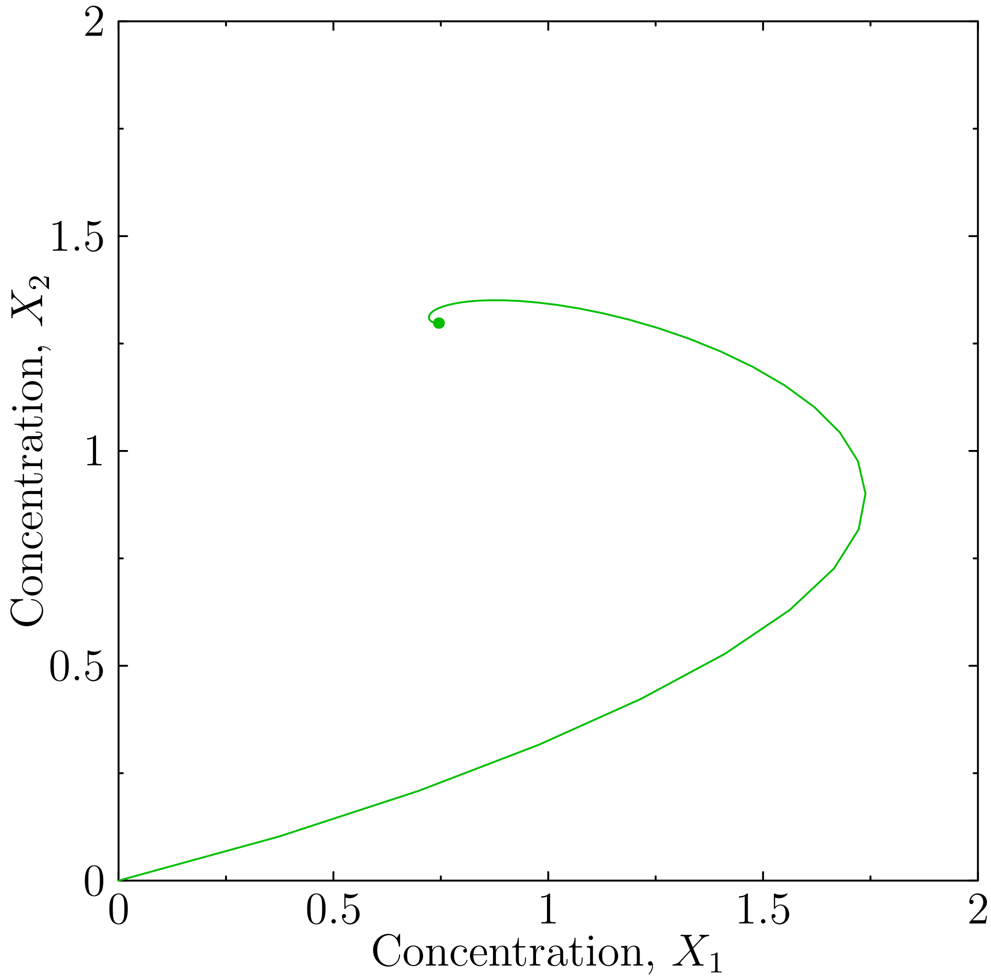

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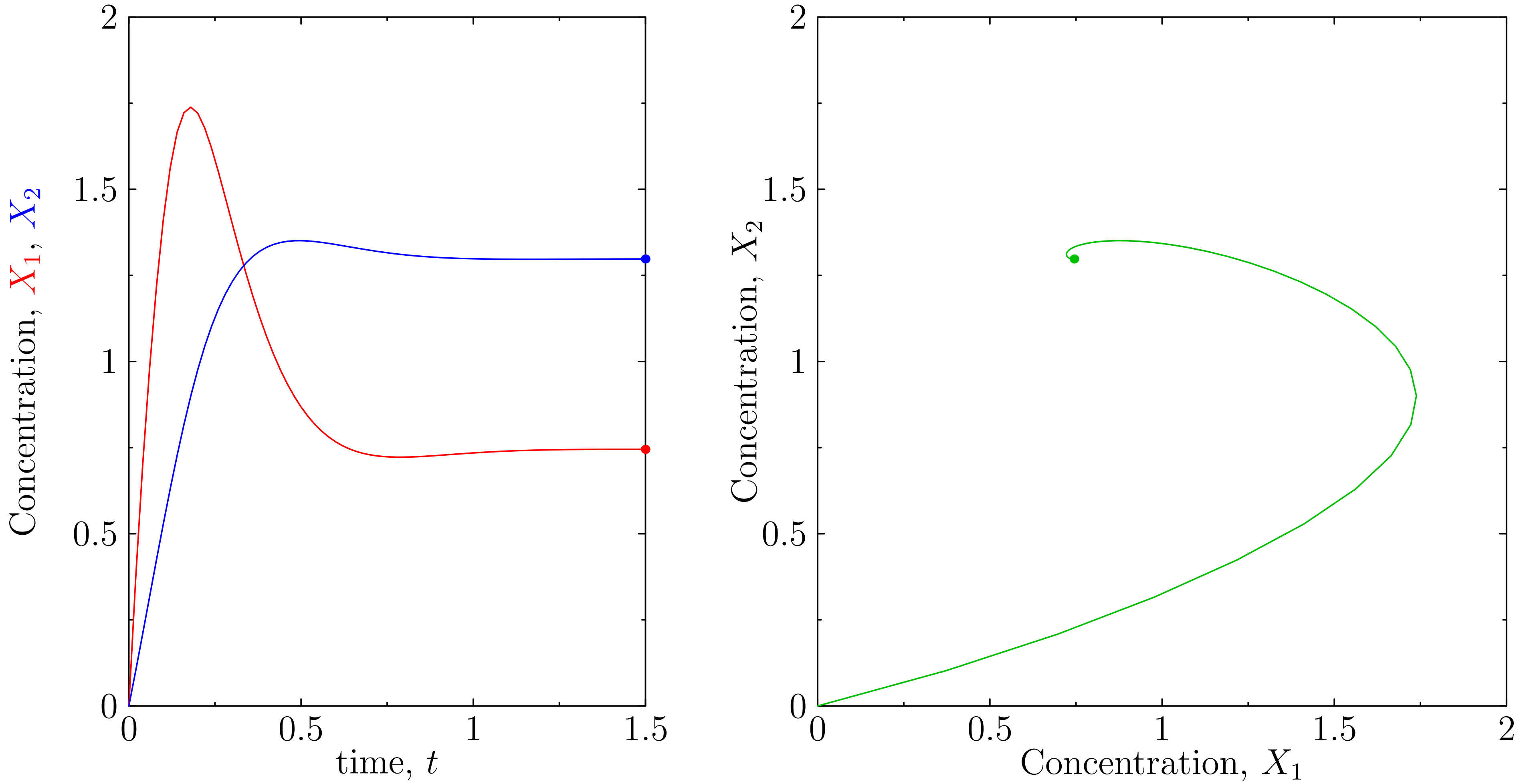

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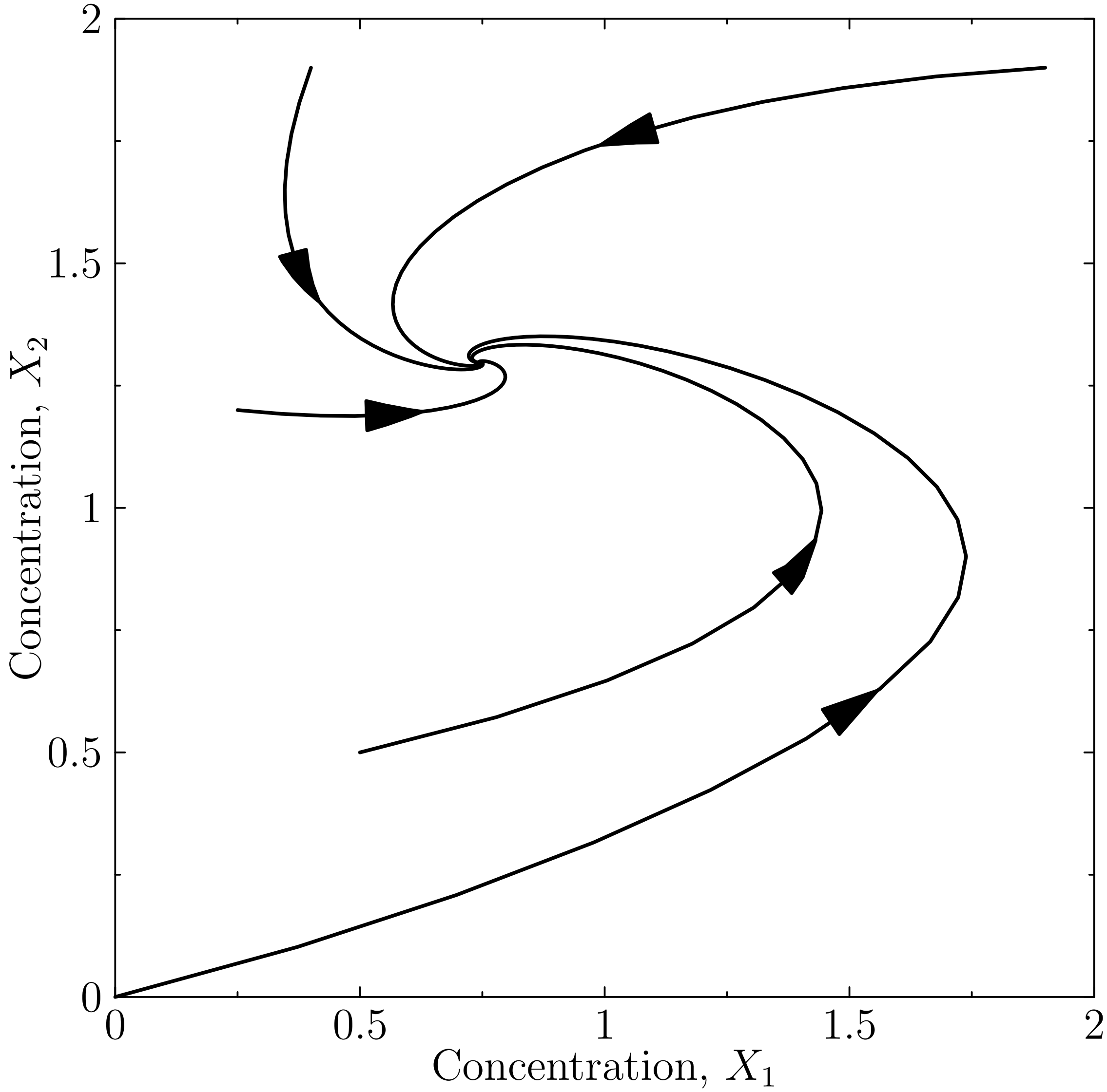
Trajectories



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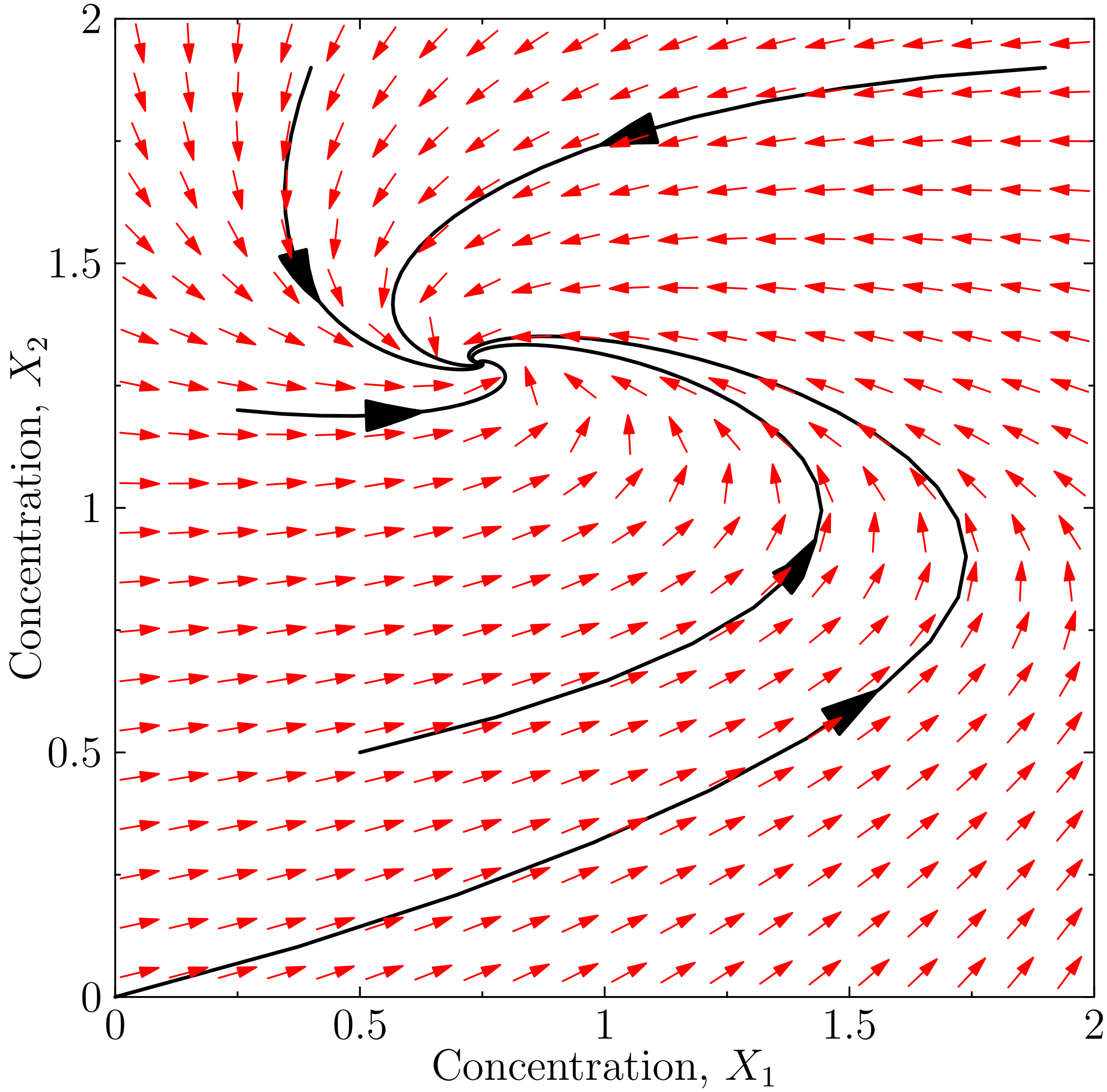
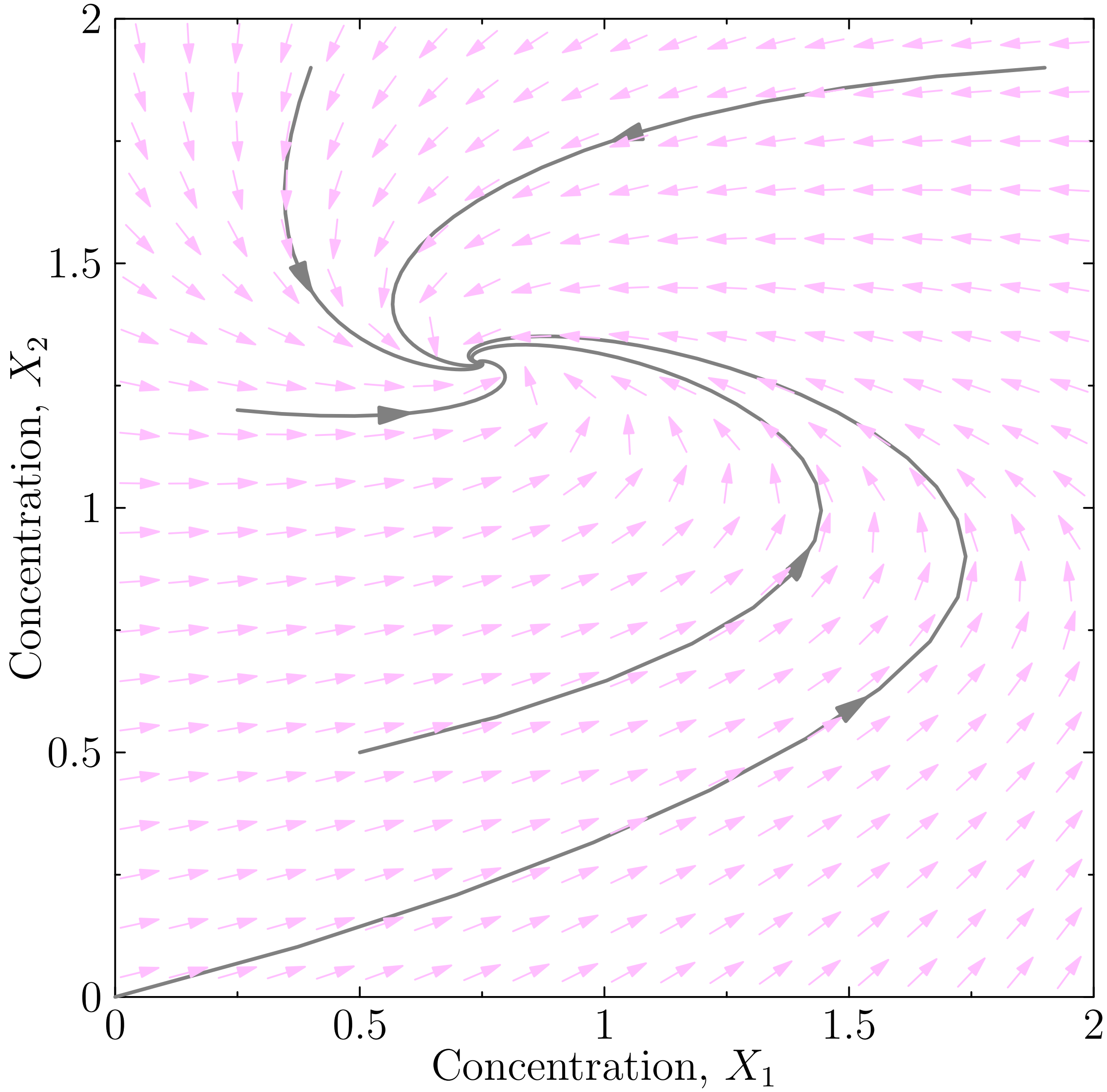


Nullclines



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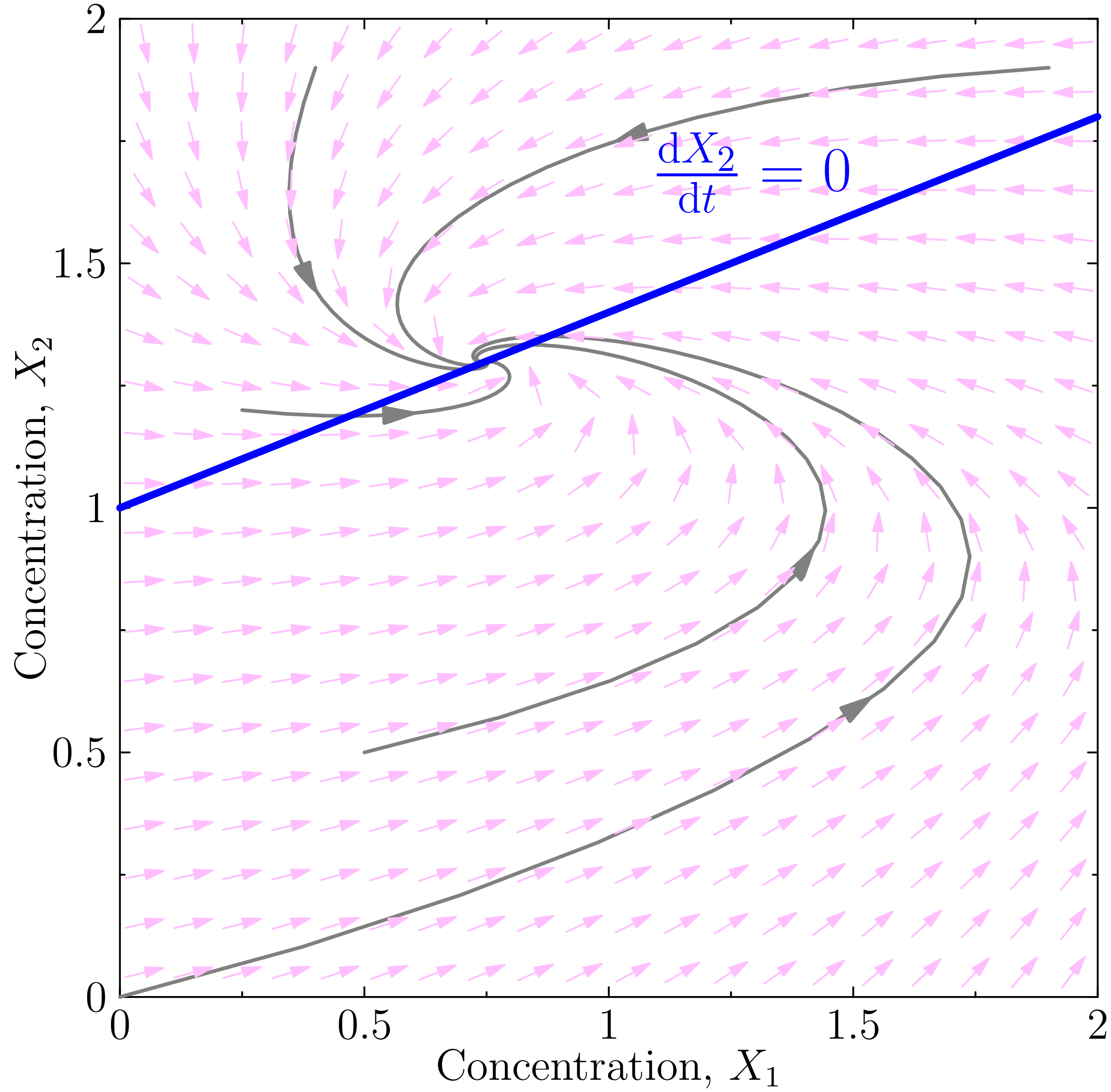

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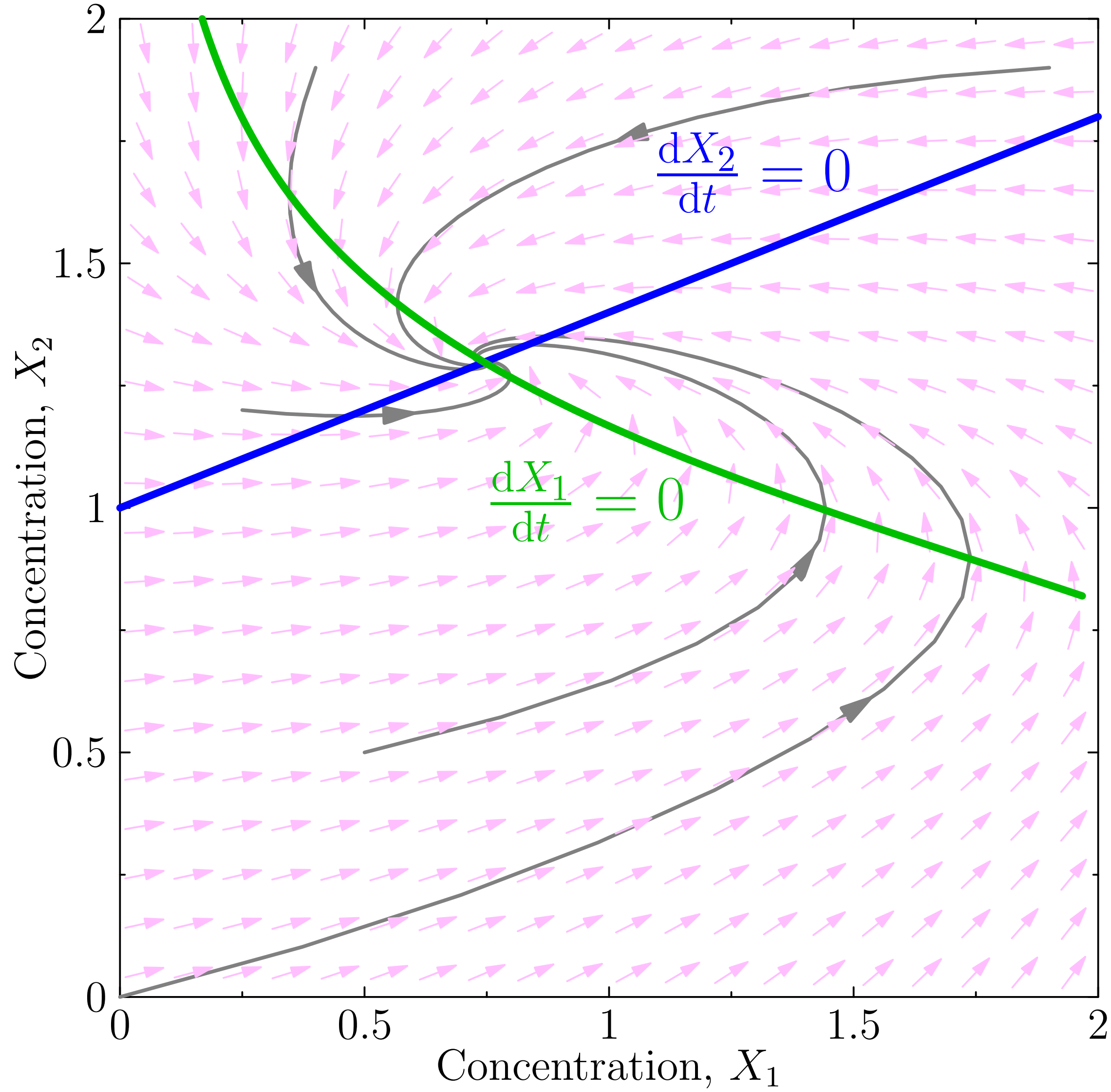
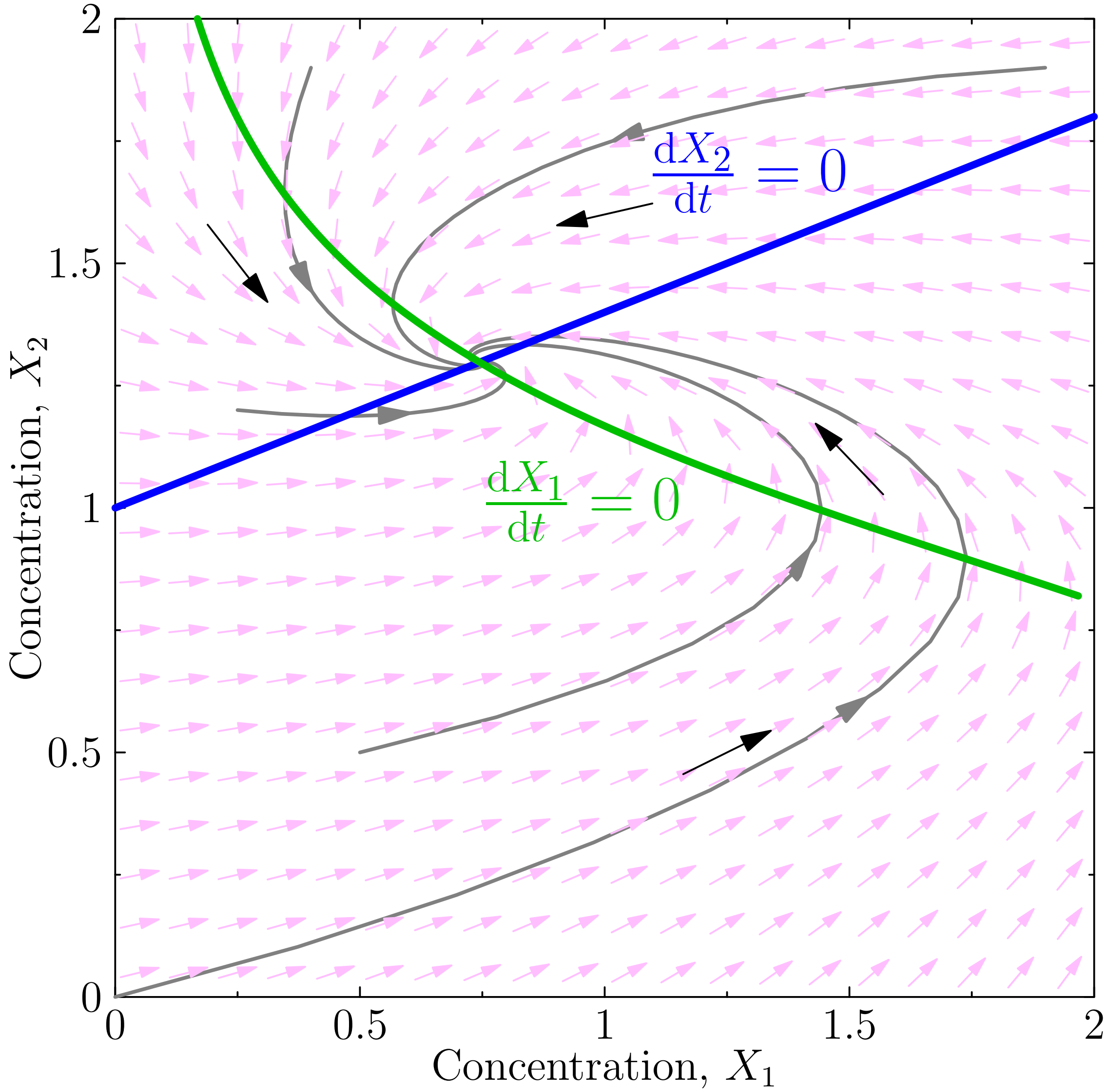


2
1.5
1
0.5
0
0
0.5
1
1.5
2
Concentration, X_1
Concentration, X_2
$\frac{dX_2}{dt} = 0$
$\frac{dX_1}{dt} = 0$

Nullclines



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Rescaling

- When we are working with ODEs we can work in any units of concentration and time

- Note that we can change our units of concentrations by a rescaling $[X_i] \to r\,[X_i]$

- If we rescale $K \to r\,K$, $t \to r\,t$, $k_i \to k_i/r$ for $i \in \{3, 4, 5\}$ this leaves the ODEs invariant

$$\frac{\mathrm{d}[X_1(t)]}{\mathrm{d}t} = \frac{k_1}{1 + ([X_2(t)]/K)^n} - k_3\,[X_1(t)] - k_5\,[X_1(t)]$$

$$\frac{\mathrm{d}[X_2(t)]}{\mathrm{d}t} = k_2 + k_5\,[X_1(t)] - k_4\,[X_2(t)]$$

- When going to numbers $[X_i(t)] = X_i/\Omega$ by choosing r we can explore systems with the same dynamics for the mean concentration but different mean numbers

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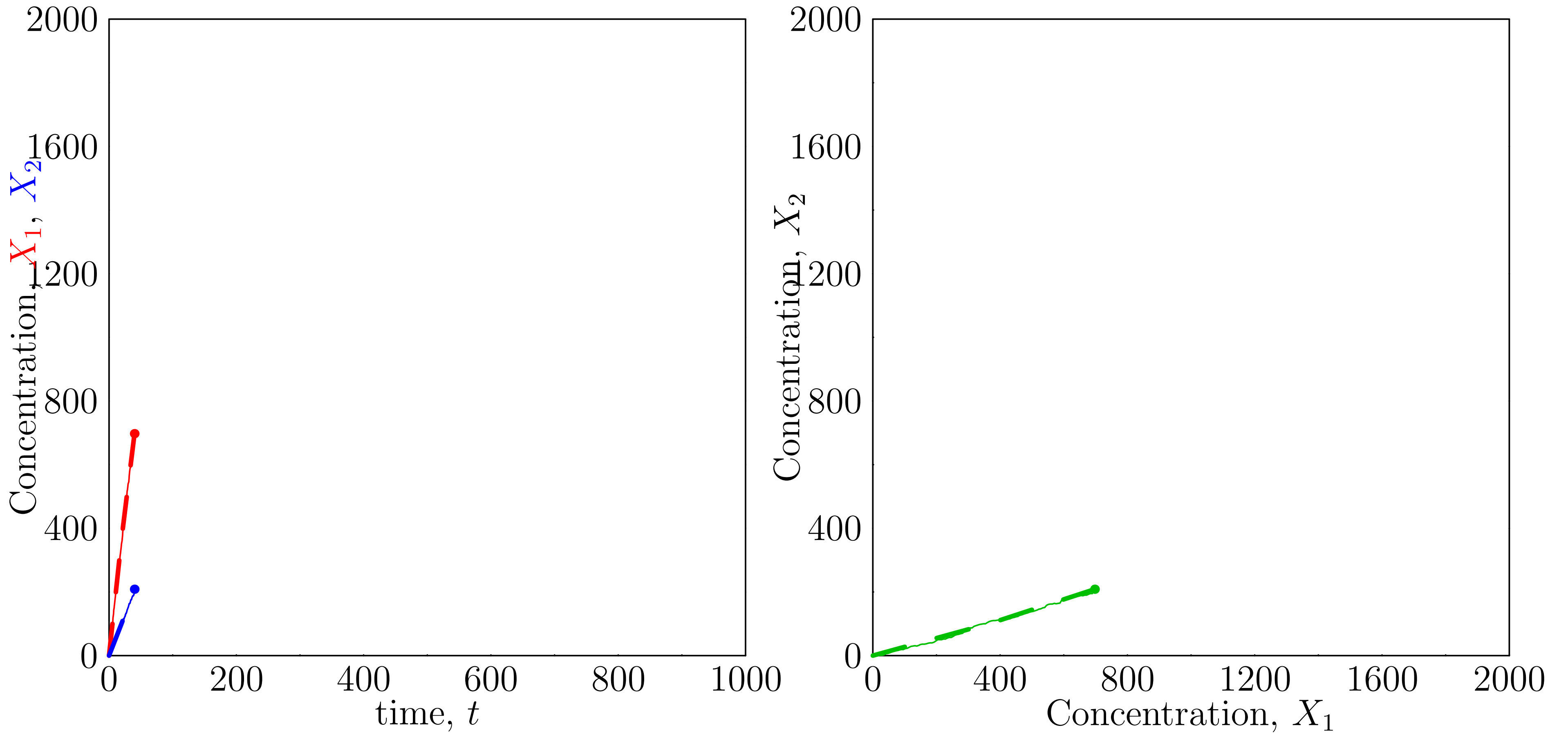
Large Numbers ($r = 1000$)



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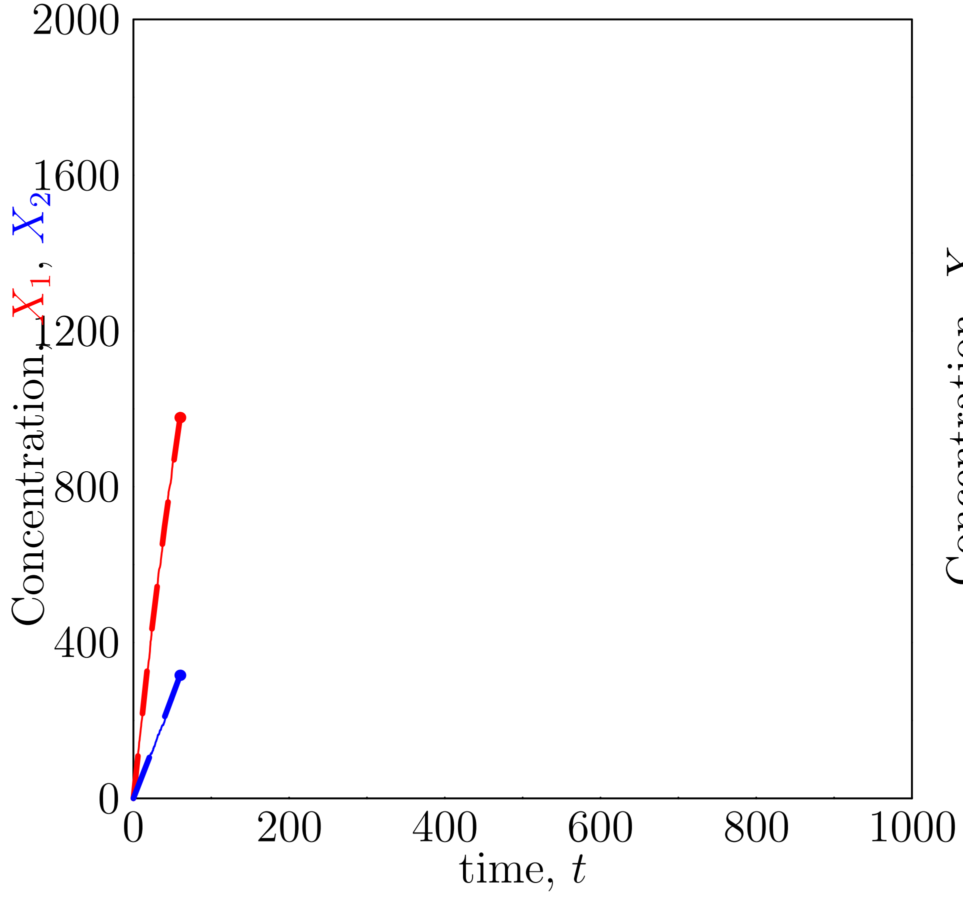

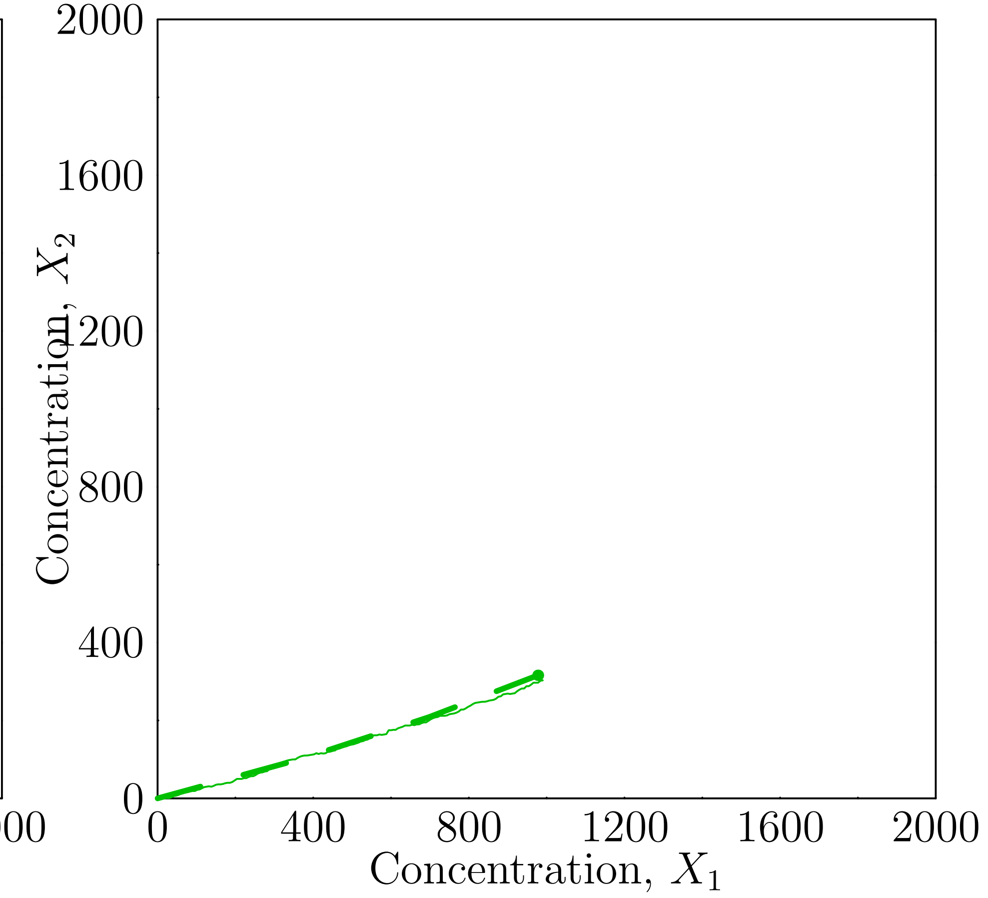

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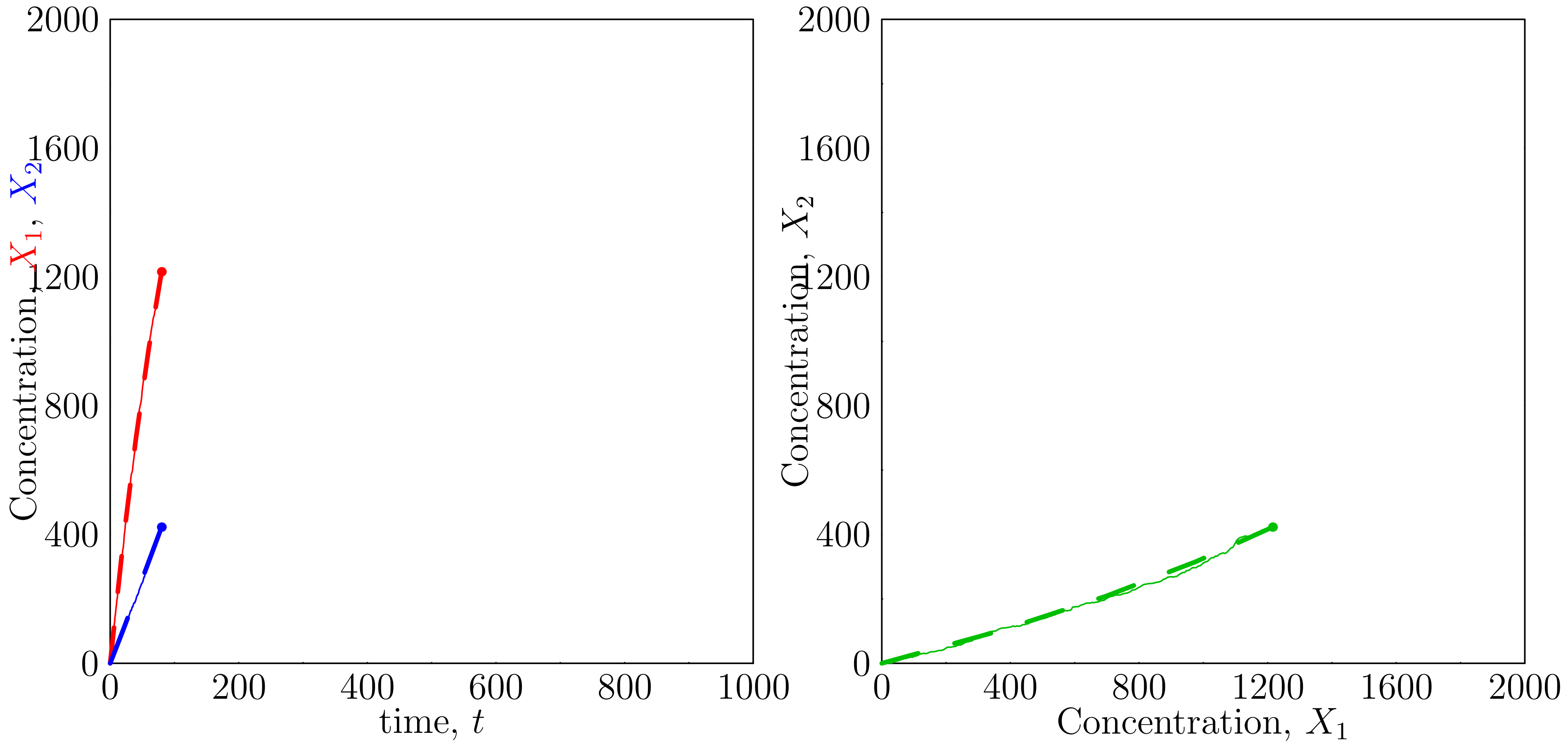

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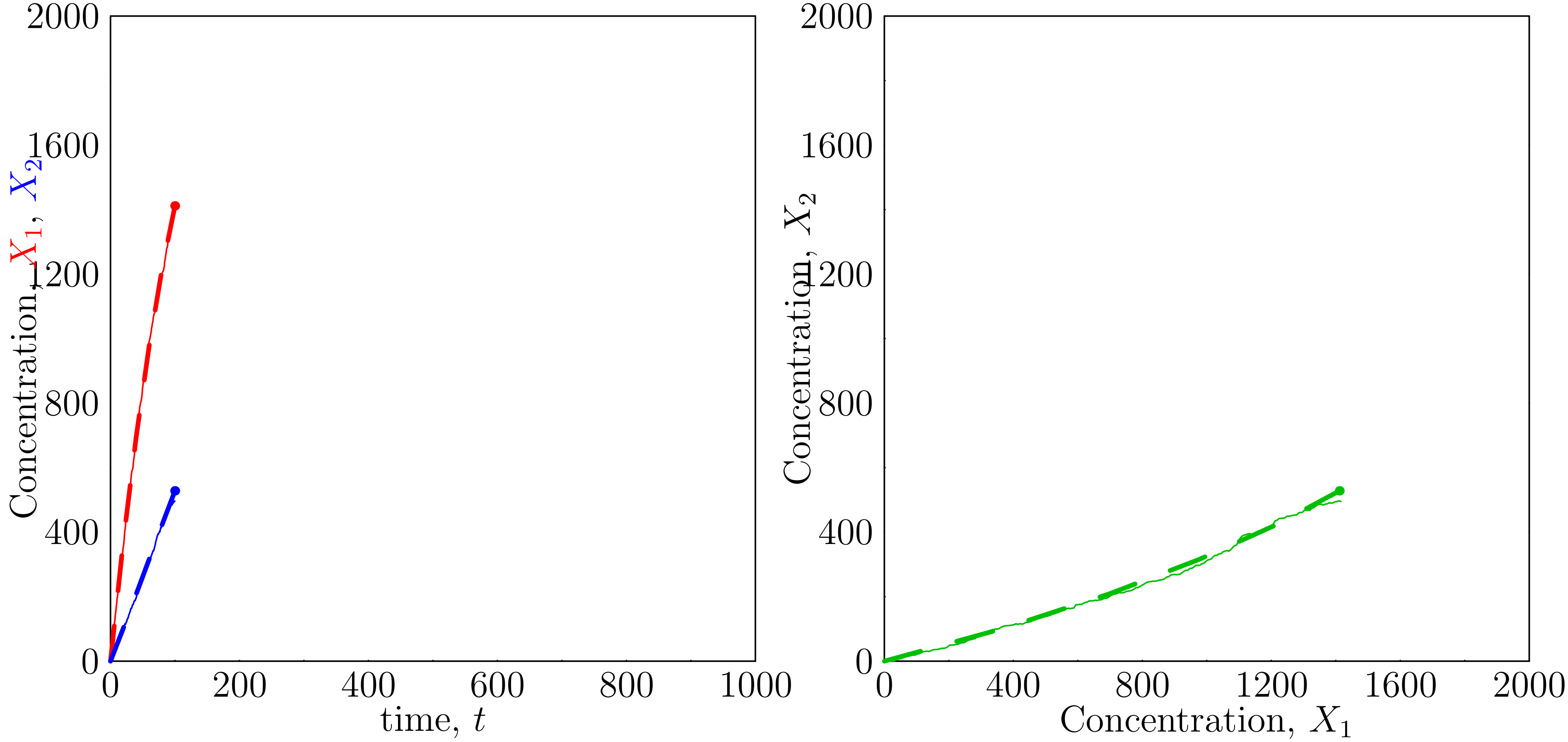

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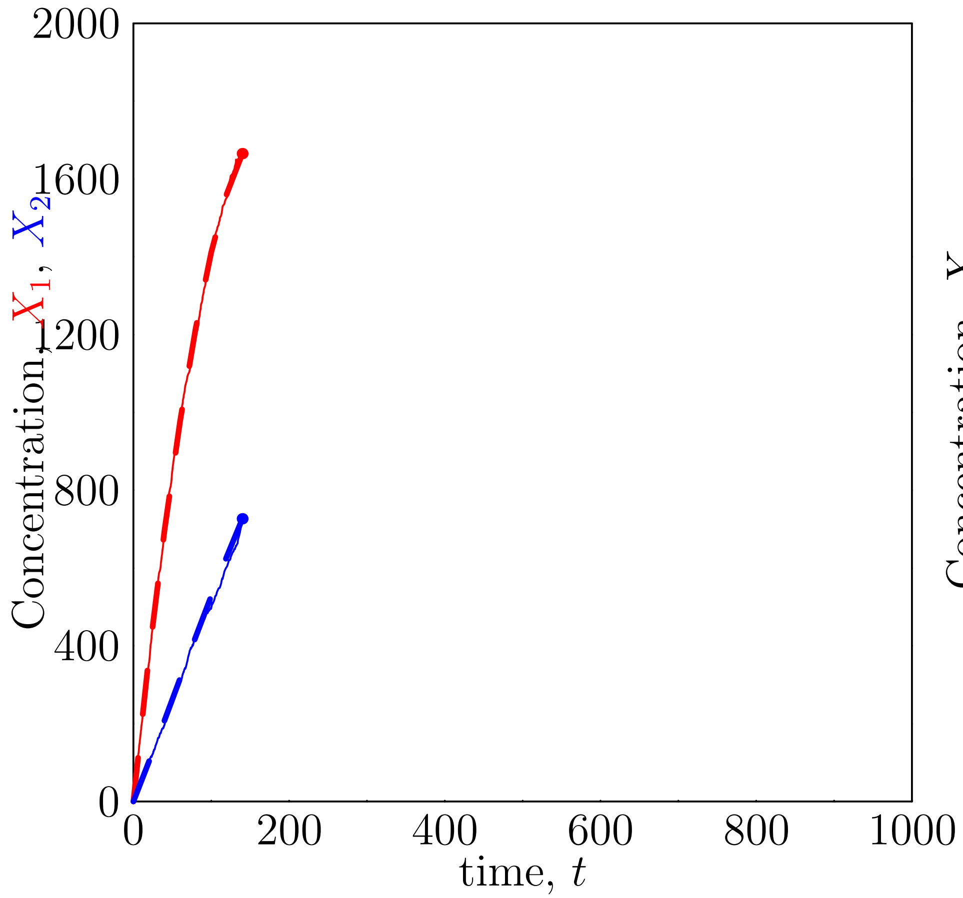

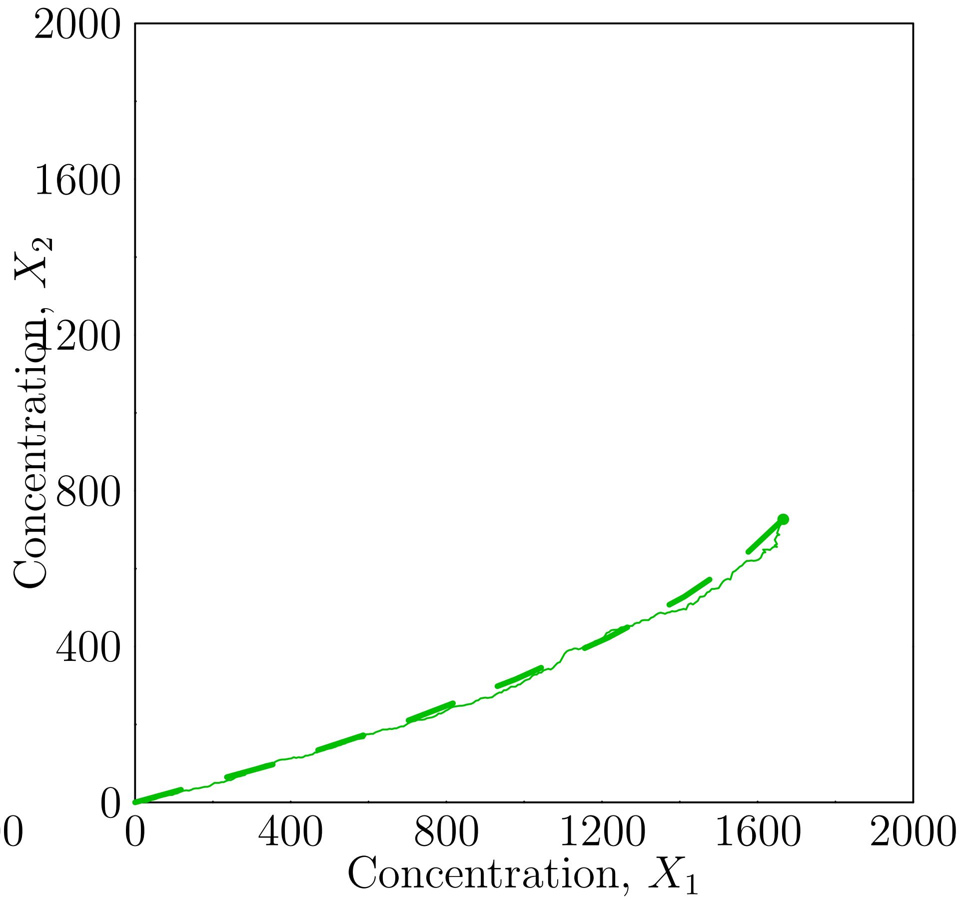

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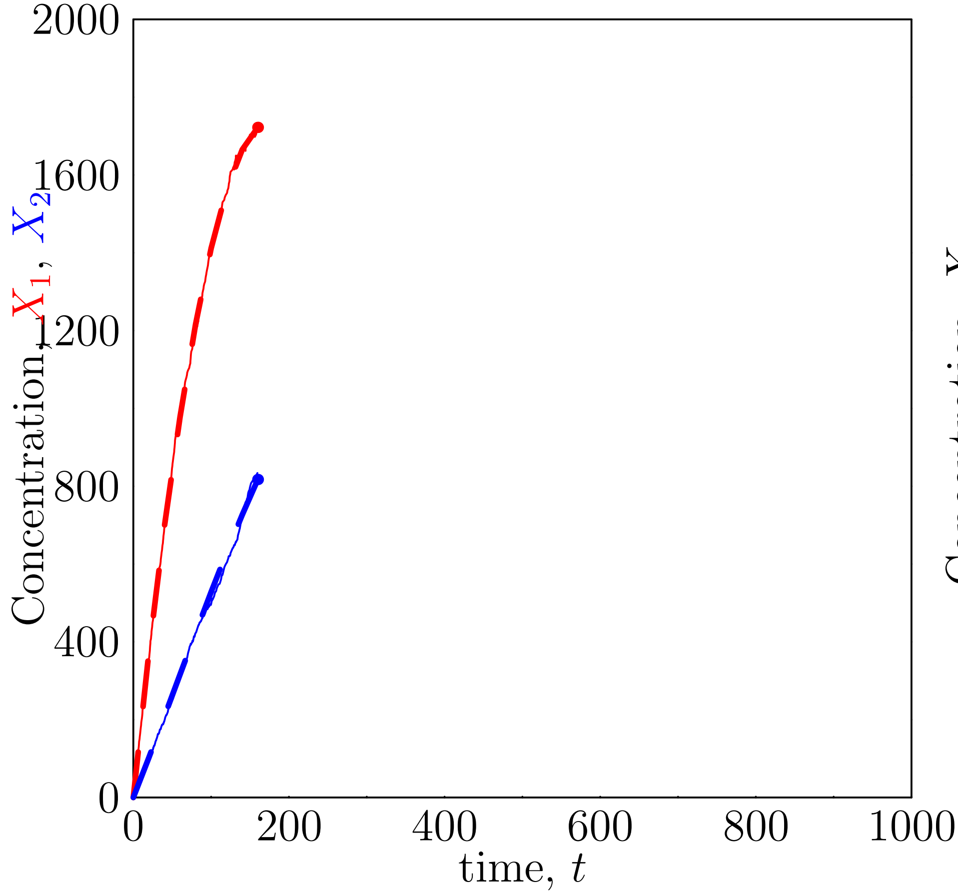

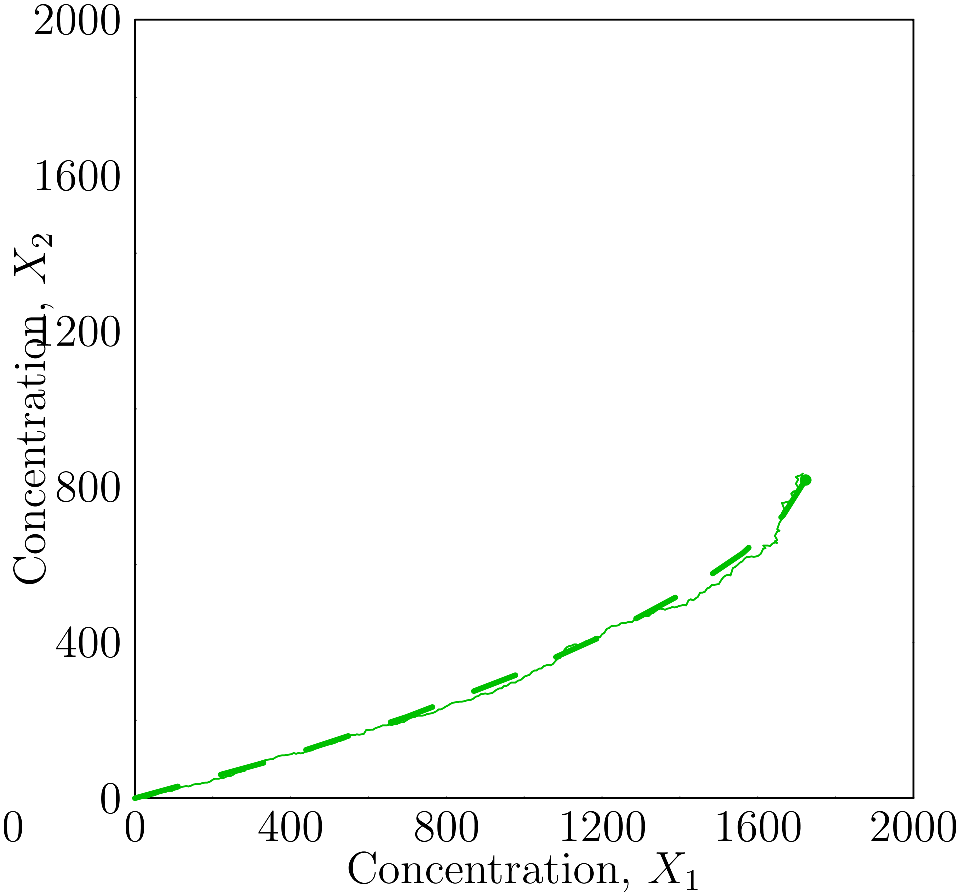

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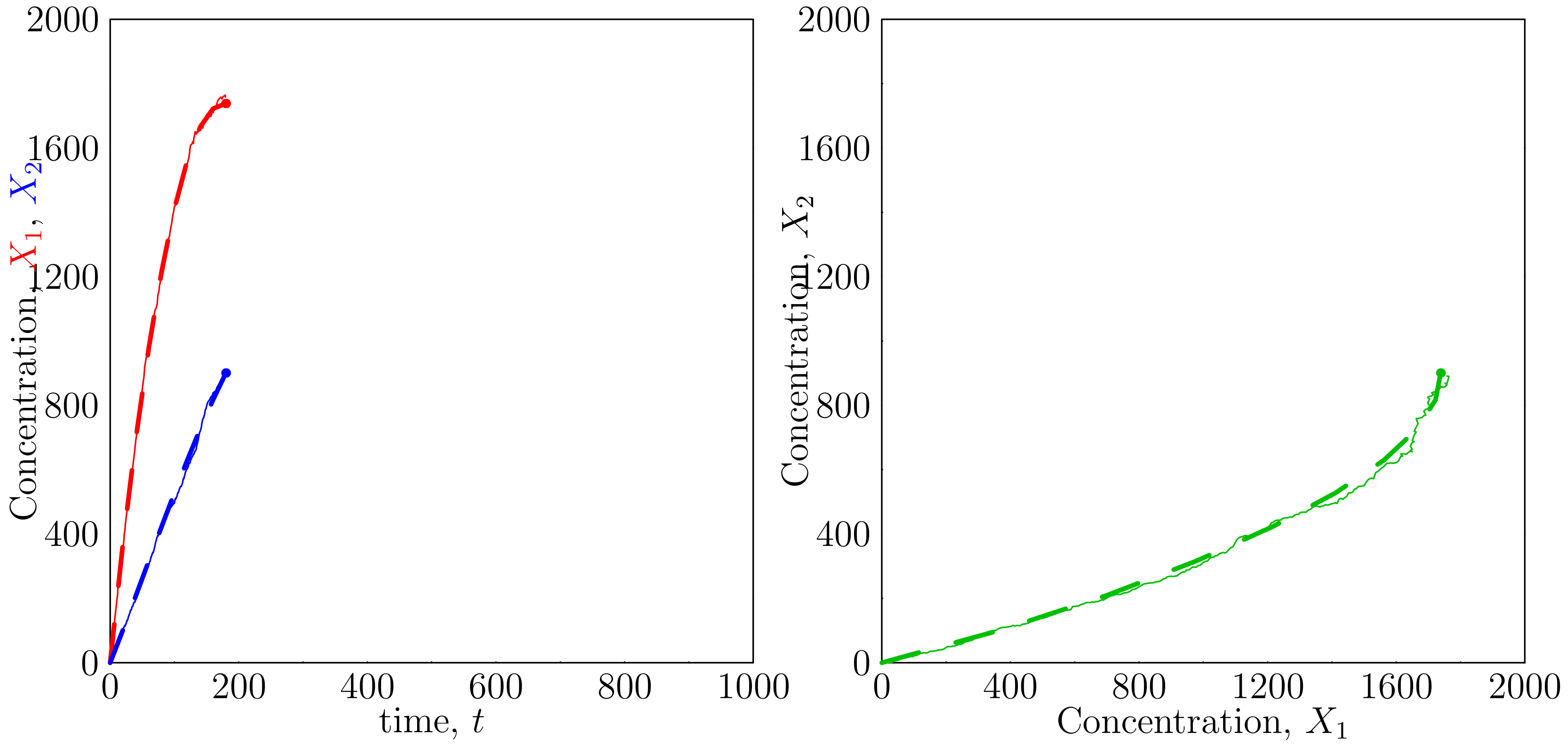

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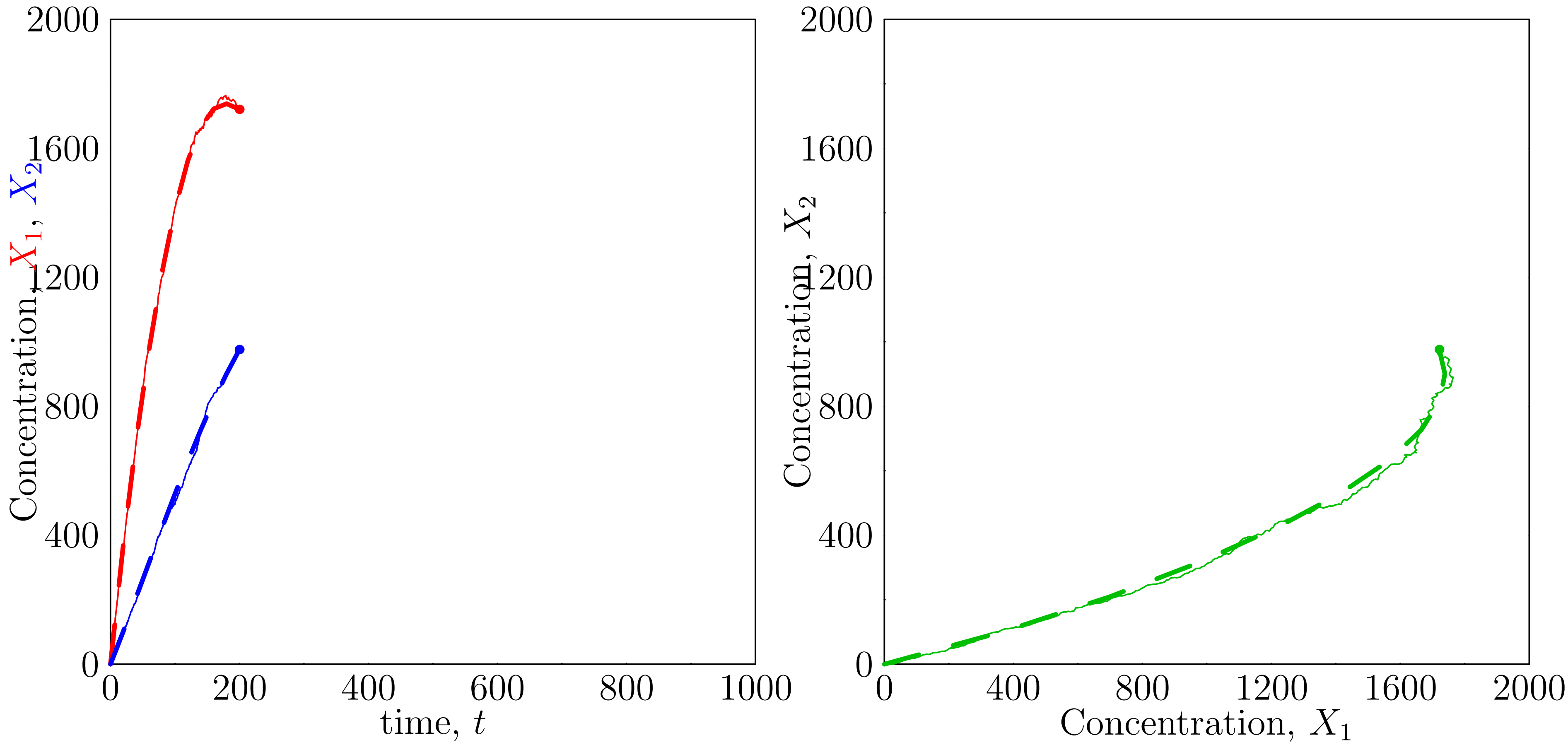

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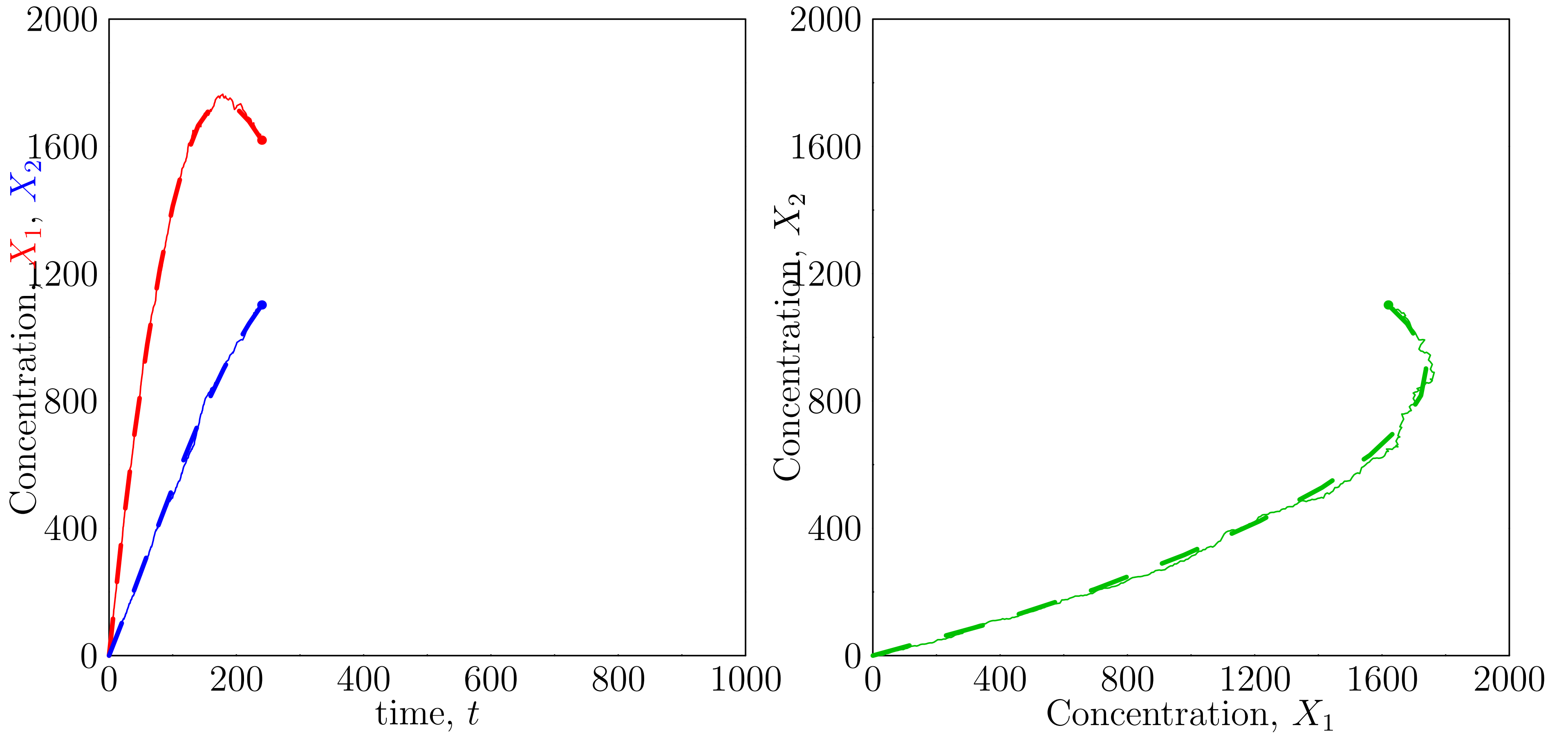

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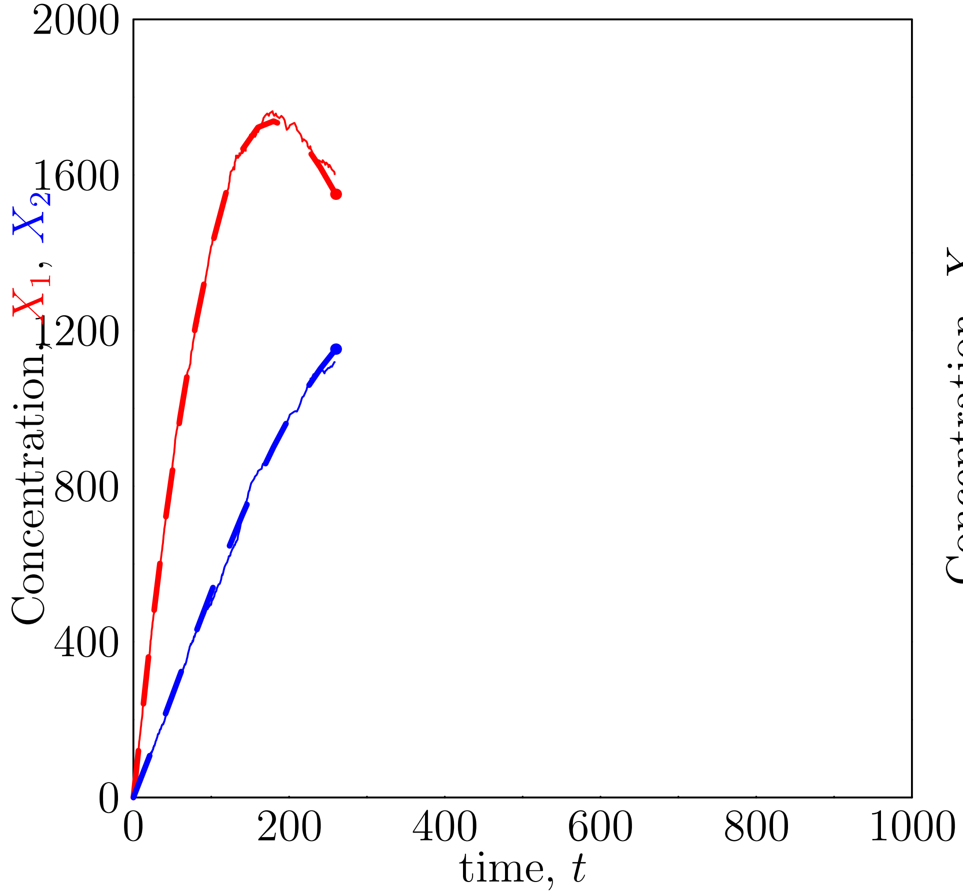

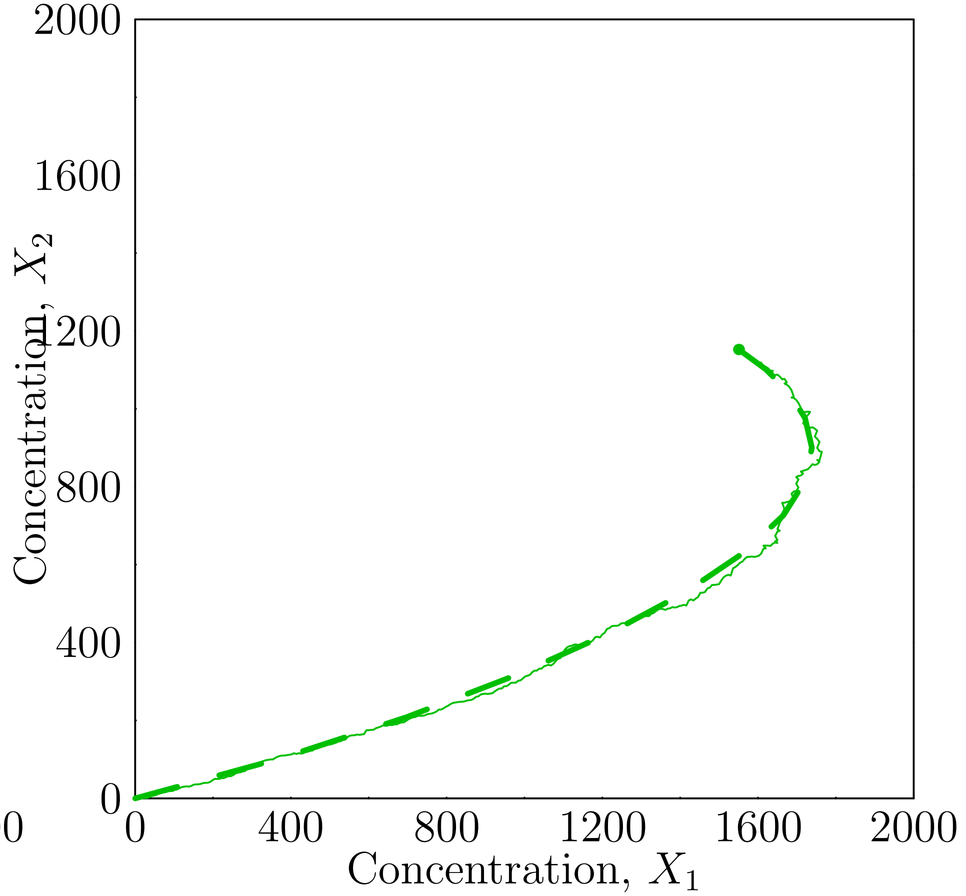

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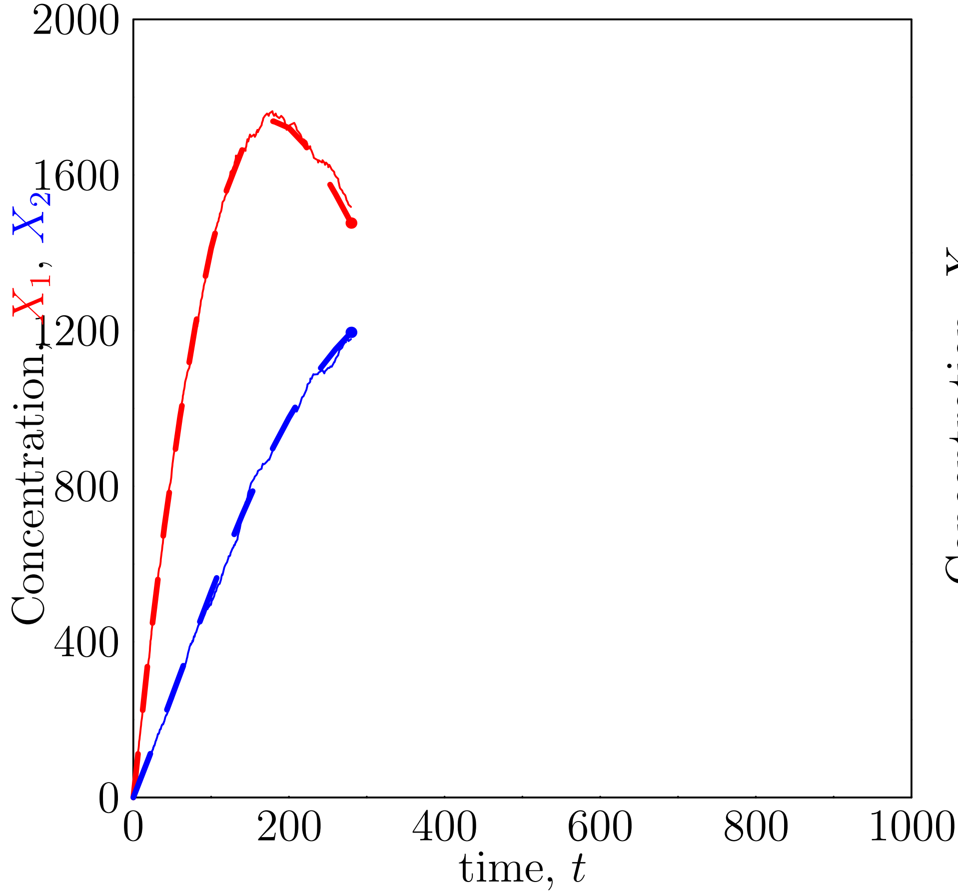

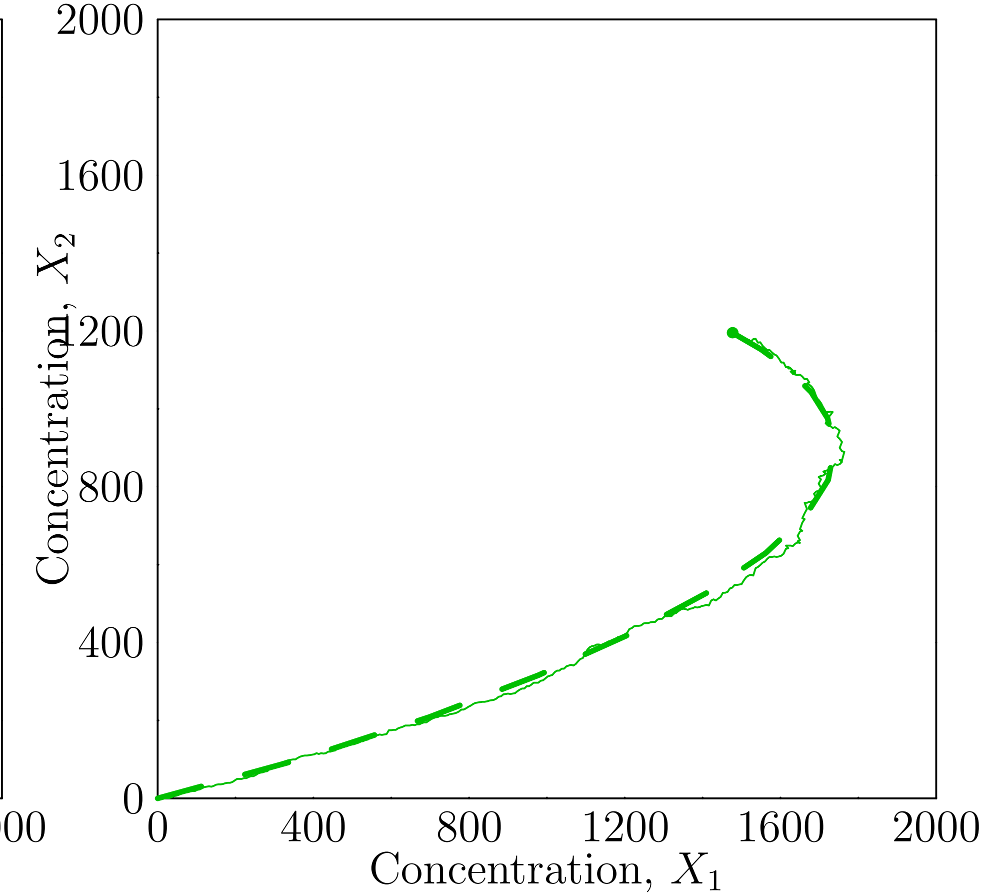

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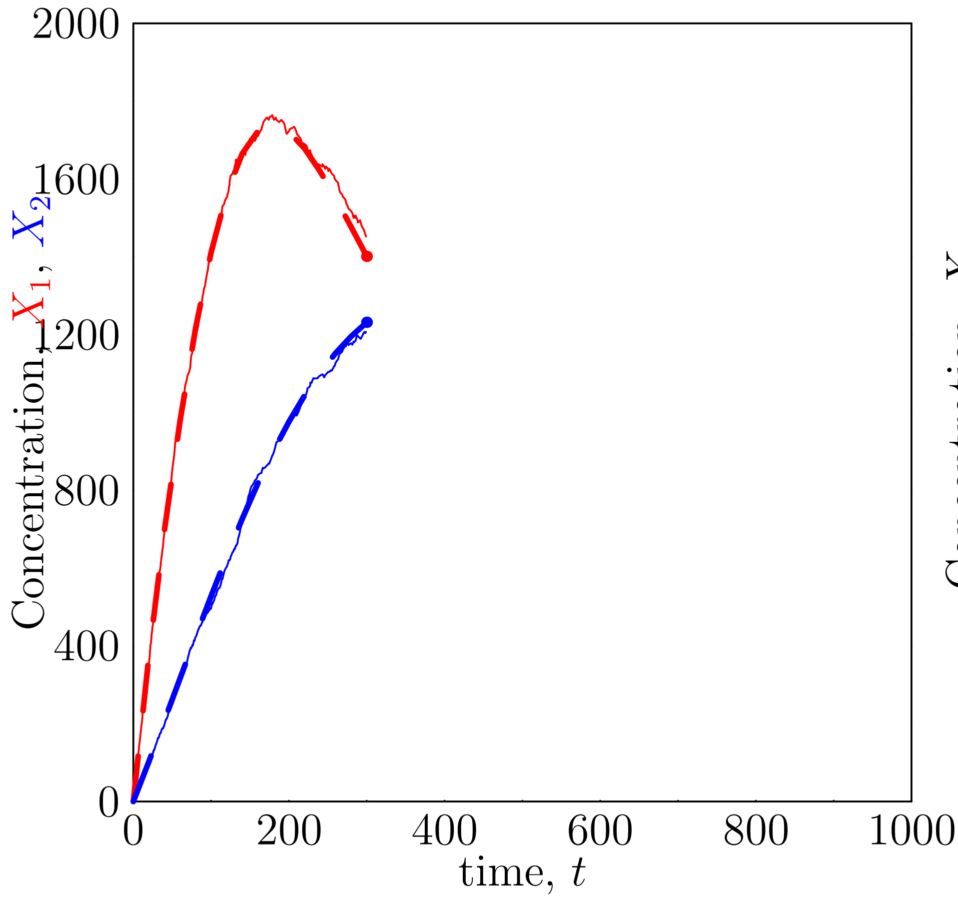

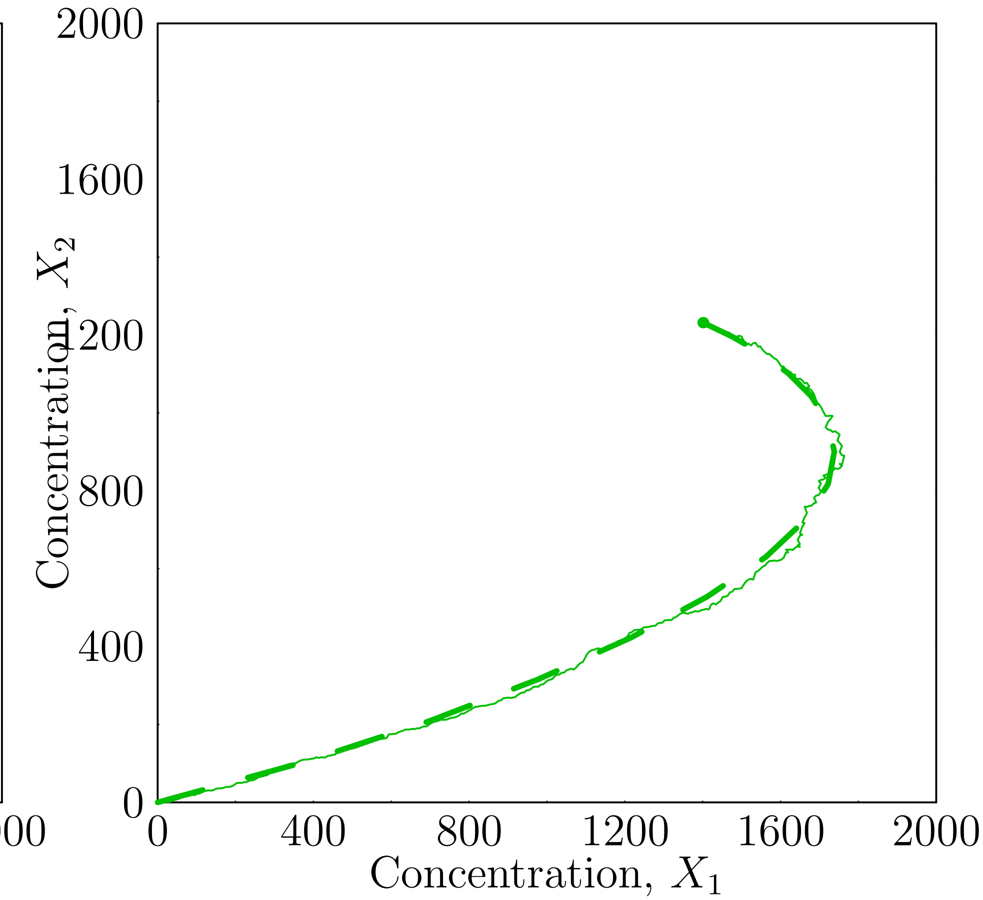

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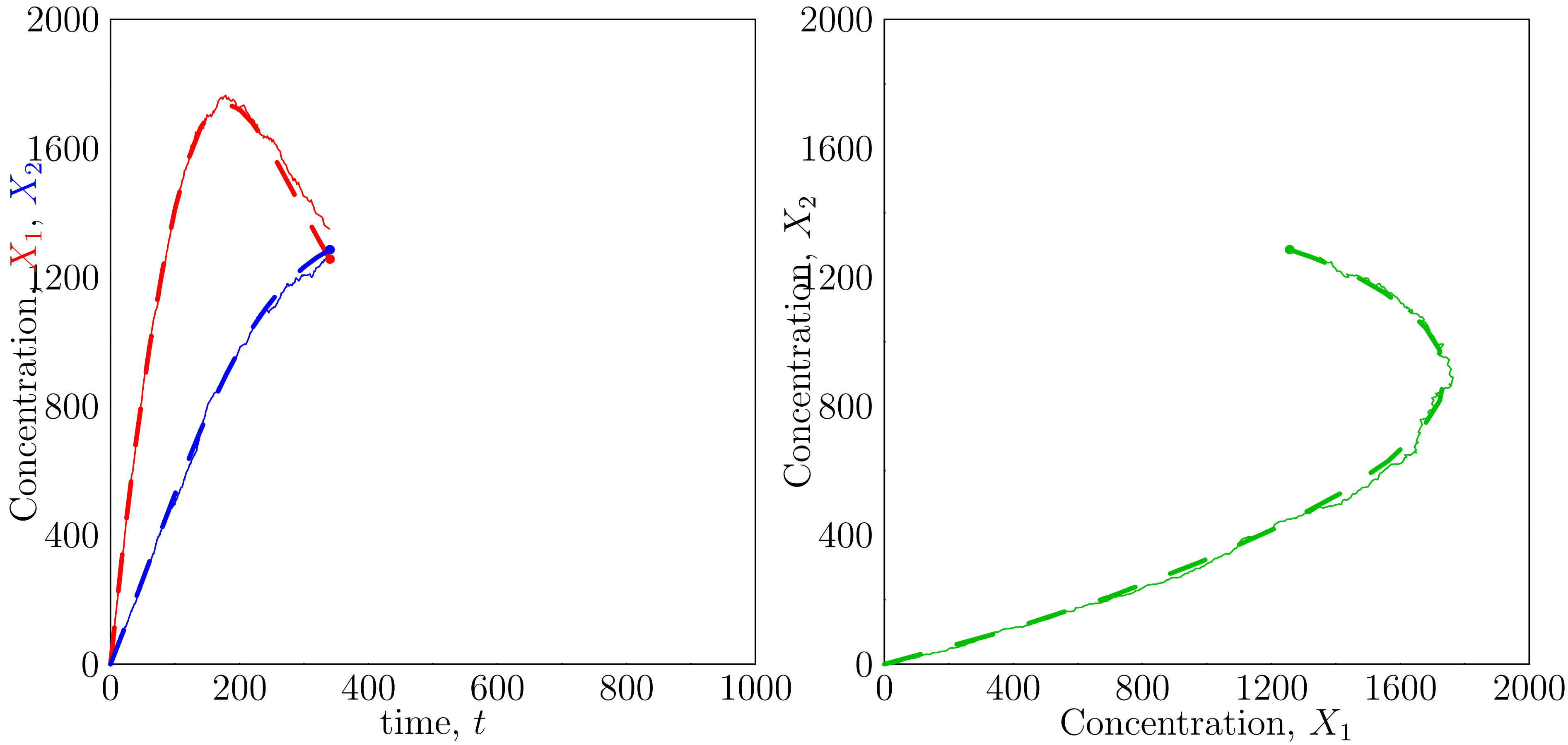

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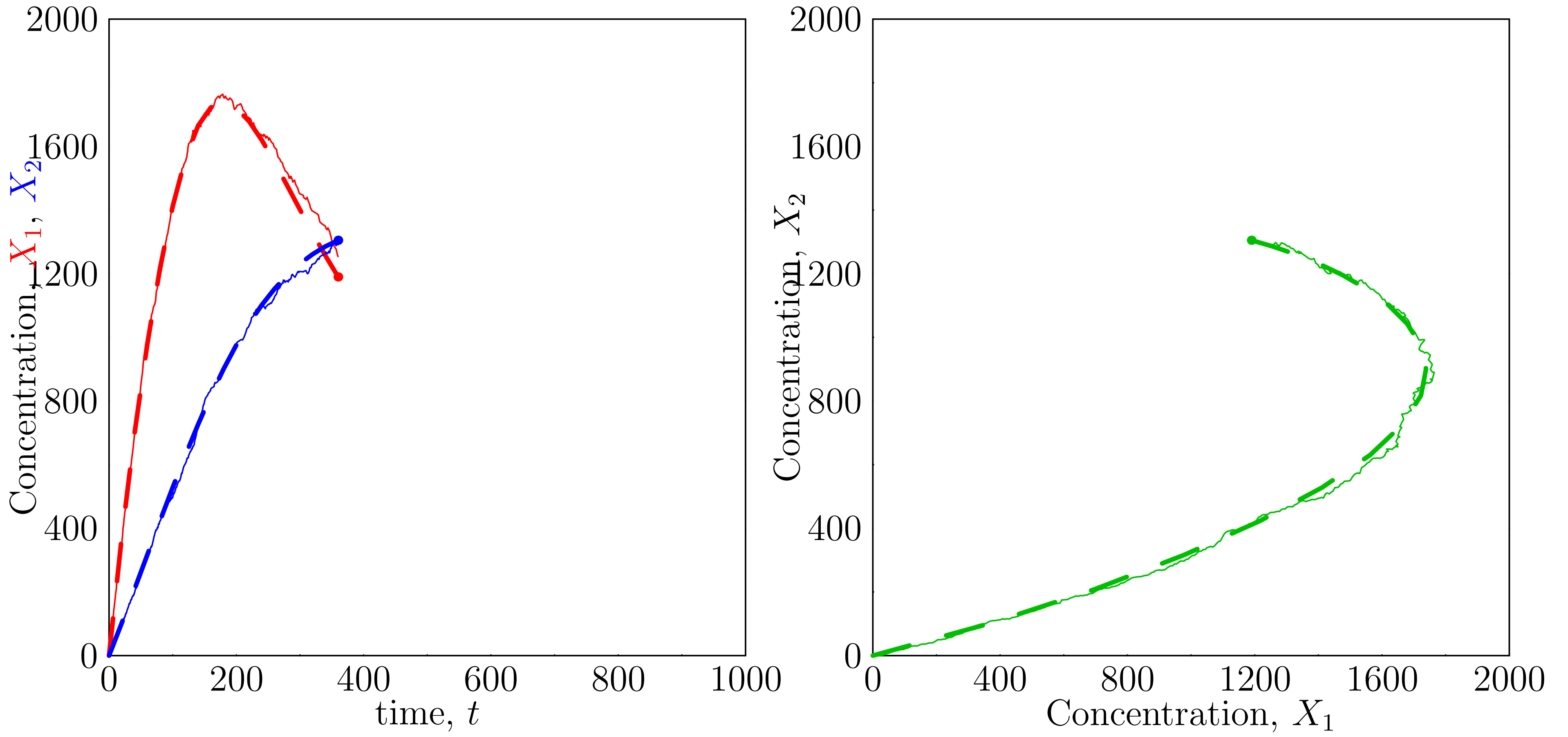

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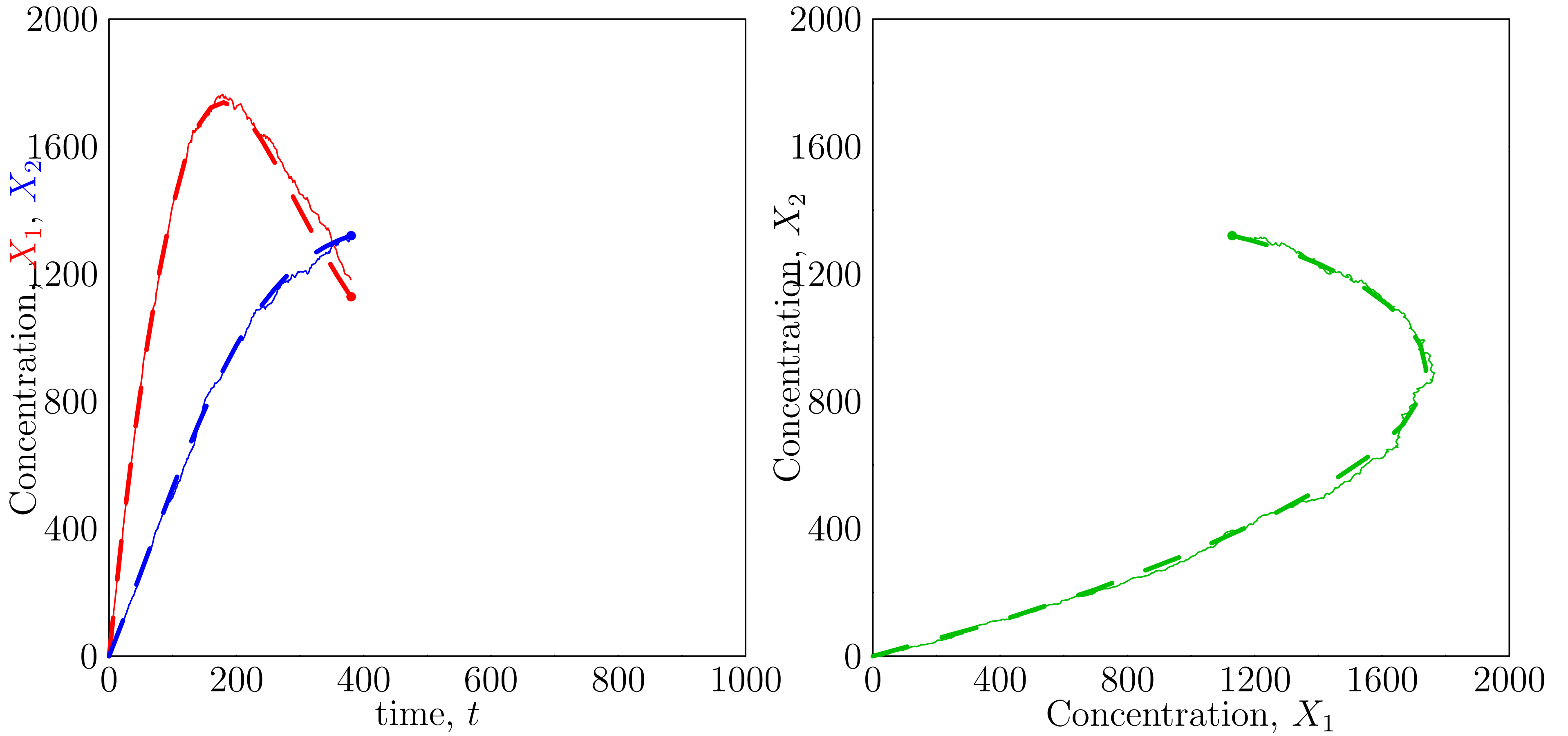

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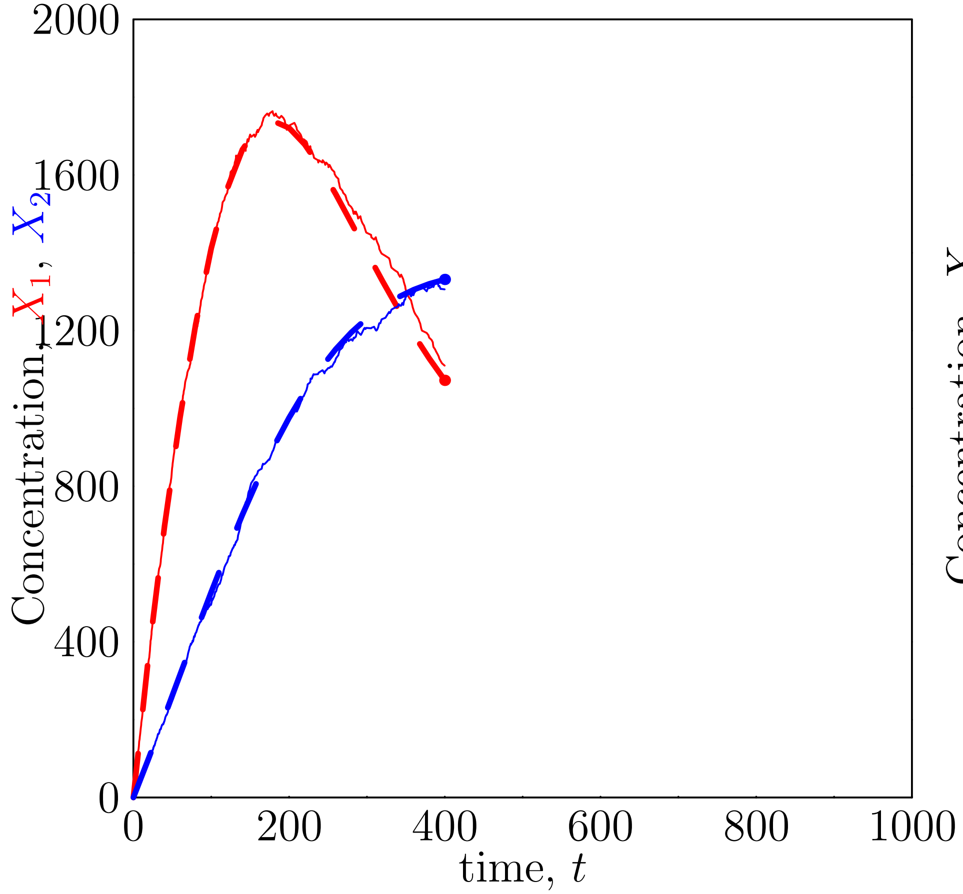

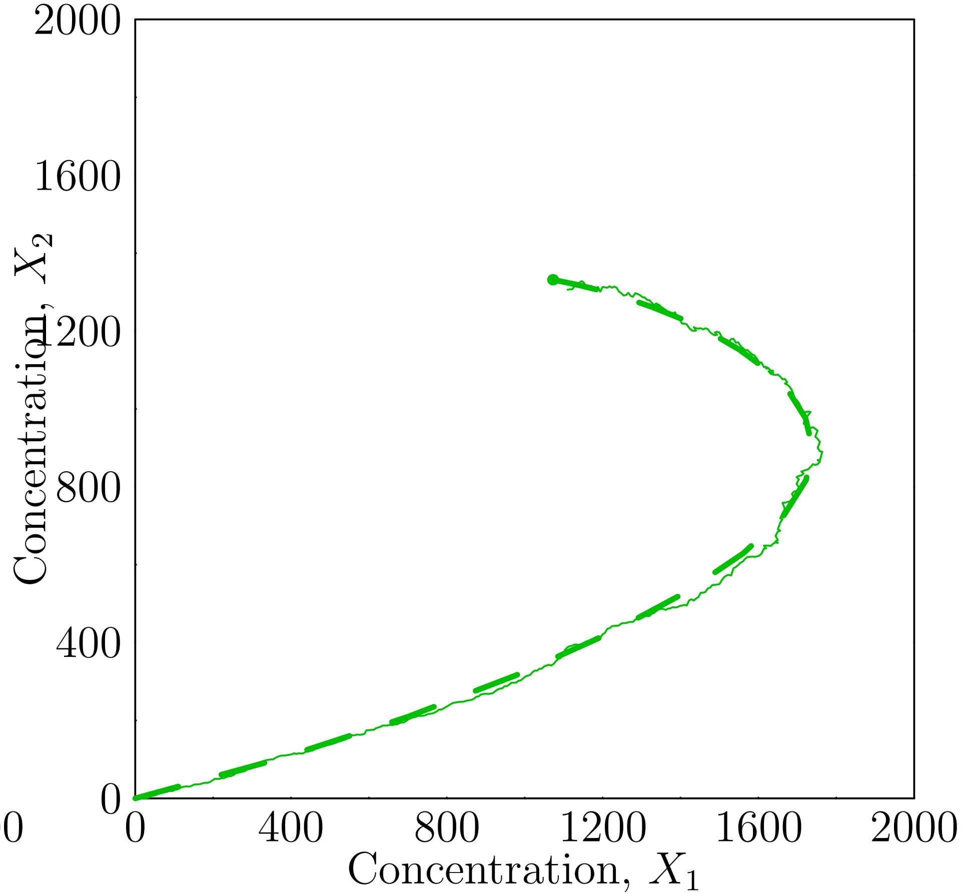

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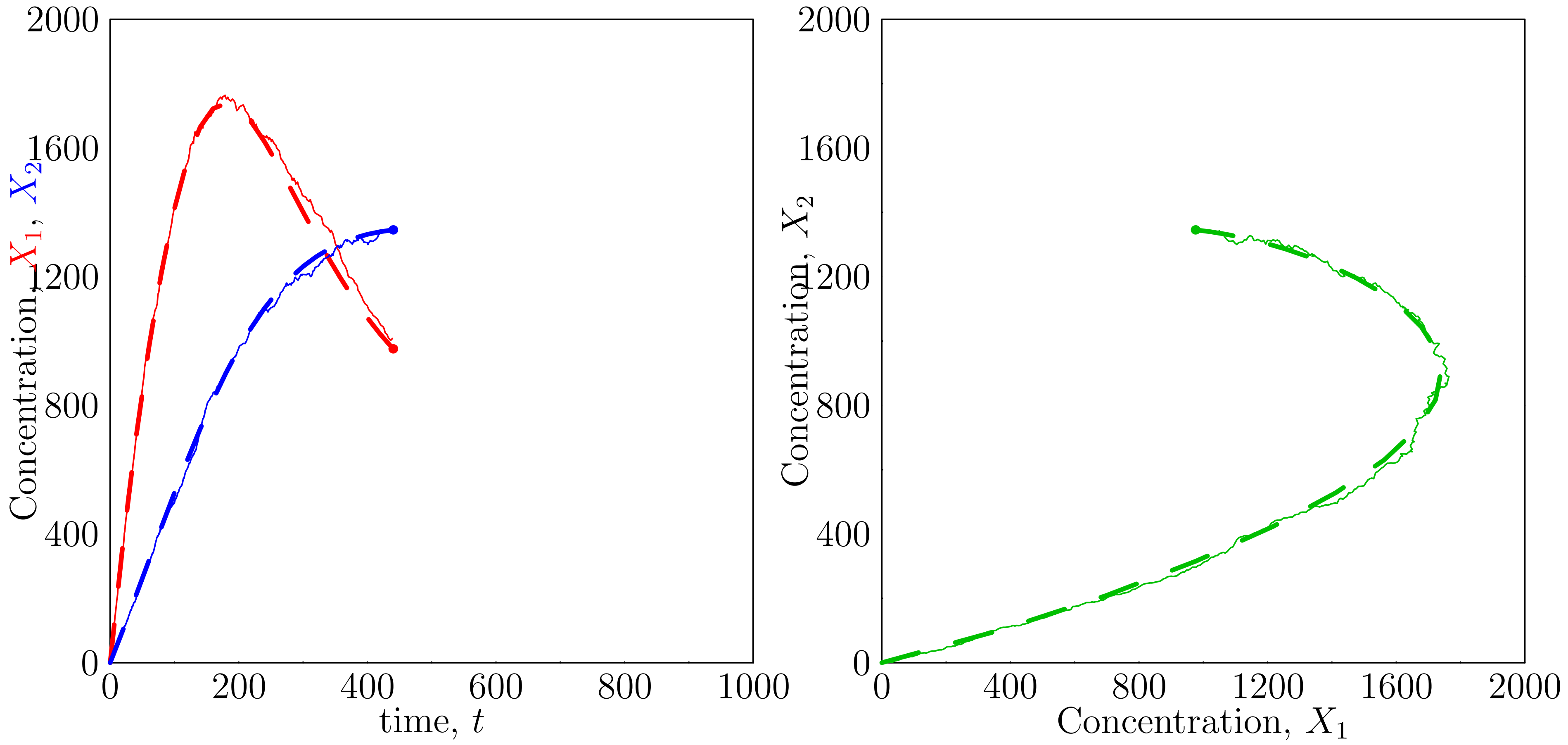

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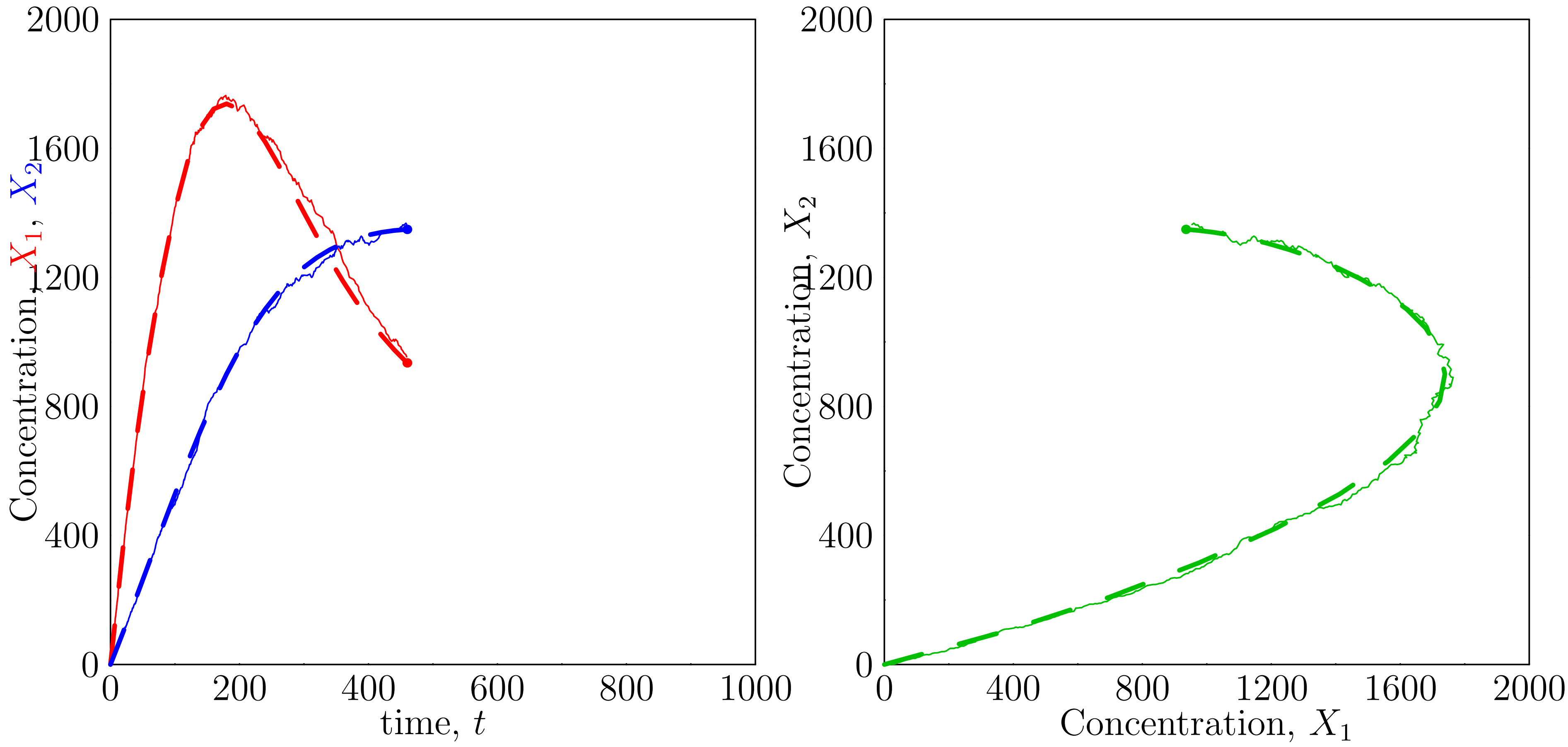

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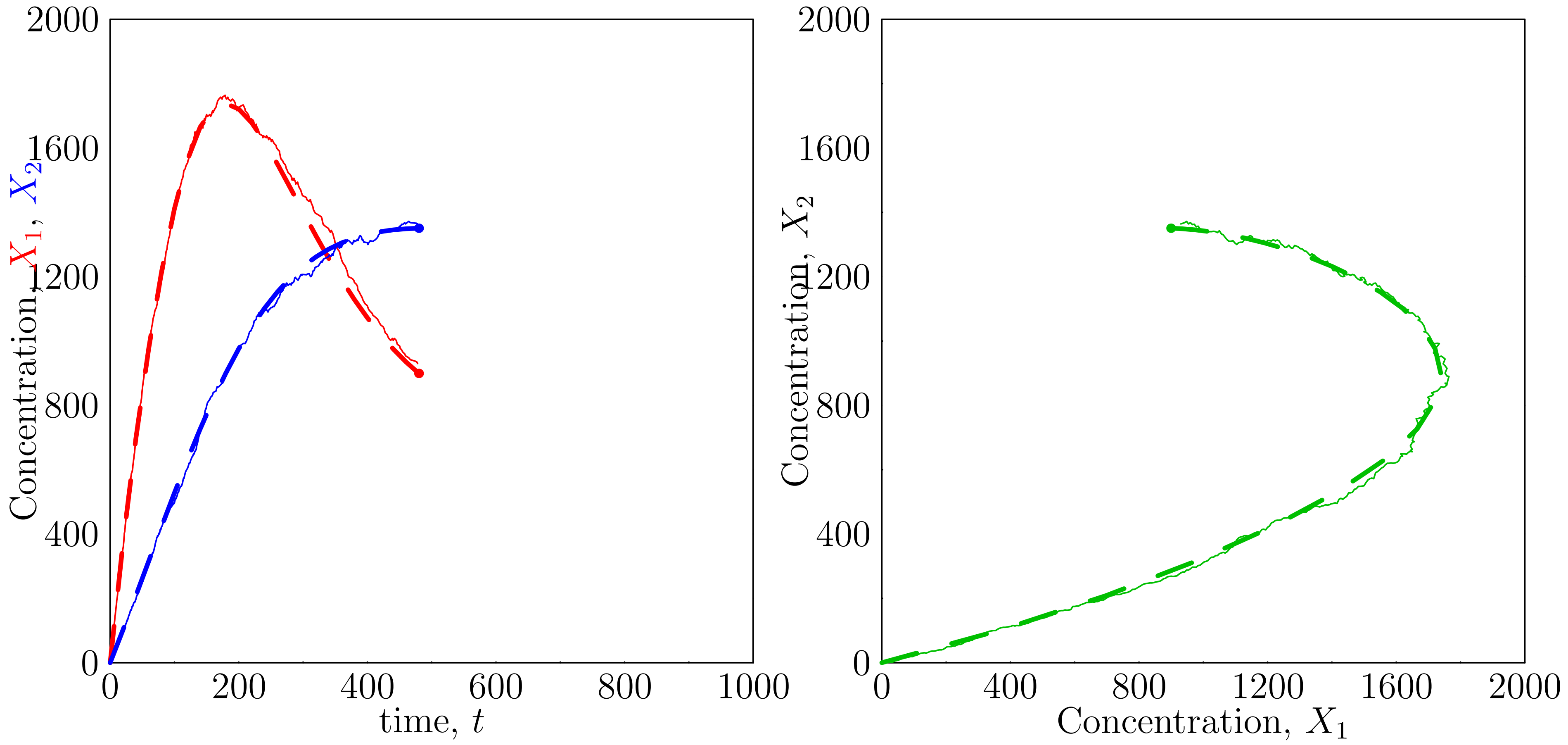

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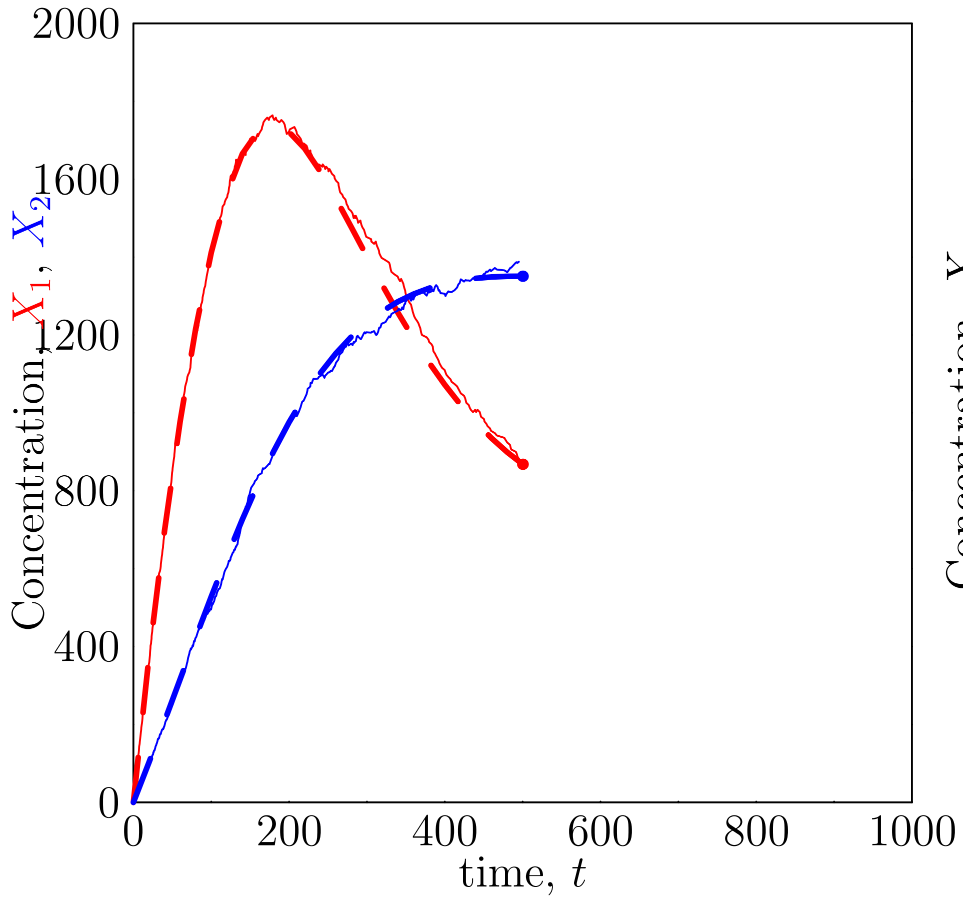

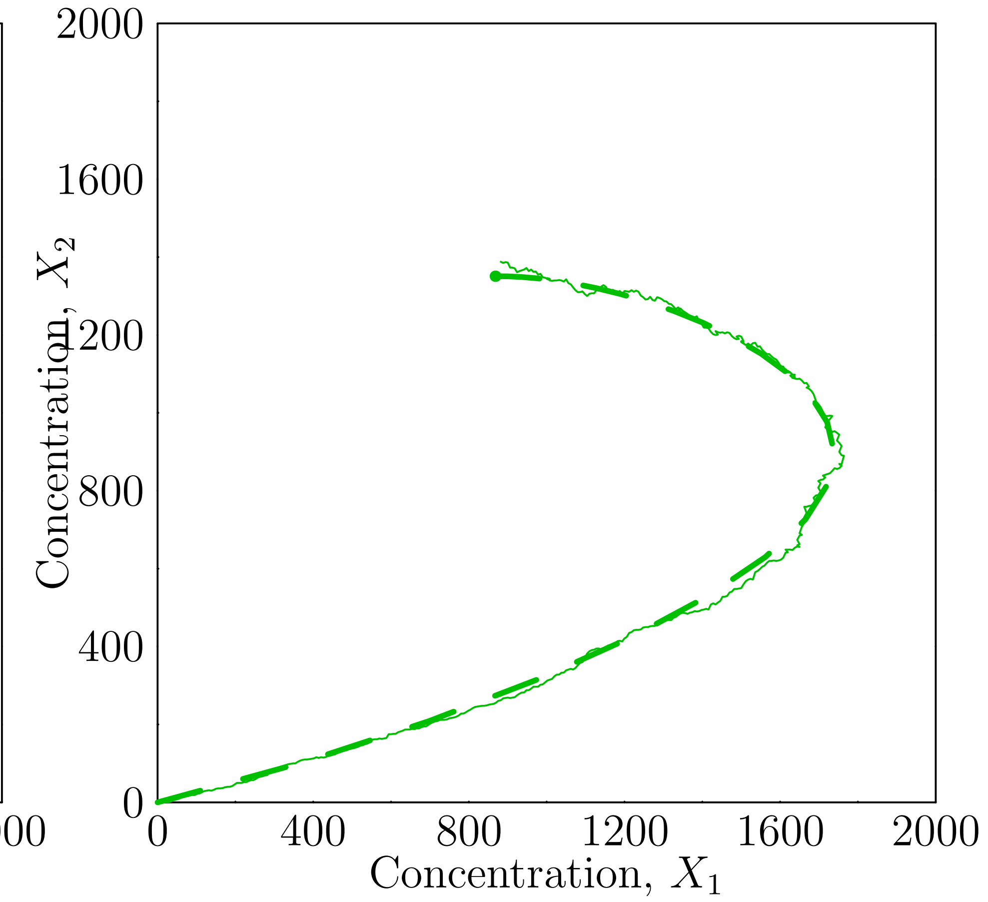

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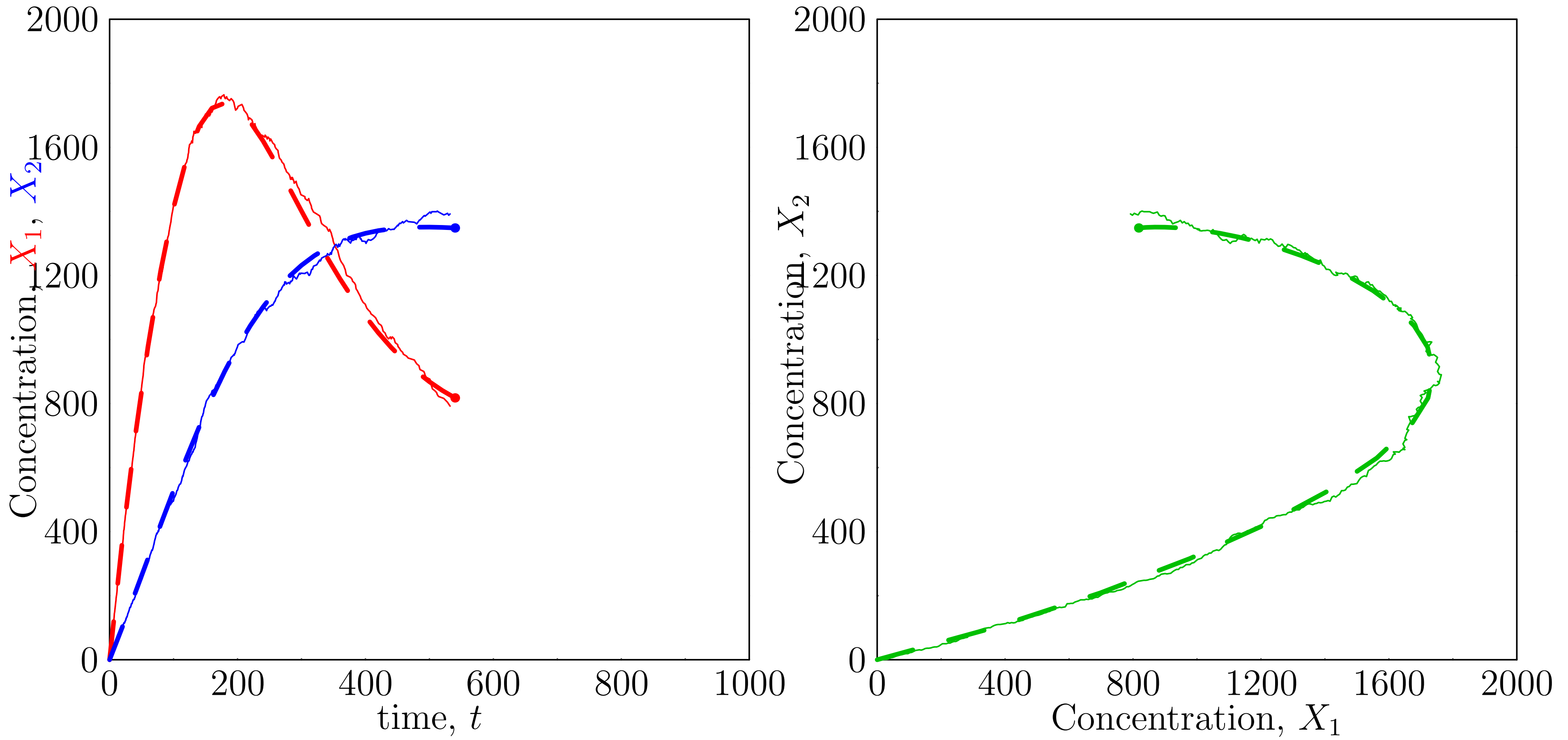

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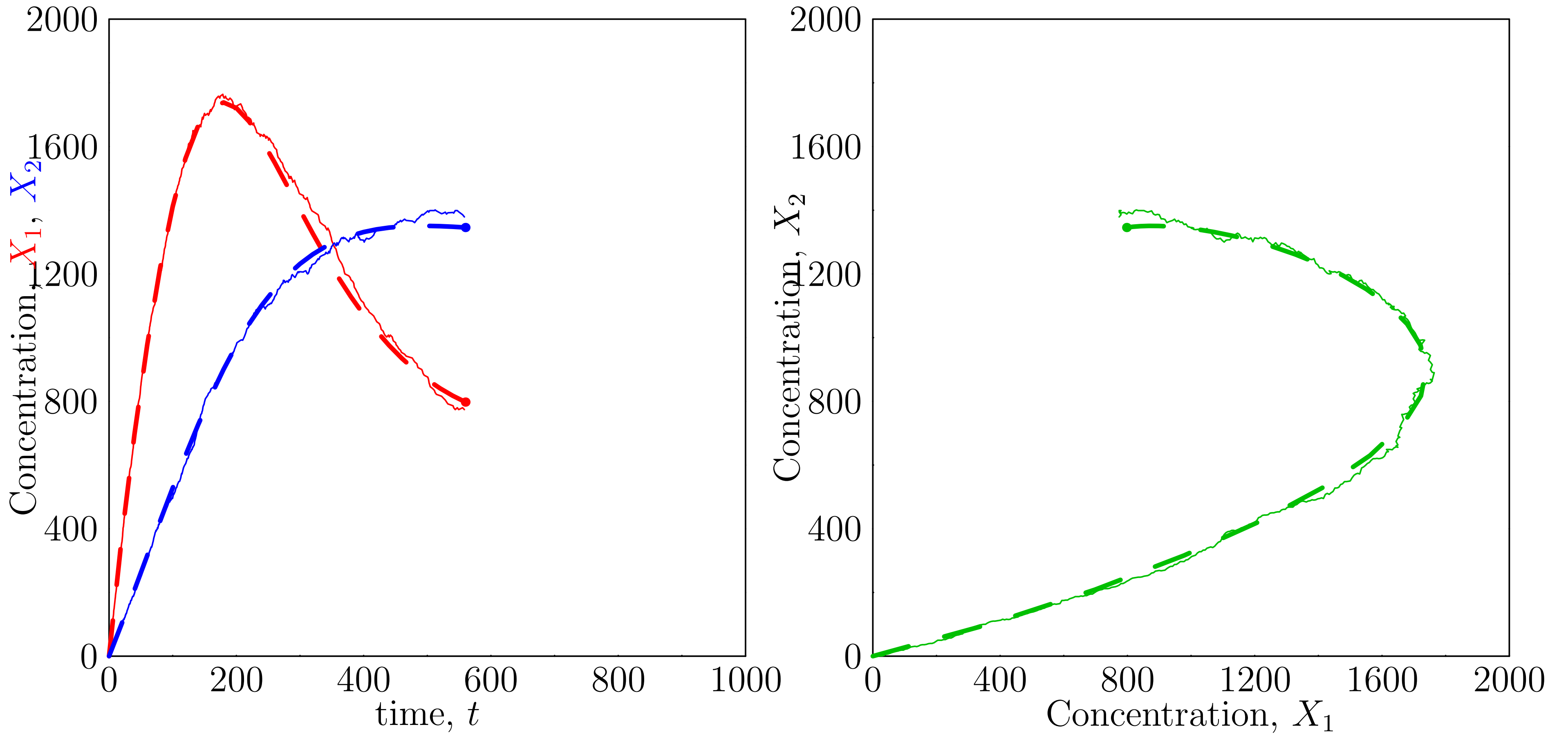

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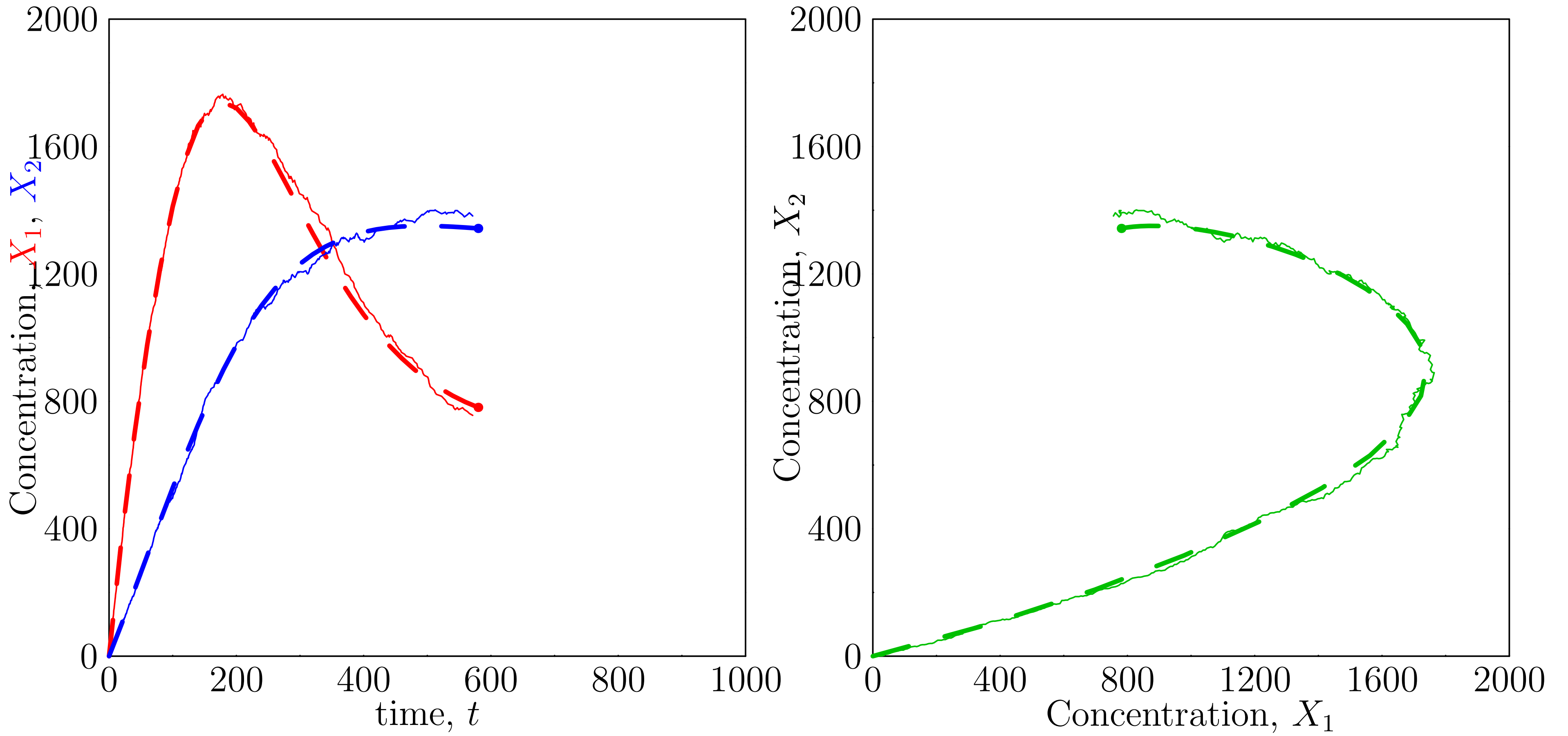

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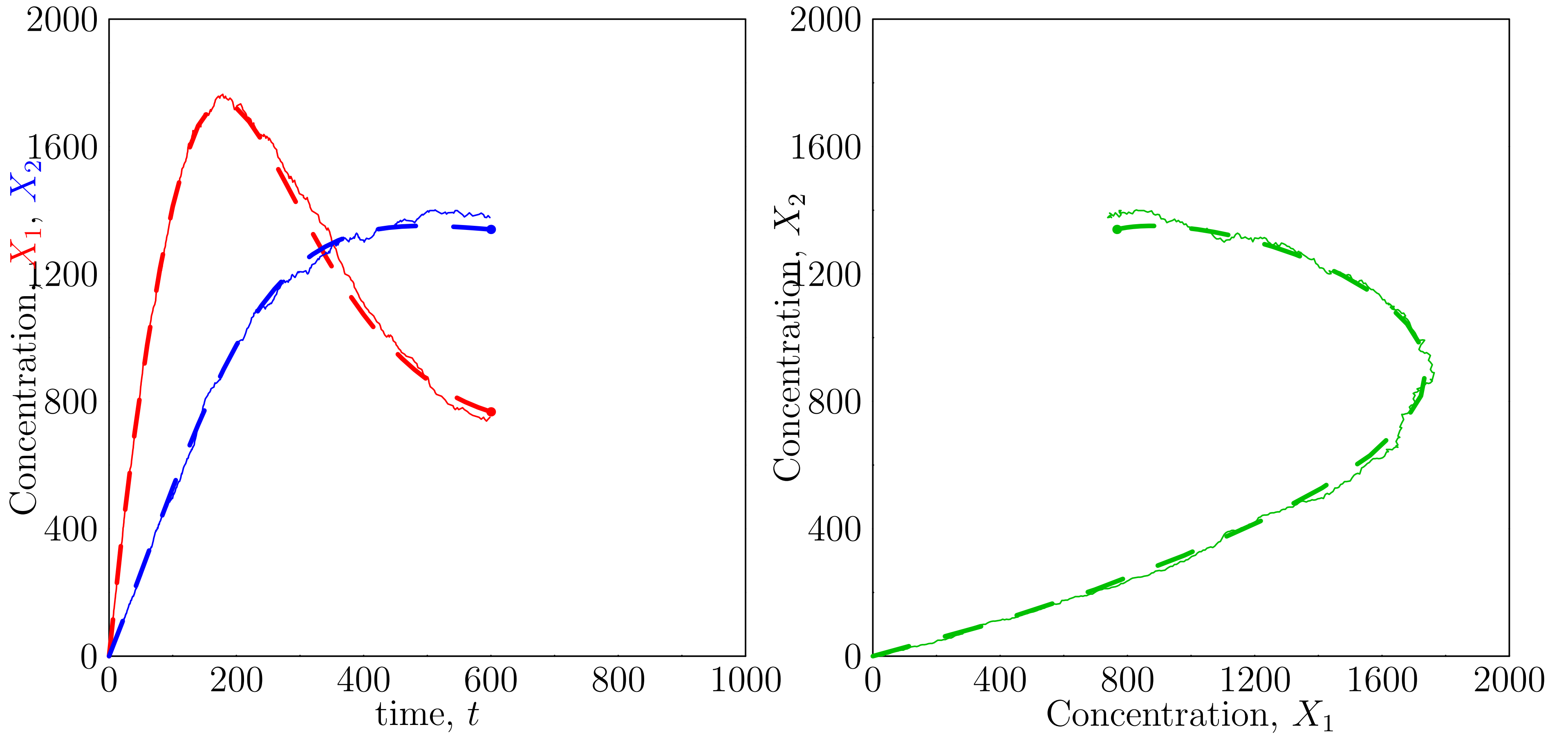

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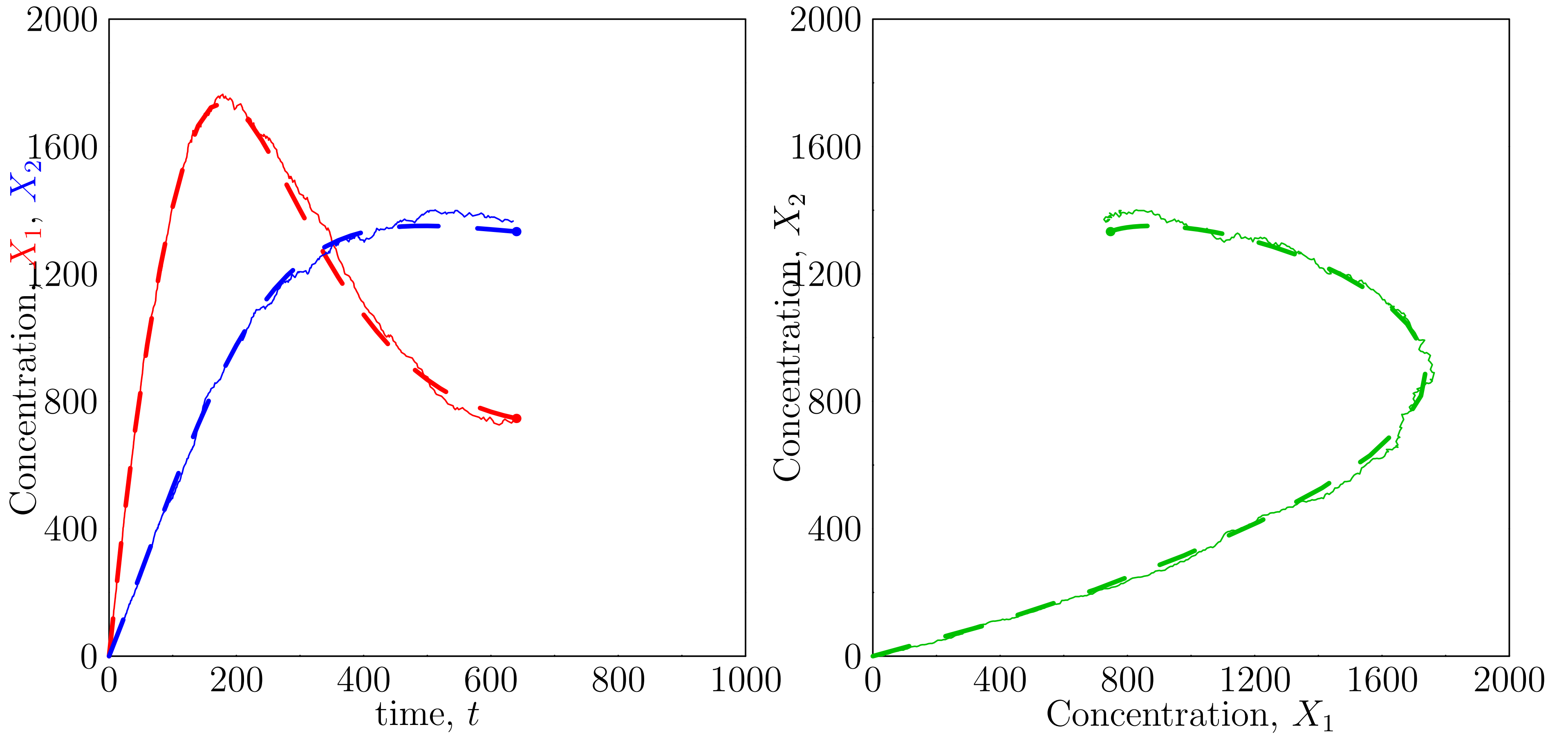

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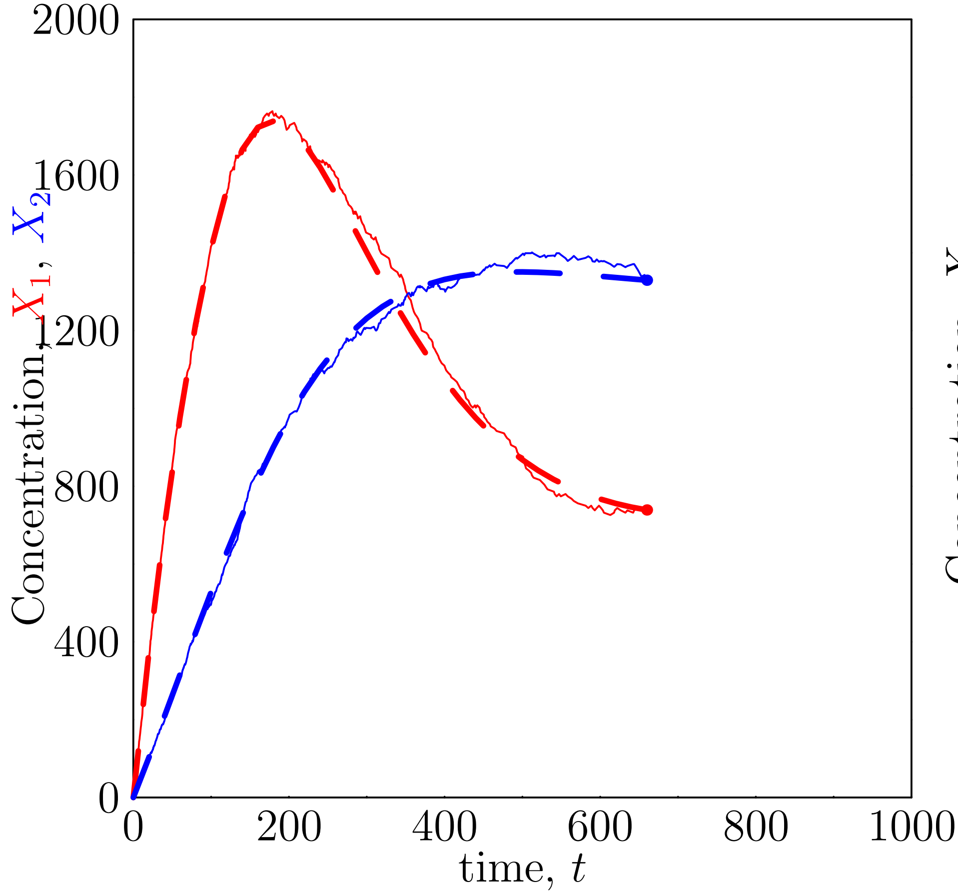

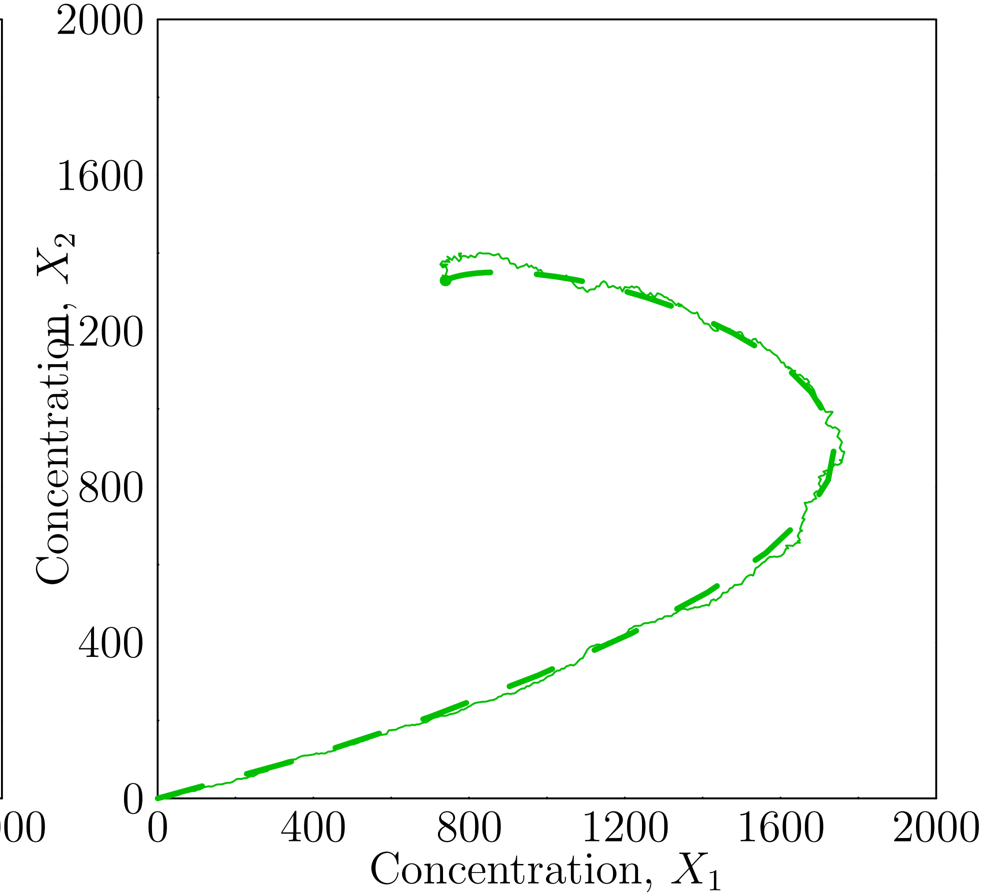

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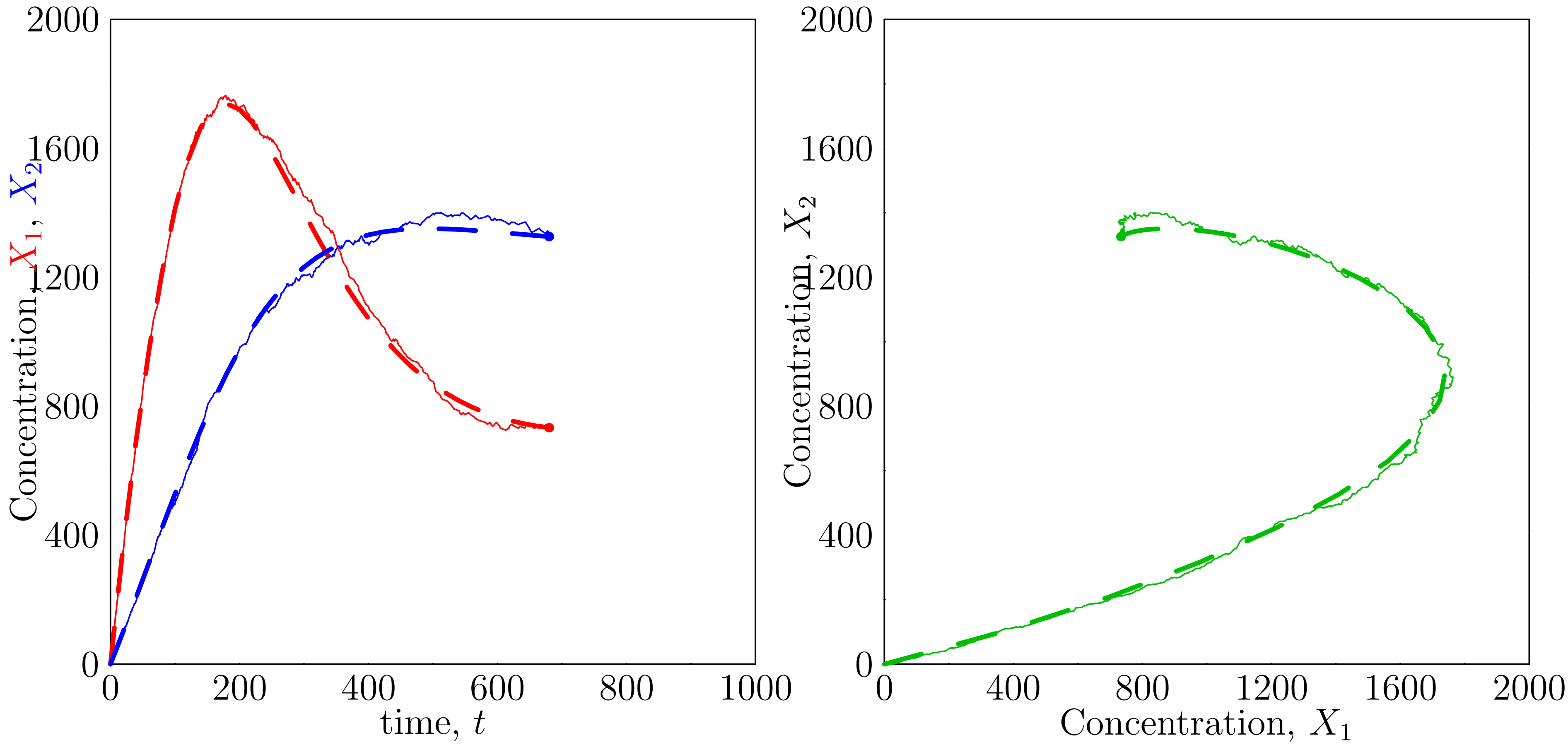

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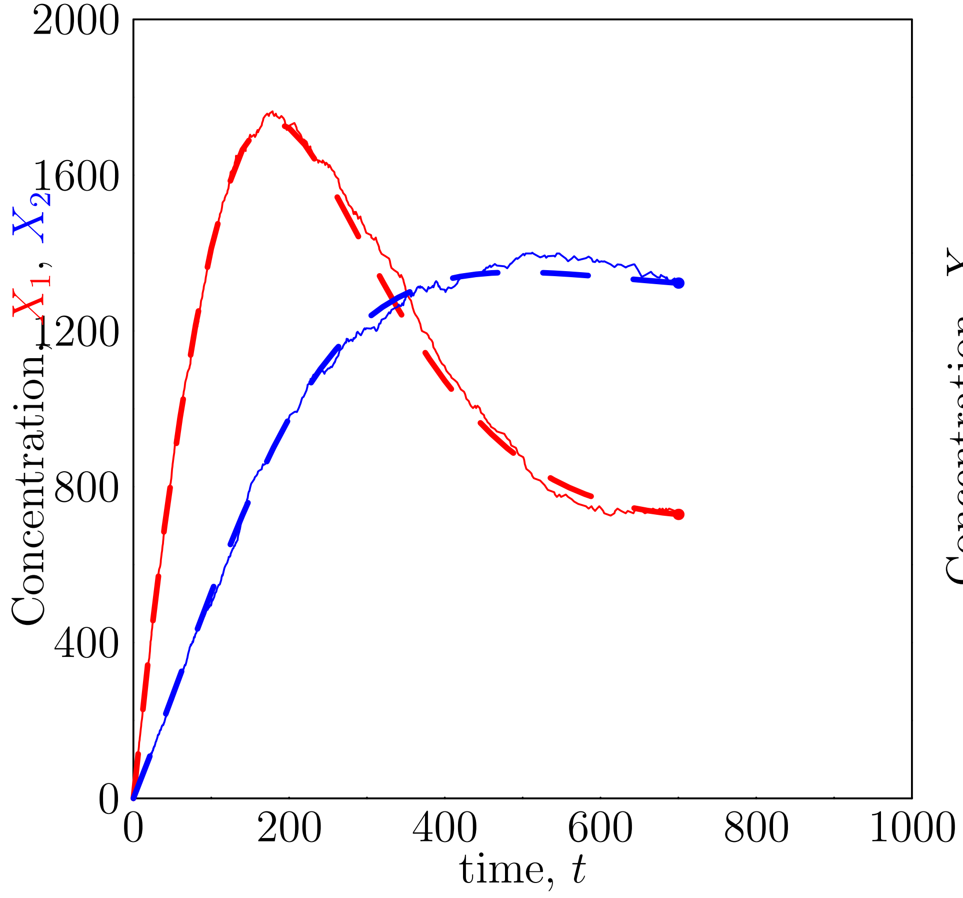

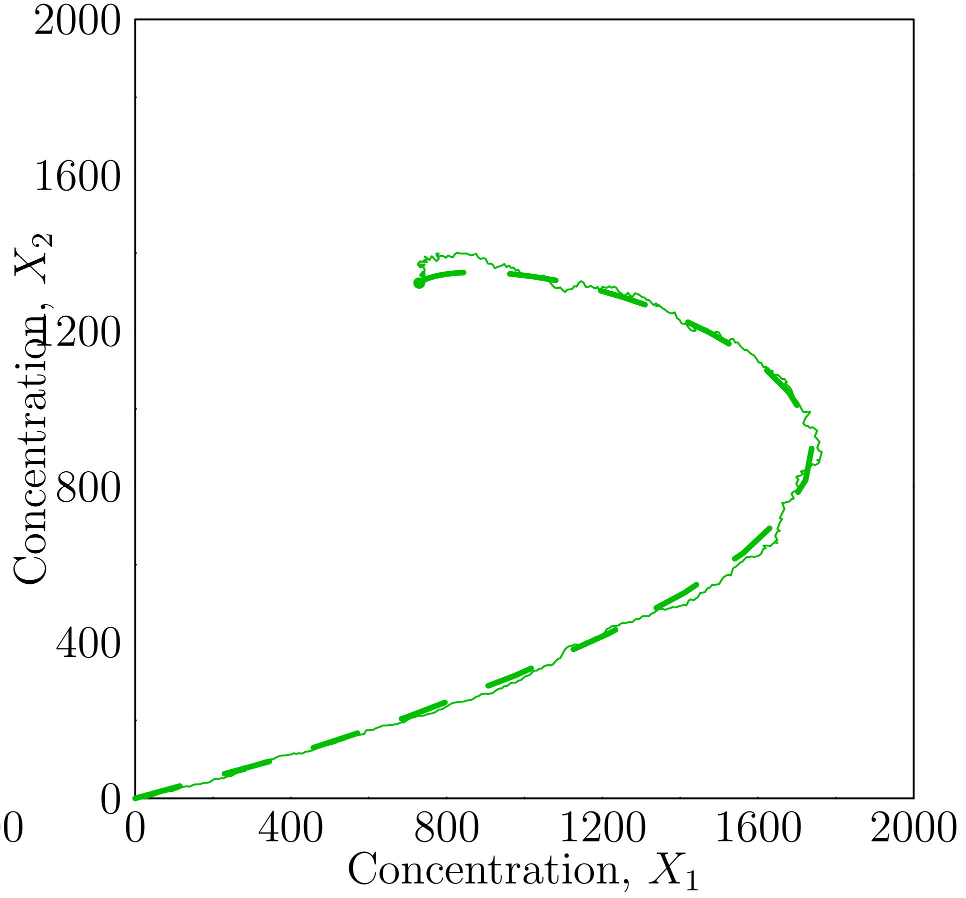

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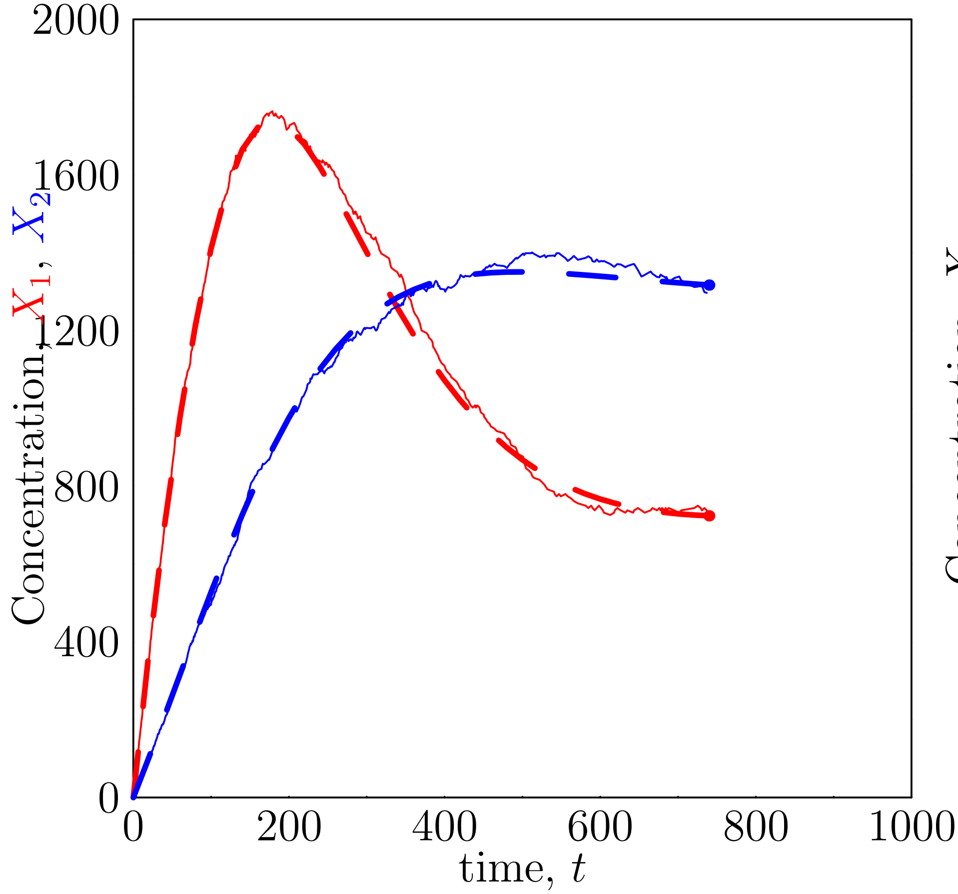

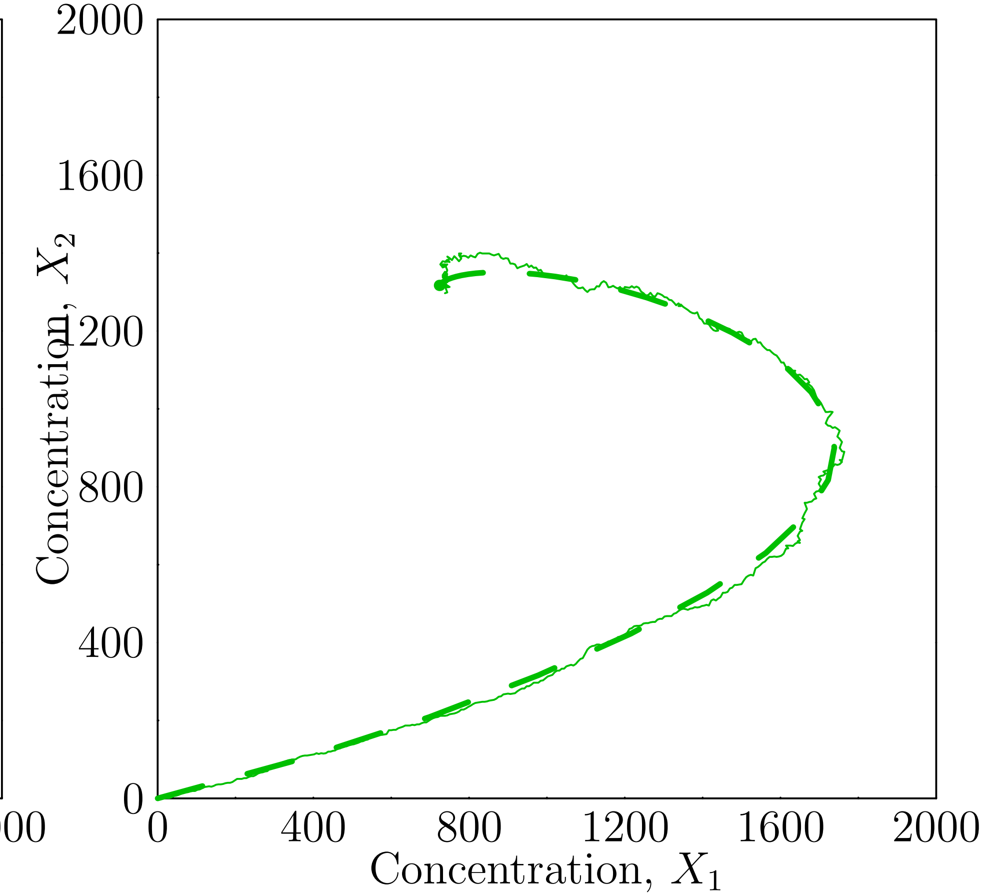

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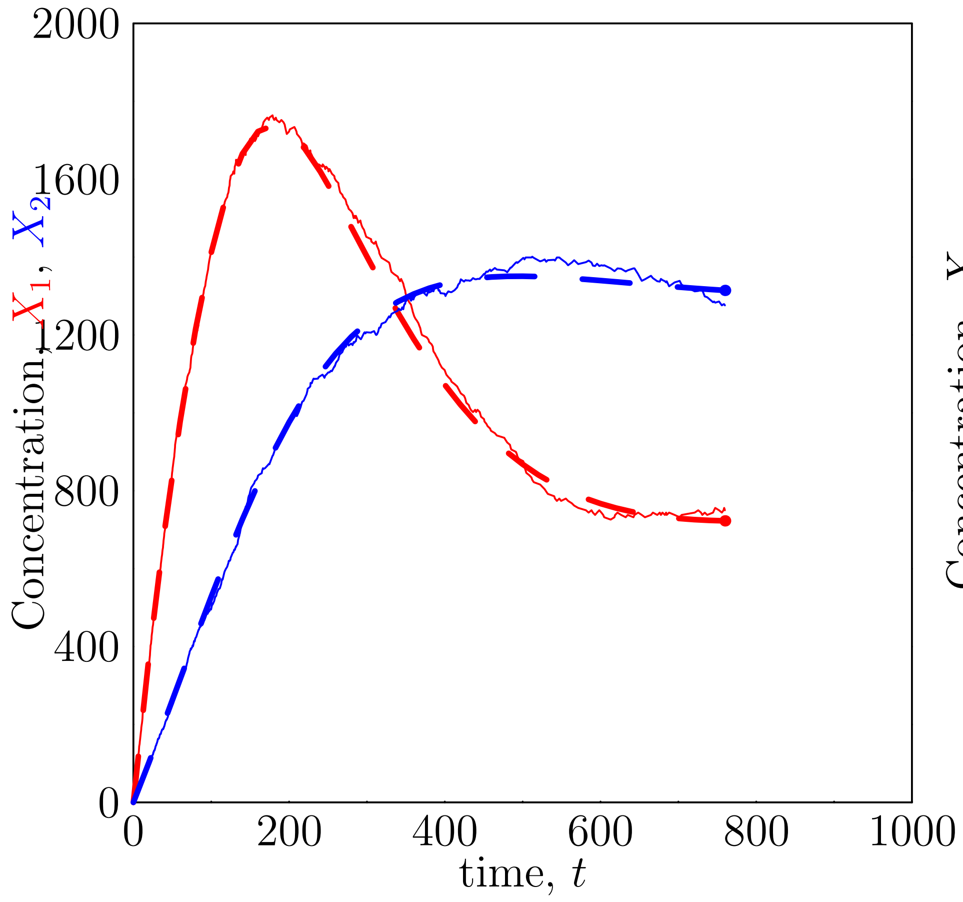
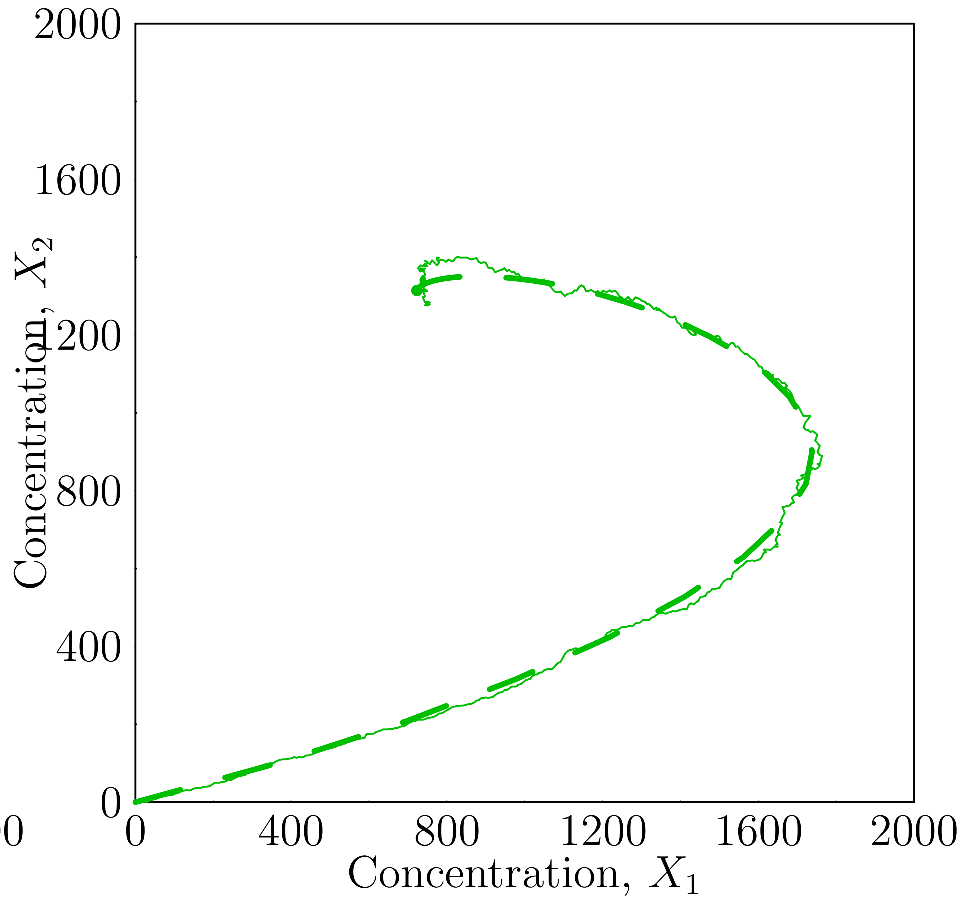
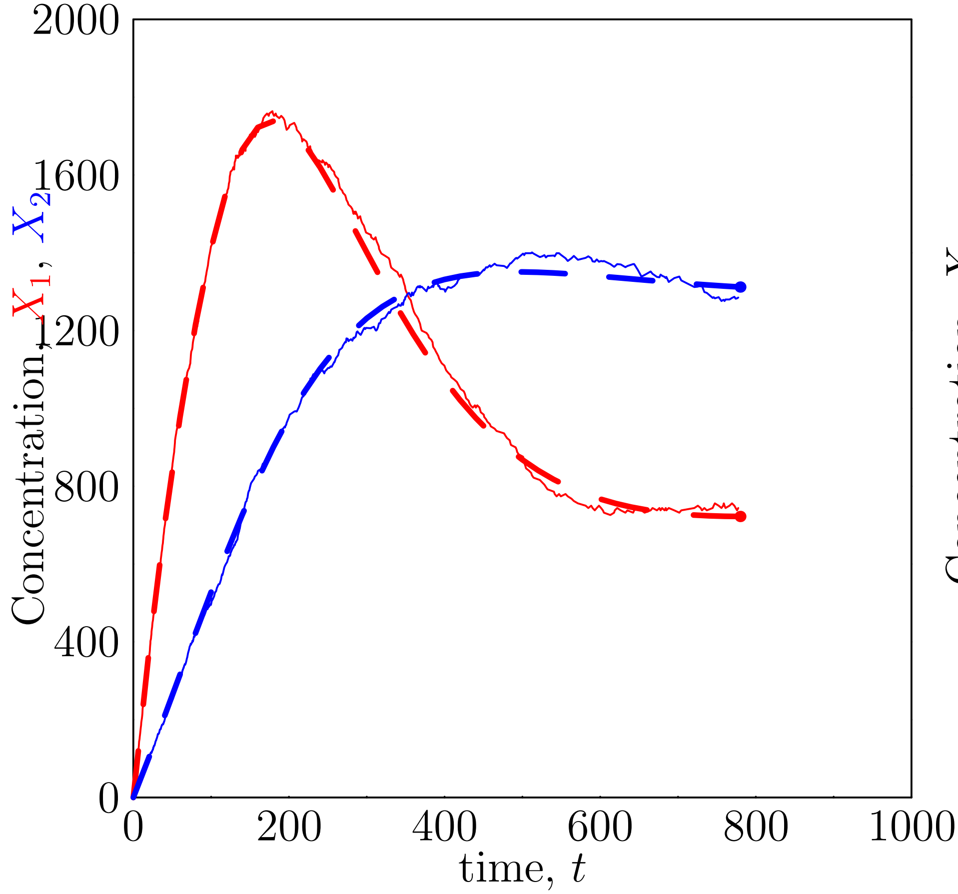
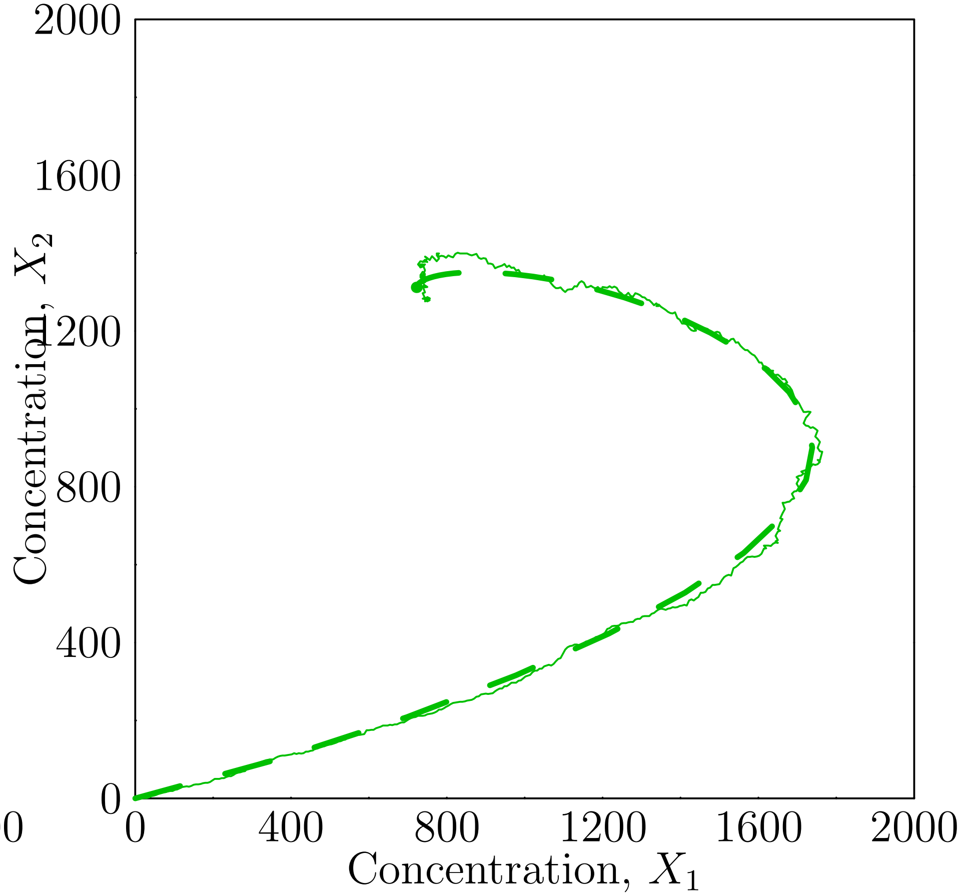




Large Numbers ($r = 1000$)





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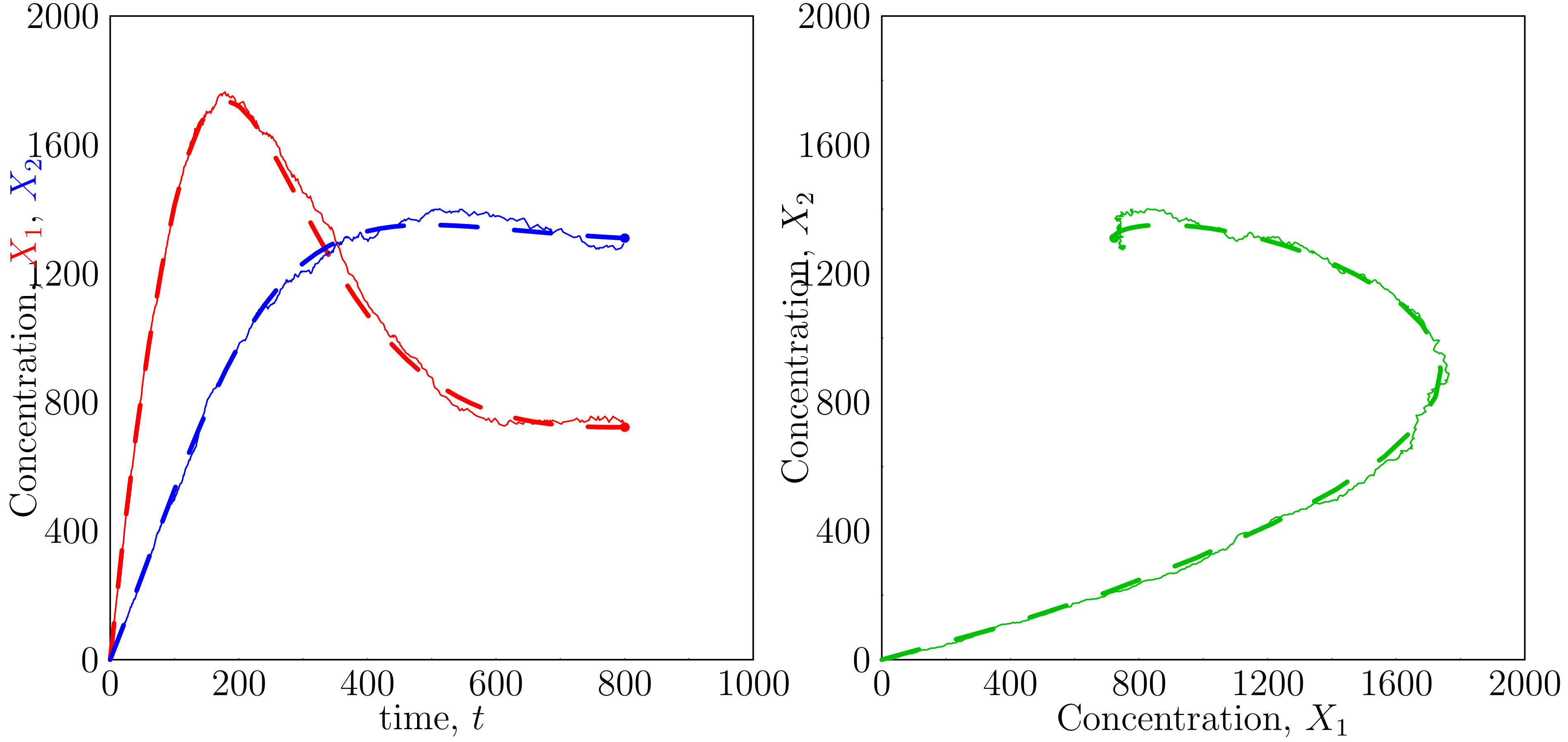

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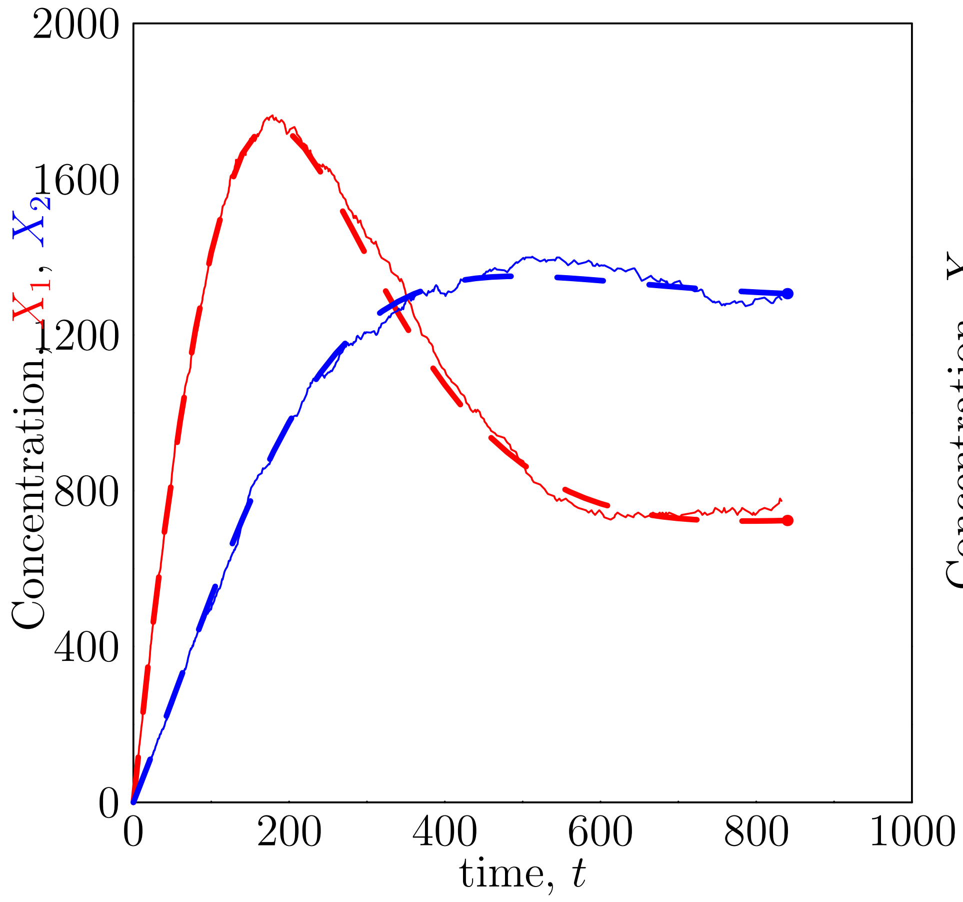

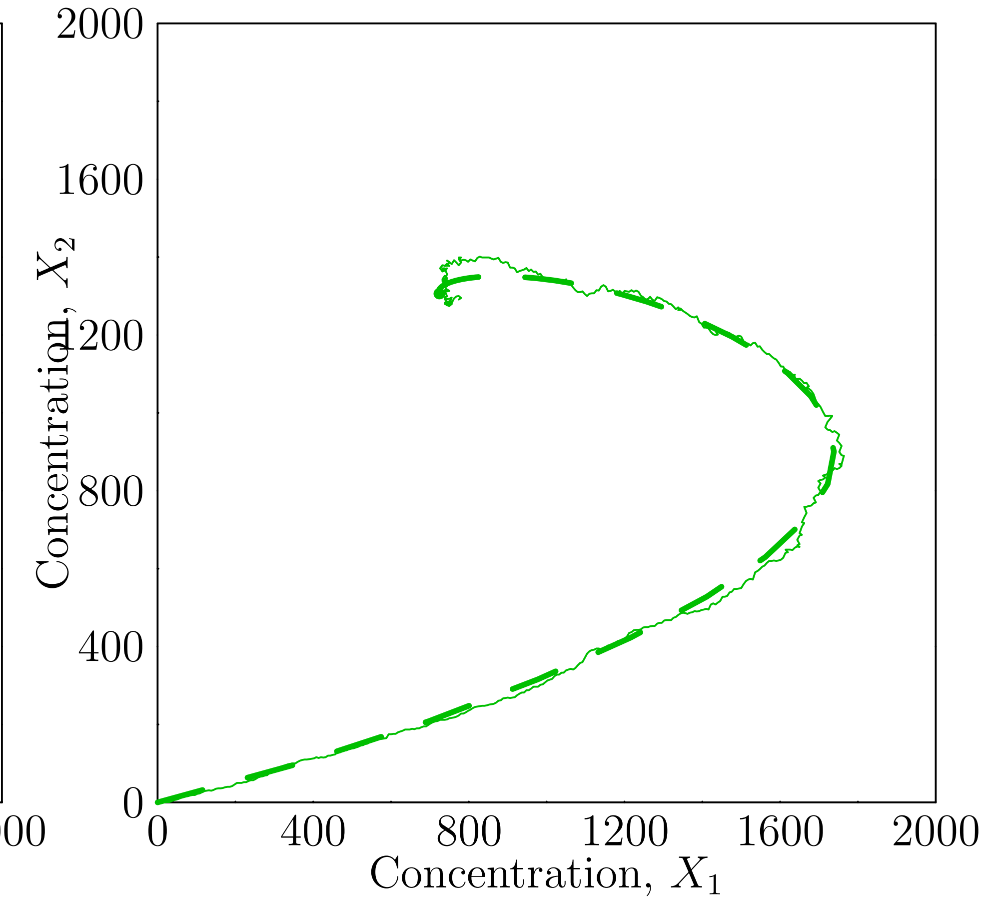

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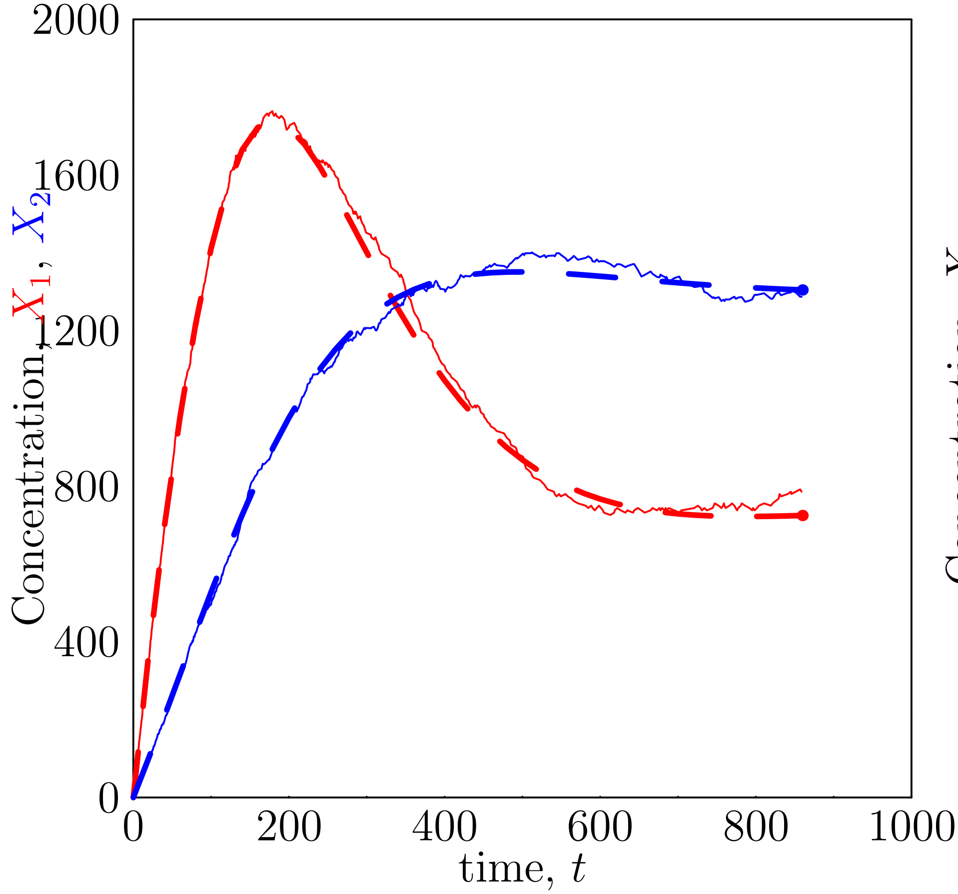

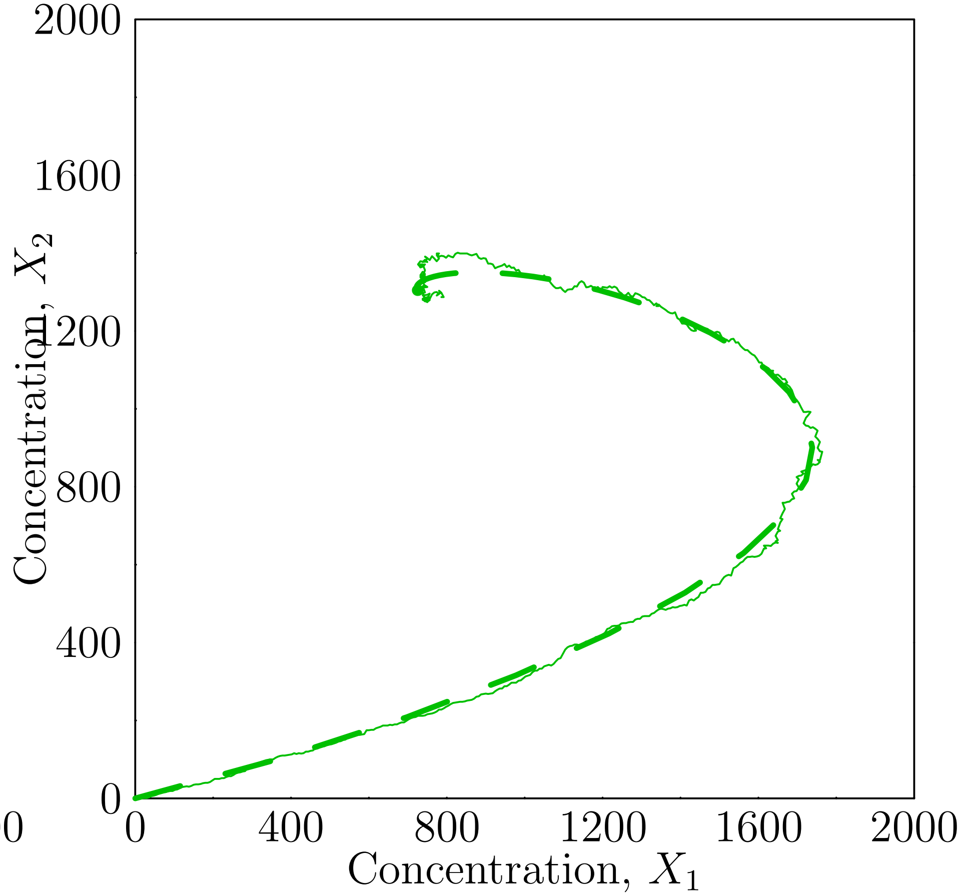

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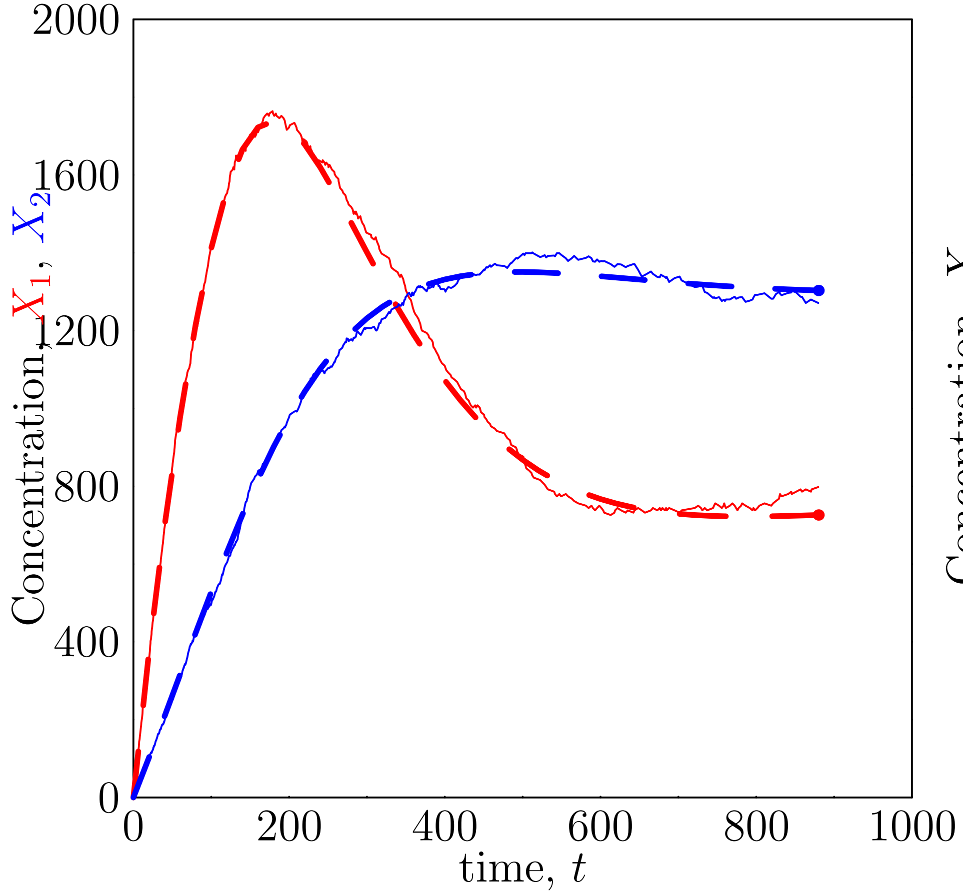

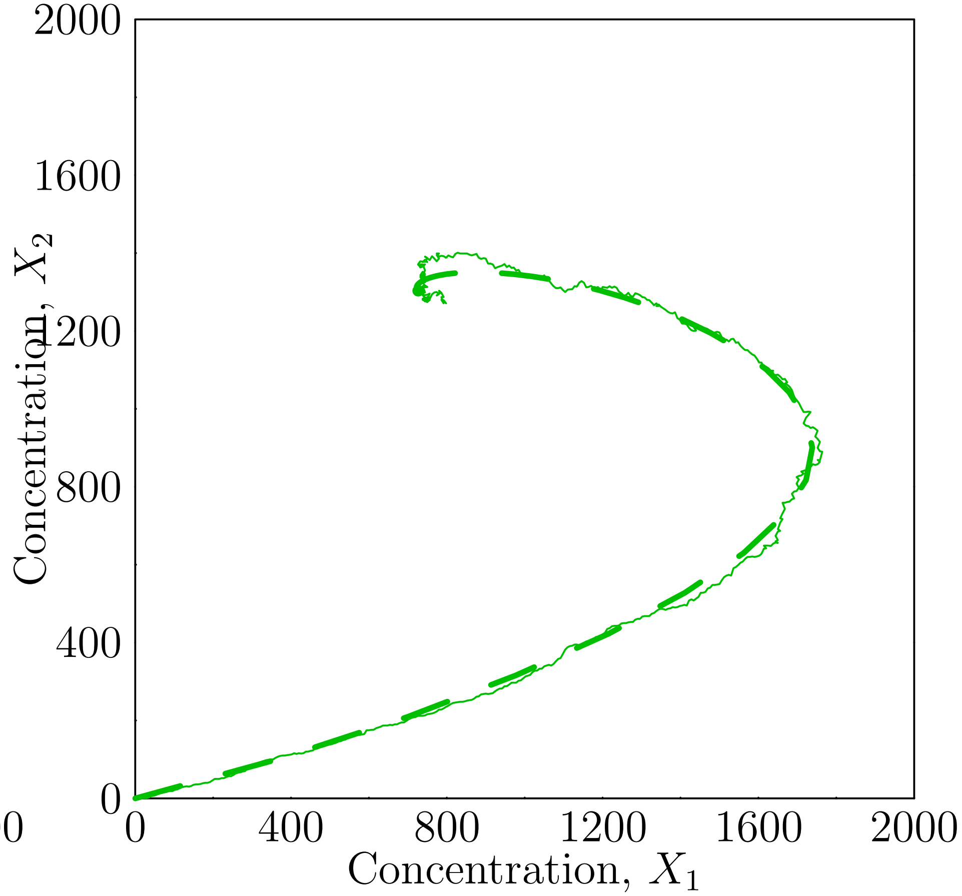

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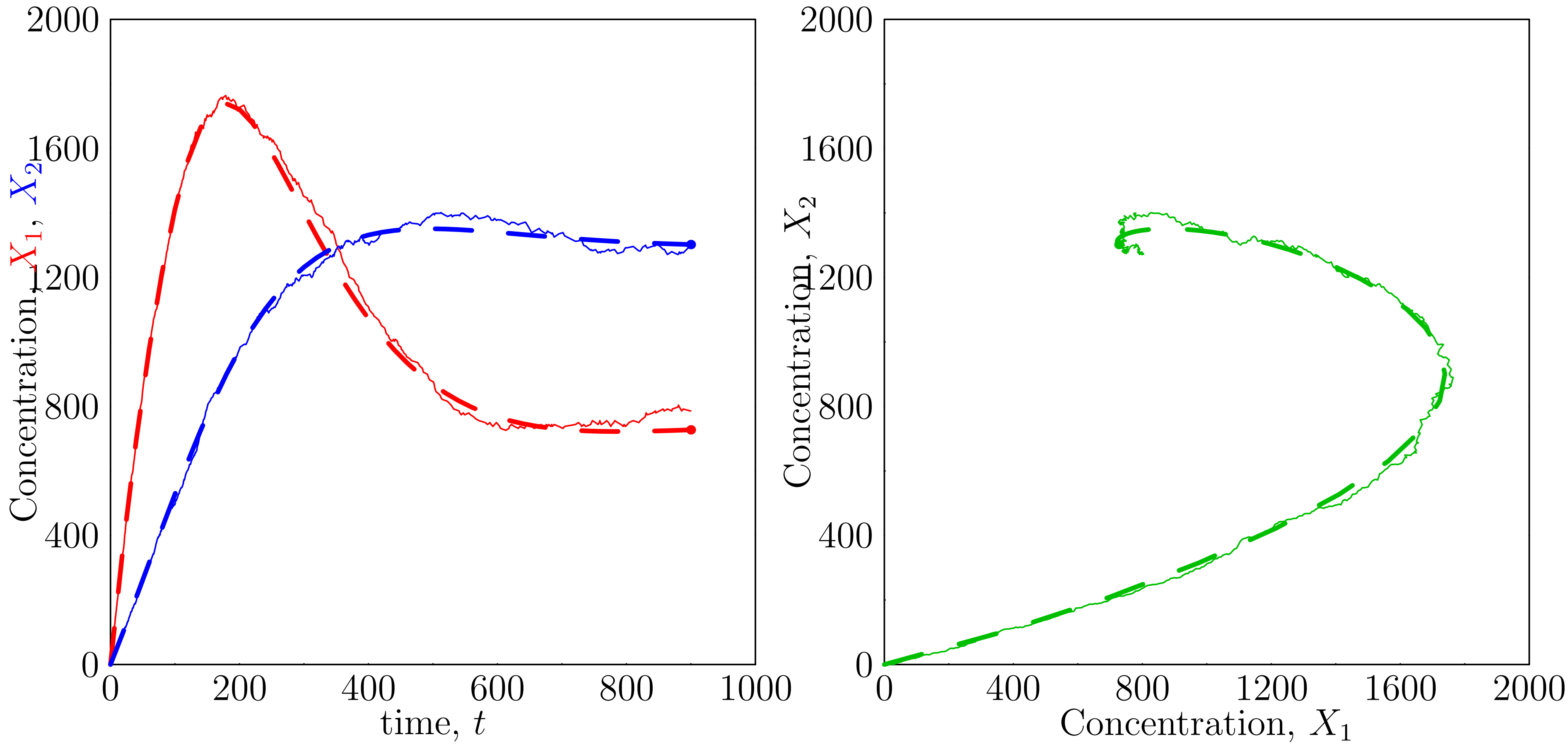

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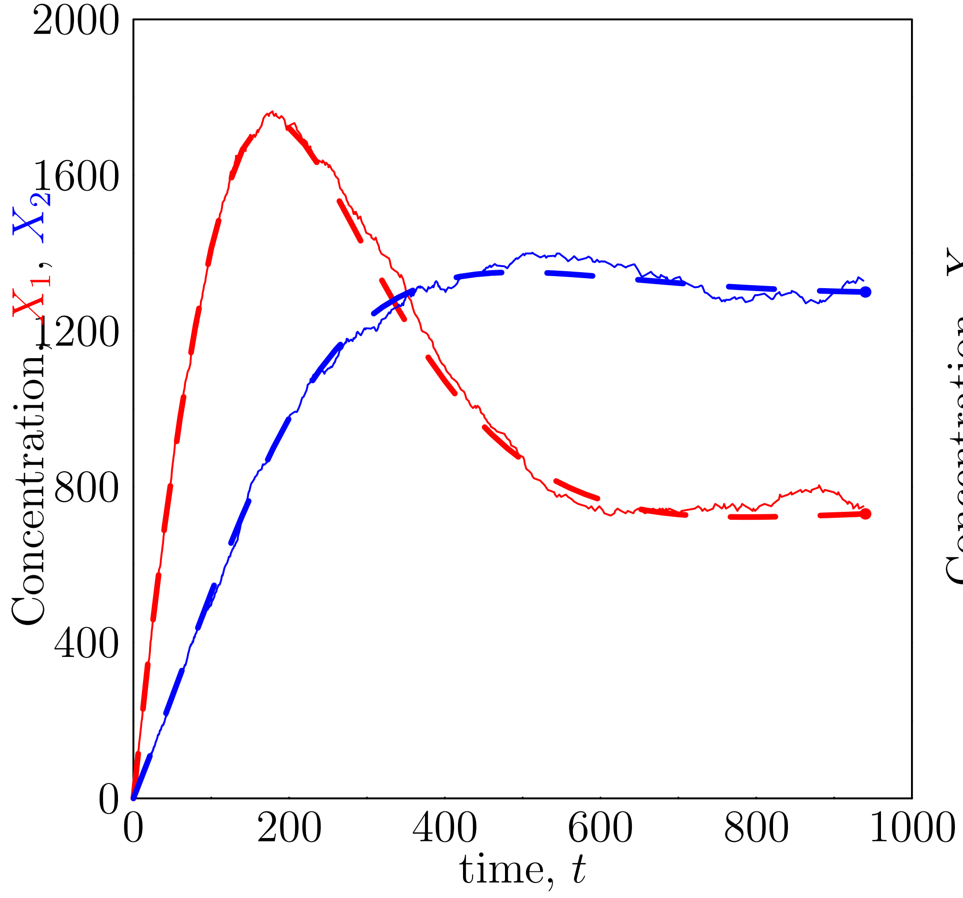

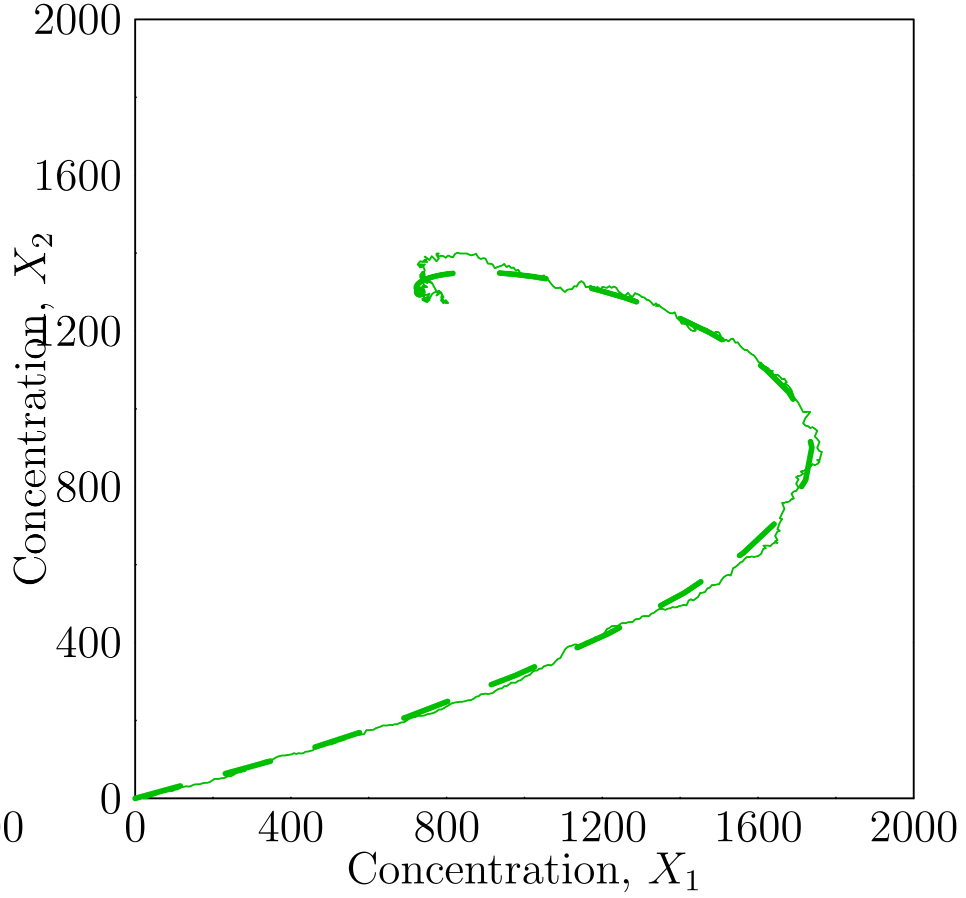

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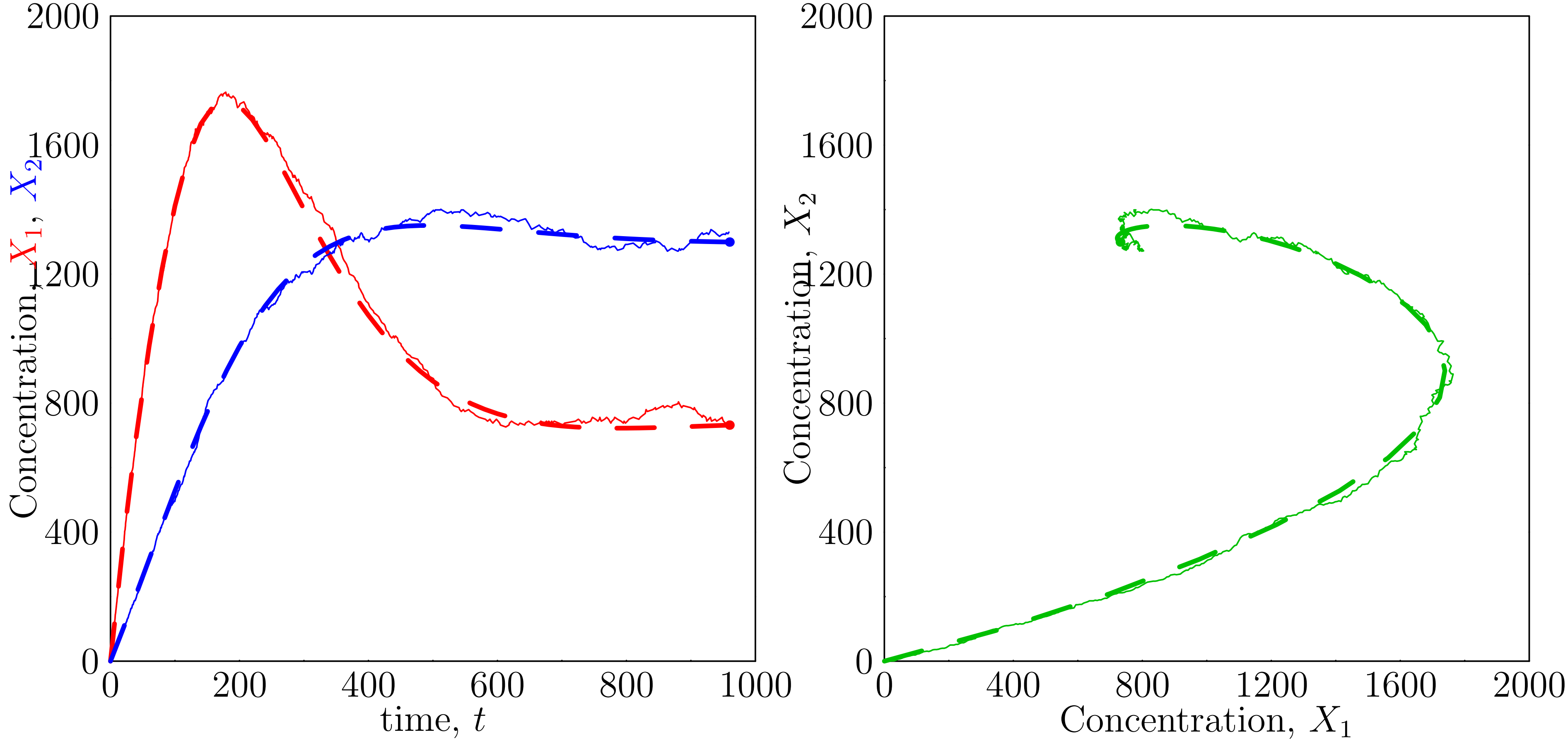

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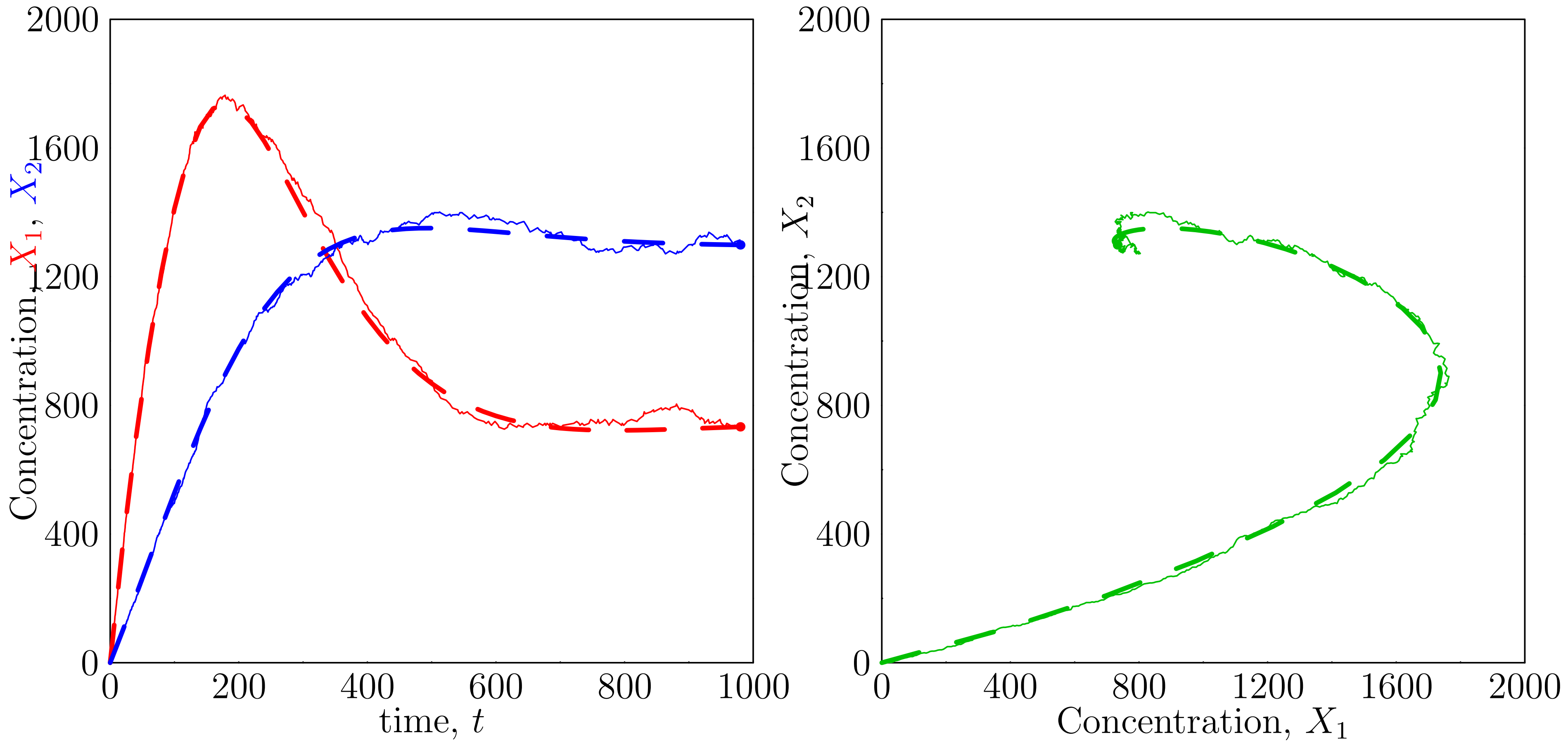

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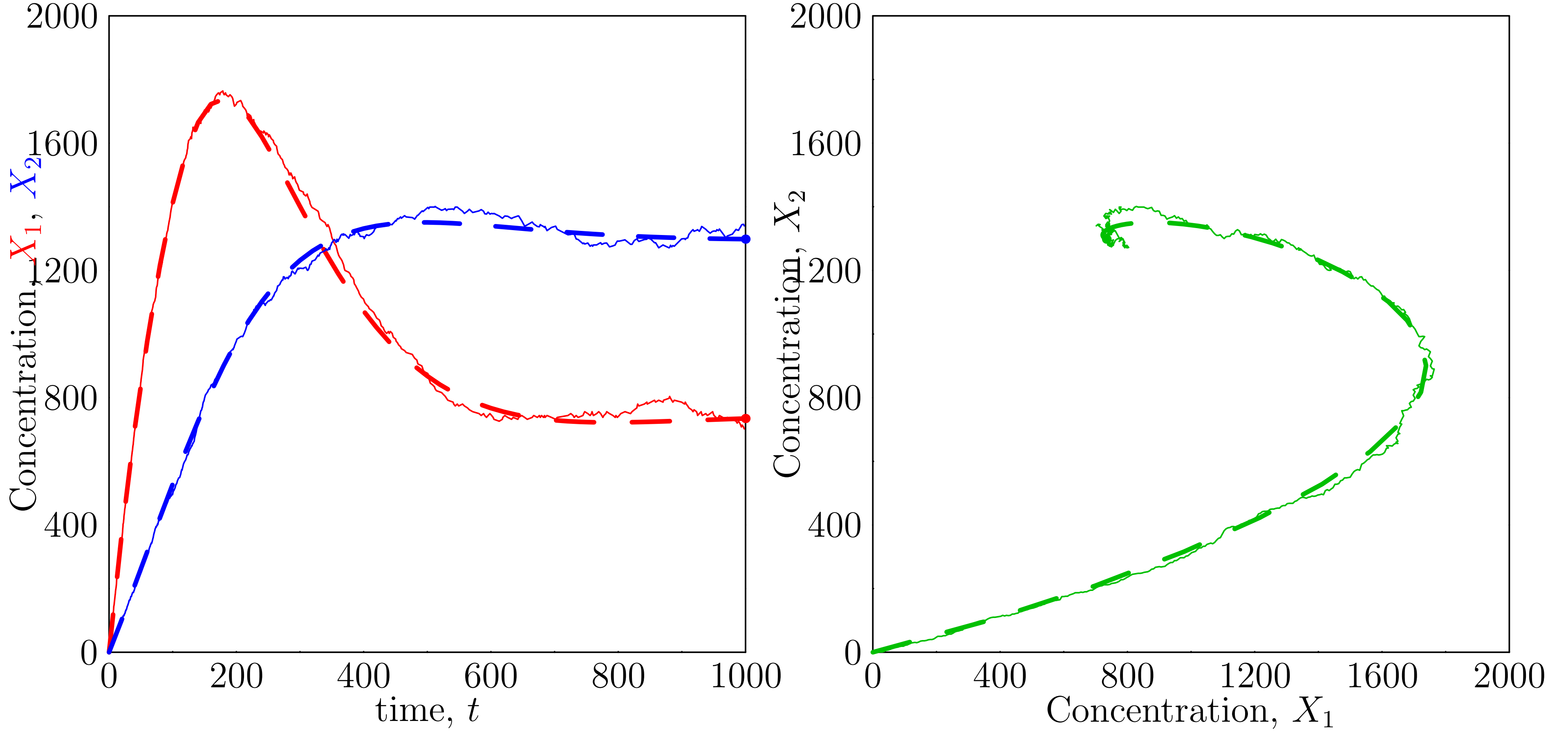

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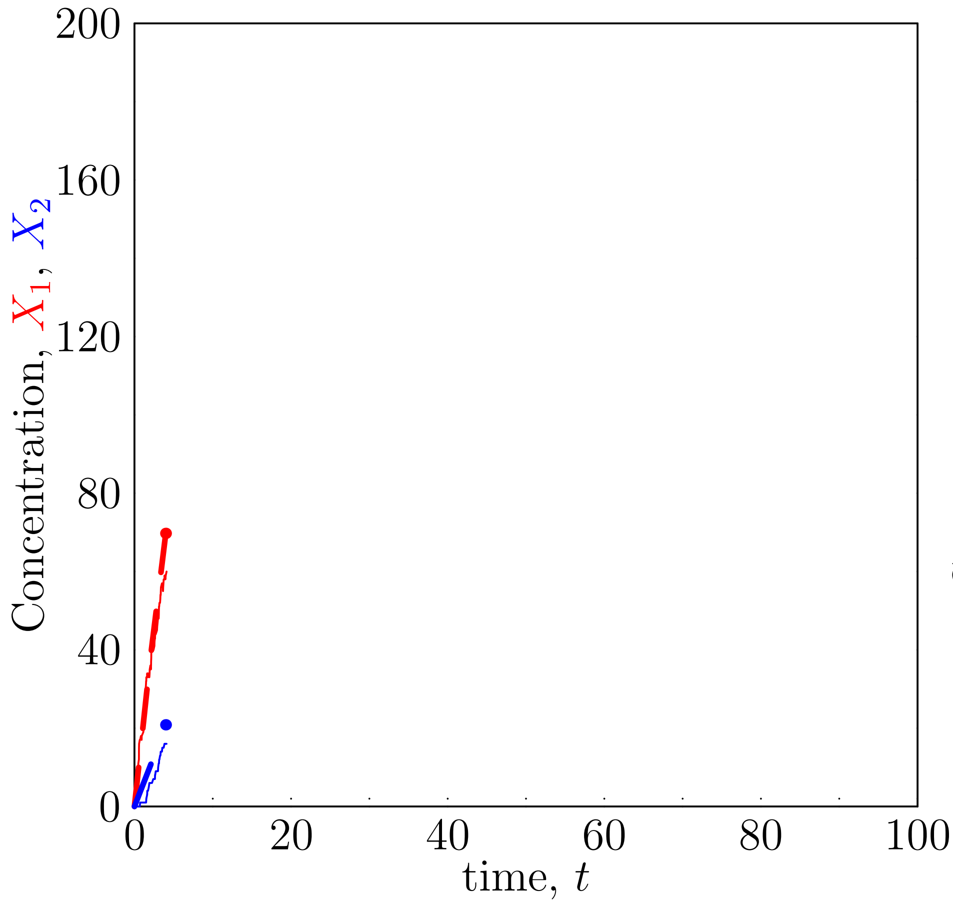

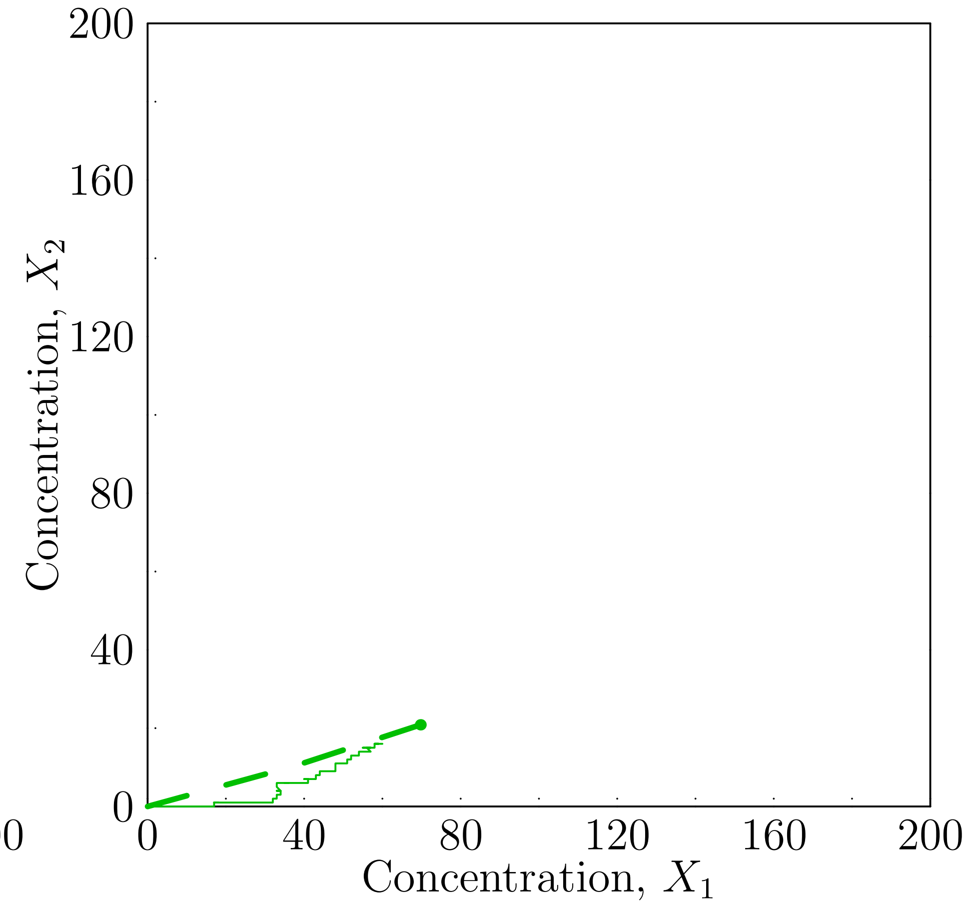

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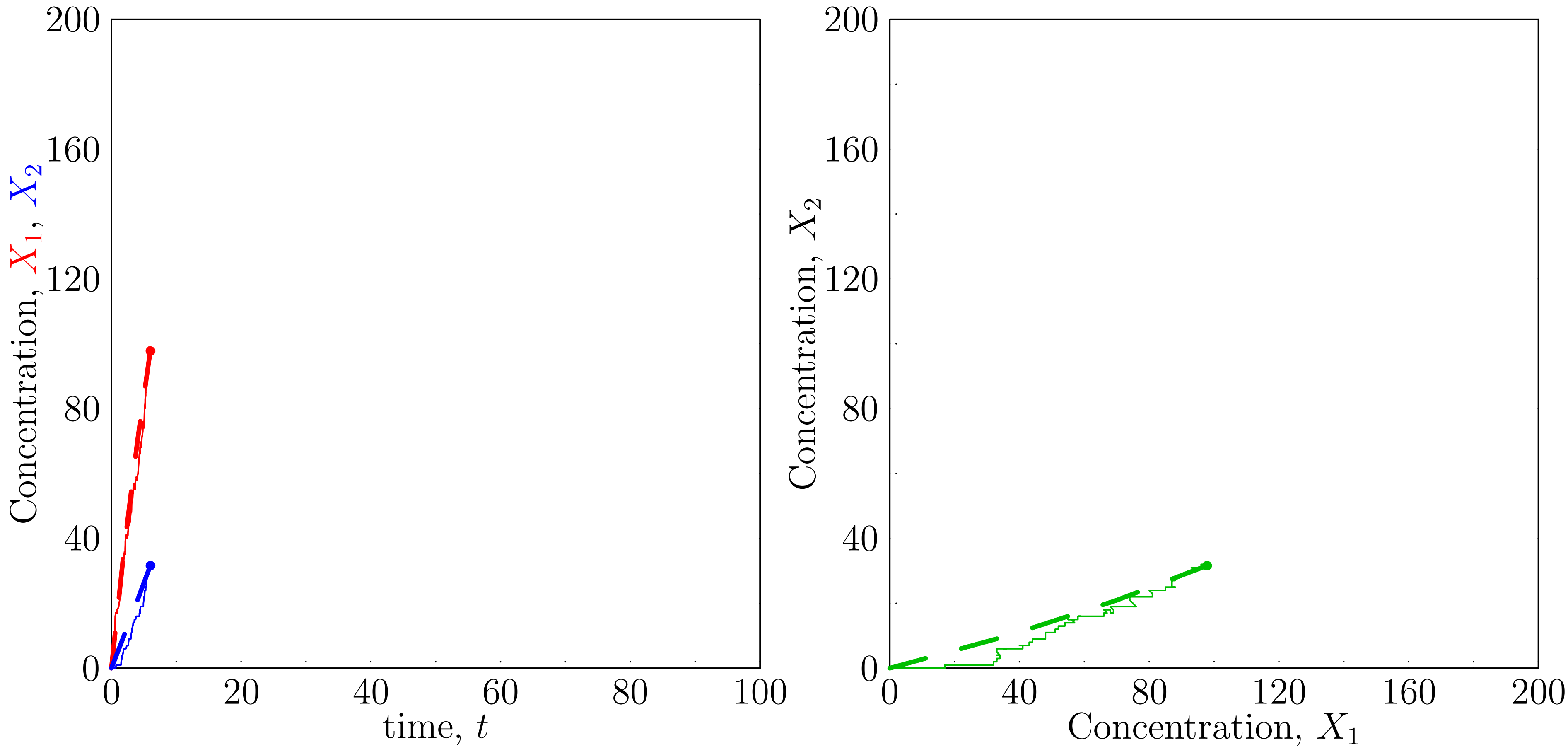

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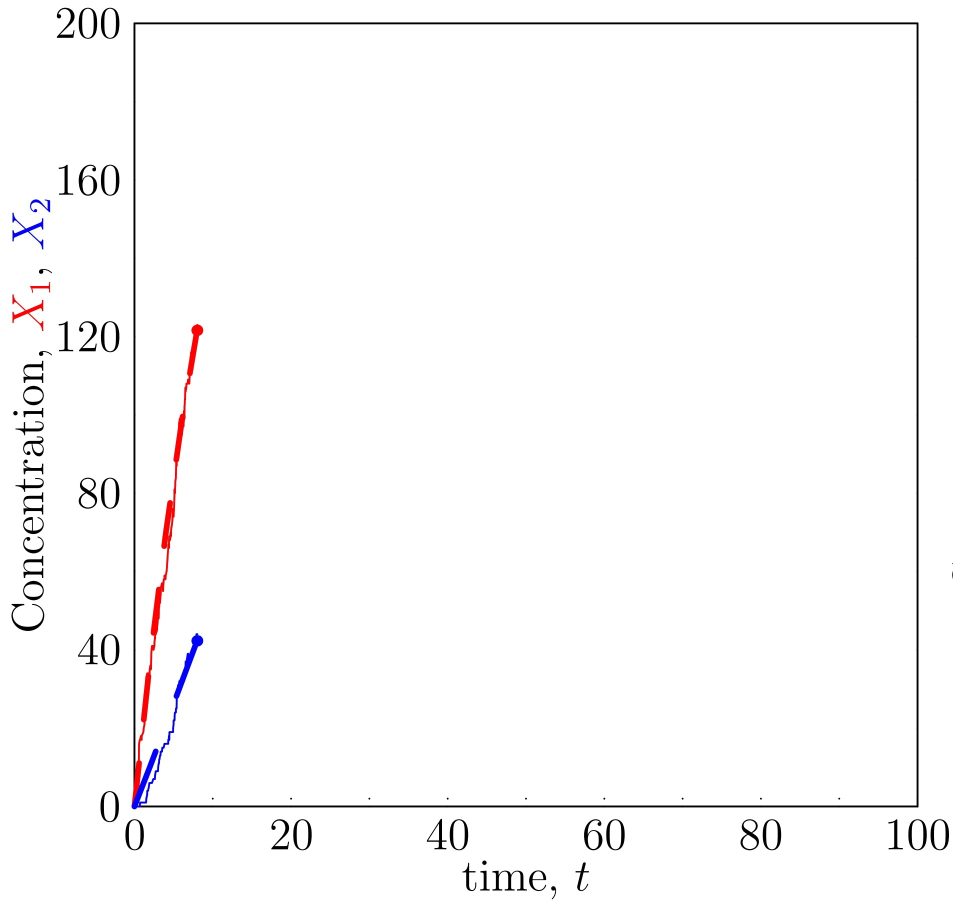

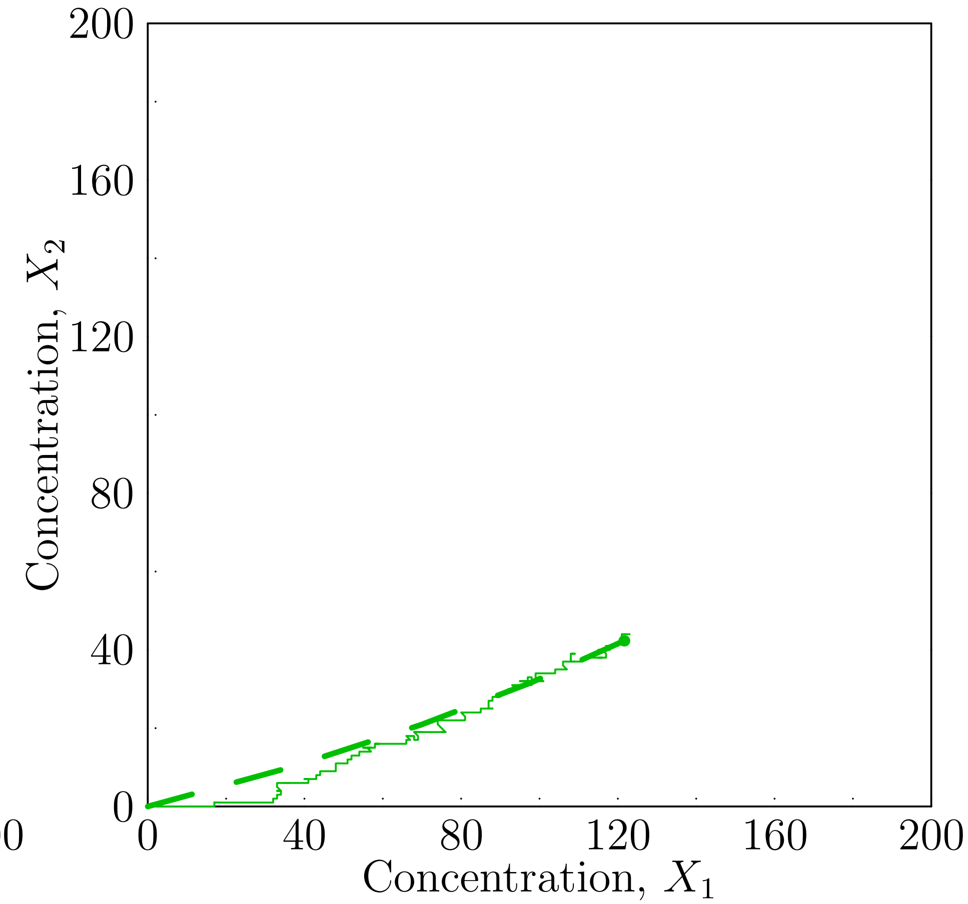

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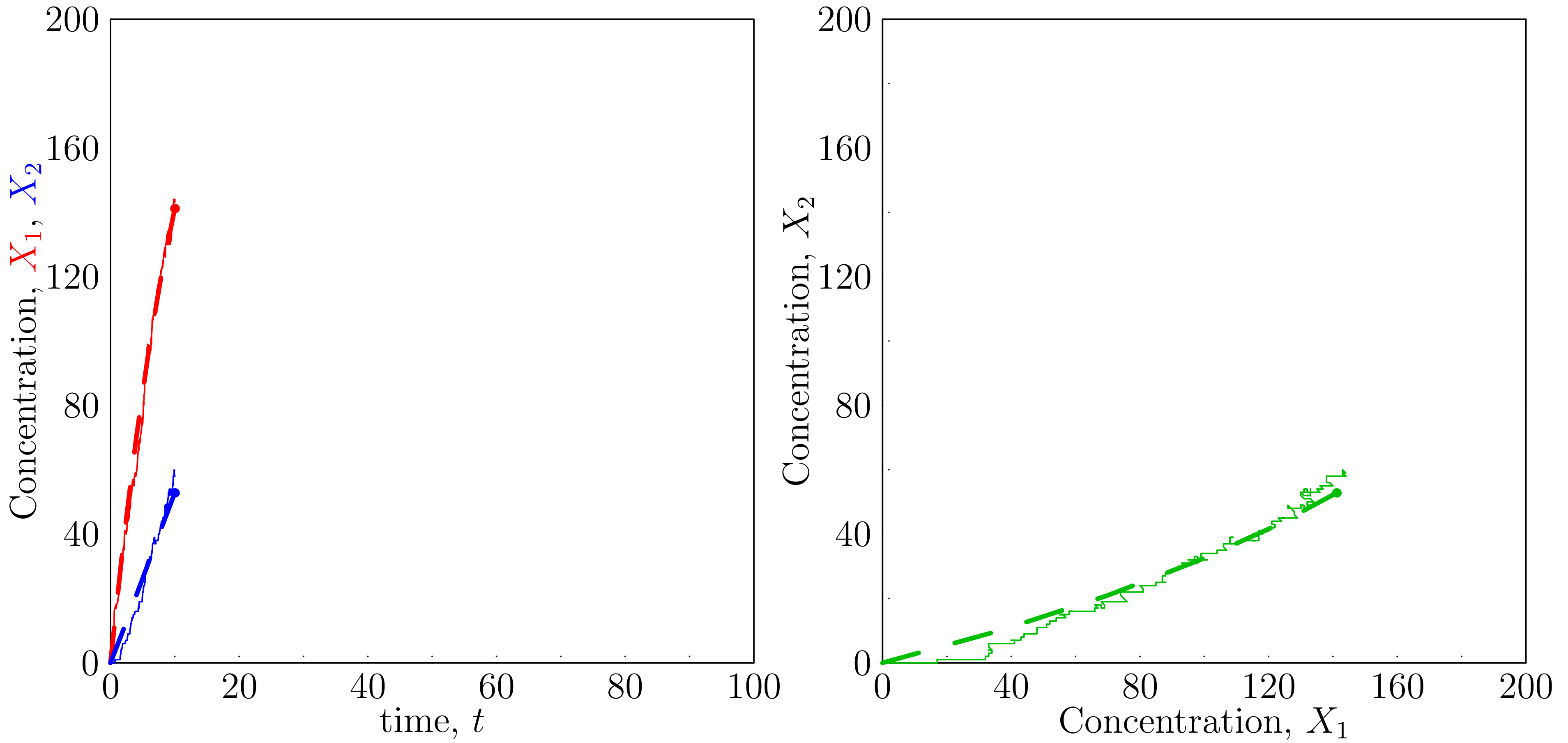

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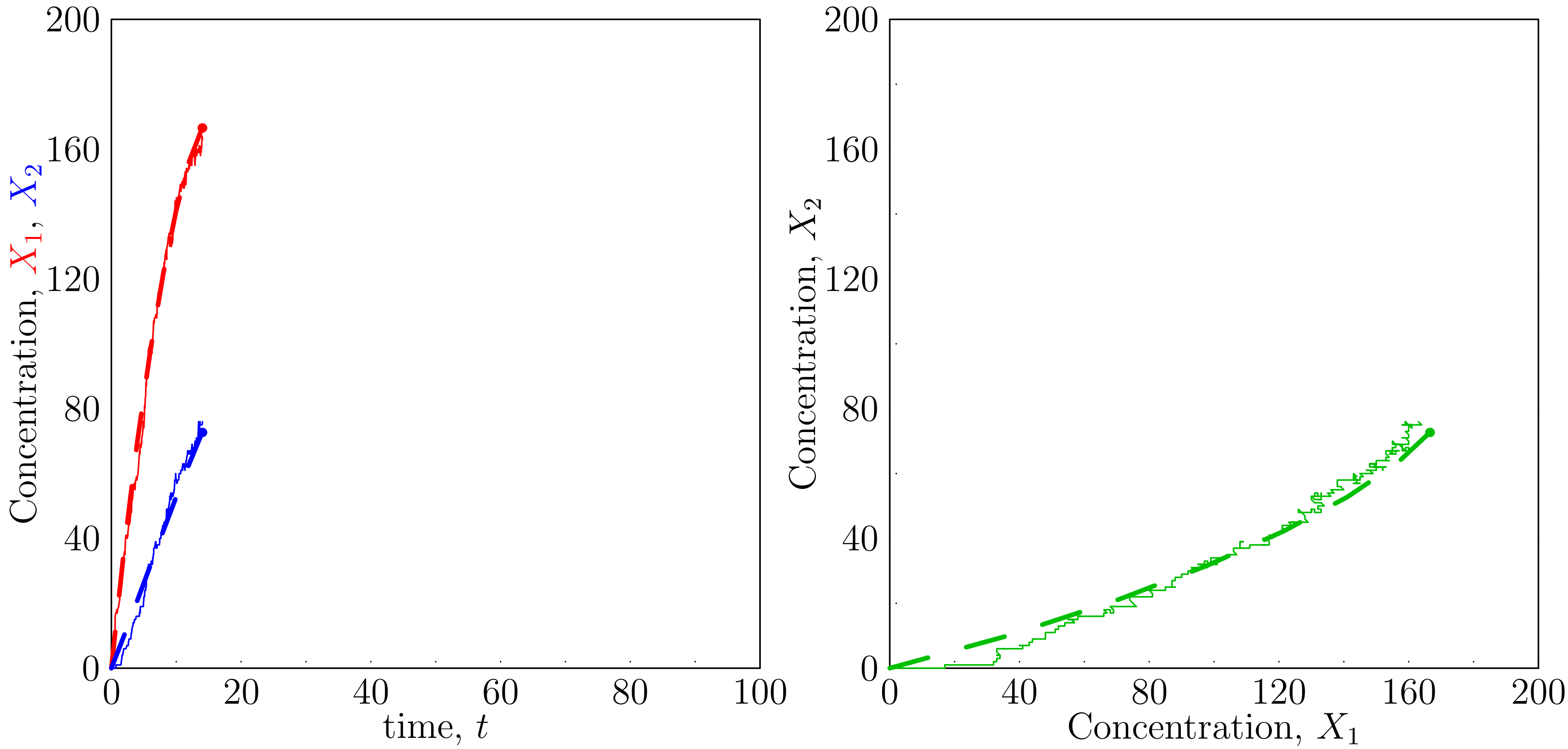

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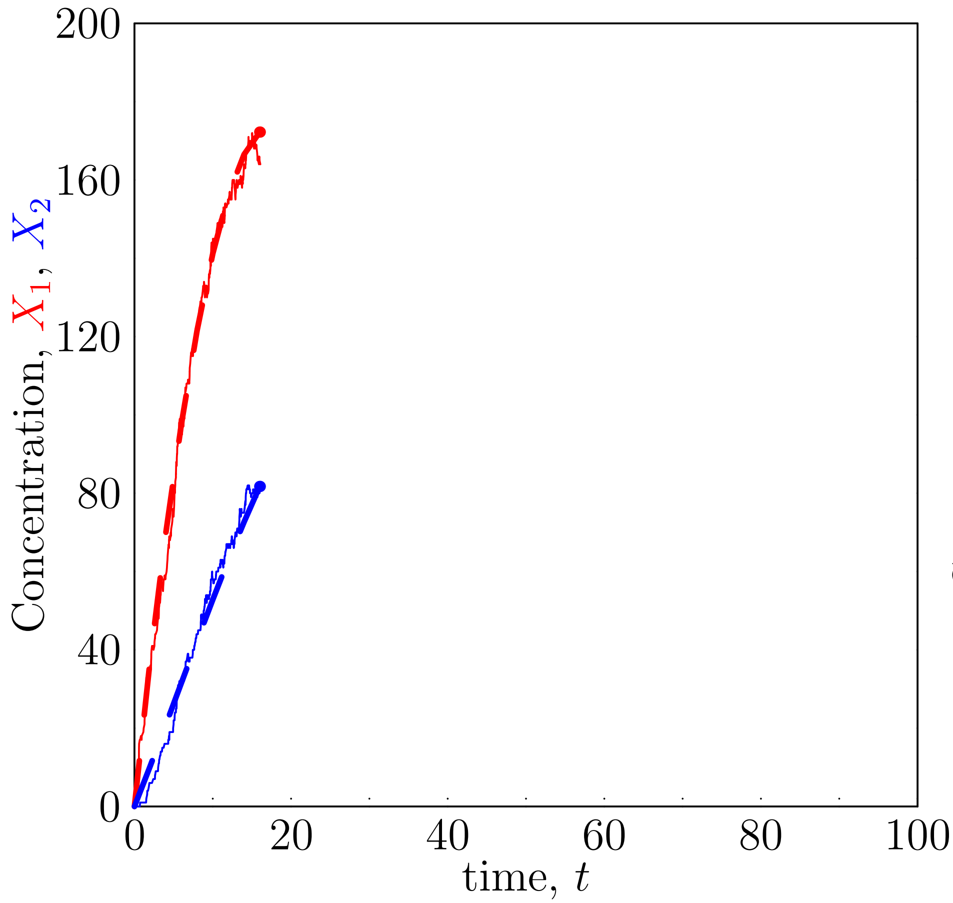

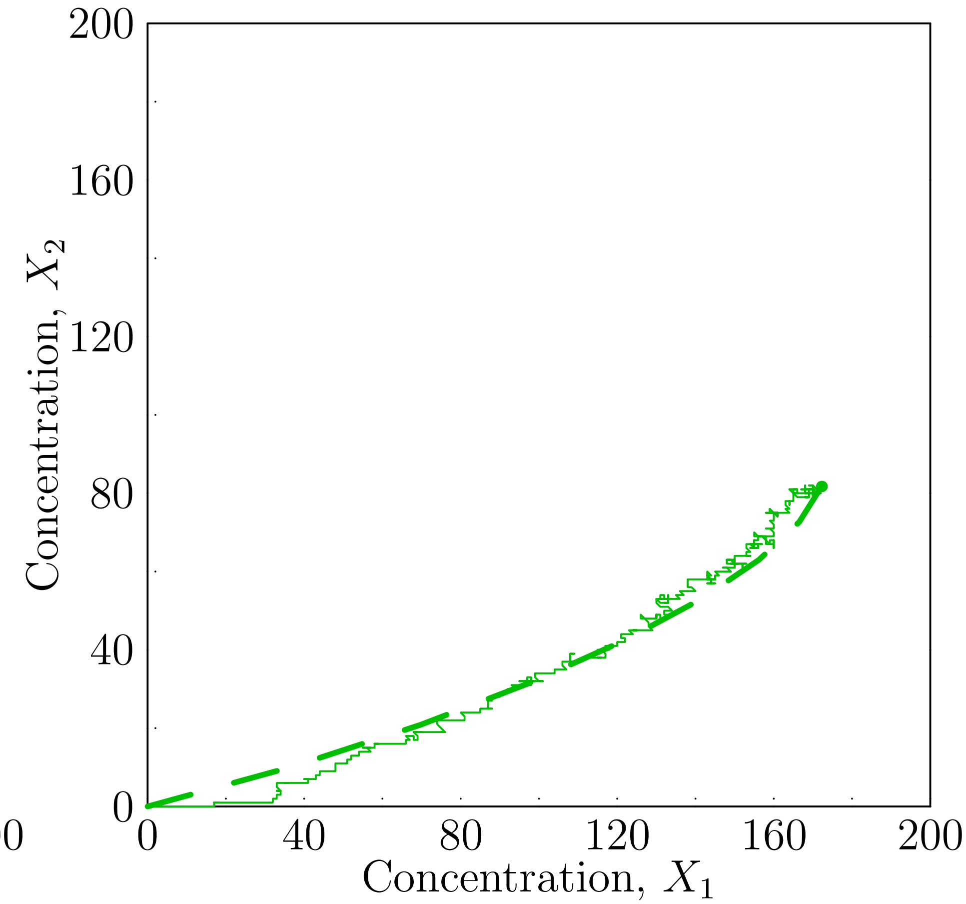

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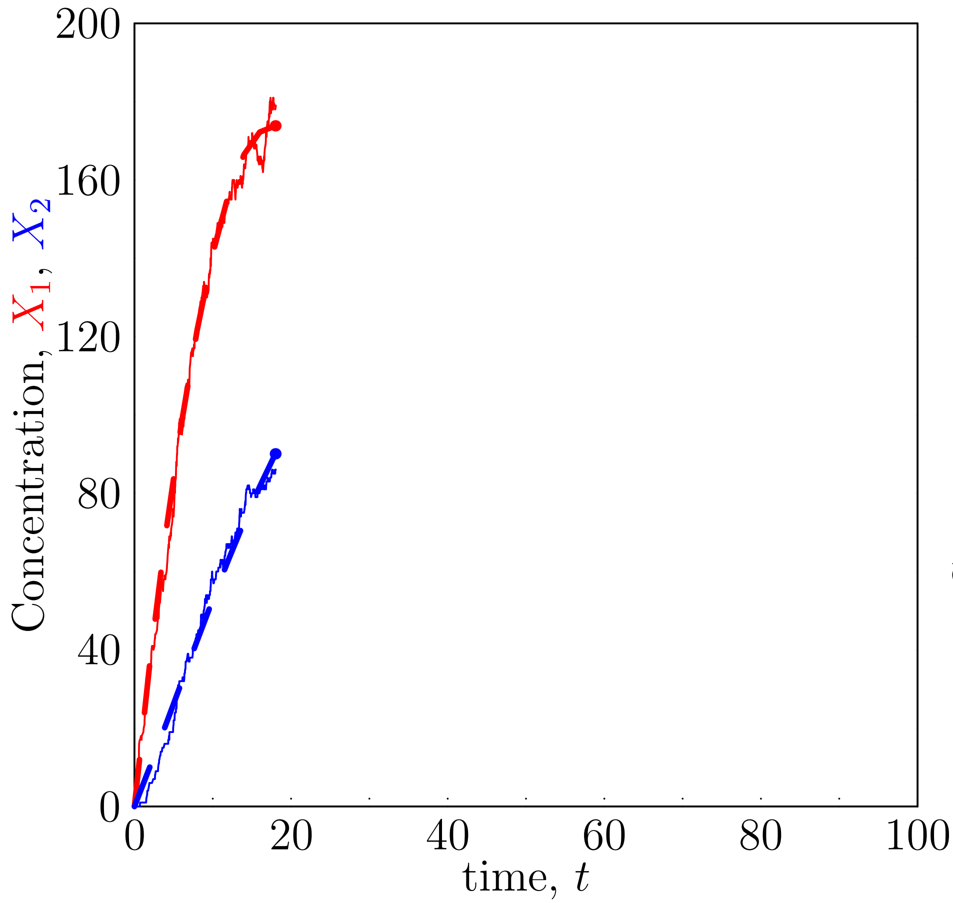

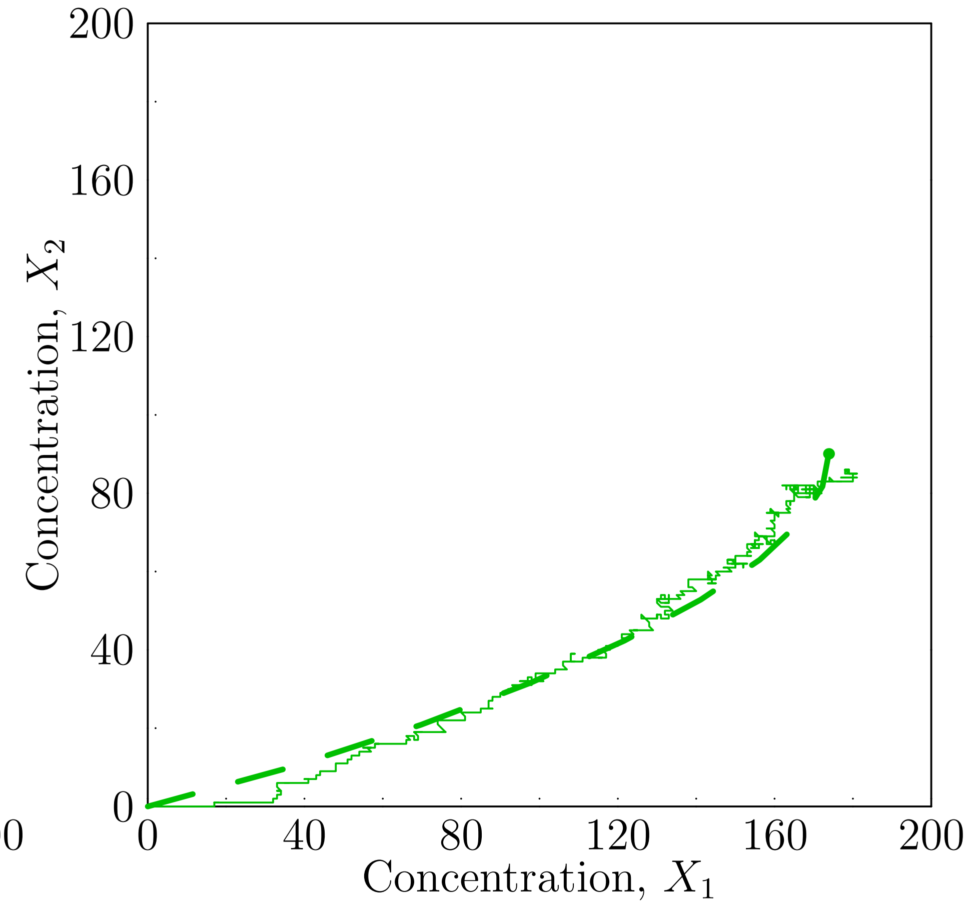

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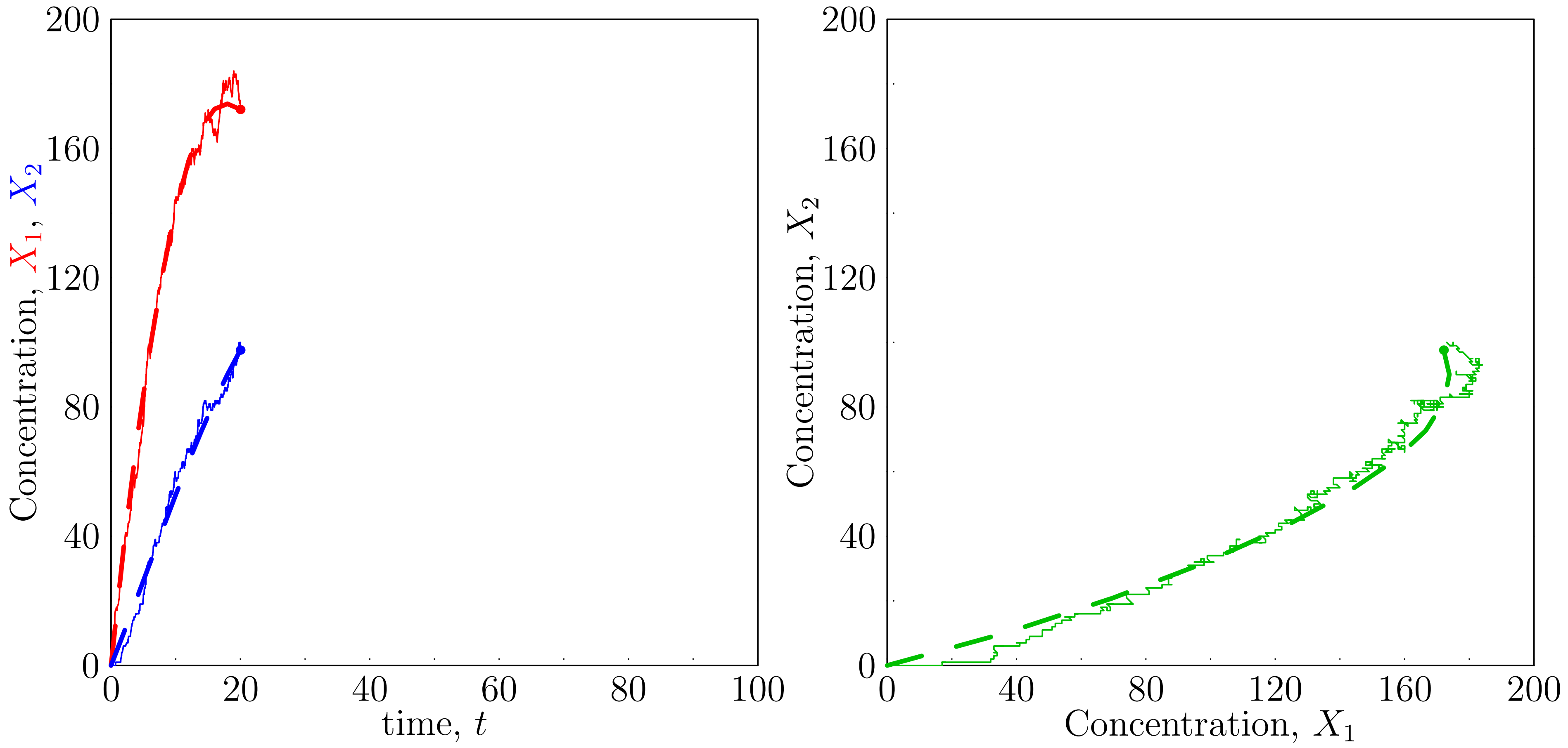

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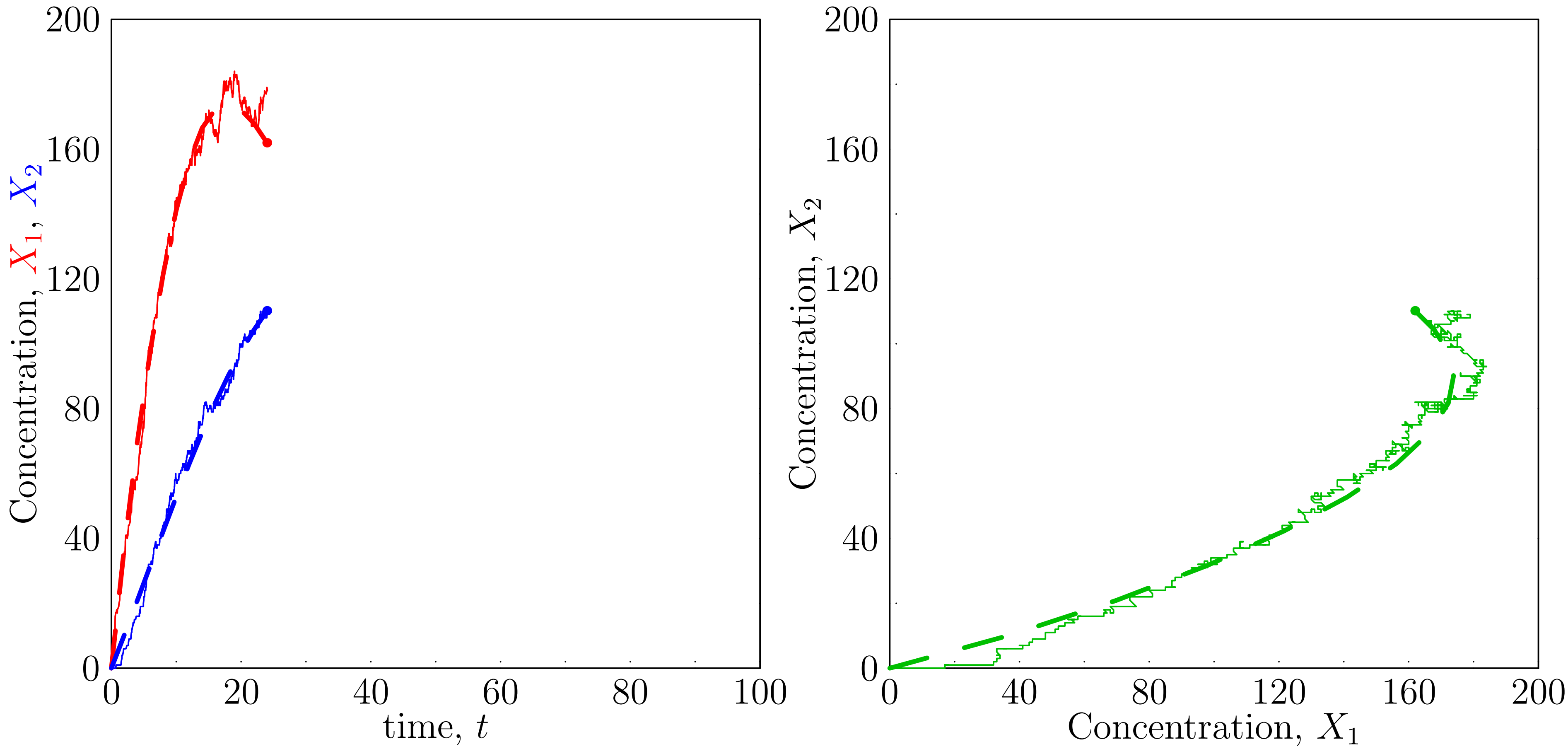

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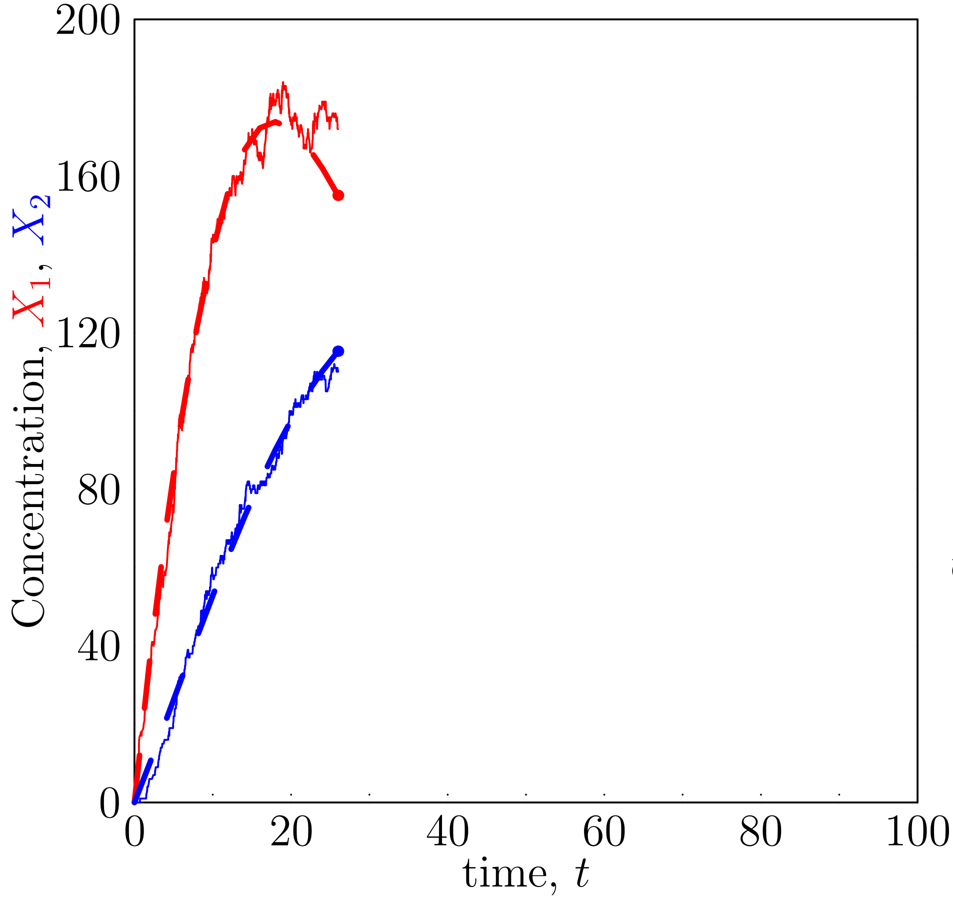

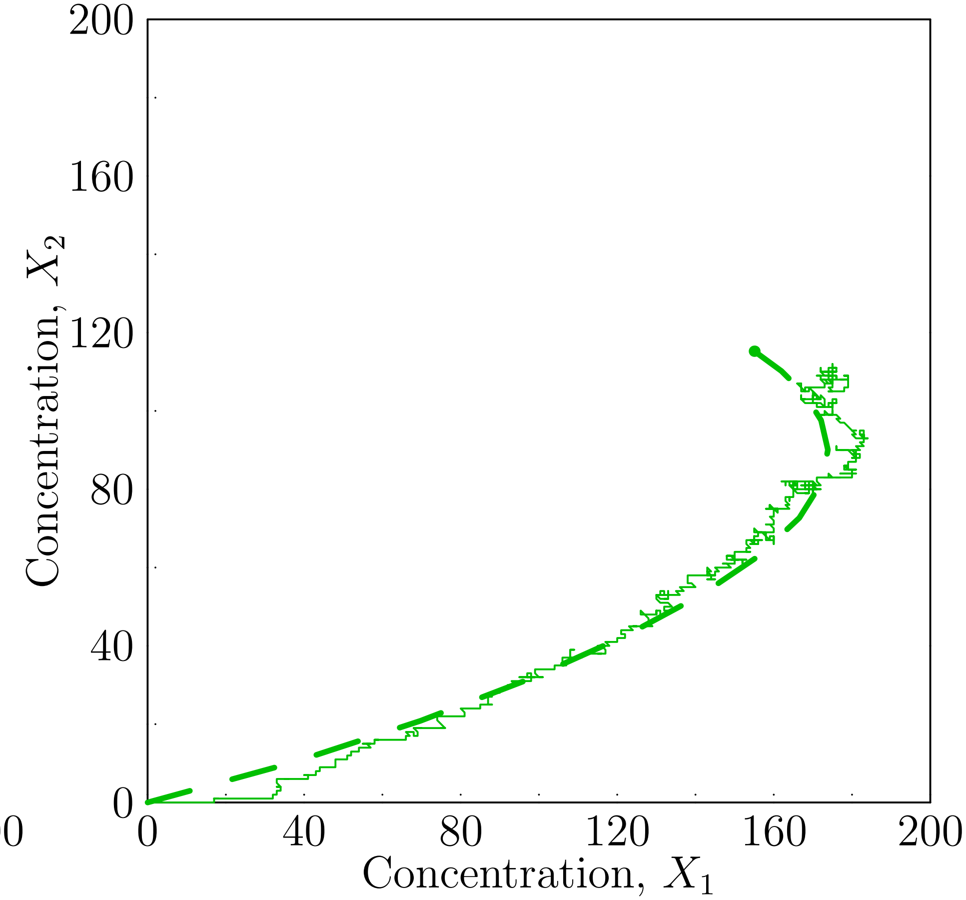

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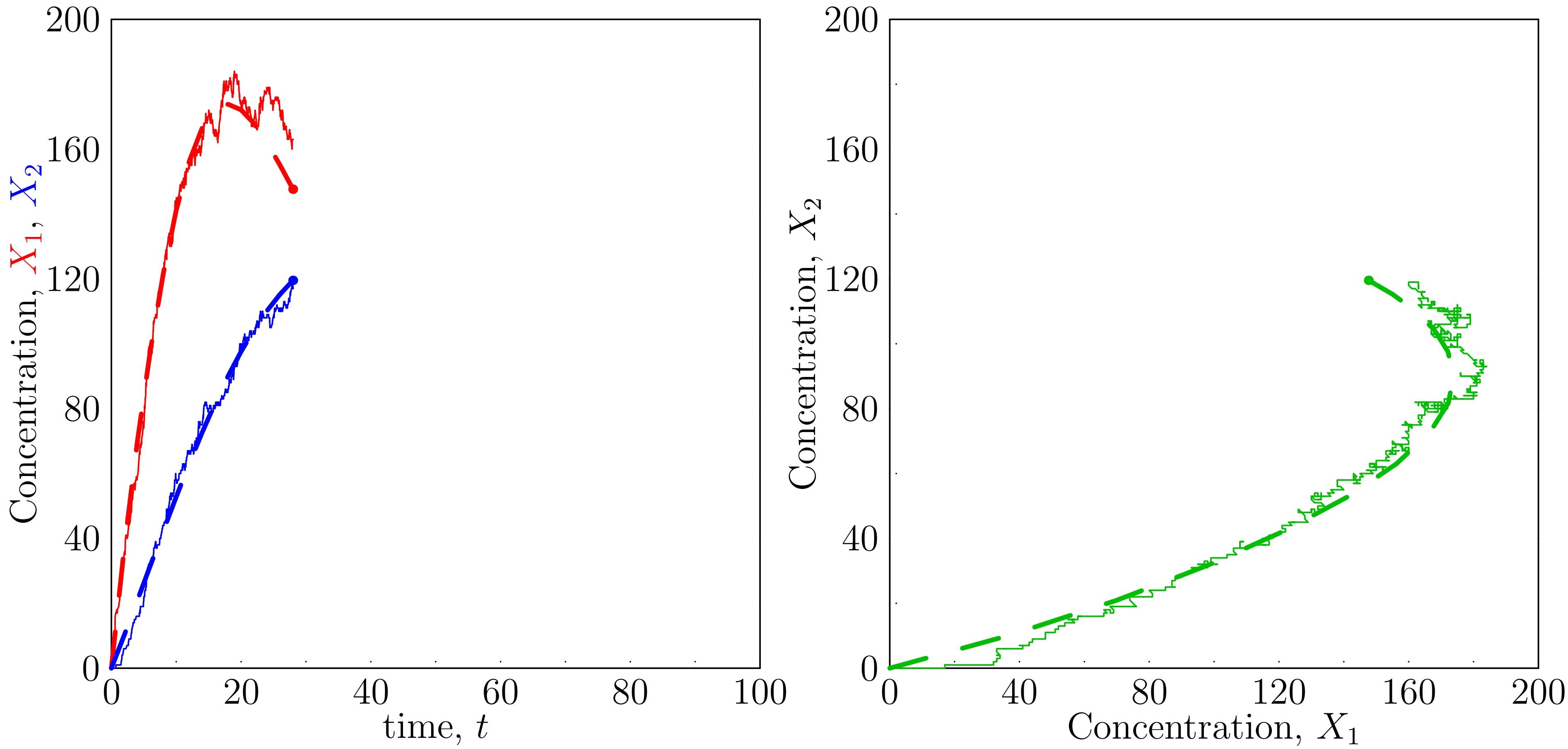

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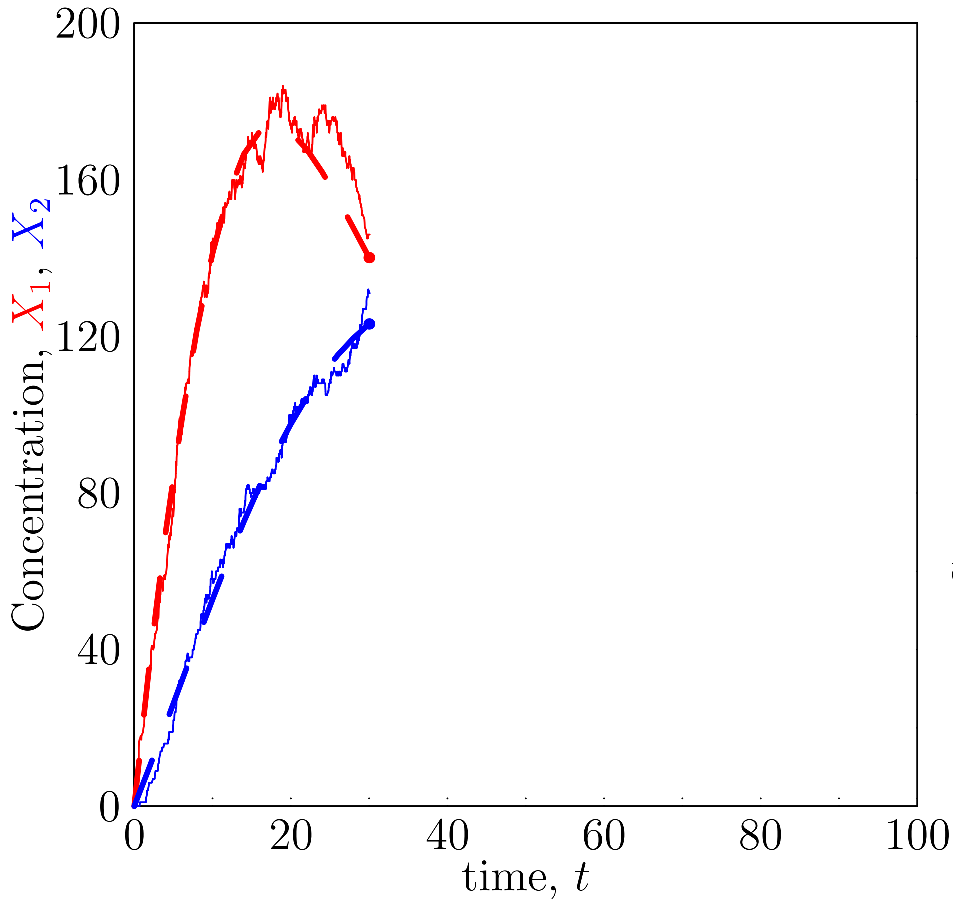

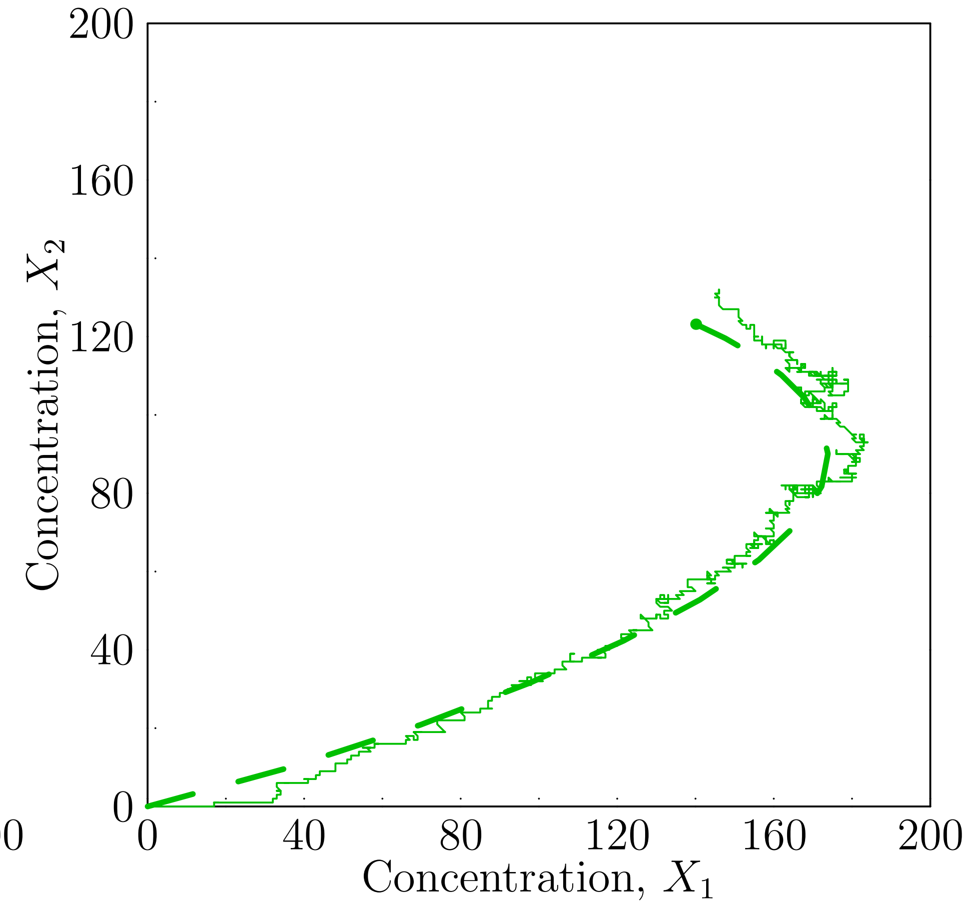

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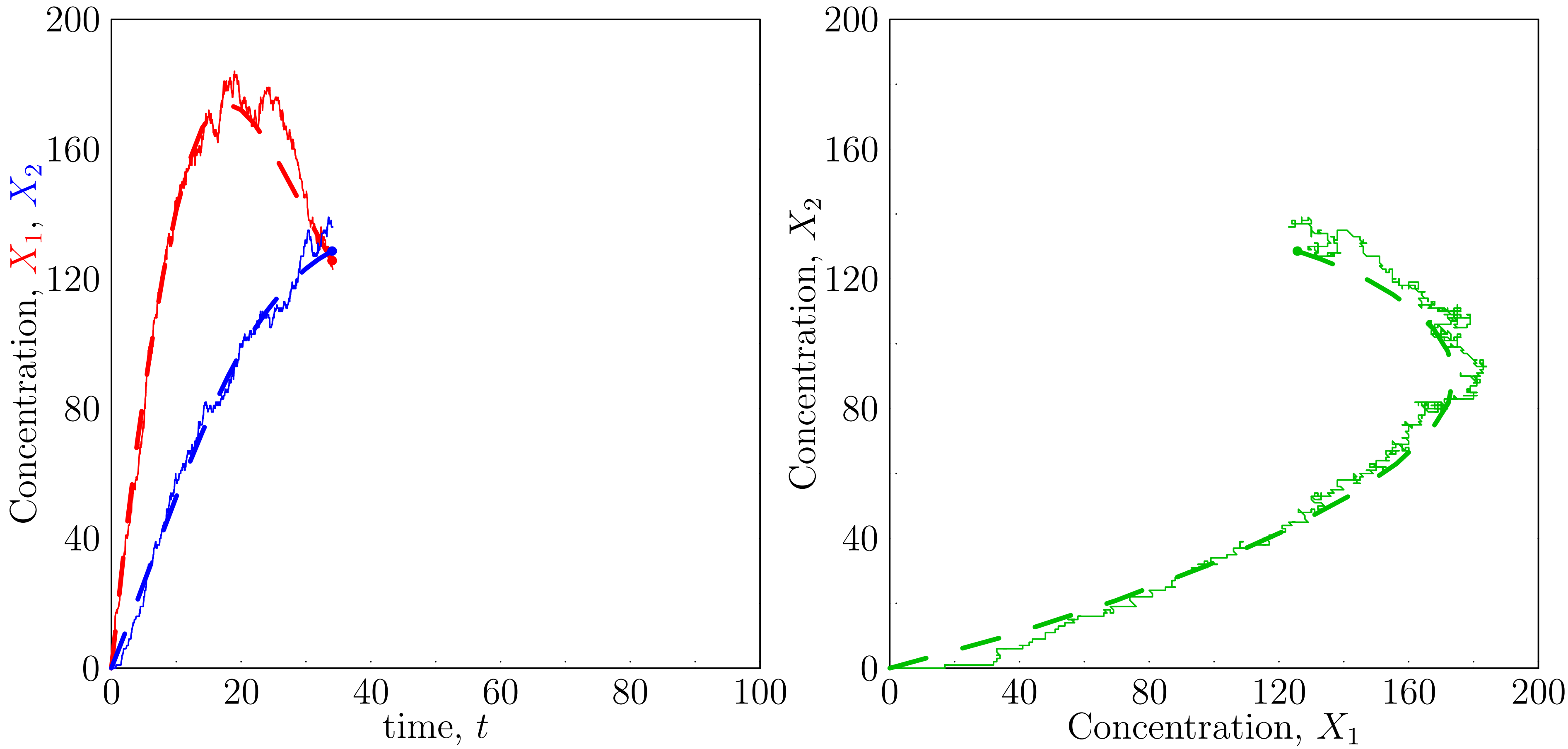

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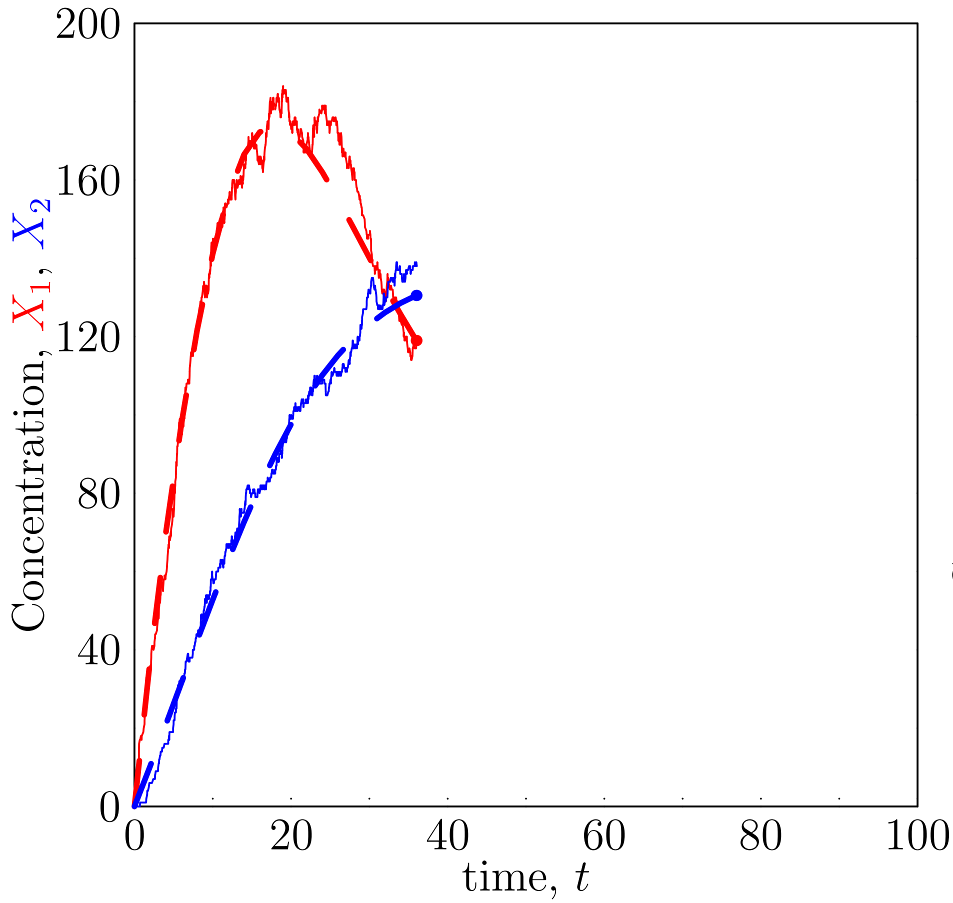

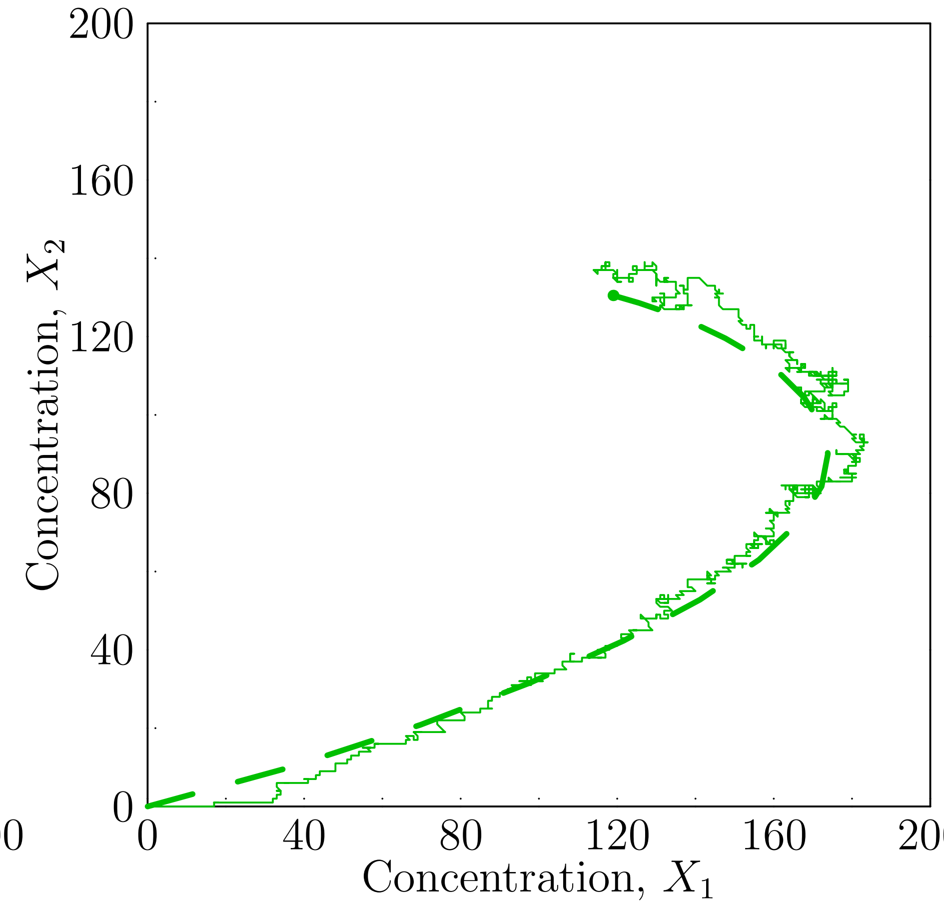

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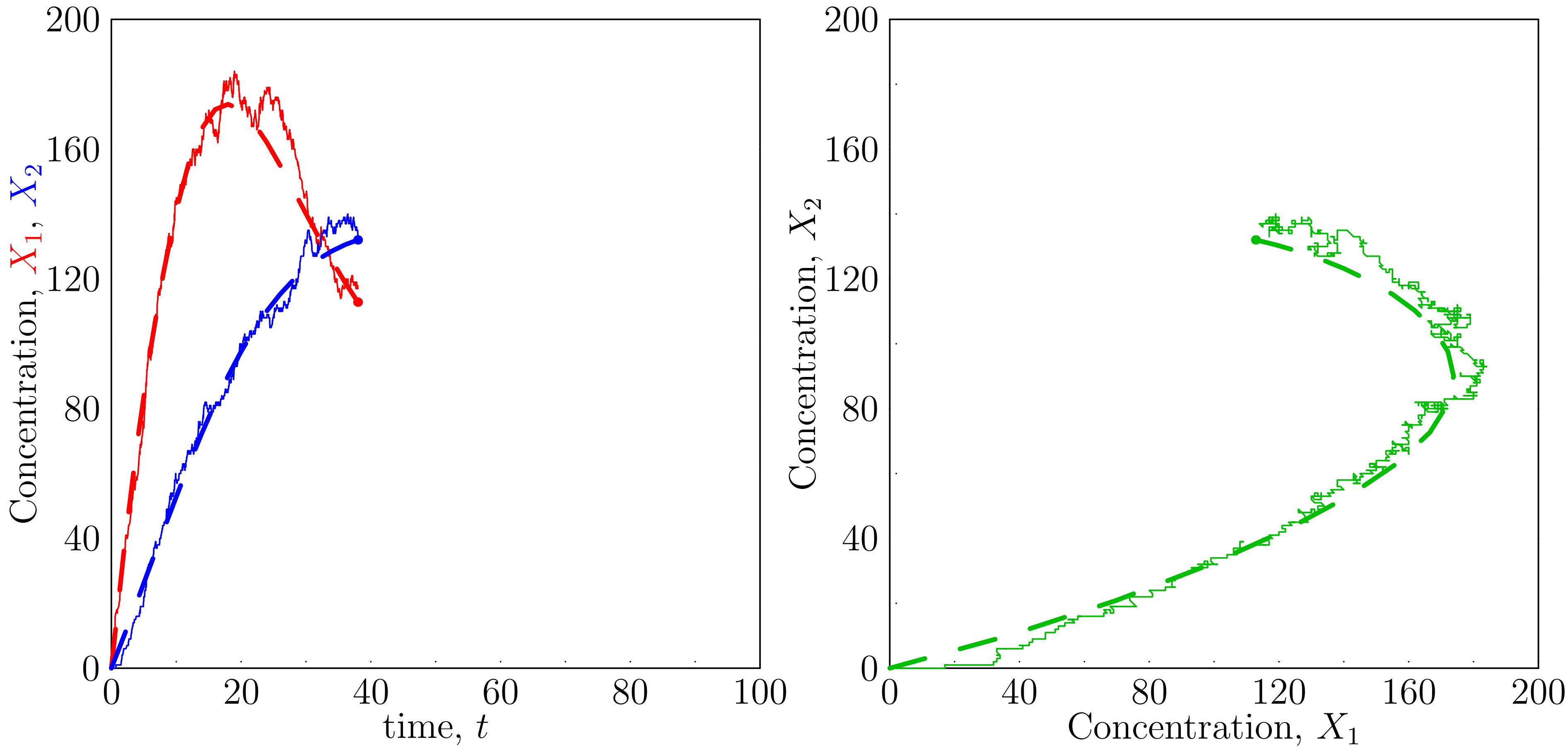

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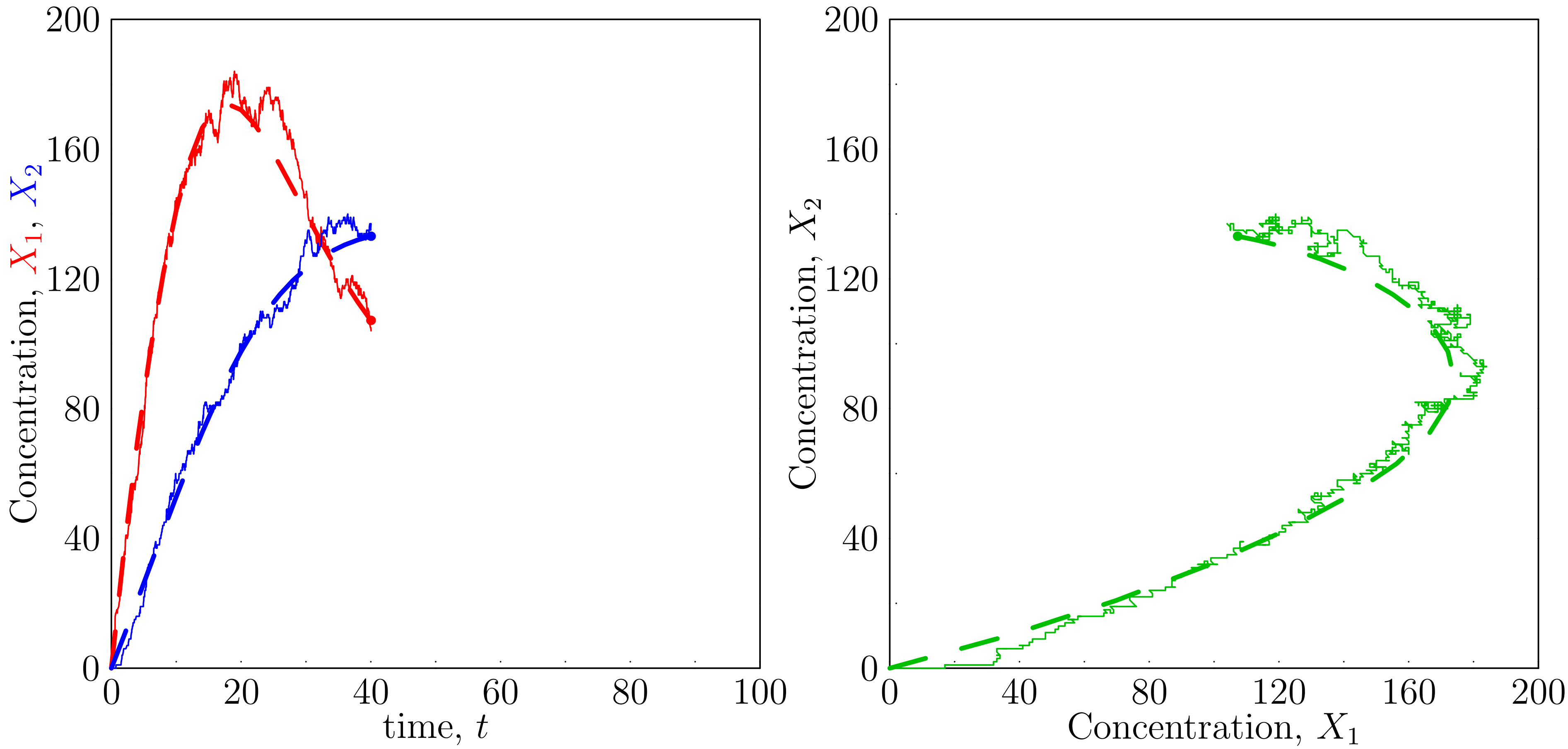

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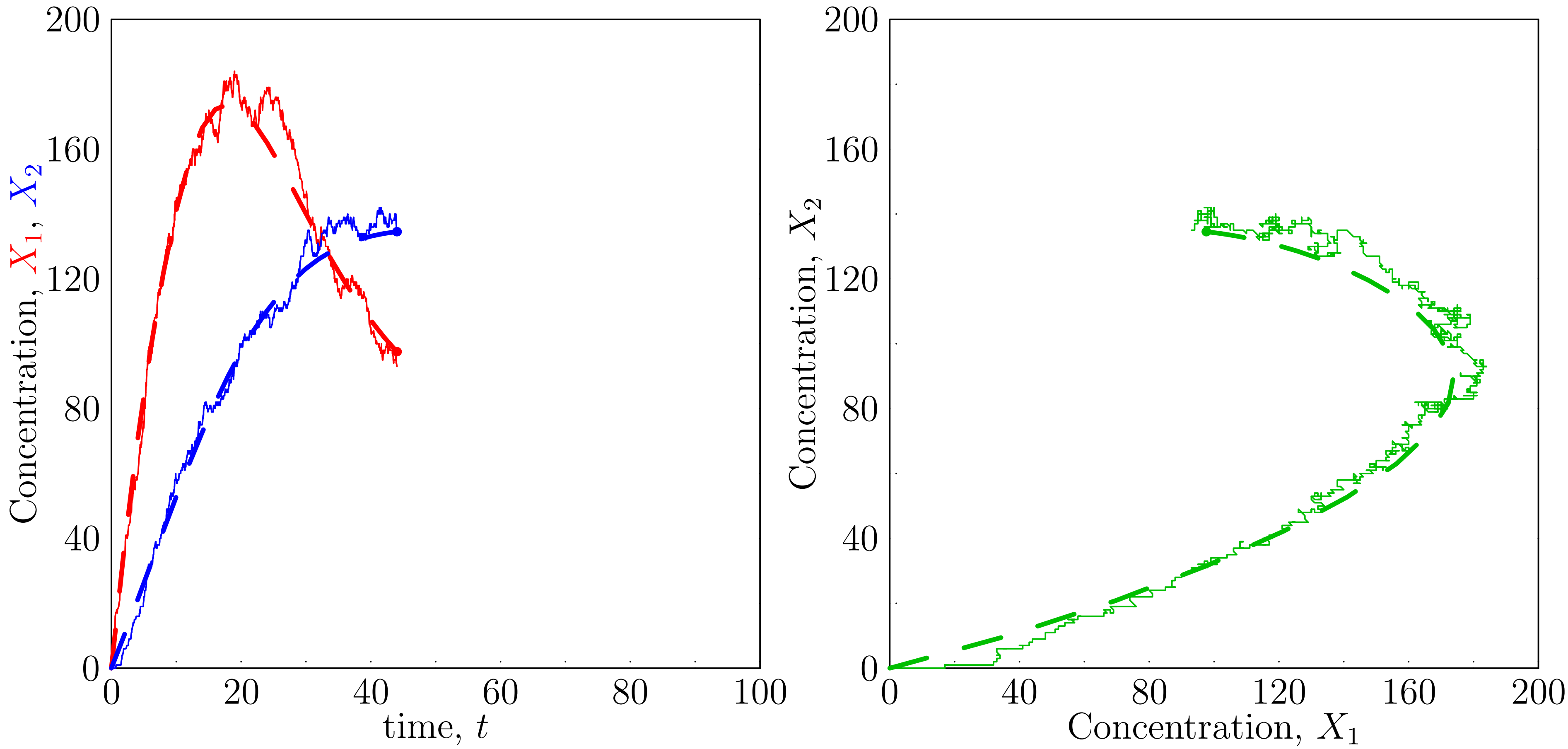

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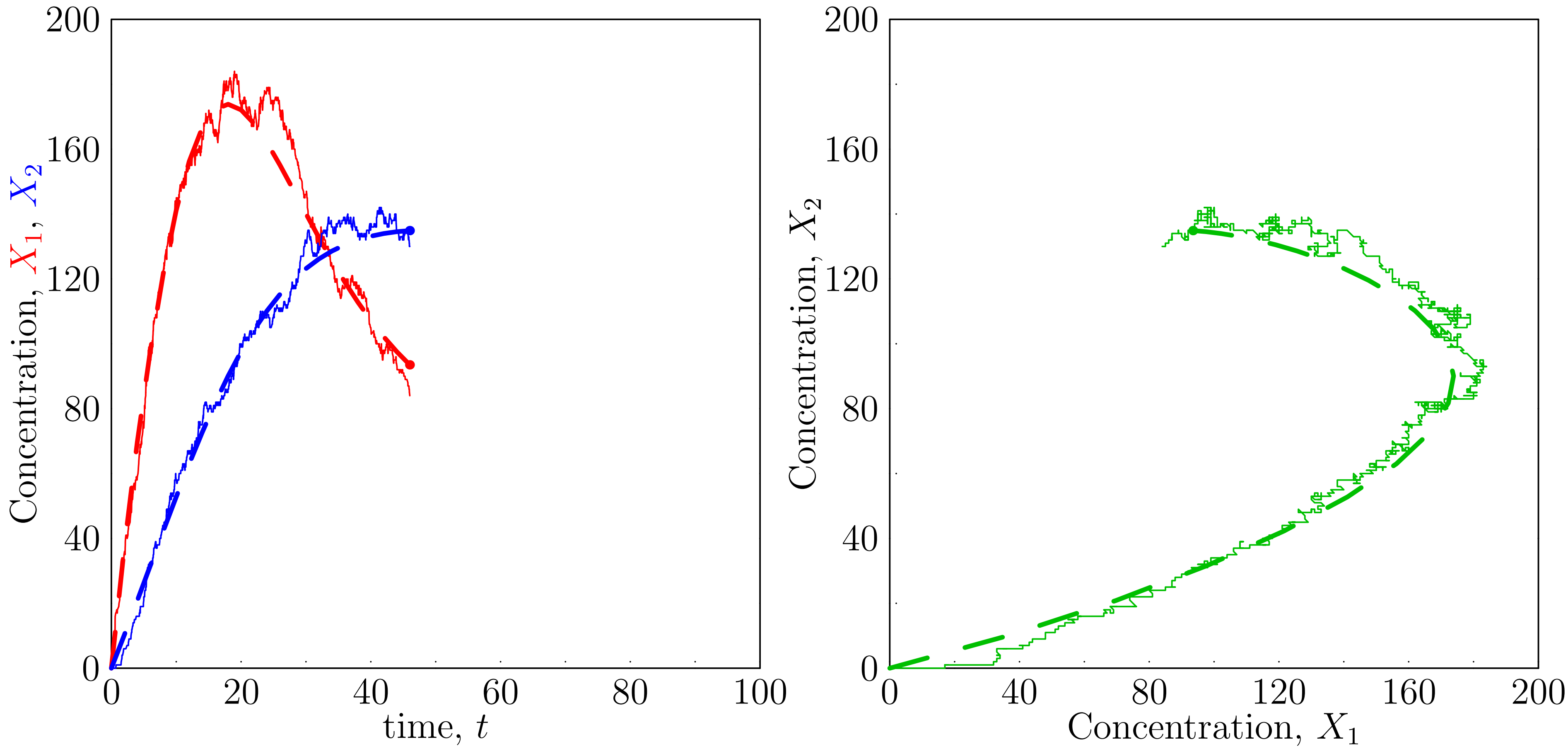

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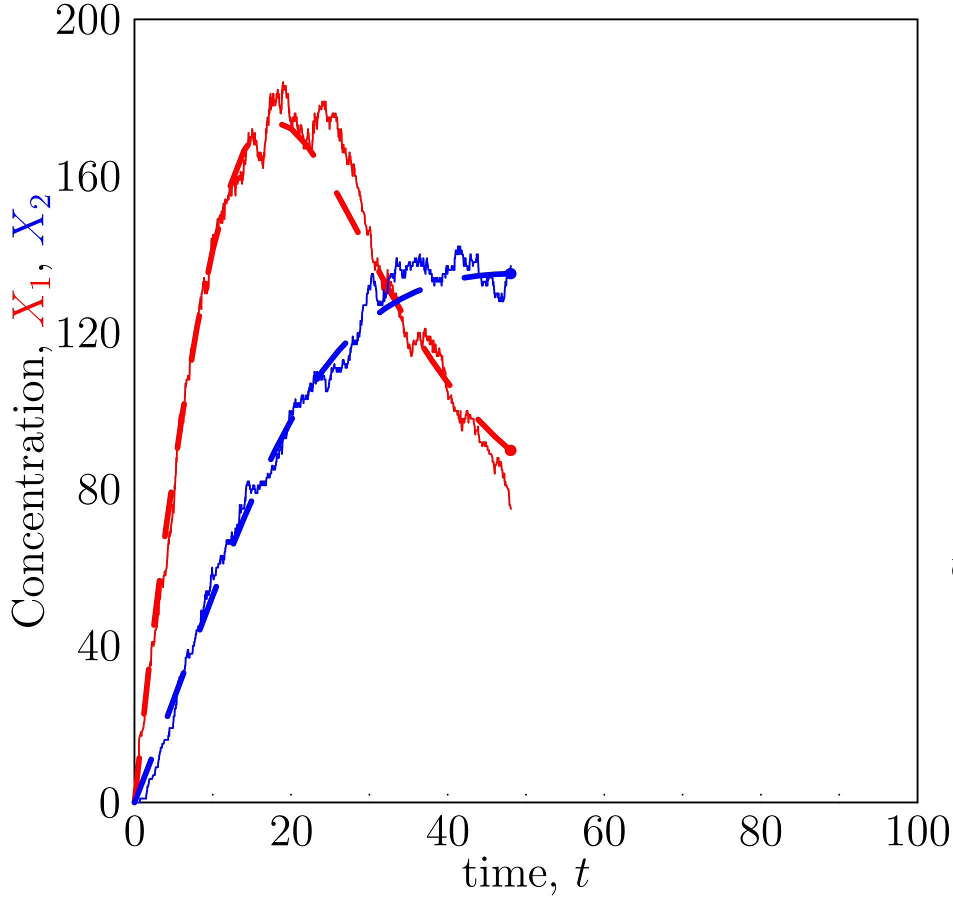

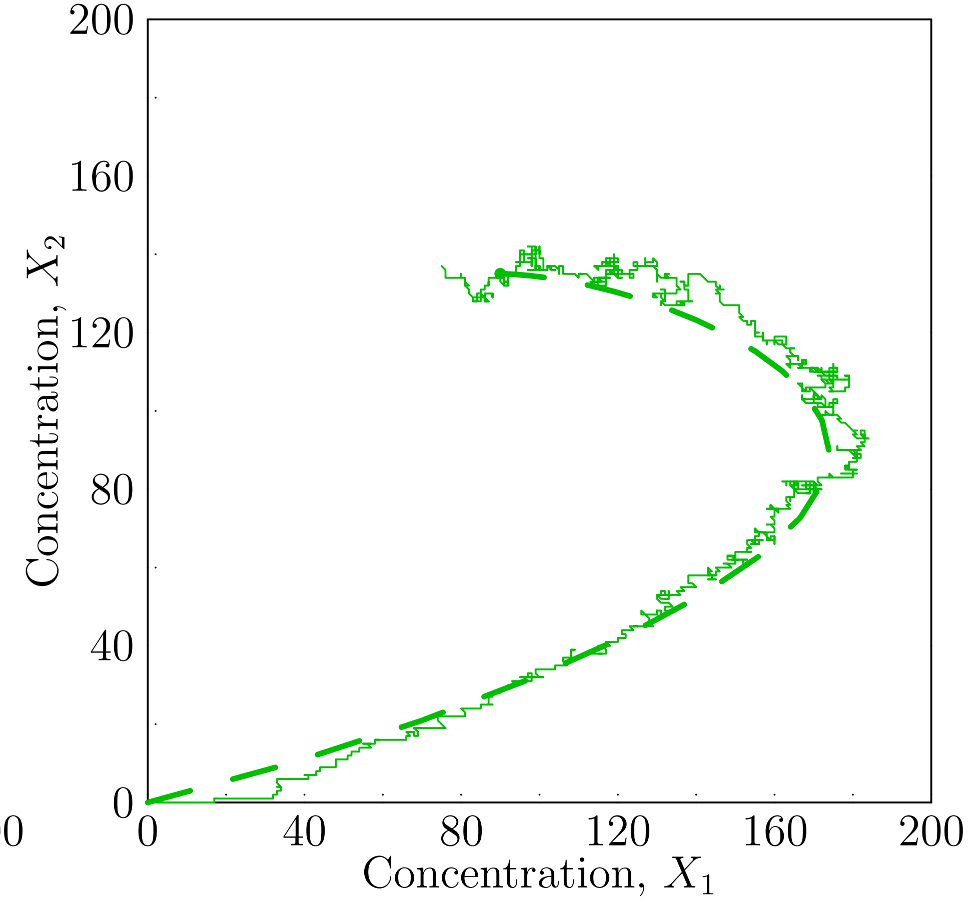

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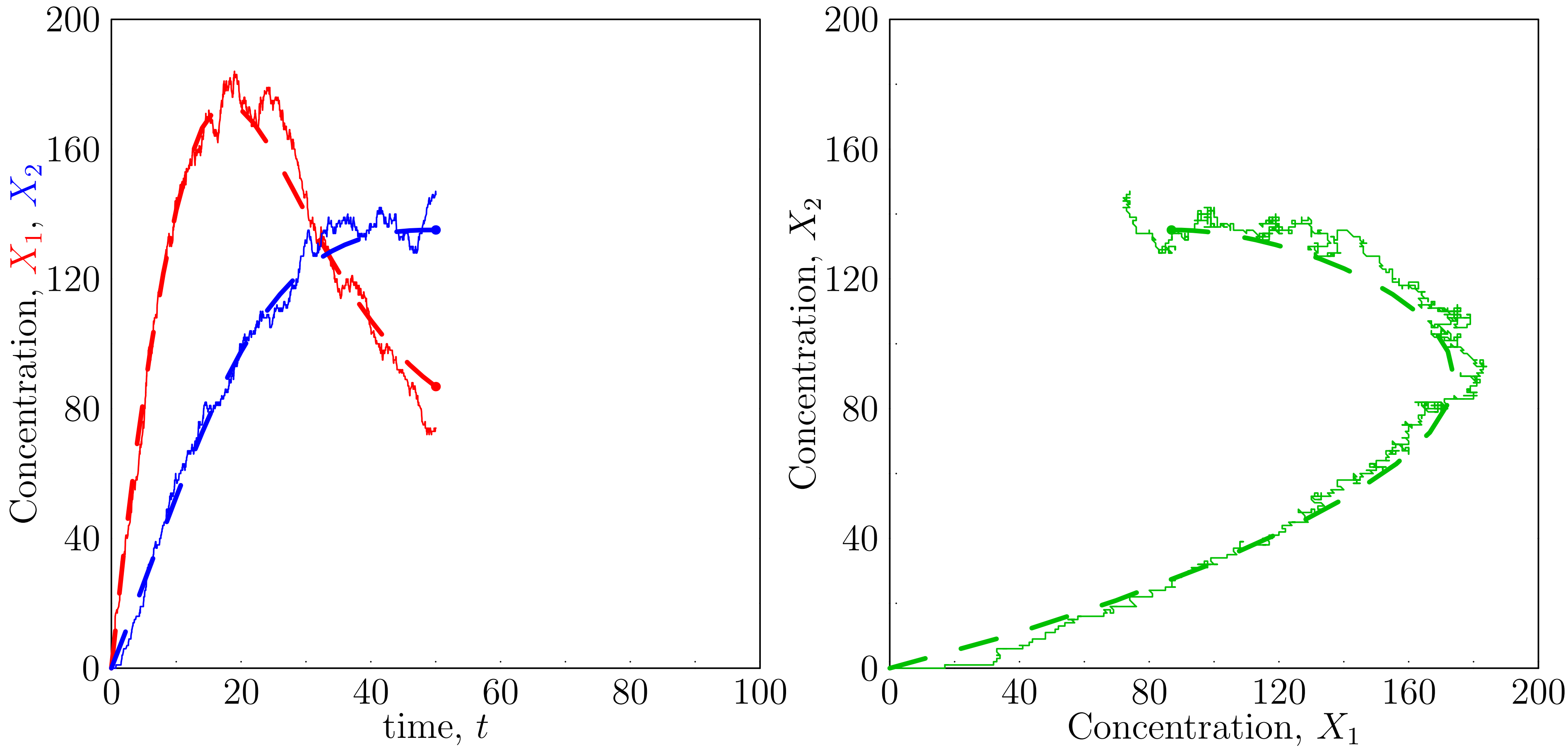

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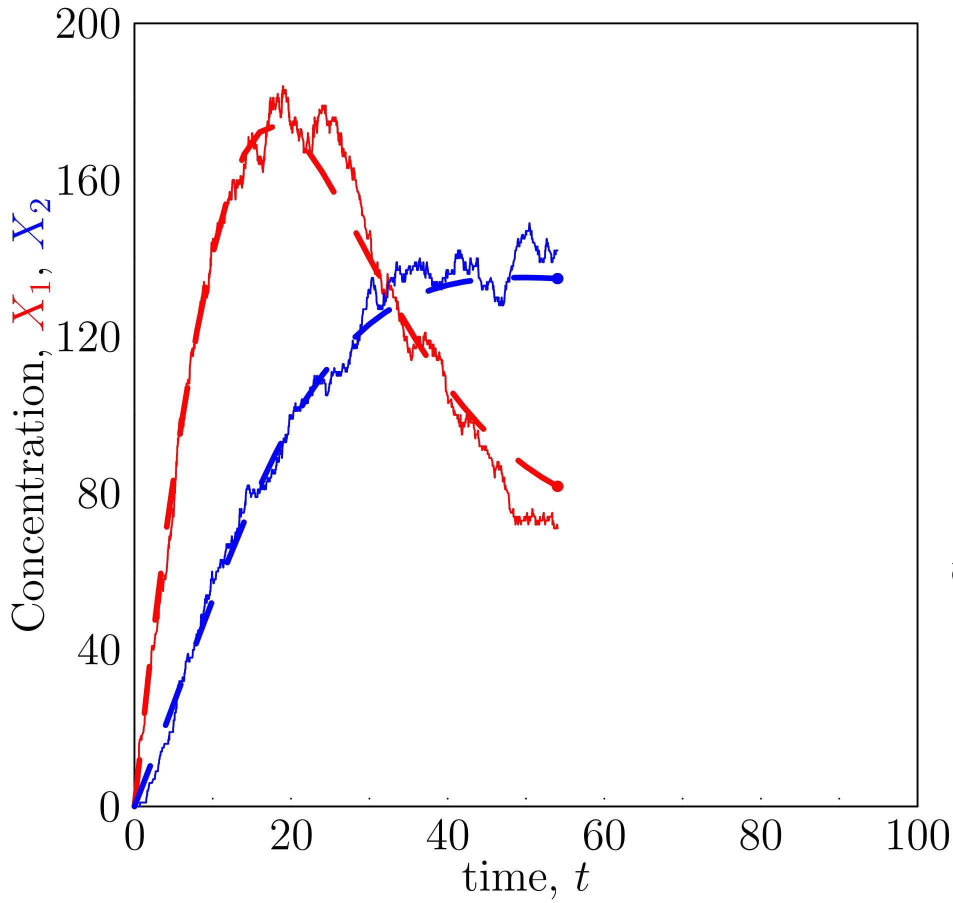

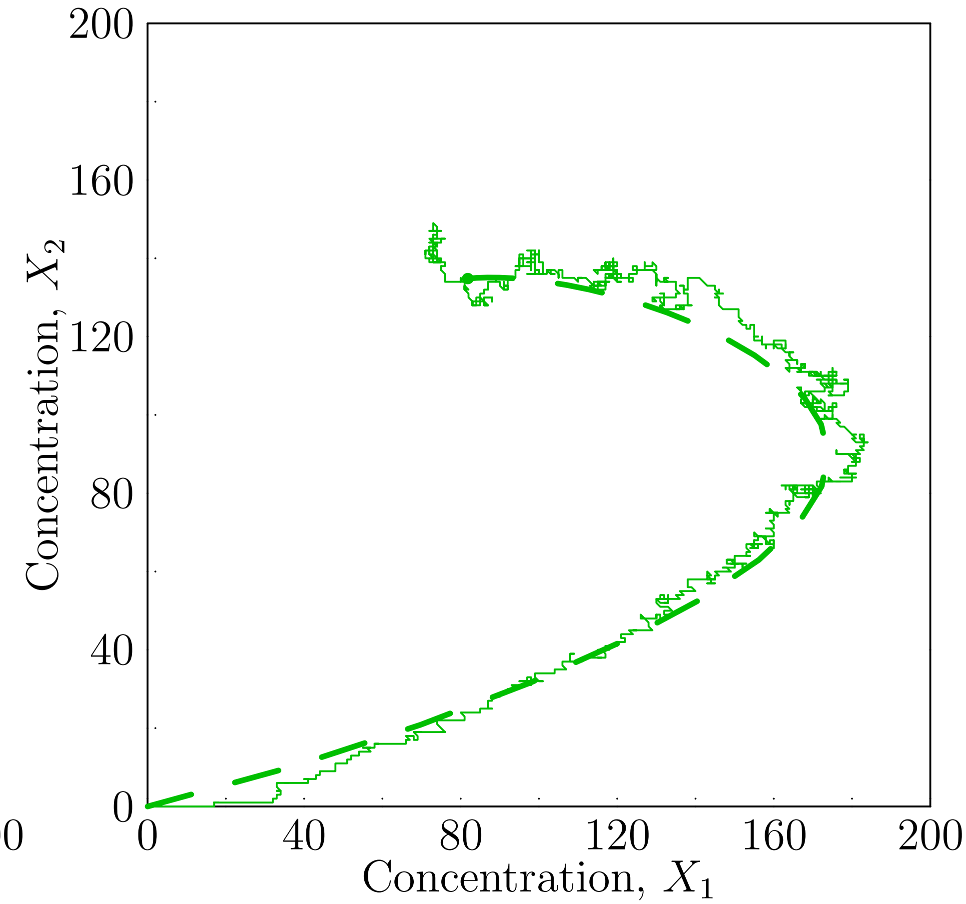

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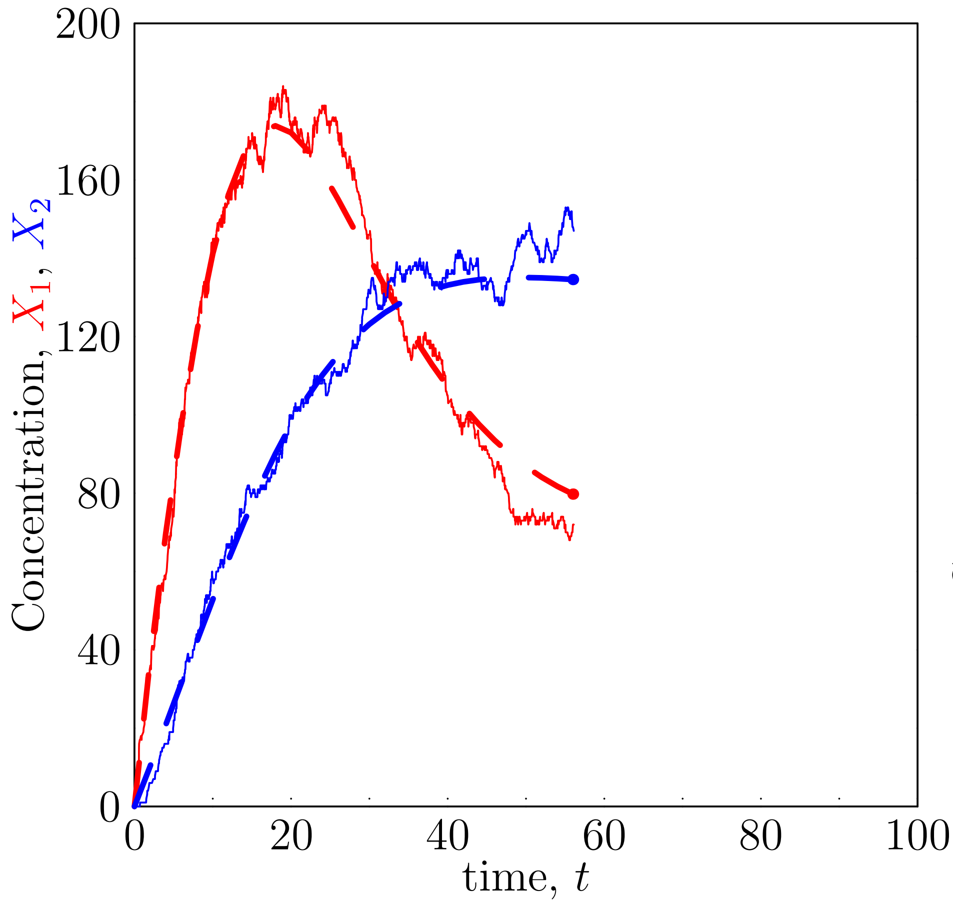

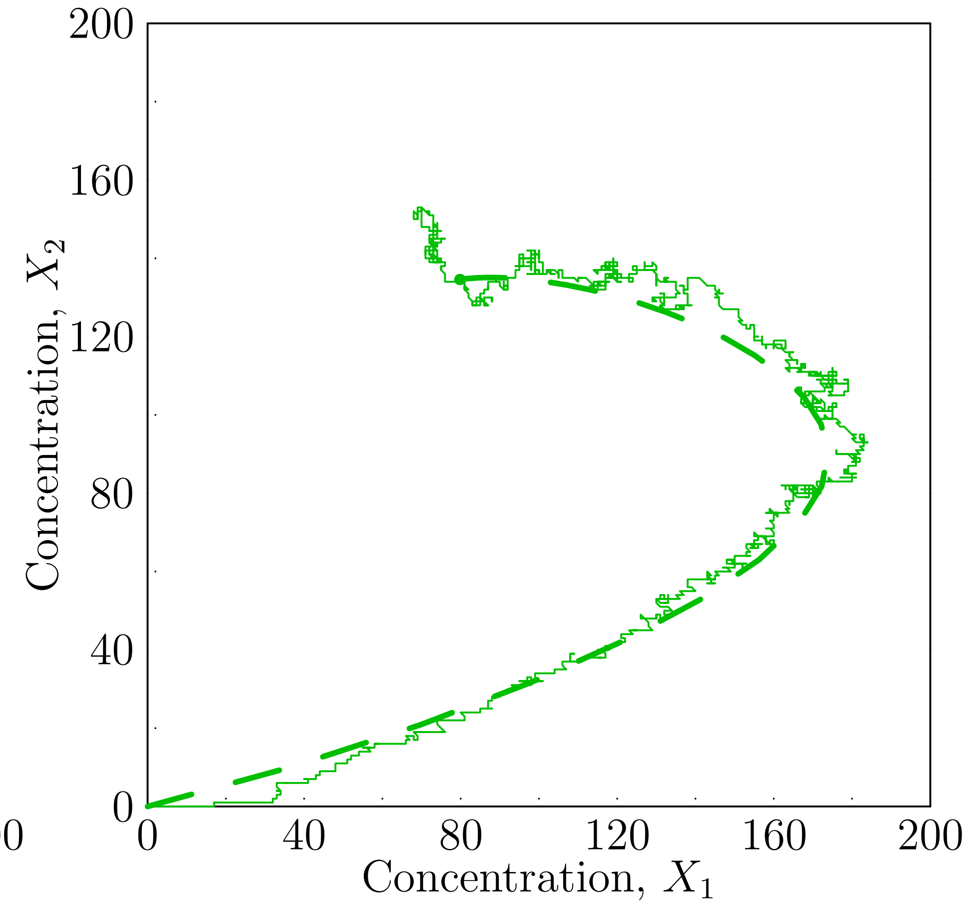

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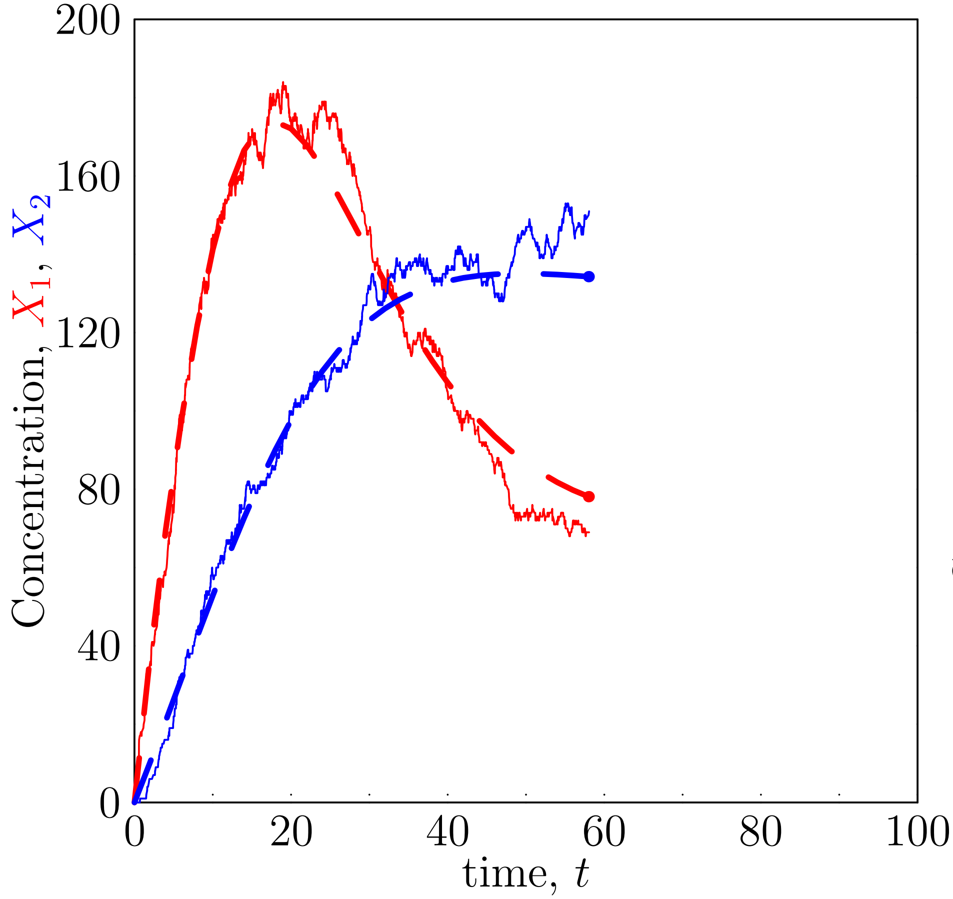

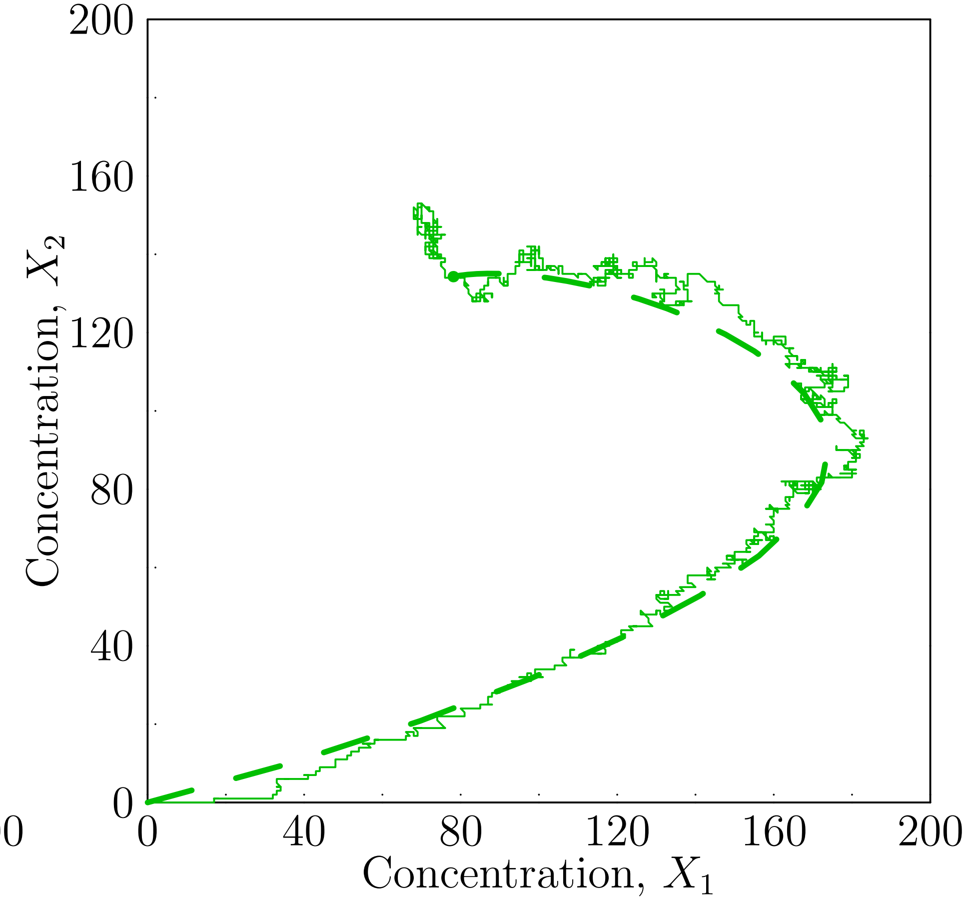

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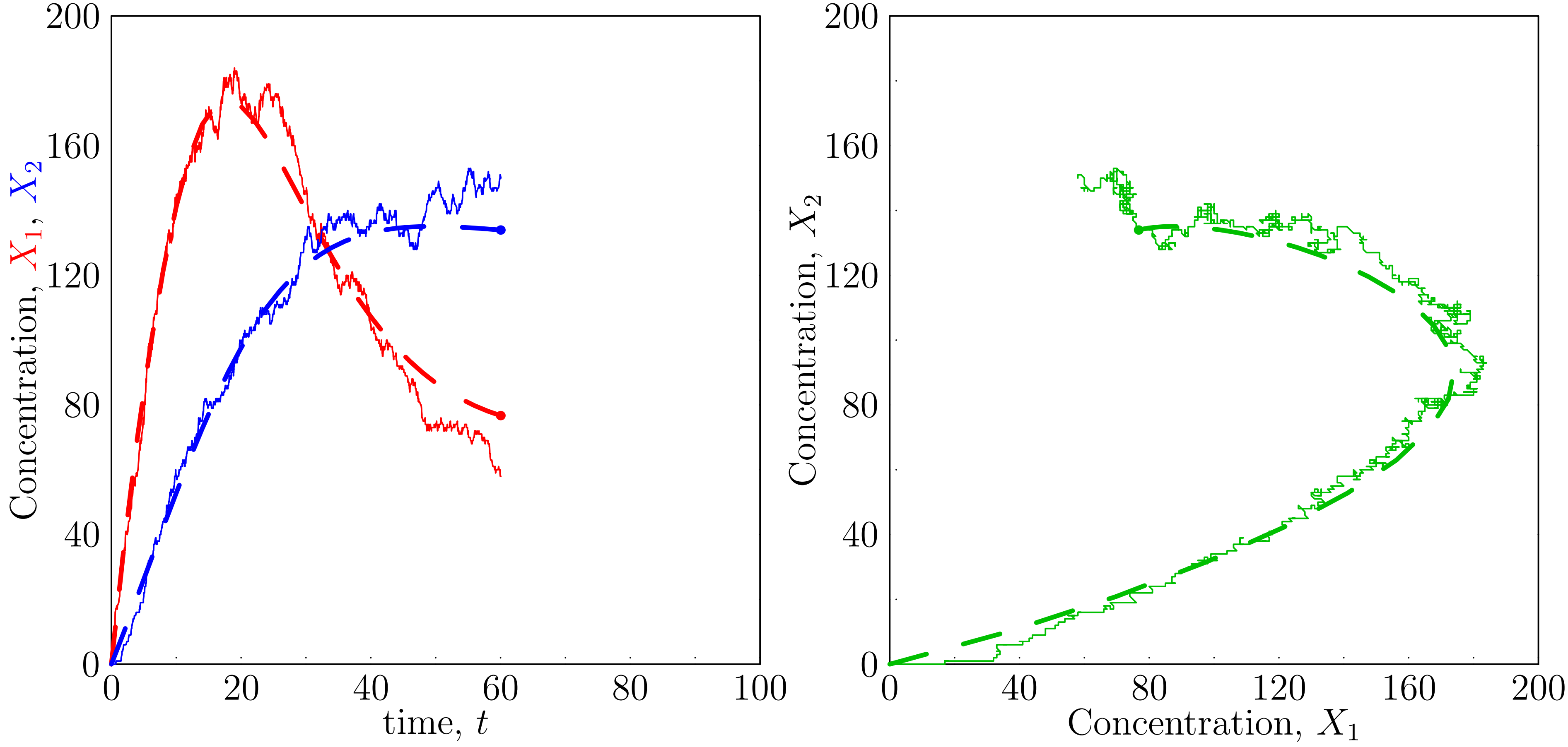

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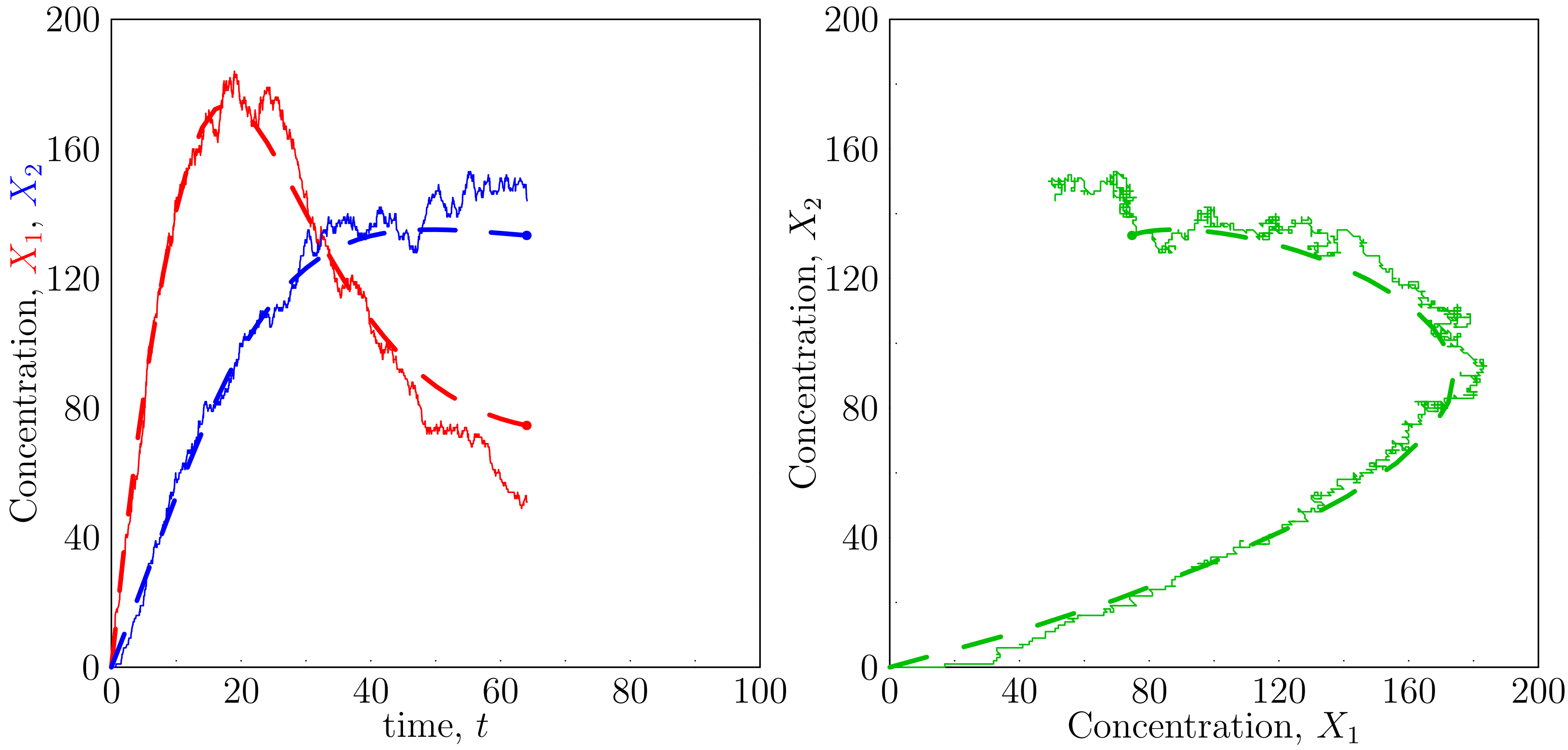

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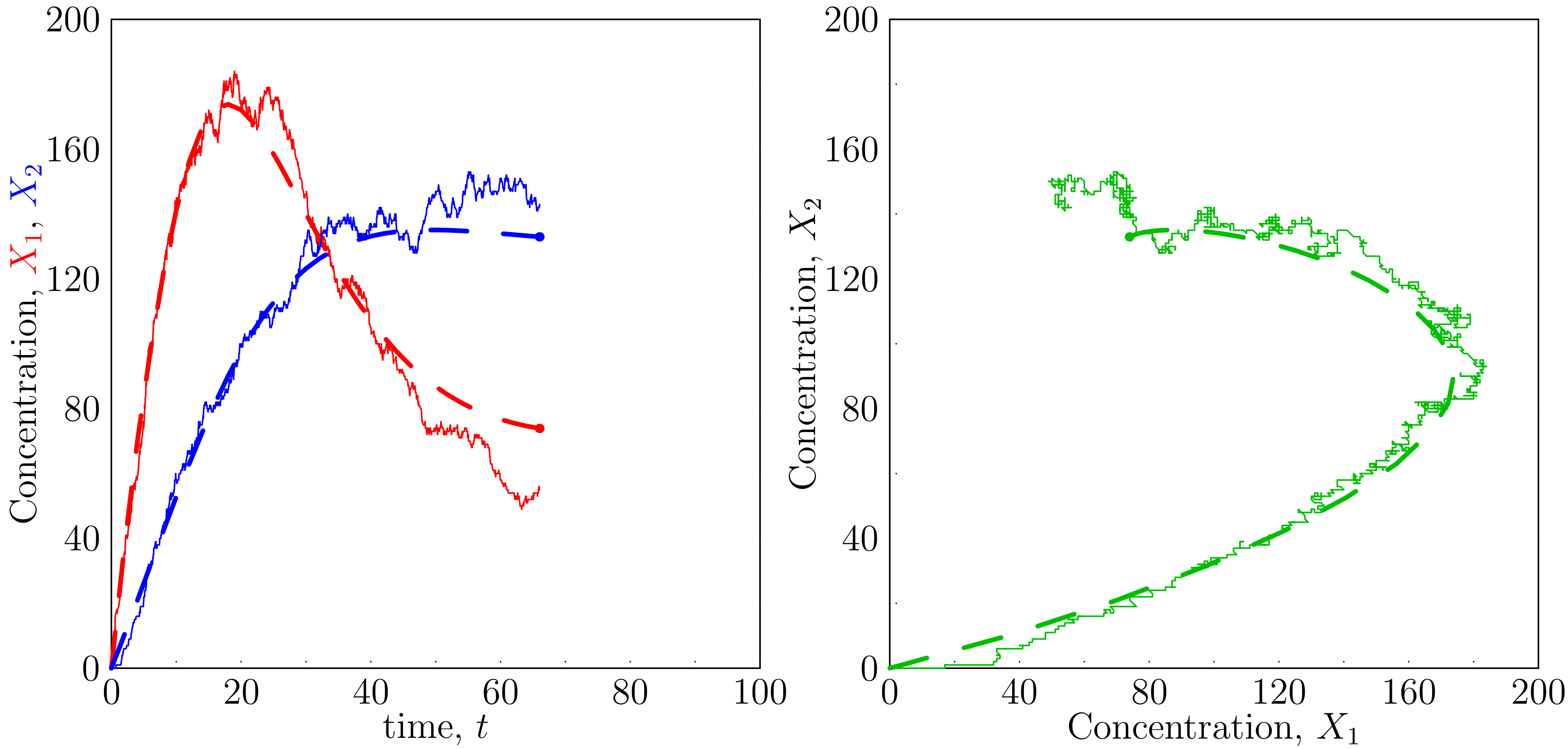

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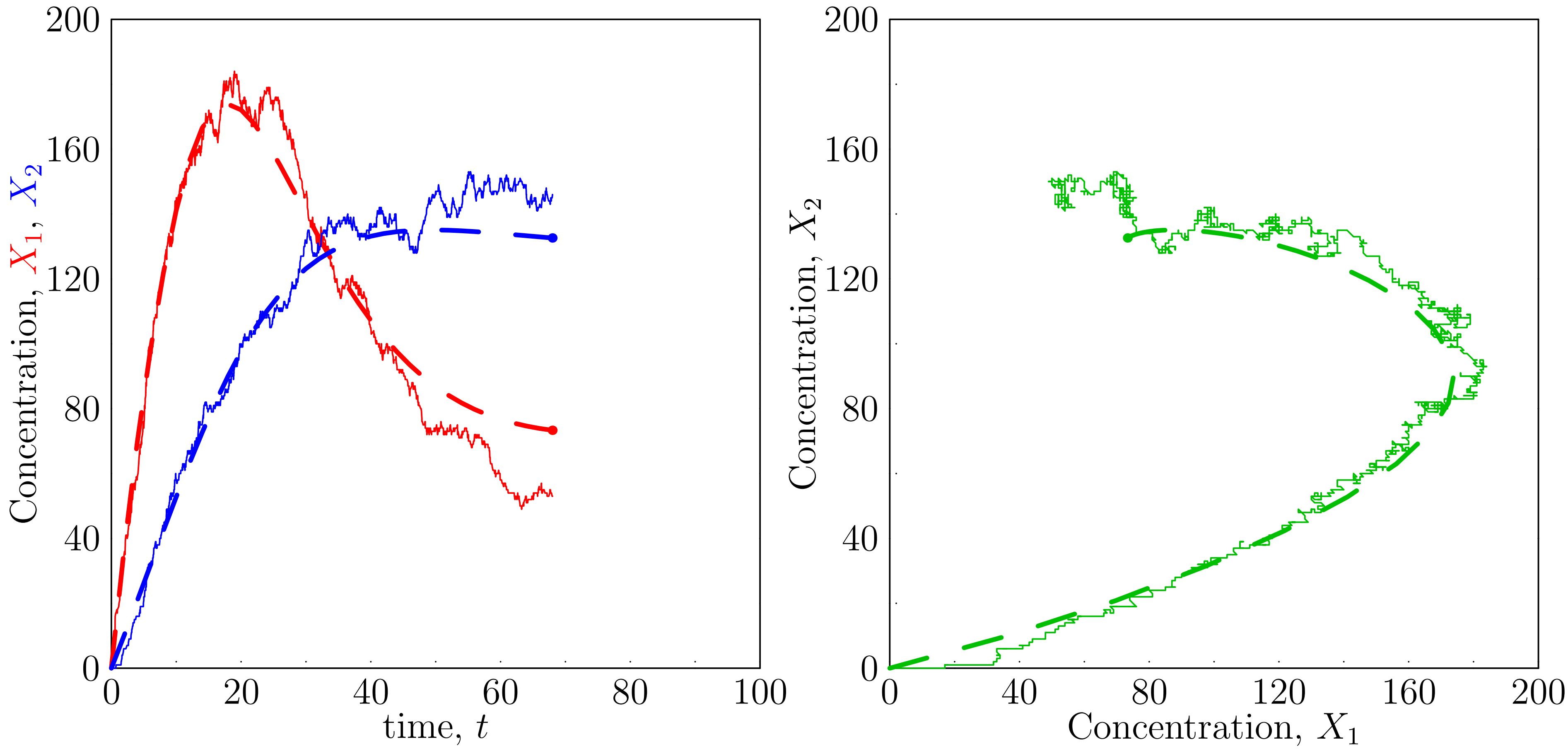

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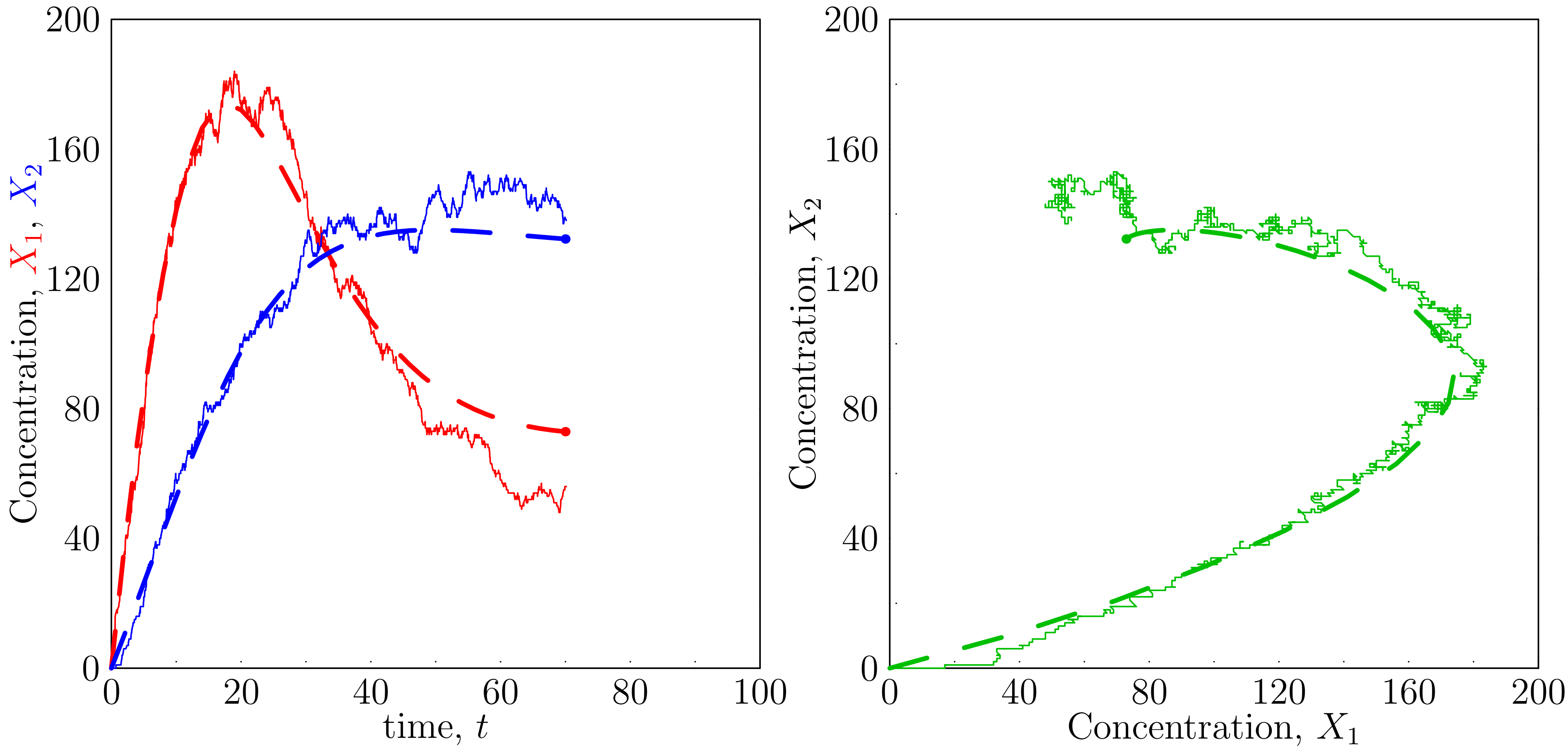

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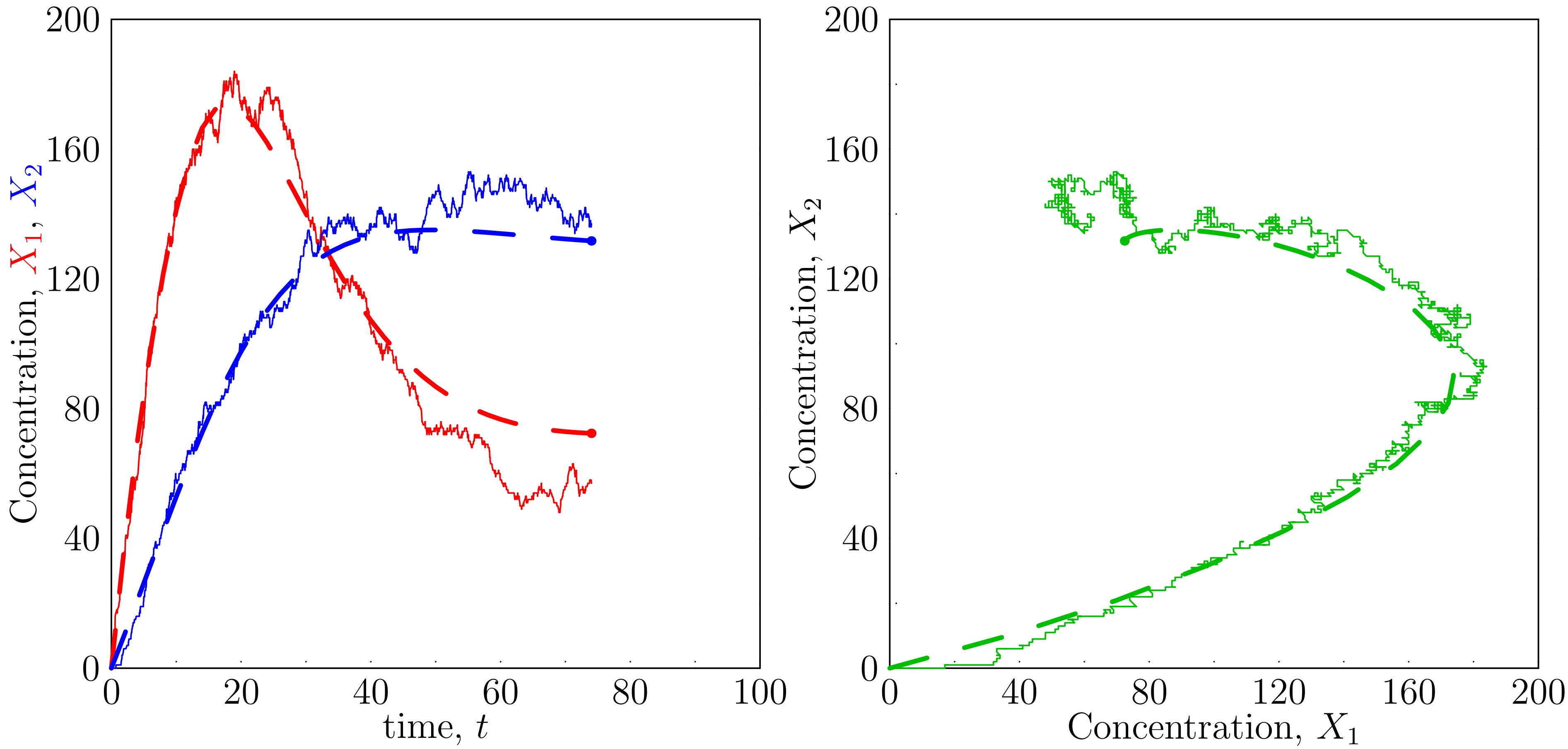

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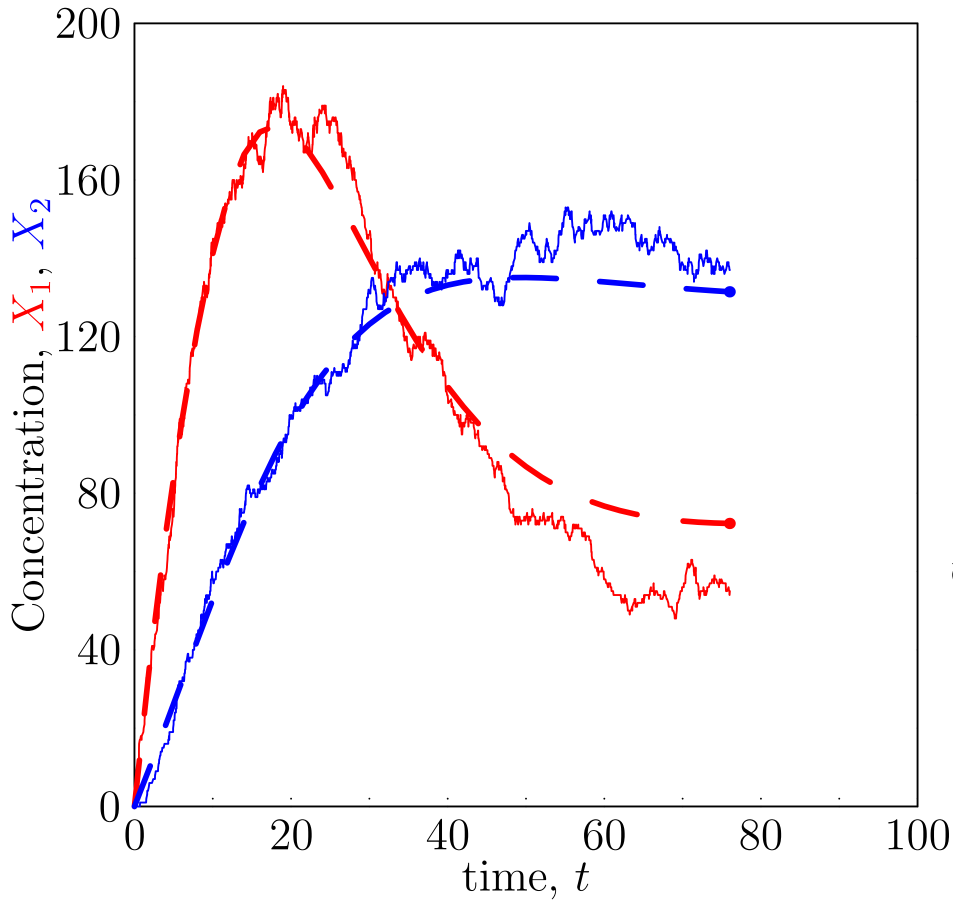

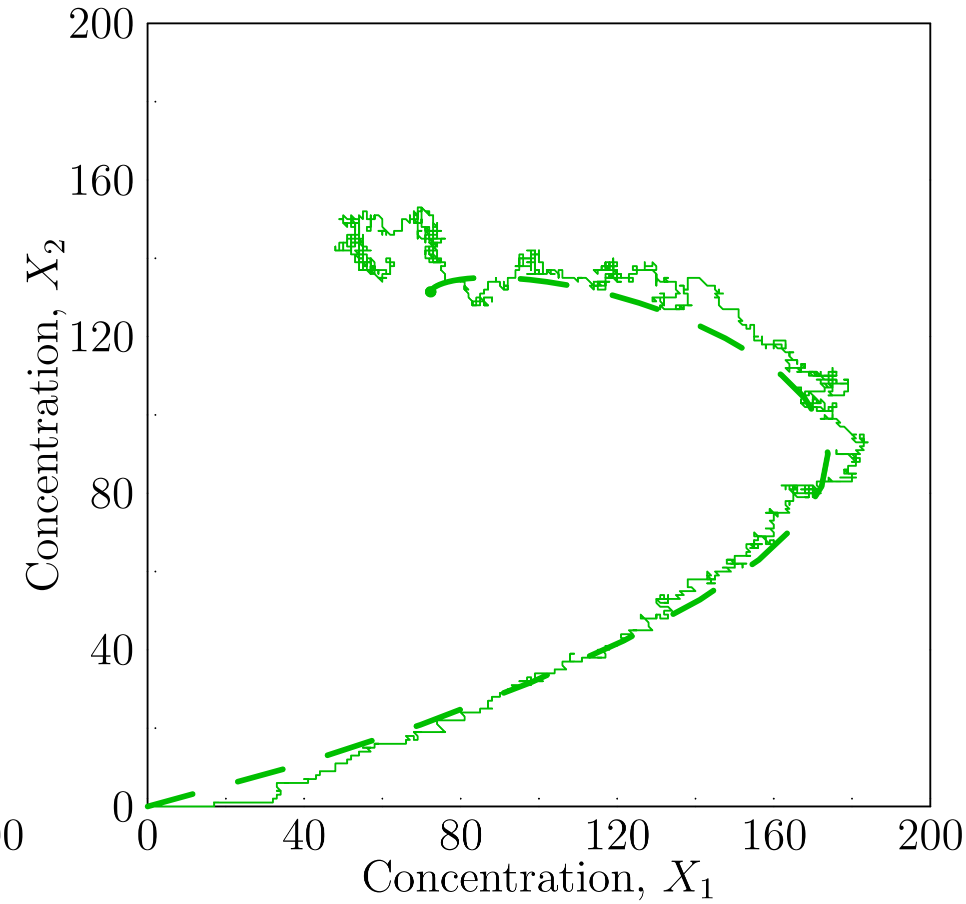

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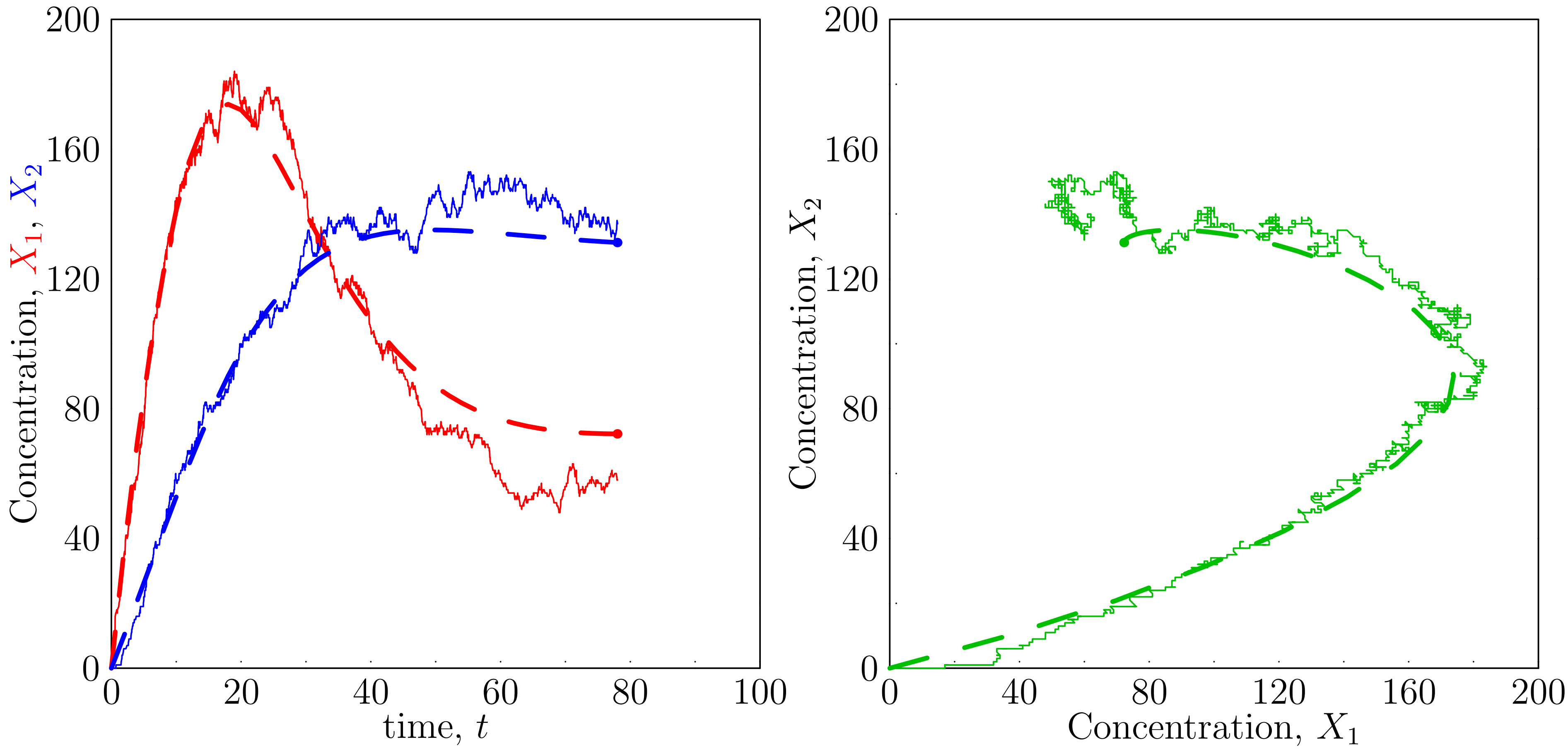

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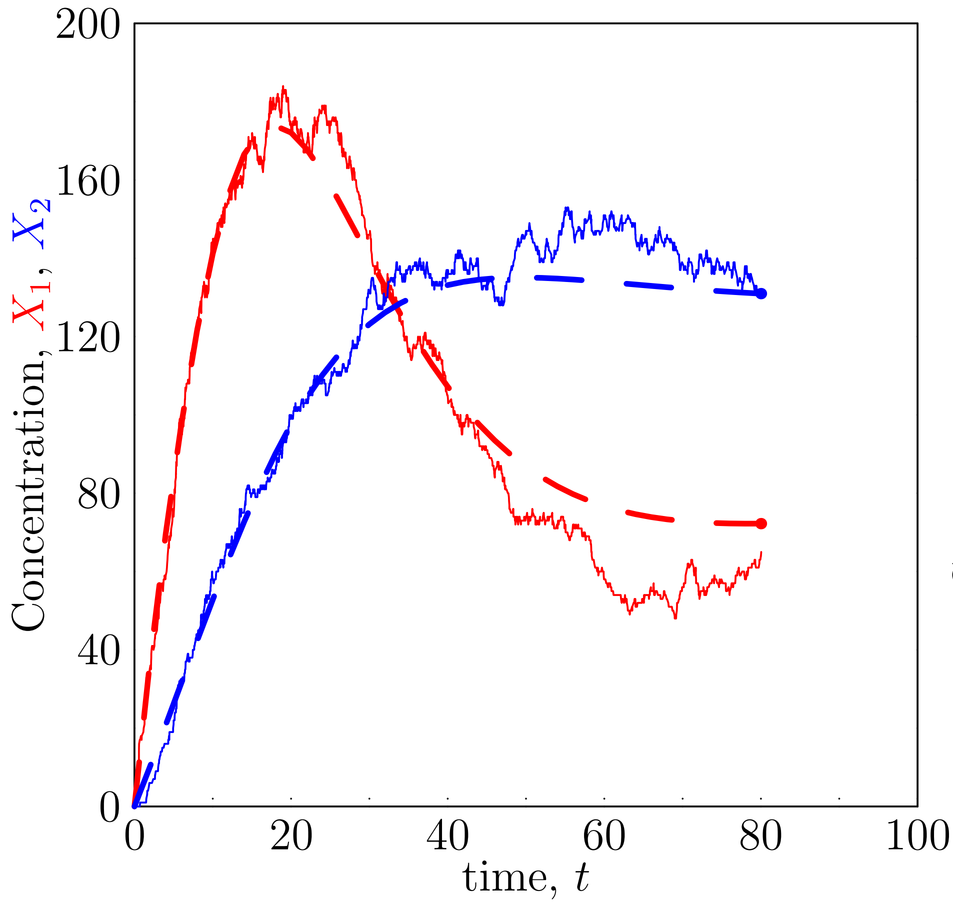

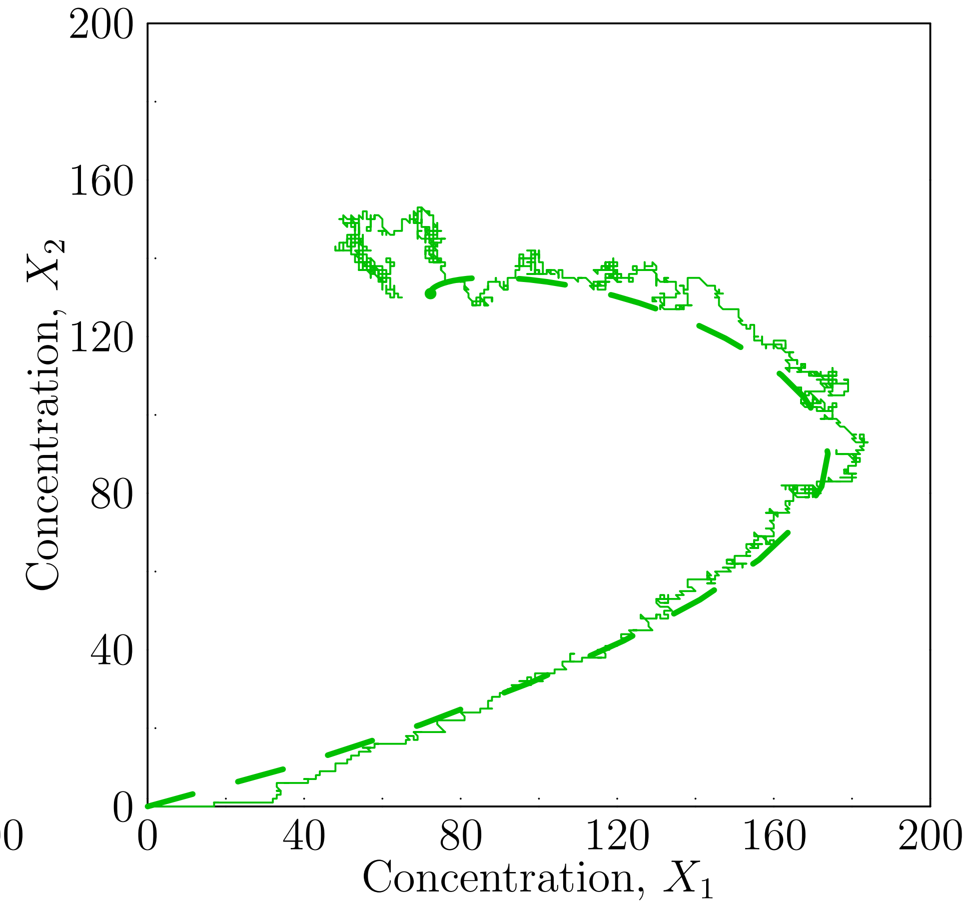

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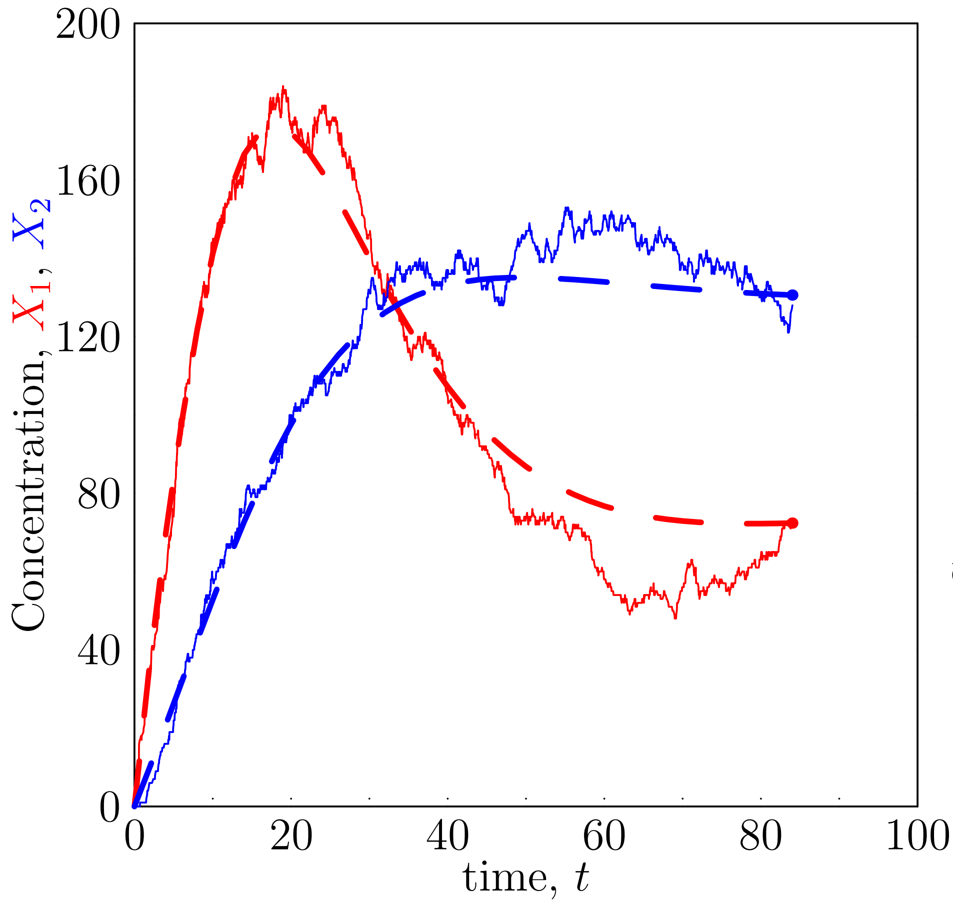

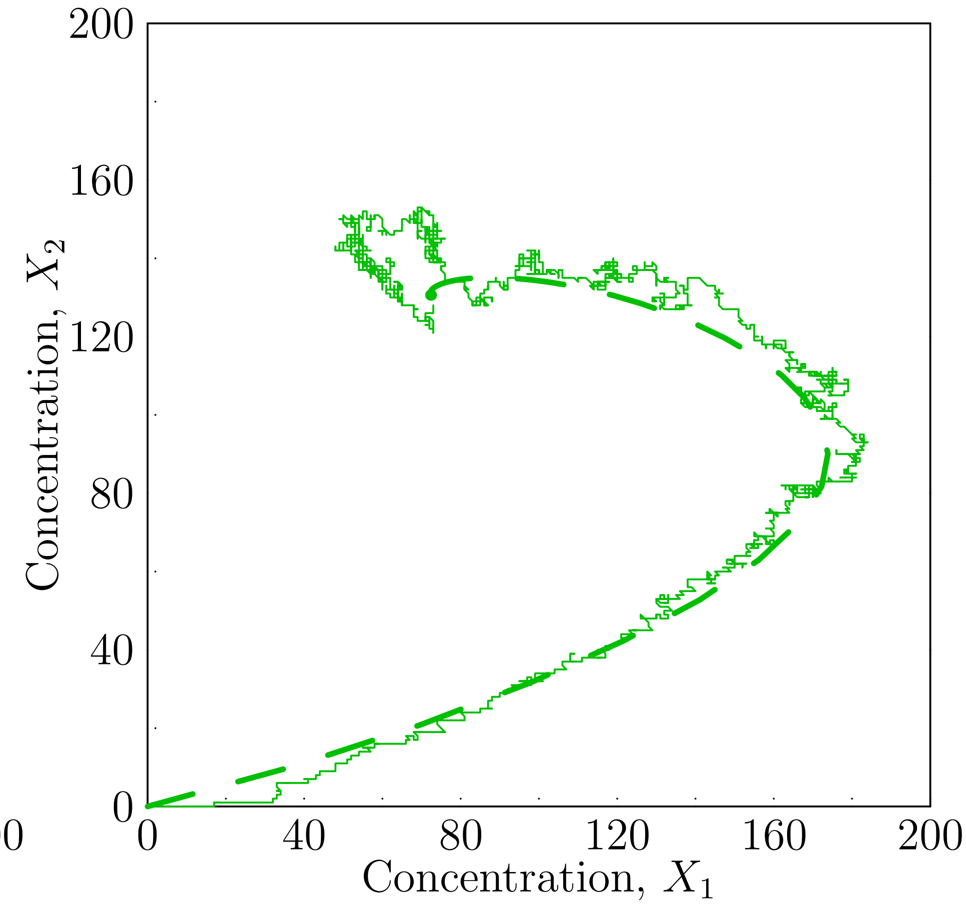

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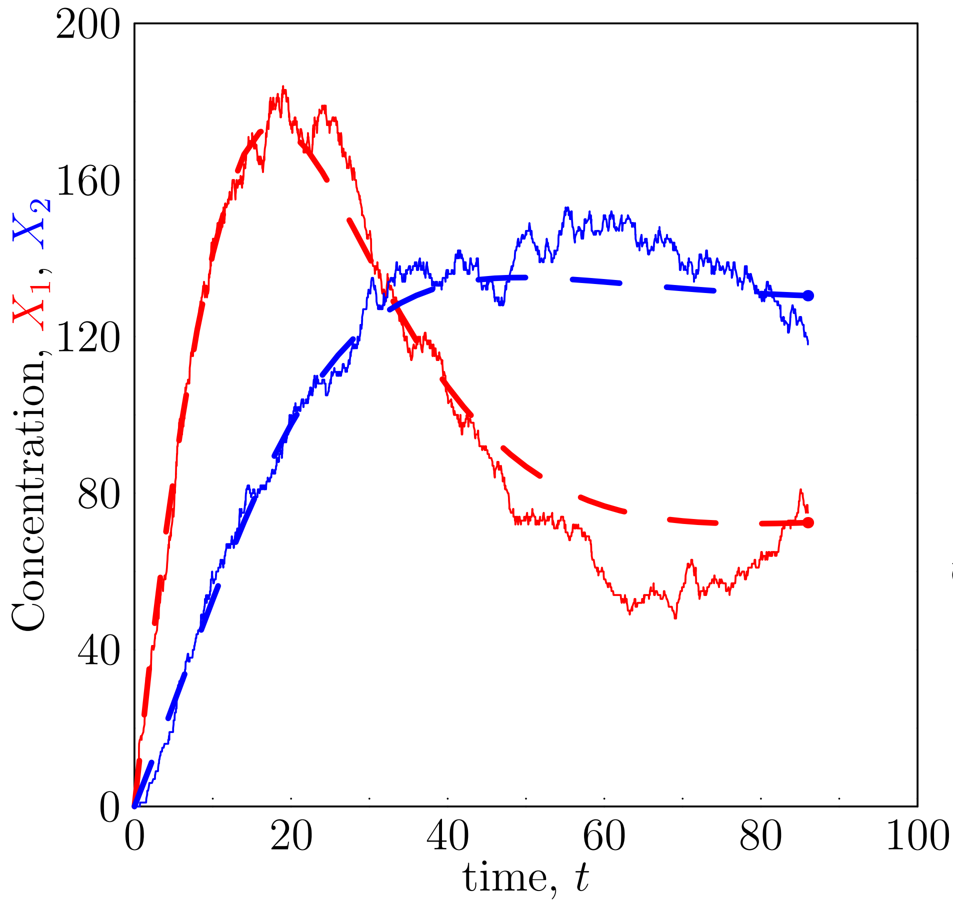

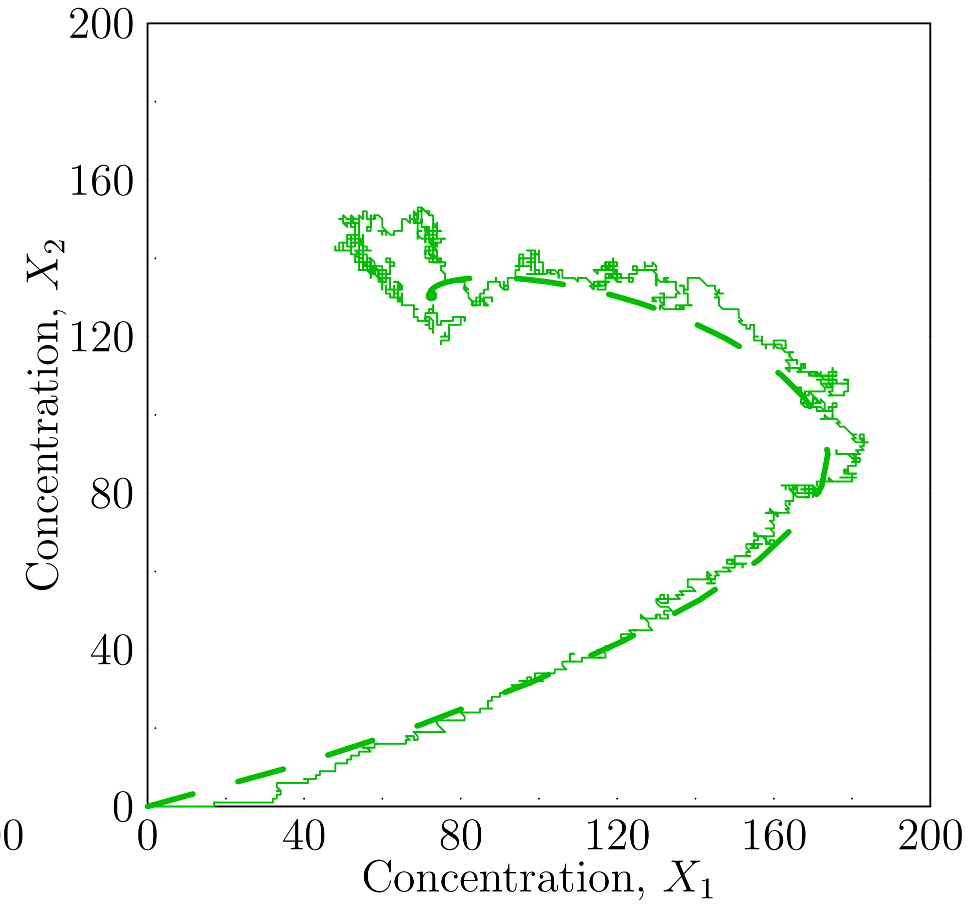

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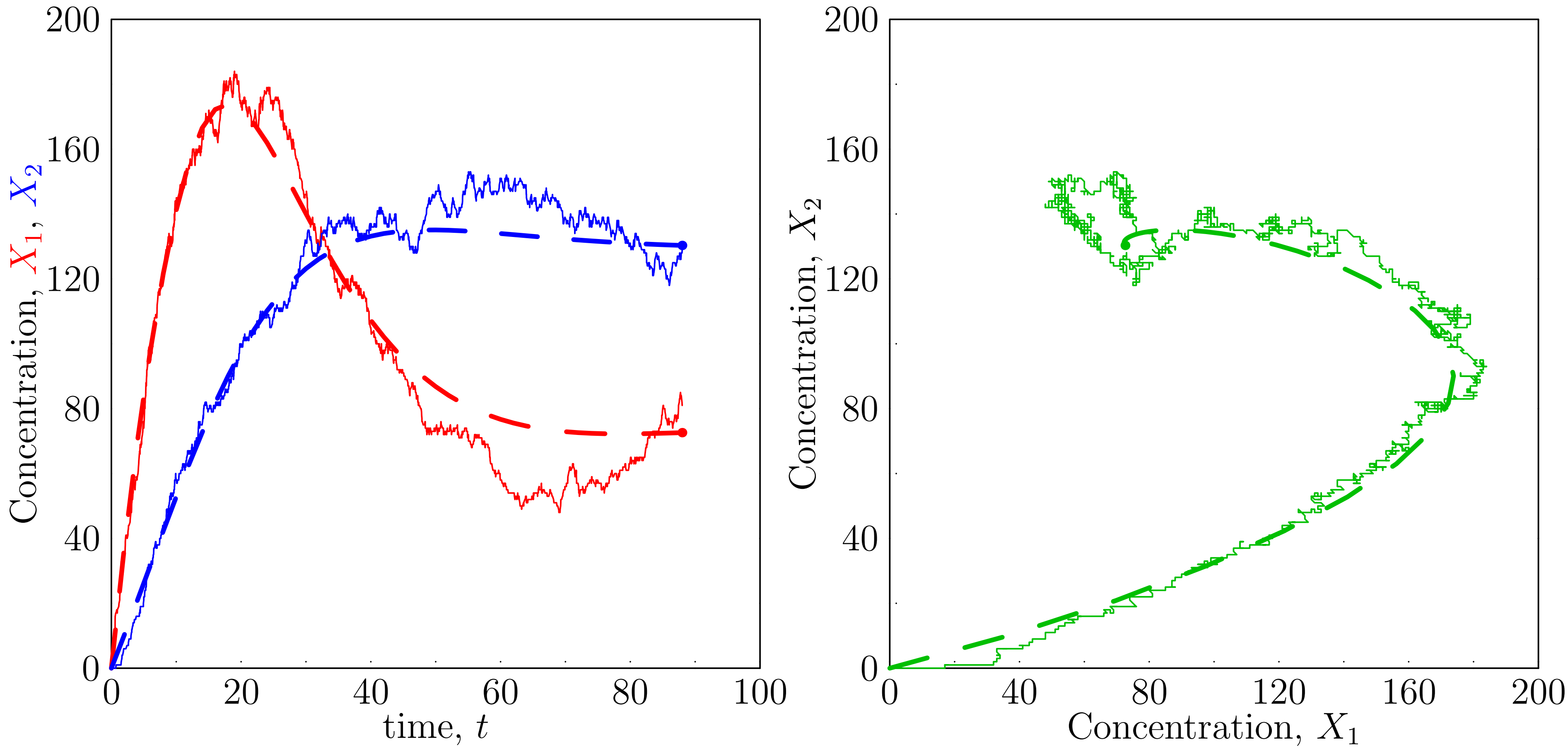

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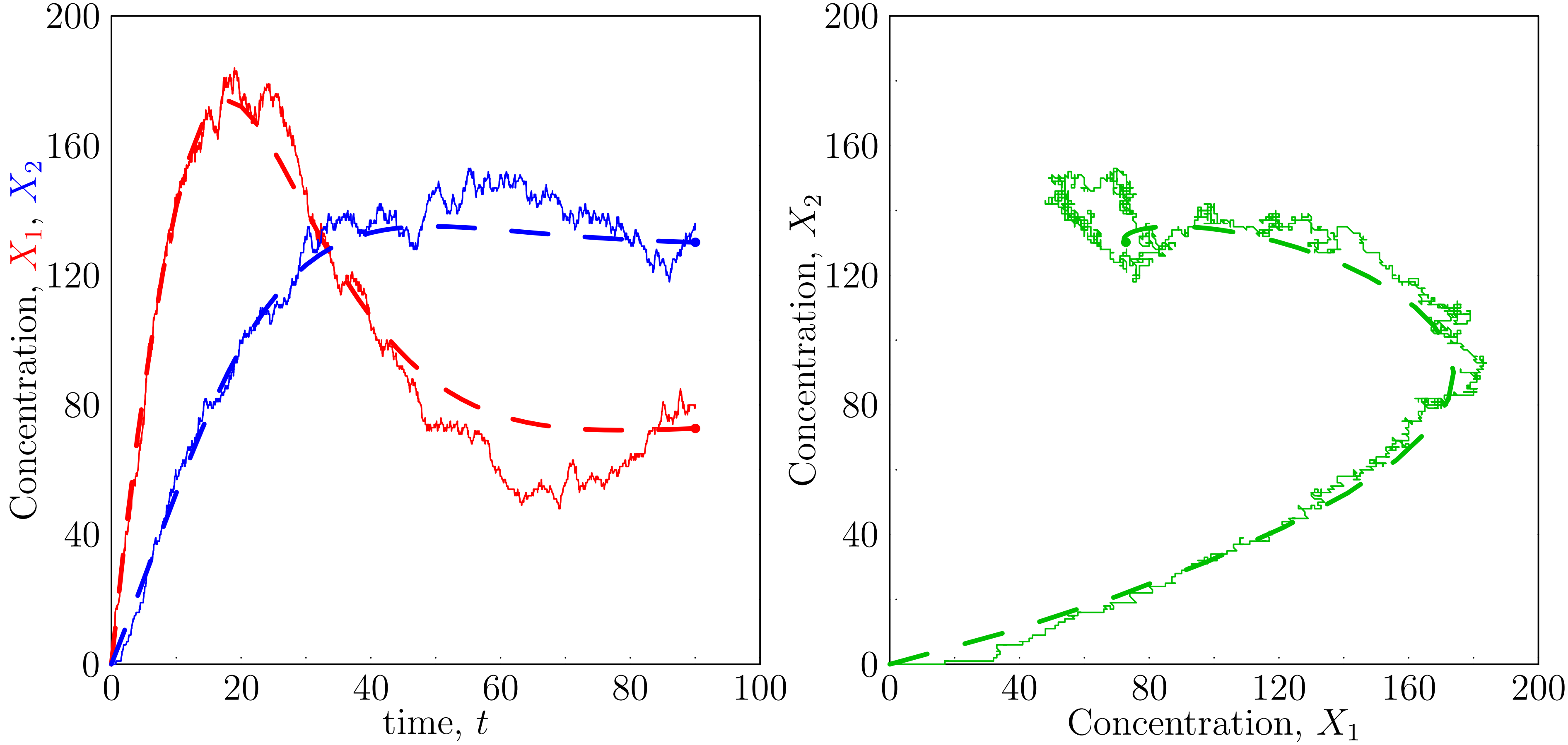

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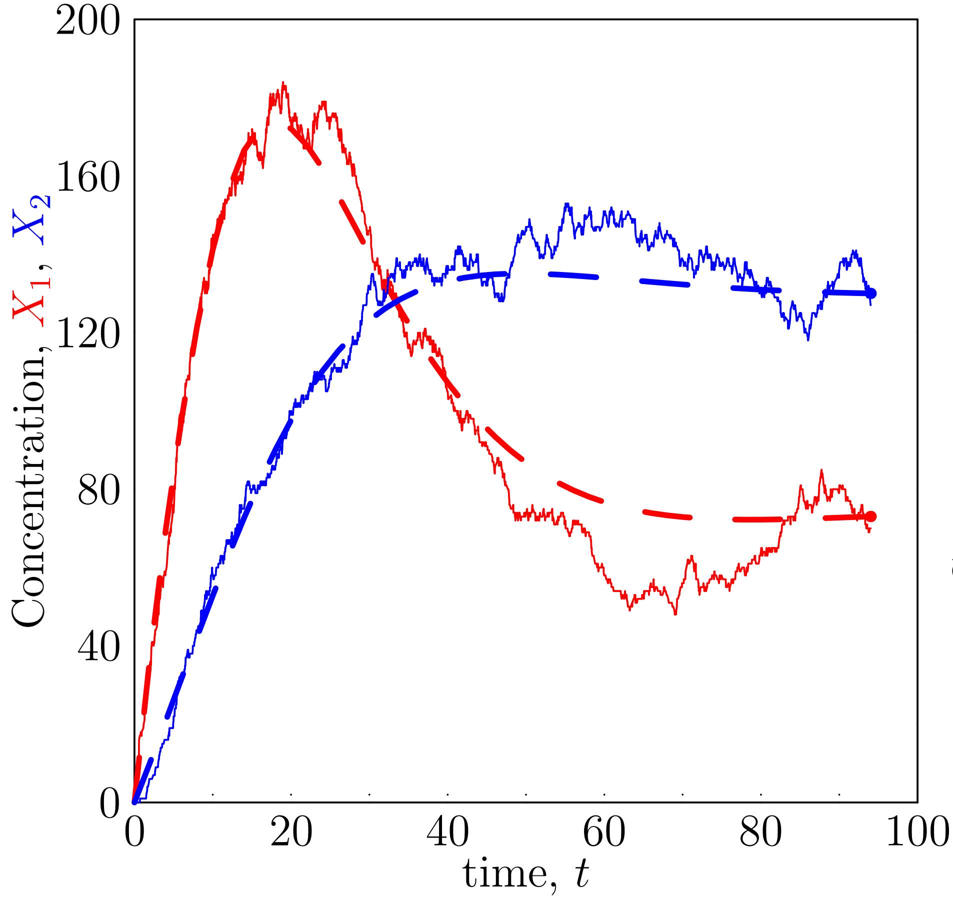

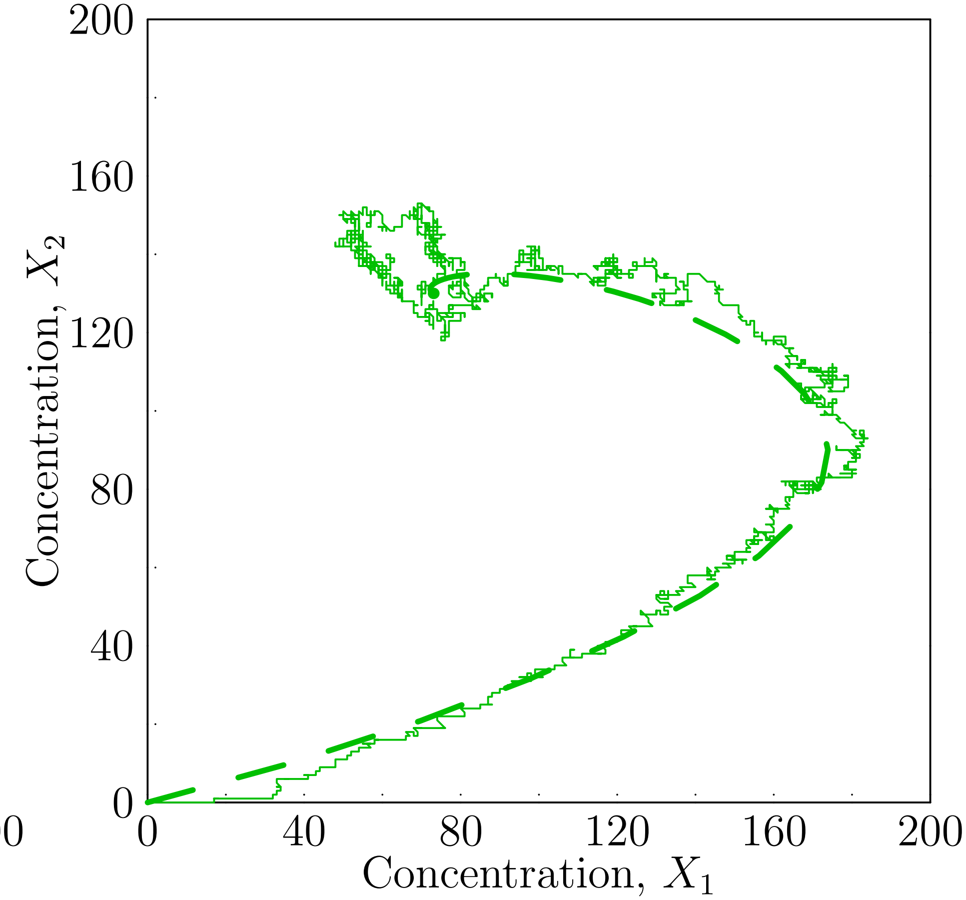

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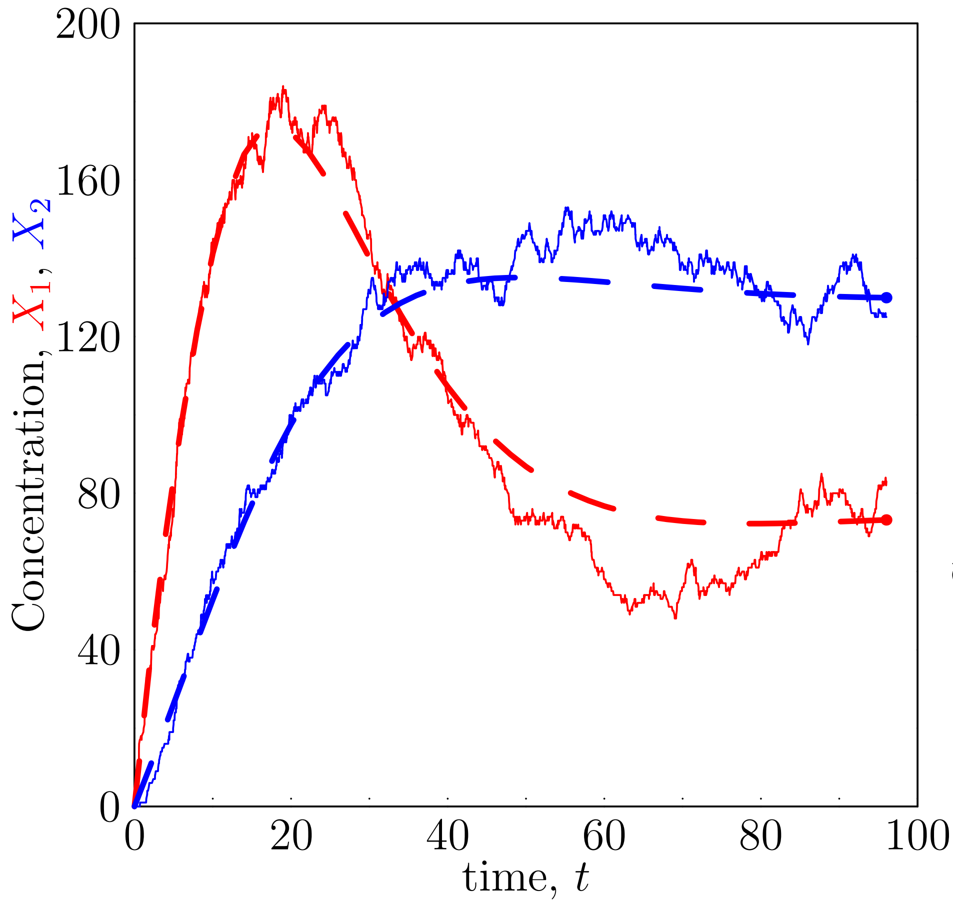

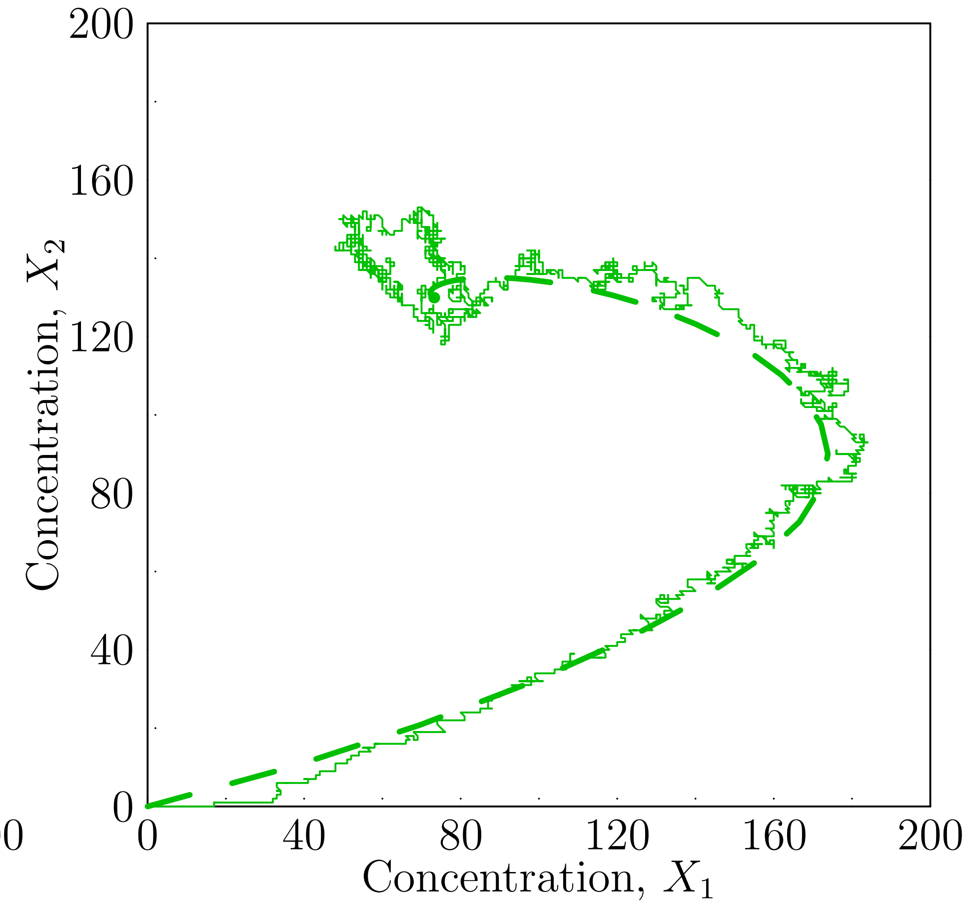

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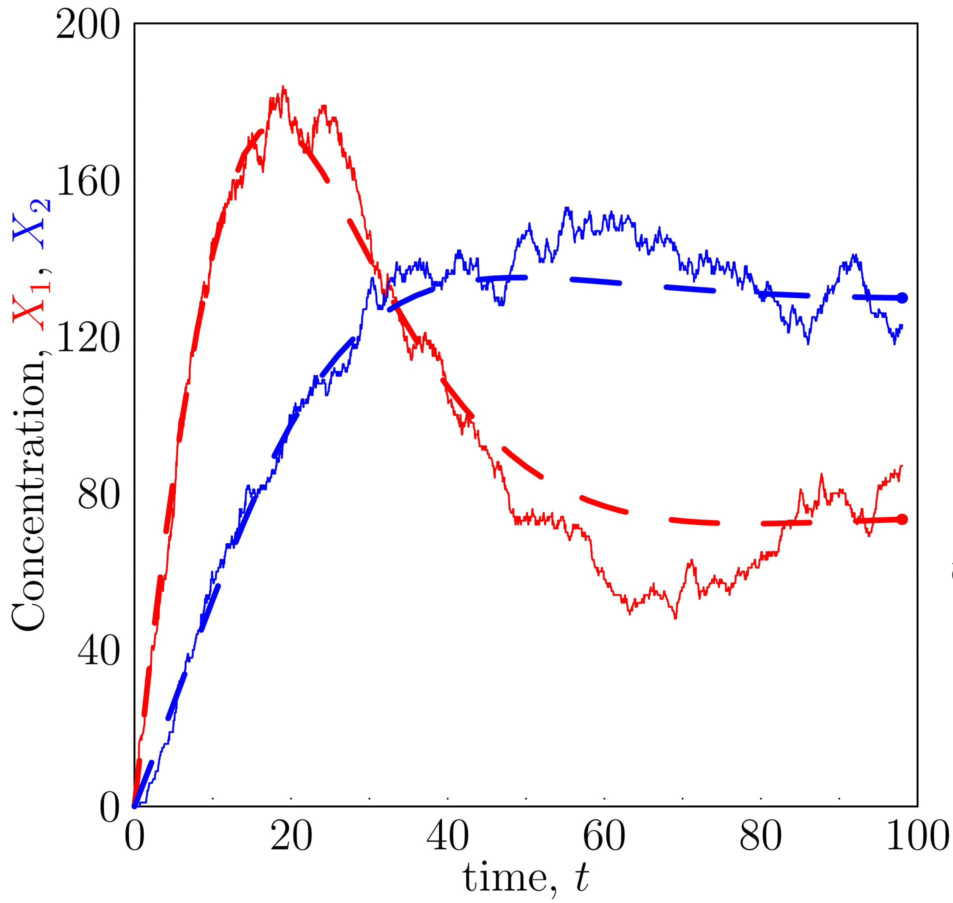

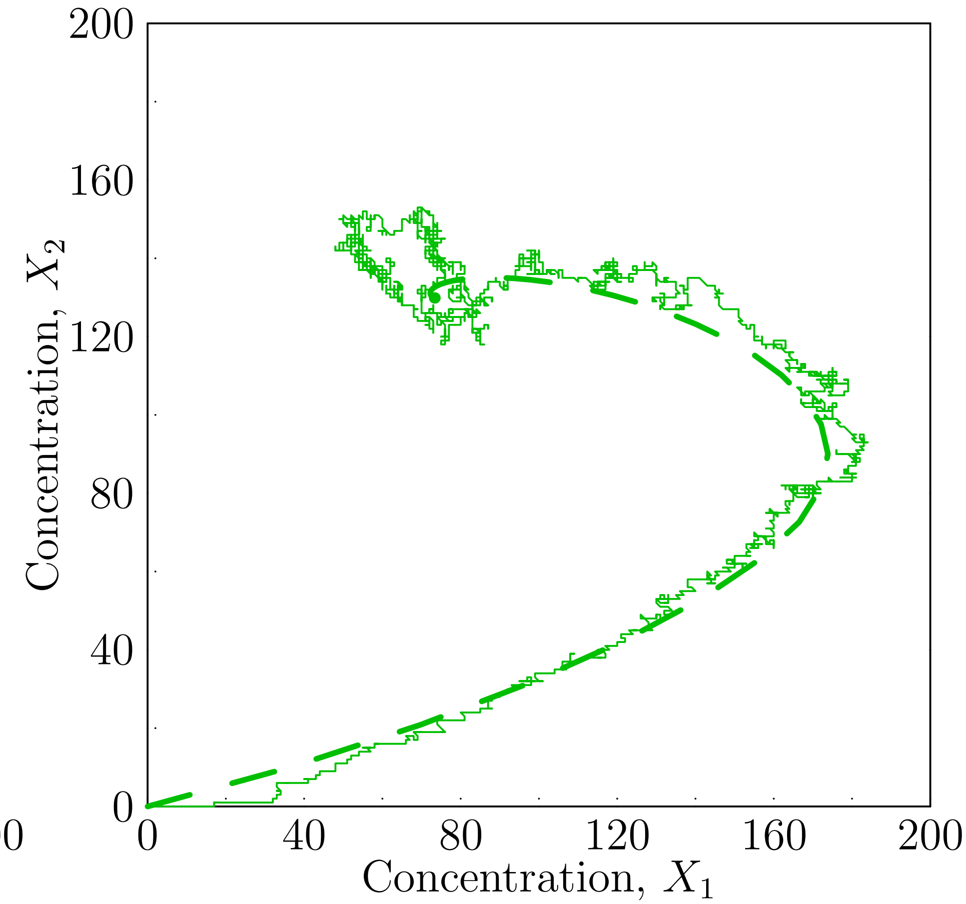

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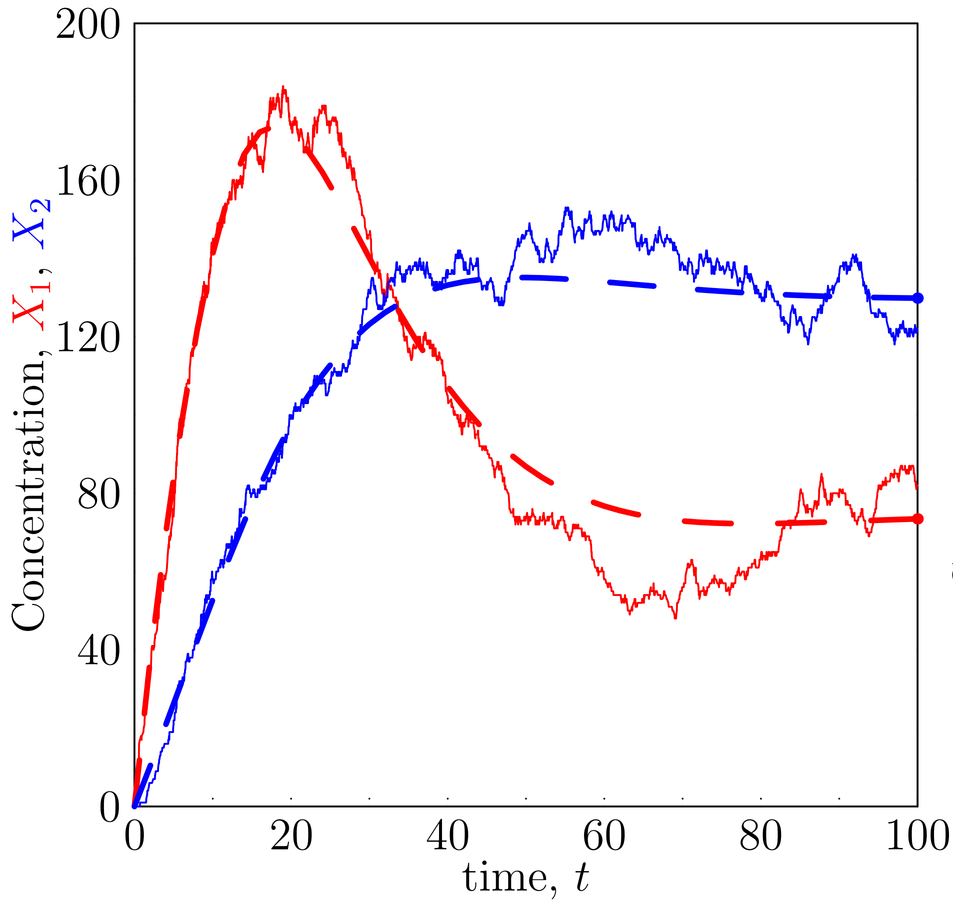
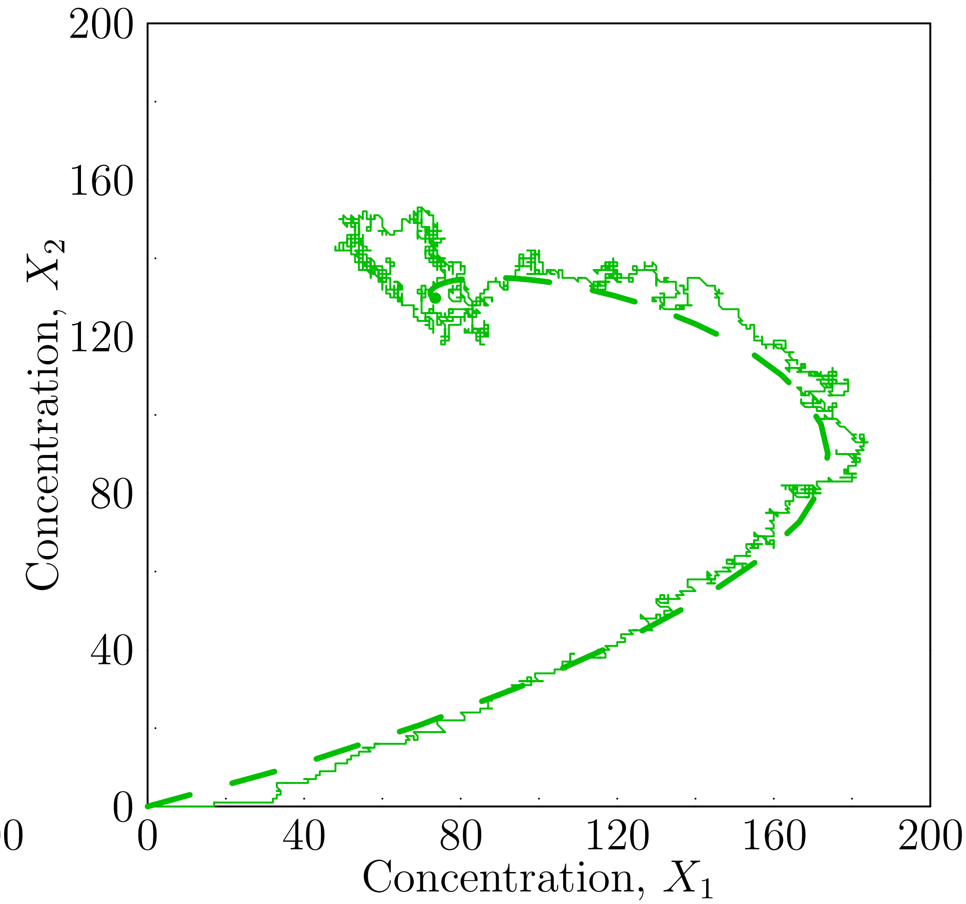
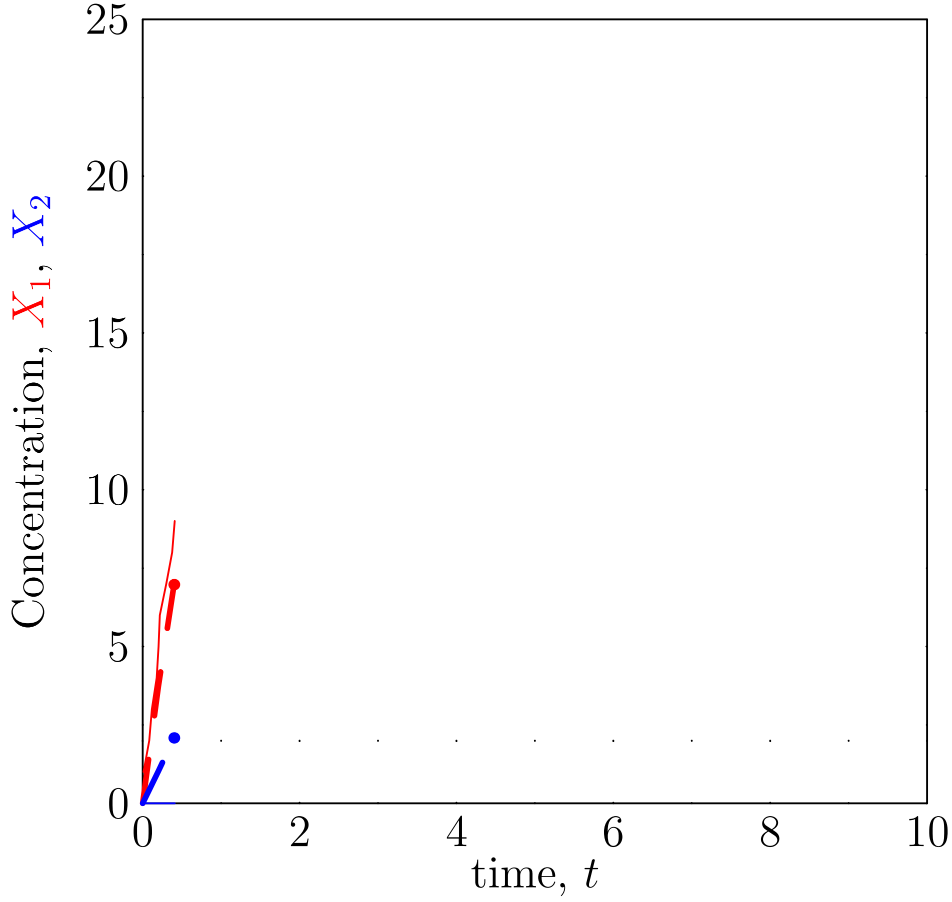
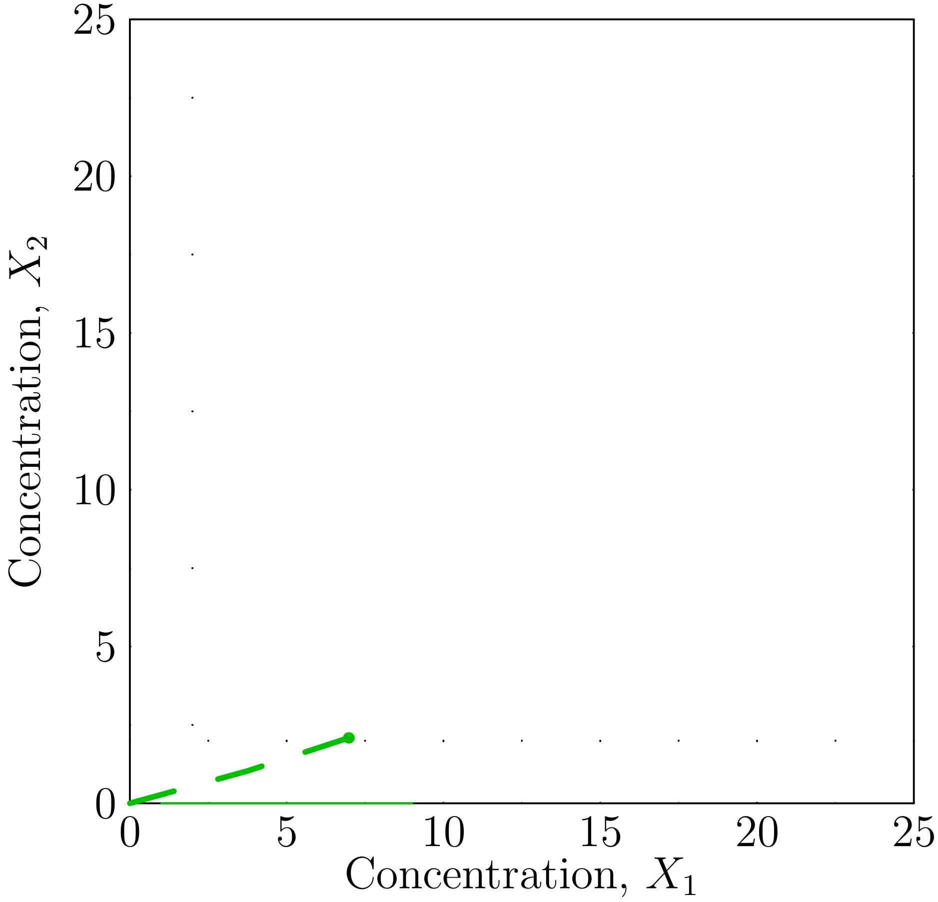


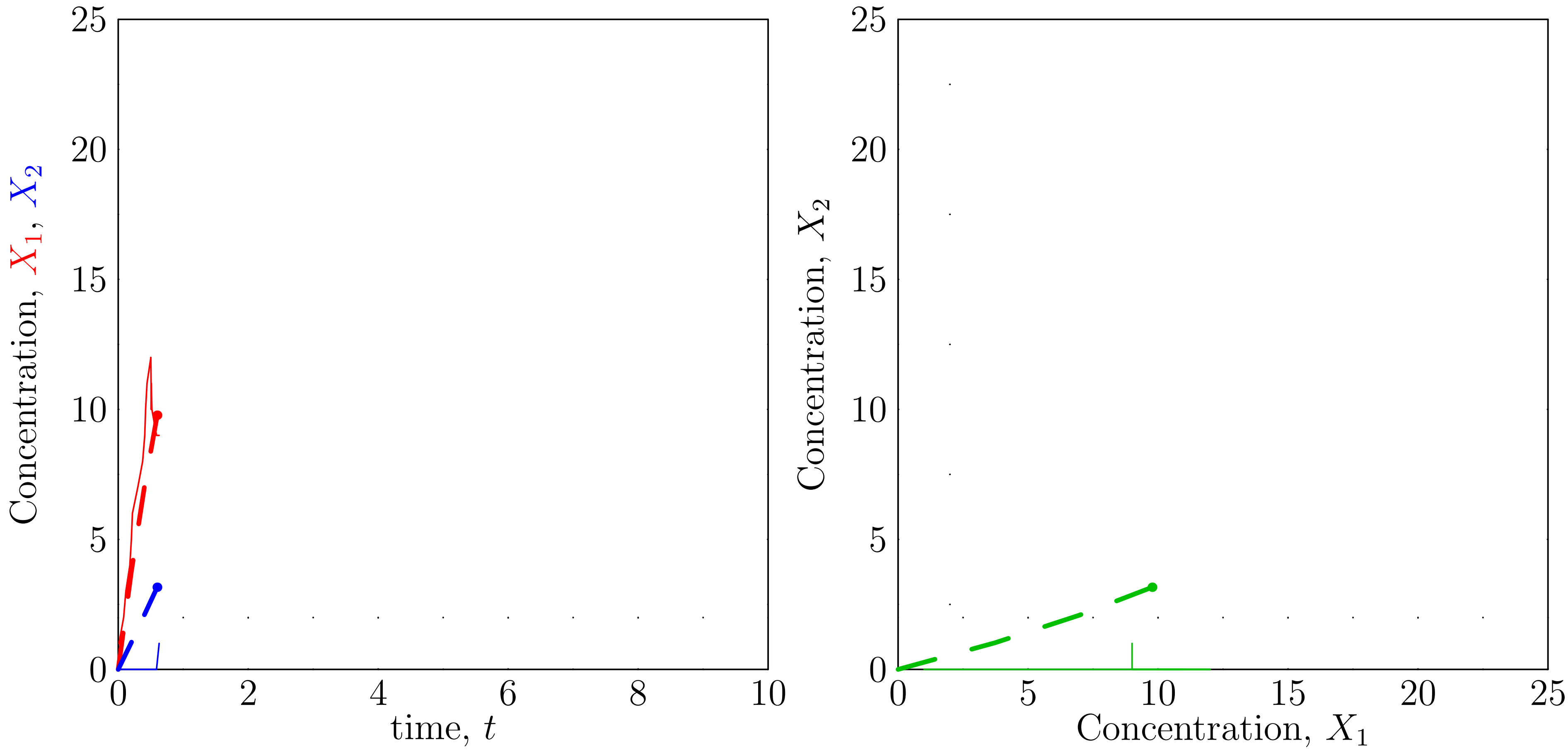


Small Numbers ($r = 10$)

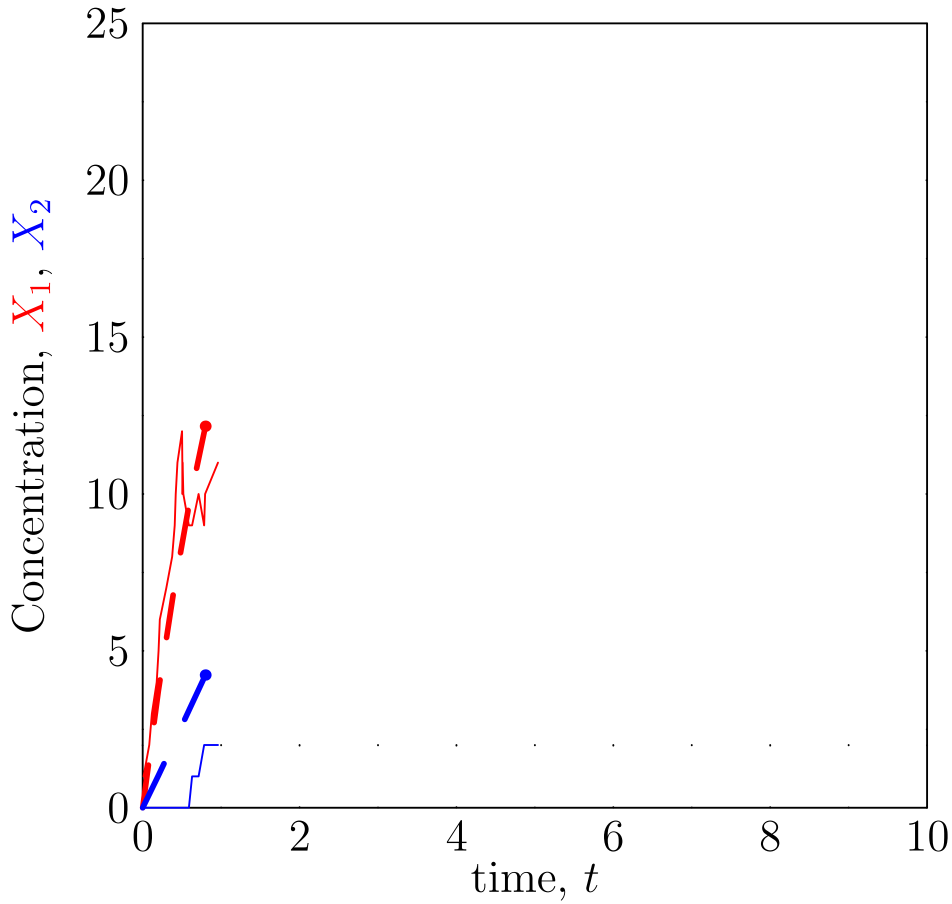


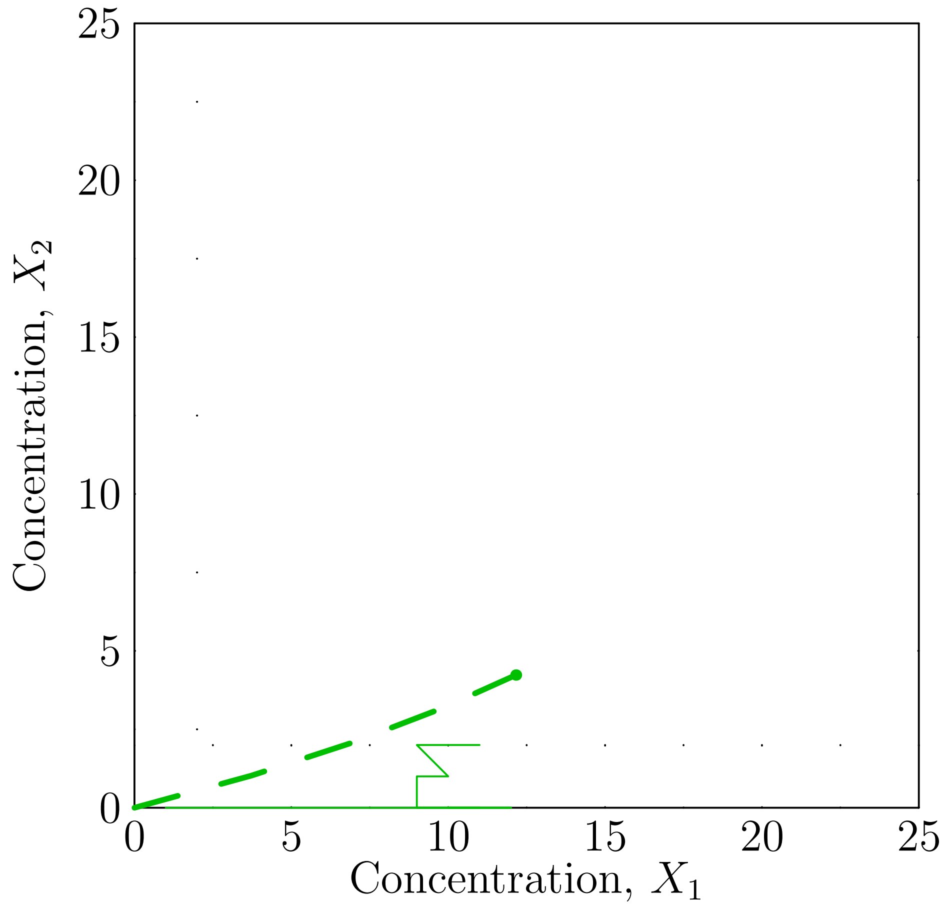


Small Numbers ($r = 10$)

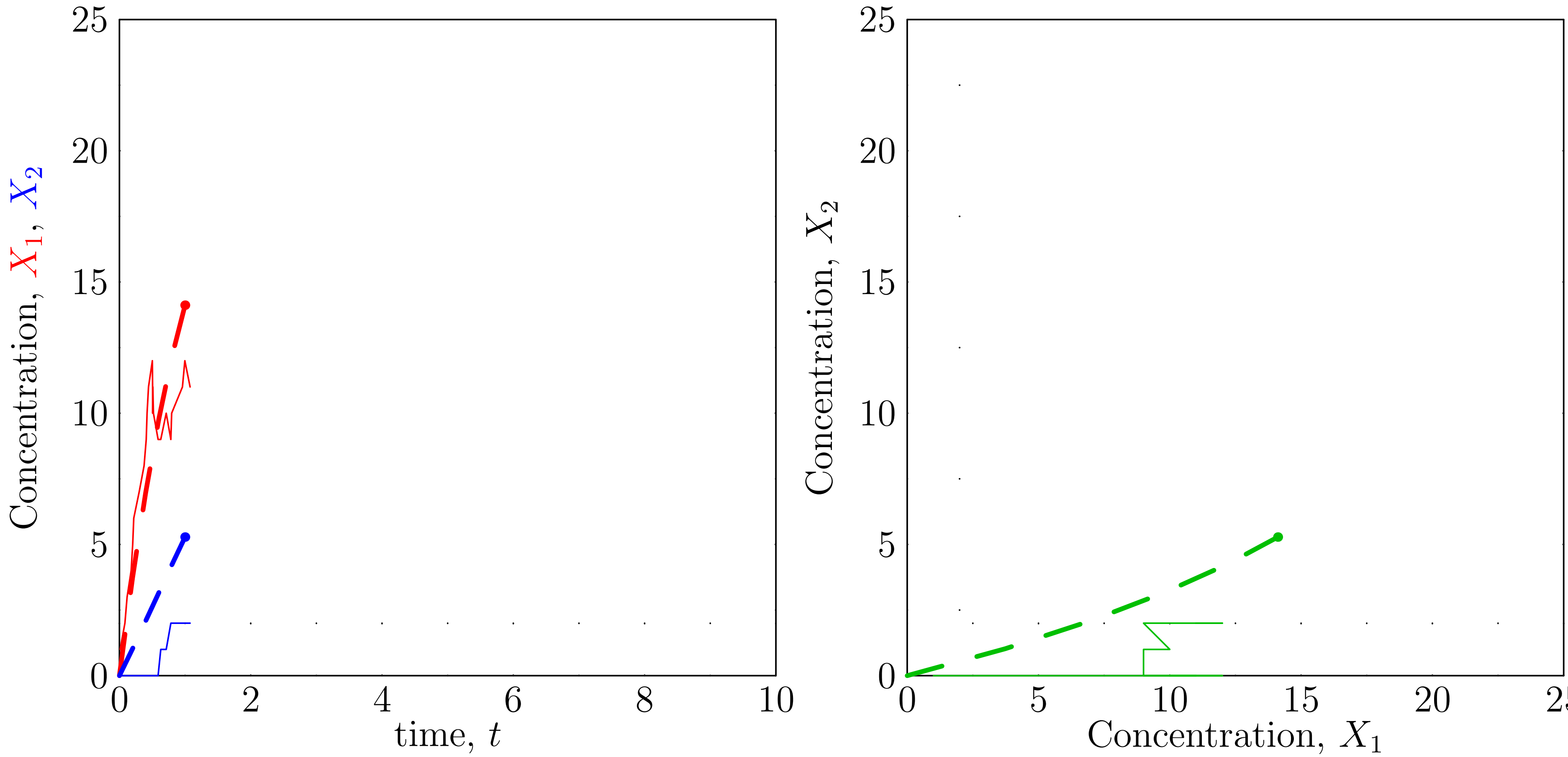


Small Numbers ($r = 10$)





Small Numbers ($r = 10$)



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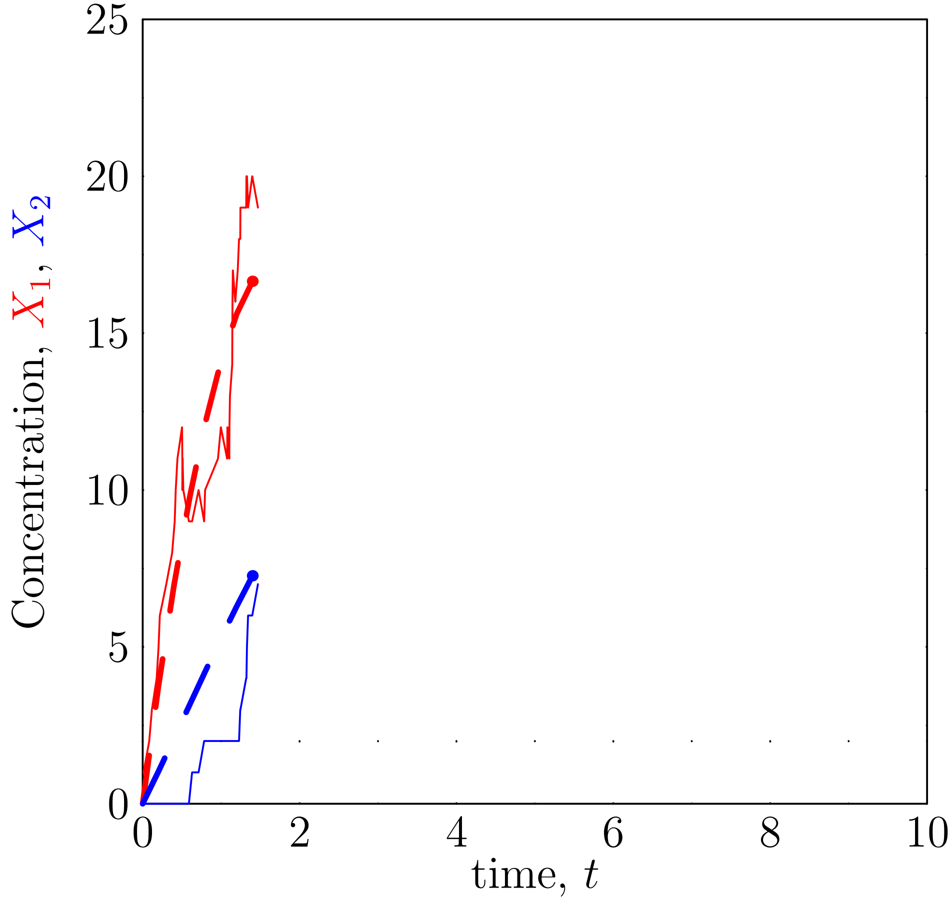

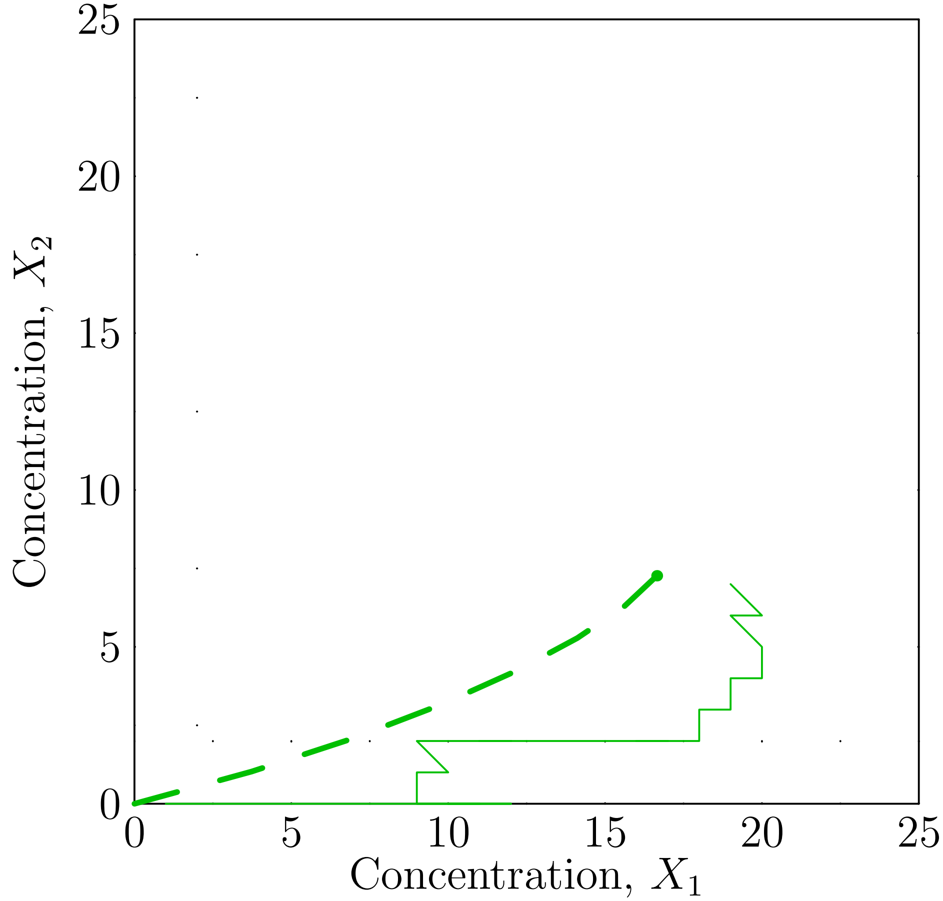

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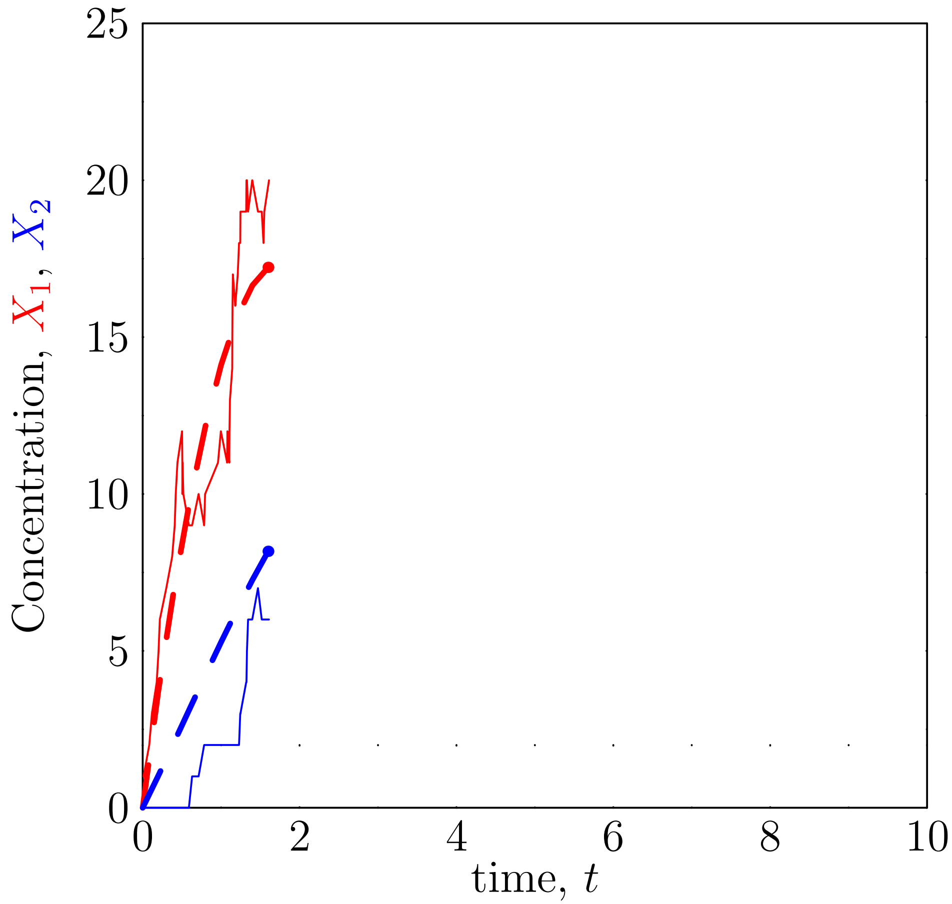

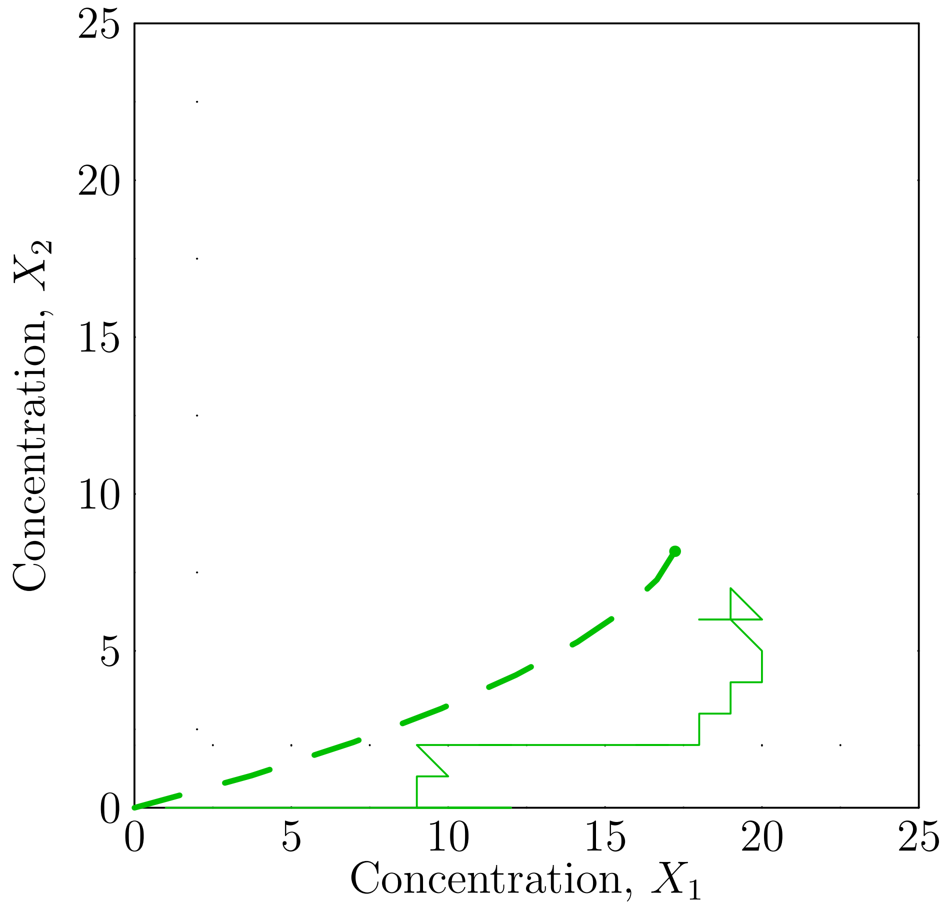

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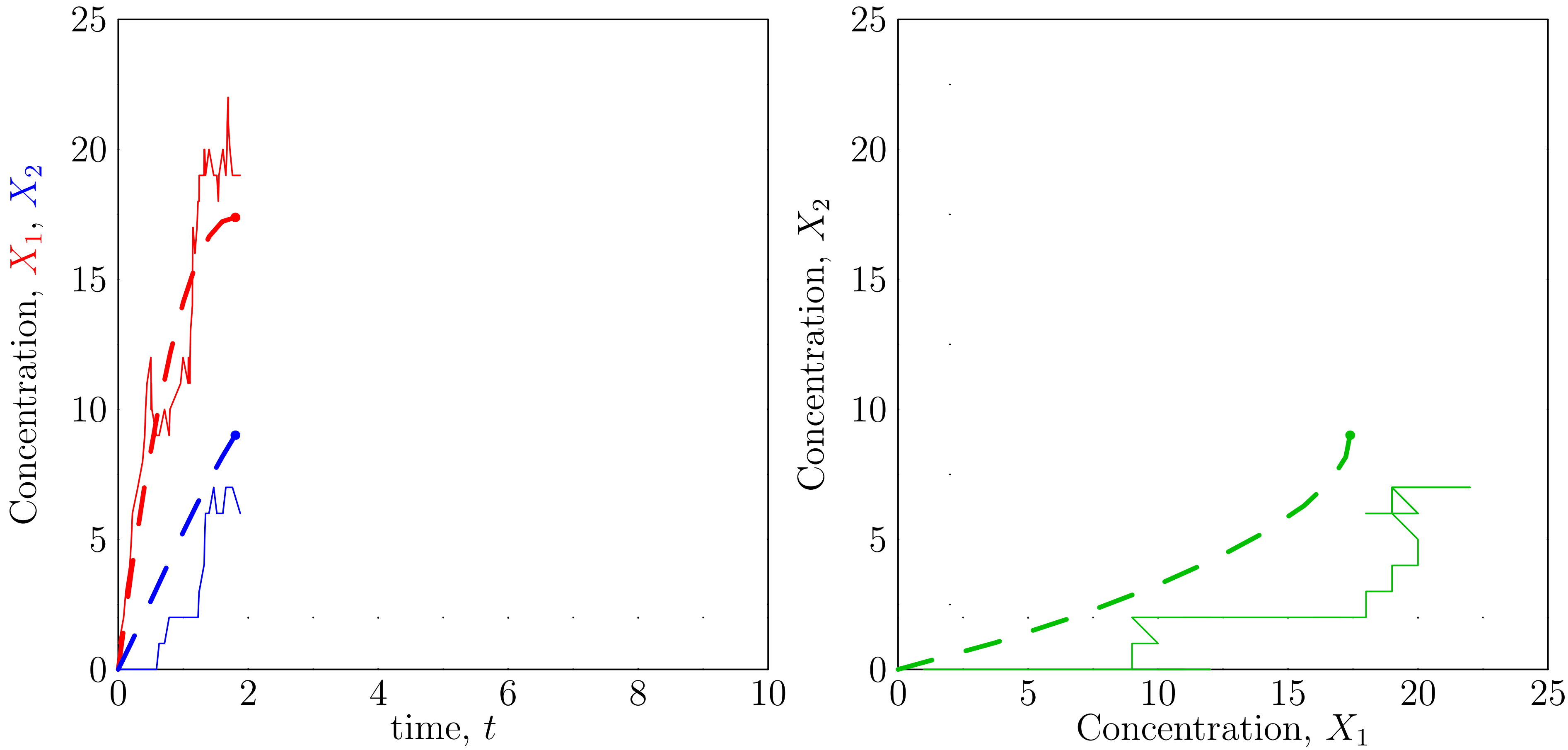

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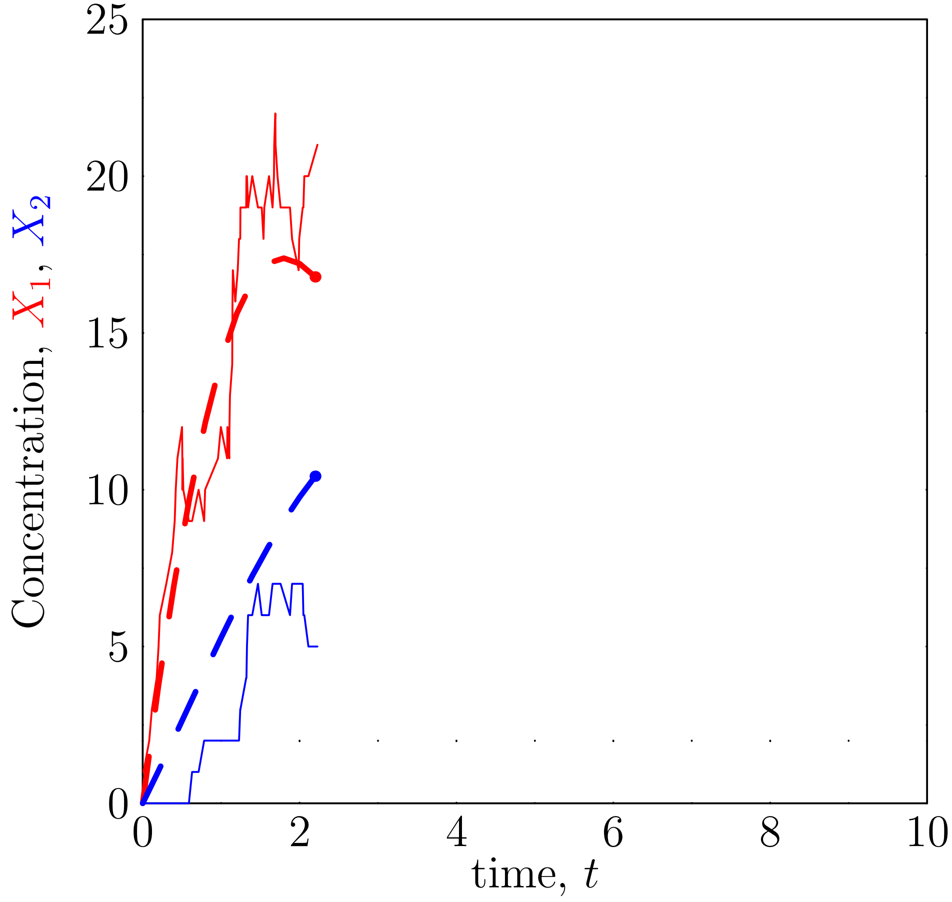

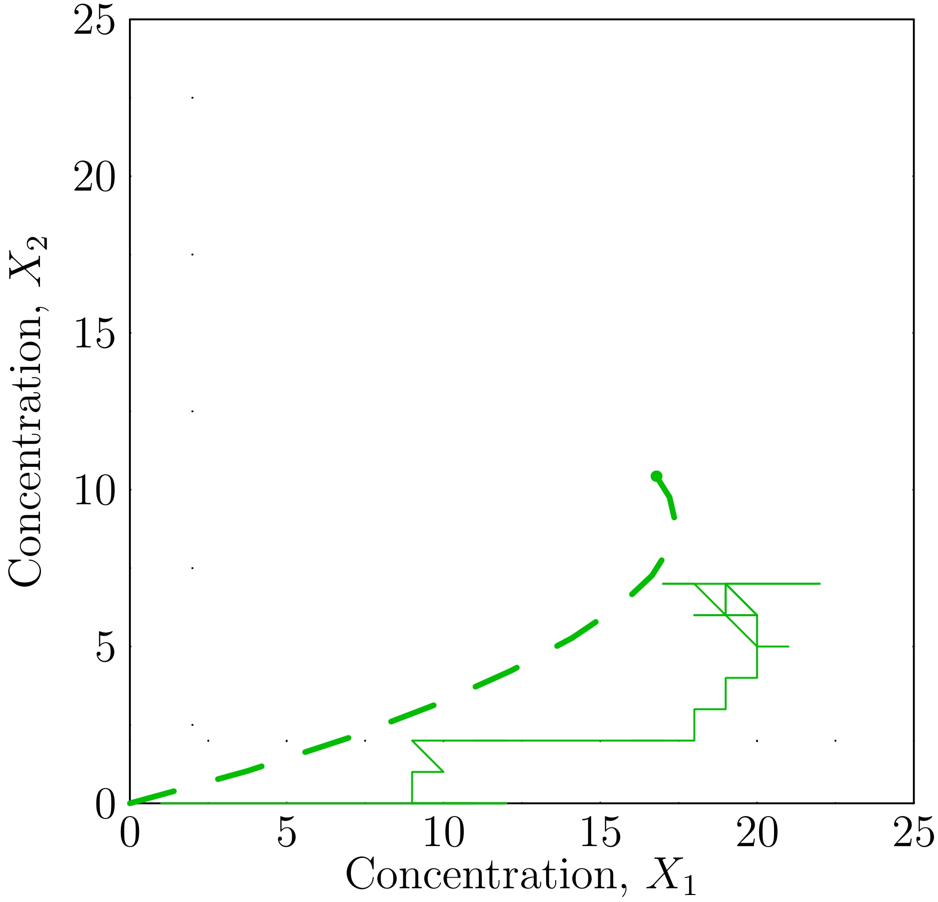

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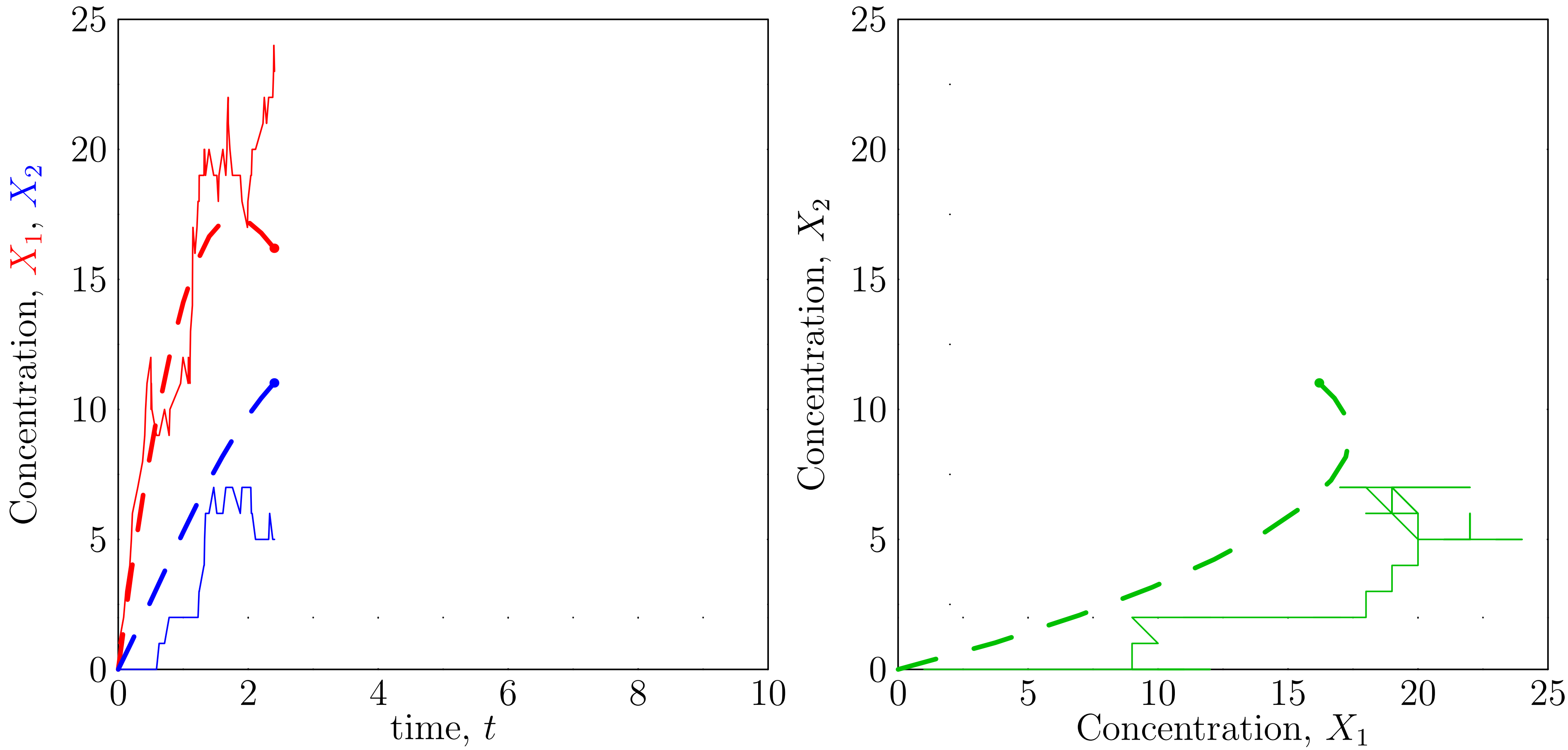

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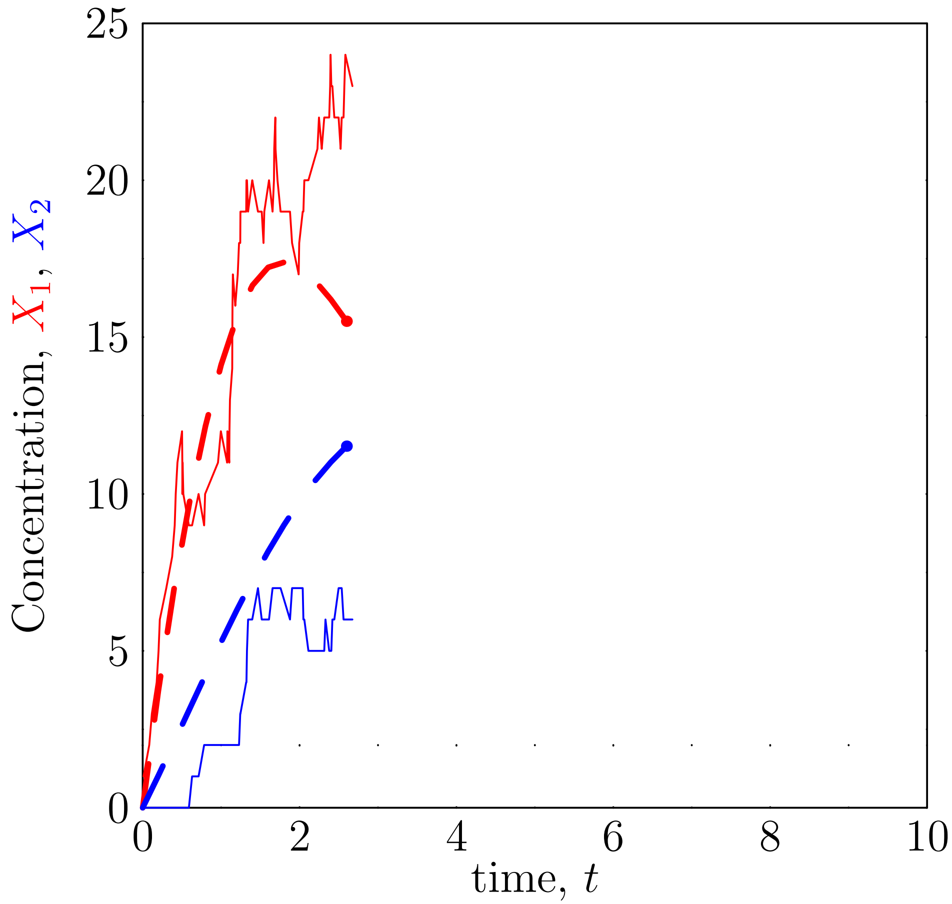

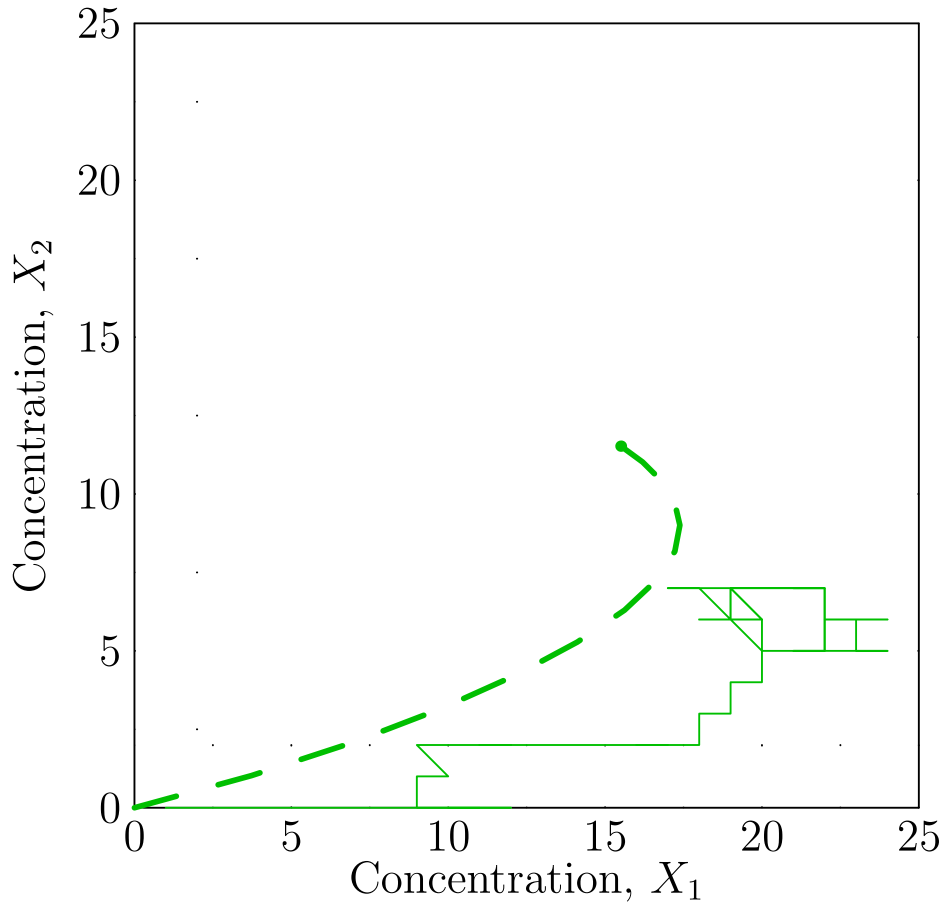

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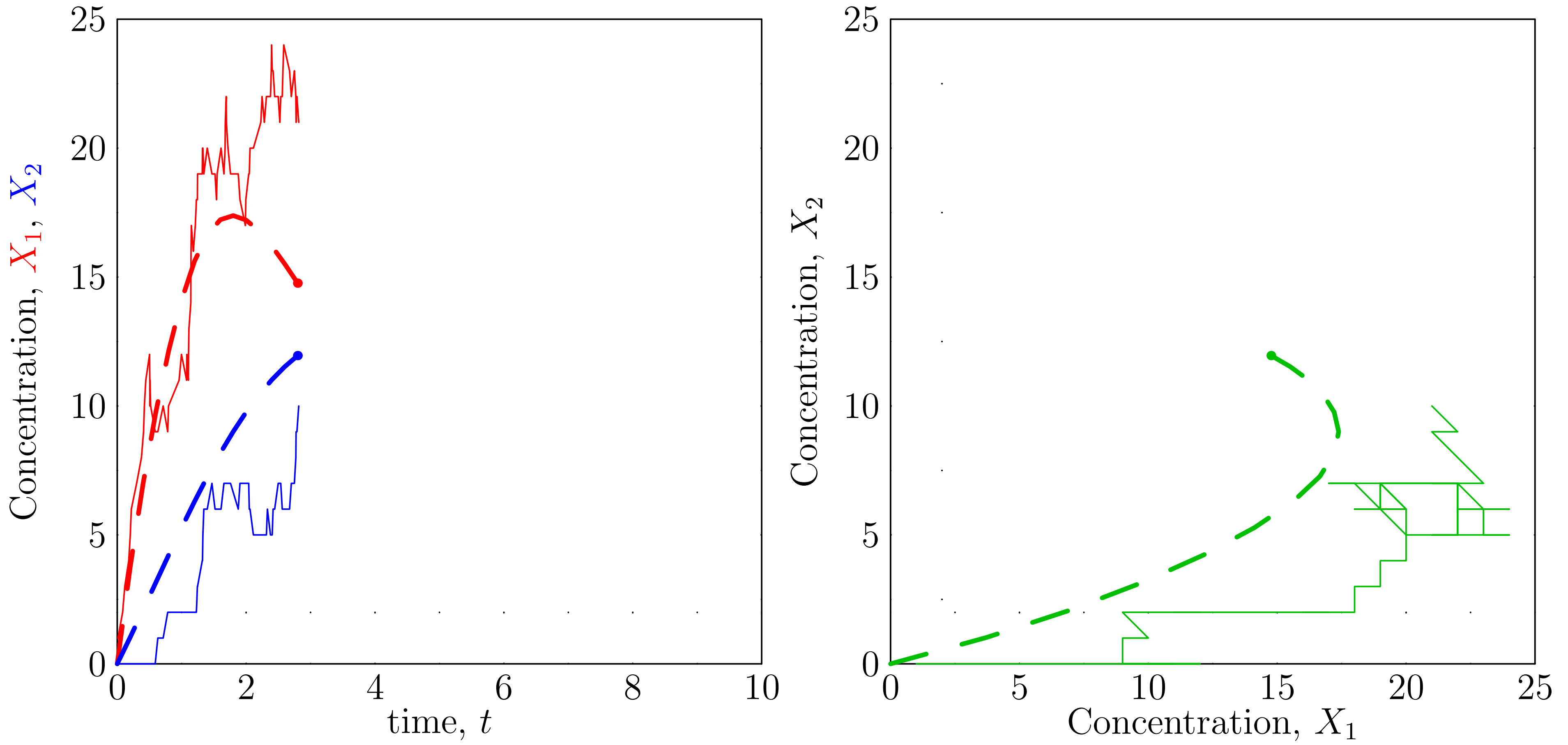

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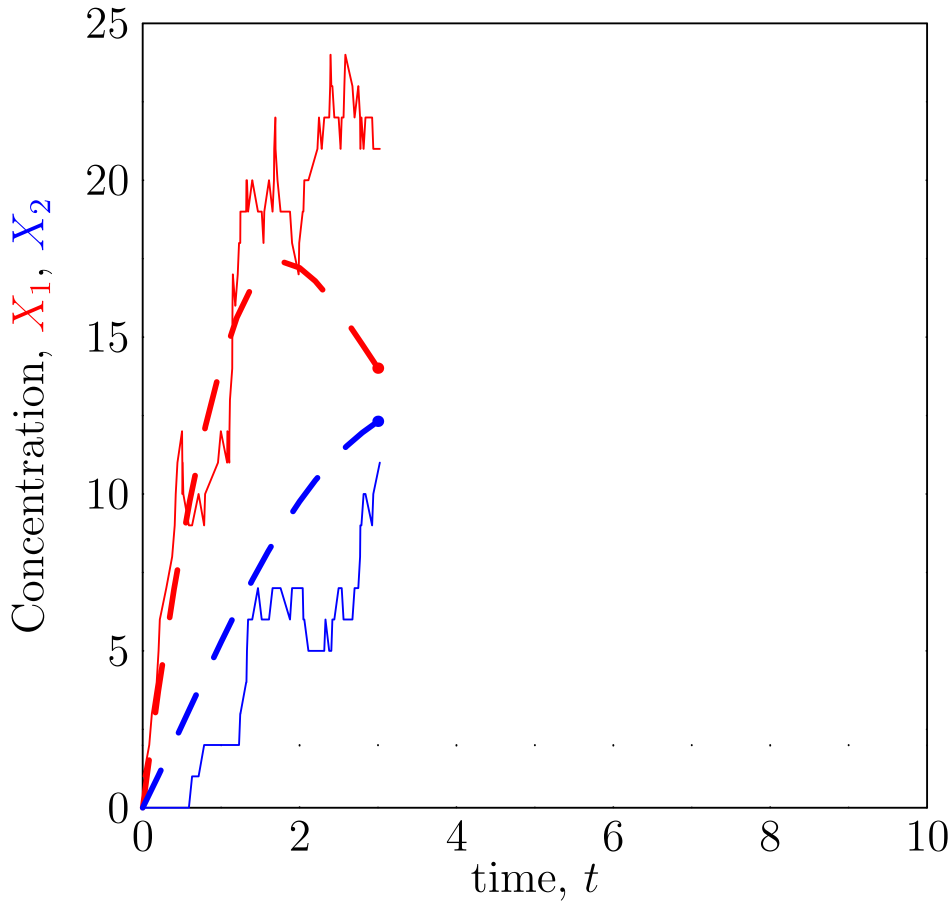

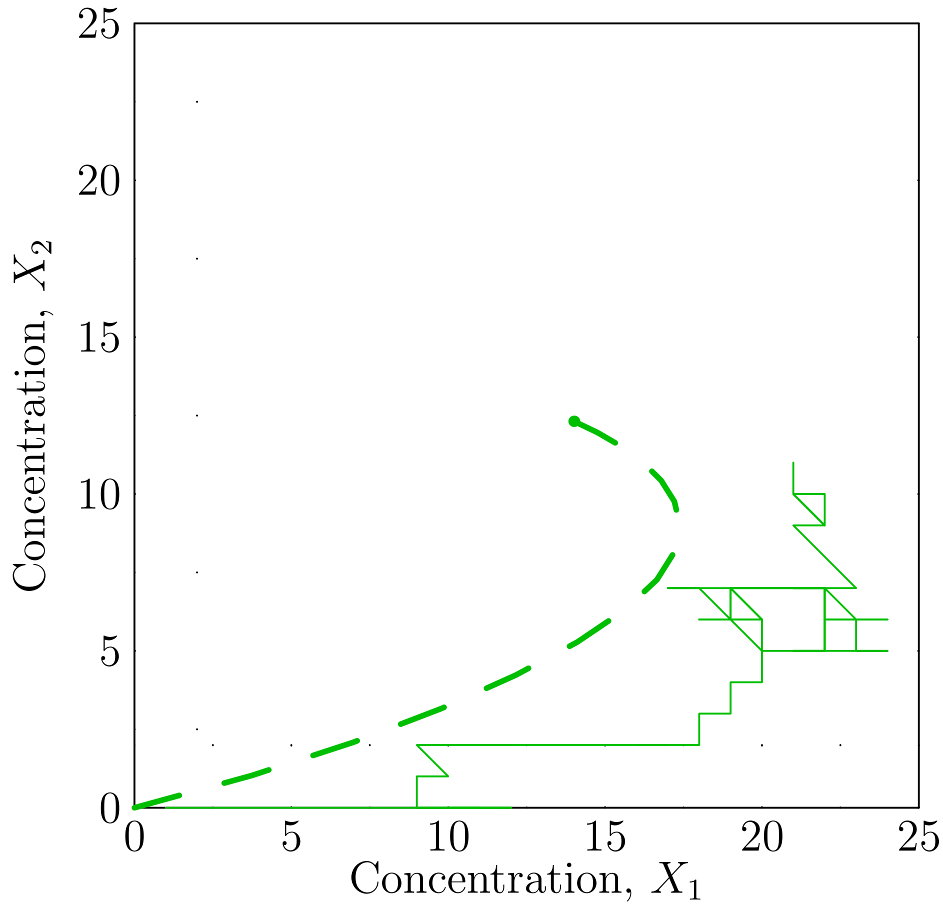

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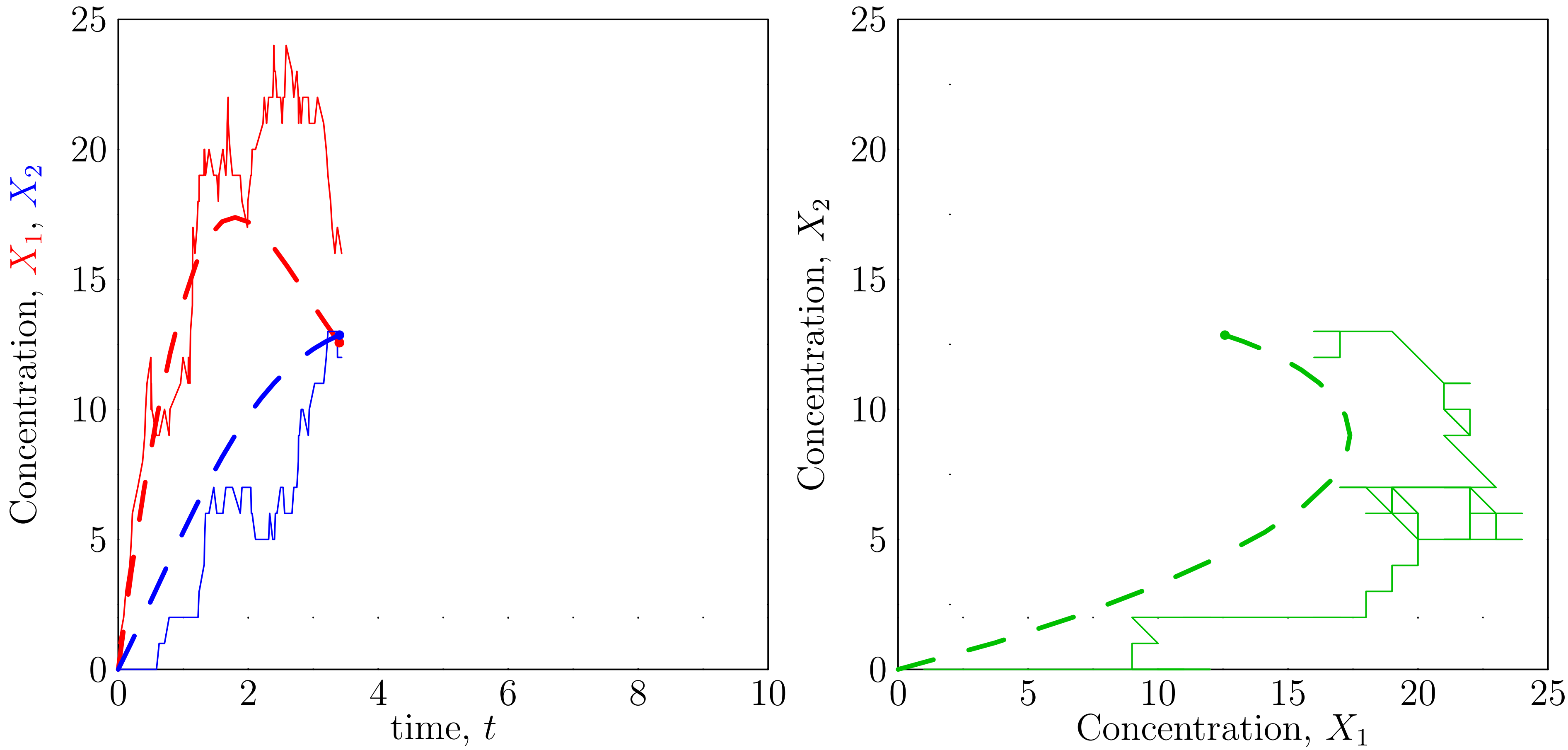

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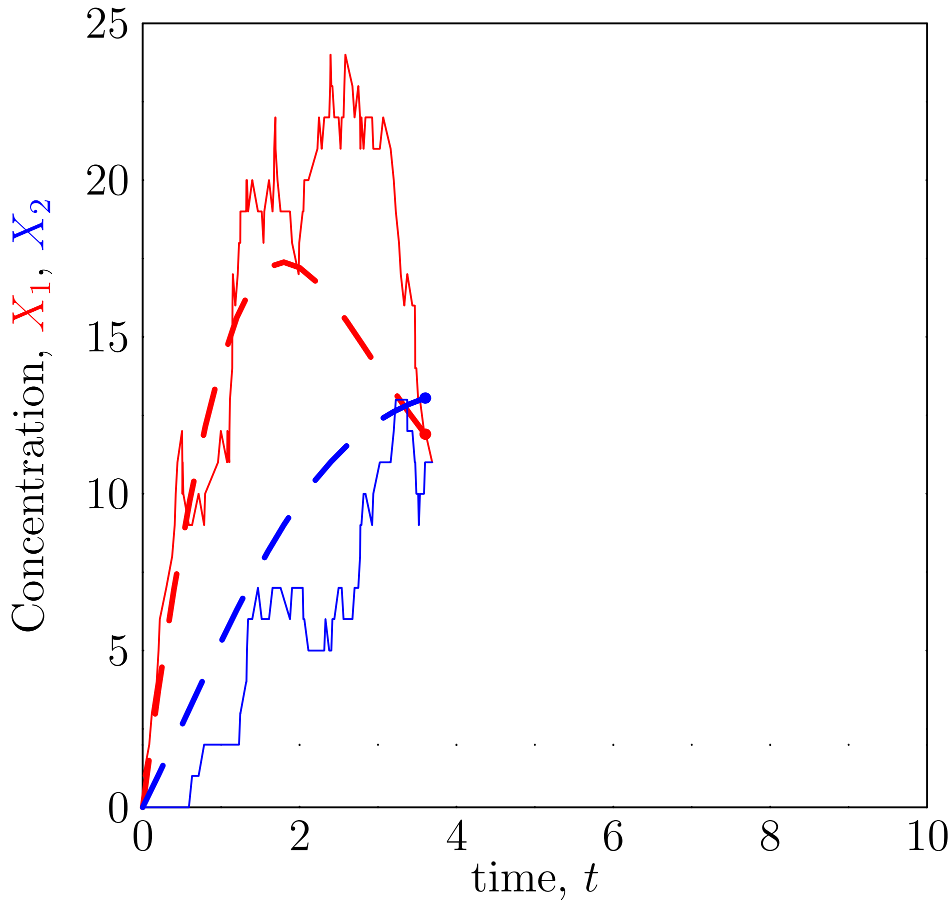

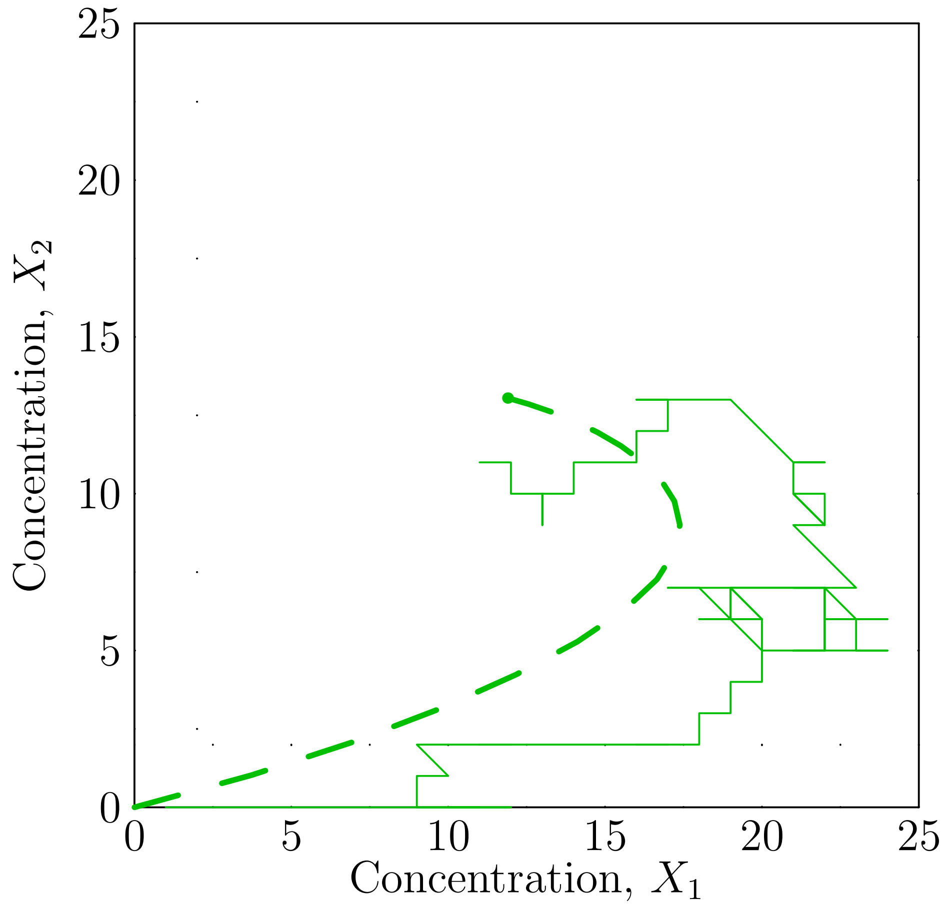

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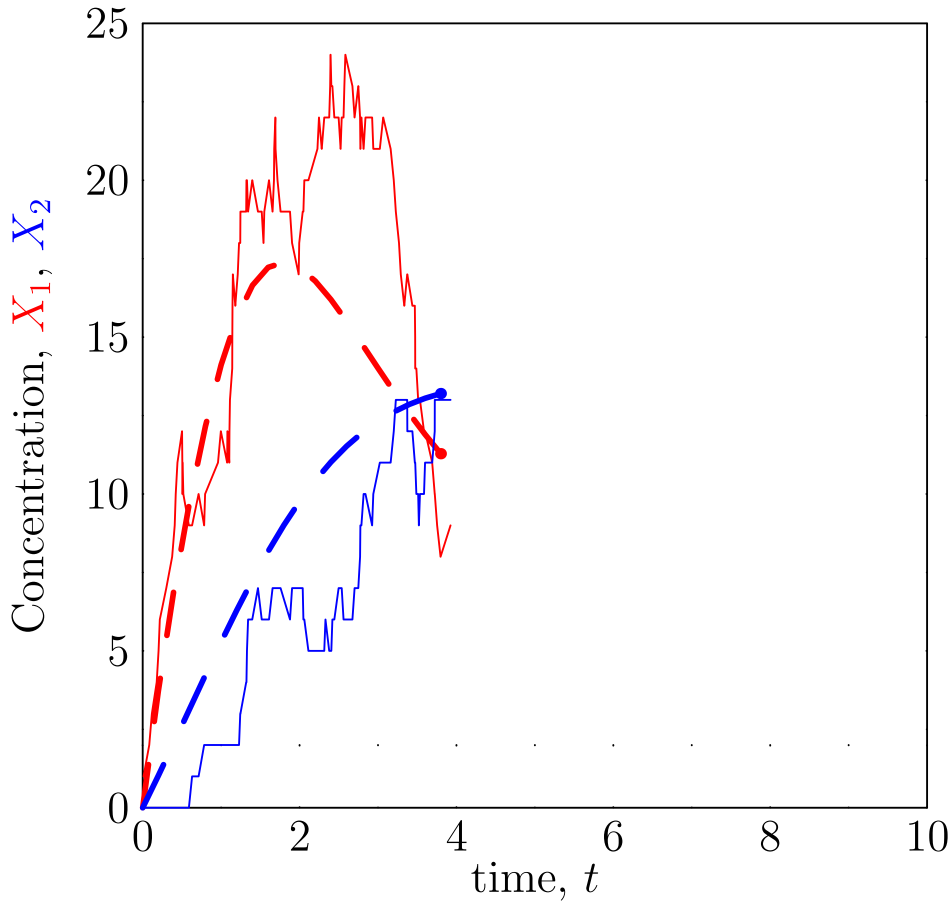

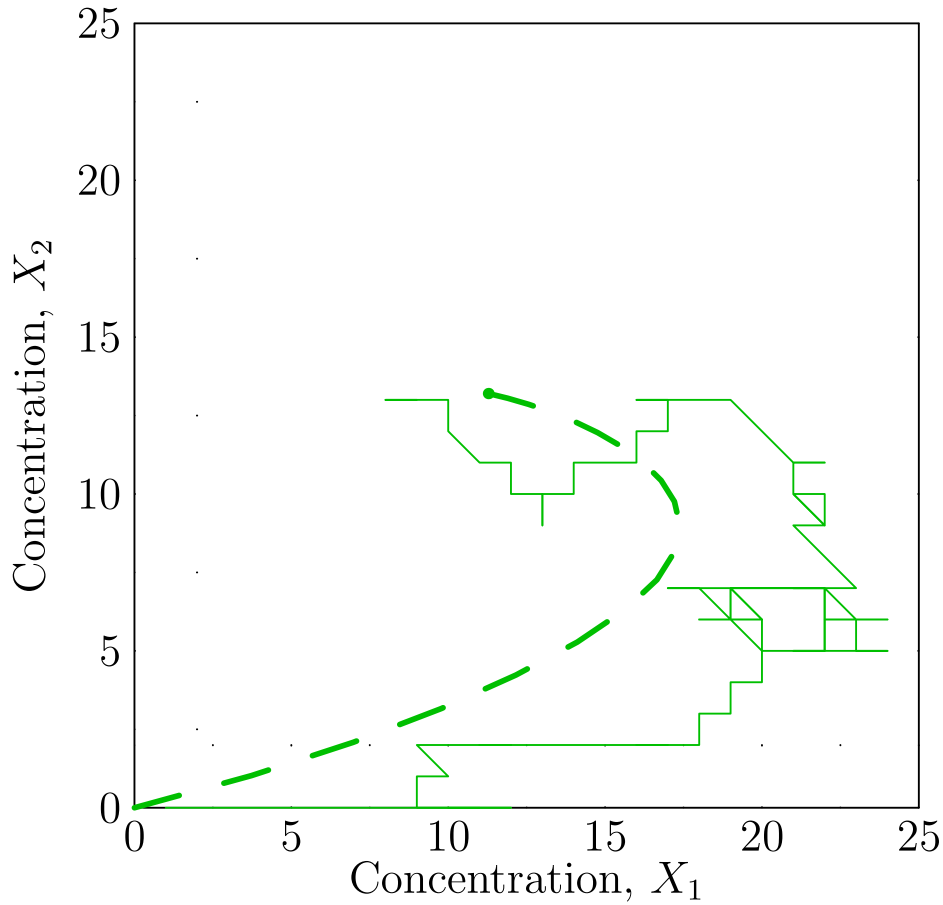

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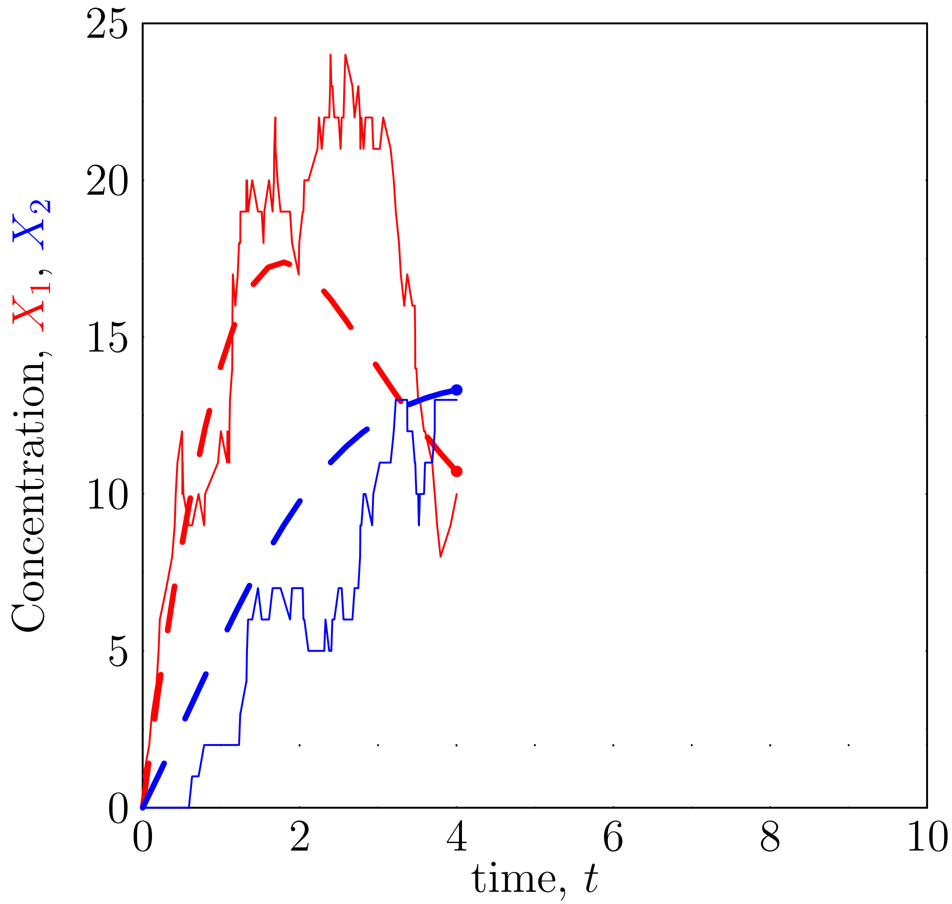

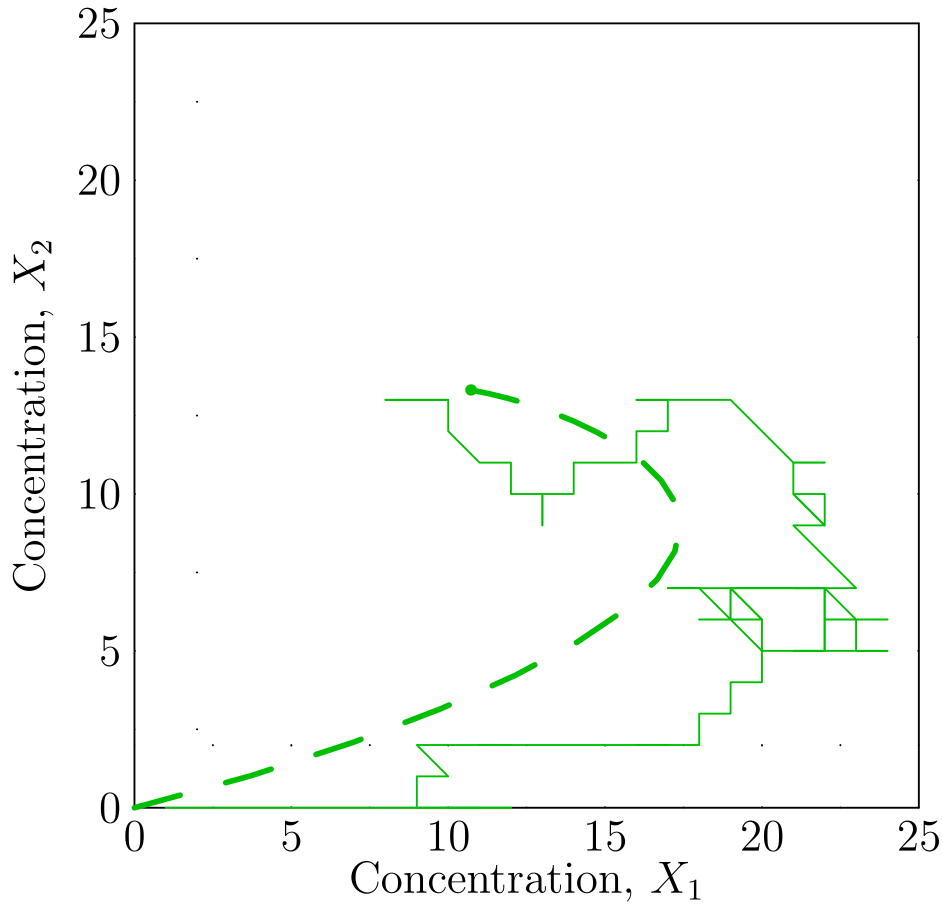

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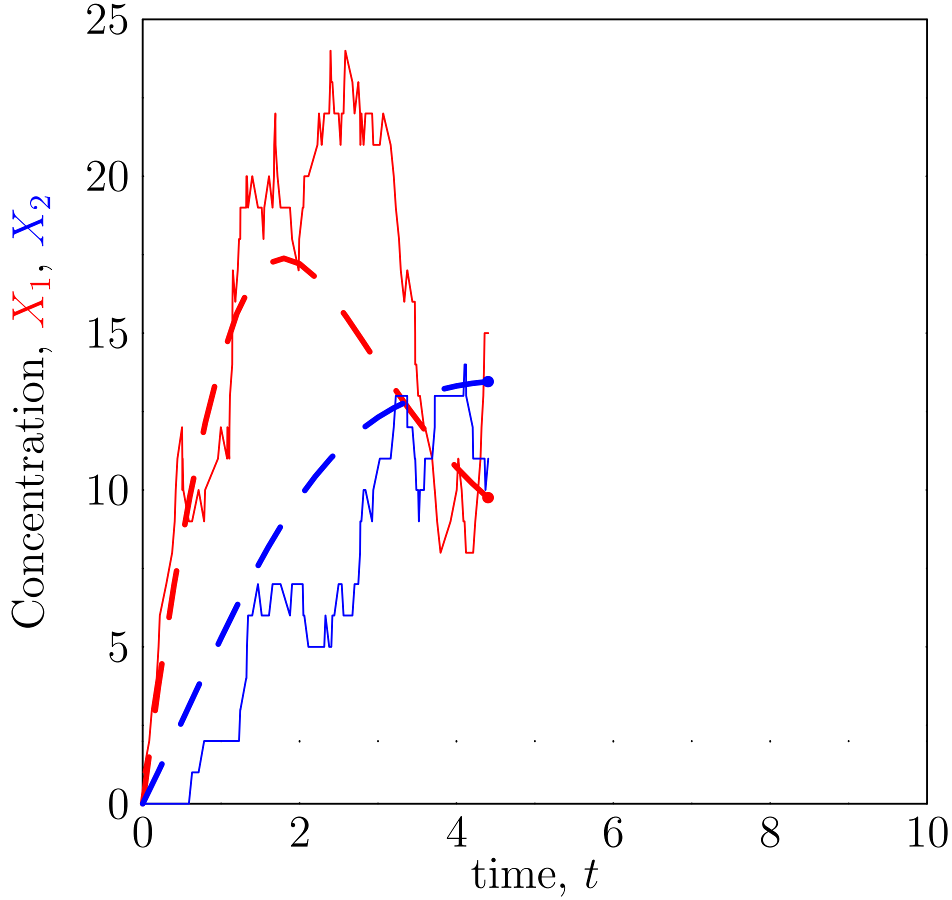

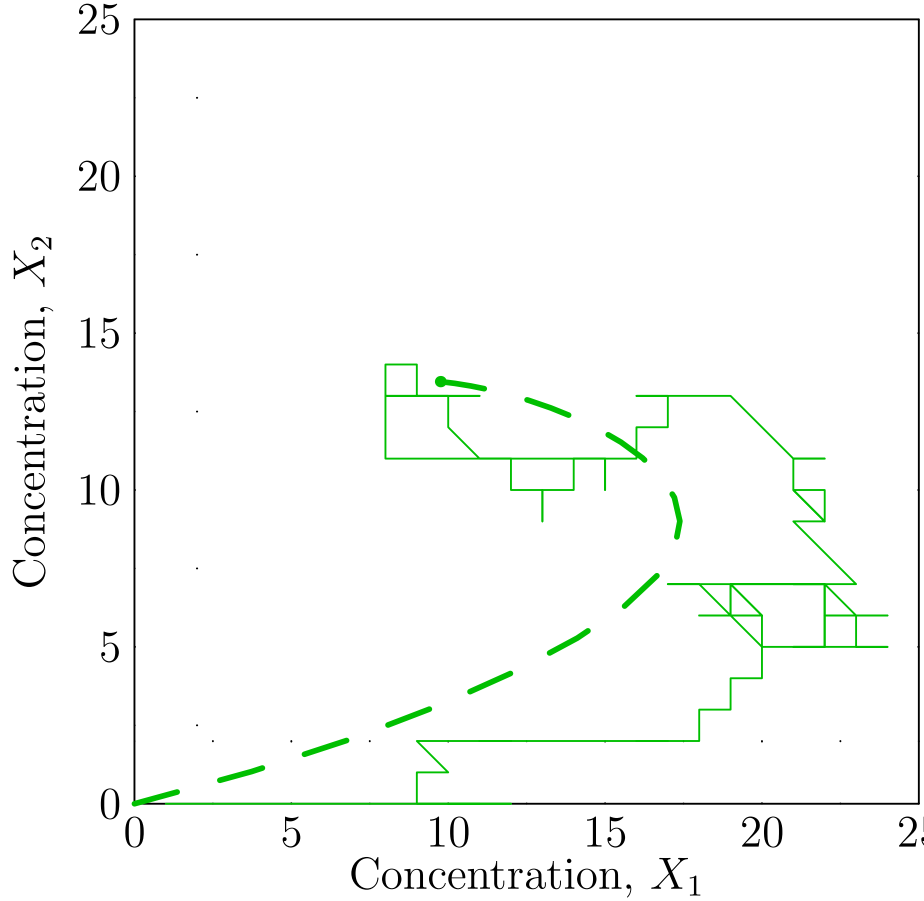

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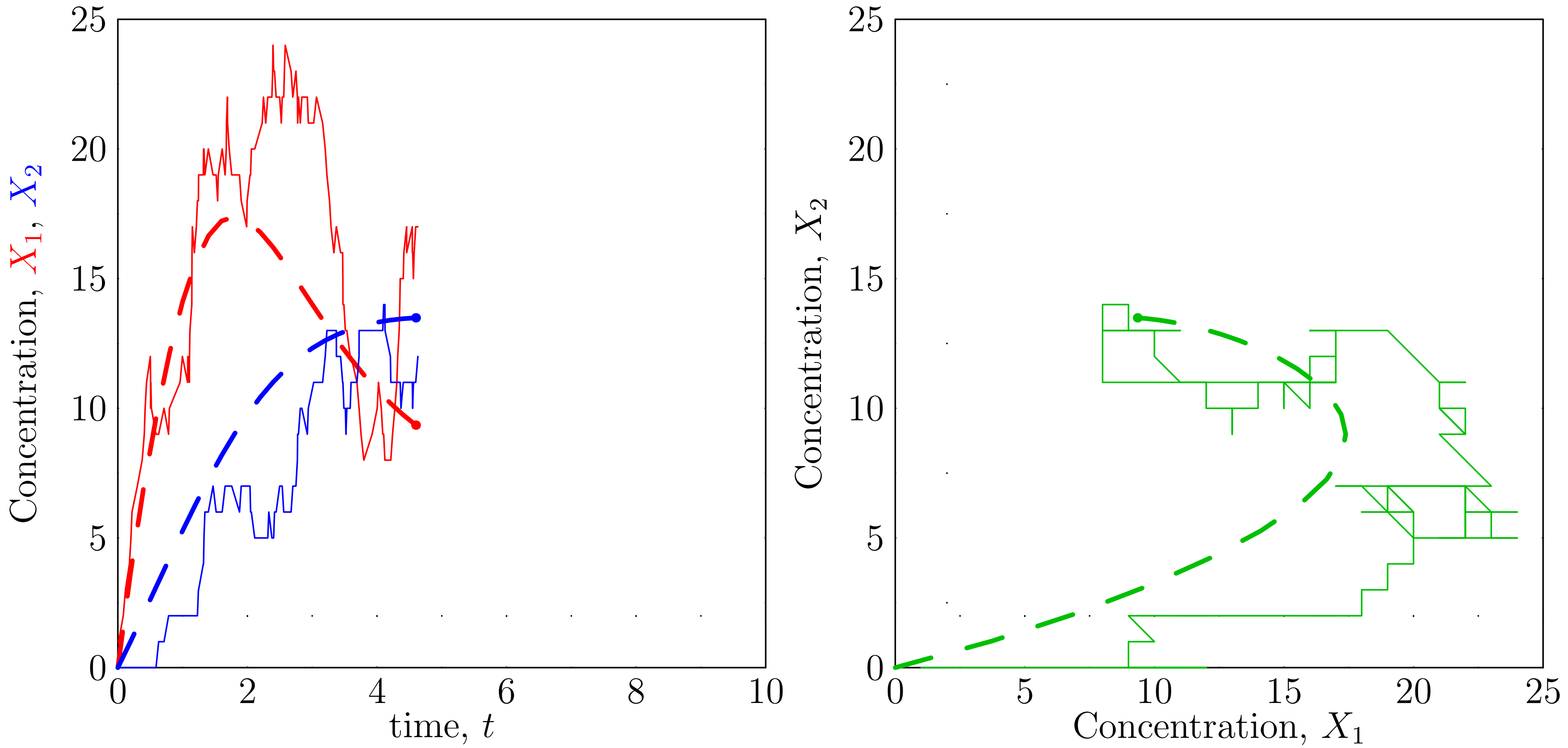

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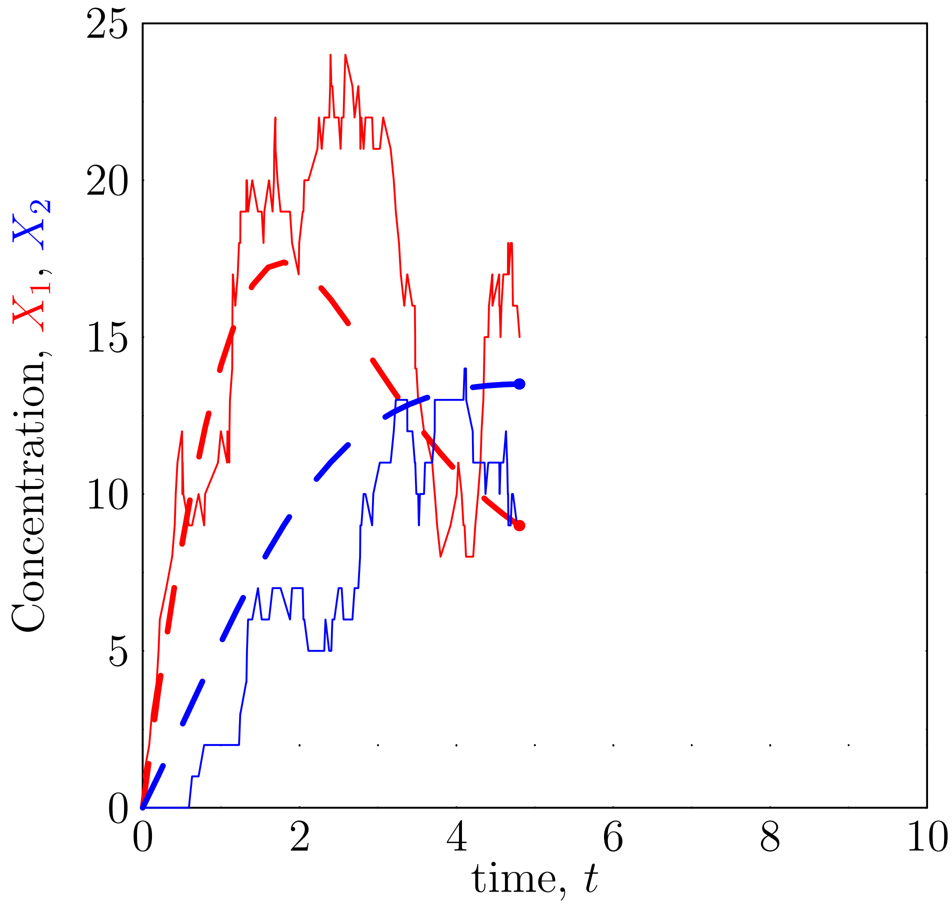

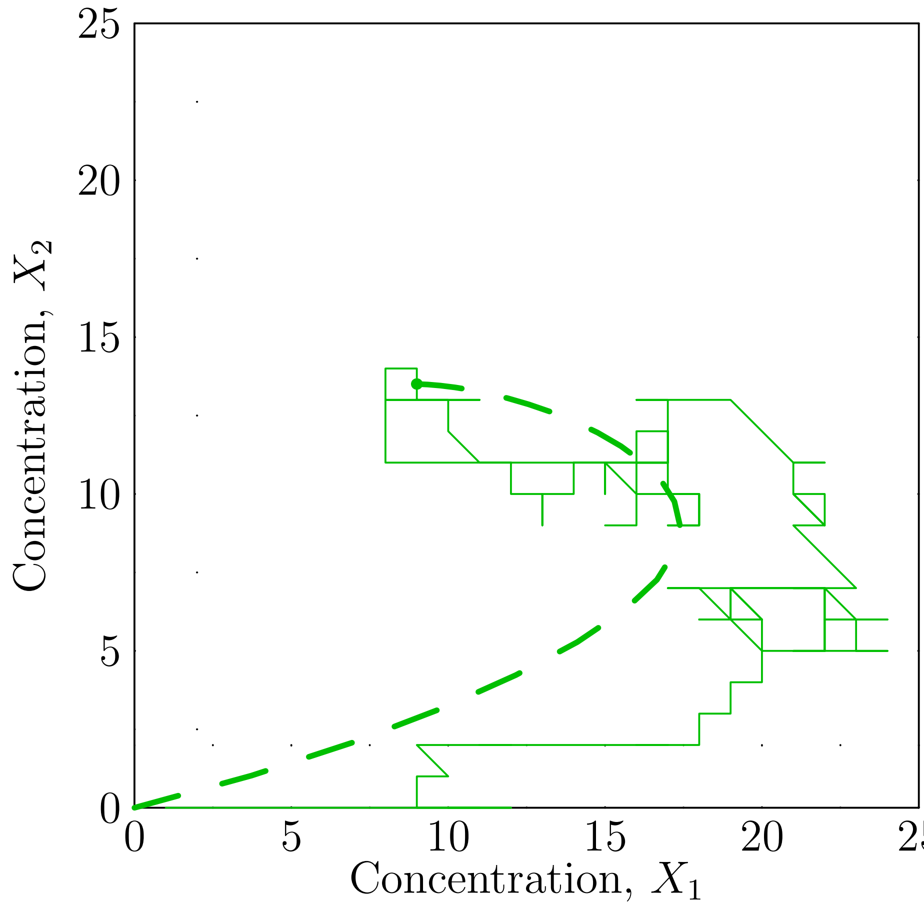

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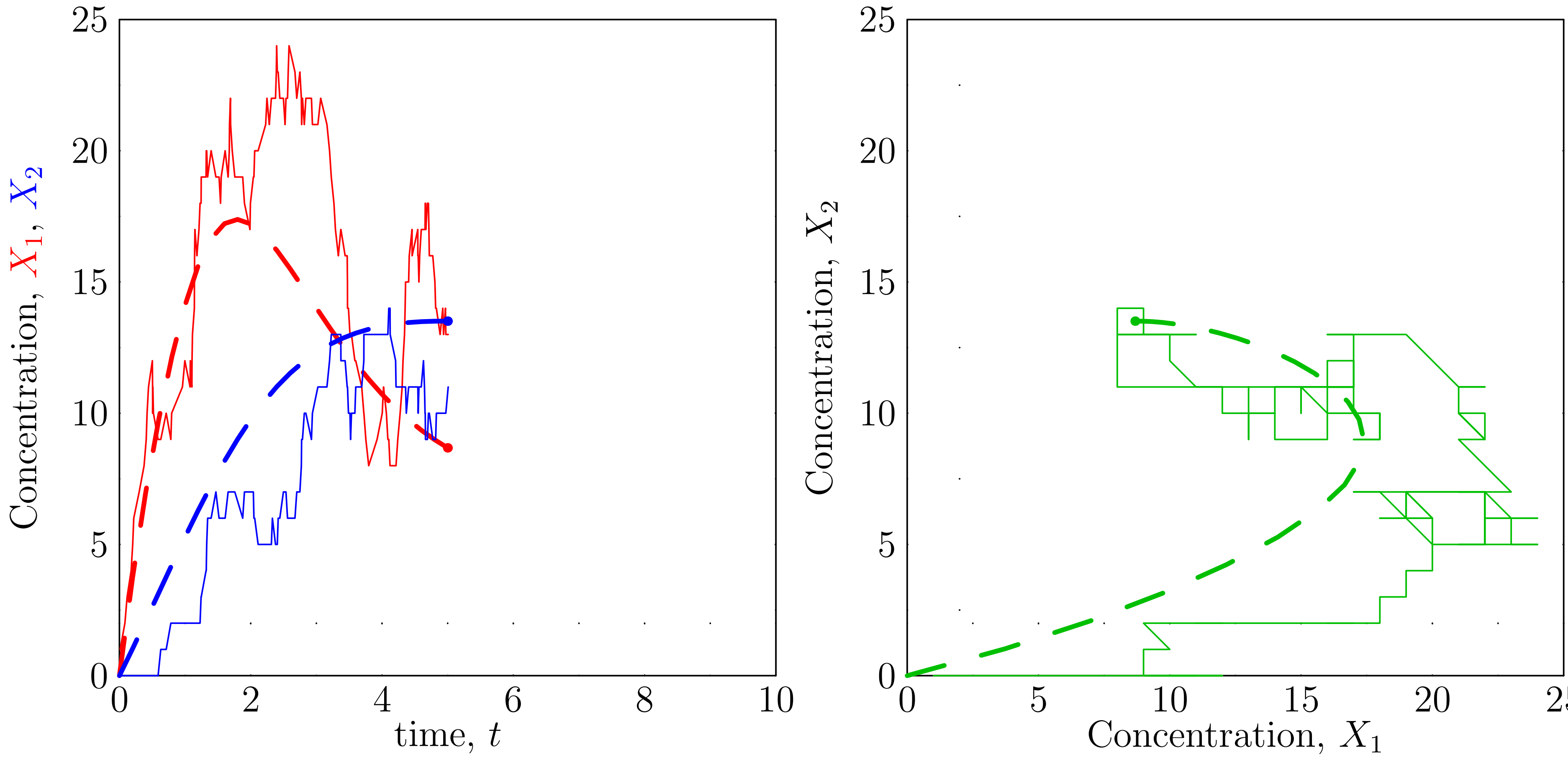

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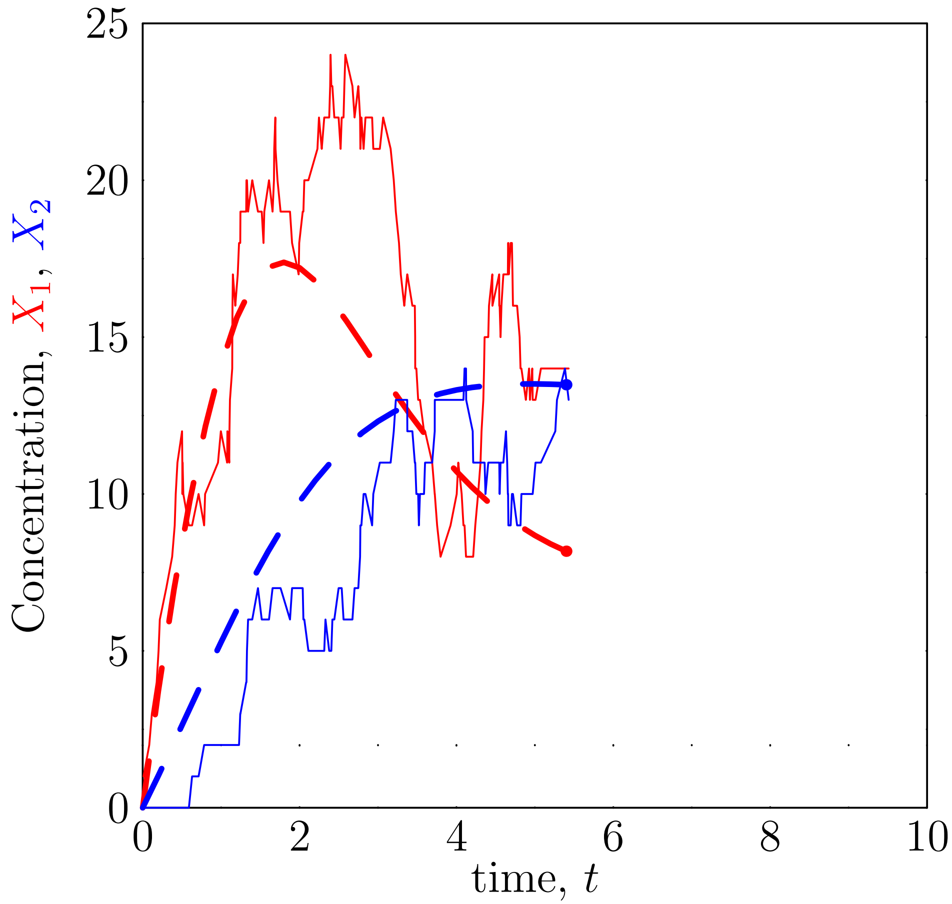

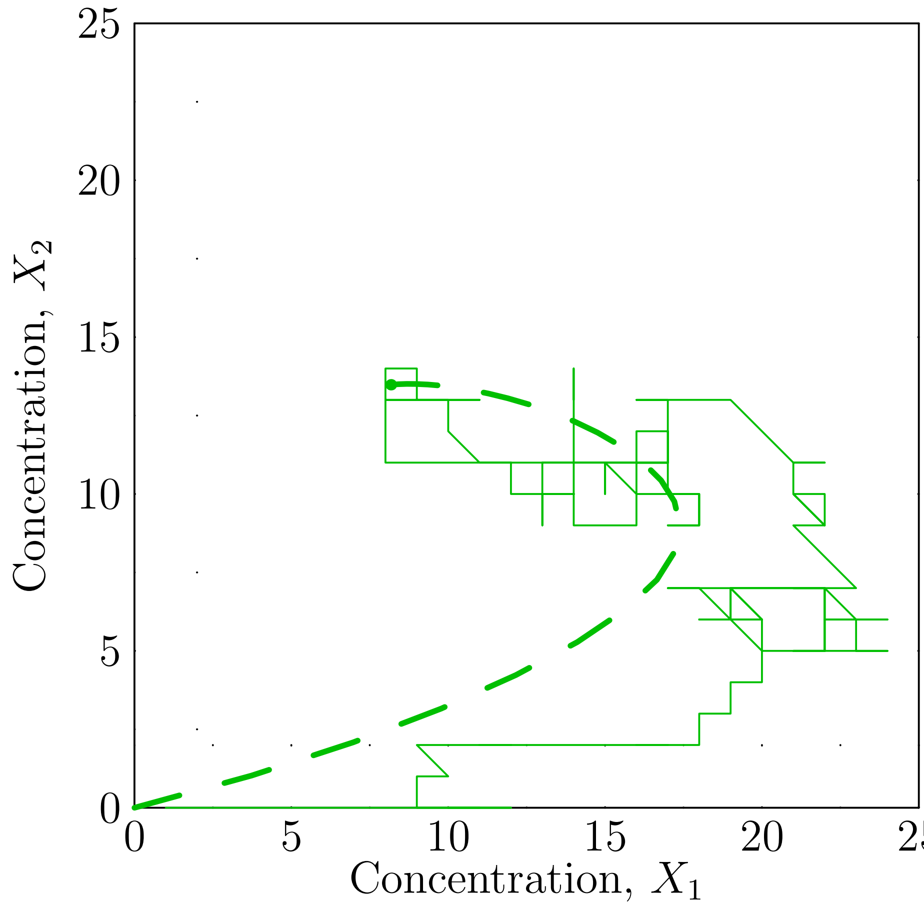

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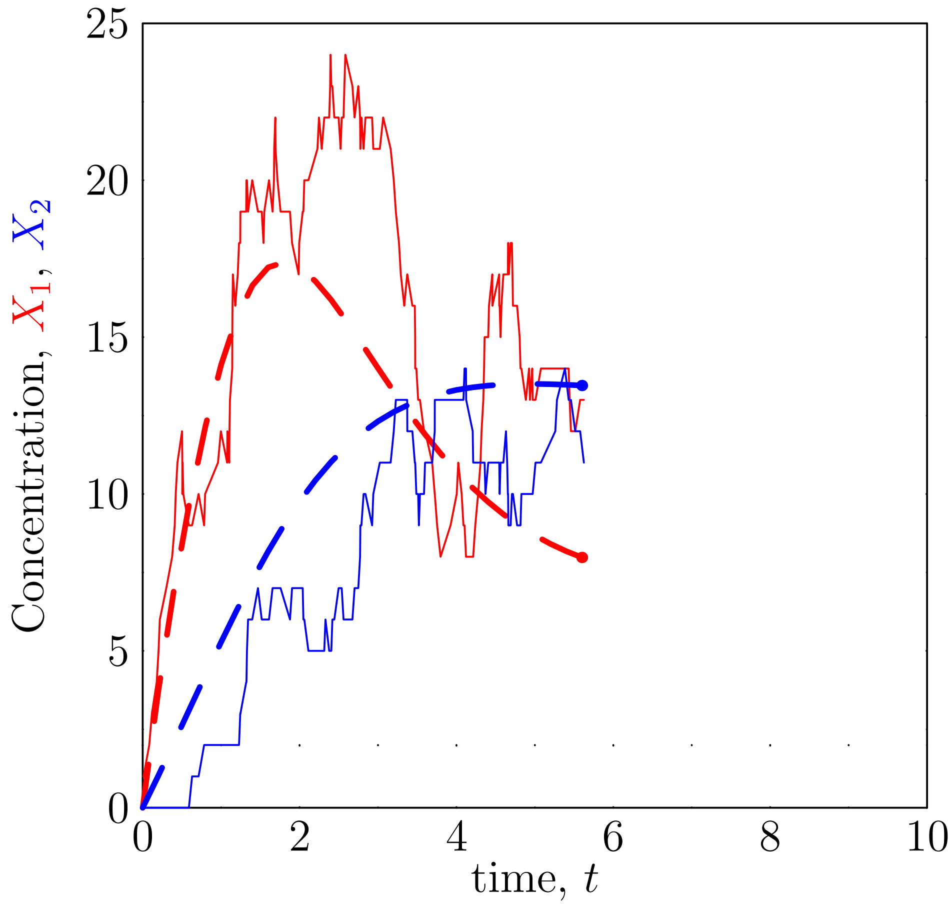

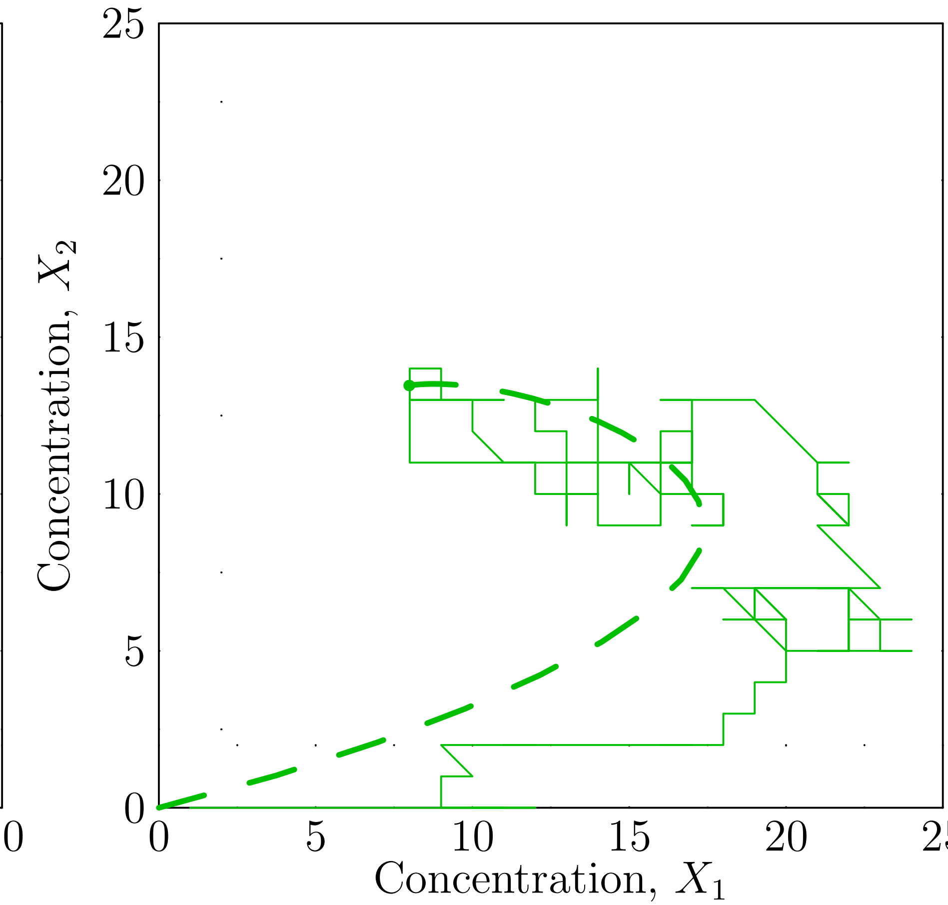

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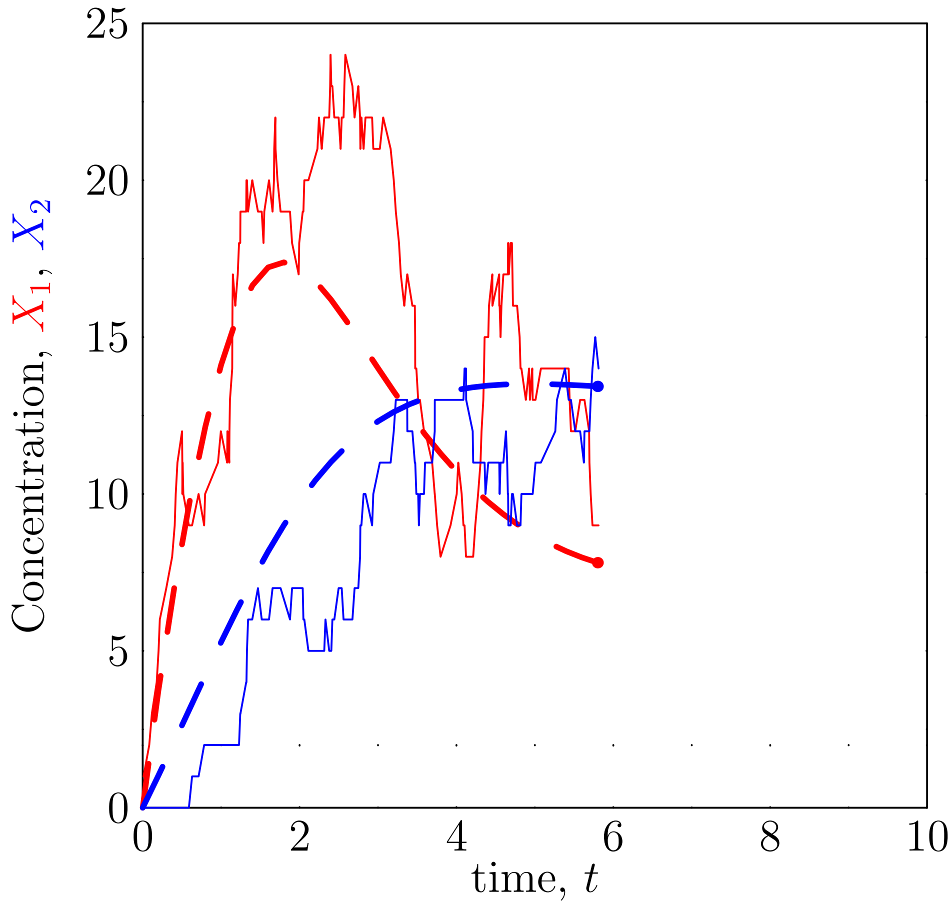

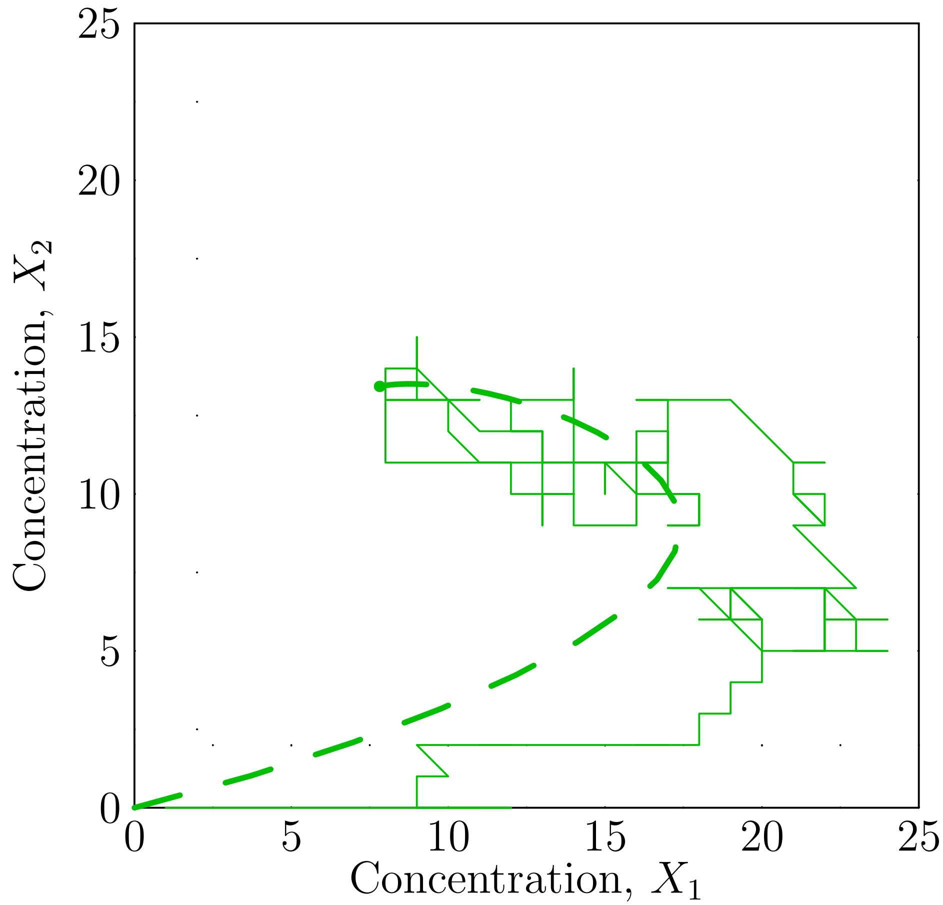

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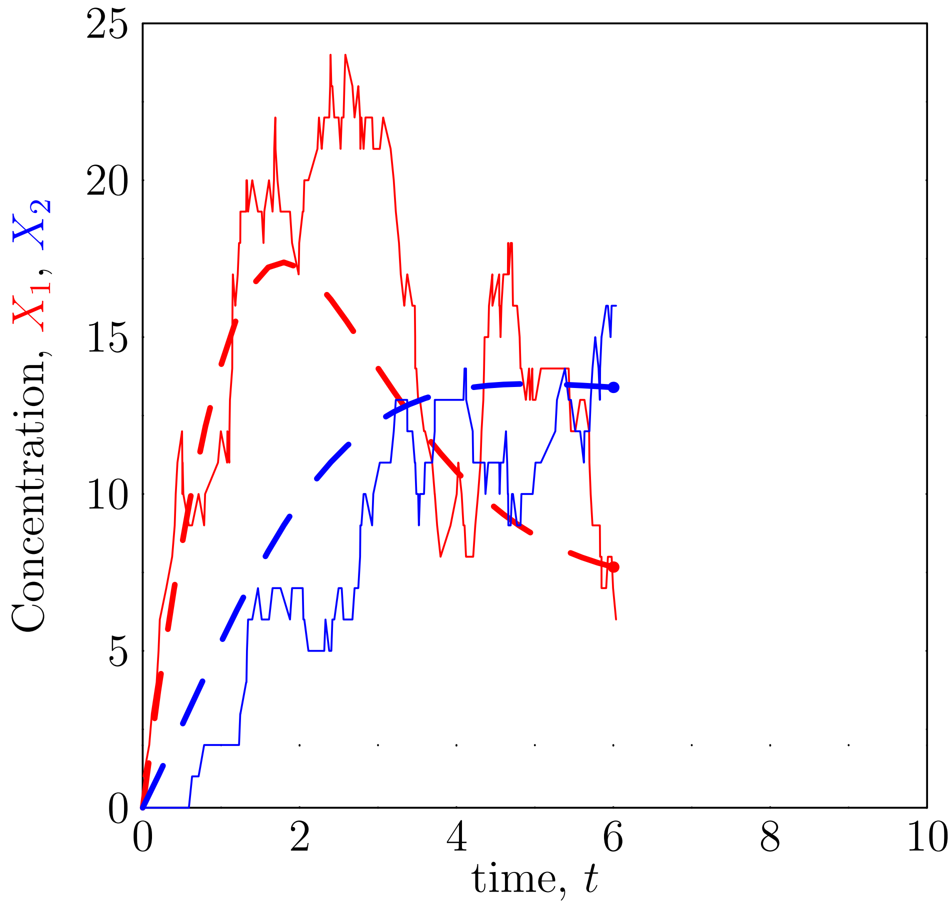

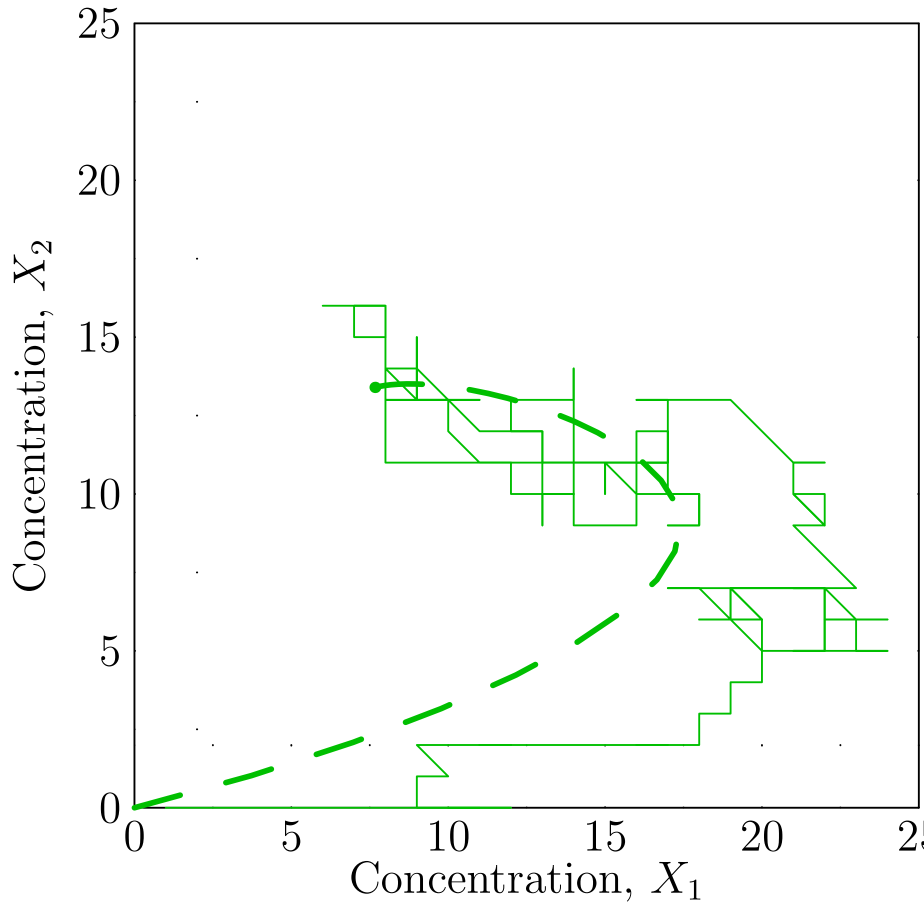

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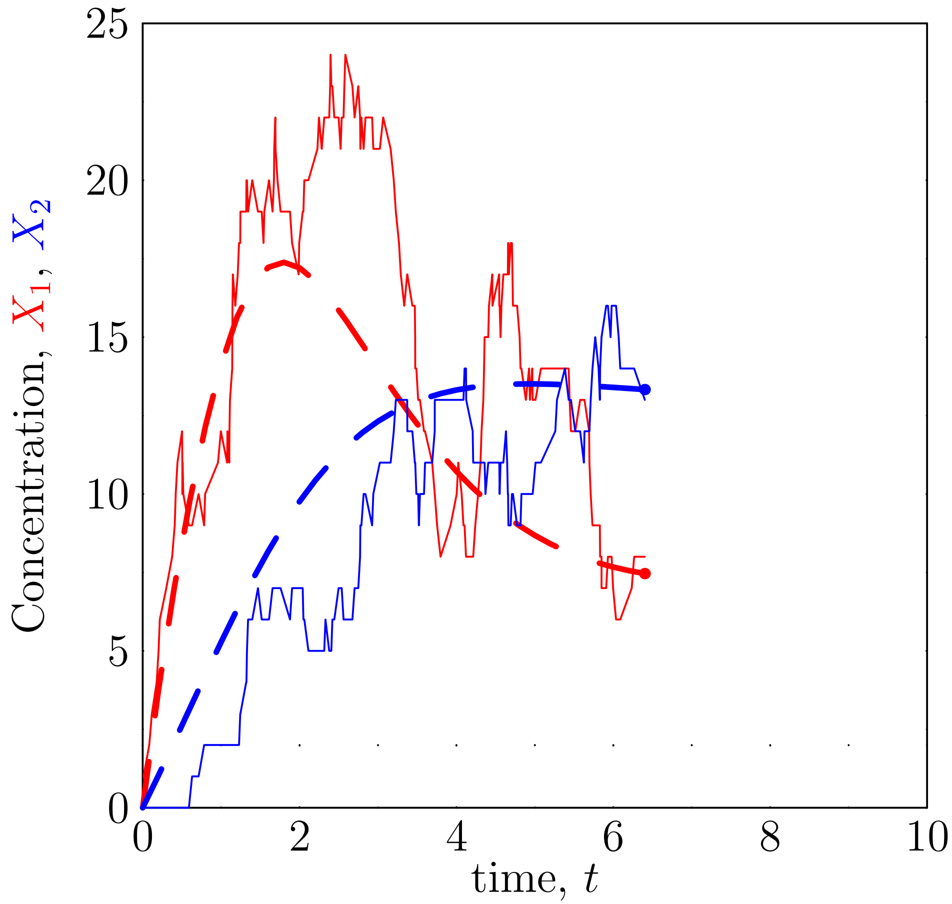

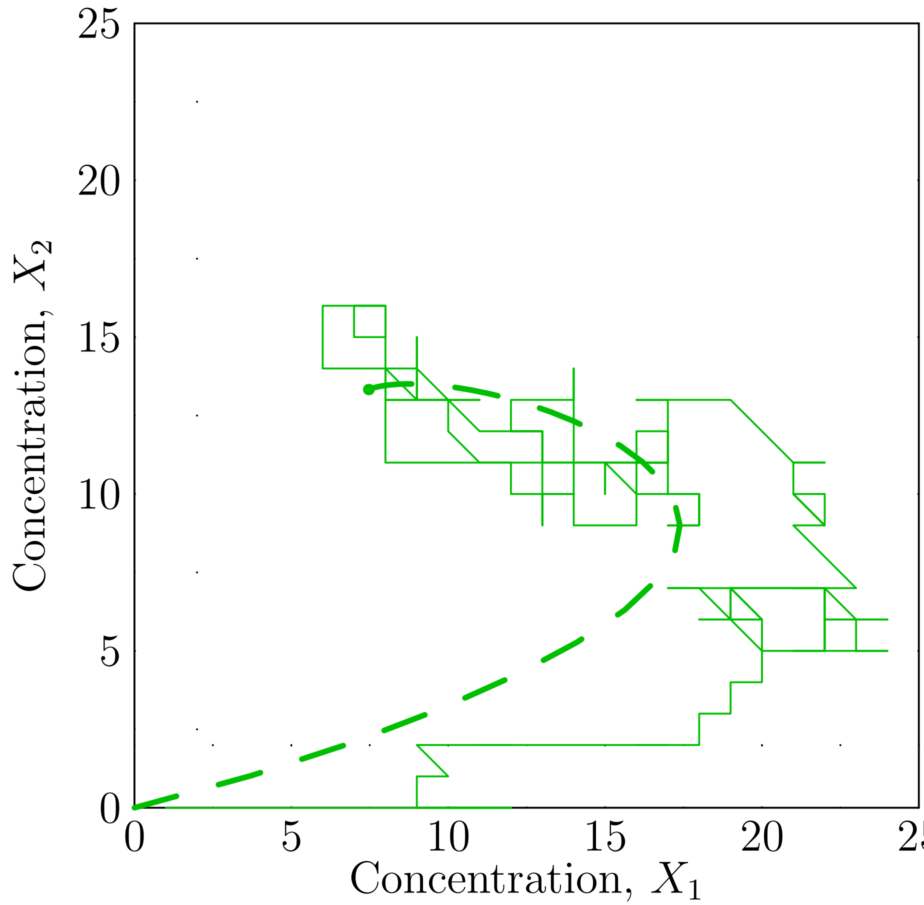

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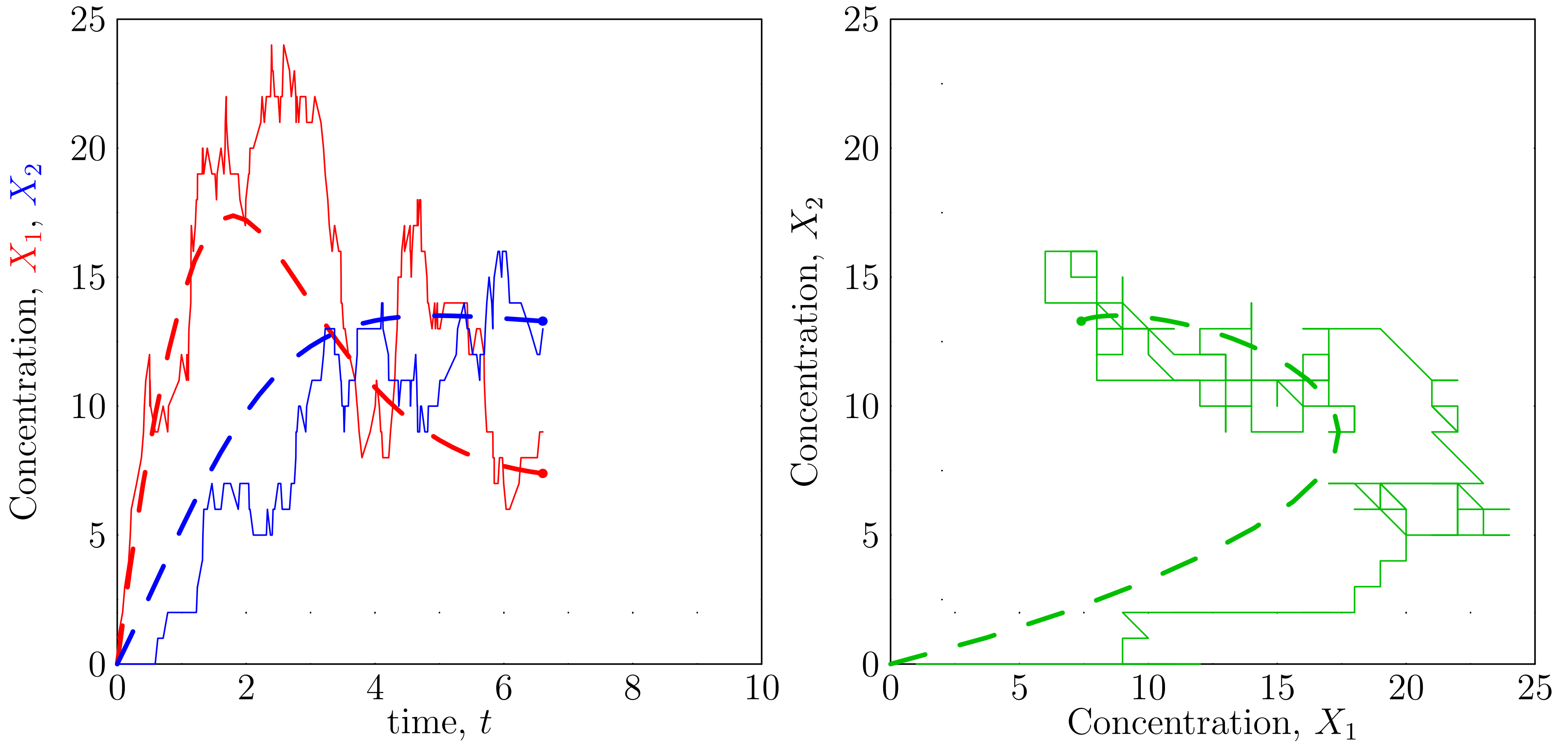

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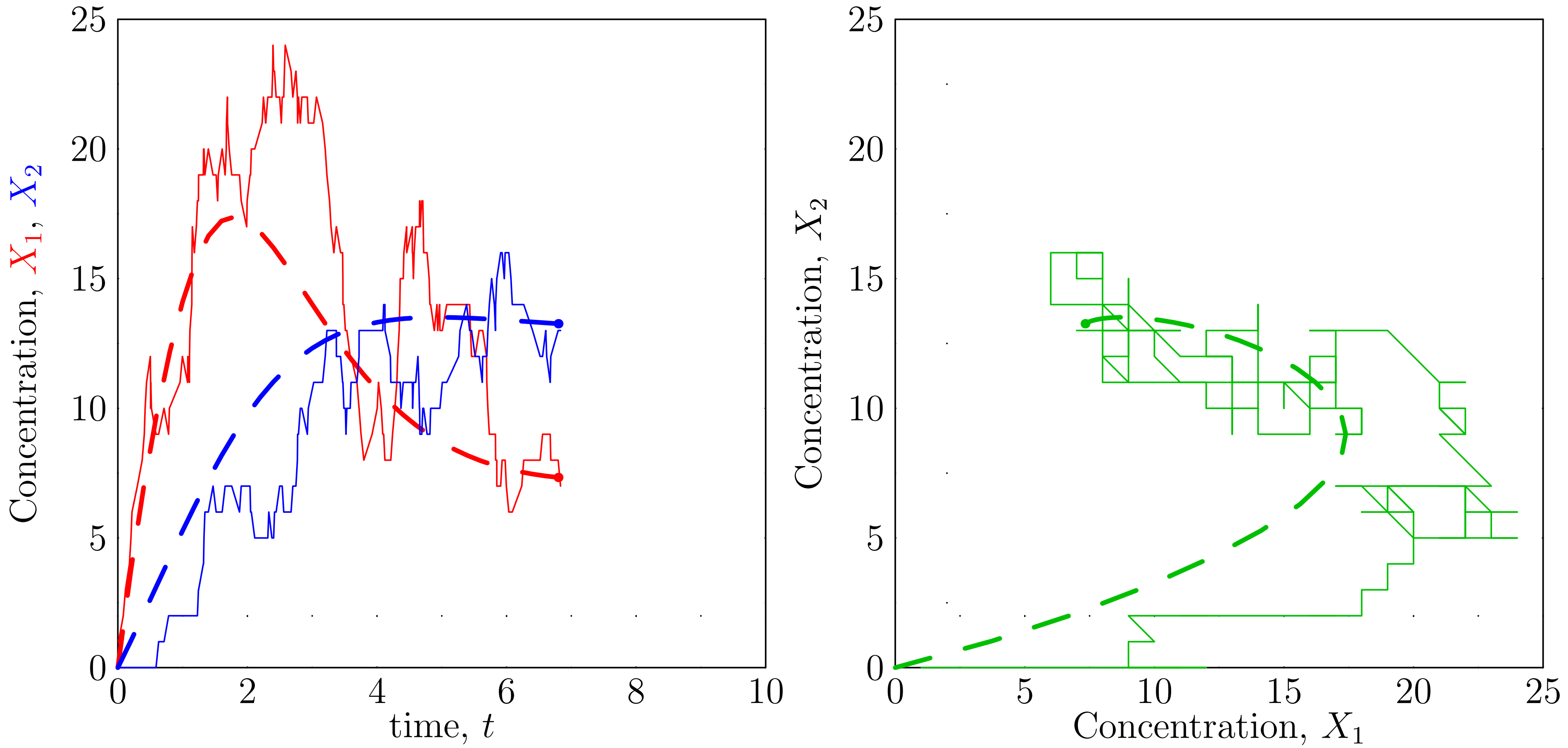

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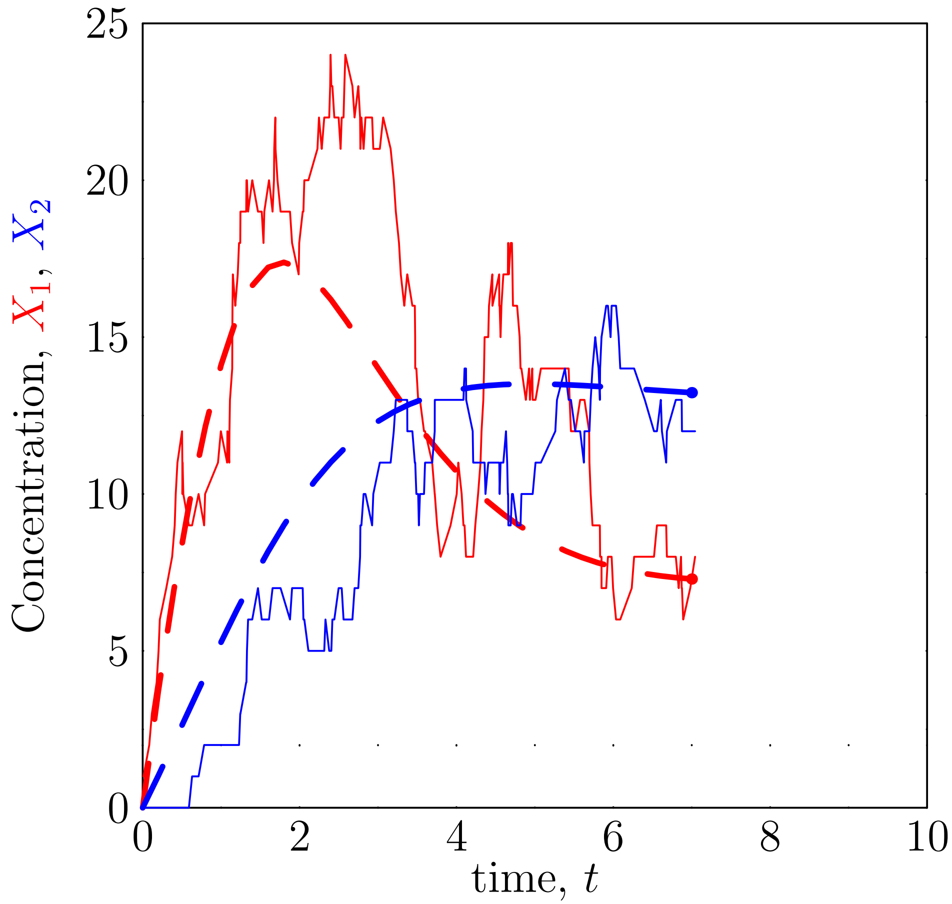

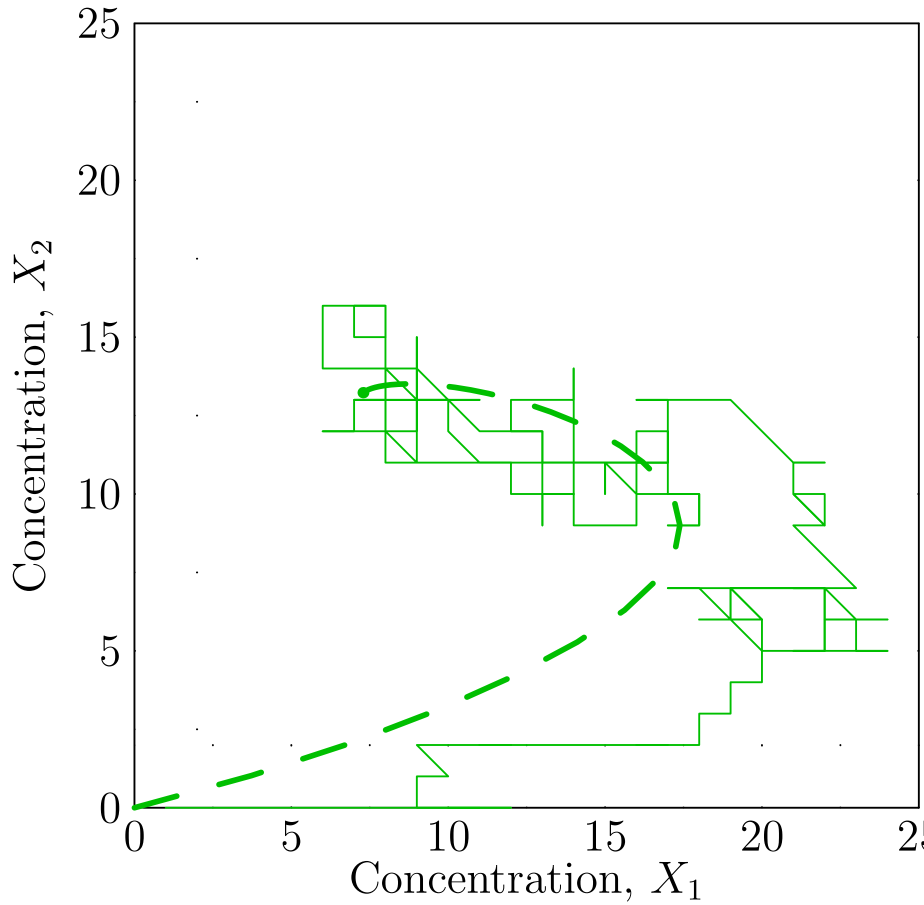

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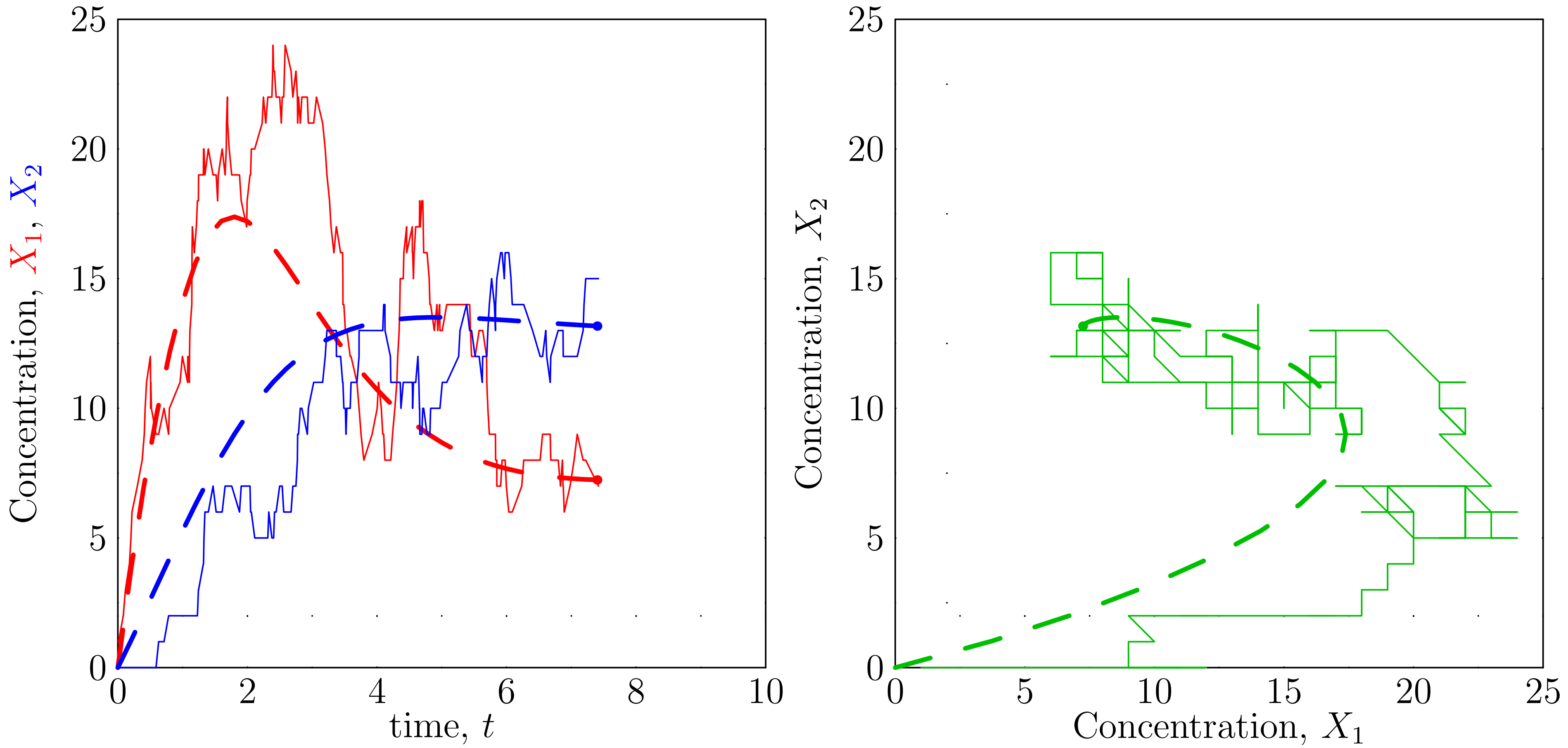

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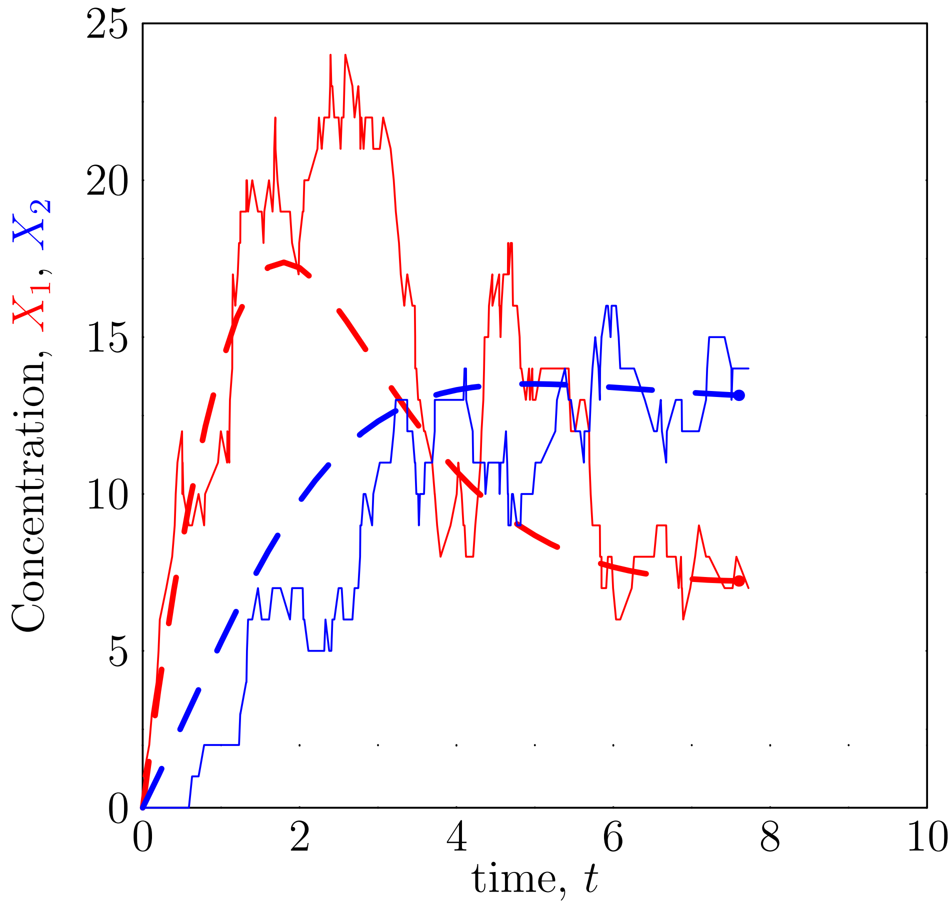

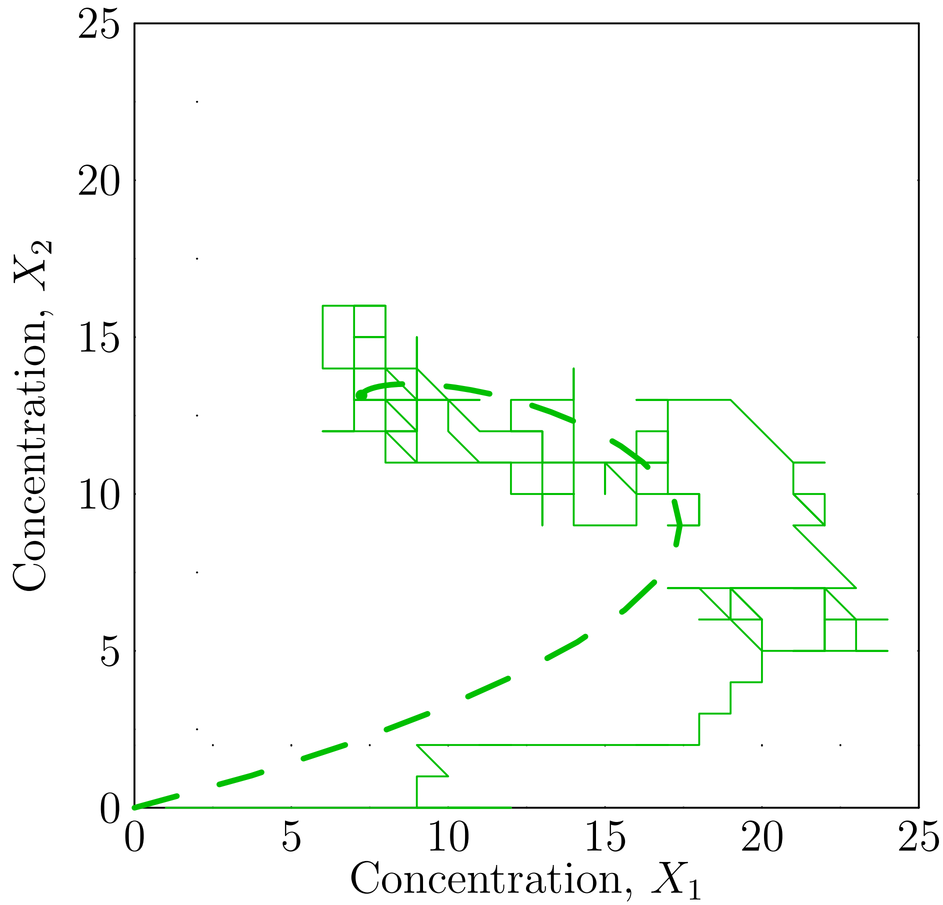

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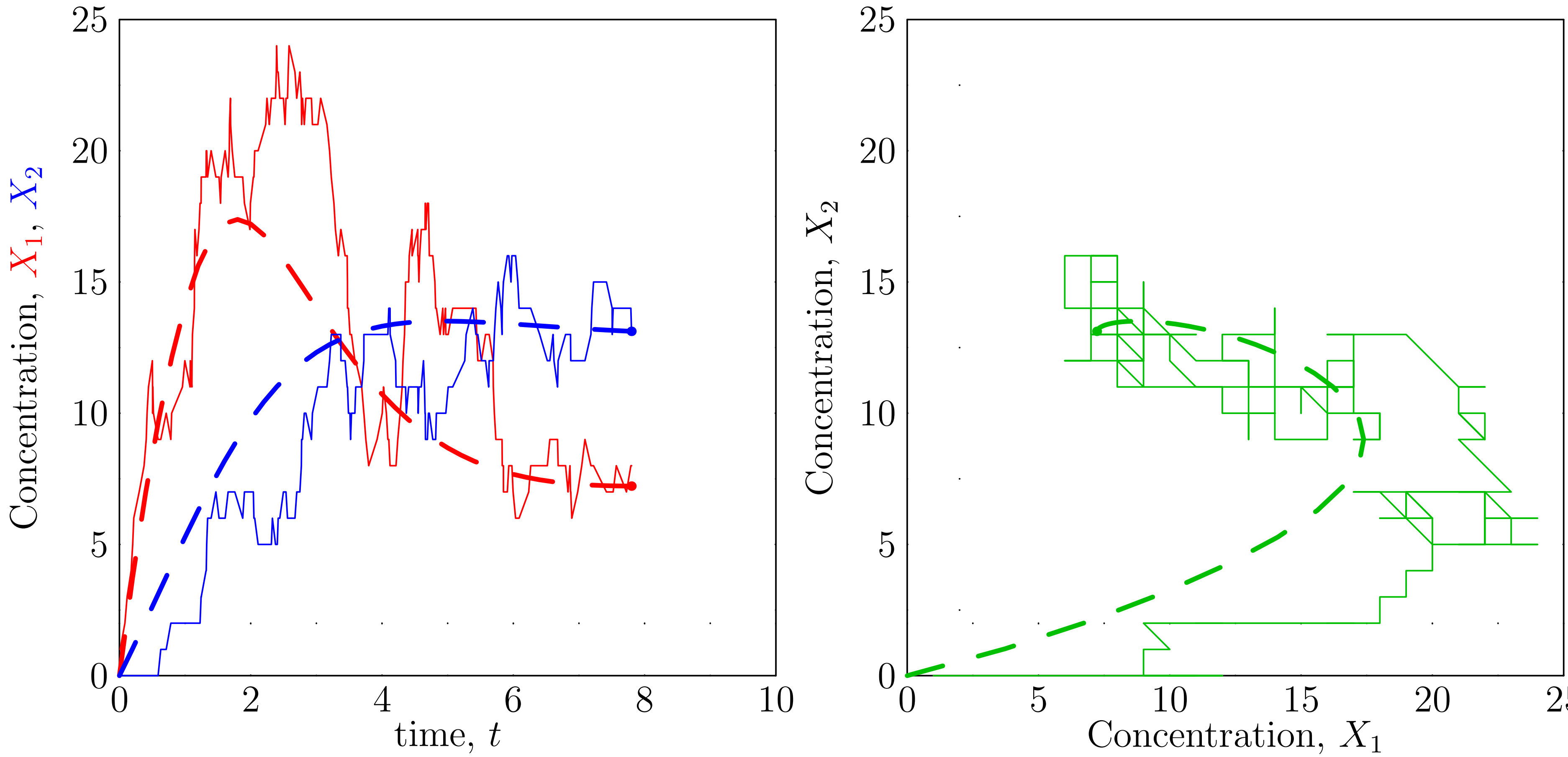

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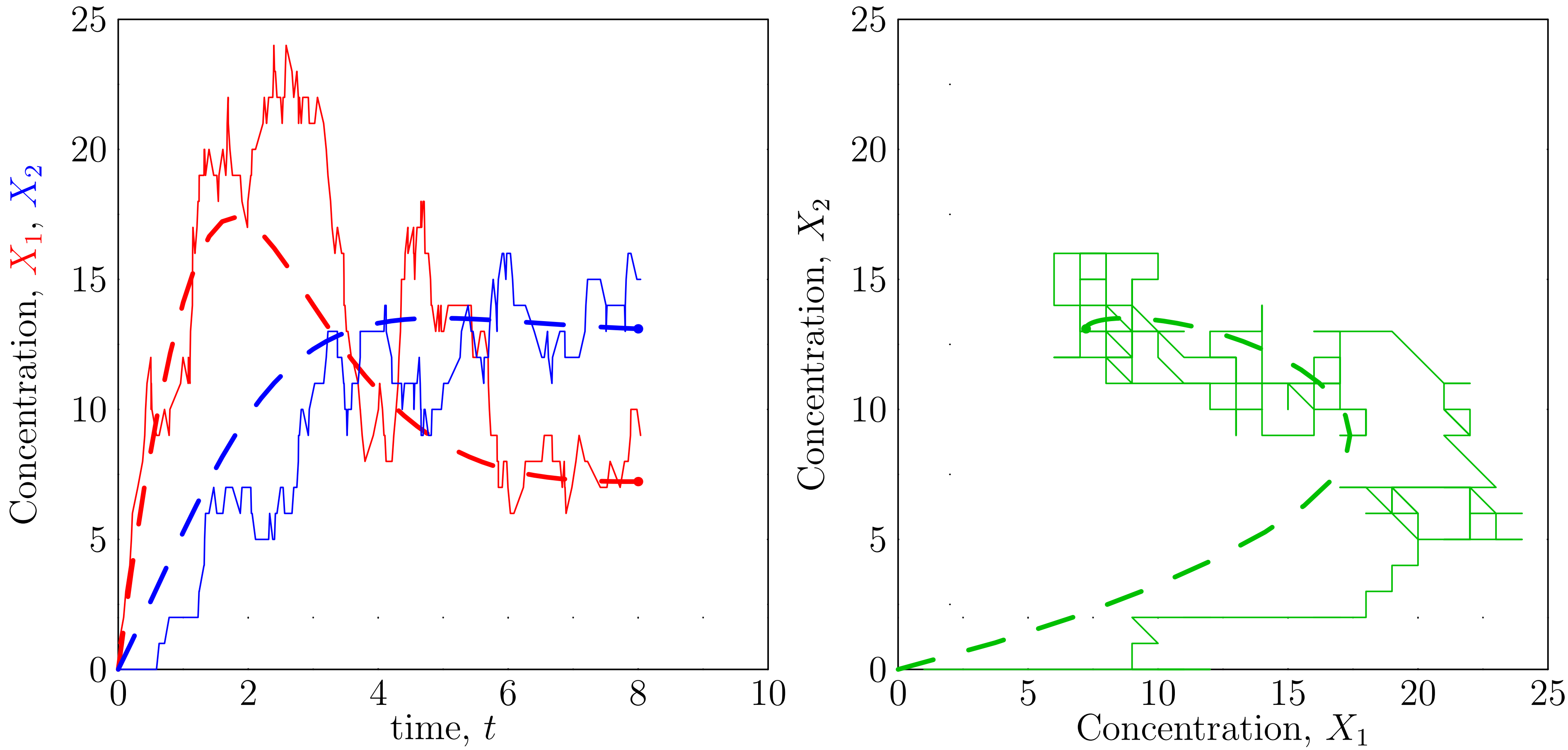

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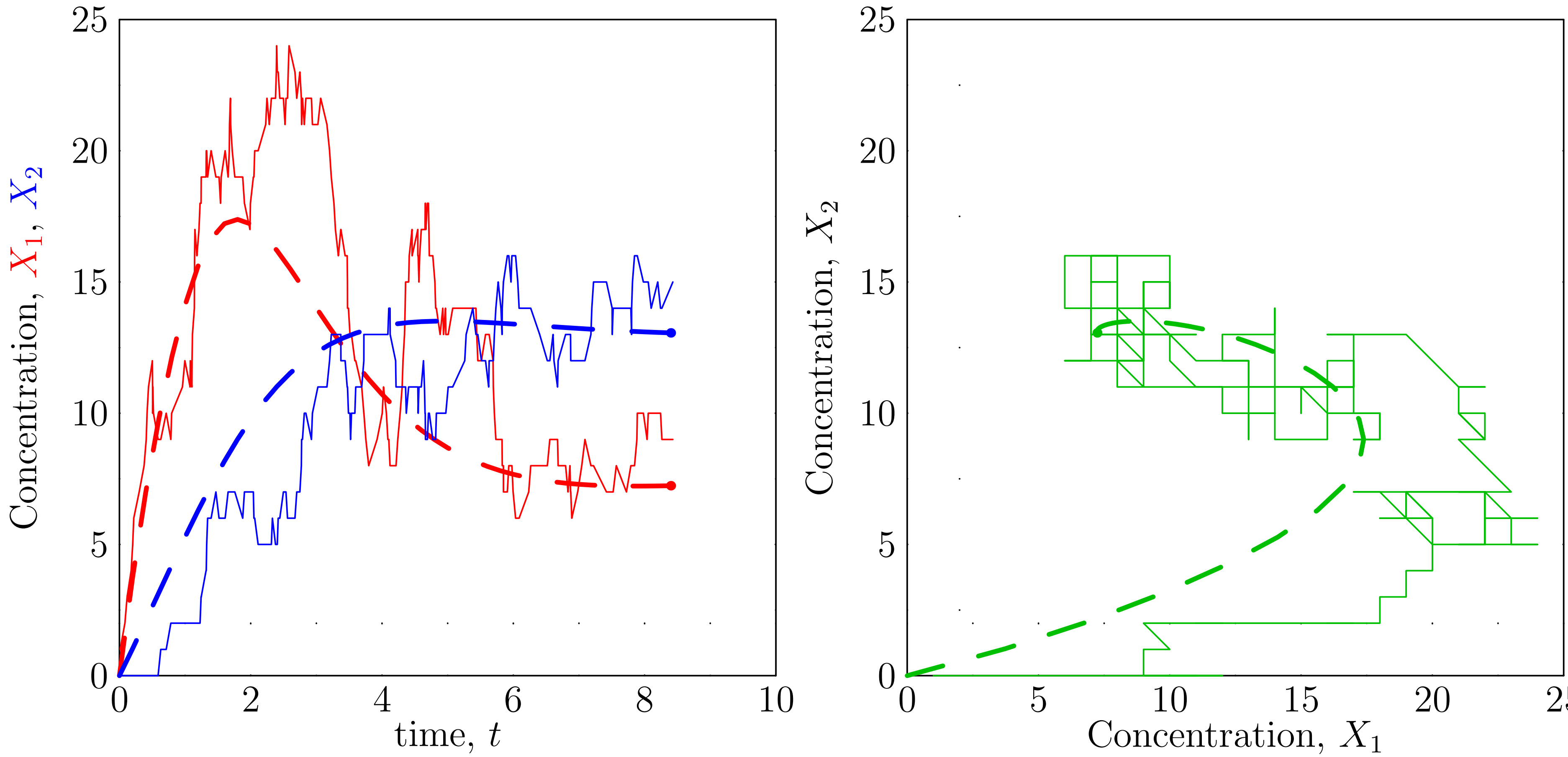

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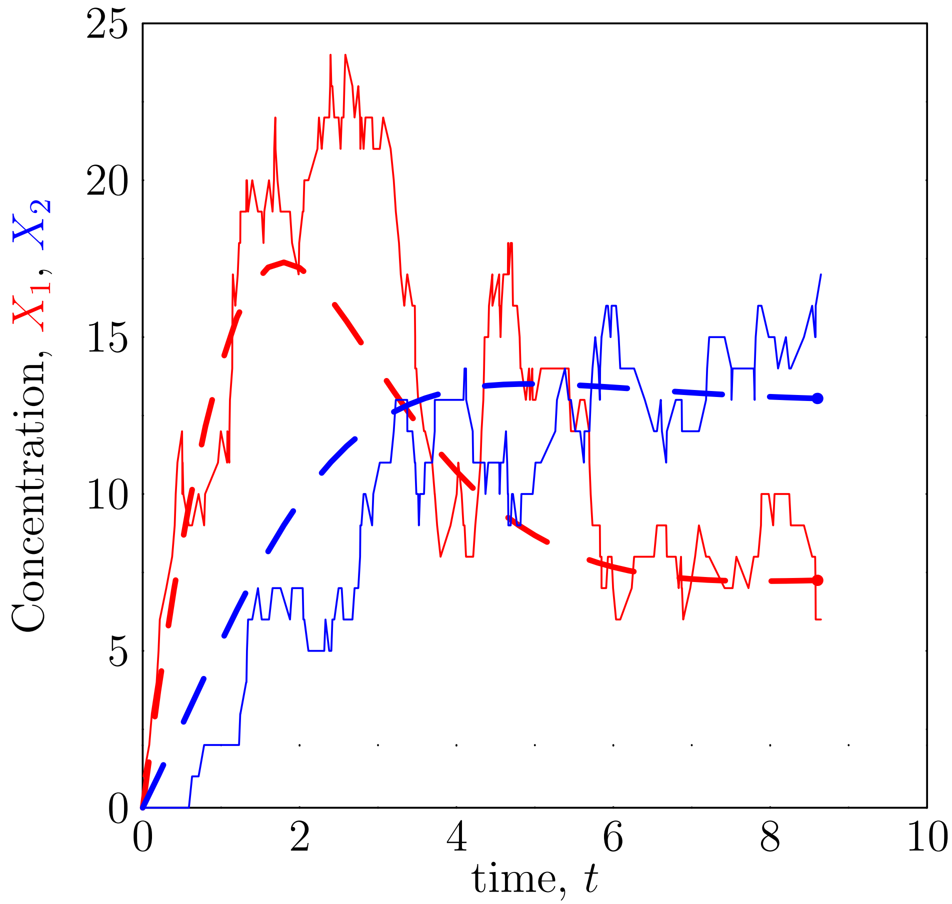

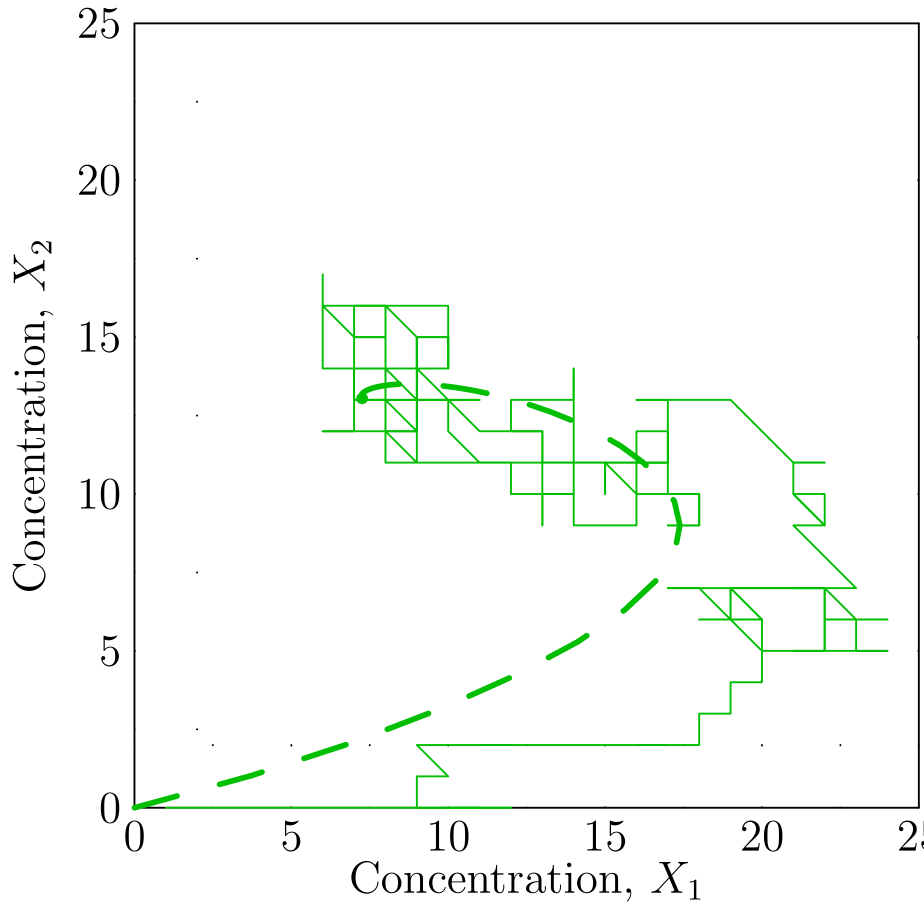

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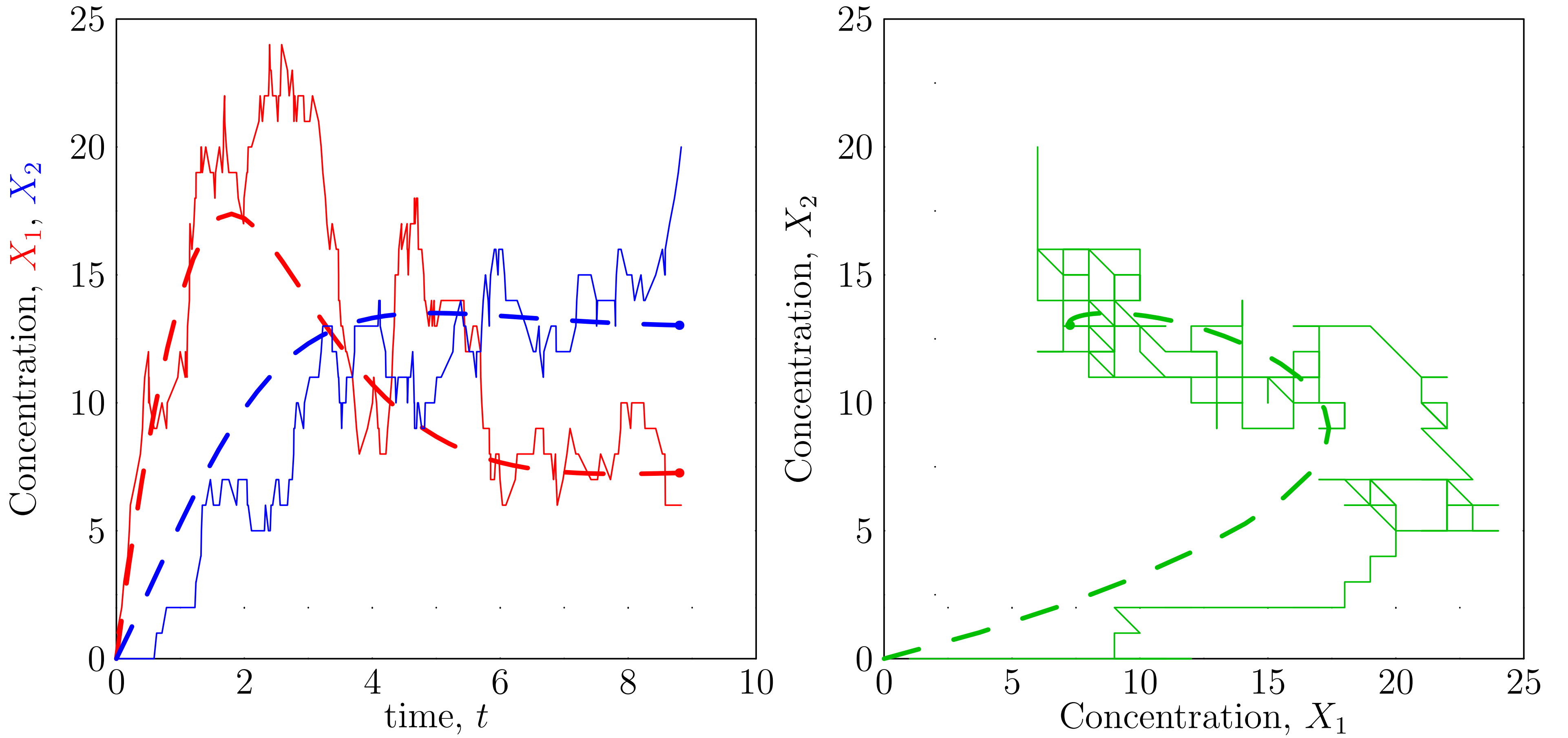

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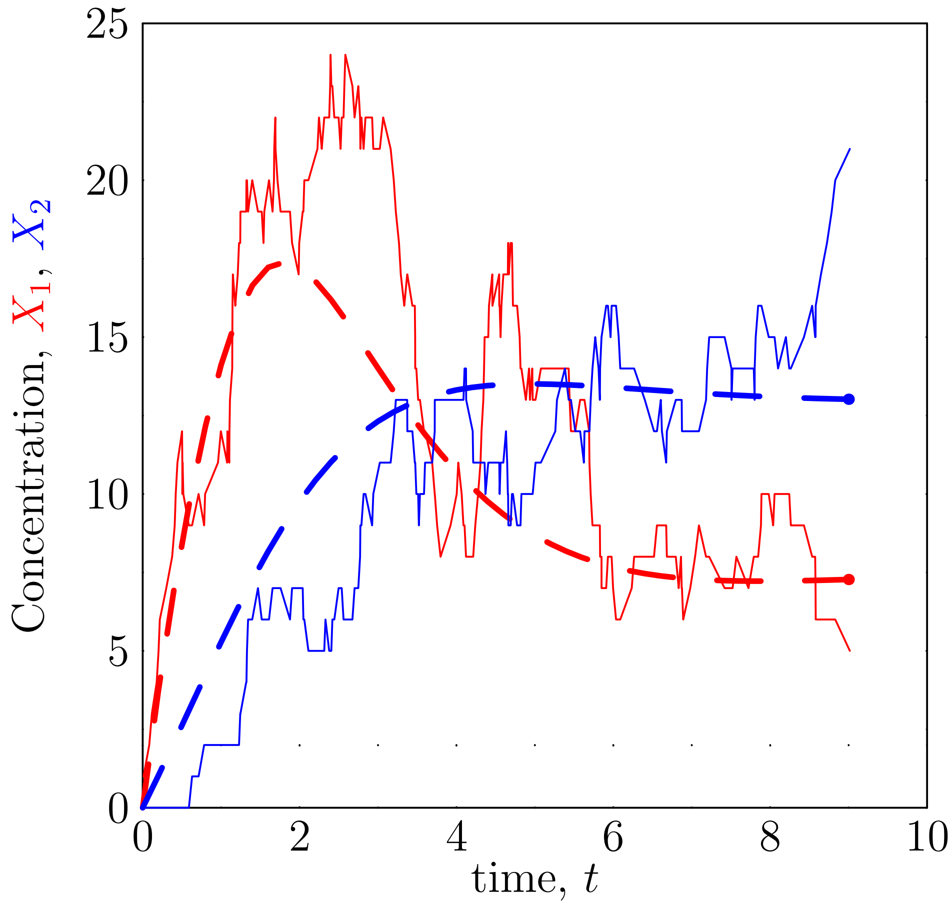

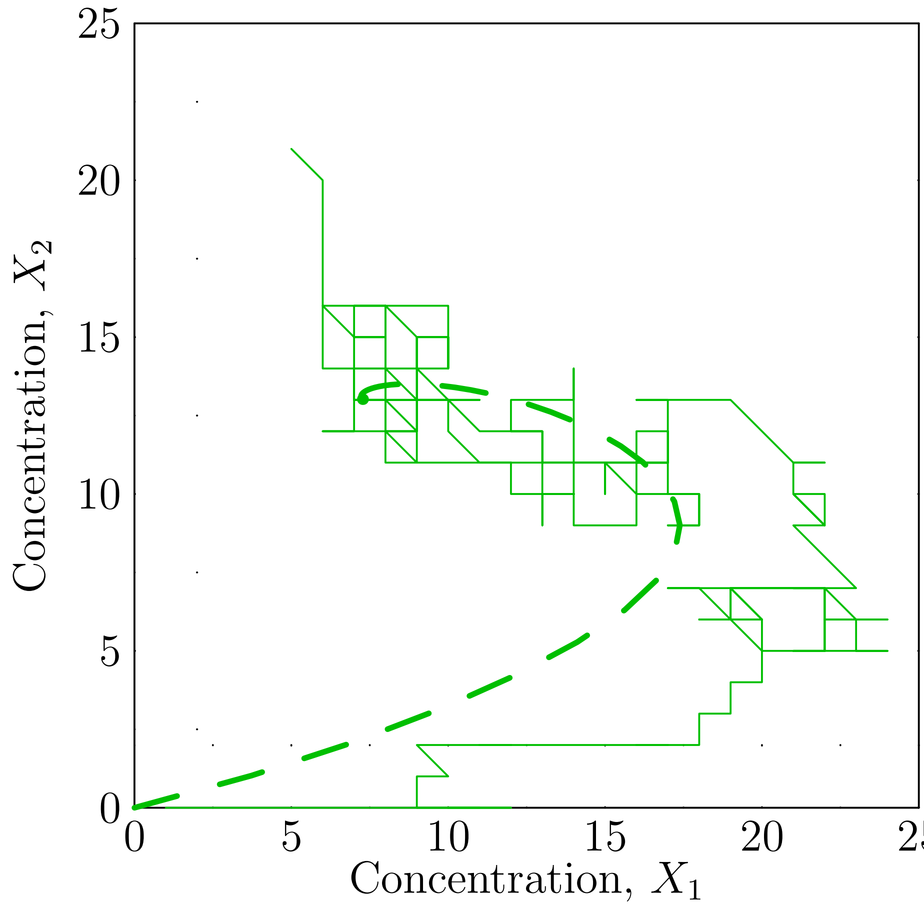

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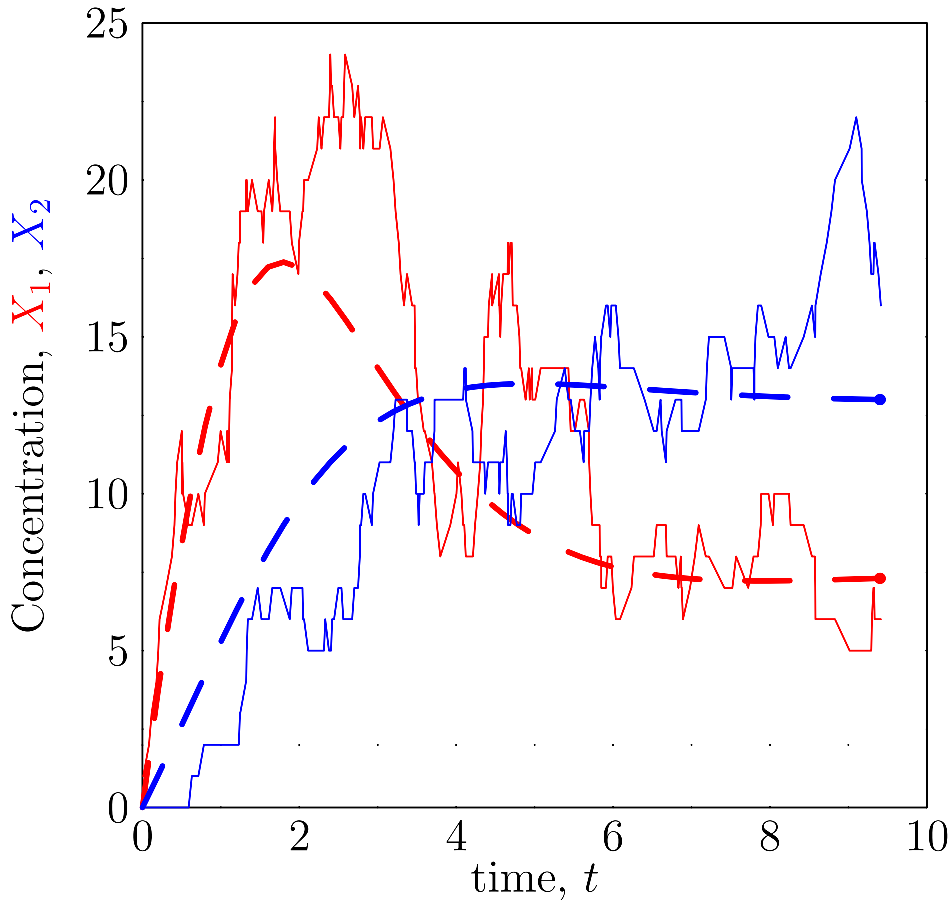

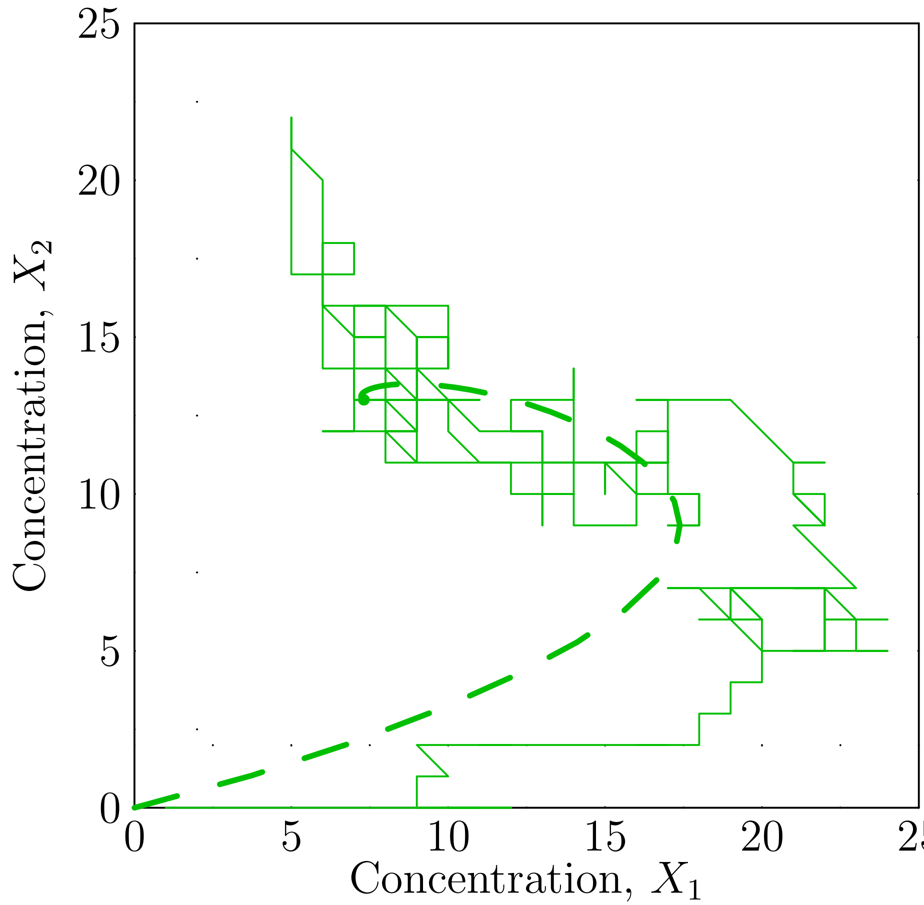

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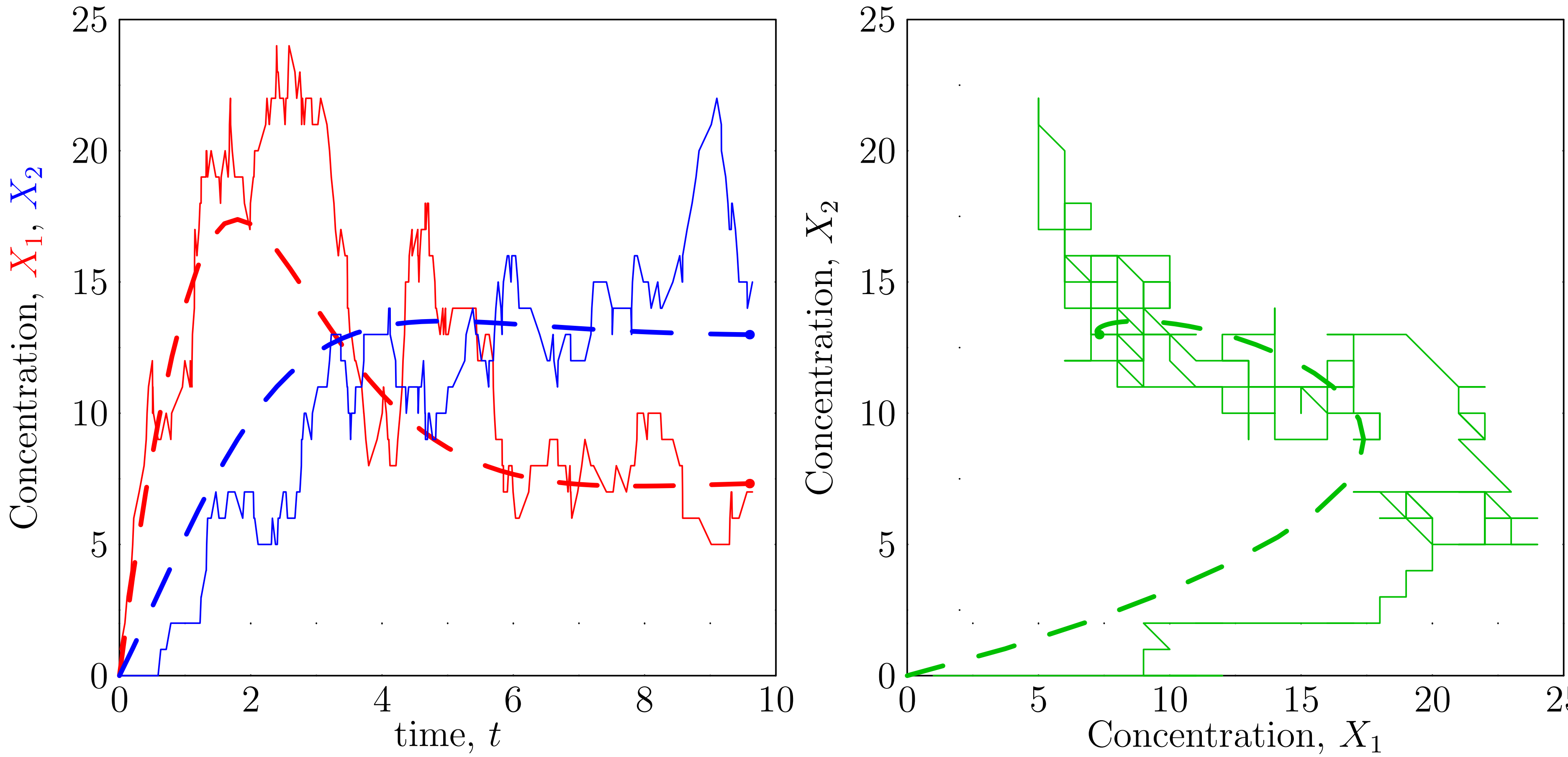

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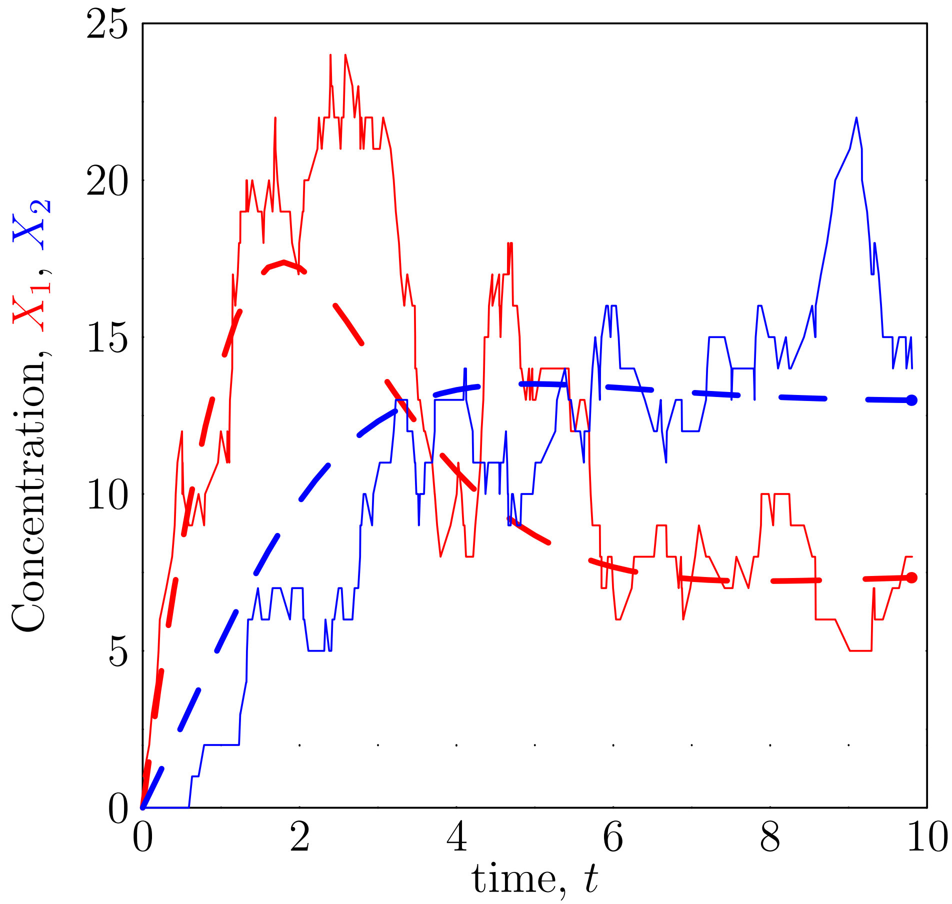

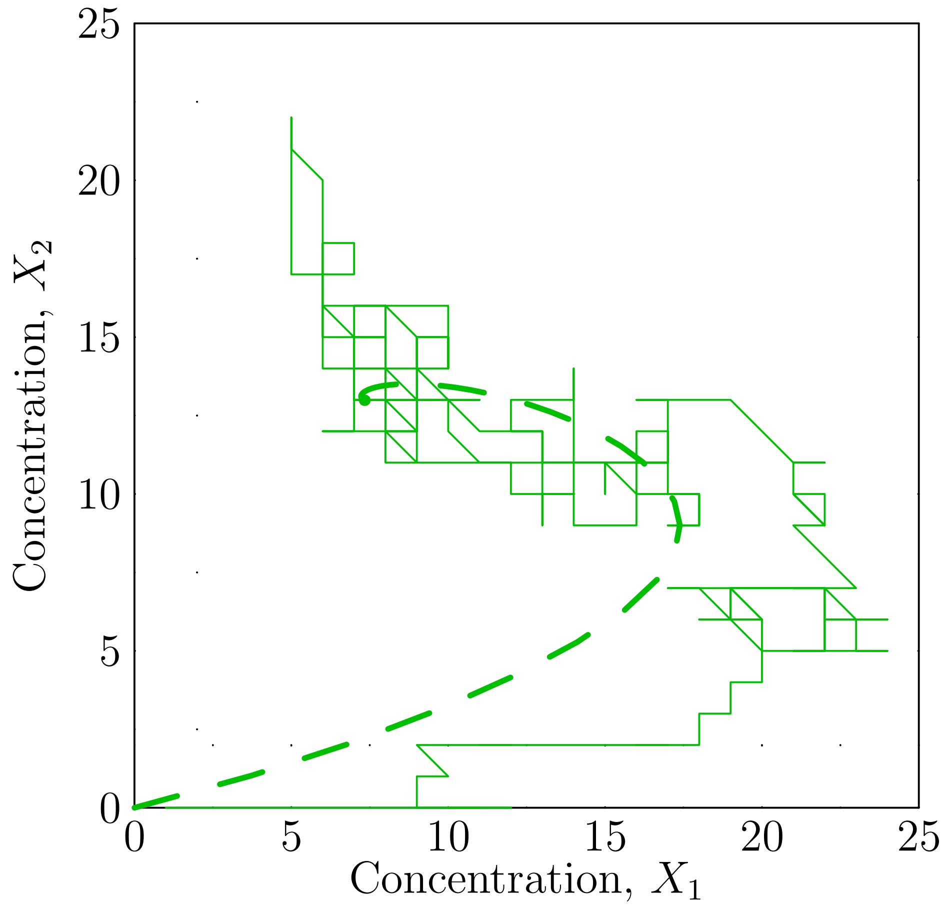

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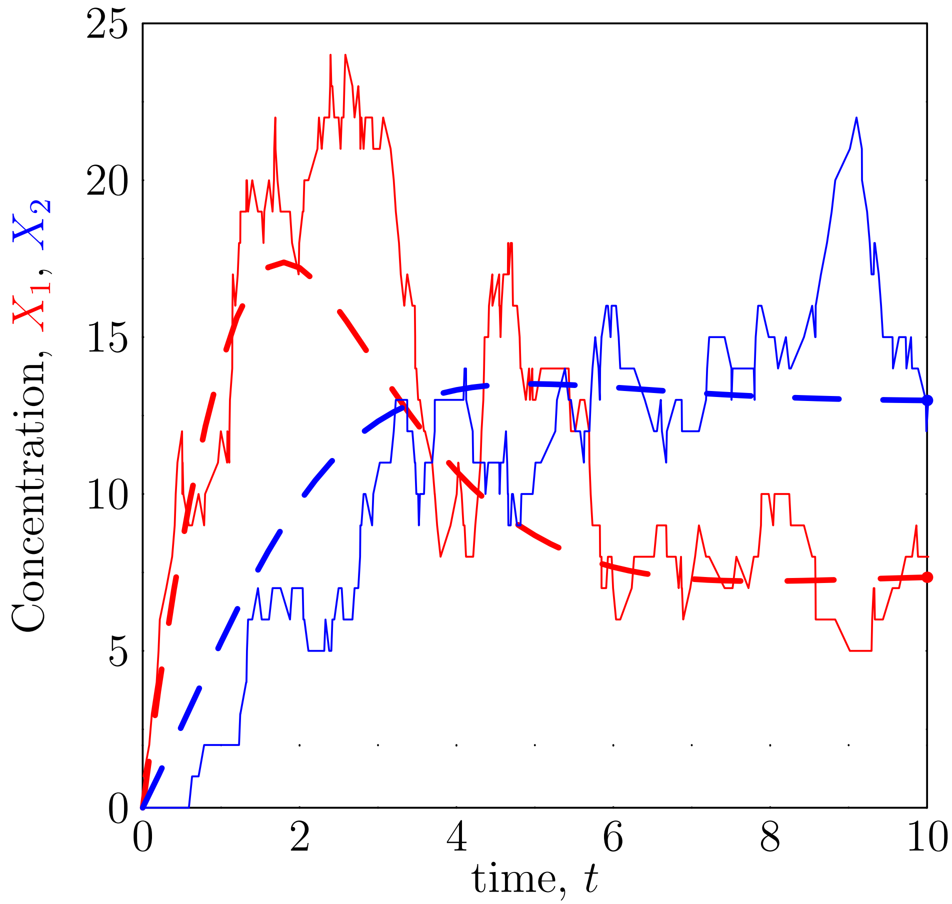

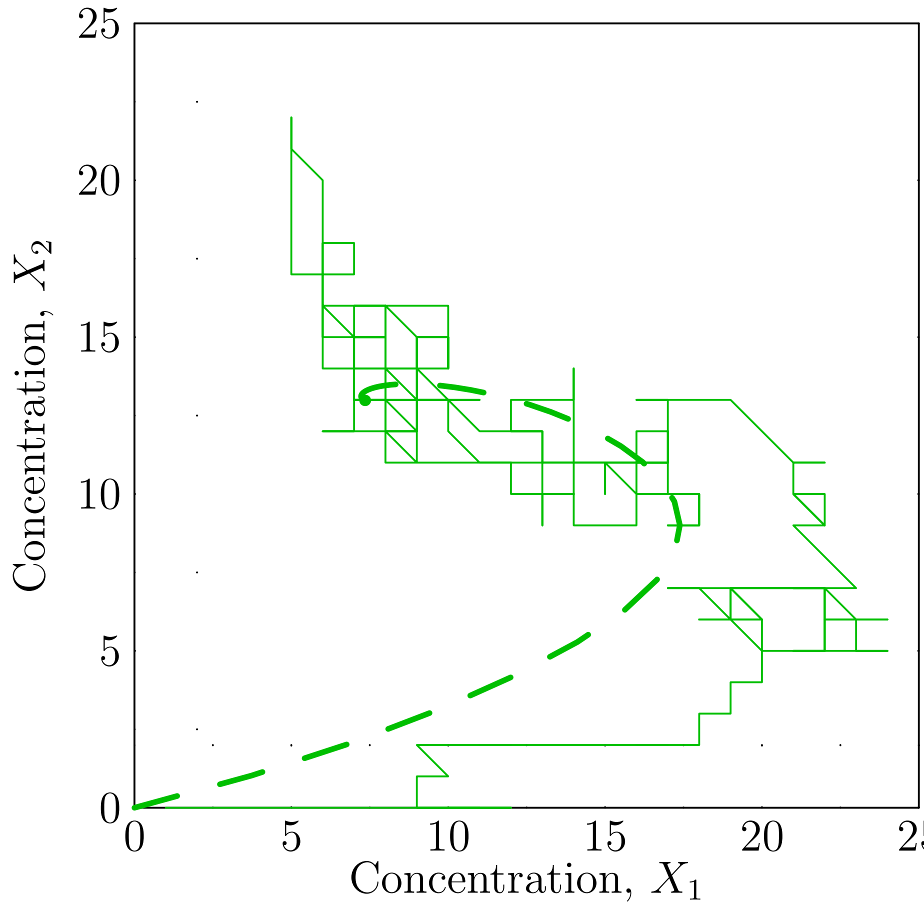

Limitations

- Computers are very fast and can happily simulate millions of steps
- However, some chemical equations can happen millions of time faster than others
- We need to run long enough to capture the slowest processes
- If we are unlucky then this can be just too slow
- We can sometimes cheat and approximate large jumps for fast moving processes
- We can also assume that the fast processes have reached a quasi-equilibrium (plus some noise)

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Using Gillespie Equations

- Gillespie algorithm provides the best approximation to the true dynamics
- Of course, being entirely numerical it doesn't provide much insight into how changes in parameters change the dynamics
- It is often used to check how accurate the ODE description is
- There is a well developed system of stochastic differential equations which bridges the gap between Monte Carlo simulation and ODEs, but it is technically challenging (beyond this course) and it is not easy to apply to these systems (it requires putting in the size of fluctuations which can be very difficult to estimate)

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Using Gillespie Equations

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Conclusions

- Chemical reactions are stochastic
- In small systems with low concentrations this can lead to large fluctuations
- Populations of cell are empirically found to have large fluctuations between individuals
- Often this is used to ensure that different cells do different things
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